



BAZIS-CNC module

User guide

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Contents

Intro	13
BAZIS-CNC module function	13
HELP use	13
Conventions of description.....	13
Technical support	14

Part I.

Documents preparation for control programs creation

Chapter1.

Module general information	16
1.1. Module start	16
1.2. BAZIS-CNC window.....	17
1.3. Librarian auxiliary panel	18
1.4. Interface elements state control	18
1.5. Toolbars arrangement	19
1.6. Desktop control	19
1.7. Skins.....	20
1.8. Document window control.....	20
1.8.1. Window state	20
1.8.2. Automatic windows arrangement.	21
1.9. BAZIS-CNC module workspace	21
1.9.1. Document types	21
1.9.2. Measurement units	21

Chapter2.

Module setup	22
2.1. General information	22

2.2.	Settings saving and restore.....	22
2.2.1.	BAZIS system settings saving	22
2.2.2.	The module current configuration saving.....	23
2.2.3.	Settings restore	24

Chapter3.

	Operations with documents	25
3.1.	Document opening	25
3.1.1.	Model from file opening	25
3.1.2.	Model from document library opening	25
3.2.	Document closing.....	26
3.2.1.	Drawings import in DXF format	26
3.2.2.	Models import in bCAD and K3 formats.....	27
3.3.	Documents closing.....	27

Chapter4.

	3D models use for control programs creation	28
4.1.	Model visualization control	28
4.1.1.	Model orientation	28
4.1.2.	Model rotation and move	28
4.1.3.	Model visualization modes	29
4.1.4.	Zooming.....	30
4.2.	Panels selection for processing.....	30
4.2.1.	General information	30
4.2.2.	Objects selection modes	30

Chapter5.

	Drawings use for control programs generation	33
5.1.	Drawing execution demands	33
5.1.1.	General information	33
5.1.2.	Layers properties	33

Chapter6.

Data preparation for control programs creation	37
6.1. General information	37
6.2. Drawing data preparation	37
6.2.1. Data types to layer name prefix mapping	37
6.2.2. Separation symbol of decimal part	38
6.2.3. Setup completion	38
6.3. Data from a model preparation	38

Part II.**Control programs setup****Chapter7.**

Preview for CNC window	40
7.1. General information	40
7.2. Control programs general information.....	40
7.2.1. Order name	40
7.2.2. Product count	40
7.2.3. Configuration choice	40
7.3. Panel list	40
7.4. Postprocessors list.....	41
7.5. Panels list from a file opening.....	42
7.6. Work with project	42
7.7. Import in DXF format	43

Chapter8.

Panels list editing.....	44
8.1. Panel to control program appending	44
8.2. List visualization	44
8.3. Holes editing.....	45
8.3.1. Hole adding	45

8.3.2.	Holes editing	45
8.3.3.	Holes deletion.....	45
8.3.4.	Holes editing completion	45
8.4.	Contour milling control	46
8.5.	Slots editing.....	46
8.6.	Mirrored view.....	48
8.7.	Panel rotation	48
8.8.	Panels in list visualization	48
8.8.1.	General information	48
8.8.2.	Edge banding visualization	48
8.8.3.	Holes and slots line colour setup	49
8.8.4.	Parameters saving and loading	50
8.8.5.	Setup completion	50

Chapter9.

	Configuration parameters setup.....	51
9.1.	Configuration common parameters	51
9.1.1.	Configuration attributes	51
9.1.2.	Program files path	51
9.1.3.	File name template	52
9.2.	Parameters saving and loading	53
9.3.	Data reading setup.....	53
9.3.1.	Common parameters	53
9.3.2.	Milling parameters	56
9.3.3.	Hole parameters	56
9.3.4.	Banding accounting	58
9.3.5.	Slots analysis	59
9.4.	Check operations realizability.....	60
9.4.1.	General information	60
9.4.2.	Checked operations	61
9.4.3.	Checked dimensions	61
9.4.4.	Reference point position.....	62
9.5.	Panel image saving parameters.....	62

Chapter10.

Process-dependent parameters setup.....	63
10.1. General information	63
10.2. Saved parameters loading	63
10.3. AES Group.....	63
10.3.1. AES Raptor.....	63
10.3.2. AES Drill Macro	70
10.3.3. AES Sirius	74
10.4. Artisan	78
10.4.1. Artisan (*.tnc, *.tn2).....	78
10.5. Beaver (Excitech)	84
10.5.1. Camdrill-Swift	84
10.5.2. OSAI Control	91
10.5.3. OSAI Control - NestConcept(E4)Pre-Labeling	97
10.5.4. OSAI Control Optimal(E3)	104
10.5.5. OSAI Control - NestConcept(E4) Syntec Pre-Labeling.....	111
10.5.6. OSAI Control Universal (E6).....	118
10.5.7. Swift HSD.....	125
10.5.8. Swift Turbo (EH, EHS, EHSW)	127
10.5.9. Syntech	134
10.5.10.TPA	141
10.5.11.VIC Control	151
10.5.12.Weihong - i-Nest, NestConcept	157
10.6. Biesse Group	163
10.6.1. BiesseWorks (*.bpp)	163
10.6.2. NC500	171
10.6.3. Biesse XNC.....	178
10.6.4. bSolid CIX	185
10.7. BJD	191
10.7.1. HiWCAM	191
10.7.2. BJD_Syntec	197
10.8. Blue Elephant.....	203
10.8.1. Blue Elephant	203

10.9. Busellato.....	205
10.9.1. Genesis Evolution.....	205
10.9.2. Wave (*.).....	212
10.10. CAD System.....	221
10.10.1.CAD System.....	221
10.11. Felder Group	223
10.11.1.F4Integrate.....	223
10.11.2.WoodFlash.....	230
10.12. Filato	240
10.12.1.ARIX-OpenCNC (*.tnc, *.tn2).....	240
10.12.2.Filato DS3.....	241
10.12.3.Filato Next	243
10.12.4.Higerman	244
10.12.5.Filato SV.....	251
10.12.6.Filato Syntec.....	257
10.12.7.Filato NC Studio	265
10.12.8.Filato RS Lab	270
10.13. Gannomat.....	277
10.13.1.GannoMat Editor (*.ascii)	277
10.14. Giben	280
10.14.1.WoodCam	280
10.15. Gorelovsky System.....	283
10.16. Griggio.....	290
10.16.1.G-Cad Lepton.....	290
10.16.2.Griggio TPA	291
10.17. Hirzt	300
10.17.1.Hirzt TPA	300
10.17.2.iLENIA	310
10.18. HOLD	312
10.18.1.CNC Drill	312
10.18.2.HOLD LNC	319
10.19. HOLZ-HER.....	325
10.19.1.Campus Hops (*.hop)	325

10.19.2.TwinCam	332
10.20. Holztek	334
10.20.1.Horizontal Drill	334
10.21. HOMAG Group	336
10.21.1.WoodWop	336
10.22. Huhua	347
10.22.1.PCDrillCAM	347
10.23. IMA	353
10.23.1.ImaWop 4.0 (*.fmc)	353
10.24. KDT	360
10.24.1.KDT Delta (*.tnc, *.tn2)	360
10.24.2.KDT Syntec	367
10.24.3.KDT TPA	374
10.24.4. CNC Drill	383
10.24.5.KDT Horizontal Drill	390
10.25. KOCH	391
10.25.1.NCStudio	391
10.26. Maggi	392
10.26.1.Maggi ICE 101	392
10.26.2.Pegasus	393
10.27. Masterwood	396
10.27.1.MW208	396
10.27.2. MW310 (*.dxf) and MW310 (*.tlf)	399
10.28. Multicam	406
10.28.1.MultiCam (*.nc)	406
10.29. Nanxing	411
10.29.1.NCB	411
10.29.2.NDC	414
10.29.3.Driller Master	415
10.29.4.Nanxing FinChinaSoft	418
10.29.5.PCDrillCAM	425
10.29.6.TPA	431

10.30. Omaksan	441
10.30.1.Omaksan Works	441
10.31. PMSK	447
10.31.1.PT2	447
10.31.2.PMSK PT6	450
10.31.3.PMSK Syntec	457
10.32. ROCTECH	464
10.32.1.Syntec	464
10.33. TechnoCNC	470
10.33.1.ATV Syntec	470
10.33.2.Suppressor Syntec	477
10.33.3.TechnoBit	484
10.34. Sandar	491
10.34.1.Sandar PA CNC	491
10.35. SCM Group	493
10.35.1.SCM H800	493
10.35.2.SCM Startech CN	494
10.35.3.SCM Startech CN V	495
10.35.4.Xilog Maestro	498
10.35.5.Xilog Plus	506
10.36. UNITeam	515
10.36.1.Albatros Cad	515
10.37. Velberg	522
10.37.1.RMZ-5D	522
10.37.2.Velberg Syntec	525
10.37.3.Weihong NCStudio	532
10.38. Vitap	538
10.38.1.Vitap TPA	538
10.38.2.Vitap Bridge ICE112	547
10.38.3.Vitap Point	548
10.39. Yiming	551
10.39.1.Syntec	551
10.40. ZaiTec	557

10.40.1.ZaiTec Delta	557
10.40.2.Zaitec TPA	564
10.40.3.Zaitec Trepan	574
10.40.4.Weihong	581
10.41. Artel	587
10.41.1.DeskCNC (*.dnc)	587
10.42. Research and Production Company «Semil»	594
10.42.1.WinKam	594
 Chapter11.	
Control program files creation	597
 Appendix I.	
Terms and definitions	599
 Index	611

Intro

BAZIS-CNC module function

BAZIS-CNC module is a part of BAZIS system and it allows to create control programs for panels processing on boring and milling CNC machines. The models can be saved in files of 3D models, fragments, sheets, libraries of BAZIS system. The module can be used as a stand-alone application and run from BAZIS-Woodworker.

HELP use

Adobe Reader window control elements – a free program for PDF documents viewing, allows to use different modes of document access.

Tab tab contains a structured list of document sections.

Editing – Find command allows to find a text string in the current document. You can also use `<Ctrl>+<F>` key combination for this purpose. For next string entry push `<F3>` button.

Editing – Search command allows to fulfill words advanced search.

On-line document contains hypertext links. They include **Tab** tab sections names, image and table numbers in a text, document underlined sections links (fig. 1, a) or with section number (fig. 1, b).

A window displaying an active document. refer to Chapter 15 on p. 71

a)

b)

Fig. 1.



If you drag a cursor to a link, a cursor will change its form. To follow a link, click it with mouse left button.

To return to the document place where you followed a link from, push `<Alt>+f` key combination, in this case use `<left arrow>`, but not a button which is also 4 number button on numeric keyboard.

Conventions of description

Usually, the application command can be activated in different ways, e.g. in Main menu by pushing the button, double click at field of context menu. One of the ways is described in User Guide.

If the commands are in Main menu sections, it is necessary to open sequentially the menu sections for a concrete command activation. E.g. in order to create a new Model document, it is necessary to fulfill the following actions:

- ▼ Open **File** menu.
- ▼ Call **Open...** command (fig. 2).

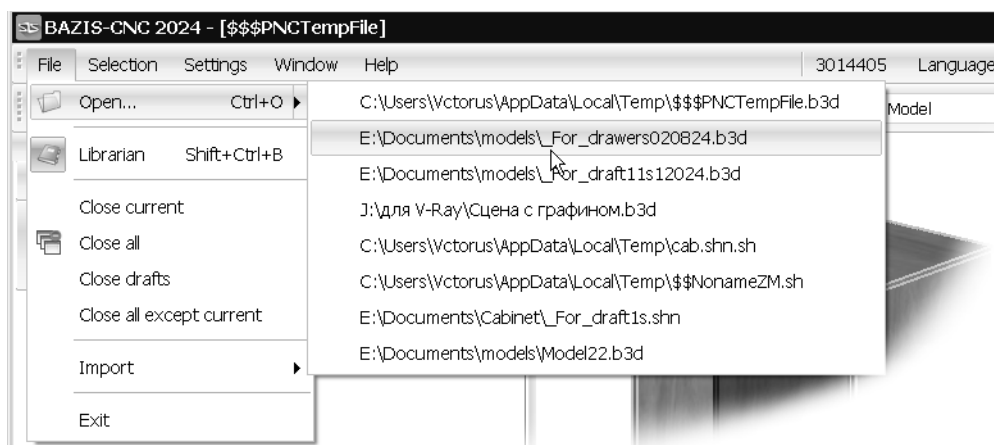


Рис. 2.

This action sequence in Help system is described in the following way: call **File – Open...** command. Module commands and operating system call is executed in the same way. If you use a command button, this button image will be moved to paragraph left field. Keyboard key names are in angle brackets and italicized. Key combinations are written in the following way. *<Key 1> + <Key 2>*. This record means, that it is necessary to press *<Key 1>*, then holding it, – *<Key 2>*.

<Key 1>, <Key 2>. This record means, that it is necessary to press *<Key 1>*, release it and push *<Key 2>*.

Notes, hints and warnings are marked with the following icons:



– Note



– Hint



– Warning

Technical support

If there are some problems with BAZIS system applications installation and use, as well as with protection keys, it is recommended to follow the actions.

- ▼ Use the system documentation and find there some information about troubleshooting.
- ▼ If there are no recommendations, consult the technical support of your supplier of BAZIS system software products (regional dealer).

Address and telephone number of regional dealer:

- ▼ If specialists couldn't help you, please contact BAZIS-Center office.

Mailing address: office 11, 15B, Shilova str., Kolomna, Moscow region, Russia

Telephone number: +7(496) 623-09-90

User account: <https://portal.bazissoft.ru>.

BAZIS-Center Internet page: <http://www.bazissoft.com>.

Before your appeal, please prepare information about problem situation, your actions, computer and peripheral equipment configurations. Indicate the serial number of protection key.

Part I

Documents preparation for control programs creation

Chapter 1. Module general information

1.1. Module start

To start BAZIS-CNC, call **Start – Programs – BAZIS – BAZIS-CNC** command. **Choose machine control system** dialogue will appear on the screen (fig. 1.1).

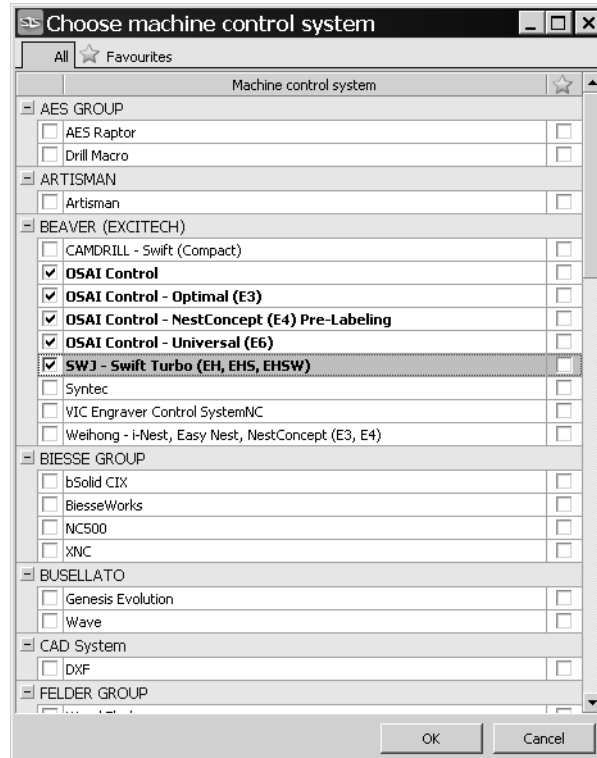


Fig. 1.1.

Activate options of machines (postprocessors) control systems in this dialogue for control programs creation, to continue the work push **OK** button. If no system is selected, a warning will appear on the screen (fig. 1.2).

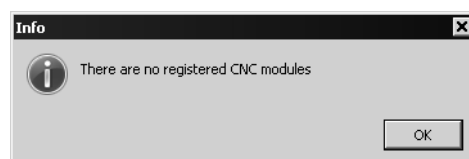


Fig. 1.2.

If module is not run, further work will be impossible. After control program selection, **BAZIS-CNC** window (see section 1.2 on p. 17) will appear on the screen. Common parameters are set in **Settings** dialogue. It will appear on the screen after **Settings – Parameters** command call. Common parameters setup can be executed from any module. It is described in detail in *BAZIS system Installation and setup*. Control programs setting is executed in **Preview for CNC** window (see Chapter 7 on p. 40). Control programs generation can be started from BAZIS-CNC as well as from other BAZIS system modules. For this purpose push **BAZIS-CNC** button in BAZIS module window.



1.2. BAZIS-CNC window

After module start and postprocessor selection **BAZIS-CNC** window will appear on the screen. Window control elements allow to operate with furniture model, control its viewing, set and create control programs for the model. The module is Windows application. Its window and its control elements are standard for operating system (fig. 1.3).

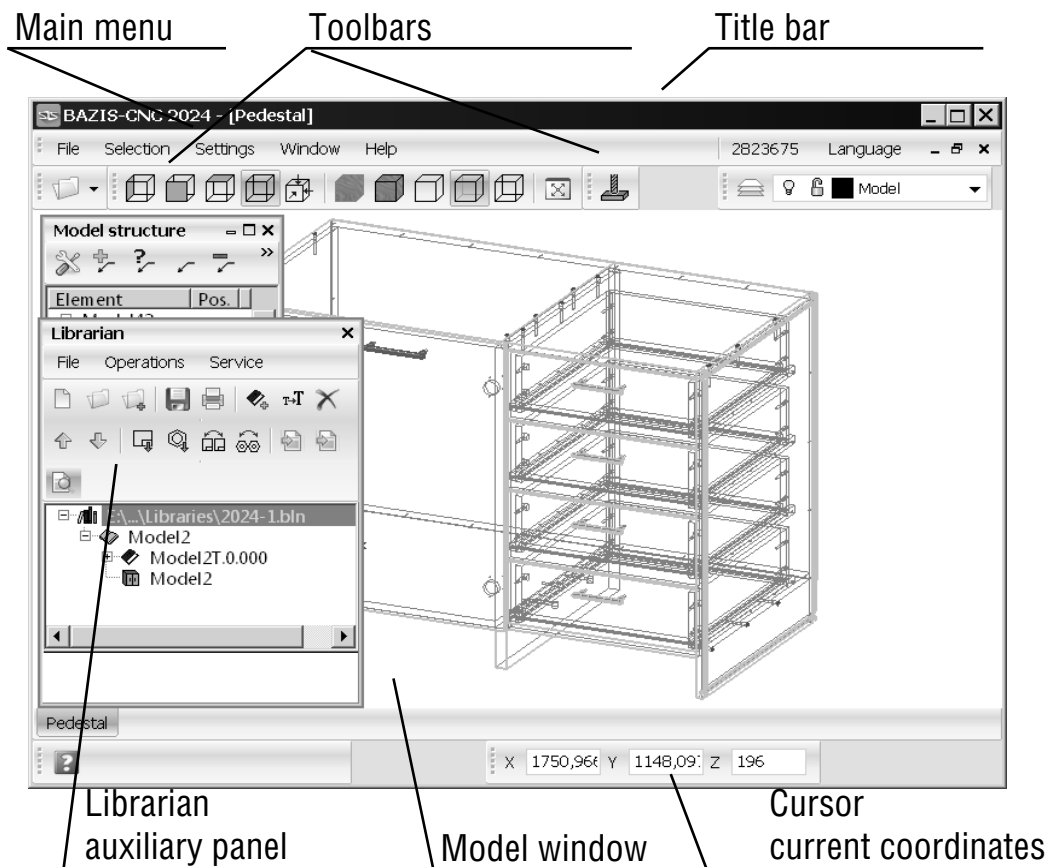


Fig. 1.3.

Interface elements concise description is shown in tab. 1.1.

Tab. 1.1. BAZIS-CNC interface elements

Name	Description
Title bar	Contains module name and a model file name.
Main menu	Contains module commands grouped in sections (menu).
Toolbars	Contain command call buttons.
Auxiliary bar Librarian	Contains control elements for BAZIS system libraries work.
Model window	Contains model image for which control programs are created.

Tab. 1.1. BAZIS-CNC interface elements

Name	Description
Cursor current coordinates	X , Y and Z fields contain current cursor coordinates in a module window. Z field is available if current model is a 3D object.

1.3. Librarian auxiliary panel

Librarian auxiliary panel control elements (fig. 1.4) allow to work with BAZIS system libraries.

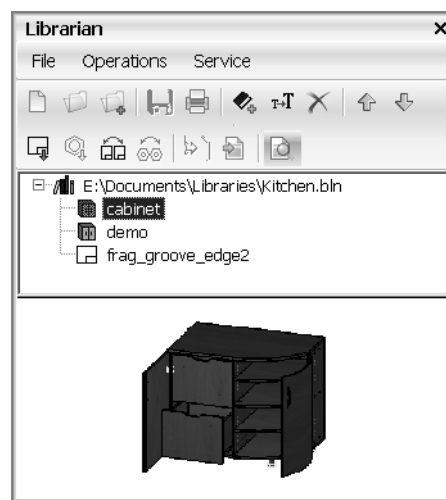


Fig. 1.4.

Library can be created in the following way:

- ▼ automatically, when you create assembly drawing or documents from a model,
- ▼ via **Librarian** auxiliary panel control elements.

BAZIS system libraries use is described in detail in *Work with libraries and projects User Guide*.

1.4. Interface elements state control

Interface elements, except Title bar and documents windows, may be in two states:

- ▼ pinned,
- ▼ floating.

In pinned state interface element is attached to module window bound. At window moving or its size changing, interface elements are moved together with attachment bound. In floating state, interface element can be moved to any place on the screen, including outside window bound. Thus, extra place for geometric constructions is free.

1.5. Toolbars arrangement



Toolbars, Main menu and Message bar arrangement is fulfilled in the same way. For setting description in this section, the only term is used – a toolbar. In pinned state toolbars are placed in the following way. Toolbars buttons and menu names are placed in one line, bars are fixed at upper or lower window bound. The title of pinned bar is not viewed. Instead of it, a vertical line appears on the left bar bound. At cursor dragging to this line, it becomes a quadrilateral arrow. To change bar placement, click left mouse button on the vertical line, and holding it, drag the bar to necessary place.

If you drag the bar to some place from the window bound, it becomes floating; it gets a title. Double click on vertical line changes the toolbar state to floating. It is untied from the window bound.



Settings – Toolbar – Fix command disables bar state changing. After command call, option next to command name will be activated; it shows the command state. Vertical line of pinned bars disappears, their dragging becomes impossible. **Fix** command can be called from context menu.

Toolbar size in floating state can be changed. The buttons and names will be automatically placed in several lines according to bar size.

For toolbar position in floating state change, it is necessary to click on its title with left mouse button, and holding it, drag the bar to necessary place. At bar dragging to upper or lower window bound, it is attracted and attached to it. Bar title disappears. At double click on a title of a floating toolbar, it is automatically attached to upper window bound. After **Fix** command call, toolbar dragging to window bound does not pin it. Double click on bar title allows to pin it.



Settings – Toolbars section commands help control bars visibility. Enabled option next to bar name shows, that bar is visualized in a window. For toolbar hiding, click its name in a menu. The bar will disappear, menu option in next to bar name will be disabled. If the toolbar is in floating state, to hide the bar push standard **Close** button in bar title.

1.6. Desktop control

The set of displayed interface elements, their sizes and position is called **desktop**. Current desktop settings are saved automatically in *UserInterface.xml* file. Desktop configuration may be saved for further use. E.g. you can create desktops for work with drawings and models, import the configuration to other desktops, etc. For configuration saving, call **Settings – Save settings** command.

A standard Windows dialogue will appear on the screen. Configuration file name should be input in this dialogue. The file automatically gets file name extension *bzconfig*. Desktop settings are saved with other current module settings. After command completion, a message will appear on the screen (fig. 1.5).

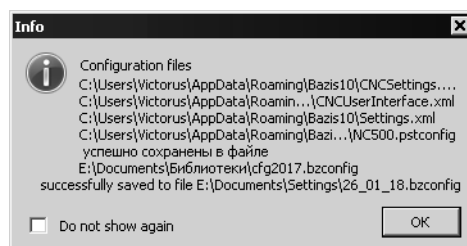


Fig. 1.5.

There are settings file names in this message saved in configuration file:

- ▼ *CNCUserInterface.xml* – interface settings.
- ▼ *CNCSettings.xml* – basic module settings fulfilled in **Settings** dialogue (see *BAZIS system Installation and setup*).
- ▼ *Settings.xml* – common settings of BAZIS system.
- ▼ *NC500.pstconfig.xml* – parameters settings of selected machine control program.

To set the module according to saved parameters, call **Settings – Restore settings** command. A standard Windows dialogue will appear on the screen. It is necessary to select a file of required configuration in this dialogue.

1.7. Skins

Skin is a set of visualization parameters of interface elements. They are colour, button visualization mode, etc. **Settings – Skin** command helps select necessary mode. Skin names submenu will appear after command call. Option next to current skin name is activated. For skin selection click on its name. Window view will be changed; option next to selected name will be activated.

1.8. Document window control

1.8.1. Window state

Each module document is opened in its window. Several documents can be opened simultaneously. One of the windows is active. Document windows can be in the following states (tab. 1.2).

Tab. 1.2. Document windows state

Window state	Description
Minimized	Windows are minimized to title size. By default, the titles of minimized windows are placed at lower window bound.
Expanded	Active window is expanded to all module's window. Active document name is in the Title bar. Other windows are below the active one and are invisible.

Tab. 1.2. Document windows state

Window state	Description
Arbitrary dimensions	Window dimensions and position are set by the user or <u>automatically</u> . The windows may partially or fully overlap each other.



For window state control, it is necessary to use standard buttons placed in title bar of each window. If it is expanded, its title bar is invisible. In this case, it is necessary to use window control buttons in Main menu.

For windows switching, it is necessary to use different means.

- ▼ Key combination `<Ctrl>+<Tab>` (`<Ctrl>+<Shift>+<Tab>`). Each pushing switches active window.
- ▼ **Window** menu commands that correspond to open windows names.
- ▼ Click on window title or tab.

1.8.2. Automatic windows arrangement

Window menu commands allow to arrange document windows automatically (tab. 1.3).

Tab. 1.3.

Command	Arrangement
Cascade	Windows are of the same size. Active window is in foreground. It overlaps next window in a way, that its title remains visible. Next windows are placed in the same way.
Tile horizontally	Windows have the same width, equal to module window width. Windows are regularly placed along module window height.
Tile vertically	The windows have the same height, equal to module window height. Windows are placed regularly along window width.

1.9. BAZIS-CNC module workspace

1.9.1. Document types

The module allows to create control programs for CNC machines. These programs are saved in files which extensions are specified by machine types.

1.9.2. Measurement units

A standard metric system is used in the module. By default, a measurement unit is a millimeter.

Chapter 2.Module setup

2.1. General information

Settings – Parameters command allows to setup the module. The setting is the same for all modules and described in *Bazis system installation and setup* document

There is **Check positions placement** option in **Positions placement** section (fig. 2.1).

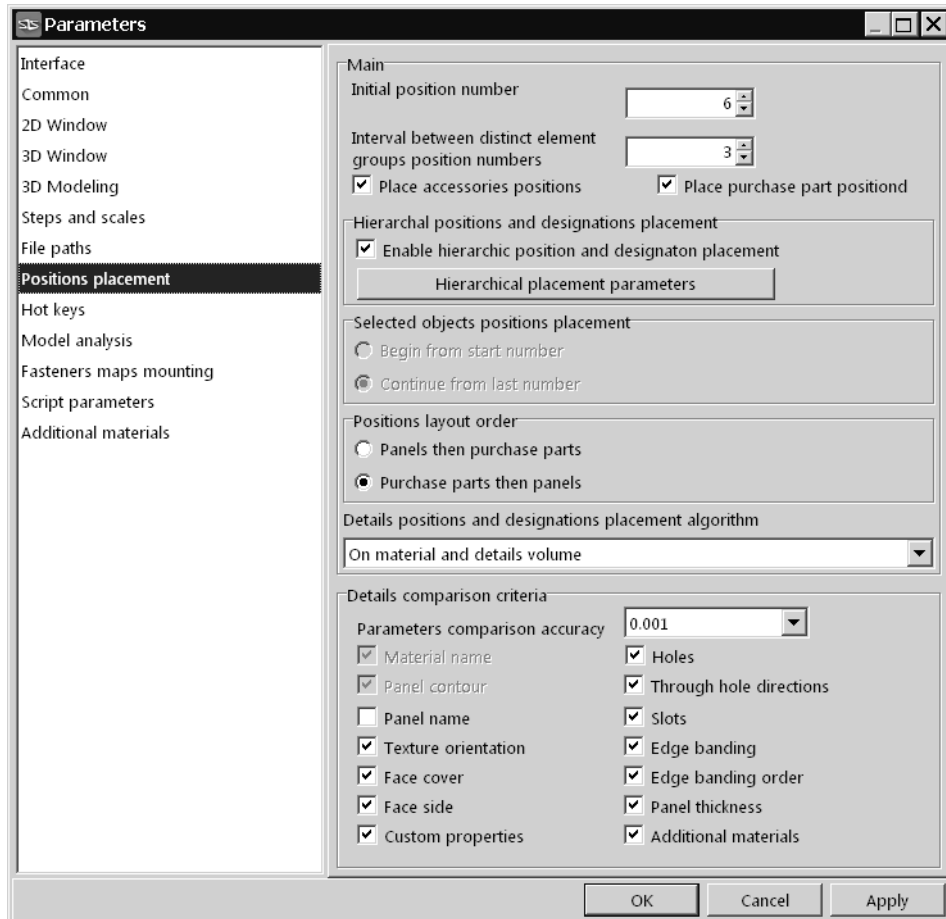


Fig. 2.1.

This option allows to control the verification of the correct positions placement of the current model before creating processing programs. It is disabled by default.

Settings – Material DB command allows to setup material DB used in BAZIS system. The setting is the same for all modules and described in *BAZIS system material database User guide* document.

2.2. Settings saving and restore

2.2.1. BAZIS system settings saving

Current settings of the BAZIS system can be saved to a file. To save configuration, call **Settings – Save settings** command. **Settings saving** dialogue will appear on the screen (fig. 2.2)

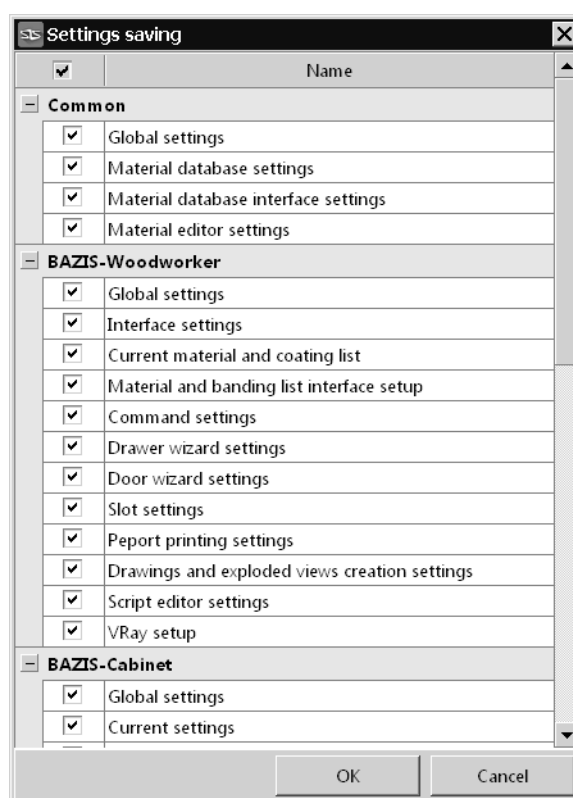


Fig. 2.2.

Enable options for the names of the settings that should be saved and push **OK** button. A standard Windows dialogue will appear on the screen. In this dialogue, enter the name of the configuration file. The file is automatically assigned the *bzconfig* extension. This file can be used, e.g. to transfer the settings of the BAZIS system to another computer.

2.2.2. The module current configuration saving

At the end of a module session, its current configuration is automatically saved in several files.

UserInterface.xml – interface settings.

Settings.xml – basic module settings fulfilled in **Settings** dialogue. This dialogue appears after **Settings – Parameters** command call.

Materials.xml – materials texture setting.

CreateDrawings.xml – drawing parameters setting.

ActionData.xml – current parameters of geometric construction commands.

To set the module according to saved parameters, call **Settings – Restore settings** command.

By default the files are saved in a folder *C:\Users\<CurrentUserName>\AppData\Roaming\Bazis*.

You can create a folder *Settings* in module installation folder, e.g. *C:\ProgramFiles (x86)\BazisSoft\Bazis\Settings*, so that configuration files will be automatically saved in this folder.

2.2.3. Settings restore

When the module is started, the configuration of the previous session is restored from automatically saved files. **Settings –Restore settings** command allows to restore the module configuration saved in the configuration file with the *bzxconfig* extension. The Restore Settings dialog will appear on the screen. **Settings restore** dialogue will appear on the screen (fig. 2.3).

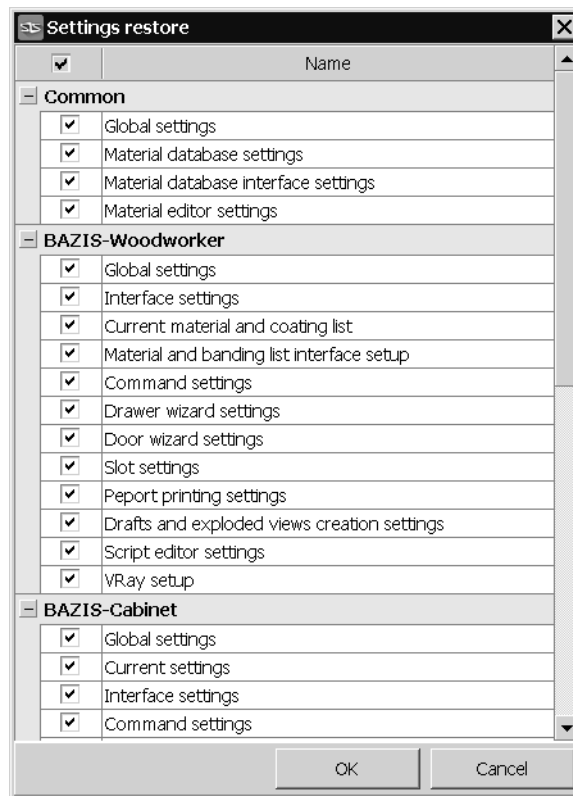


Fig. 2.3.

Enable the options for the names of the settings that should be saved and push **OK** button. A message will appear on the screen.

Chapter 3. Operations with documents

3.1. Document opening

BAZIS-CNC module is a multiwindow application. Several models can be opened in it simultaneously for control programs generation. Models can be opened from model files or BAZIS libraries and from the projects. If module is started from BAZIS-Woodworker or BAZIS-Cabinet modules, current model from these modules will be opened in it automatically.

3.1.1. Model from file opening



To open furniture model for control programs generation, call **File – Open** command. A standard Windows dialogue will appear on the screen. Control programs data can be generated for models saved in files. These types are listed in tab. 3.1.

Tab. 3.1. File types for panels list creation

File name extension	Description
b3d	Furniture models.
f3d	Fragments.
fr3d	3D fragments.
ldw	Drawing sheets.
bln, blf	BAZIS system libraries
shn	Cabinets



In this dialogue select a model file or model libraries of BAZIS system.



Viewing button allows to control model icon viewing. Opened files list is saved in the module. List size is set at setup. Button with downward arrow allows to open a list for file selection. After file opening a document window will appear on the screen.

3.1.2. Model from document library opening

Models for control programs generation can be opened from BAZIS system libraries. Libraries use is described in detail in *Work with libraries and projects User Guide*. To open library use **Librarian** auxiliary panel control elements or **File – Open** command.

After library file opening its content will be shown on **Librarian** auxiliary panel (fig. 3.1).

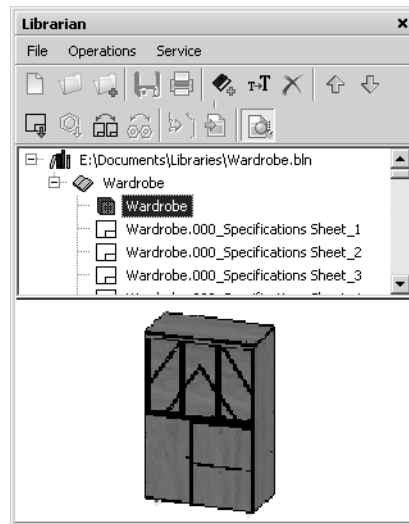


Fig. 3.1.



Viewing button allows to control model icon viewing.



To open a model select a file name in a list and push **Open for editing** button. The document will be opened in BAZIS-CNC module window.

3.2. Document closing

3.2.1. Drawings import in DXF format

For perform the import, call **File – Import – Import from DXF format** command. A standard Windows dialogue will appear on the screen. Select a file in the dialogue. After file selection, **Import from DXF format** dialogue will appear on the screen (fig. 3.2).

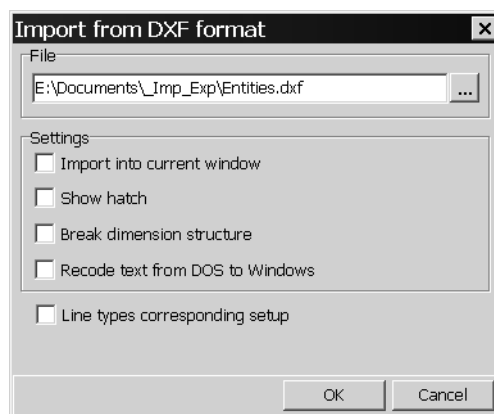


Fig. 3.2.

The dialogue control elements allow to setup import parameters. To complete the setup, push **OK** button. If **Line types corresponding setup** option is enabled, **Line types corresponding setup** dialogue will appear on the screen (fig. 3.3).

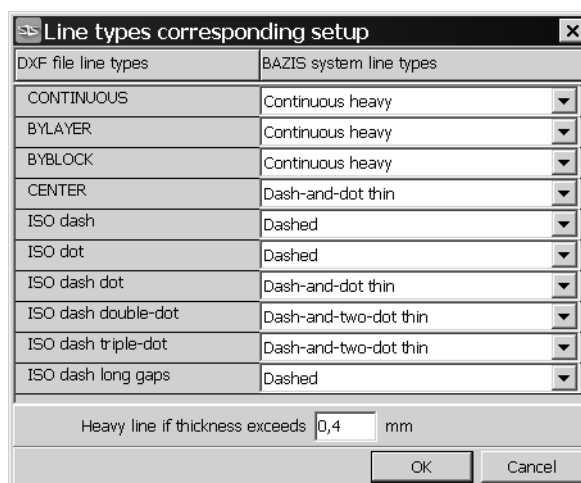


Fig. 3.3.

In this dialogue map line types of imported file shown in **DXF file line types** column to BAZIS system line types. To complete the setup, push **OK** button. The drawing saved in DXF format will be opened in new window.

3.2.2. Models import in bCAD and K3 formats

File – Import – Data import from K3 project and **File – Import – Import data from bCAD project** commands allow to import the models, saved in corresponding formats. After command call, a standard Windows dialogue will appear on the screen. In this dialogue select a file for opening. Work with imported files is similar to work with model files of BAZIS system.

3.3. Documents closing

File – Close current... command allows to close active document.

File – Close all command allows to close all open documents.

File – Close drawings command allows to close all open drawings.

File – Close all except current command allows to close all open documents except active one.

Chapter 4.3D models use for control programs creation

4.1. Model visualization control

4.1.1. Model orientation

Views toolbar buttons allow to control model orientation in a window.



Left-hand view, **Top view**, **Front view** buttons allow to view the model in corresponding main views.



If at module setup **Switch views by mouse middle button** option was enabled, views can be selected via mouse middle button.



Change to projection view button allows to transform the model in a view, which is the most parallel to the screen plane at the moment.

In fig. 4.1, a) the model is viewed in axonometric projection.

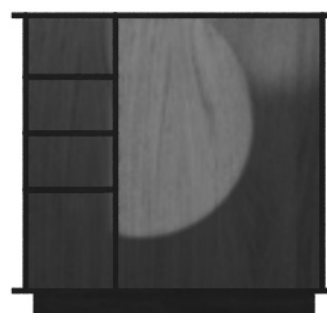
In fig. 4.1, b) the model is arbitrarily rotated. Coordinate plane XOY is the most parallel to the screen plane. After pushing **Change to projection view** button, model is viewed on front view (fig. 4.1, c).



a) axonometry



b) rotated picture



c) front view

Fig. 4.1.

4.1.2. Model rotation and move

Model image rotation

Objects axonometric images can be rotated in model window. For this purpose push left mouse button or a scroll wheel, move the cursor. The model orientation will be arbitrary changed. If you need to view the model in one of the standard views, use **Views** toolbar buttons (see section 4.1.1 on p. 28).

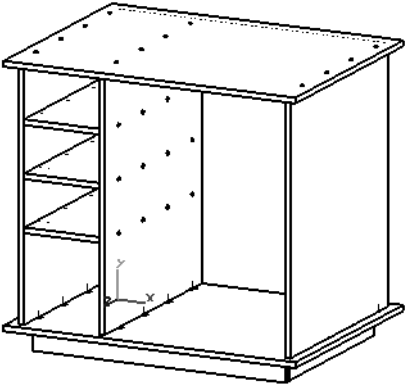
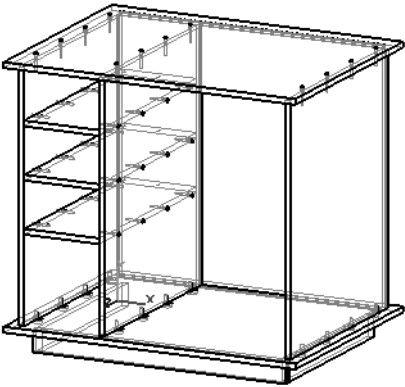
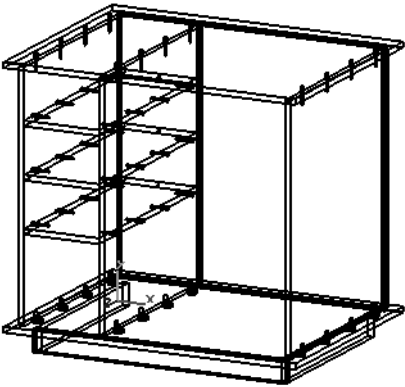
View move

The picture in the model window can be moved in a way, that its required part will be viewed. Push right mouse button and holding it move the cursor. The cursor view will be changed.

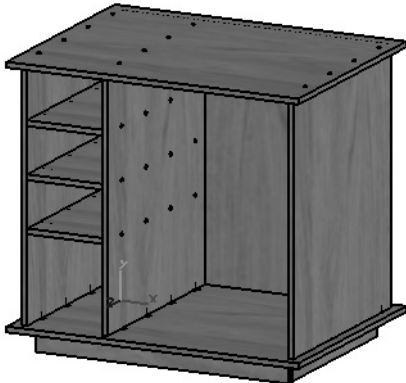
4.1.3. Model visualization modes

For each type of model orientation you can use different visualization modes (tab. 4.1).

Tab. 4.1. Model visualization modes

Type	Description	Visualization
 Visible lines	The model contour and visible lines at the current model edge orientation are viewed.	
 Show visible and hidden lines	Invisible edges and edges parts are viewed with thinner lines.	
 Wireframe	Outline and all edges are viewed with lines of the same thickness.	

Tab. 4.1. Model visualization modes

	Type	Description	Visualization
	Textured	The model texture is viewed.	
	Textured with edges	Combination of textured and hidden lines removed visualization modes.	

4.1.4. Zooming

For objects convenient viewing in a model window you can zoom. Zooming allows to «zoom in» and «zoom out» the objects. You can zoom via a scroll wheel or key combination **<Ctrl>+<->** and **<+>** on a numeric keyboard. Zoom factor specifies image dimension changing. This parameter setting is described in *BAZIS system Installation and setup*. **Show all** button on **View** toolbar allows to set zoom in the document window in a way, that all geometrical objects will be viewed on the screen in maximum zooming.



4.2. Panels selection for processing

4.2.1. General information

By default control programs can be created for all model panels. If some panels are selected in the model, you can activate only these panels processing (see section 6.3 on p. 38).

4.2.2. Objects selection modes

You can select the panels in different ways. Selected objects in a document window are coloured.

Cursor selection

Single objects cursor selection can be customized in the module settings (see *BA-ZIS system Installation and setup*). **Cursor** or **Shift+LeftMouse Click** selection modes should be chosen as most convenient. Select an object by left mouse click. To select several objects with a cursor hold <Ctrl> button pushed.

Frame selection

If you need to select a group of objects with a frame, push <Shift> button and holding left mouse click in blank space of the drawing. Thus you specify frame first vertex position. Holding the buttons drag the cursor in the drawing window. Frame selection phantom will appear on the screen. Then specify the second vertex position in a way, that the rectangle will embrace all selected objects.

Objects of specified material selection

If you need to select the objects made of some specified material, call **Selection – Select by material** command. **Objects selection** dialogue will appear on the screen (fig. 4.2).

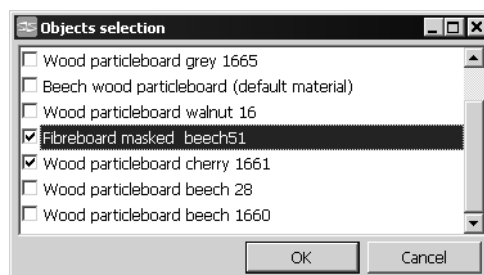


Fig. 4.2.

The dialogue contains material list used in the product. Option left to the material name allows to choose material for selection. To select the objects with chosen materials push **OK** button. **Cancel** button closes the dialogue without changes.

Selection by objects names and their properties

Document - model objects can be selected by their names or positions. To select the objects this way call **Selection – Select by mask** command. **Object name mask** dialogue will appear on the screen (fig. 4.3).

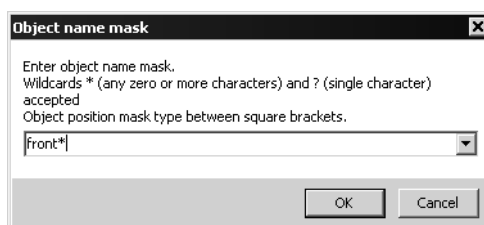


Fig. 4.3.

Enter object name in the field. It is possible to select the objects with standard masks use. <*> (asterix) sign replaces any number of symbols, <?> sign replaces one symbol. Mask text is case-insensitive.

When you set selection mask, all the objects with their names beginning with *front* will be selected. If you need to select the objects by their positions, enter these figures within square brackets (fig. 4.4).

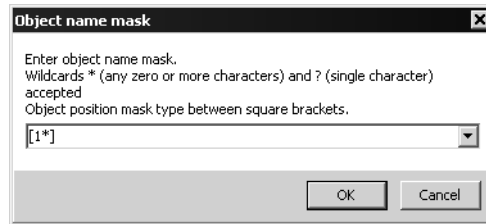


Fig. 4.4.

For example, objects with position numbers from 10 to 19 will be selected. Entered selection condition text will be saved. You can choose from combo box the strings that had already been used in further search. To select the objects push **OK** button. **Cancel** button allows to close the dialogue without selection.

All objects selection

Selection – Select all command allows to select all objects of active document.

Unselection

Object unselection mode depends on the system setup. If **Cursor** variant is selected, to unselect the object click it with mouse left button again. If **Shift+LeftMouse Click** variant is selected, click with mouse left button outside the object contour. If several objects are selected, to unselect them call **Unselect** command.

Chapter 5. Drawings use for control programs generation

5.1. Drawing execution demands

5.1.1. General information

Control programs can be created for models in BAZIS system sheets format and saved in *ldw* file. Panel drawing should be executed in particular way – panel objects should be arranged on separate layers. Separate layers should be created for the following objects:

- ▼ panel contour,
- ▼ face holes,
- ▼ edge holes,
- ▼ slots on a face,
- ▼ edge slots.

If there are several objects of the same type in the panel, e.g. face holes, but these holes have different radius values, a separate layer should be created for each radius. Layer names should consist of the following parts.

- ▼ Prefix – arbitrary symbols. It is reasonable to set prefix that corresponds to layer object type, e.g. *CONTOUR* layer will contain panel contour image, *HOLE_FACE* layer – face holes images, etc.
- ▼ Layer object parameter value, e.g. panel thickness, hole depth, slot width and depth.



Layer name prefix for the same objects should be the same, i.e. for face holes of 4 and 10 mm diameter it is necessary to create layers with names *HOLE_FACE4* and *HOLE_FACE10*.

5.1.2. Layers properties

Contour

Contour parameter is panel thickness in millimeters. Panel contour may contain line, arc and circle segments. Contour should be closed. Contour lines are visualized with continuous line. Name example of panel layer of 16 mm thickness: *CONTOUR16*.

Face holes

Hole parameter is its depth value. If hole is through, it is reasonable to use value that exceeds panel thickness by 2...4 mm. Face holes layer number should correspond to different holes depth values. Holes are visualized with circles of different diameter. Face holes are visualized with continuous line, back side holes – with dashed line (fig. 5.1).

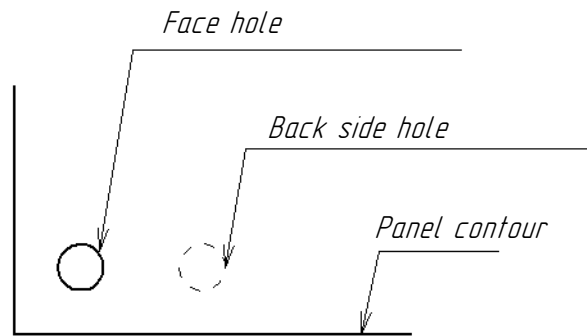


Fig. 5.1.

Name example of face blind holes layer of 8 mm depth: *HOLE_FACE8*. Name example of panel through holes layer of 16 mm thickness: *HOLE_FACE20*.

Edge holes

Edge hole parameter is distance between front face and Z hole axis (fig. 5.2).

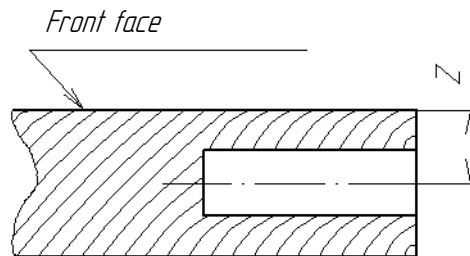


Fig. 5.2.

Edge holes layer number should correspond to Z parameter values number. Hole is visualized with a rectangle, one of its side coincide with contour line. Sides length are equal to hole diameter and depth. Rectangle is visualized with dashed line (fig. 5.3).

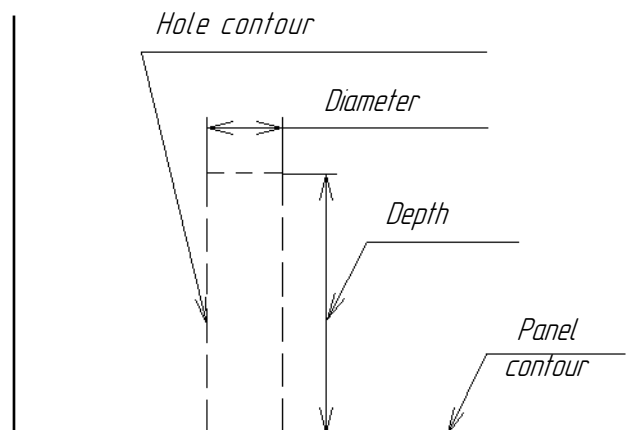


Fig. 5.3.

Name example of edge holes layer which distance between axis and front face is equal to 8 mm: *HOLE_EDGE8*.

Slots on a face

Slot parameters are width and depth values. Parameters in layer name are separated with symbol / (<width>/<depth>). Face slot layer number should correspond to slots of different sizes number. Slots are visualized with closed or open-ended contour. It may contain line, arc and circle segments. Contour lines on front side are visualized with continuous line, on back side – with dashed line (fig. 5.4).

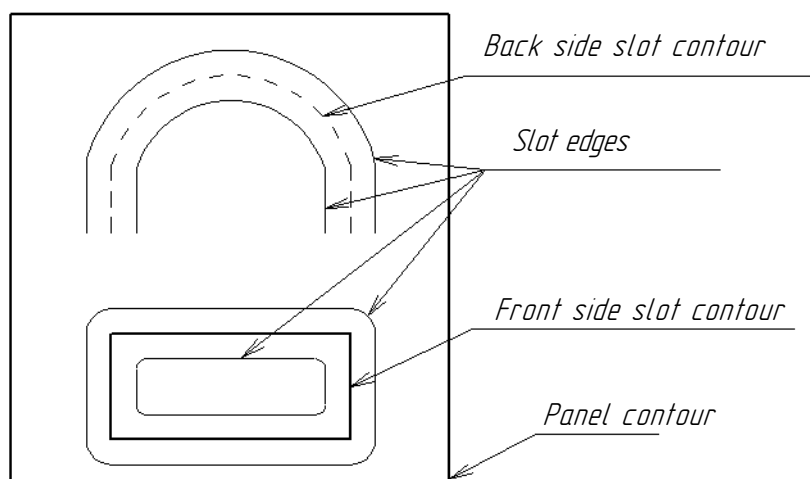


Fig. 5.4.

Name example of slot on a face layer of 10 mm width and 5 mm depth: *SLOT_FACE10/5*.

Edge slots

Slot parameters are width value and distance between front face and the nearest Z slot bound (fig. 5.5).

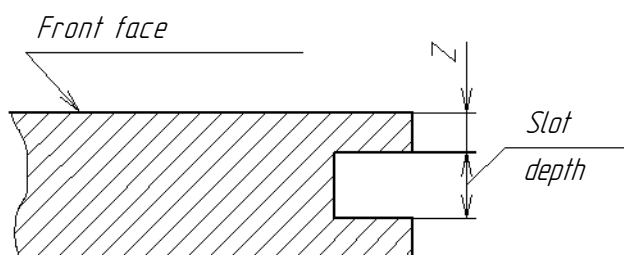


Fig. 5.5.

Parameters in layer name are separated with symbol / (<width>/<Z value>). Face slot layer number should correspond to slots of different size number. Slots are visualized with closed or open-ended contour. It may contain line, arc and circle segments. Slot depth is constant, thus slot contour should be equidistant from panel contour. The distance between contour lines corresponds to slot depth. Slot contour is visualized with continuous line (fig. 5.6).

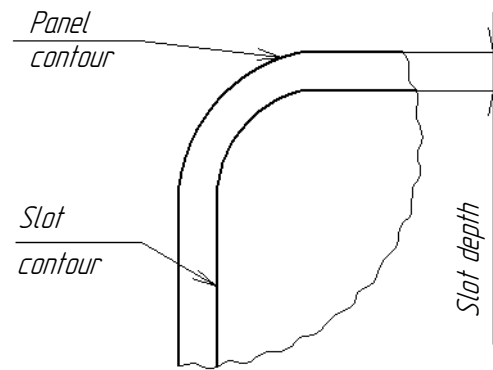


Fig. 5.6.

Name example of edge slot layer of 4 mm width, distance between front face and the nearest bound is 8 mm: *SLOT_EDGE4/8*.

Chapter 6. Data preparation for control programs creation

6.1. General information



To create control programs for current model push **BAZIS-CNC** button on **Modules** toolbar in **BAZIS-CNC** window. To create control programs for the current model, set input data format. Settings are specified by document type.

6.2. Drawing data preparation



After **BAZIS-CNC** button pushing, **Input data** dialogue will appear on the screen (fig. 6.1).

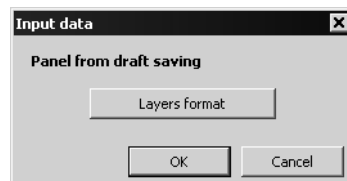


Fig. 6.1.

To prepare data push **Layers format** button. **Layers format under drawing reading** dialogue will appear on the screen (fig. 6.2).

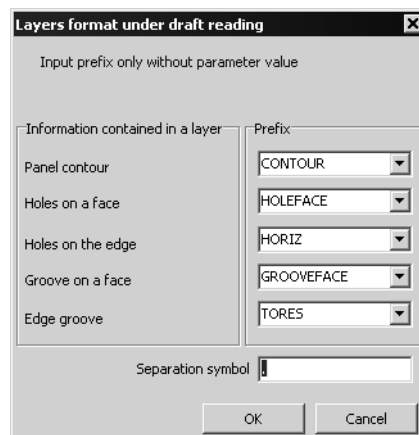


Fig. 6.2.

6.2.1. Data types to layer name prefix mapping

Information contained in a layer group contains data types that are used in a module for control programs creation. Each data type should be mapped to layer name prefix that contains corresponding data.



Only layer name prefix will be mapped. Data from all layers with the same type objects will be grouped, e.g. face holes with different diameter.

Prefix group contains layer names combo boxes. These names are imported from a drawing. To map layer prefix to data type do the following.

1. Select layer name from a combo box that contains the objects that correspond to current data type.

Several layers with the same name prefix and different parameters may correspond to one data type. You can select any name. E.g. blind holes of 8 mm depth and through holes should be bored in panel face. Layers with names are created for their visualization, *HOLE_FACE8* and *HOLE_FACE20*. Let *HOLE_FACE8* layer be selected for instance.

2. Click with mouse left button in a layer name line.

Text cursor will appear in a string.

3. Delete parameter values so that prefix will remain, in the example, *HOLE_FACE*.

Then map all data types to drawing layers prefixes in the same way.

6.2.2. Separation symbol of decimal part

Separation symbol field allows to set a sign that separates integer part from fractional part in layer parameter designation. Comma is used as a separator in Russian designations, in English – a point. Thus, to set a layer parameter that contains holes of 6,5 mm diameter, add 6,5 string to prefix. In some CAD, e.g. AutoCAD, no commas and points are allowed in layer names. Thus layers imported from such systems will contain a string, e.g. 6_5. To make the module interpret parameter in layer name correctly, specify a symbol as decimal separator. Thus, if drawing is imported from AutoCAD and an underscore <_> character is used in layer names as a separator, this very sign should be input in **Separation symbol** field.

6.2.3. Setup completion

To complete layer prefixes setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Layers format under drawing reading** dialogue will be closed.

6.3. Data from a model preparation



After pushing **BAZIS-CNC** button and if separate panels were selected in a model, **Input data** dialogue will appear on the screen (fig. 6.3).



Fig. 6.3.

If none of the objects was selected, this message will not appear. **Save panels from a model** group variants allow to select the panels for processing.

Part II

Control programs setup

Chapter 7. Preview for CNC window

7.1. General information

Control programs are generated in **Preview for CNC** window (fig. 7.1 on p. 41). Current model file name is shown in window title bar.



If model or drawing are transferred from other modules, temporary file will be created, its name is viewed in a title bar.

This window will appear on the screen after layers or drawing prefixes setup or model panels selection completion, and pushing **OK** button in **Input data** dialogue.

7.2. Control programs general information

7.2.1. Order name

Order field allows to input order name. Name may contain arbitrary text. When you open a project, project file name will appear in a field automatically.

7.2.2. Product count

Product count field allows to set panels count. By default one panels set will be processed. If you change field value, **Count** fields values will be changed automatically.

7.2.3. Configuration choice

Configuration combo box contains configuration names. Each configuration is a set of the following parameters:

- ▼ model reading parameters,
- ▼ technological parameters, that are specified by machine control system type,
- ▼ panels list that require control programs creation.

Several configurations can be created for each control system. They may differ in e.g. operations (boring, milling) or processed panels list. Configurations are created for each control system chosen in **Choose machine control system** dialogue (see section 1.1 on p. 16). When you choose one of them, this very configuration data will appear on window tabs.

7.3. Panel list

When you open **Preview for CNC** window, panel lists for all configurations will be generated automatically. Current configuration list is viewed on **Panel list** tab (fig. 7.1).

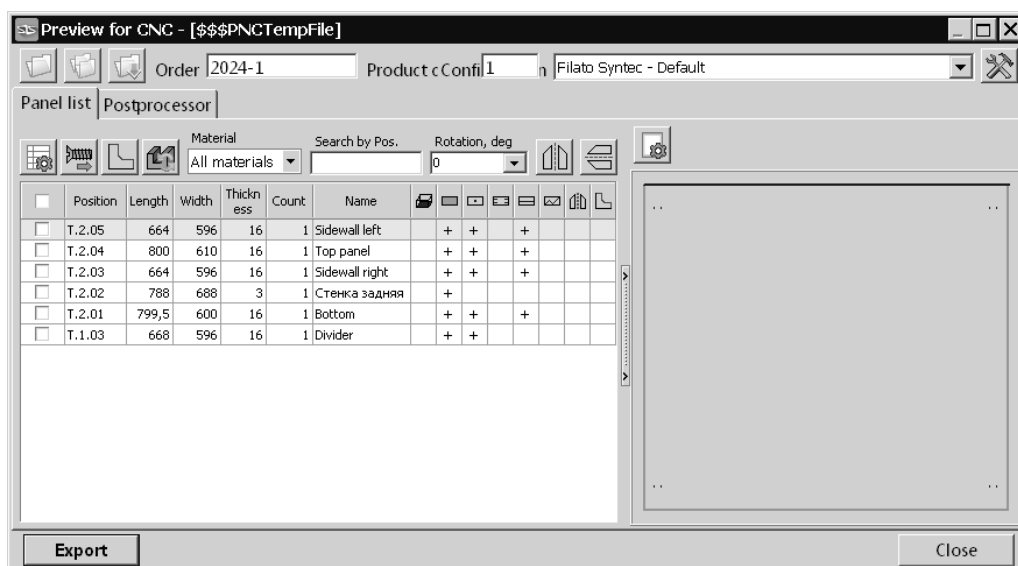


Fig. 7.1.

By default there are all panels in a list. **Material** combo box allows to filter panels list. When you choose material name, panels made of this material will appear in the list. **Search by Pos.** field allows to move selection cursor to the panel which position coincides with entered value. One of the panels is marked. It can be moved along the list via a mouse or cursor control buttons. Selected panel contour is shown on preview panel. Holes, slots and contour segments for processing will be highlighted. Panels list setup is described in Chapter 8 on p. 44. Each panel parameters are viewed: position, dimensions, copy count. Symbolic representations are used for holes in face, edge, slots designation. Contour segments milling set by the user is also designated (see section 8.4 on p. 46) and mirrored views as well (see section 8.6 on p. 48).

7.4. Postprocessors list

There is configuration list on **Postprocessors** tab (fig. 7.2). Configuration setup is described in Chapter 9 on p. 51 and Chapter 10 on p. 63.

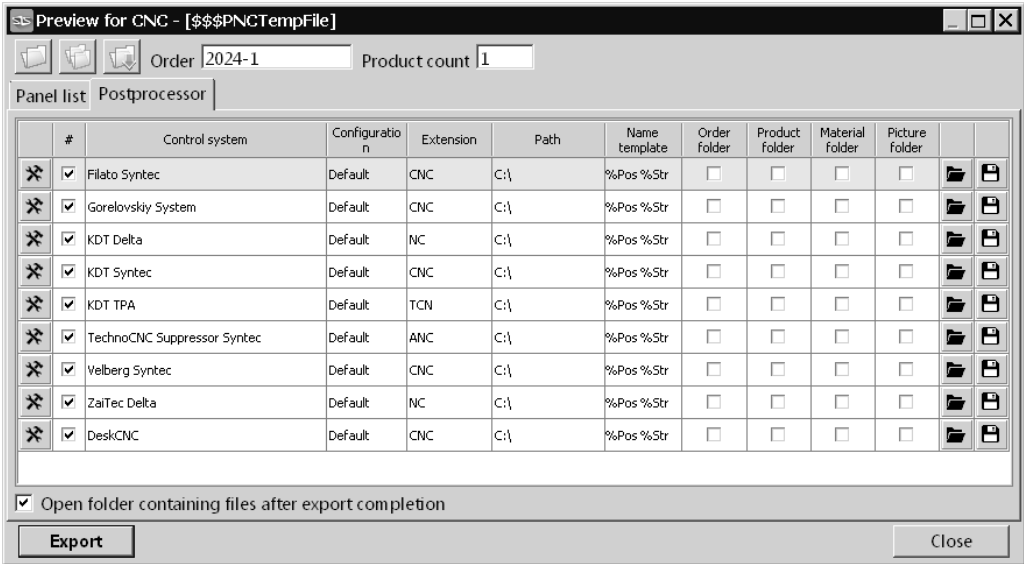


Fig. 7.2.

7.5. Panels list from a file opening



Open button allows to create a new panels list from a file. After pushing the button, a standard Windows dialogue will appear on the screen. Panels list can be created from files as in tab. 7.1.

Tab. 7.1. File types for panels list creation

File extension	Description
b3d	Furniture model.
f3d	Fragments.
fr3d	3D fragments.
ldw	Drawing sheets.
bln, blf	BAZIS system libraries
shn	Cabinets

7.6. Work with project



Work with project button allows to execute cutting not for separate products, but for whole furniture sets. Such furniture set is called a project. After pushing the button **Project creation** dialogue will appear on the screen (fig. 7.3).

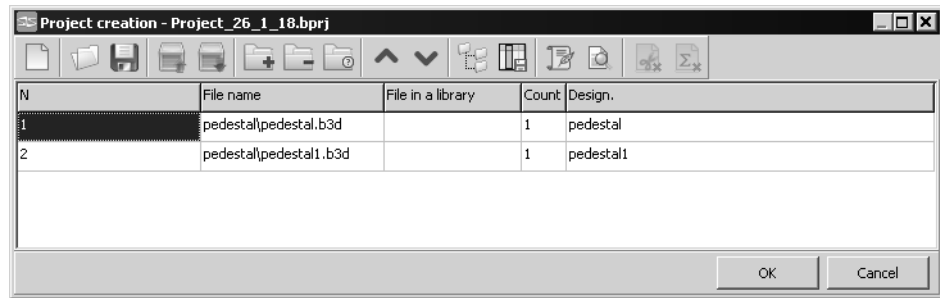


Fig. 7.3.

In this dialogue create a project for the models to be cut. Projects use is described in *Work with libraries and projects User Guide*.

To complete project creation and transfer its models to control programs, push **OK** button. **Project creation** dialogue will be closed. Project panels list will be created automatically. **Cancel** button allows to close the dialogue without changes.

7.7. Import in DXF format



Import button allows to import a list of panels from a file in DXF format. After pushing the button, a standard Windows dialogue will appear on the screen. In this dialogue select the required file.

Chapter 8.Panels list editing

8.1. Panel to control program appending

Enabled option in panel line shows that this panel should be appended to CNC job, control program will be created. Context menu commands allow to execute group operations of panel choice for export (tab. 8.1).

Tab. 8.1. Panels list context menu commands

Command name	Description
Select all	Appends all list panels to CNC job.
Select back sides	Appends all back sides to job.
Unselect all	Exclude all list panels from CNC job.
Inversion choice for export	Reverse all panels appending to CNC job flag.
Reset choice flag after export	After control programs creation it unselects appending to job flag for all panels.
Rotate back side automatically	If both panel sides are being executed, and one of them is rotated to some angle, the other one will be automatically rotated to the same angle.
Invert "Back face"	It changes back and face sides for selected panel.

8.2. List visualization



Panels list visualization setup button allows to set list colour style. After pushing the button **View parameters** dialogue will appear on the screen (fig. 8.1).

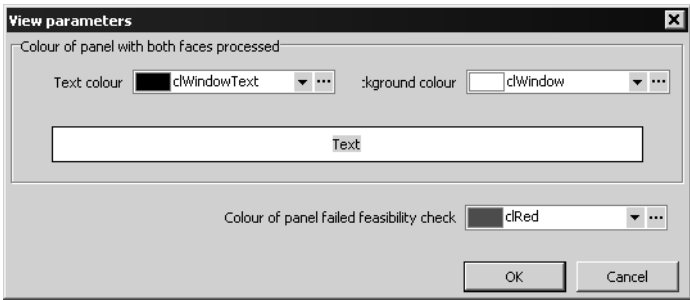


Fig. 8.1.

Text, background and panels colour which realizability had not been checked (see section 9.4 on p. 60) is set in the same way. Colour can be selected from a standard Windows combo box. You can also push **Browse** button and select colour in a standard Windows **Colour** dialogue. To complete setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.



8.3. Holes editing



Holes editing button allows to change holes parameters on selected panel. After pushing the button, **Holes editing** dialogue will appear on the screen (fig. 8.2).

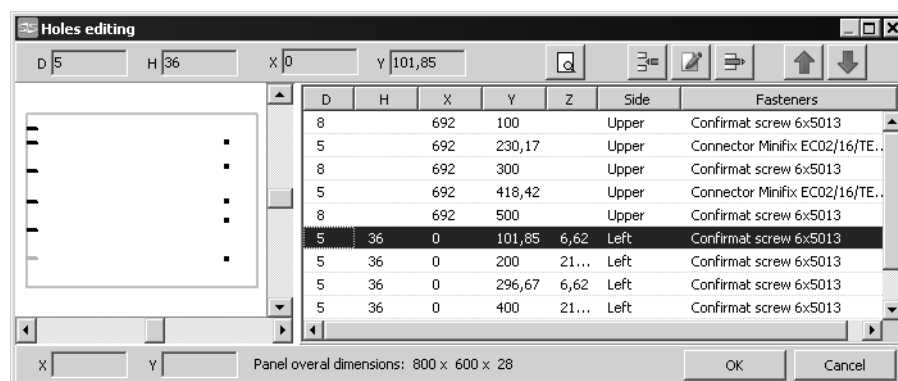


Fig. 8.2.

There is edited panel and holes list in this dialogue. When you select a list string, hole image will be highlighted. The following values will be shown for each hole: diameter, depth, center coordinates, layout side, belonging to accessories.

8.3.1. Hole adding



Add button allows to add a hole. After pushing the button diameter, depth and new hole center coordinates fields will become available. Fill the fields, to finish hole adding, push **OK** button to the right of entry fields. Hole string will appear in a list, its image will appear on a panel. **Cancel** button allows to cancel editing.

8.3.2. Holes editing



Edit button allows to edit hole diameter and selected hole depth. To finish editing push **OK** button to the right of entry fields. **Cancel** button allows to cancel editing.

8.3.3. Holes deletion



Delete button allows to delete selected hole. After pushing the button, a warning will appear on the screen (fig. 8.3).

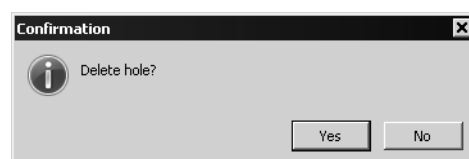


Fig. 8.3.

Message buttons allow to confirm deletion or reject it.

8.3.4. Holes editing completion

To complete holes editing push **OK** button of **Holes editing** dialogue. **Cancel** button allows to close the dialogue without changes.

8.4. Contour milling control



Point contour segments for milling button allows to specify panel contour segments for milling and set milling order. After pushing the button **Point contour segments for milling** dialogue will appear on the screen (fig. 8.4).

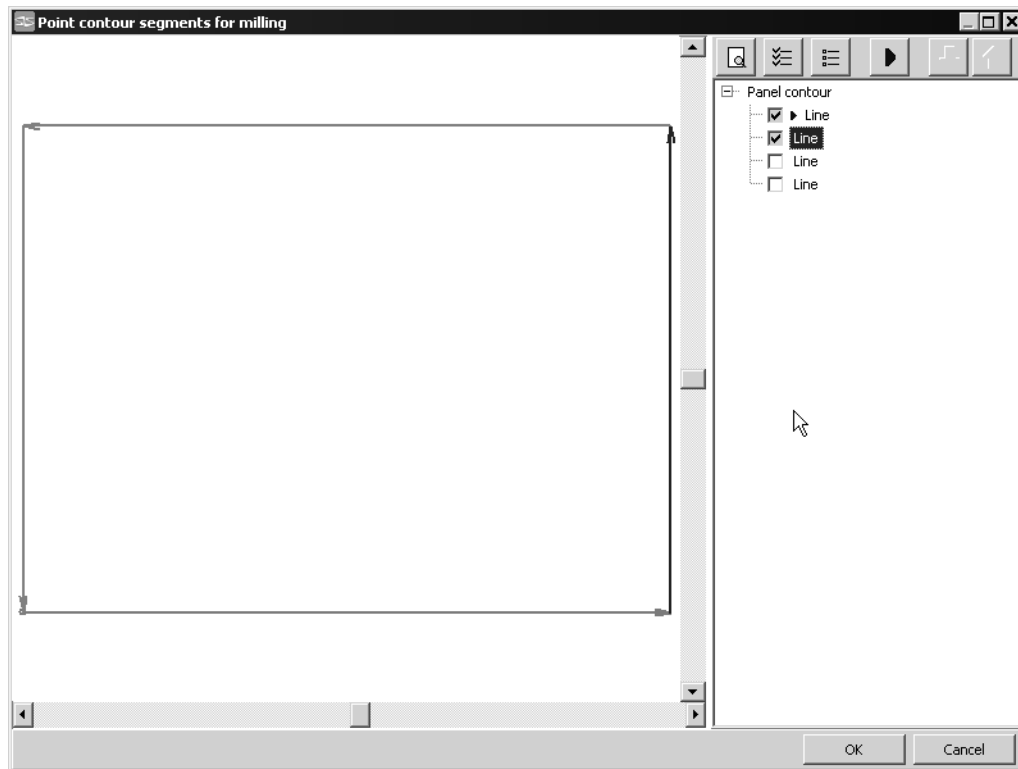


Fig. 8.4.



To specify a contour segment for milling, click its image with mouse left button. Milling segment in **Panel contour** list is designated with an option. **Select all** and **Undo all** buttons allow to control all contour segments selection and unselection.



Processing beginning button allows to set contour segment for milling start. Selected segment will be automatically moved to list beginning and marked with an icon.



Scroll wheel allows to scale the panel image. **Show all** button allows to view a panel in a way, that it will fit a preview panel in maximum scale.



To complete contour segments milling setup push **OK** button. **Point contour segments for milling** dialogue will be closed. A dagger will be upper in **Machining contour custom setting** cell. **Cancel** button allows to close the dialogue without changes.

8.5. Slots editing



Trajectory direction and work sequence of slots button allows to operate with selected panel slots. After pushing the button **Trajectory direction and work sequence of slots** dialogue will appear on the screen (fig. 8.5).

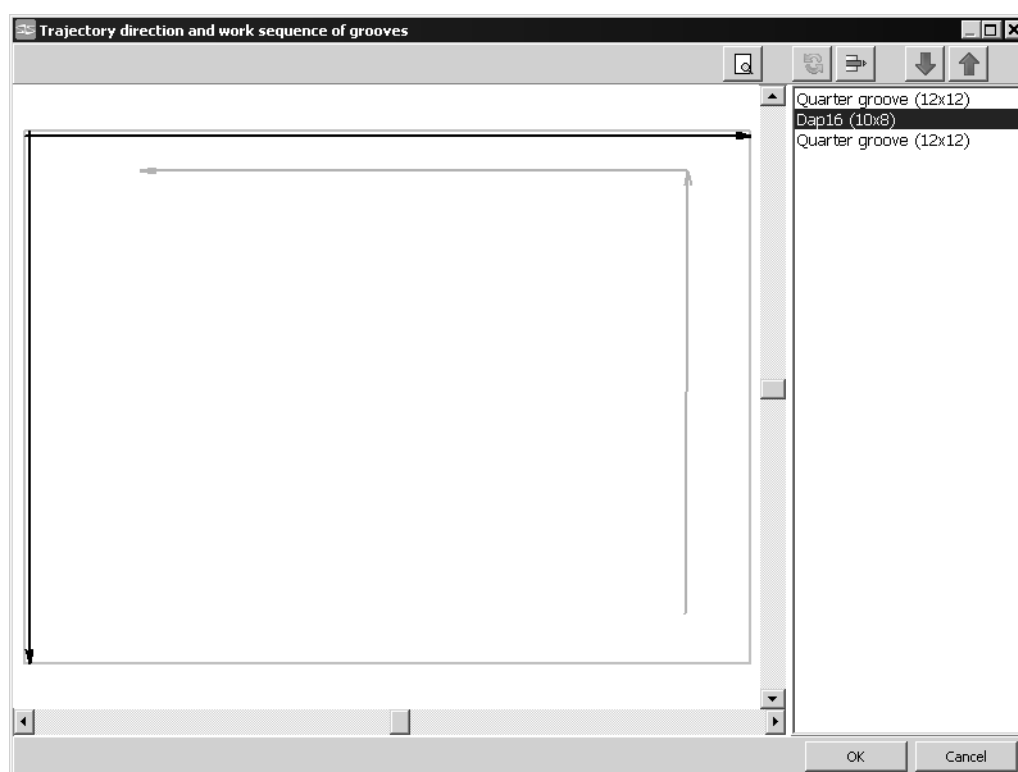


Fig. 8.5.

There is a current panel on dialogue viewing panel. Slot names are in the list. One of the names can be selected with mouse left button. Slot image will be also selected.



Up and **Down** buttons allow to move selected name along the list. Slots processing will be changed correspondingly.



Scroll wheel allows to scale the panel image. **Show all** button allows to view the panel in a way, that it will fit the viewing panel in maximum scale.



Processing direction is viewed by an arrow on a slot line. **Change direction** button allows to change processing direction of selected slot to the opposite.



Delete button allows to delete selected slot. After pushing the button, a warning will appear on the screen (fig. 8.6).

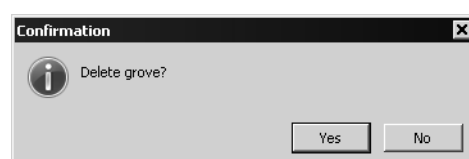


Fig. 8.6.

Message buttons allow to confirm deletion or reject it. To complete slots editing push **OK** button. **Slot trajectory direction setup** dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

8.6. Mirrored view



Transform symmetrically about horizontal axis and **Transform symmetrically about vertical axis** buttons allow to get current panel mirrored view about horizontal and vertical axes respectively.

In panel **Mirrored view** cell a dagger will appear.

8.7. Panel rotation

Panel image in viewing area and a panel on machine work area can be rotated. Enter angle value in **Rotate board on** field. Combo box variants allow to rotate in 0°, 90°, 180° and 270°. If rotation angle is set by the user, the following additional values will appear in a combo box:

- ▼ user angle negative value allows to restore panel initial position,
- ▼ value that completes the angle to 90°.

Angles combo box for user angle of 60° is in fig. 8.7.

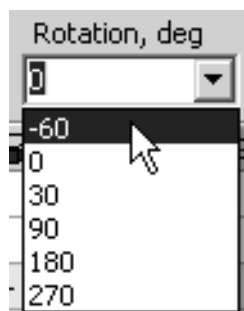


Fig. 8.7.

In some cases, rotated panel processing is impossible. To define such situations, set operations checking (see section 9.4 on p. 60).

8.8. Panels in list visualization

8.8.1. General information



Selected panel is viewed on viewing panel as a map. **Panel chart view parameters** button allows to set this image. After pushing the button **Panel chart view parameters** dialogue will appear on the screen.

8.8.2. Edge banding visualization

Banding section control elements (fig. 8.8) allow to set edge banding visualization parameters.

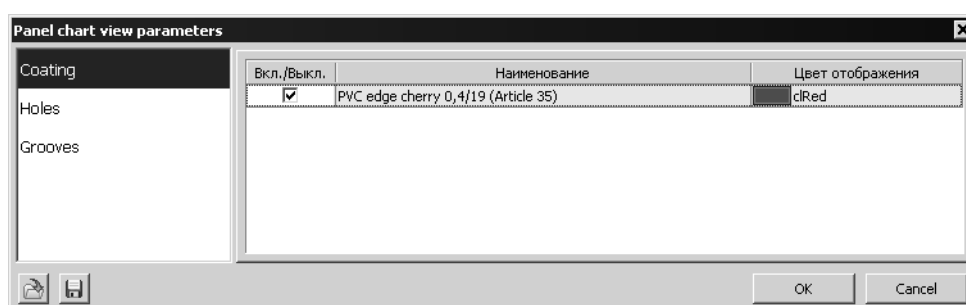


Fig. 8.8.

Banding is visualized with a colour line. A line colour can be set by banding material thickness or its name.

Colour selection by thickness

If **By thickness** variant is selected, banding line colour will be specified by banding material thickness. For each thickness range you should select the colour from a combo box. **Browse** button allows to select additional colours. After pushing the button, you can select the colour in a standard Windows dialogue.

Colour selection by material name

If **By name** variant is selected, banding line colour will be specified by banding material name. The section table contains banding material names, that are used in the current model. **On\Off** option allows to control particular materials visualization. To specify the line colour for the current material, click the rectangle with mouse left button in **Display colour** field and select it from a combo box of standard colours. If it is necessary to set specification colour, push **Browse** button and select the colour in a standard Windows dialogue.

8.8.3. Holes and slots line colour setup

Holes and slots line colour setup control elements are placed in corresponding **Panel chart view parameters** dialogue sections. These settings function in the same way and they are described in holes setting example (fig. 8.9).

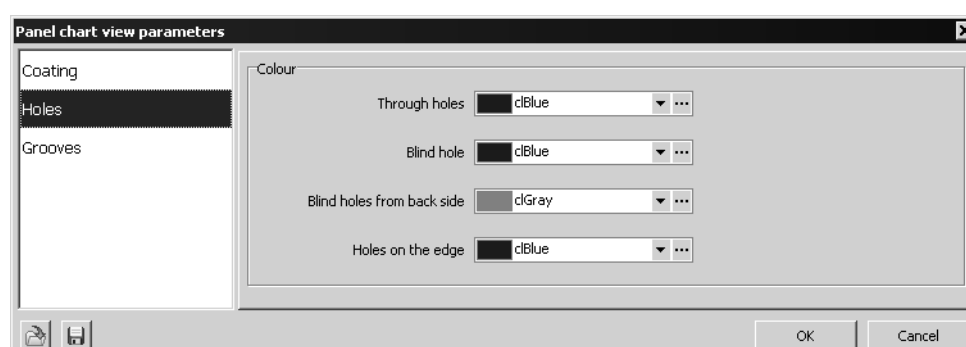


Fig. 8.9.

Line colour of each hole type is viewed in a rectangle. To change it, select it from standard colours combo box. If you need to set specific colour, push **Browse** button and select colour in a standard Windows dialogue.

8.8.4. Parameters saving and loading



Current parameters set can be saved. For this purpose push **Save** button. A warning will appear on the screen (fig. 8.10).

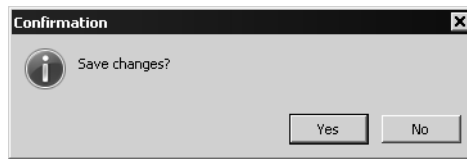


Fig. 8.10.

Message buttons allow to confirm saving or reject it. Set file name in a standard Windows dialogue. The file will have *pvconfig* extension.



To load parameters from a file, push **Load** button and select a file with *pvconfig* extension in a standard Windows dialogue. Setting parameters will be loaded from a file. To apply these settings reload the panels list. A hint will appear on the screen (fig. 8.11).

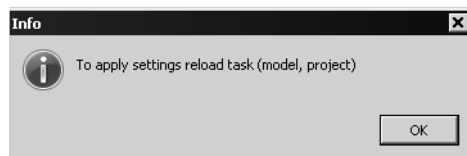


Fig. 8.11.



After parameters loading **Panel chart view parameters** dialogue will be closed automatically.

8.8.5. Setup completion

To complete panel visualization setup push **OK** button. **Panel chart view parameters** dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Chapter 9. Configuration parameters setup

Configuration parameters setting is on **Postprocessors** tab (fig. 9.1).

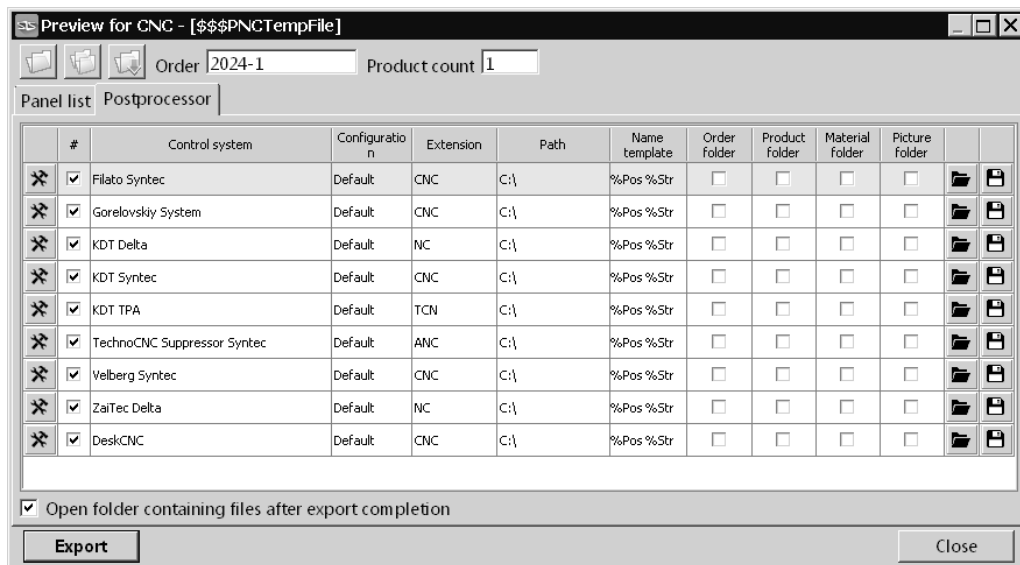


Fig. 9.1.

9.1. Configuration common parameters

9.1.1. Configuration attributes

Machine control system name and configuration name is shown for each configuration.

9.1.2. Program files path



You can view and edit root folder path for control program files saving in **Path** field. To set a path push **Browse** button and select a folder in **Browse** dialogue (fig. 9.2).



Fig. 9.2.

To put files storing in order you can create folders for files distribution by orders, products, materials inside root folder. Separate folder can be created for image files storing. For this purpose enable **Orders folder**, **Product folder** etc. options.

9.1.3. File name template



Control program file name is created automatically according to set template. Template content is shown in **Name template** field. To set a template push **Browse** button. **File name template** dialogue will appear on the screen (fig. 9.3).

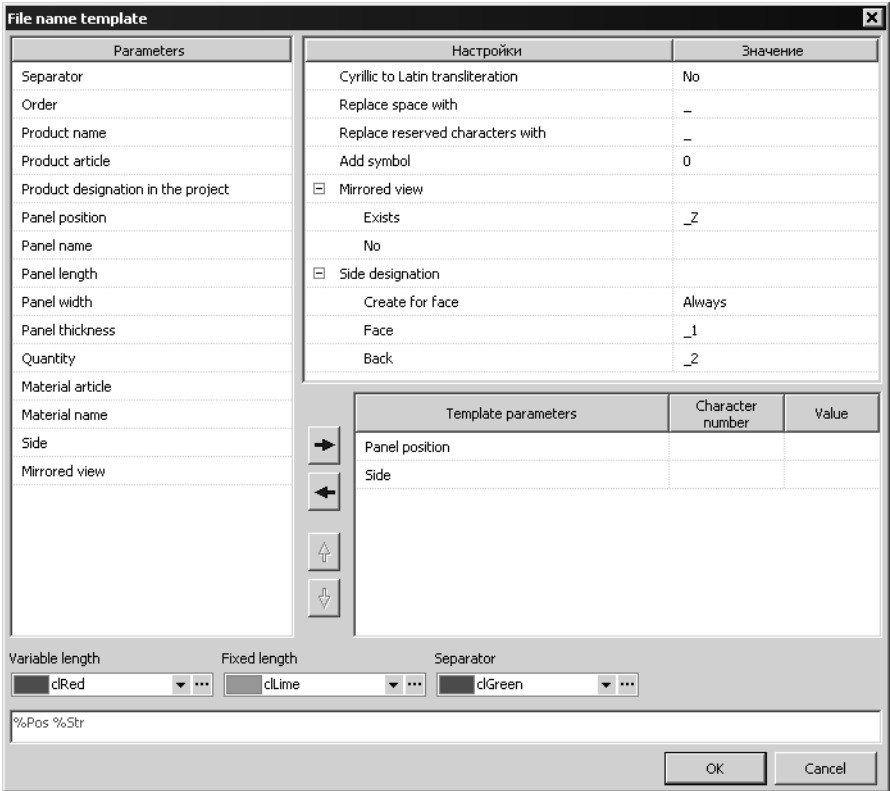


Fig. 9.3.

Template elements



Element names that can be included into a file name are shown in **Parameters** list. When you save files, these parameters values from a model will be used instead of them. To add an element to the template, select it in the list and push **Add parameter to template** button. An element will appear in **Template parameters** column. Set character count in **Character count** field.



Delete parameter from template button allows to delete selected element from a template.



Originally the elements are placed in **Template parameters** list in addition order. **Move up** and **Move down** buttons allow to move selected element along the list.



For *Separator* element **Value** field is available. You can input a symbol that separates other elements. *Serial Number* parameter is used for control programs file names creation taking into account the number of parts of one item.

Template setup

Cyrillic symbols can be automatically changed to Latin. For this purpose check in **Transliteration to Latin** field. Space in file name is unwanted, some symbols can not be used because of Windows file system restrictions. **Replace spaces with** and

Replace reserved characters with fields allow to set the symbols for their replacement. If symbols count is set for parameter saving and parameter value length is less than this count, additional symbols will be added to a file name. By default zeros will be added. **Add symbol** field allows to set another additional symbol.

If it is necessary to process both panel faces, two files of control programs will be created. Side designation may be one of file name elements. **Create for face** combo box variants in **Side designation** group allow to choose designation creation condition. Set corresponding designations in **Face** and **Back** fields. There is created template viewing string in the lower dialogue part. To can choose colour for template elements with different parameters visualization. Individual colour can be set for variable length parameters, fixed length (with set **Character count** parameter) and for separator. Select colour in corresponding combo boxes. If necessary you can select additional colours. For this purpose push **Browse** button and select colour in a standard Windows dialogue. To complete file name template setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.



9.2. Parameters saving and loading



A set of current configuration parameters can be saved in a file. Push **Save configuration parameters** button and set a file name in a standard Windows dialogue. A file will have *cncconfig* extension.



To load a set of parameters from a file, push **Load configuration parameters** button and choose a file with *cncconfig* extension in a standard Windows dialogue.

9.3. Data reading setup



Configuration parameters button allows to set current configuration parameters. Data from a model reading parameters are the same for all control systems. Technological parameters setup is specific for each control system and it is described in Chapter 10 on p. 63. Model reading parameters are set on **Reading parameters** tab of **Configuration parameters** dialogue.

9.3.1. Common parameters

Common section control elements allow to set panels reading parameters (fig. 9.4).

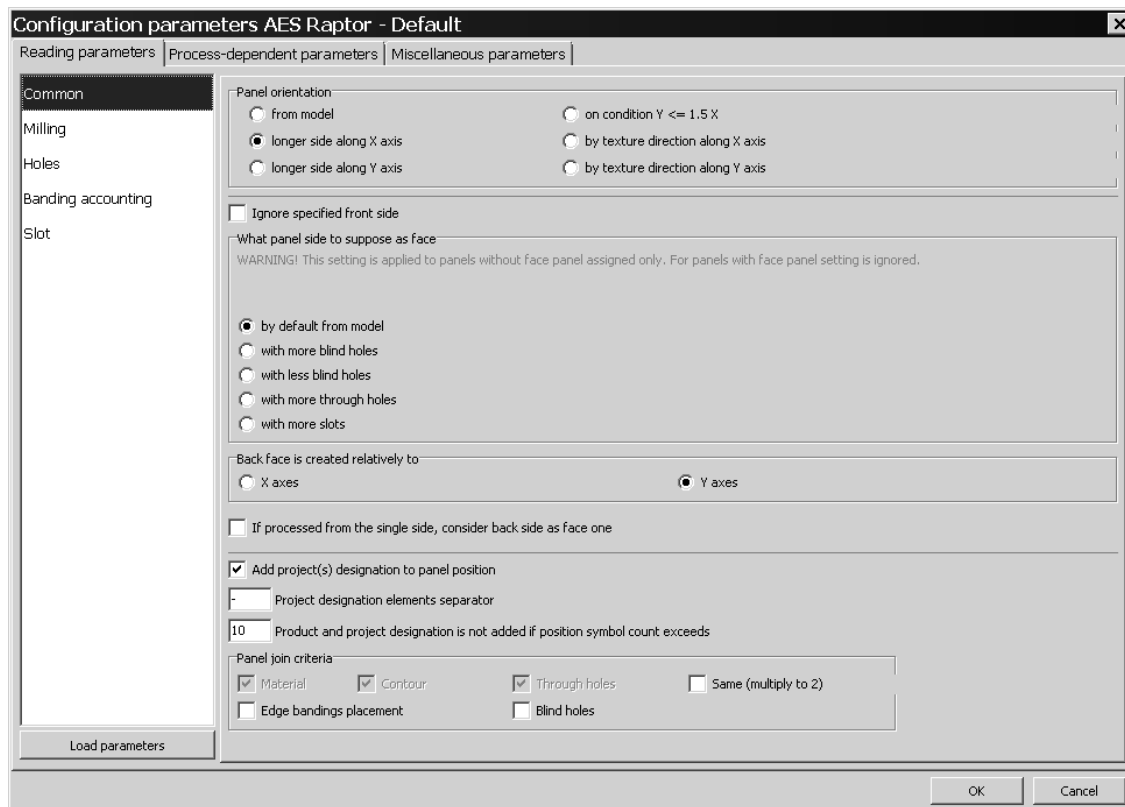


Fig. 9.4.

Panel orientation

Control program is created for particular panel position on machine fences. It is necessary to take into account machine desktop dimensions and configuration and panel overall size. To set panel position use **Panel orientation** group variants.



Setting data work for all panels of the model. Panels orientation can be changed (see section 8.7 on p. 48).

If **From model** variant is selected, panel position on a desktop will correspond to its position in a model.

Longer side along X axis variant allows to place the panel in a way, that its long side will be horizontally placed.

Longer side along Y axis variant allows to place the panel in a way, that long side will be vertically placed.

On condition $Y \leq 1.5 X$ variant allows to place the panel in a way, so that the side that is one and a half times larger than the other side is positioned vertically.

By texture direction along X axis and **along Y** variants allow to take material texture direction into account. The panel will be placed in a way, that its texture will be parallel to selected axis.

Front side selection

The panel is arranged on machine desktop in a way, that processed surfaces will be available for a tool. If slots or blind holes are placed on both panel faces, two control

programs will be created for them. Control programs design job will contain two images of such panel. One of the panel faces is called front.

Panel face side choice criteria group variants allow to assign the panels that will be considered front panels automatically.

If **Face side specified** variant is selected, then front side is assigned the area that is assigned in the panel properties. If **By default from model** variant is selected, front side will be the face that is arranged in the following ways in a model:

- ▼ horizontal – above,
- ▼ vertical – from the right,
- ▼ front – from the front.

More blind holes, Less blind holes, More through holes, More slots variants allow to assign a front panel according to selected criterion. **Up** and **Down** buttons allow to move criteria through the list. The panel sides are checked for compliance with the criteria in turn, according to the list. The check is terminated if the side meets the next criterion.



The side of through hole boring is set at accessories mounting and editing commands. Accessories work is described in *BAZIS-Woodworker module User Guide*.

Back face position



If there are two panel images in panels list, panel back side image will be marked with a sign in **Back face** cell. The panel should be rotated to the user from the processed side. Initially the panel is arranged in a way as it is placed in a model. If processed side is turned up, no extra actions are required. If it is turned down, the panel should be turned over, i.e. visualize its opposite side. **X axis** and **Y axis** variants of **Back face is created relatively to** group allow to select an axis for panel rotation. **If processed from the single side, consider back side as face one** option allows to exclude back face symbol from side designations, even if this side is such.

Panel joining

Parts that are mirror image of each other can be joined for processing. One of the panels is placed on another. **Panel join criteria** group control elements allow to select parameters, if they coincide the panels will be joined automatically. **Same (multiply 2)** option allows to join the same panels, if their number is a multiple of two.

Project panels designation

If control program is created for project with several products, it is reasonable to add project and product name to it at automatic panel designation creation. For this purpose use **Add project(s) designation to panel position** option. If you disregard it, panels list will contain panels with the same designations.

Project designation elements separator field allows to select corresponding symbol. Too long position designation can be rather inconvenient. In this case you can reject supplementary information automatic adding to its designation. **Product and project designation is not added if position symbol count exceeds** field allows to set position designation maximum size.

9.3.2. Milling parameters

Milling section control elements (fig. 9.5) allow to set milling parameters.

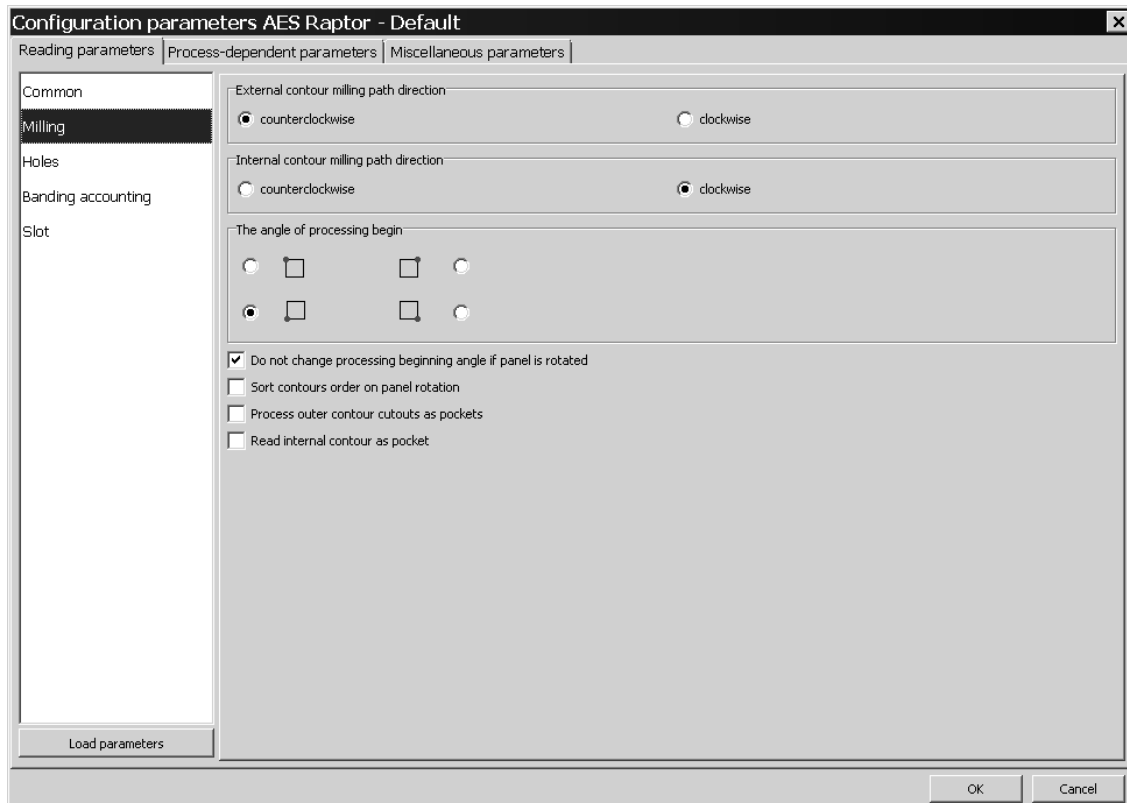


Fig. 9.5.

Path direction

External contour milling path direction and **Internal contour milling path direction** groups variants allow to select the corresponding milling directions.

Processing beginning

The angle of processing begin group variants allow to specify a panel angle for milling begin.

If the panel is rotated or mirrored, angles position will be changed. **Do not change processing beginning angle if panel is rotated** option allows to control selected angle fixing.

9.3.3. Hole parameters

Holes section control elements allow to set panel holes processing (fig. 9.6).

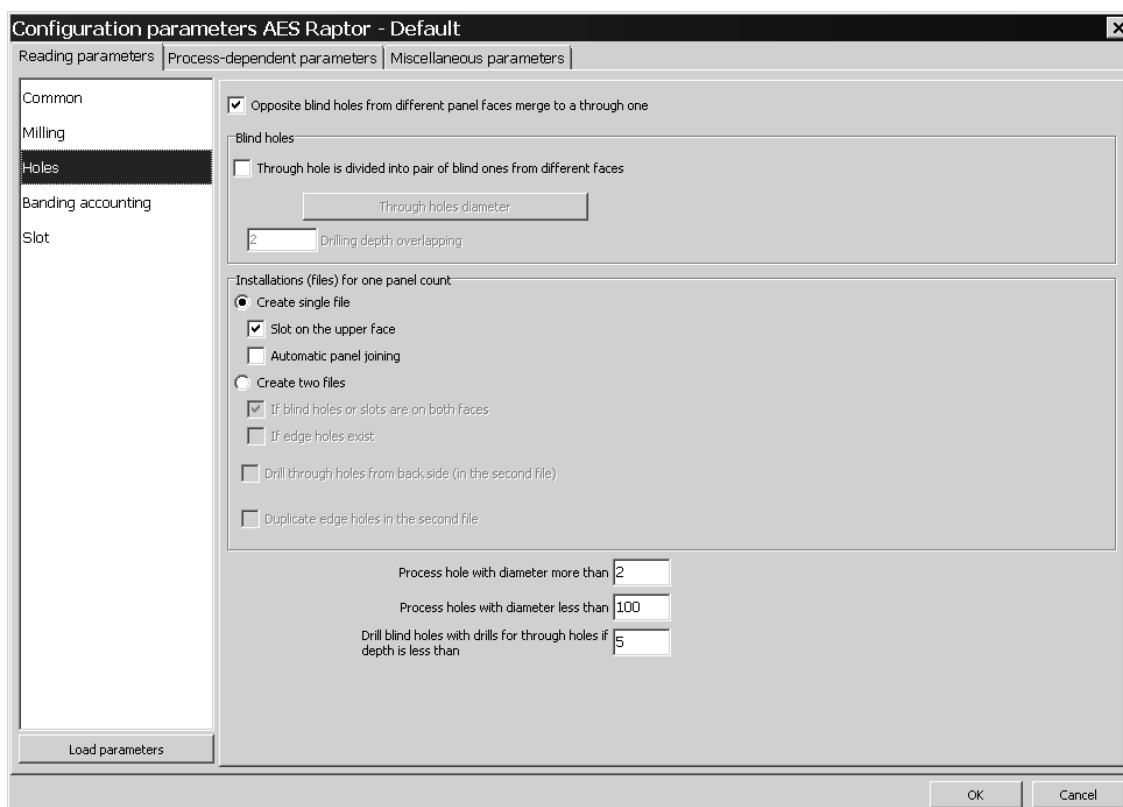


Fig. 9.6.

If **Opposite blind holes from different panel faces merge to a through one** option is enabled, coaxial blind holes that intersect by depth will be merged to through hole. It slows to bore only from one face. Boring may lead to splinters on the face back side, especially if a drill is dull. **Through hole is divided into pair of blind ones from different faces** option allows to bore different faces by-turn without damage. Through hole division into two blind holes is not obligatory for all holes. To specify the diameter of holes for boring from both sides push **Through holes diameter** button. It is available if the option is enabled. After pushing the button, **Through holes diameter** dialogue will appear on the screen (fig. 9.7).

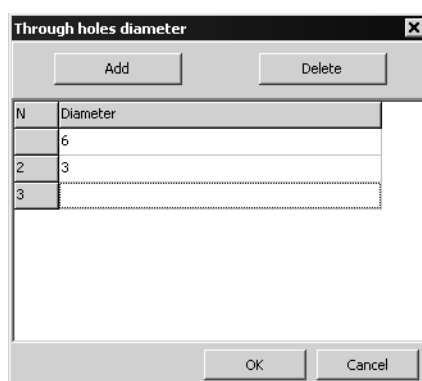


Fig. 9.7.

Diameter field allows to set diameter value of divided hole. To add a blank line to the table push **Add** button. **Delete** button allows to delete the current diameter table line. To complete diameter list of divided holes creation push **OK** button. **Cancel**

button allows to close the dialogue without changes. **Drilling depth overlapping** field allows to set holes bore depth overlapping that ensures through hole creation. Machines may bore from one side as well as from both panel sides. These peculiarities specify control programs creation order. If boring is executed from one side, it is necessary to create control program for boring from both sides. Two separate files will be created. To set such control programs creation order select **Create two files** variant in **Installations (files) for one panel count** group. If you enable **If edge holes exist** option, additional condition of two files creation will be set.

If two files creation variant is selected, **Drill through holes from back side (in the second file)** option will become available. It allows to add operations of through holes boring to the second program part, i.e. from back panel face. **Duplicate edge holes in the second file** option allows to duplicate edge holes.

Machine may stop the operation so that the operator can turn over the panel for second side processing. In this case one program for both faces processing will be created. To set such order of control programs creation, select **Create single file** variant. This variant should be selected if machine may process both faces at once.

There can be holes of different diameters in a model. Some of them, e.g. nail holes, are designated on a drawing, but they are not bored. **Process hole with diameter more than** field allows to set minimum holes diameter value. **Process holes with diameter less than** field allows to set holes diameter maximum value. **Drill blind holes with drills for through holes if depth is less than** field allows to set blind holes depth for drills for through holes use.

9.3.4. Banding accounting

Banding accounting section control elements (fig. 9.8) allow to control panel edge banding programs accounting.

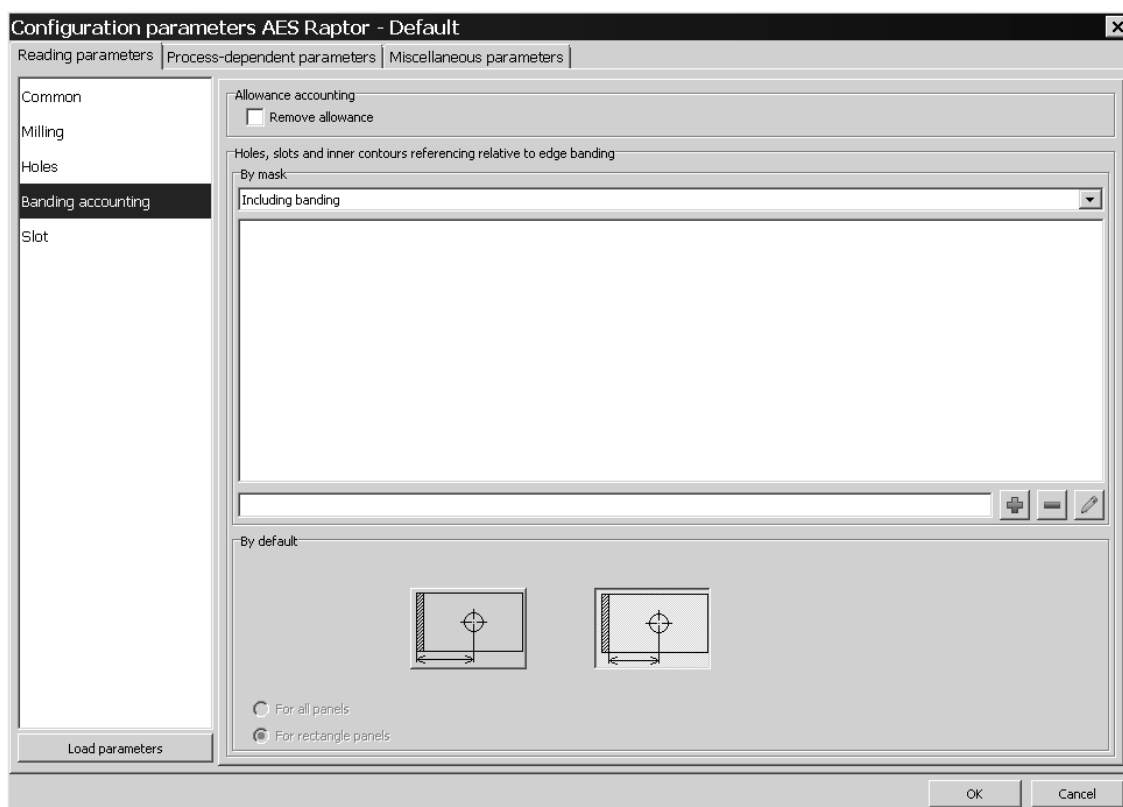


Fig. 9.8.

Remove allowance option allows to add allowance remove on panel edges operations to control programs. If it is enabled, allowance will be removed in further technological operations. Panel can be processed on CNC machine with banded edges. Thus, holes, slots and panel internal contours may be referenced from panel bound or from banded edge bound. To select a necessary variant push a button with a corresponding icon in **By default** group. If referencing is executed from banded bound, you can specify the panel types for this setting application. For this purpose use **For all panels** or **For rectangle panels** variants.

9.3.5. Slots analysis

Slots section control elements (fig. 9.9) allow to set slots trajectory construction on a panel.

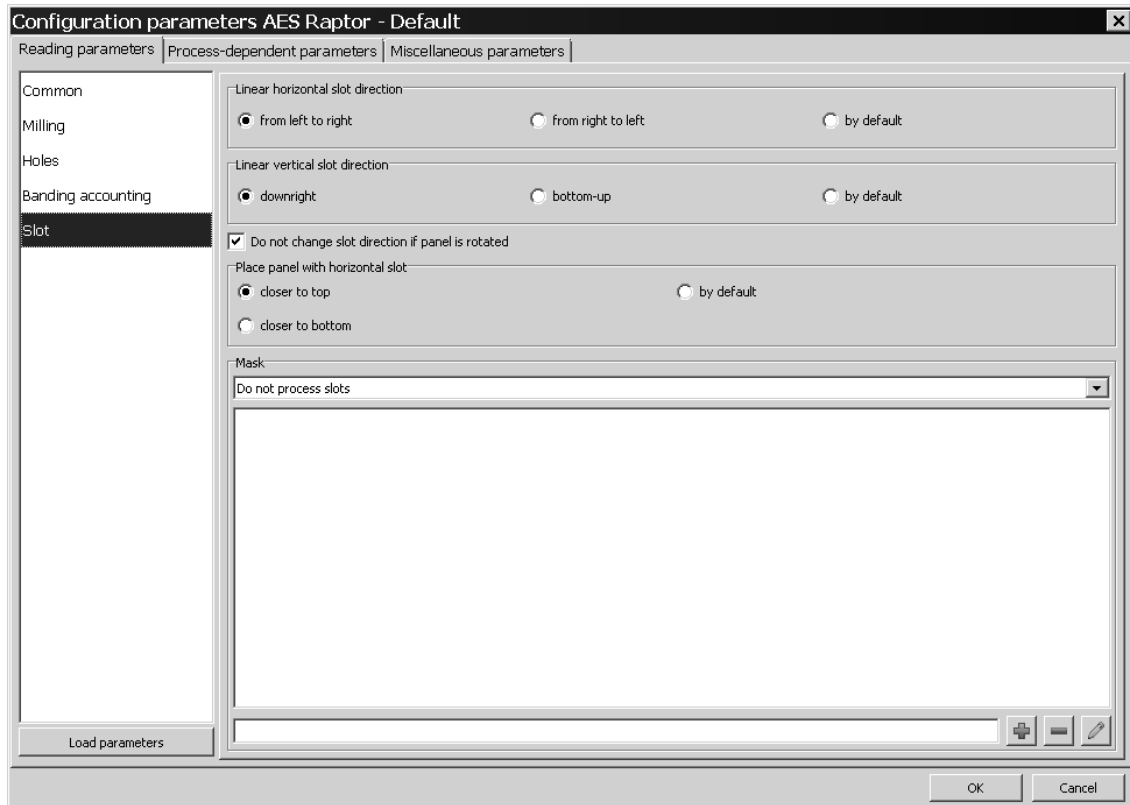


Fig. 9.9.

Linear horizontal slot direction and **Linear vertical slot direction** groups variants allow to select corresponding slots processing direction. If the panel is rotated or mirrored, slot processing direction will be changed. **Do not change slot direction if panel is rotated** option allows to control selected direction fixing. The panel can be rotated in consideration of slot position. **Place panel with horizontal slot** group variants allow take this parameter into account. If **by default** variant is selected, slot position will not be taken into account, the panel will be placed according to other reading settings.

Slots can be excluded from processing or included into it by slot name mask. To create a mask of names selection input it in a field. You can use substitutes in mask. Asterix <*> replaces any number of symbols. There can be several masks. To add a mask to the list, push **Add** button.

Delete button allows to exclude a mask from the list.

To edit selected mask push **Edit** button.



9.4. Check operations realizability

9.4.1. General information

Operations realizability check setup relates to other parameters and is performed in **Satisfiability check** section on **Miscellaneous parameters** tab (fig. 9.10). This setup is the same for all control systems.

The check allows to specify the ability to create control programs depending on configuration of machine work area and panel size. If control programs creation for current parameters is impossible, a warning will appear in module window, and panels designations that cause the errors will be highlighted. To eliminate the errors, change the corresponding parameter. For example, if the current position of the panel does not fit on the desktop, rotate it in the corresponding way (see section 8.7 on p. 48).

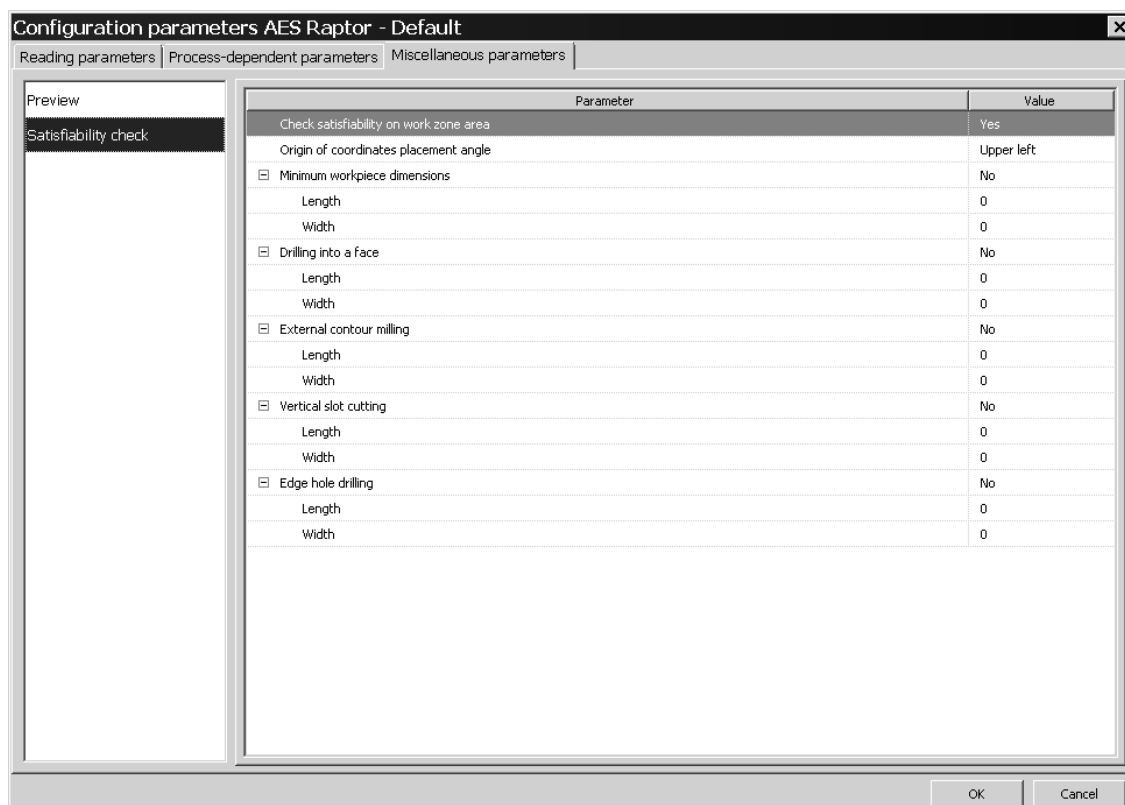


Fig. 9.10.

9.4.2. Checked operations

Check satisfiability on work zone area option allows to control check performing. Options in operations names, e.g. **Drilling into a face**, allow to include operations in the number of checked ones.



Checked operations content depends on the design and technical capabilities of a particular machine. That is, if the machine is not designed for milling, this operation will be omitted from the list.

9.4.3. Checked dimensions

Panel size

Length and **Width** fields of **Minimum workpiece dimensions** group allow to set minimum panel dimensions that can be processed on the machine. To check the panel, you should enable the corresponding option.

9.4.4. Reference point position

Reference point position of machine work area is determined by its design. It can be located at one of the vertexes of overall rectangle. During control programs creating dimensions are based from this very point. The position of origin of coordinates is taken into account when checking the satisfiability of operations. To set its position, select one of **Origin of coordinates placement angle** group variants.

9.5. Panel image saving parameters

Panel images can be saved to files. Saving parameters setup is performed in **Pre-view** section on **Miscellaneous parameters** tab (fig. 9.11). This setting is the same for all control systems.

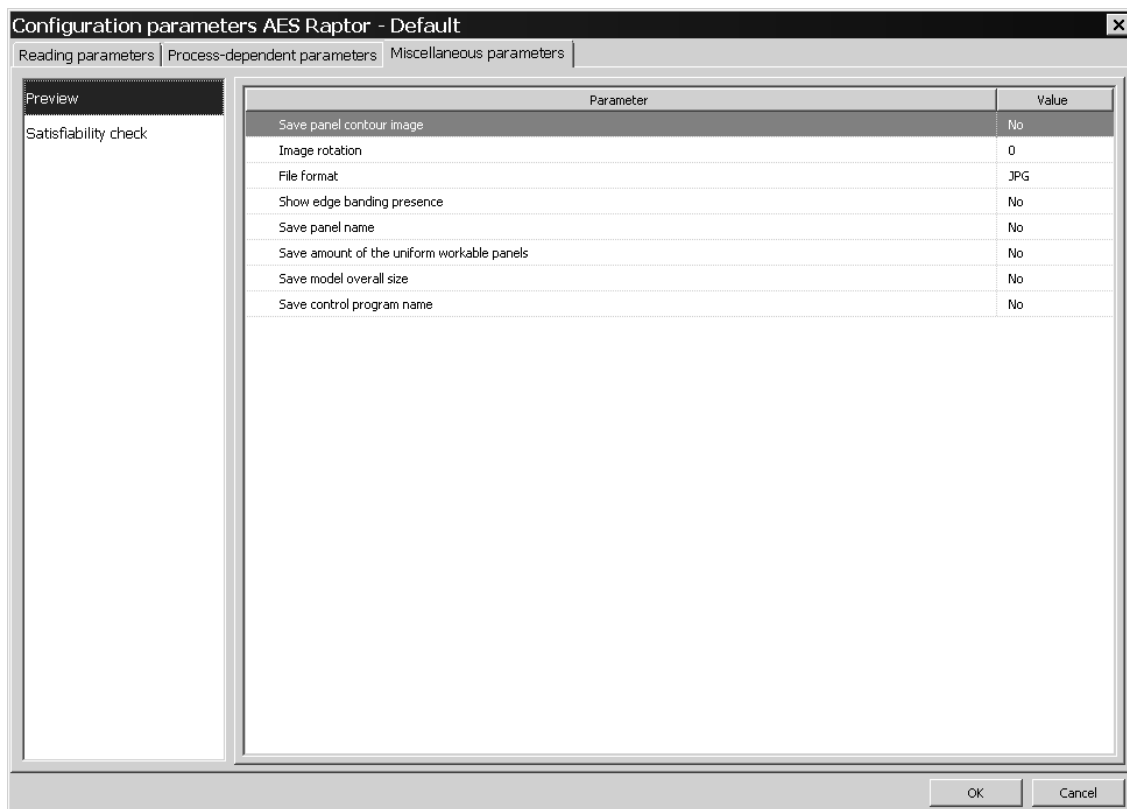


Fig. 9.11.

To save panel images, enable **Save panel contour image** option. **Image rotation** combo box allows to select image rotation angle, 0° or 90°. **File format** combo box allows to select image saving format, JPEG or WMF. **Show edge banding** option allows to control banding visualization in saved images. **Save control program name** option allows to add control program file name to panel image.

Chapter 10. Process-dependent parameters setup

10.1. General information

Set parameters for each machine for automatic control programs creation. Programs for some machine types can be created in two formats, e.g. in machine control program format and in AutoCad format. This format can be read by control system but some parameters can be absent, e.g. correction type, etc. To set program parameters use control elements on **Process-dependent parameters** tab of **Configuration parameters** dialogue.

10.2. Saved parameters loading

Load parameters button is available for all control systems. It allows to load configuration parameters generated in BAZIS-CNC of previous version. After pushing the button select a file with *npr* extension in a standard Windows dialogue.

10.3. AES Group

10.3.1. AES Raptor



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.1) allow to specify the operations that can be used in control program.

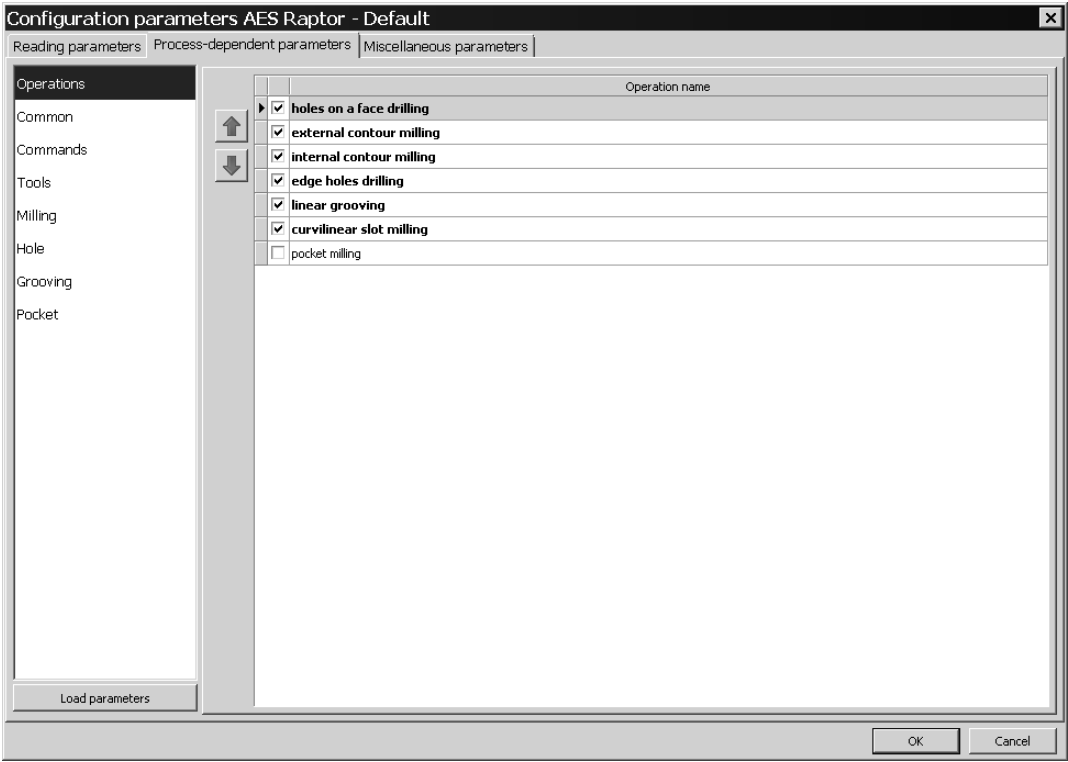


Fig. 10.1.

Common parameters

Common section control elements (fig. 10.2) allow to set common parameters of control programs generation.

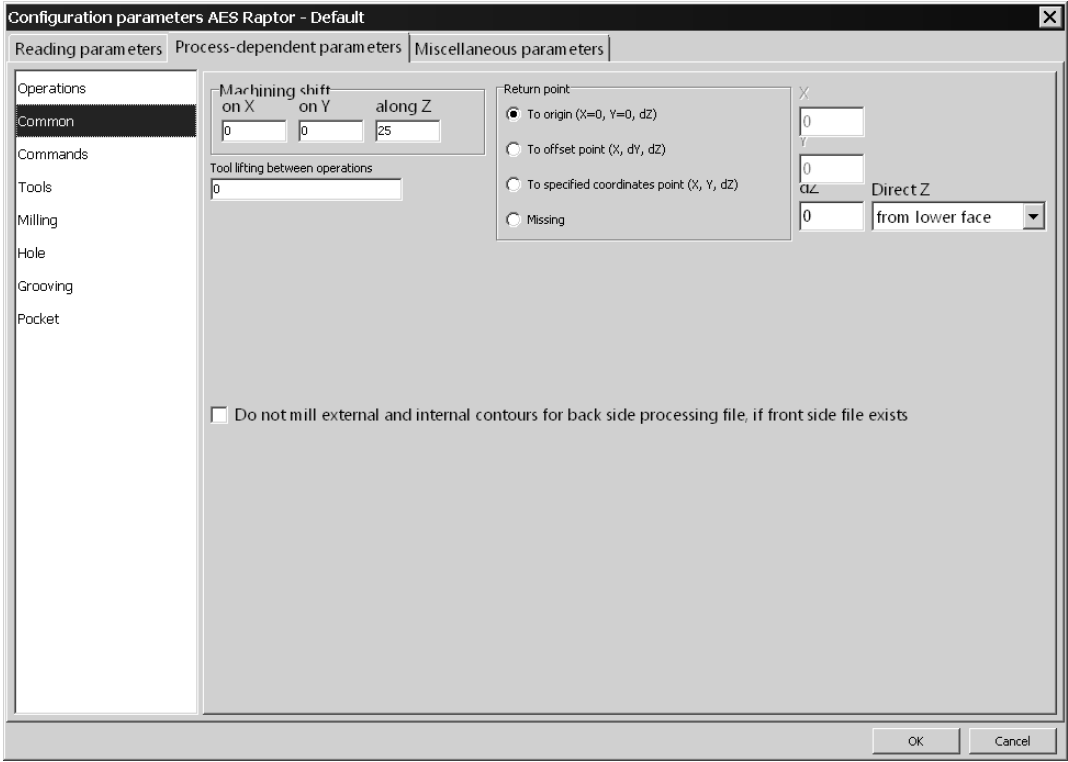


Fig. 10.2.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height.

Usable commands

Commands section control elements (fig. 10.3) allow to control separate commands to control program inclusion.

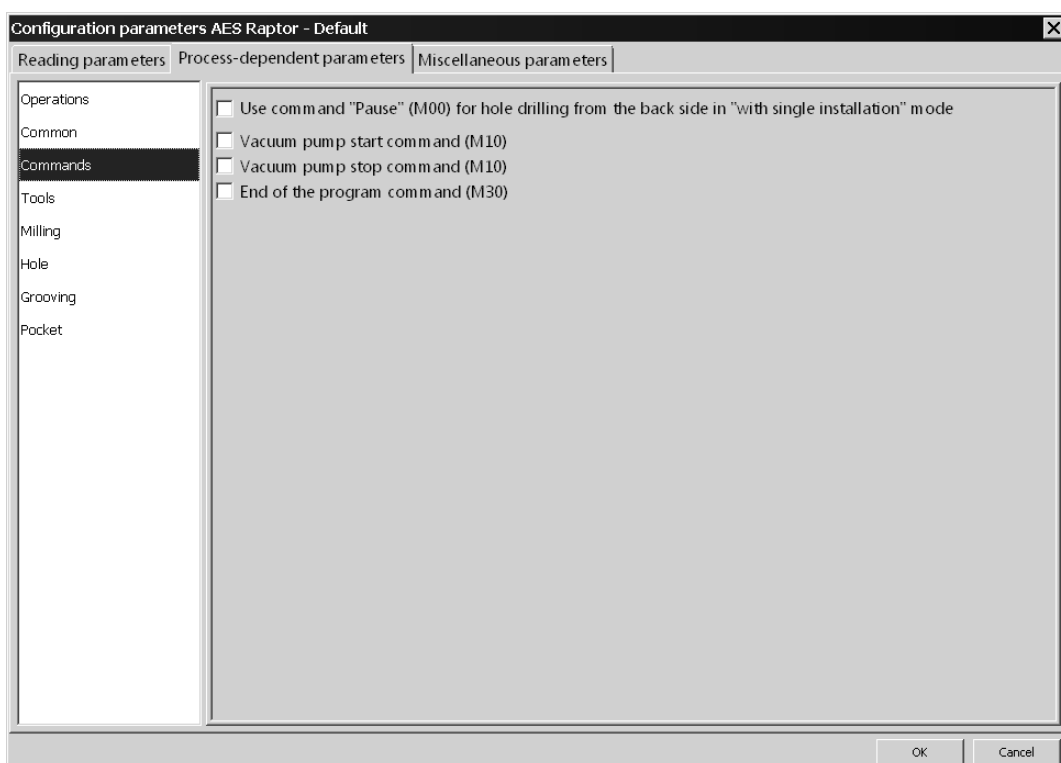


Fig. 10.3.

Tools parameters

Parameters setup of usable tools is performed in **Tools** section (fig. 10.4).

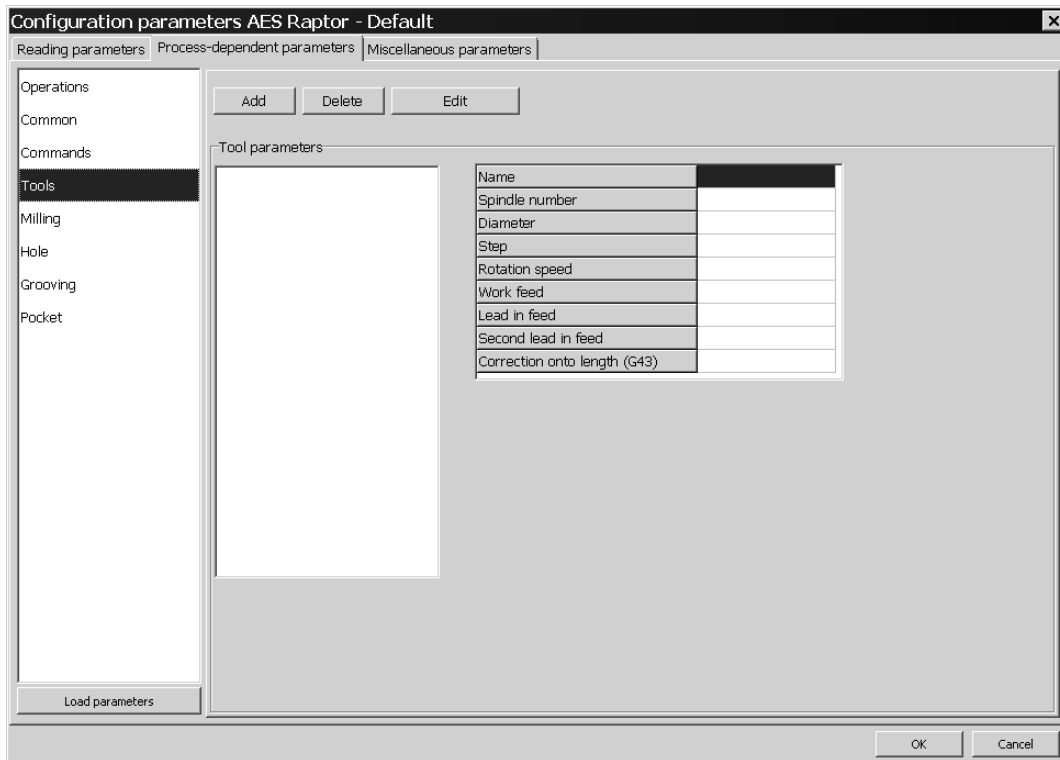


Fig. 10.4.

Dialogue table contains parameters values of selected tool. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.5).

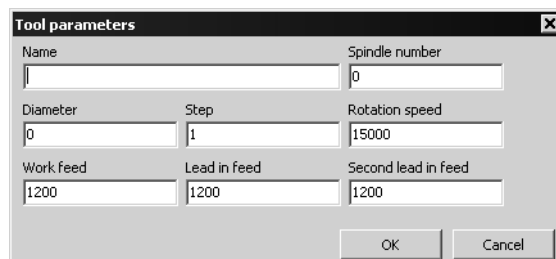


Fig. 10.5.

The dialogue control elements allow to set tool parameters values. To complete the adding, push **OK**. Tool name and its diameter should be sure input. Otherwise a warning will appear on the screen. The message example is shown in fig. 10.6.

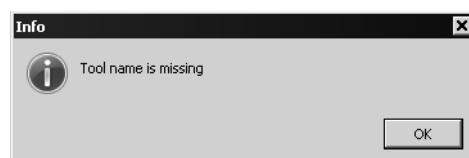


Fig. 10.6.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.5). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.7).

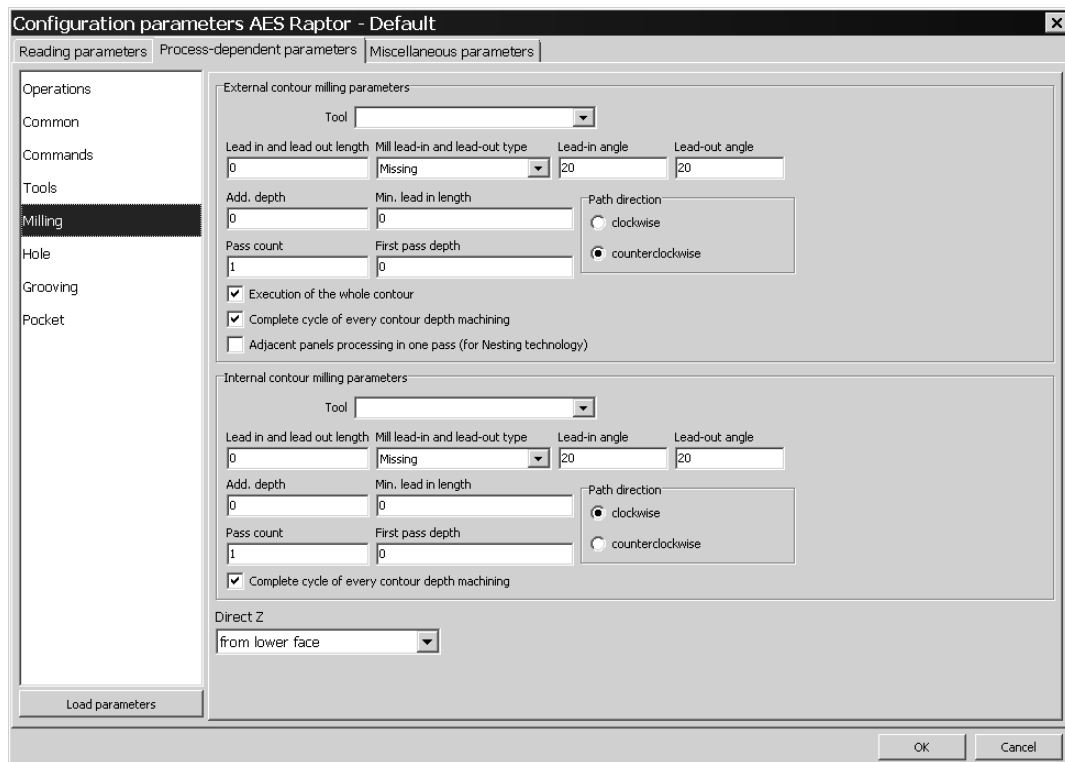


Fig. 10.7.

Tool combo box allows to select a tool milling from the list prepared on **Tools** tab. Tool lists should be created in **Tools** section. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Path direction** group variants of outer and internal contour allow to select tool moving direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.8).

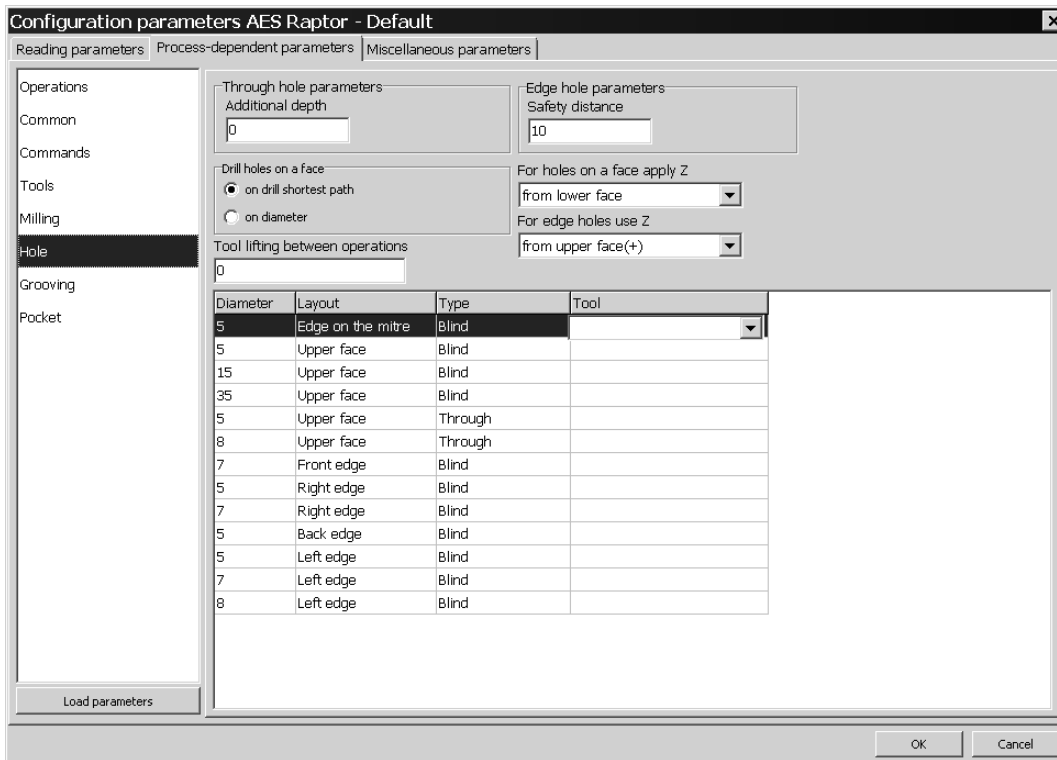


Fig. 10.8.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Slots processing parameters

Grooving section control elements (fig. 10.9) allow to set slots processing parameters.

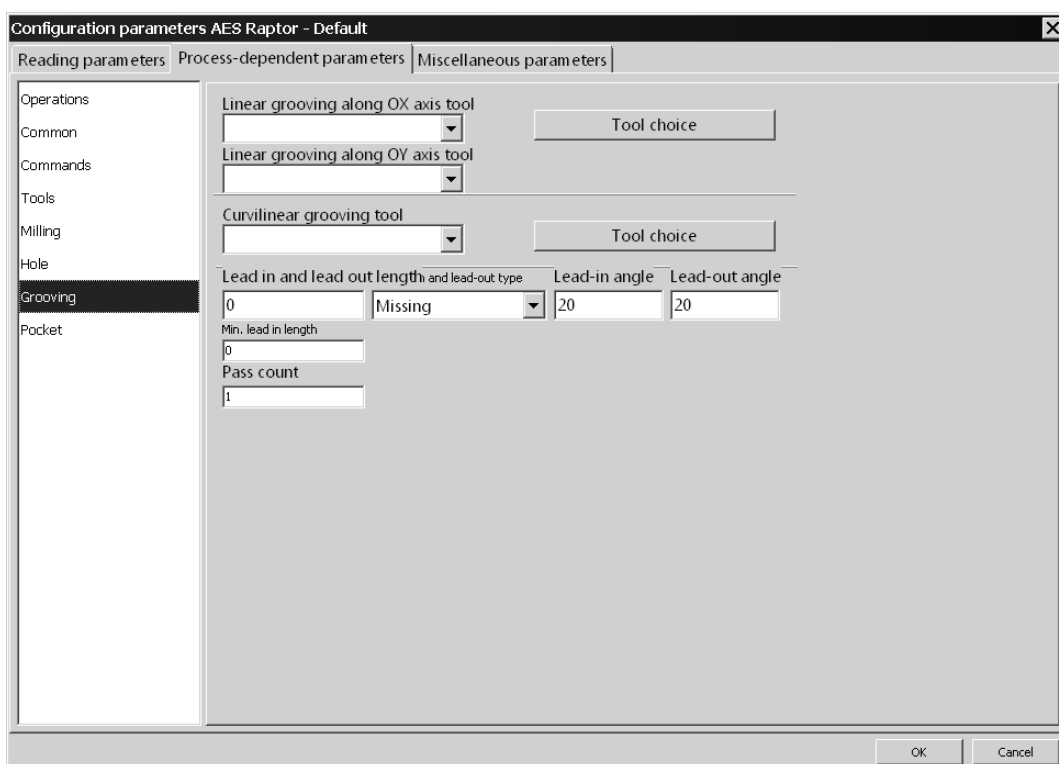


Fig. 10.9.

Combo boxes in this section allow to select tool names that will be automatically used for linear slots of X and Y axes processing and curvilinear slots. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to assign tools that will be used for particular slots. After pushing the button, **Tool choice** dialogue will appear on the screen (fig. 10.10).

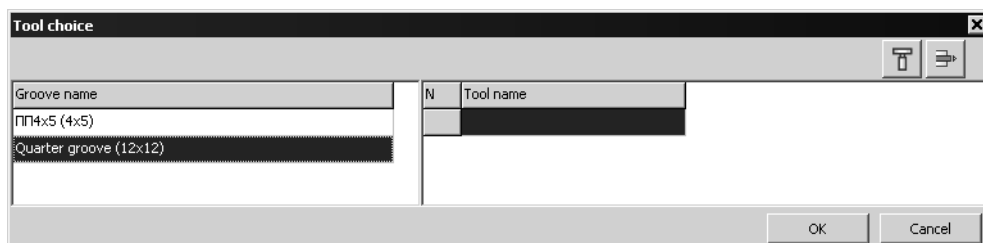


Fig. 10.10.



Slot name table contains the names of all slots that are present in the model. For each slot processing you can assign particular tools. **Add mill** button allows to assign a tool for slot from the list created in **Tools** section.



Delete button allows to delete selected tool from the list. To finish tools assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.11) allow to setup parameters of pocket processing.

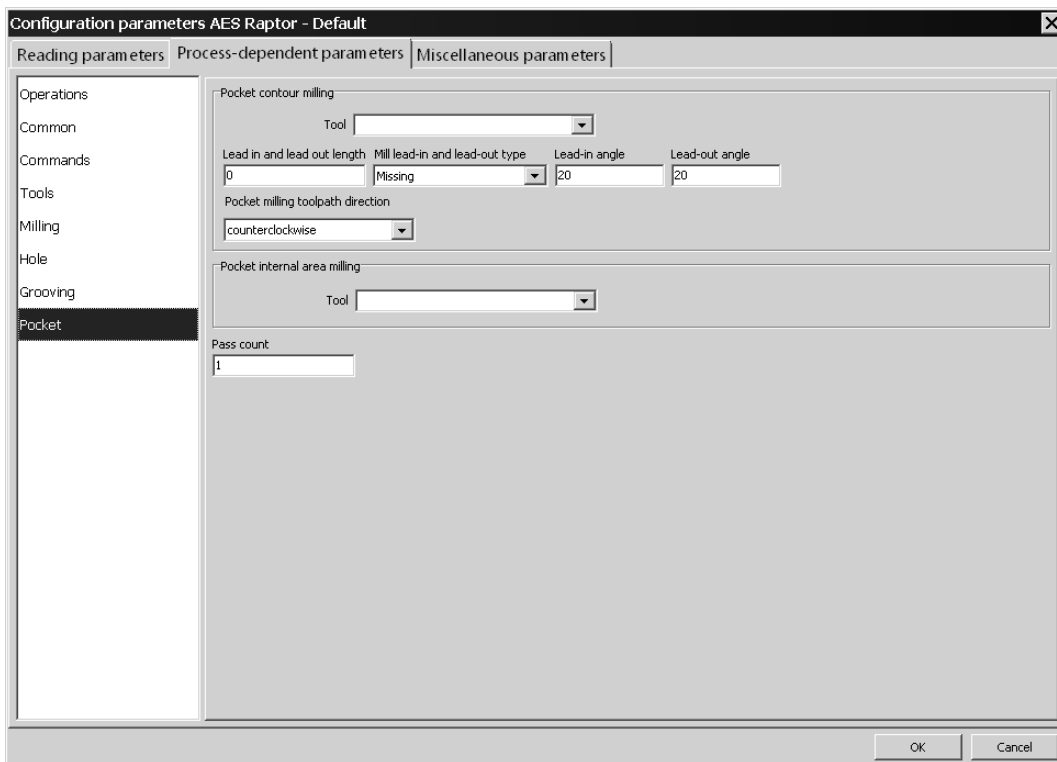


Fig. 10.11.

Section control elements allow to set generation parameters of control programs for contour and outer and internal pocket processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour processing and pocket internal area are selected from the list, created in **Tools** section.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.3.2. AES Drill Macro



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.12) allow to choose the operations that can be used in control program.

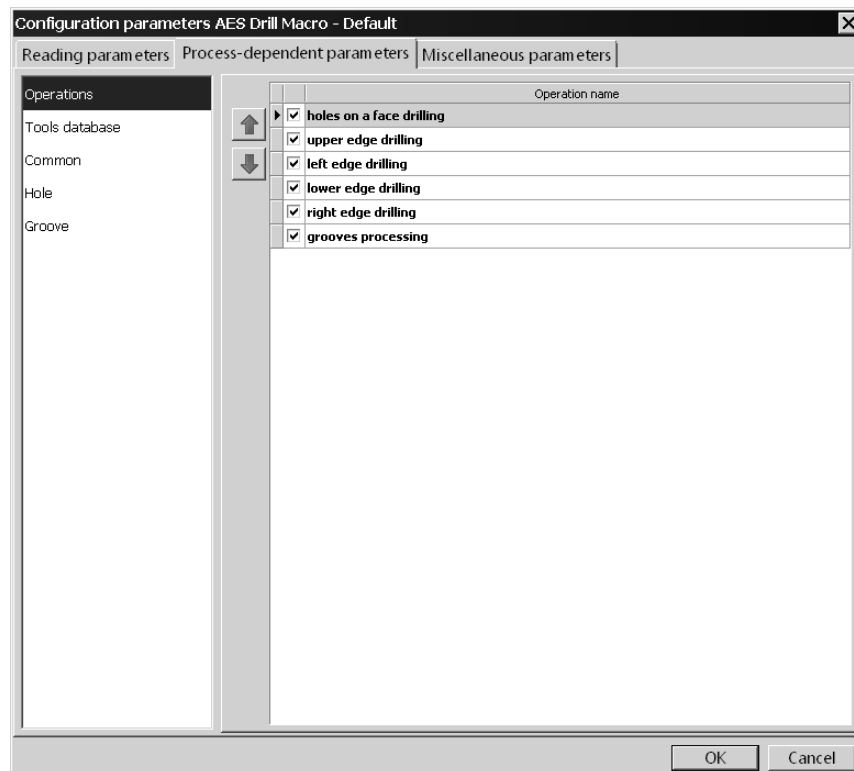


Fig. 10.12.

Usable tools

Tools database section control elements (fig. 10.13) allow to create a list of tools available for use in control programs.

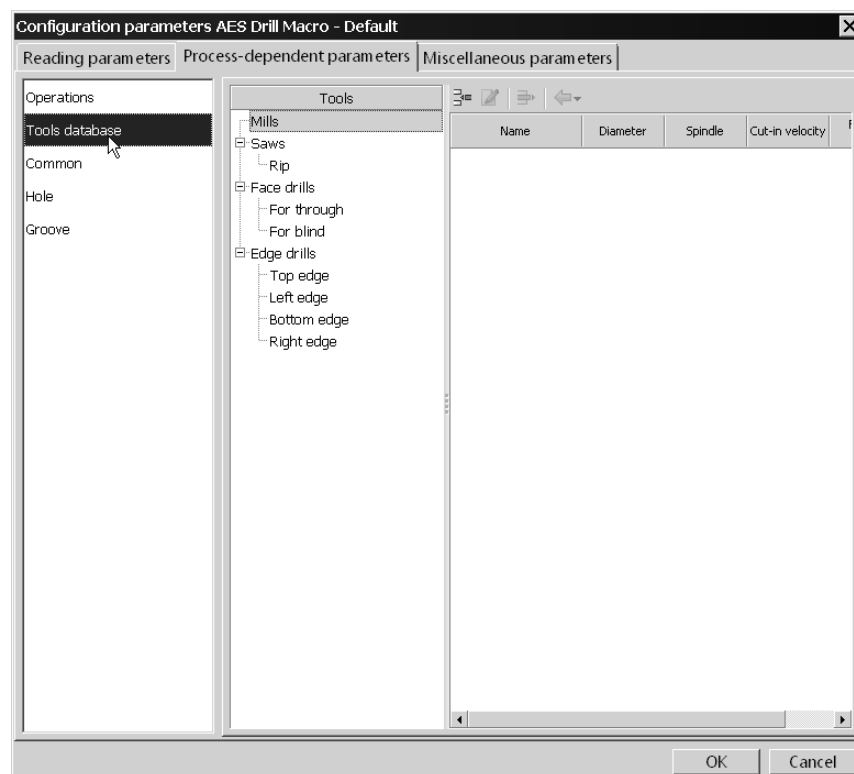


Fig. 10.13.



To add a tool to the list push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.14).

Tool	
Name	
Diameter	0
Width	0

OK Cancel

Fig. 10.14.

In the dialogue enter a tool name and set its parameters values. To finish tool adding, push **OK** button. The dialogue will be closed, a tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change selected tool parameters.



Delete button allows to delete selected tool from the list.



Move to group button allows to move selected tool to the group. After pushing the button a tool group tree will appear on the screen (fig. 10.15).

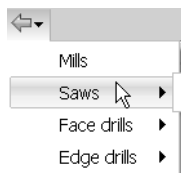


Fig. 10.15.

To move a tool to the group, doubleclick on its name with mouse left button.

Common parameters

Common section control elements (fig. 10.16) allow to set control programs common parameters.

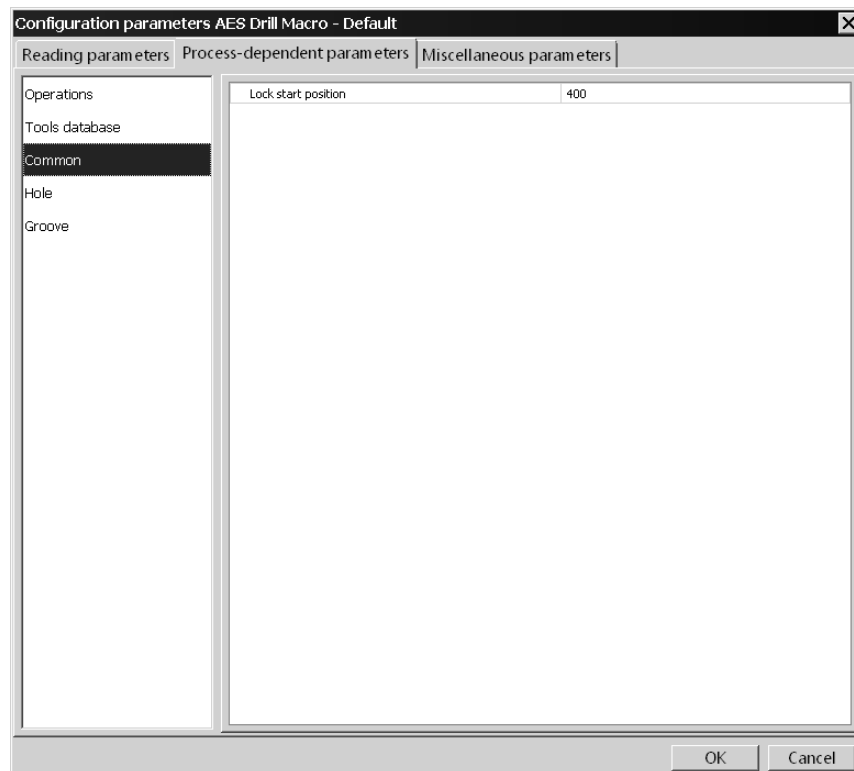


Fig. 10.16.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.17).

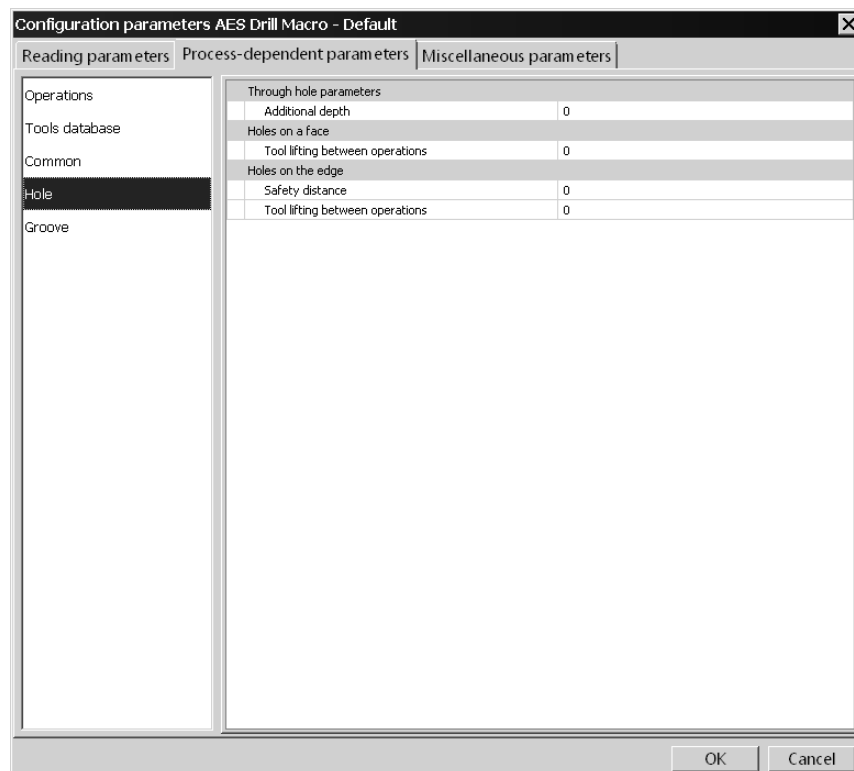


Fig. 10.17.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove parameters

Groove processing setup is performed in **Grooving** section (fig. 10.18).

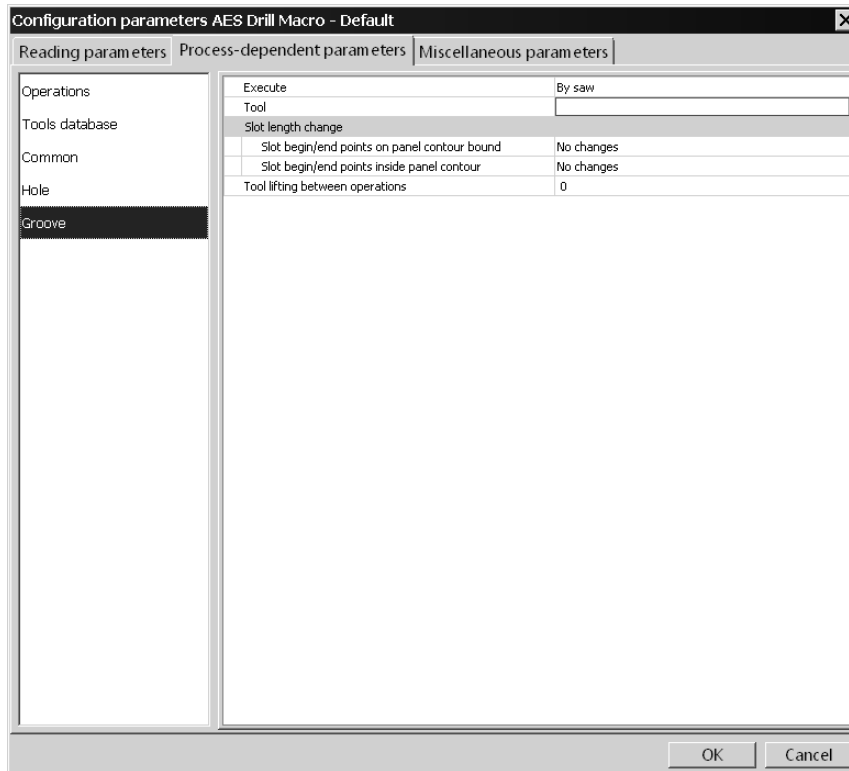


Fig. 10.18.

Section control elements allow choose groove processing tool, a tool from database and tool parameters.

10.3.3. AES Sirius



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.19) allow to specify operations that can be used in control program.

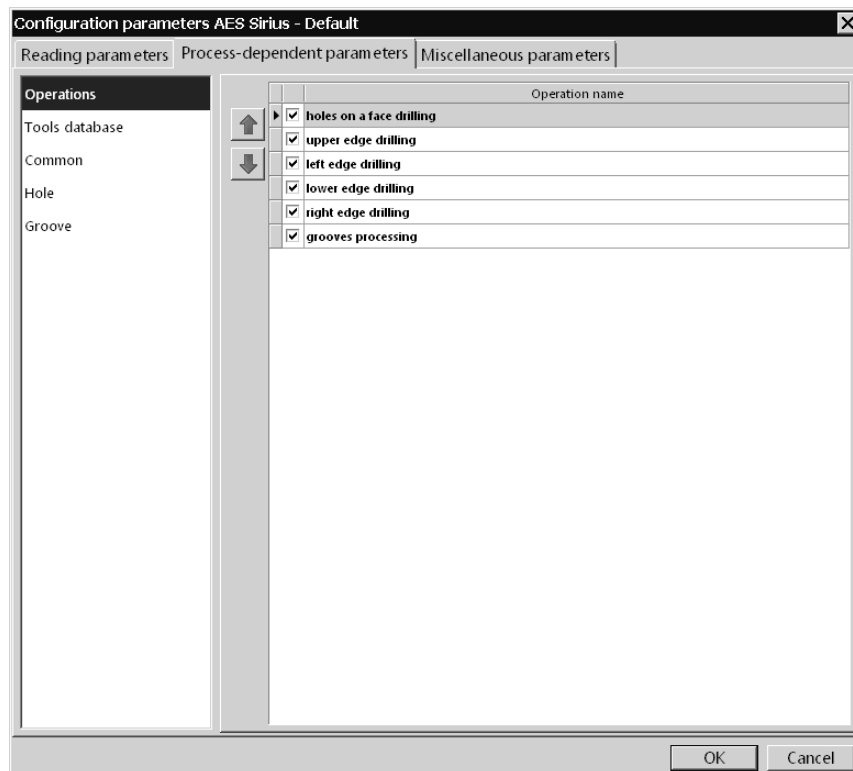


Fig. 10.19.

Usable tools

Tools database section control elements (fig. 10.20) allow to create a list of tools available for use in control programs.

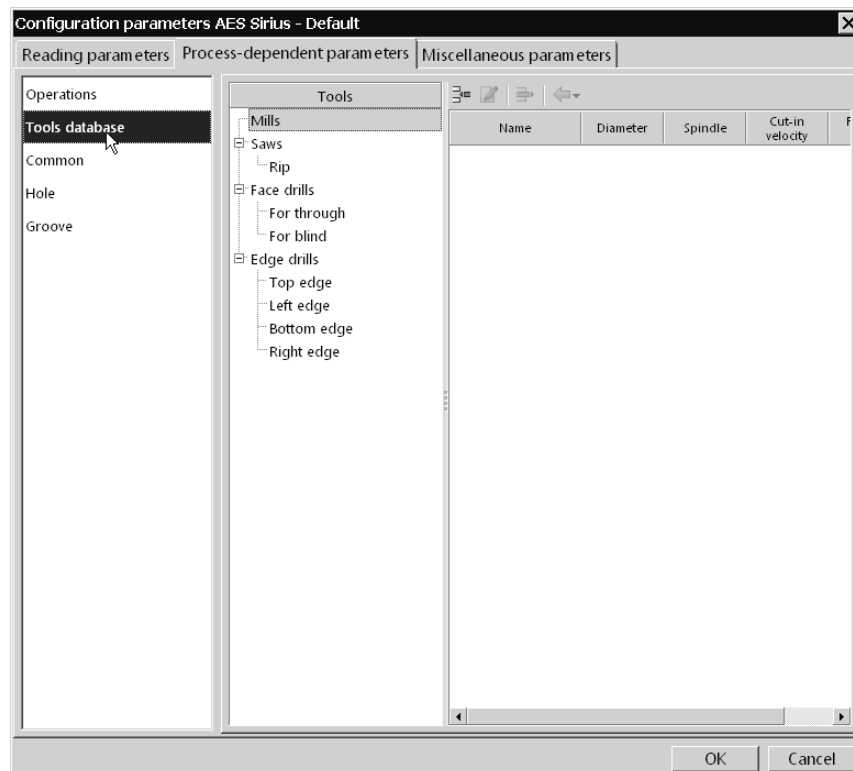


Fig. 10.20.



To add a tool to the list push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.21).

Name	
Diameter	0
Spindle	0
Cut-in velocity	0
Feeding speed	0

Fig. 10.21.

In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes.



Edit button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.



Move button allows to move selected tool to another folder. After pushing the button, a tools tree will appear on the screen (fig. 10.22).

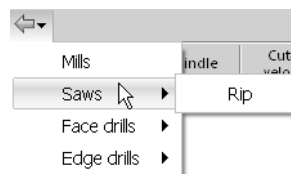


Fig. 10.22.

To move a tool to the folder, doubleclick on its name.

Common parameters

Common section control elements (fig. 10.23) allow to setup common parameters of control programs.

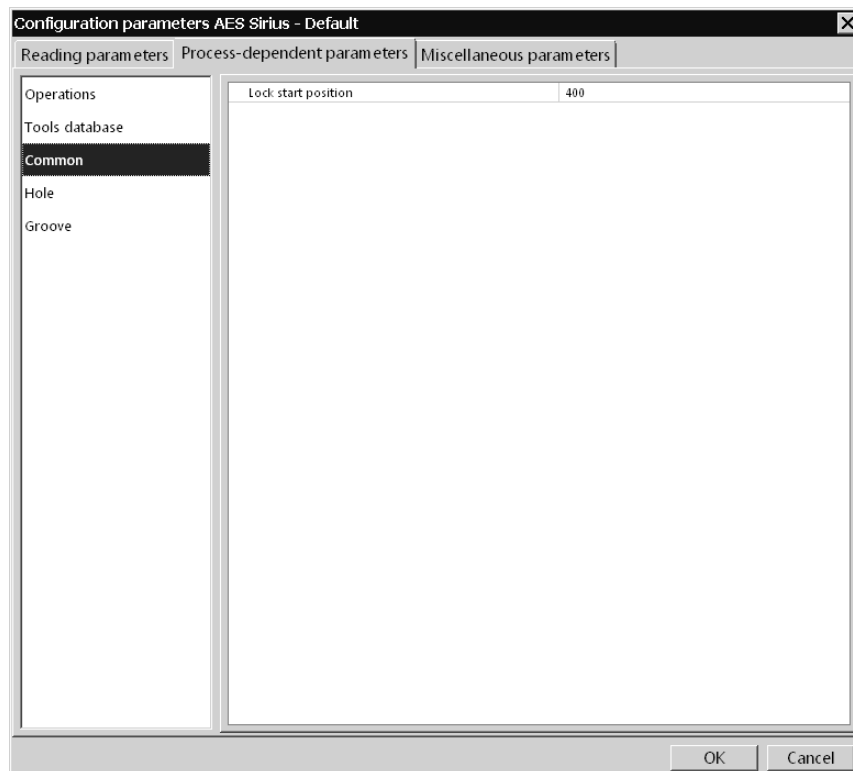


Fig. 10.23.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.24).

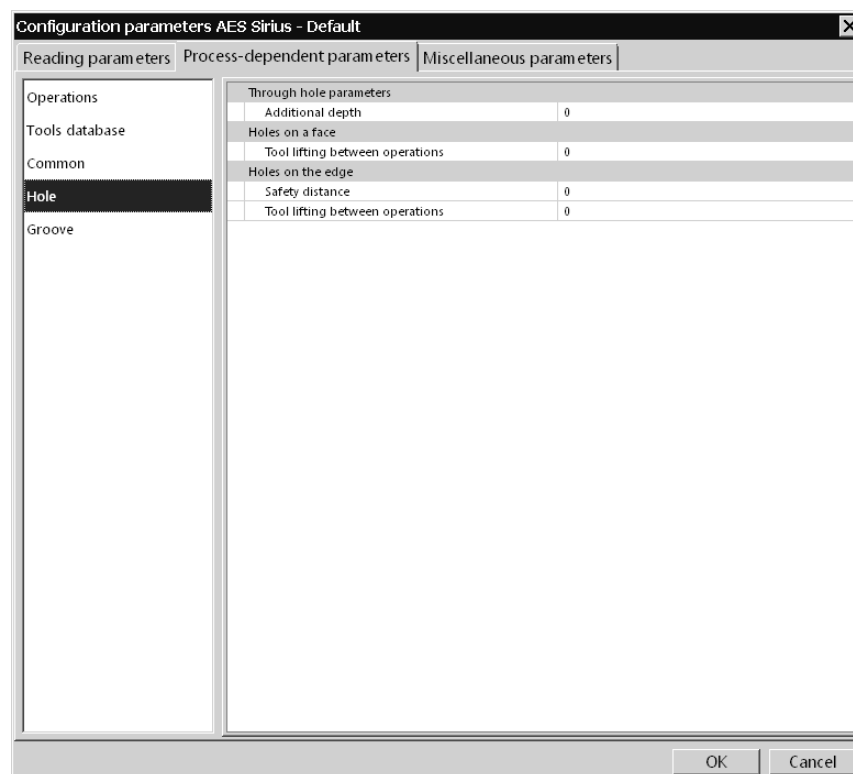


Fig. 10.24.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove parameters

Groove processing setup is performed in **Groove** section (fig. 10.25).

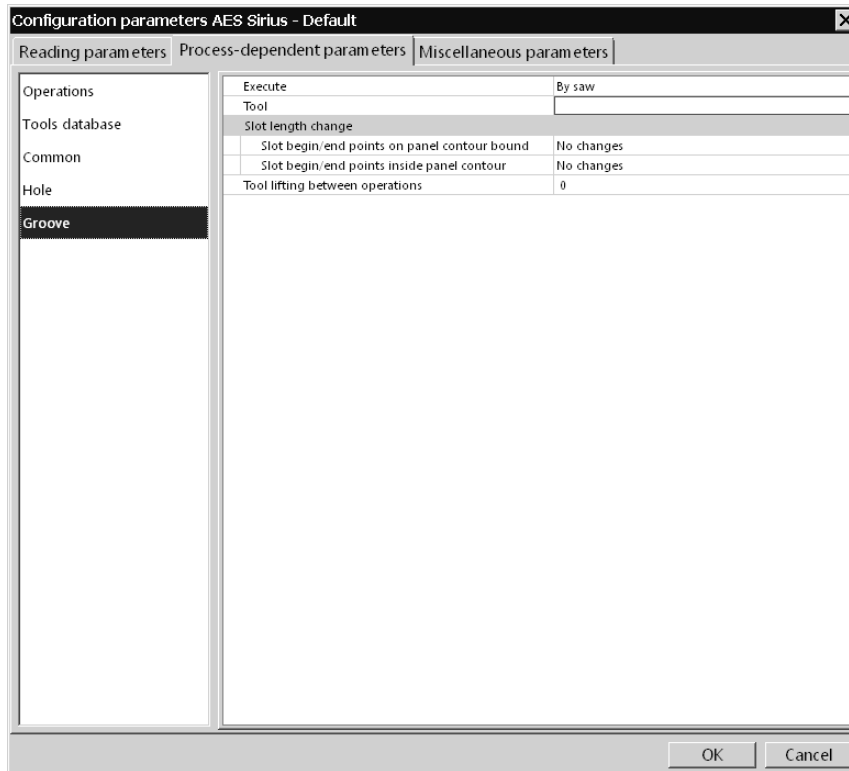


Fig. 10.25.

Section control elements allow to choose a grooving tool type, database tool and tool parameters.

10.4. Artisan

10.4.1. Artisan (*.tnc, *.tn2)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.26) allow to set control programs creation common parameters.

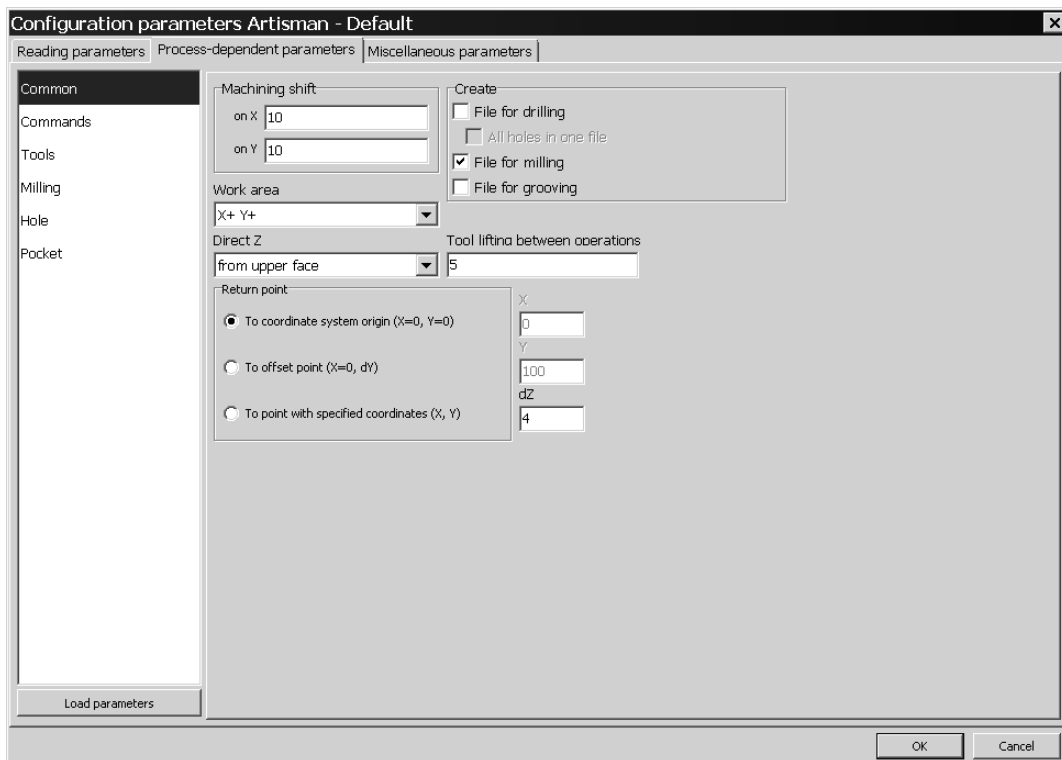


Fig. 10.26.

On X and **on Y** fields of **Machining shift** group allow to set work area initial point moving. **Create** group options allow to control programs creation for separate operations. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Work area** combo box variants allow to set work area coordinate axes direction.

Usable commands

Commands section control elements (fig. 10.27) allow to control separate commands to control program inclusion.

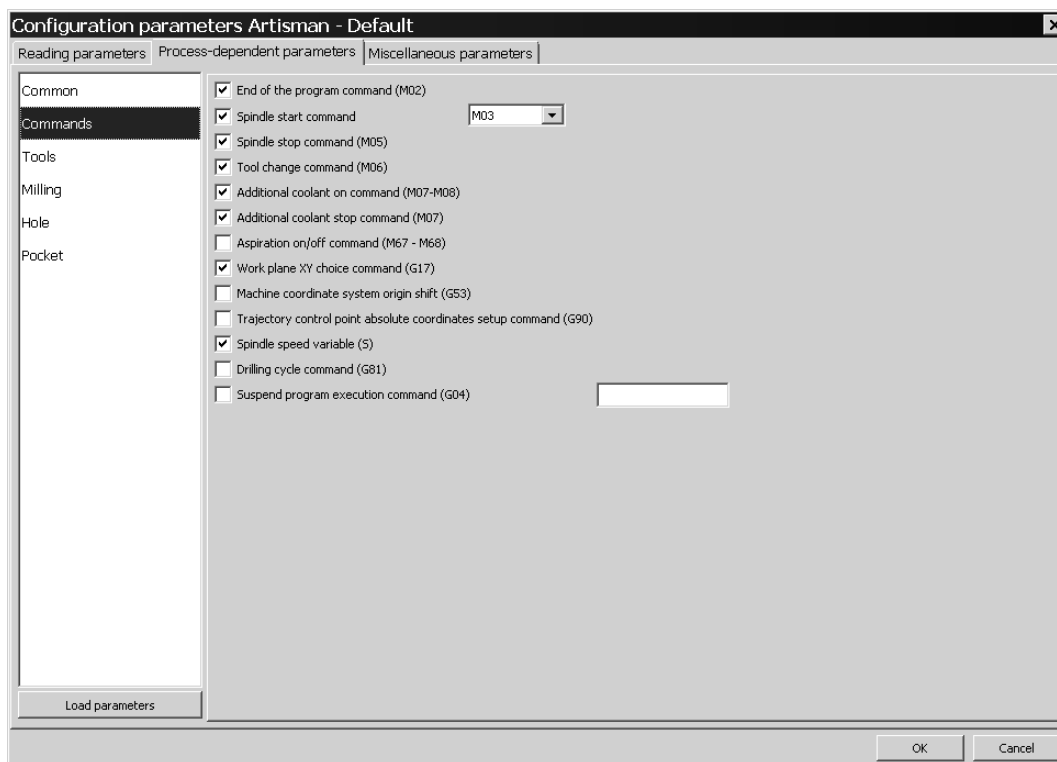


Fig. 10.27.

Spindle start command can be selected from variants in a combo box. If program execution delay command is activated (G04) a delay setup field will become available.

Usable tools

Tools section control elements (fig. 10.28) allow to create a list of tools available for use in control programs.

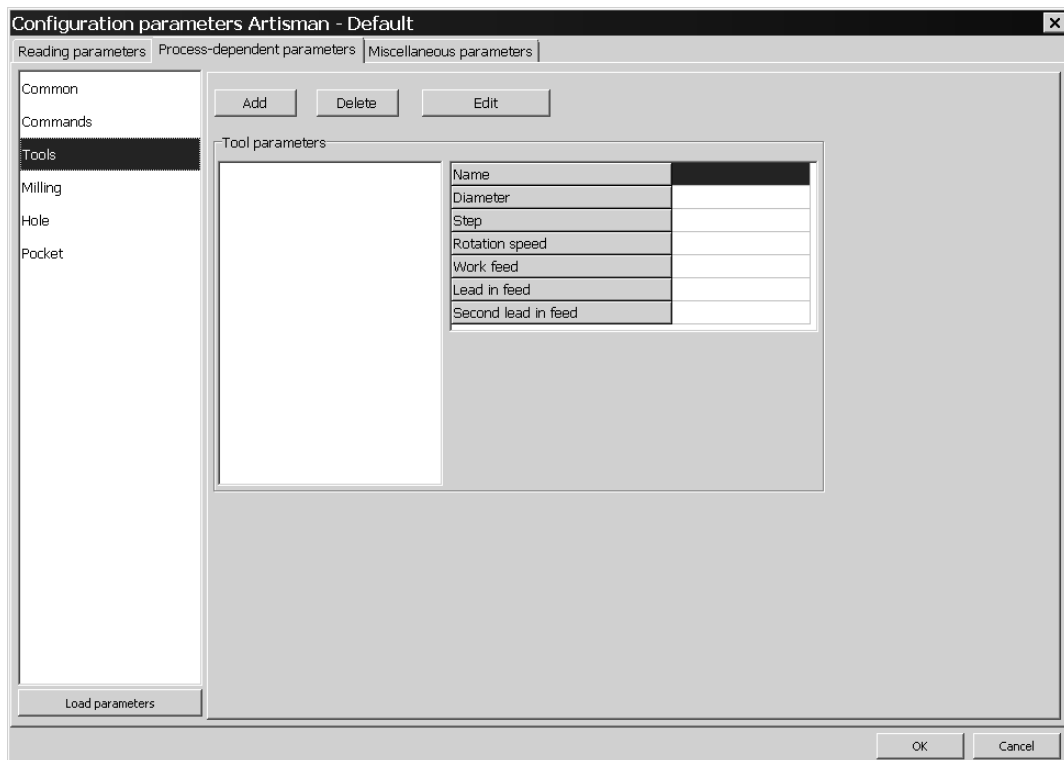


Fig. 10.28.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.29).

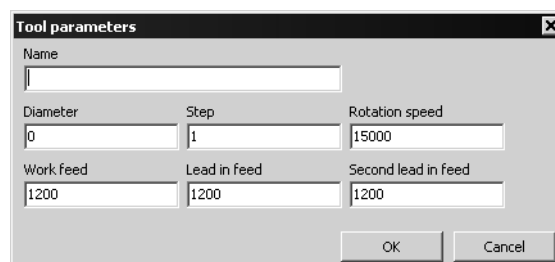


Fig. 10.29.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.30) allow to set milling parameters.

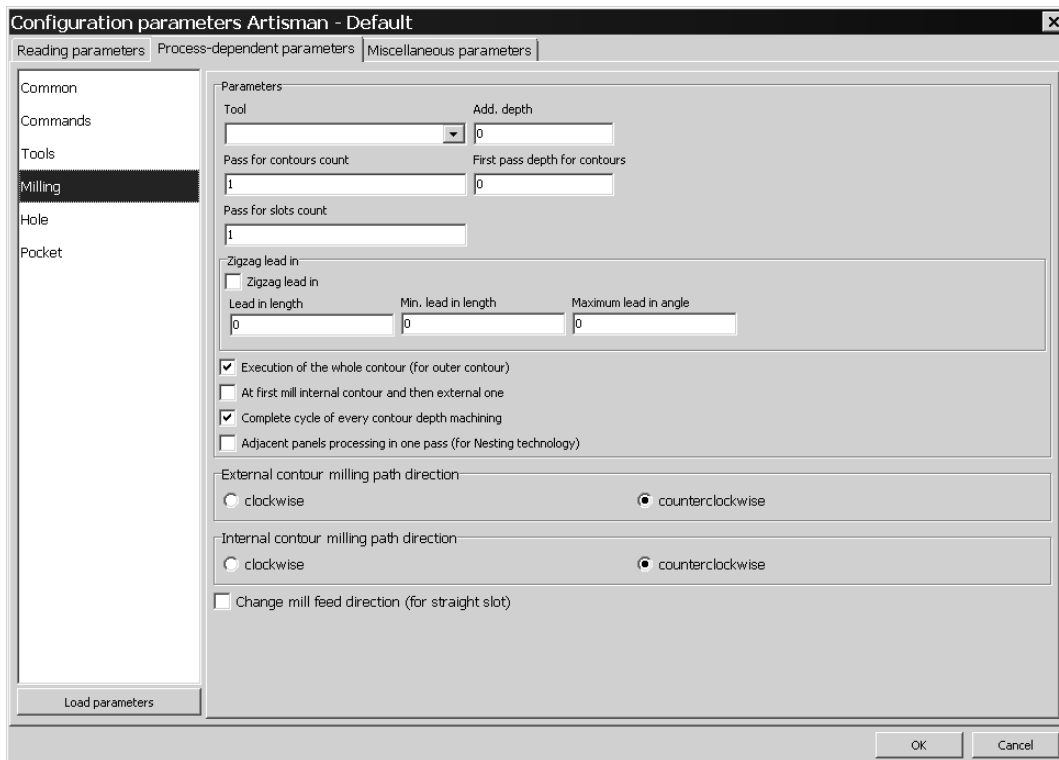


Fig. 10.30.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Zigzag lead in** option allows to activate this operation. Group fields allow to set option parameters. **Milling path direction** group variants allow to select tool moving direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.31).

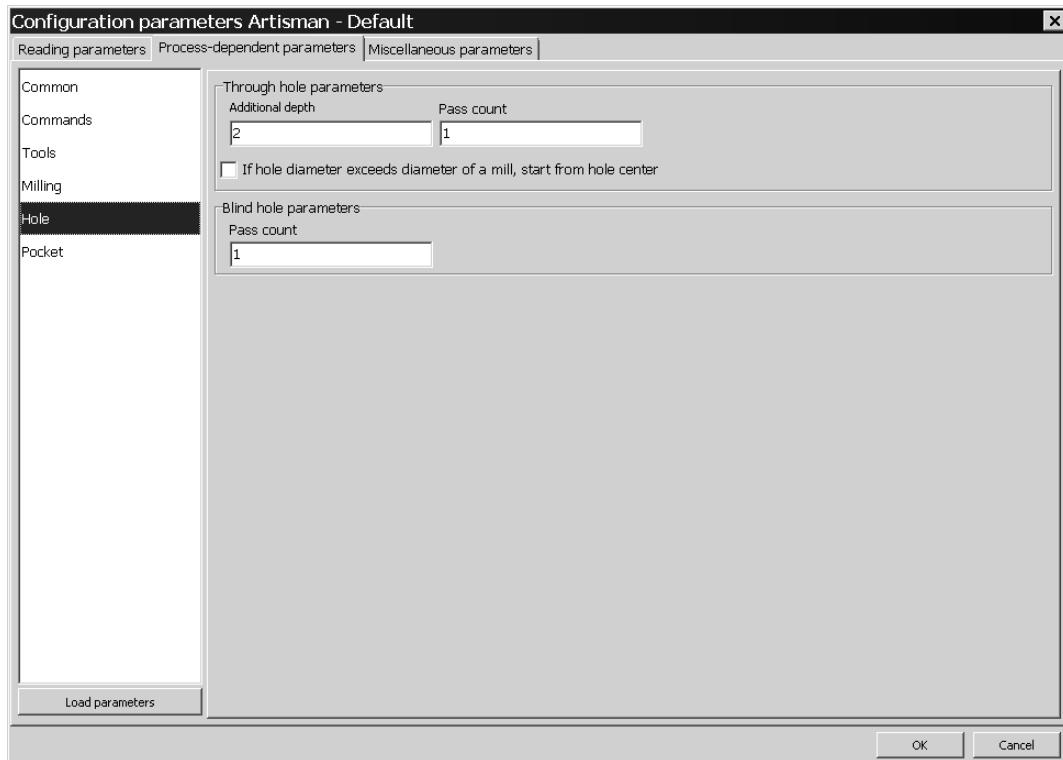


Fig. 10.31.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Pocket parameters

Pocket section control elements (fig. 10.32) allow to set pocket processing parameters.

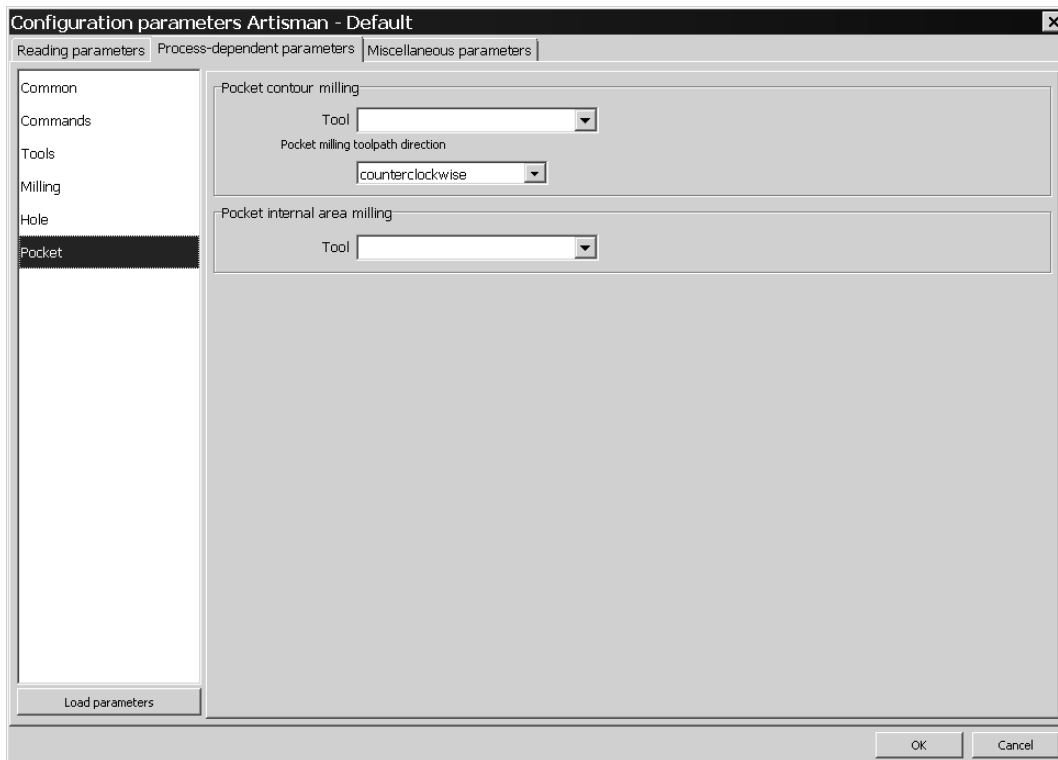


Fig. 10.32.

Tool combo boxes in **Pocket contour milling** and **Pocket internal area milling** groups allow to select tool name for corresponding objects milling. Tools should be created beforehand in **Tools** section. It is necessary to select milling execution direction for pocket contour in corresponding combo box.

Setup completion

To complete control programs for Artisan setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5. Beaver (Excitech)

10.5.1. Camdrill-Swift



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.33) allow to specify operations, that can be used in control program.

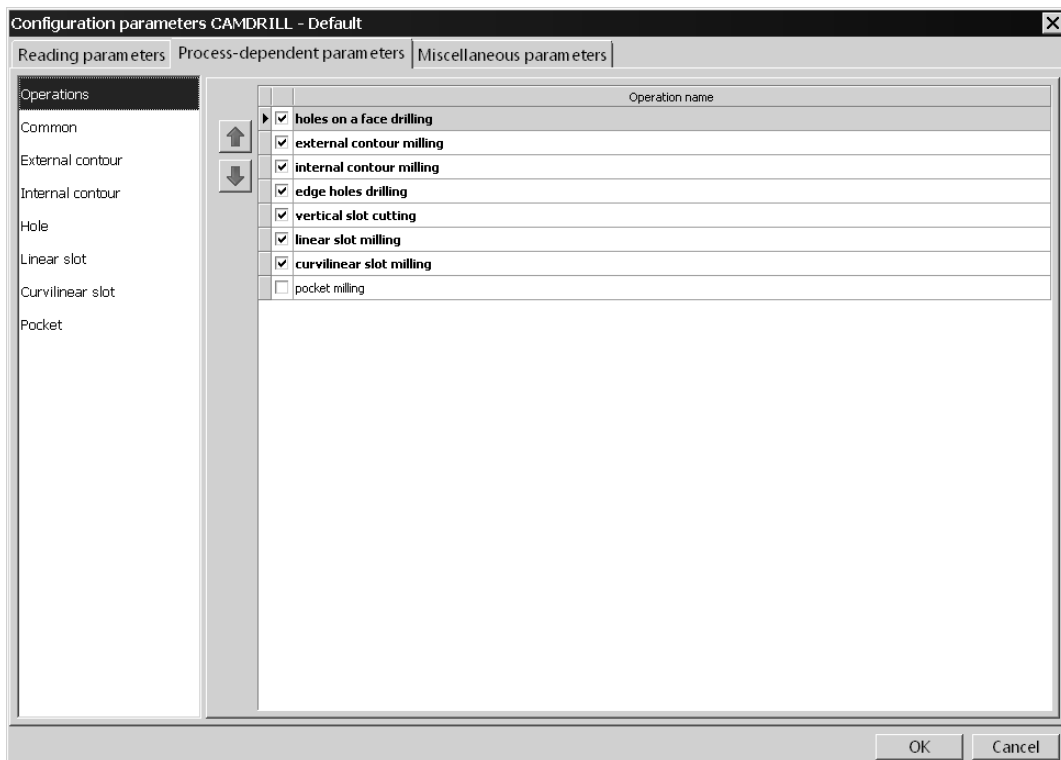


Fig. 10.33.

Common parameters

Common section control elements (fig. 10.34) allow to setup common parameters of control programs generation.

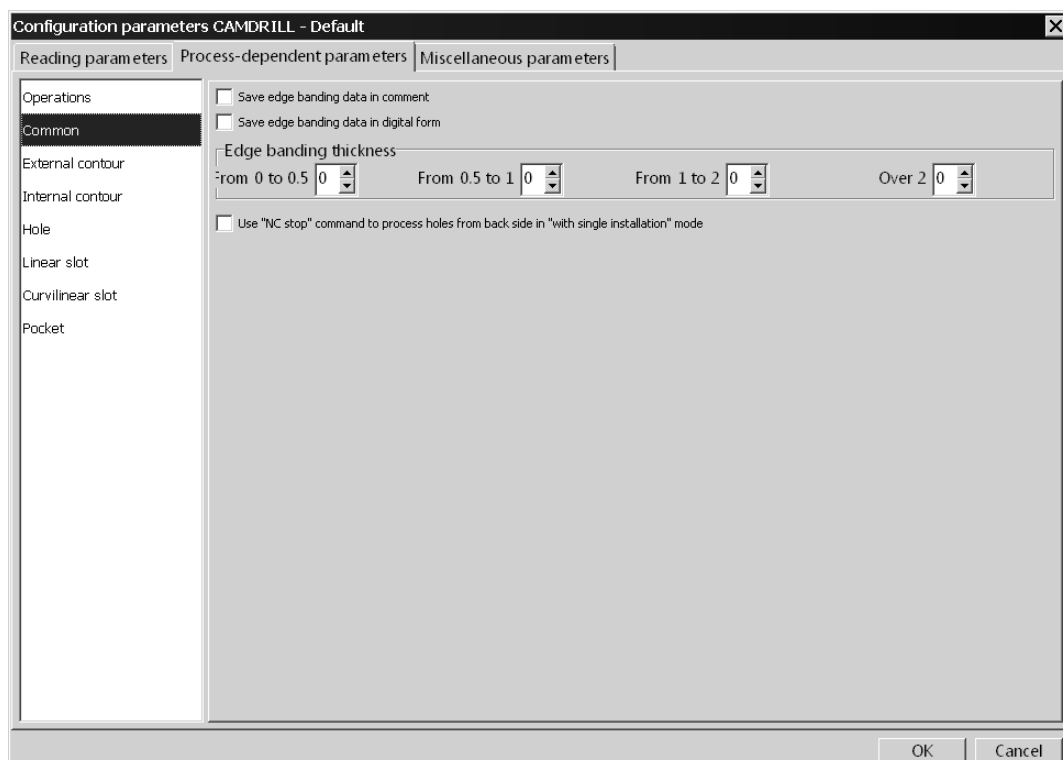


Fig. 10.34.

Save edge banding data in comment, Save edge banding data in digital form, Use «NC stop» command to process holes from back side in «with single installation» mode options assignment corresponds to their names.

Contour parameters

Contour milling parameters are set in **Outer contour** (fig. 10.35) and **Internal contour** sections of **Process-dependent parameters** tab.

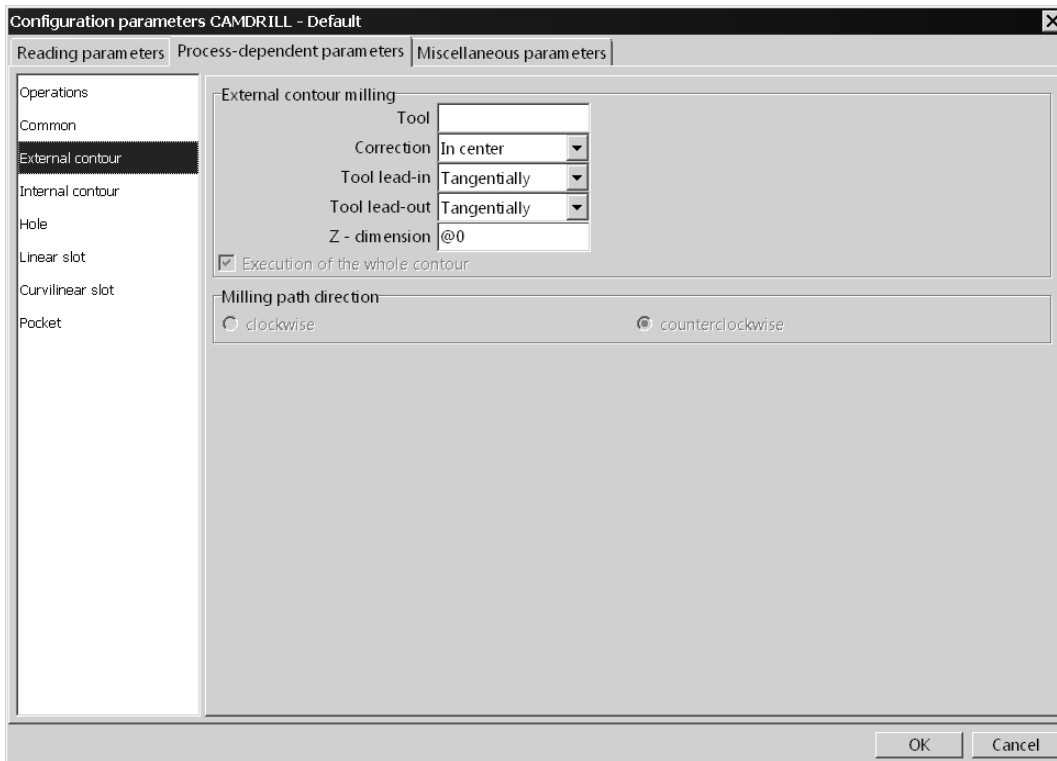


Fig. 10.35.

Tab control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. Tab control elements sets are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Holes

Hole section control elements (fig. 10.36) allow to setup milling parameters.

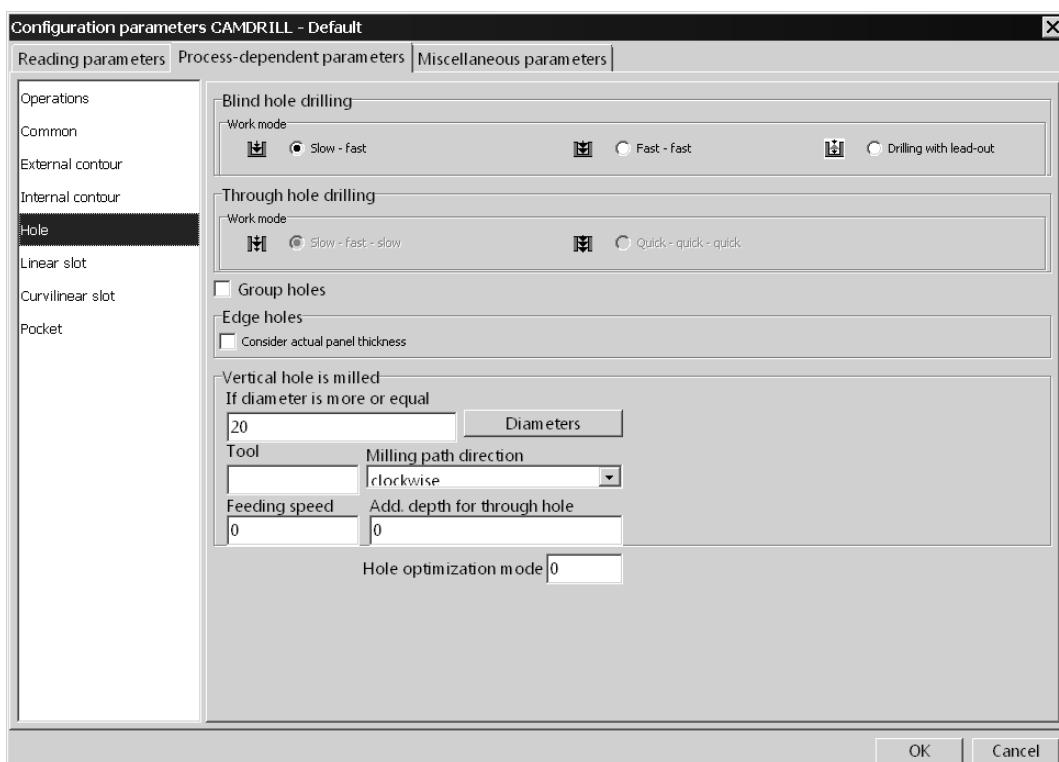


Fig. 10.36.

Blind hole drilling and **Through hole drilling** groups variants allow to select drilling speed. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Holes can be bored with a drill as well as with a mill. Hole diameter usually specifies tool selection. **If diameter is more or equal** field allows to set hole diameter, if it exceeds, the hole will be milled automatically, mill name is shown in **Tool** field. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.37).

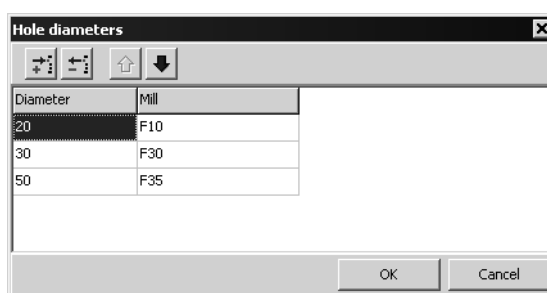


Fig. 10.37.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To complete setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

Linear slot

Linear slot section control elements (fig. 10.38) allow to setup linear slots processing parameters.

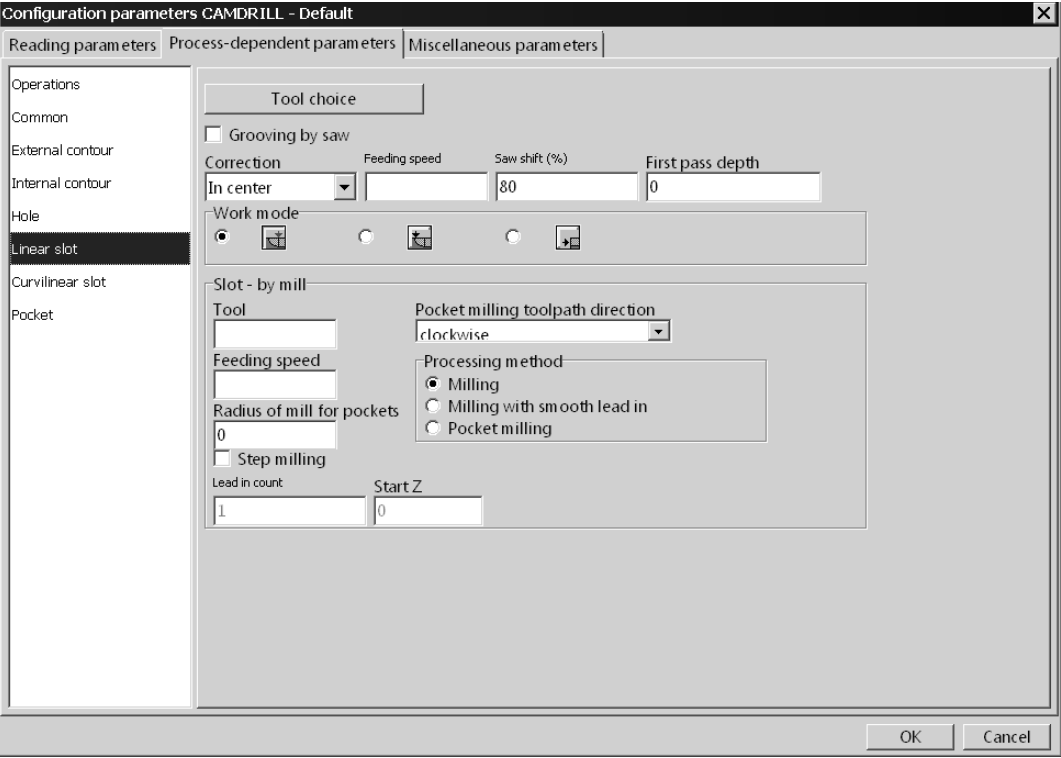


Fig. 10.38.

Tool choice button allows to setup the tools for slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.39).

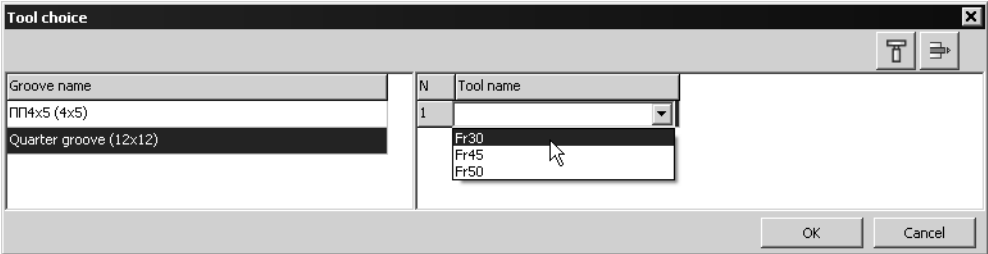


Fig. 10.39.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.40).

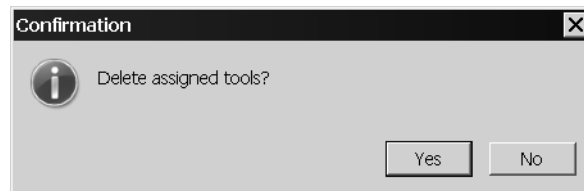


Fig. 10.40.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To finish tools assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. The slots can be processed with a saw or a mill. If you use a saw from **Correction** combo box, you can choose correction type.

Curvilinear slot

Curvilinear slot section control elements (fig. 10.41) allow to setup parameters of curvilinear slots processing operations.

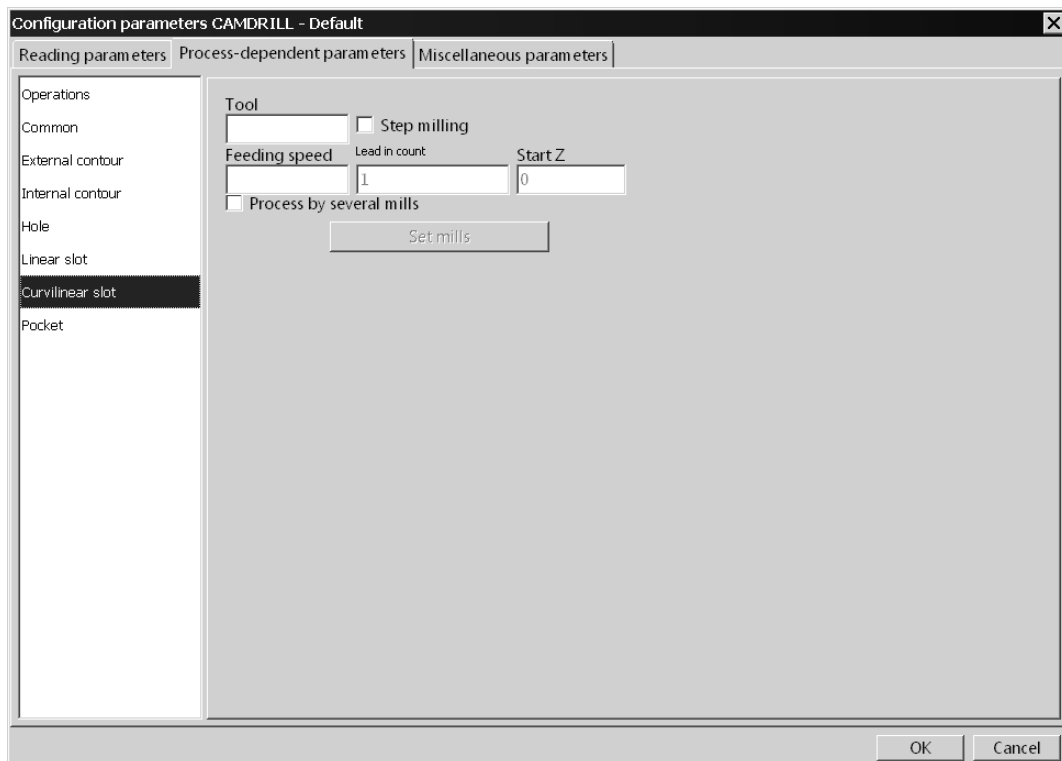


Fig. 10.41.

Tab control elements allow to setup control programs generation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.42).

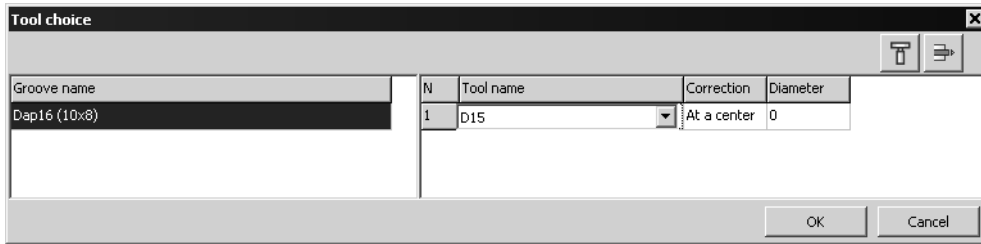


Fig. 10.42.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.43) allow to setup parameters of pocket processing.

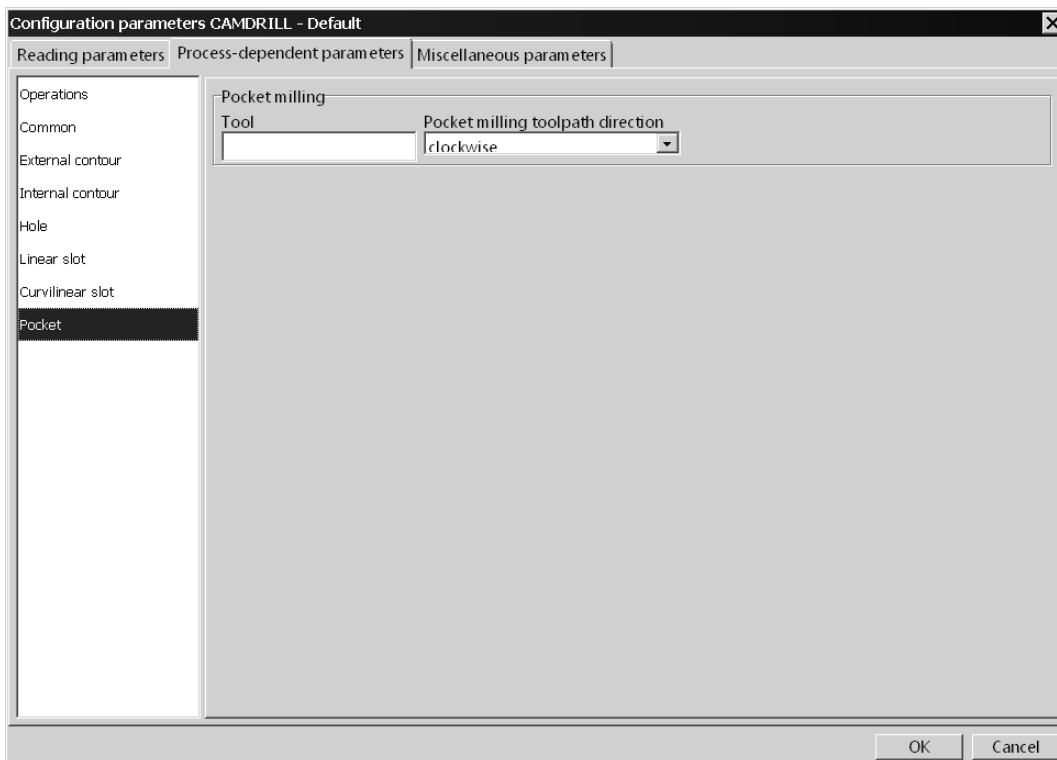


Fig. 10.43.

Tab control elements allow to set control programs creation parameters for pocket processing: tool name, path direction, feeding speed, etc.

Setup completion

To complete control program parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.2. OSAI Control



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.44) allow to specify operations that can be used in control program.

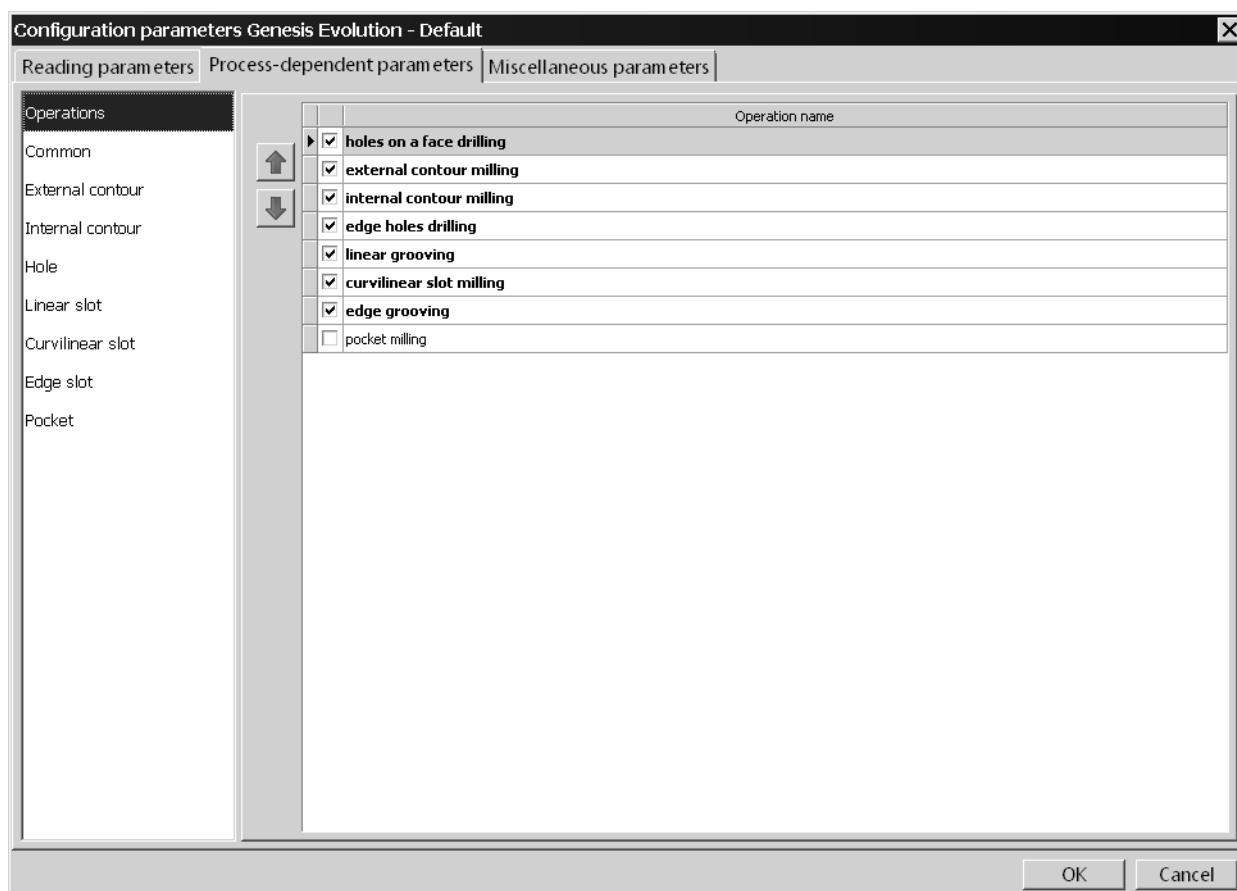


Fig. 10.44.

Common parameters

Common section control elements (fig. 10.45) allow to set control programs common parameters.

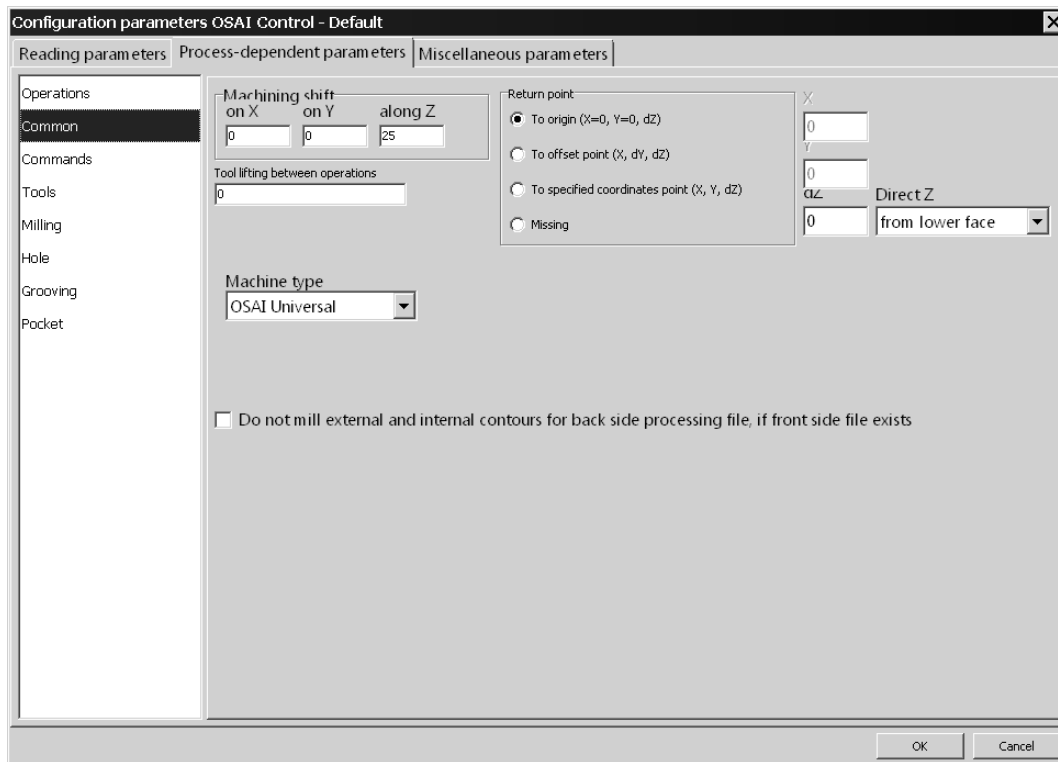


Fig. 10.45.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** variant is selected, tool will be returned to this point. If **To offset point (X, dY)** or **To point with specified coordinates** variant is selected, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Machine type** combo box allows to select a machine type for program creation.

Usable commands

Commands section control elements (fig. 10.46) allow to control separate commands to control program inclusion.

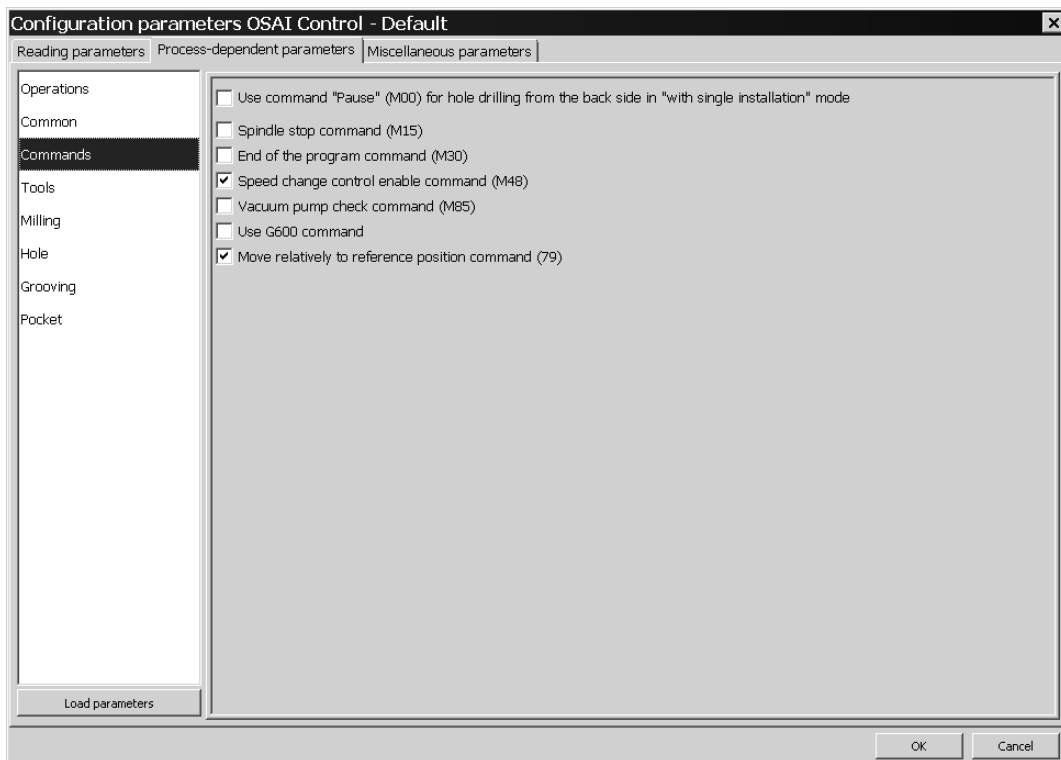


Fig. 10.46.

Usable tools

Tools section control elements (fig. 10.47) allow to create a list of tools available for use in control programs.

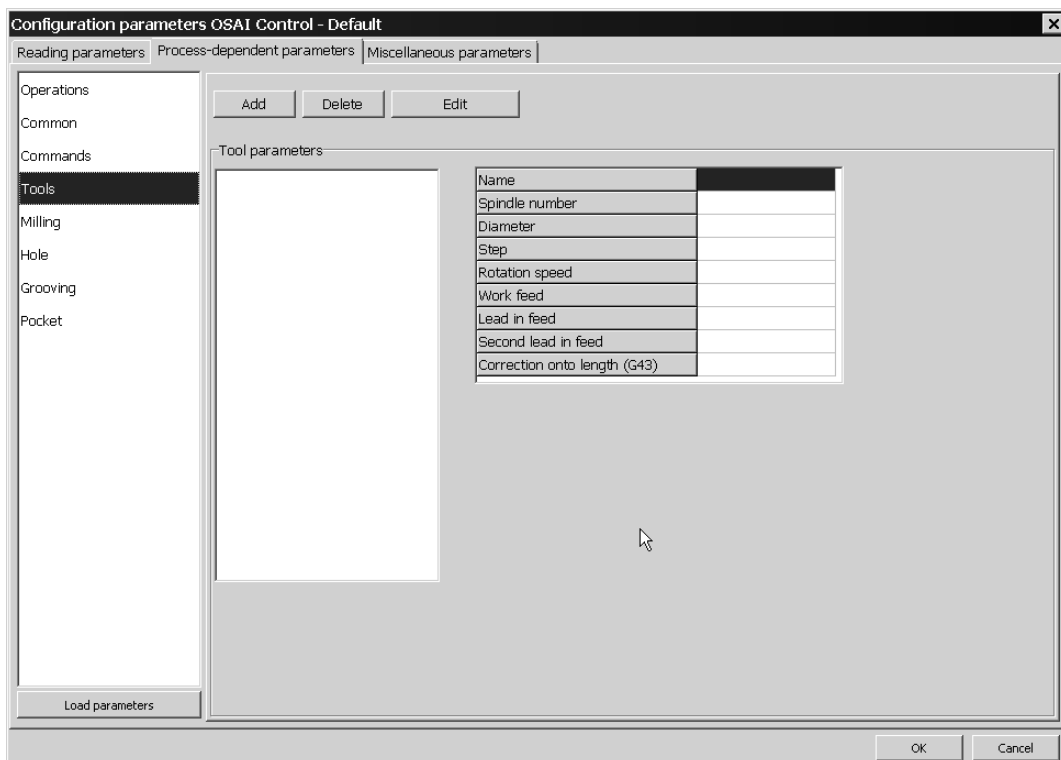
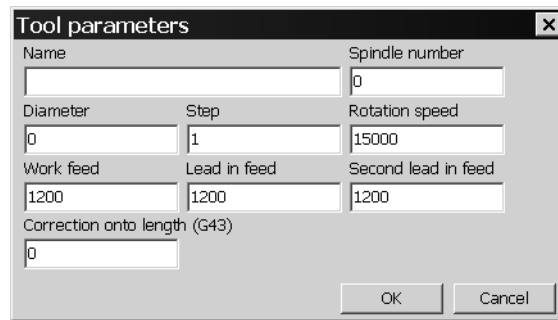


Fig. 10.47.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.48).



The 'Tool parameters' dialog box contains the following fields and controls:

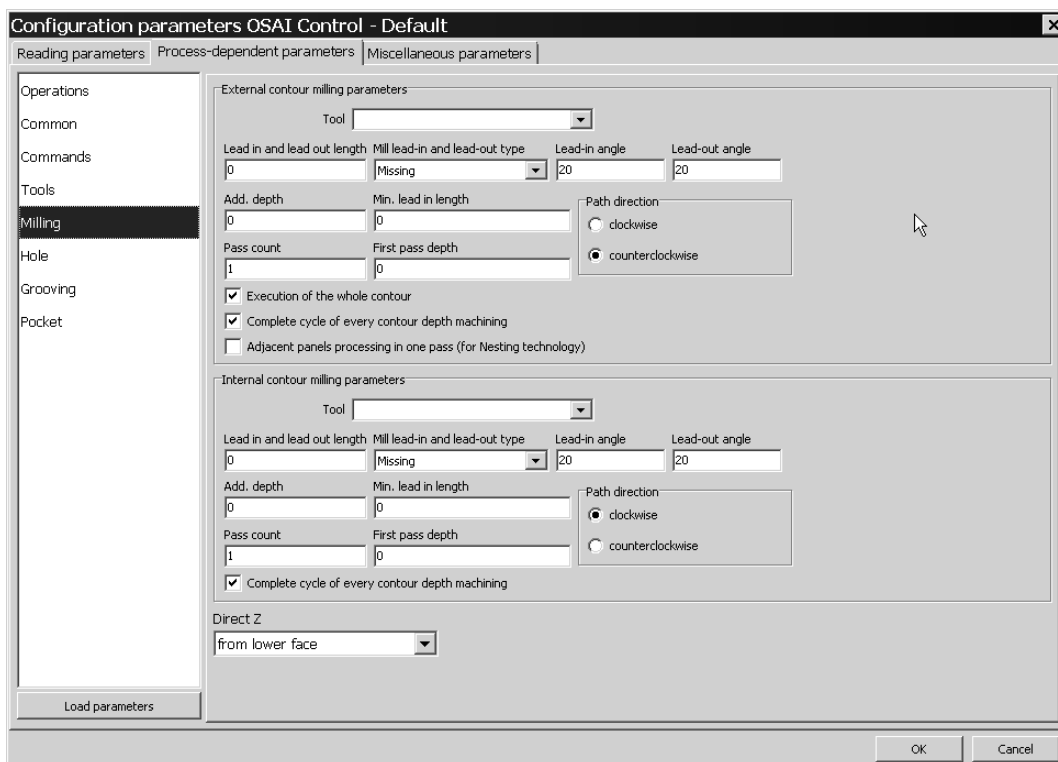
- Name:** Text input field.
- Spindle number:** Text input field with value '0'.
- Diameter:** Text input field with value '0'.
- Step:** Text input field with value '1'.
- Rotation speed:** Text input field with value '15000'.
- Work feed:** Text input field with value '1200'.
- Lead in feed:** Text input field with value '1200'.
- Second lead in feed:** Text input field with value '1200'.
- Correction onto length (G43):** Text input field with value '0'.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Fig. 10.48.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.49) allow to set contour milling parameters.



The 'Configuration parameters OSAI Control - Default' dialog box has three tabs: 'Reading parameters', 'Process-dependent parameters', and 'Miscellaneous parameters'. The 'Process-dependent parameters' tab is active.

Left sidebar (Navigation):

- Operations
- Common
- Commands
- Tools
- Milling** (selected)
- Hole
- Grooving
- Pocket

Main content area:

- External contour milling parameters:**
 - Tool:** Dropdown menu.
 - Lead in and lead out length:** Text input field with value '0'.
 - Mill lead-in and lead-out type:** Dropdown menu with value 'Missing'.
 - Lead-in angle:** Text input field with value '20'.
 - Lead-out angle:** Text input field with value '20'.
 - Add. depth:** Text input field with value '0'.
 - Min. lead in length:** Text input field with value '0'.
 - Pass count:** Text input field with value '1'.
 - First pass depth:** Text input field with value '0'.
 - Path direction:** Radio buttons for 'clockwise' (selected) and 'counterclockwise'.
 - ☒ Execution of the whole contour
 - ☒ Complete cycle of every contour depth machining
 - ☐ Adjacent panels processing in one pass (for Nesting technology)
- Internal contour milling parameters:**
 - Tool:** Dropdown menu.
 - Lead in and lead out length:** Text input field with value '0'.
 - Mill lead-in and lead-out type:** Dropdown menu with value 'Missing'.
 - Lead-in angle:** Text input field with value '20'.
 - Lead-out angle:** Text input field with value '20'.
 - Add. depth:** Text input field with value '0'.
 - Min. lead in length:** Text input field with value '0'.
 - Pass count:** Text input field with value '1'.
 - First pass depth:** Text input field with value '0'.
 - Path direction:** Radio buttons for 'clockwise' (selected) and 'counterclockwise'.
 - ☒ Complete cycle of every contour depth machining
- Direct Z:** Text input field with value 'from lower face'.

Buttons: 'Load parameters' (bottom left), 'OK' (bottom right), and 'Cancel' (bottom right).

Fig. 10.49.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that

do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool shift direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.50).

Diameter	Layout	Type	Tool
5	Edge on the mitre	Blind	
5	Upper face	Blind	
15	Upper face	Blind	
35	Upper face	Blind	
5	Upper face	Through	
8	Upper face	Through	
7	Front edge	Blind	
5	Right edge	Blind	
7	Right edge	Blind	
8	Right edge	Blind	
5	Back edge	Blind	
5	Left edge	Blind	
7	Left edge	Blind	

Fig. 10.50.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **For edge holes use Z** and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height. Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slot parameters

Slots processing is set in **Grooving** section (fig. 10.51).

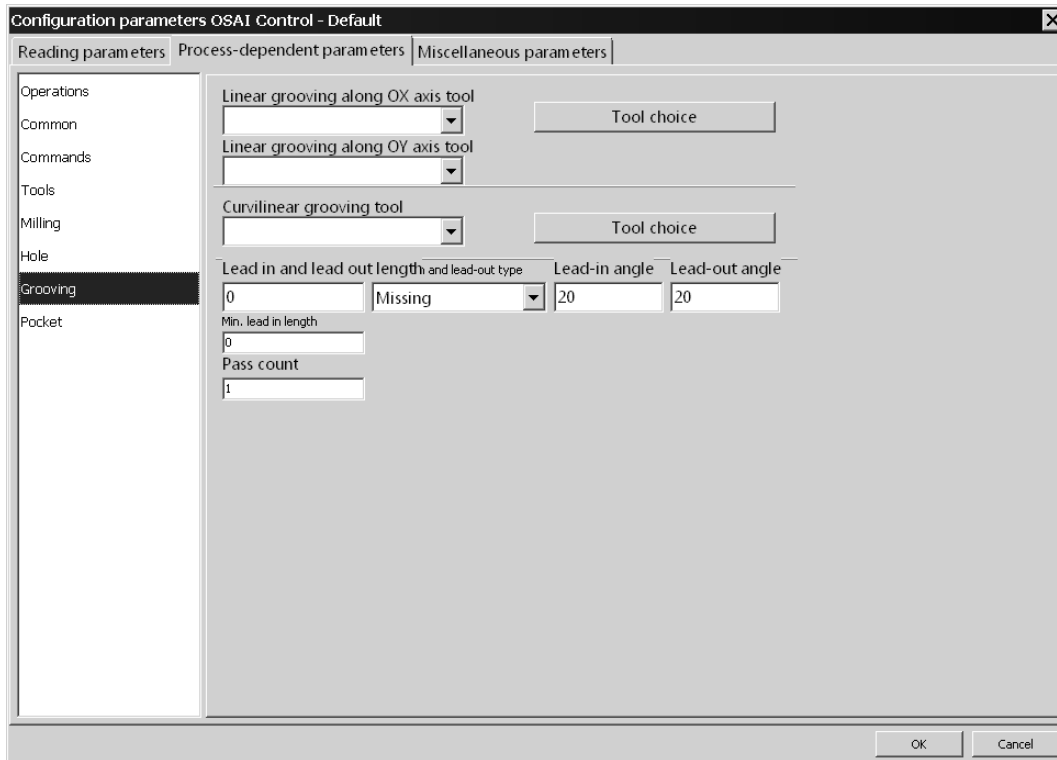


Fig. 10.51.

Linear grooving along OX axis tool, **Linear grooving OY axis tool**, **Curvilinear grooving tool** combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.52).

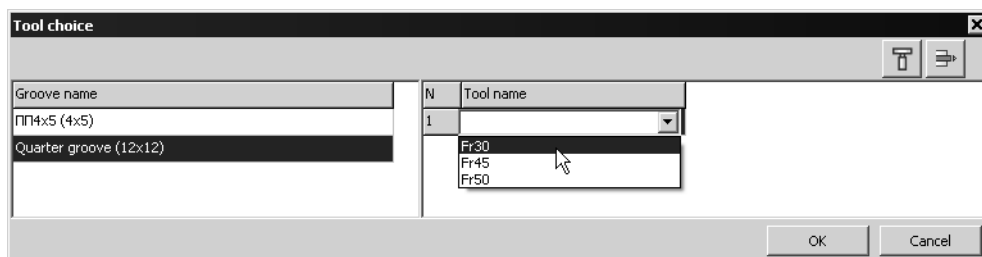


Fig. 10.52.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length**, **Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.53) allow to set pocket processing parameters.

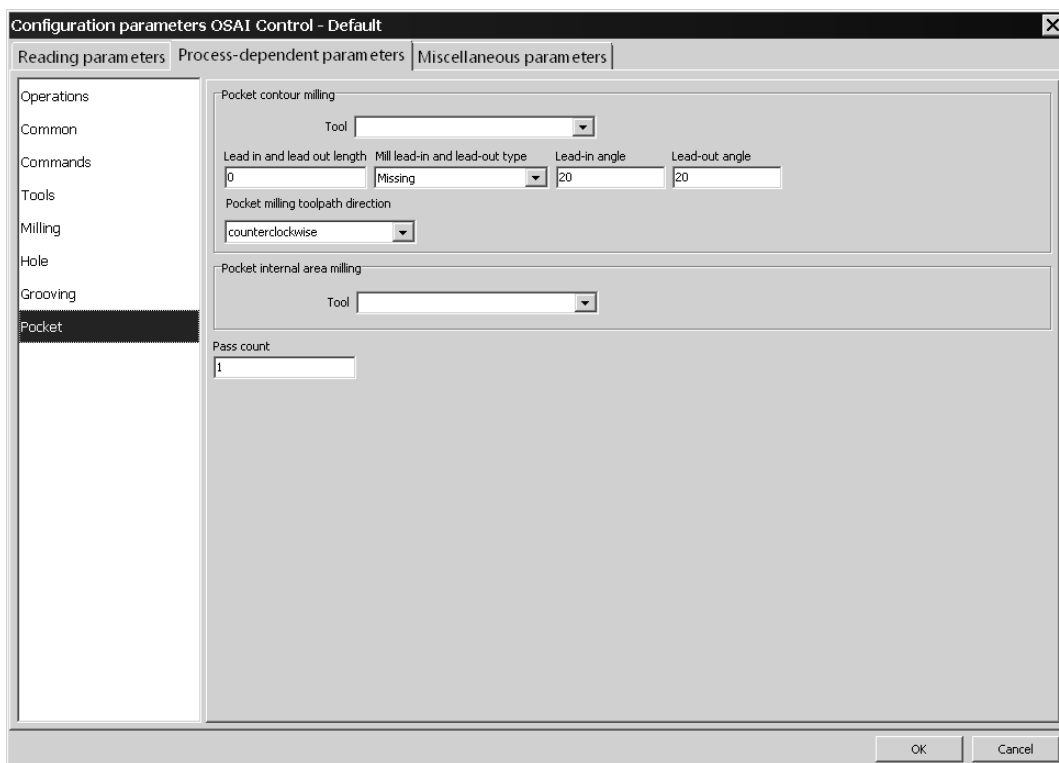


Fig. 10.53.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.3. OSAI Control - NestConcept(E4)Pre-Labeling



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.54) allow to specify operations that can be used in control program.

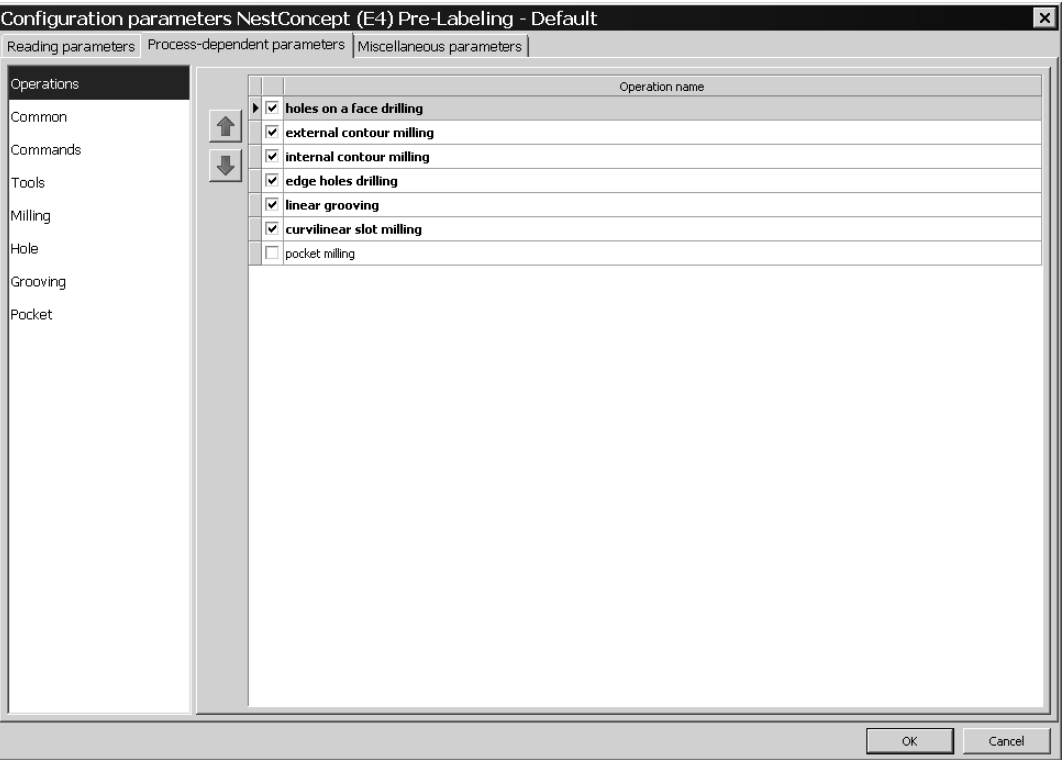


Fig. 10.54.

Common parameters

Common section control elements (fig. 10.55) allow to set control programs common parameters.

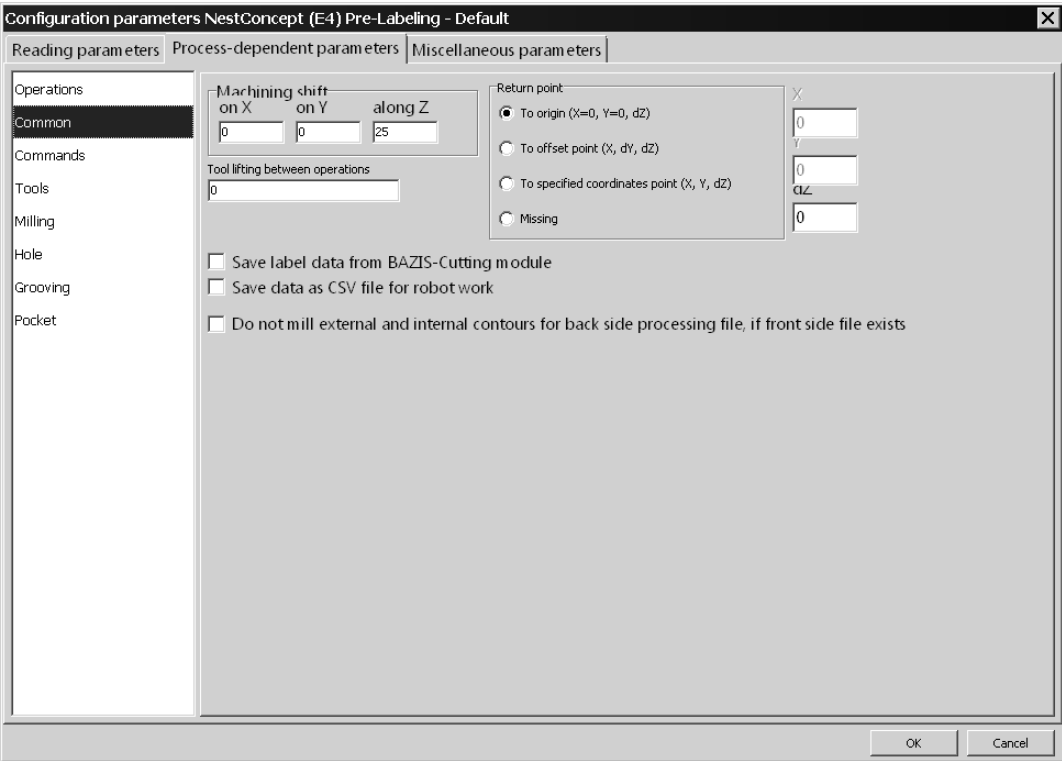


Fig. 10.55.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** variant is selected, tool will be returned to this point. If **To offset point (X, dY)** or **To point with specified coordinates** variant is selected, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height.

Usable commands

Commands section control elements (fig. 10.56) allow to control separate commands to control program inclusion.

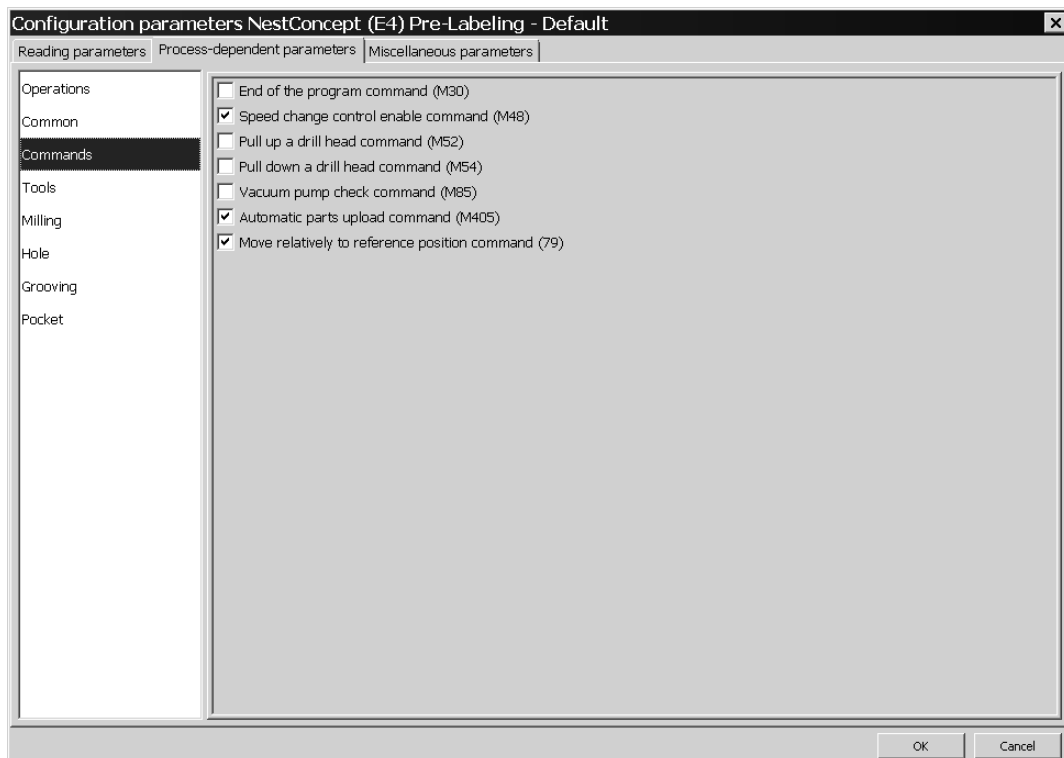


Fig. 10.56.

Usable tools

Tools section control elements (fig. 10.57) allow to create a list of tools available for use in control programs.

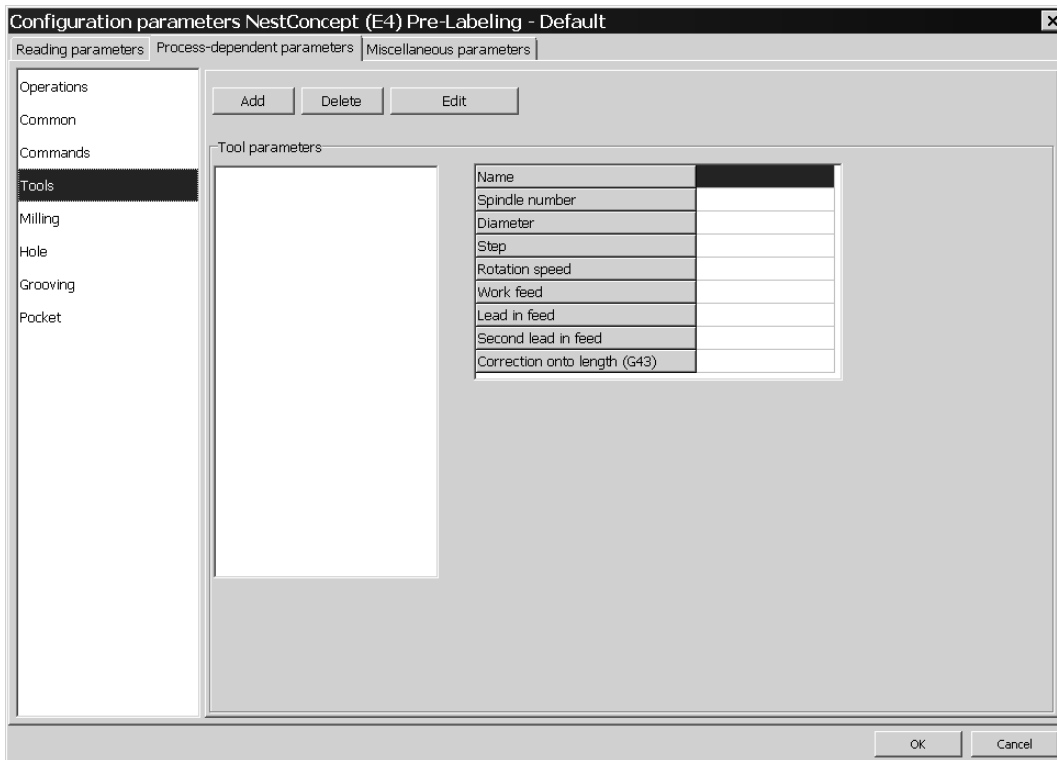


Fig. 10.57.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.58).

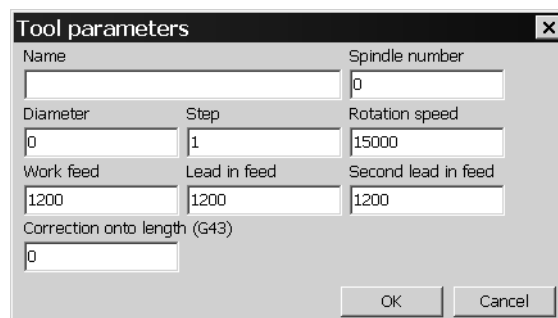


Fig. 10.58.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.59) allow to set contour milling parameters.

Fig. 10.59.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and it is disabled, contour will not be milled. **Path direction** group variants allow to select tool shift direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.60).

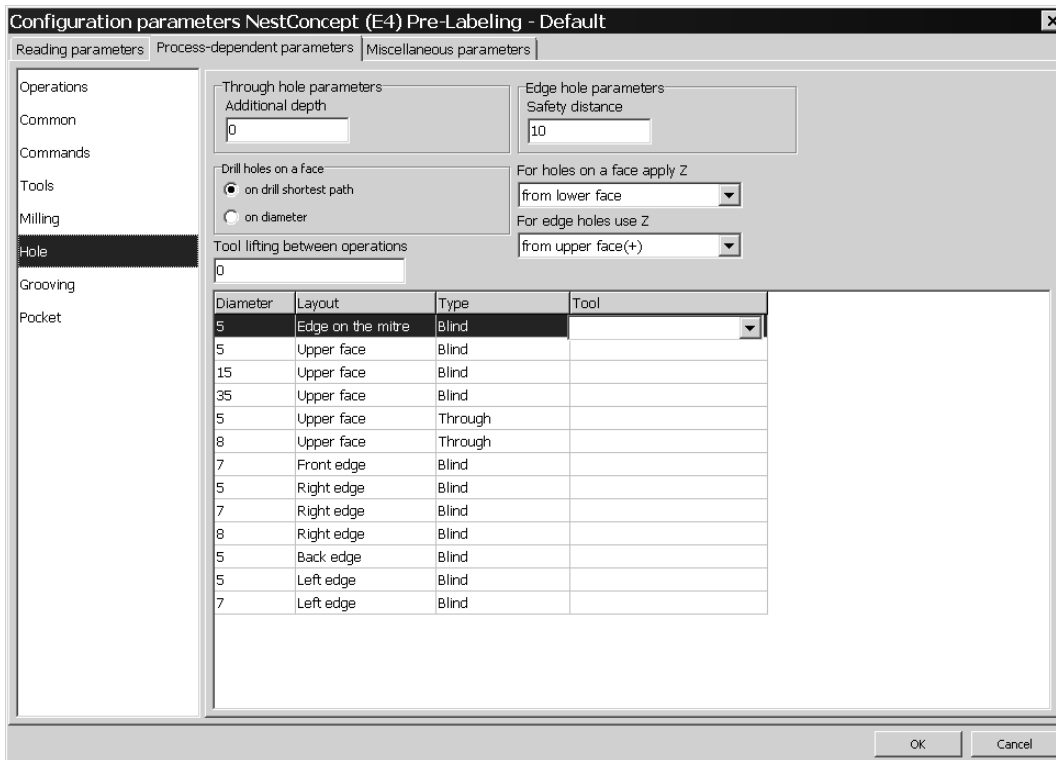


Fig. 10.60.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

For edge holes use Z and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height. Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

Slot parameters

Slots processing is set in **Grooving** section (fig. 10.61).

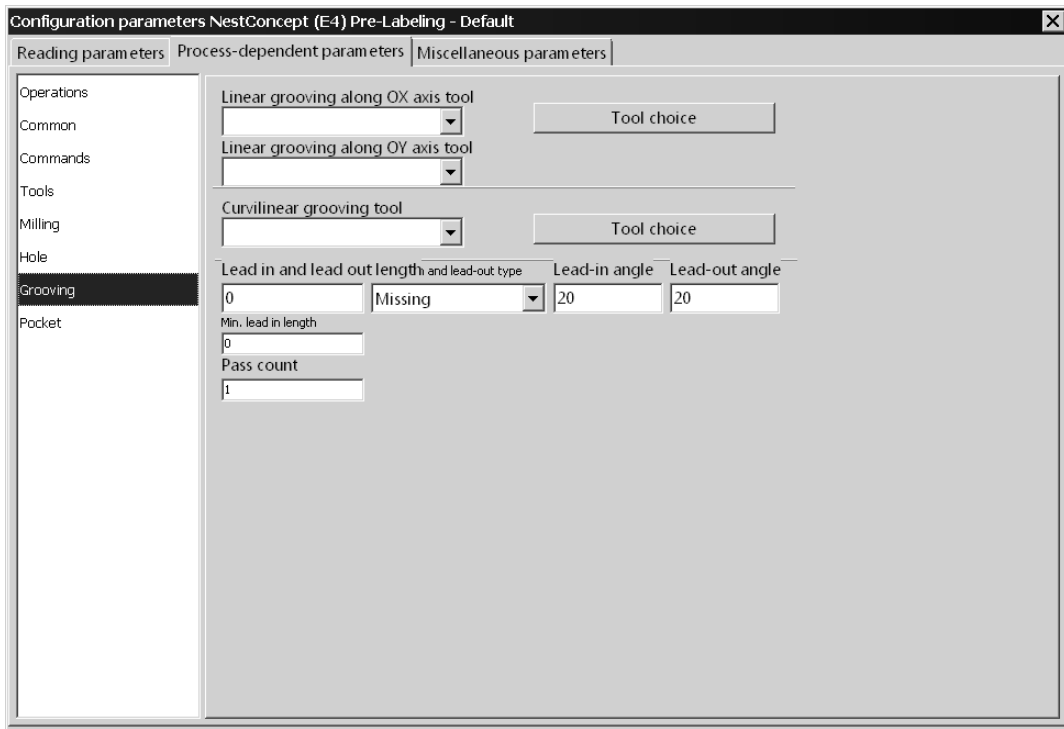


Fig. 10.61.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.62).

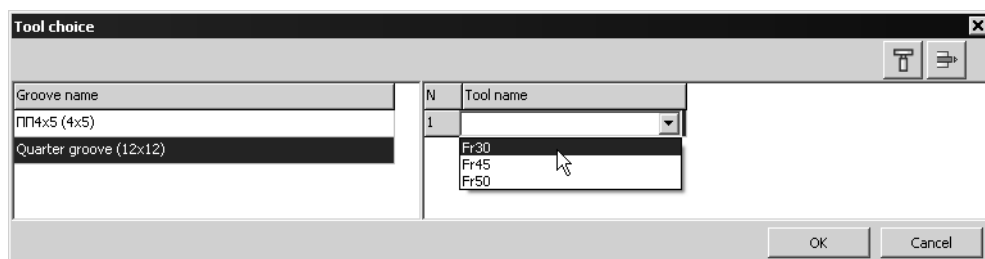


Fig. 10.62.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.63) allow to set pocket processing parameters.

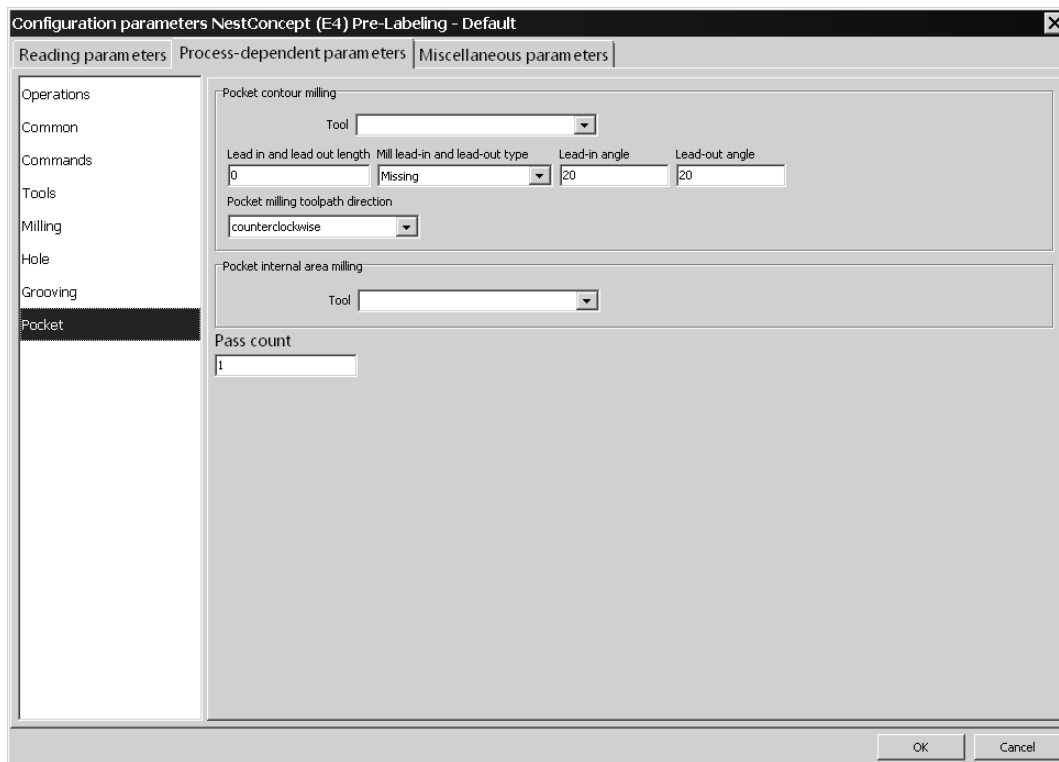


Fig. 10.63.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.4. OSAI Control Optimal(E3)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.64) allow to specify operations that can be used in control program.

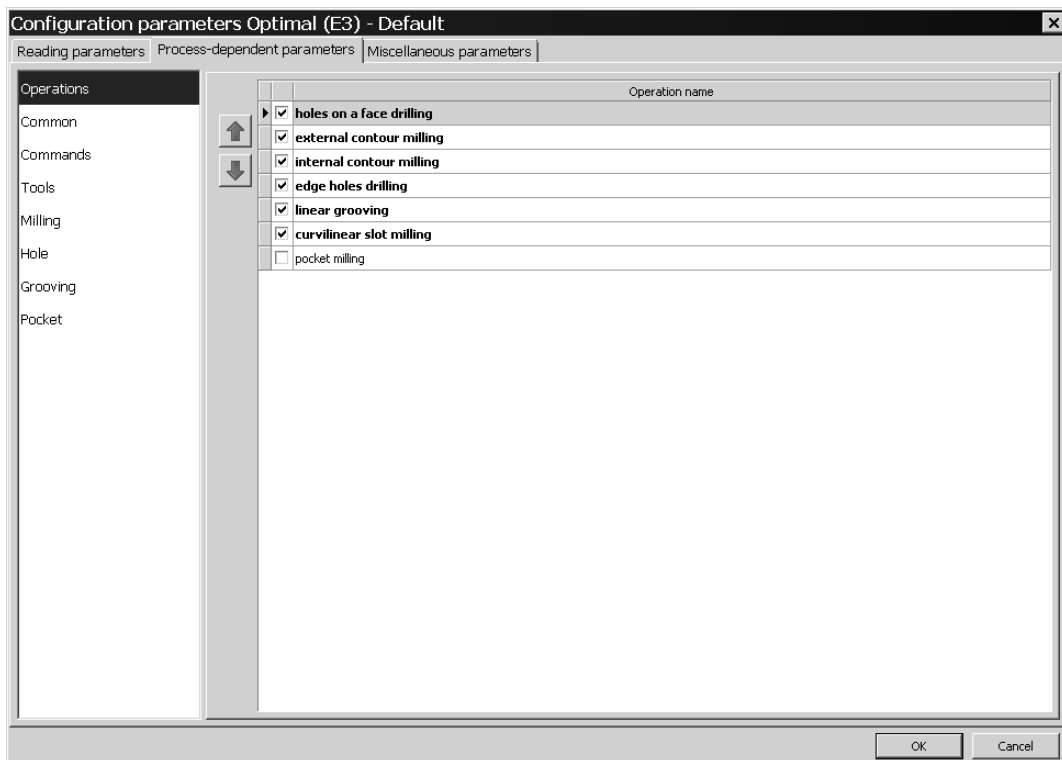


Fig. 10.64.

Common parameters

Common section control elements (fig. 10.65) allow to set control programs common parameters.

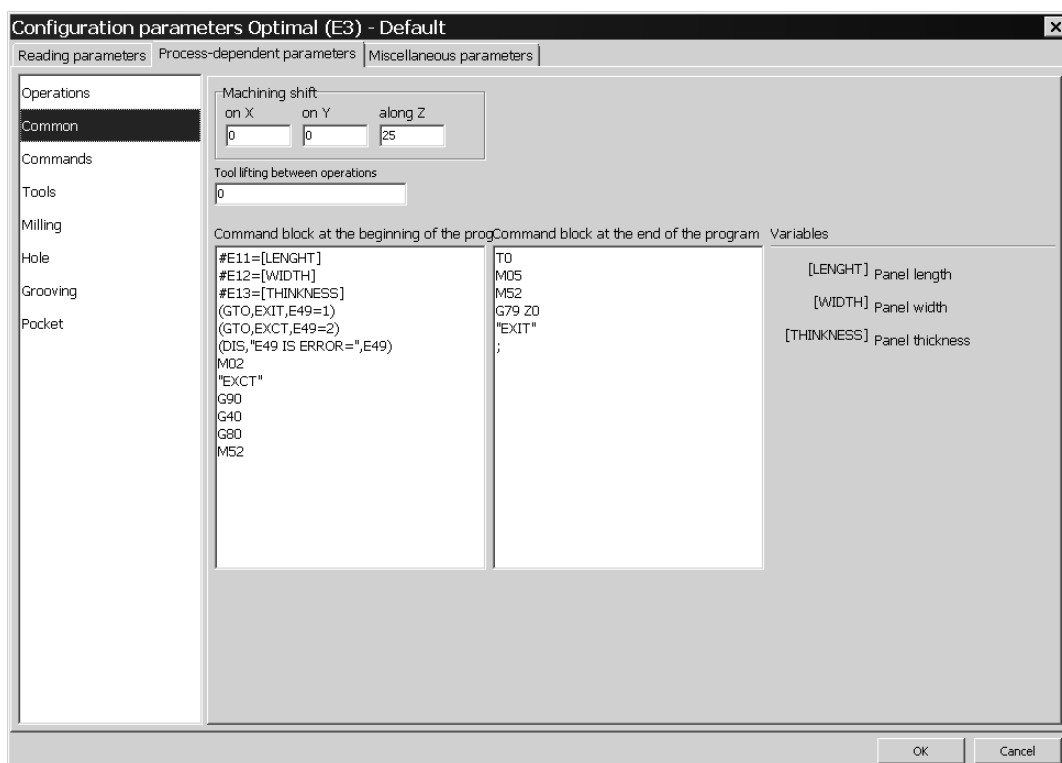


Fig. 10.65.

Machining shift group fields allow to set work area initial point moving. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create commands list that will be automatically placed in specified program points.

Usable commands

Commands section control elements (fig. 10.66) allow to control separate commands to control program inclusion.

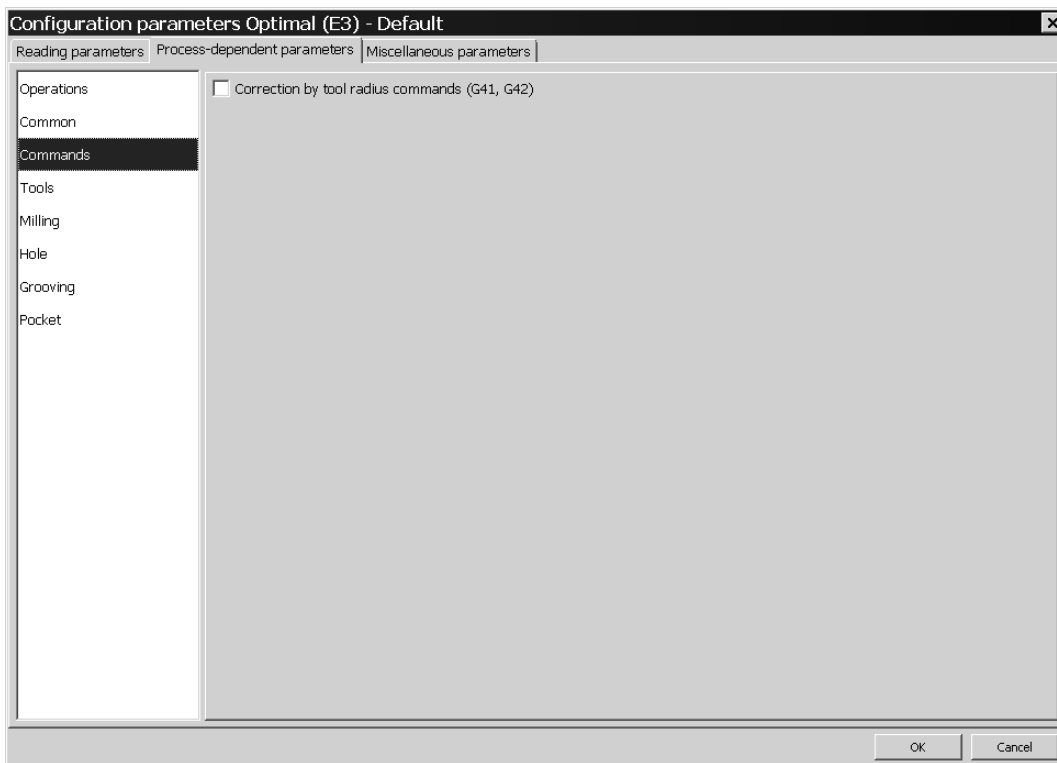


Fig. 10.66.

Usable tools

Tools section control elements (fig. 10.67) allow to create a list of tools available for use in control programs.

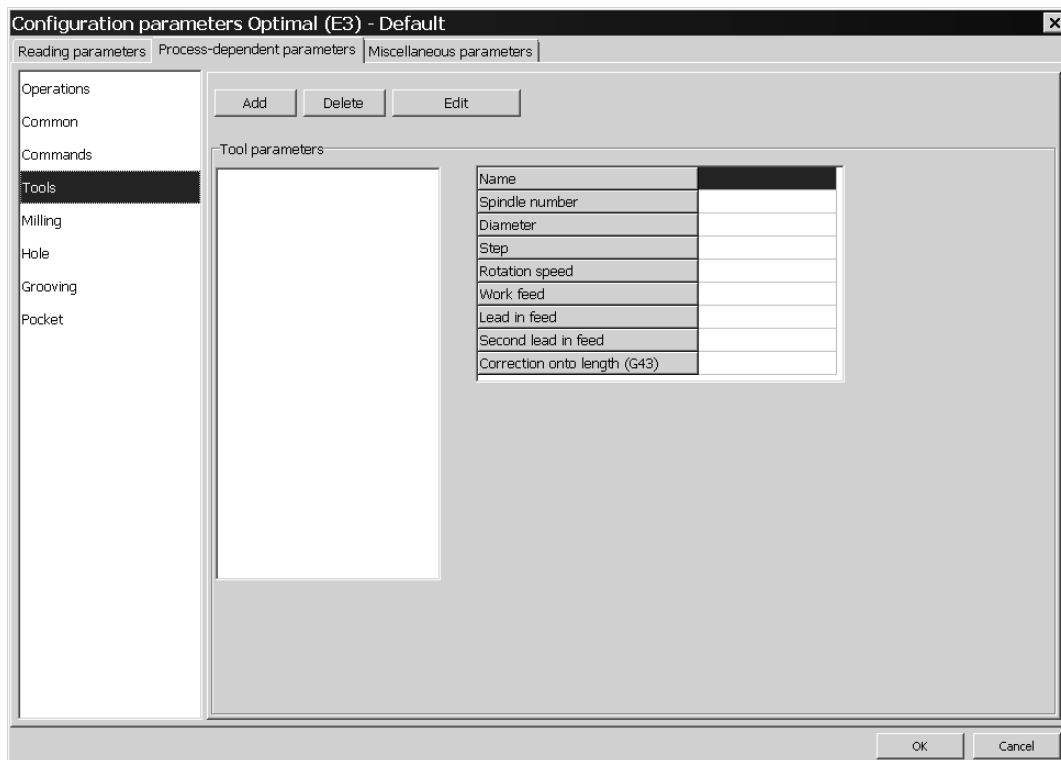


Fig. 10.67.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.68).

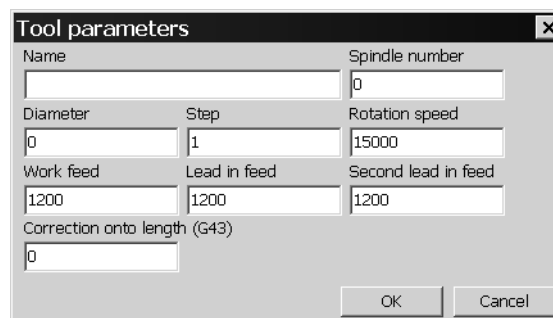


Fig. 10.68.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.69) allow to set contour milling parameters.

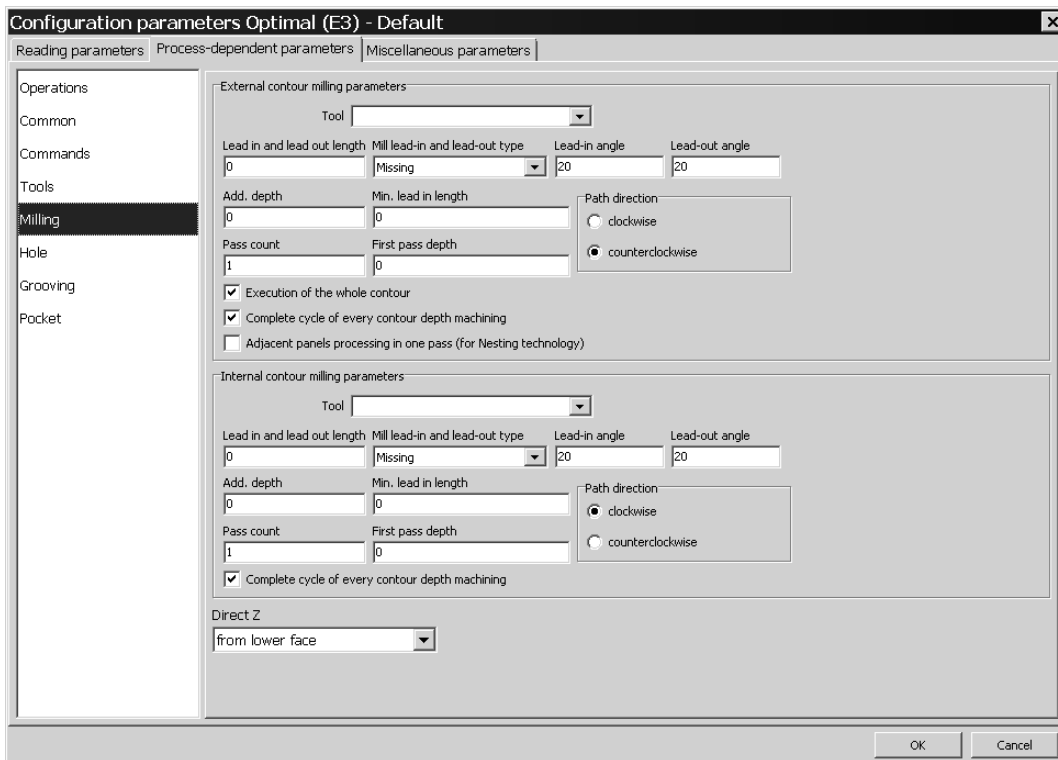


Fig. 10.69.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section (see section Usable tools on p. 93). Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and it is disabled, contour will not be milled. **Path direction** group variants allow to select tool shift direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.70).

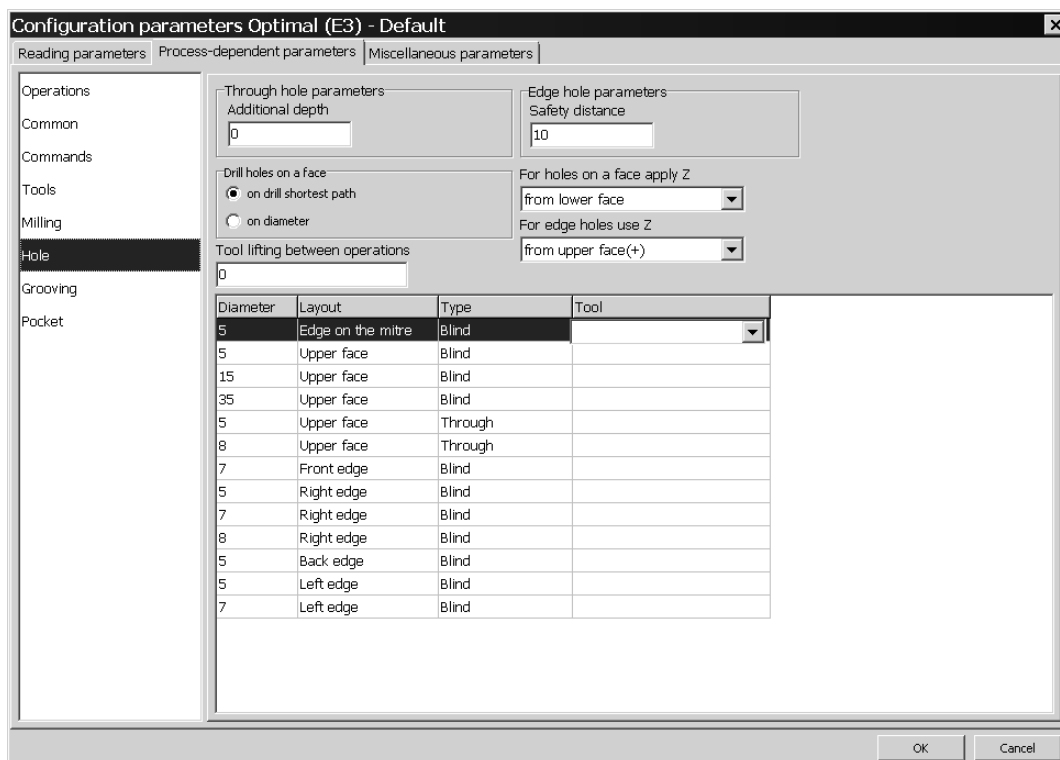


Fig. 10.70.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

For edge holes use Z and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height. Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

Slot parameters

Slots processing is set in **Grooving** section (fig. 10.71).

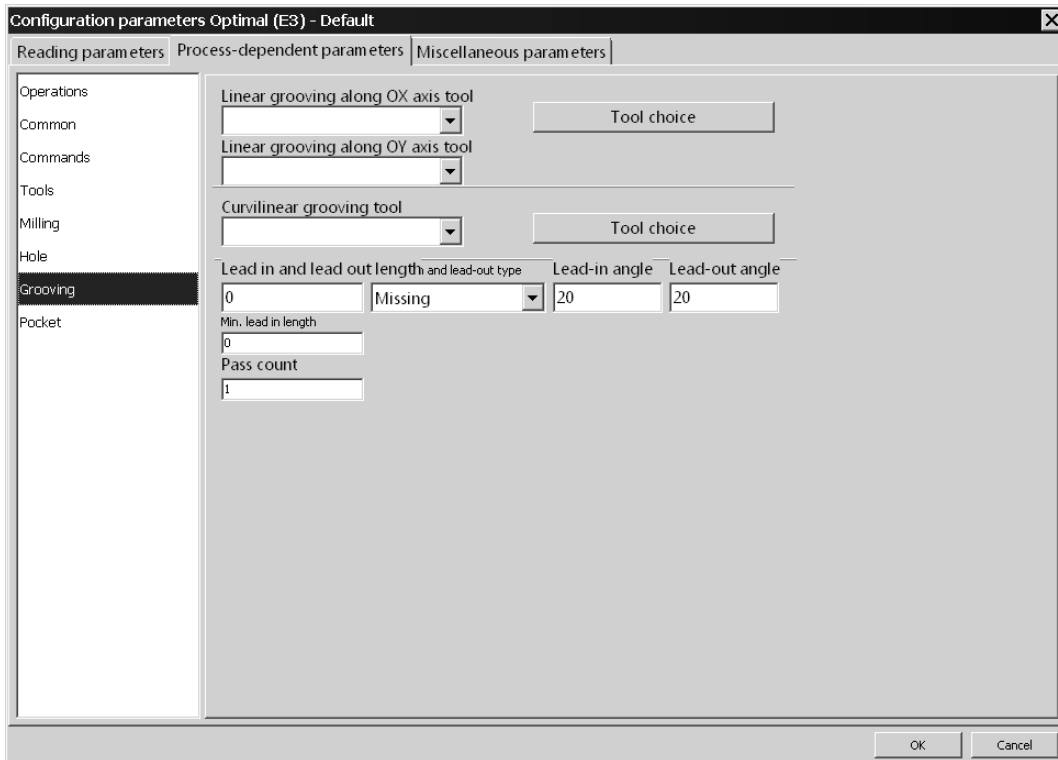


Fig. 10.71.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.72).

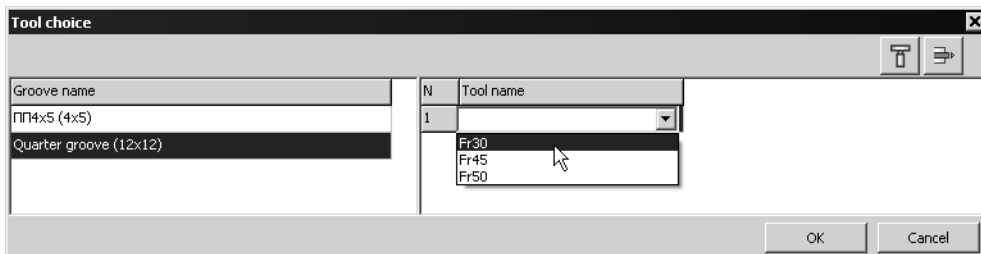


Fig. 10.72.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.73) allow to set pocket processing parameters.

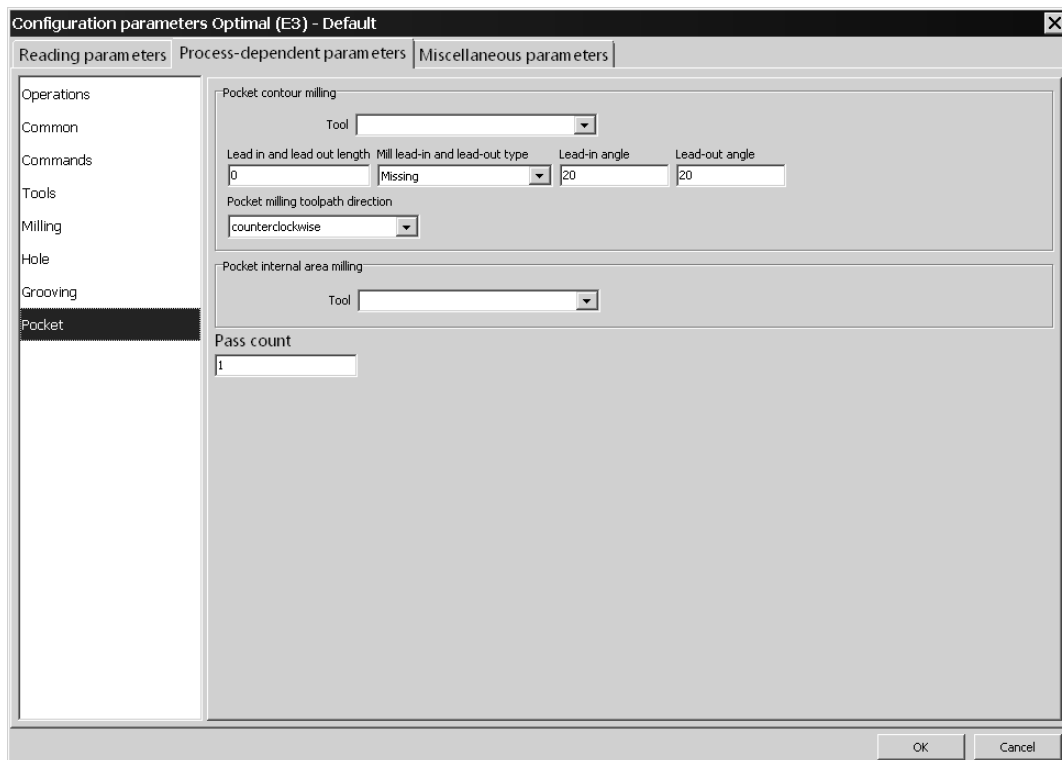


Fig. 10.73.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.5. OSAI Control – NestConcept(E4) Syntec Pre-Labeling



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide

Operations

Operations section control elements (fig. 10.74) allow to specify operations that can be used in control program.

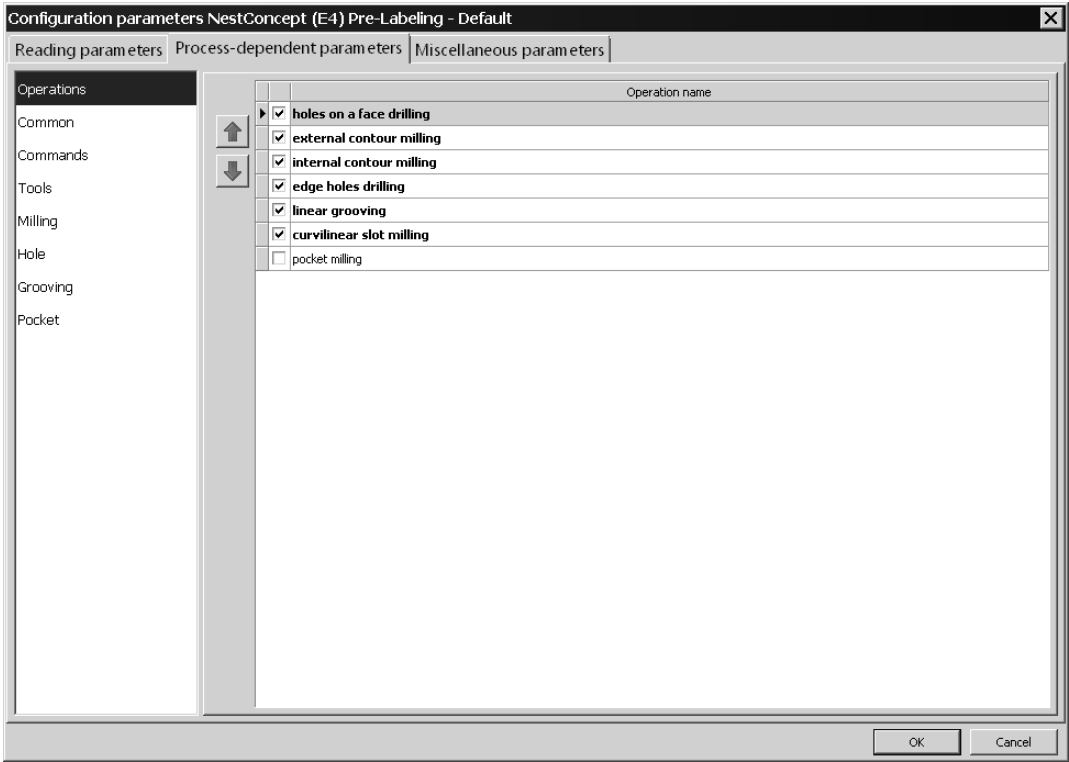


Fig. 10.74.

Common parameters

Common section control elements (fig. 10.75) allow to set control programs common parameters and labels information from BAZIS-Cutting module.

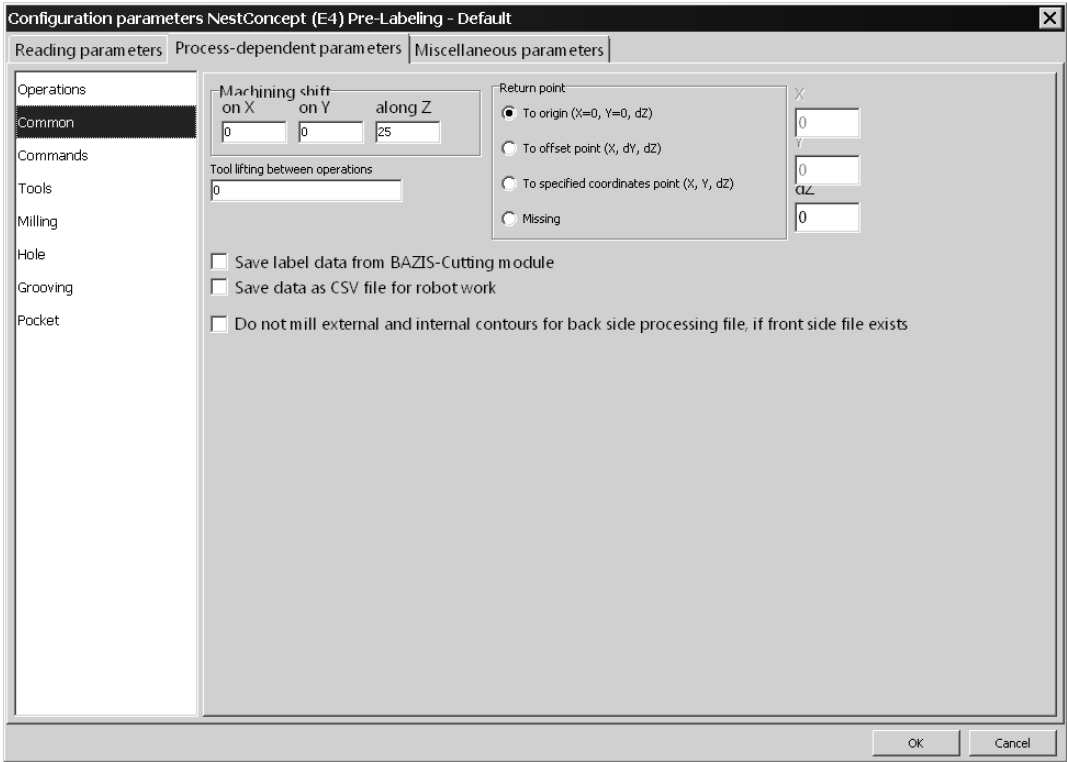


Fig. 10.75.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Base reference area** combo box variants allow to set base reference area position. **Y base point** field allows to set base point position.

Usable commands

Commands section control elements (fig. 10.76) allow to control separate commands to control program inclusion.

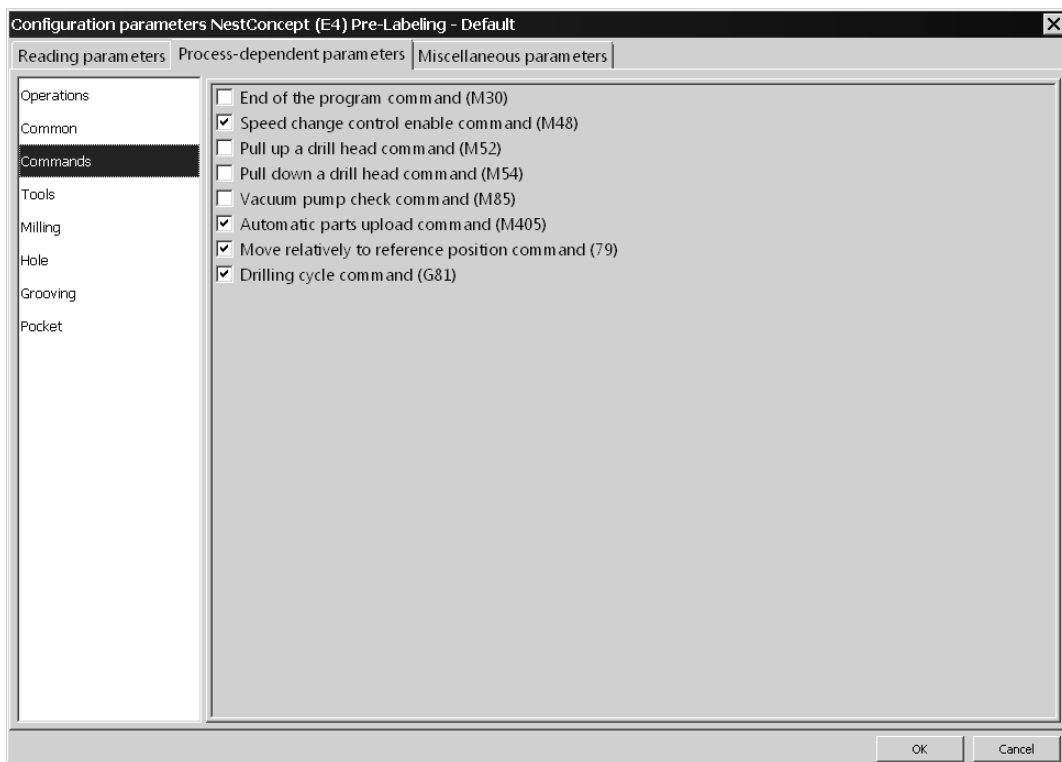


Fig. 10.76.

Usable tools

Tools section control elements (fig. 10.77) allow to create a list of tools available for use in control programs.

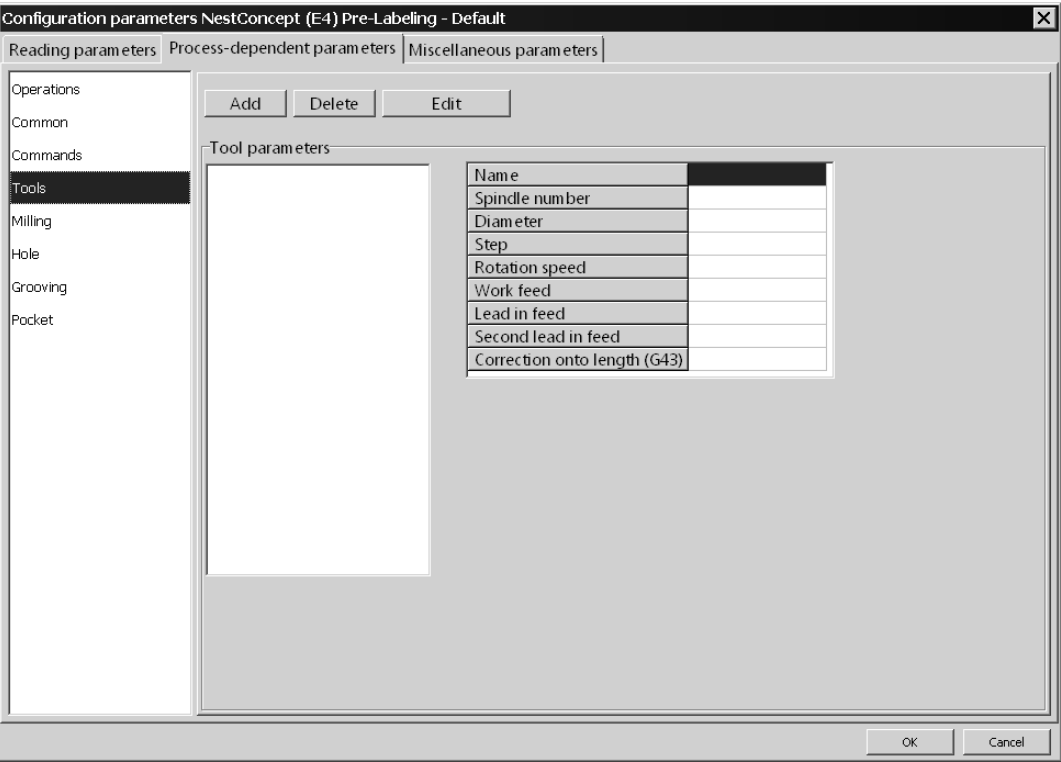


Fig. 10.77.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.78).

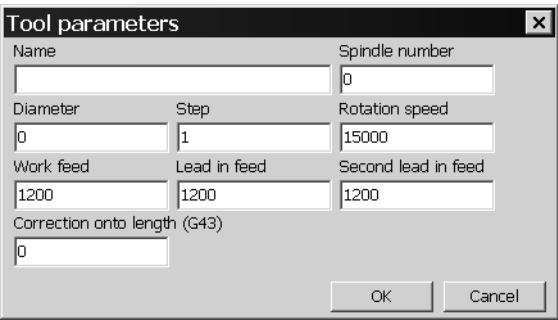


Fig. 10.78.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements allow to set contour milling parameters (fig. 10.79).

Fig. 10.79.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and it is disabled, contour will not be milled. **Path direction** group variants allow to select tool shift direction.

Holes parameters

Holes processing is set in **Hole** section (fig. 10.80).

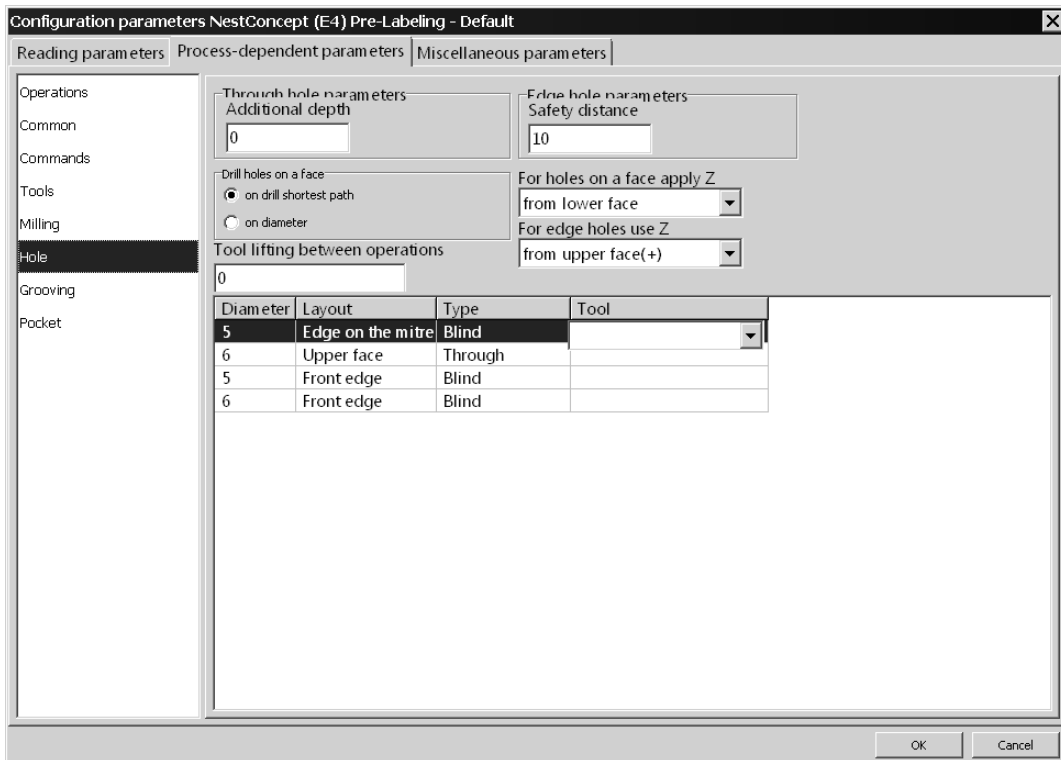


Fig. 10.80.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

For edge holes use Z and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height. Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

Slots parameters

Slots processing is set in **Grooving** section (fig. 10.81).

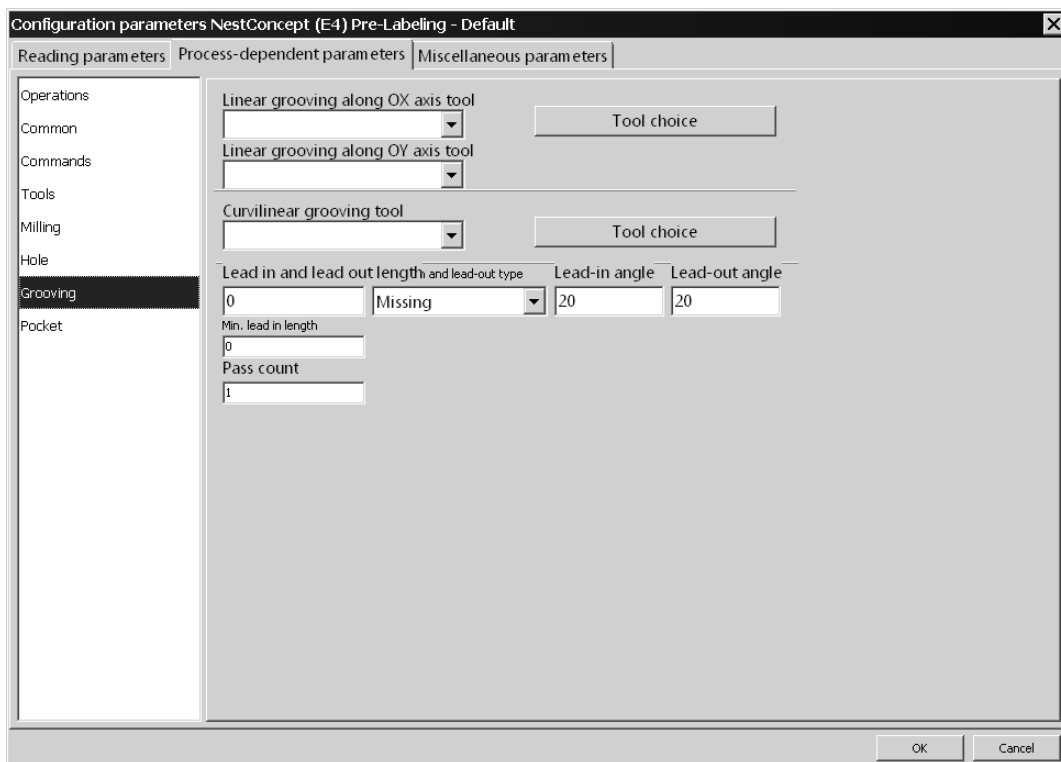


Fig. 10.81.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.82).

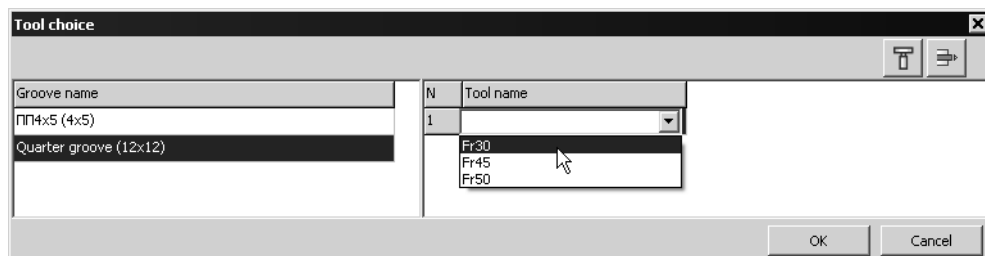


Fig. 10.82.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements allow to set pocket processing parameters (fig. 10.83).

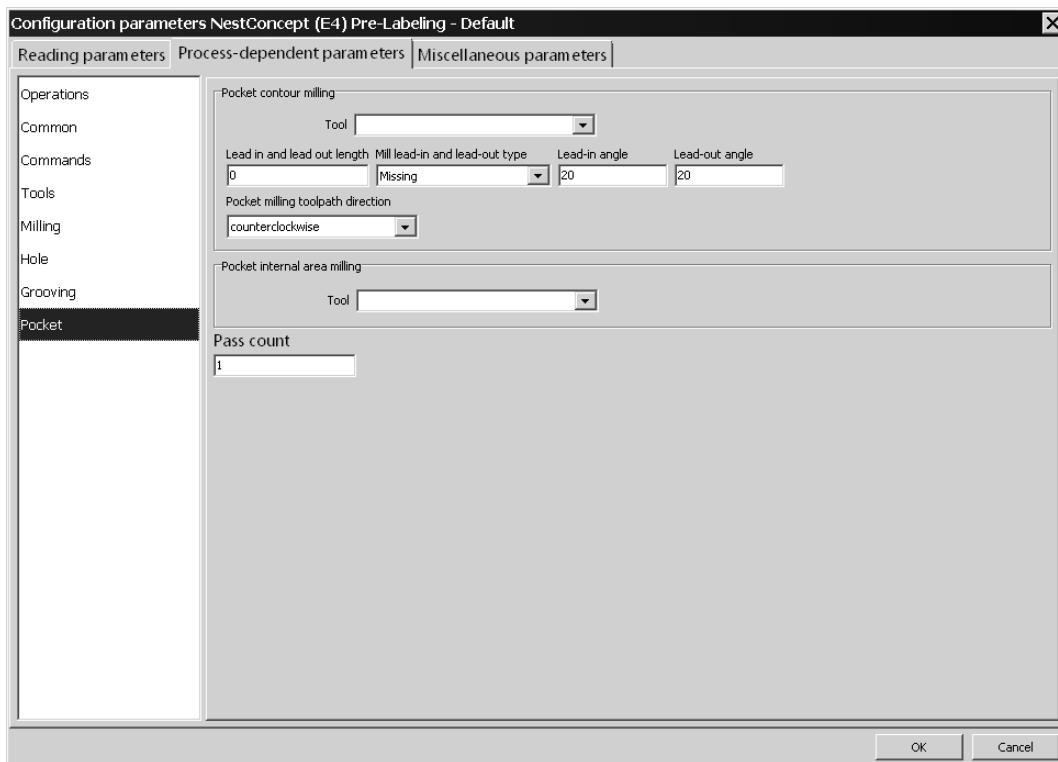


Fig. 10.83.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.6. OSAI Control Universal (E6)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.84) allow to specify operations that can be used in control program.

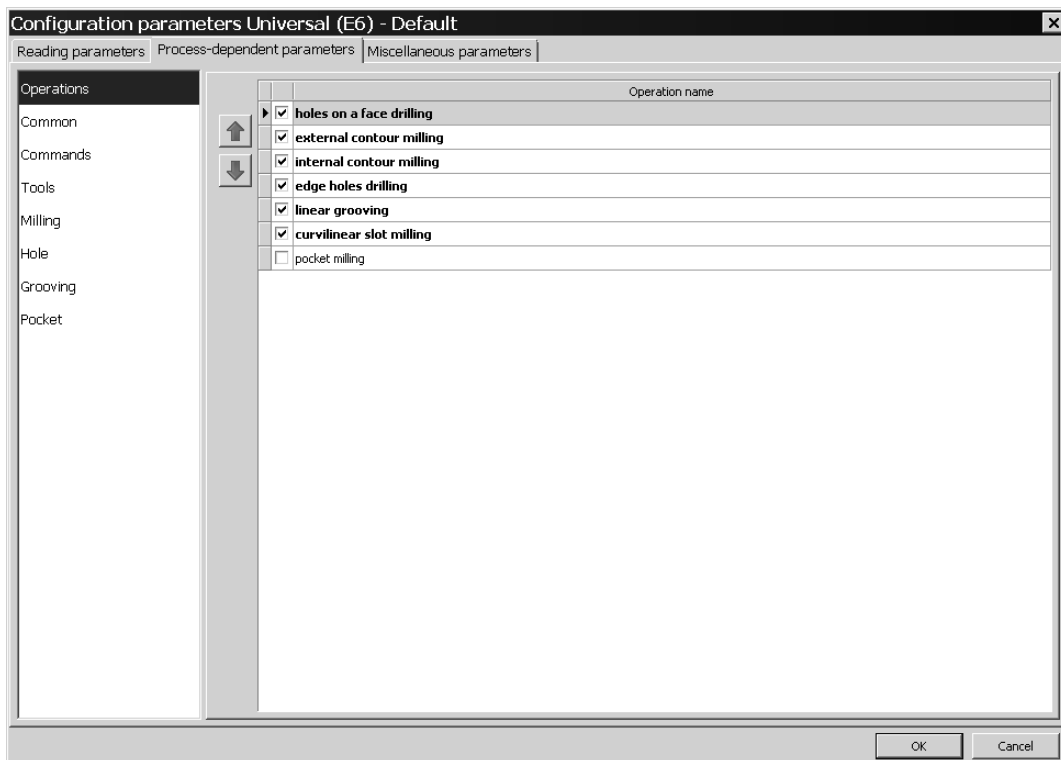


Fig. 10.84.

Common parameters

Common section control elements (fig. 10.85) allow to set control programs common parameters.

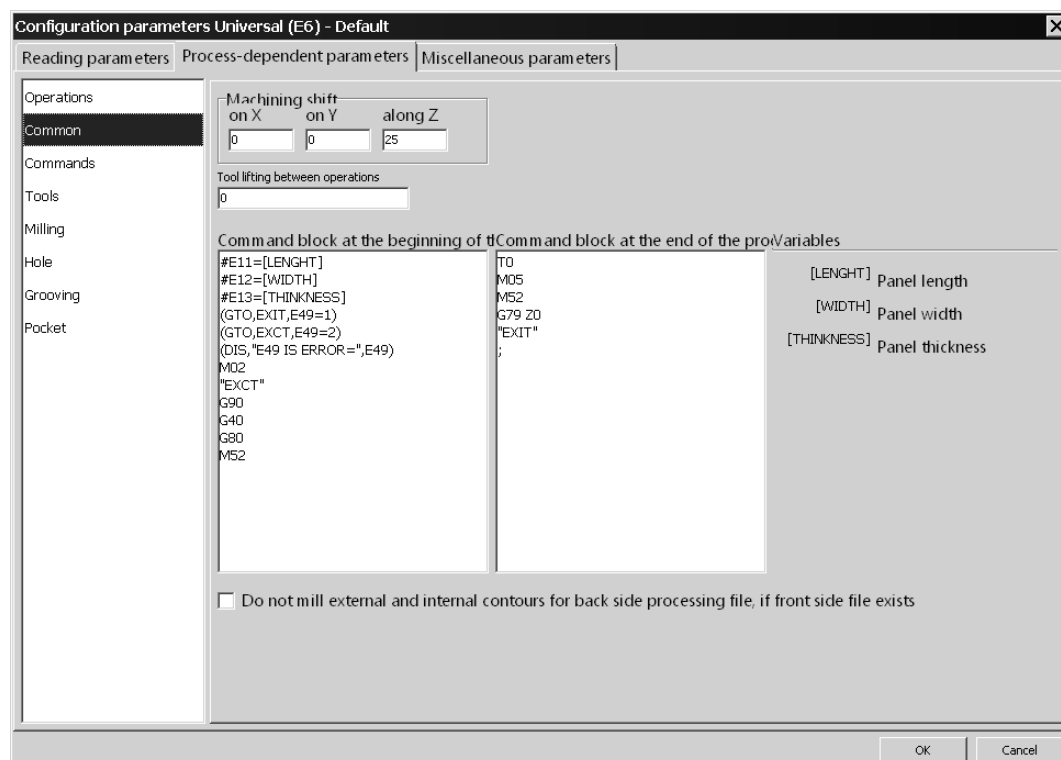


Fig. 10.85.

Machining shift group fields allow to set work area initial point moving. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create commands list that will be automatically placed in specified program points.

Usable commands

Commands section control elements (fig. 10.86) allow to control separate commands to control program inclusion.

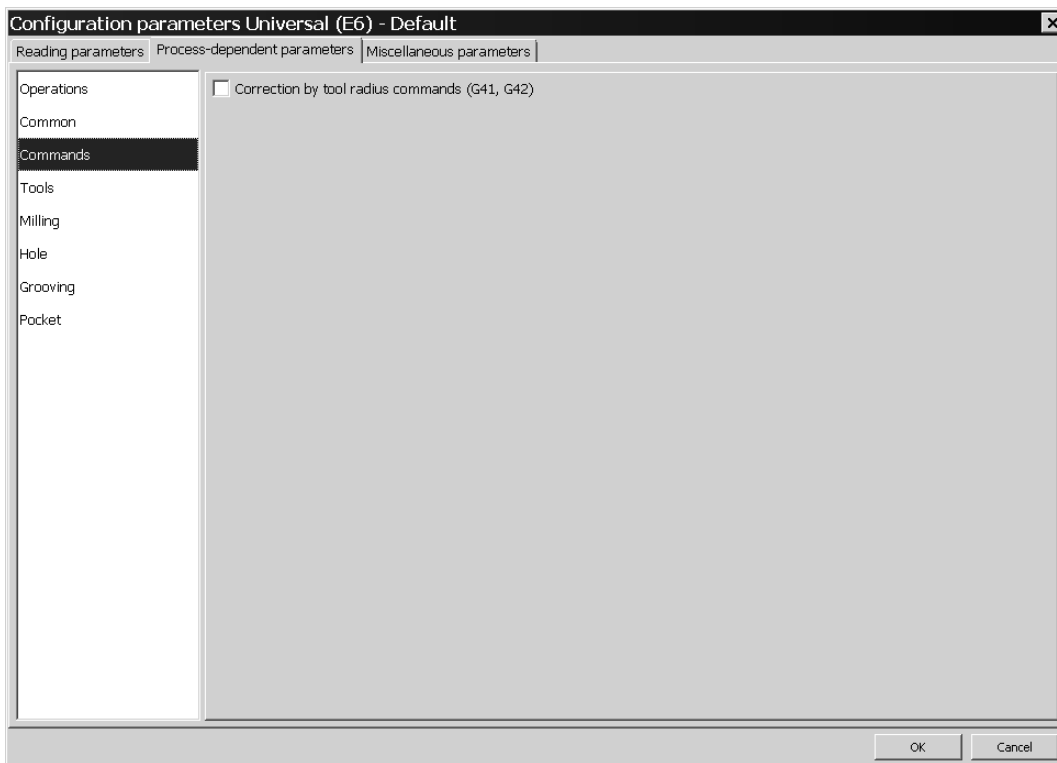


Fig. 10.86.

Usable tools

Tools section control elements (fig. 10.87) allow to create a list of tools available for use in control programs.

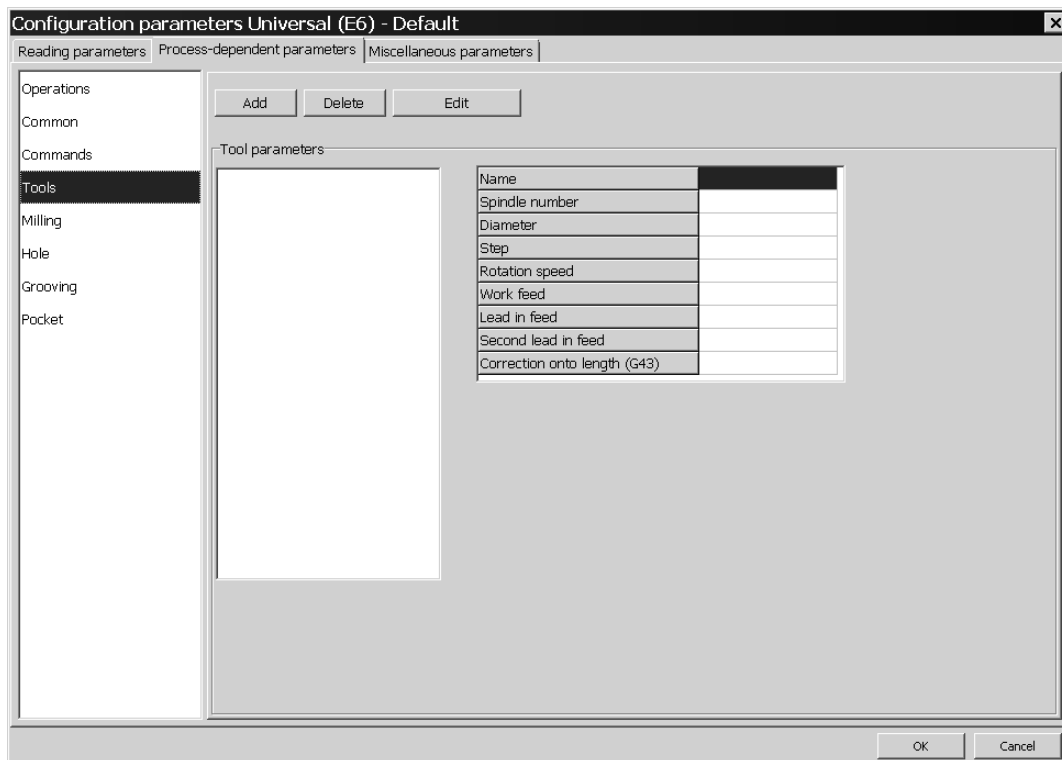


Fig. 10.87.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.88).

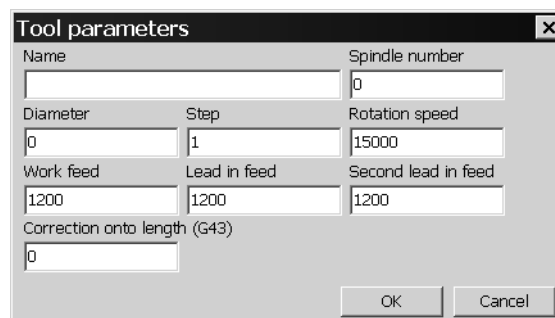


Fig. 10.88.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.89) allow to set contour milling parameters.

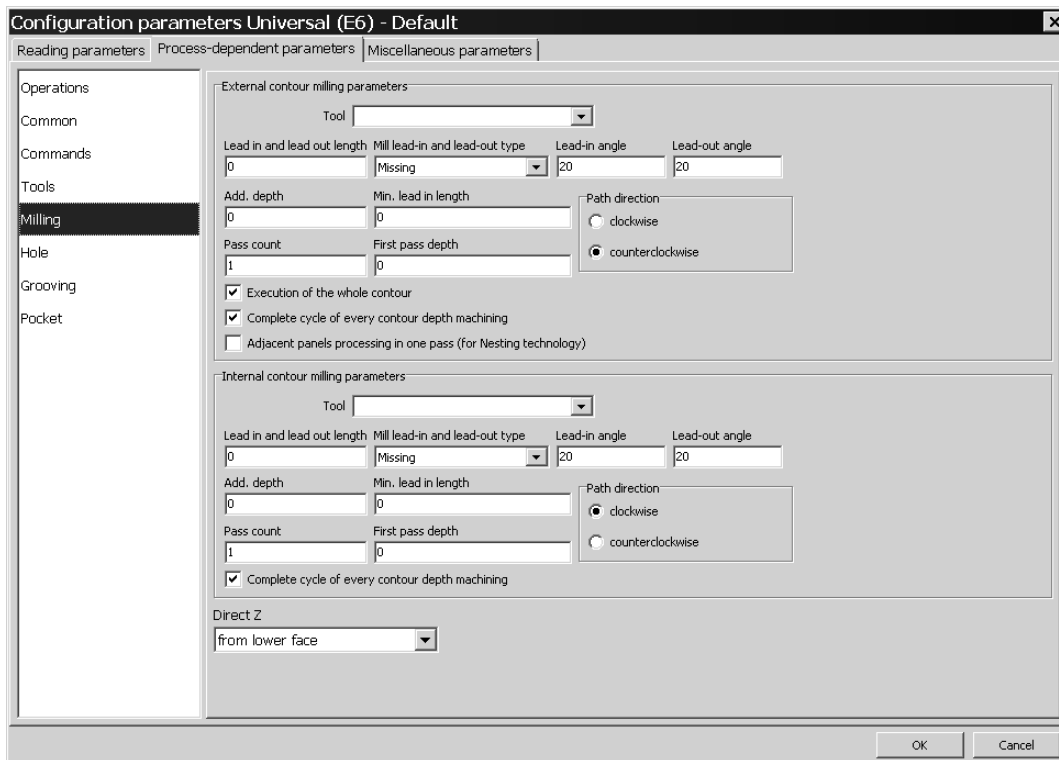


Fig. 10.89.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool shift direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.90).

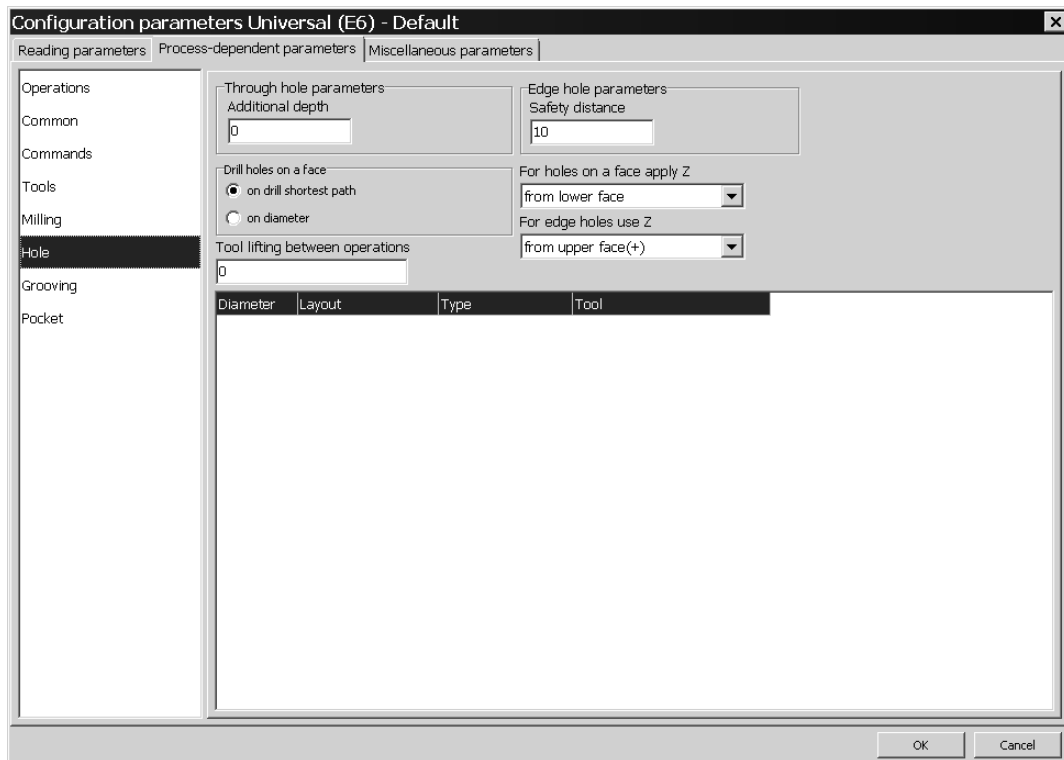


Fig. 10.90.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

For edge holes use Z and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height. Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

Slot parameters

Slots processing is set in **Grooving** section (fig. 10.91).

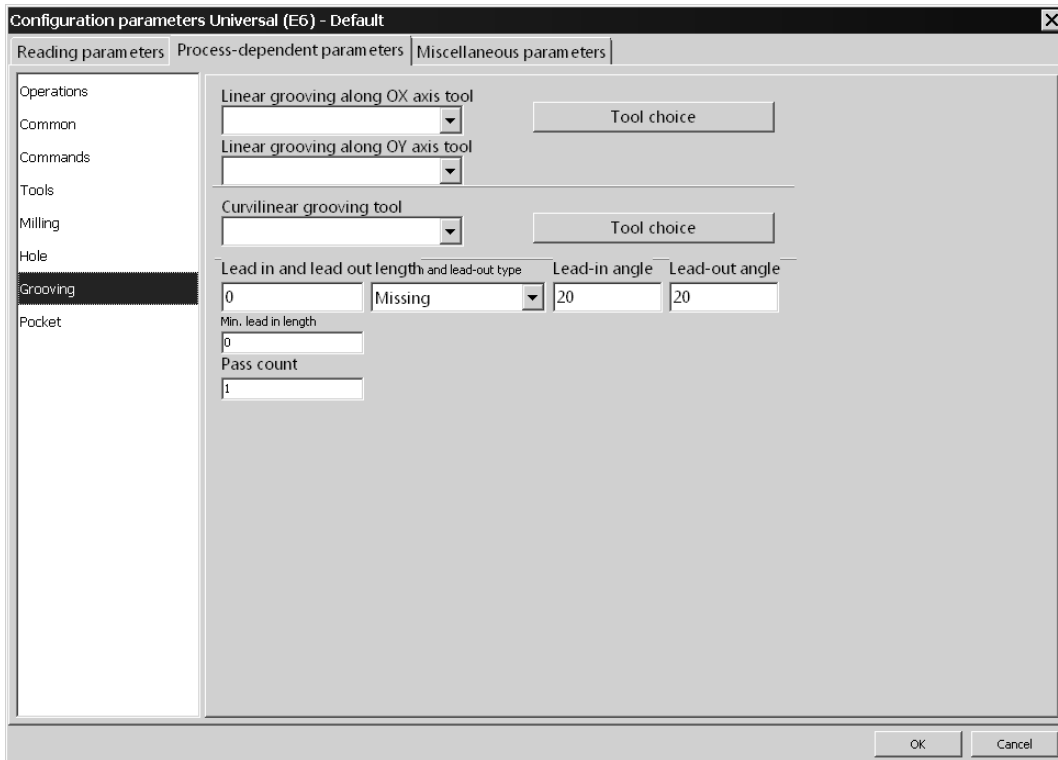


Fig. 10.91.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.92).

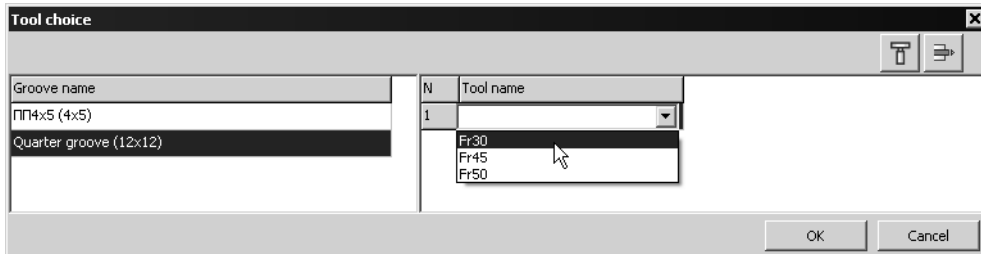


Fig. 10.92.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.93) allow to set pocket processing parameters.

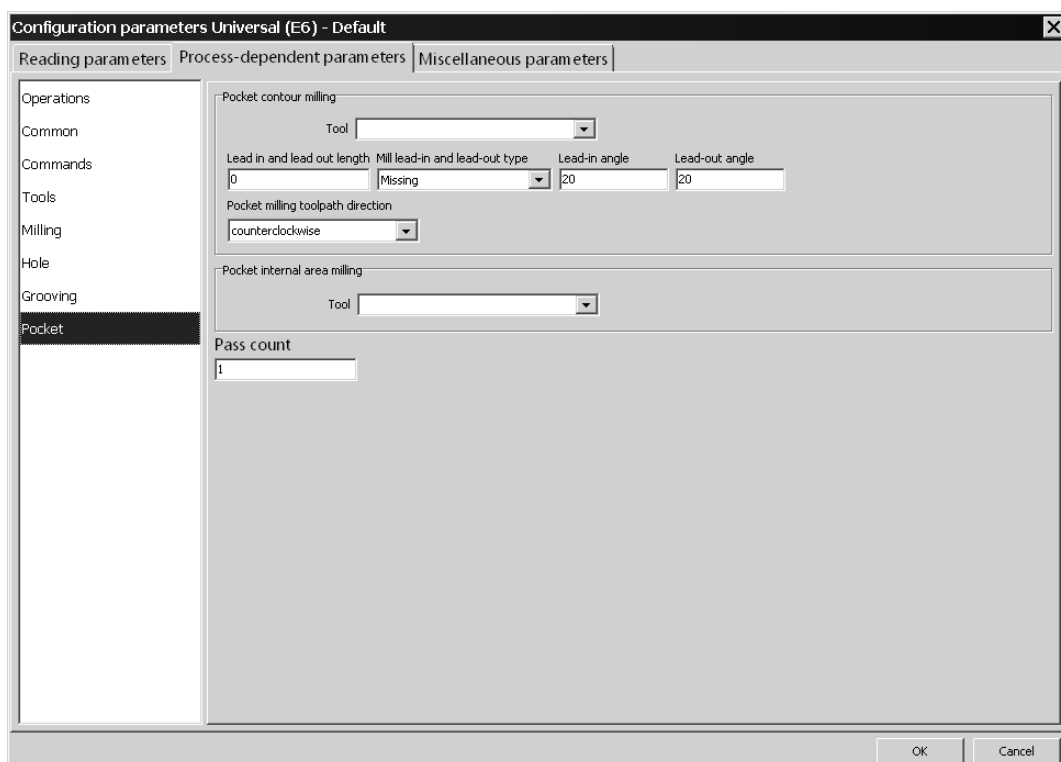


Fig. 10.93.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.7. Swift HSD



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.94) allow to specify operations that can be used in control program.

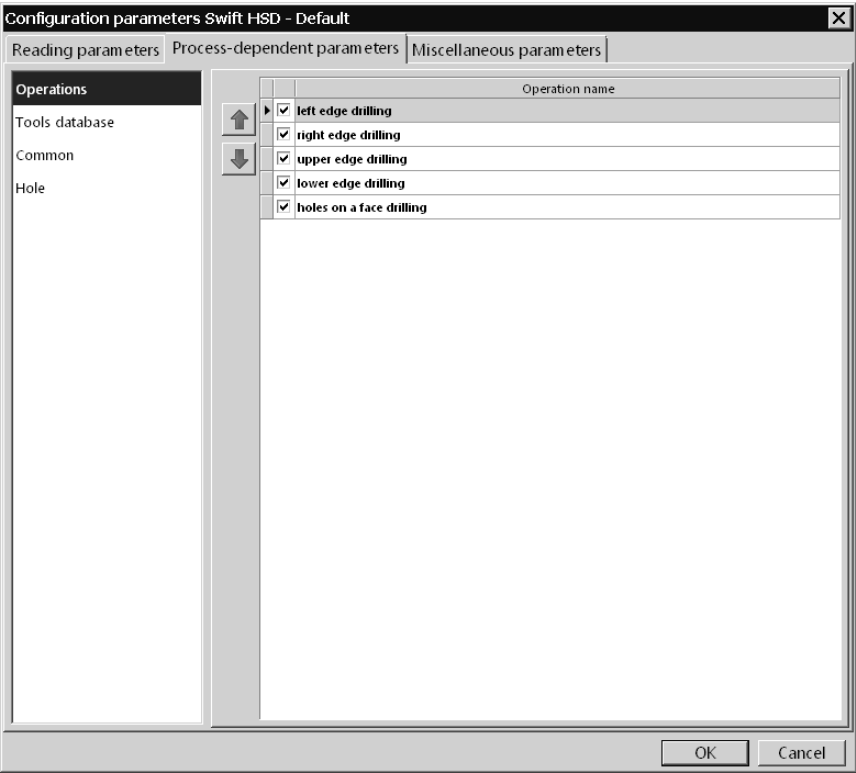


Fig. 10.94.

Usable tools

Tools database section control elements (fig. 10.95) allow to create a list of tools available for use in control programs.



Fig. 10.95.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.96).

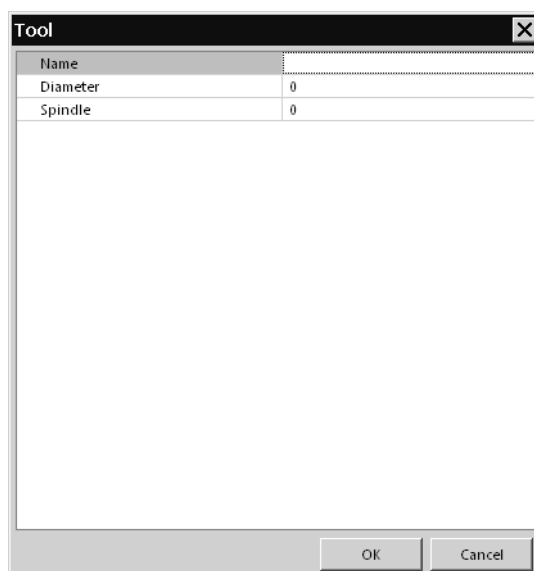


Fig. 10.96.

this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes.



Edit button allows to change tool parameters selected in the list. **Delete** button allows to delete selected tool from the list.



Move button allows to move selected tool to another group. After pushing the button, a group tree will appear on the screen (fig. 10.97).

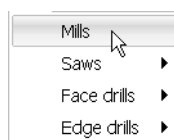


Fig. 10.97.

To move a tool to the folder, doubleclick on its name.

10.5.8.Swift Turbo (EH, EHS, EHSW)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.98) allow to specify operations that can be used in control program.

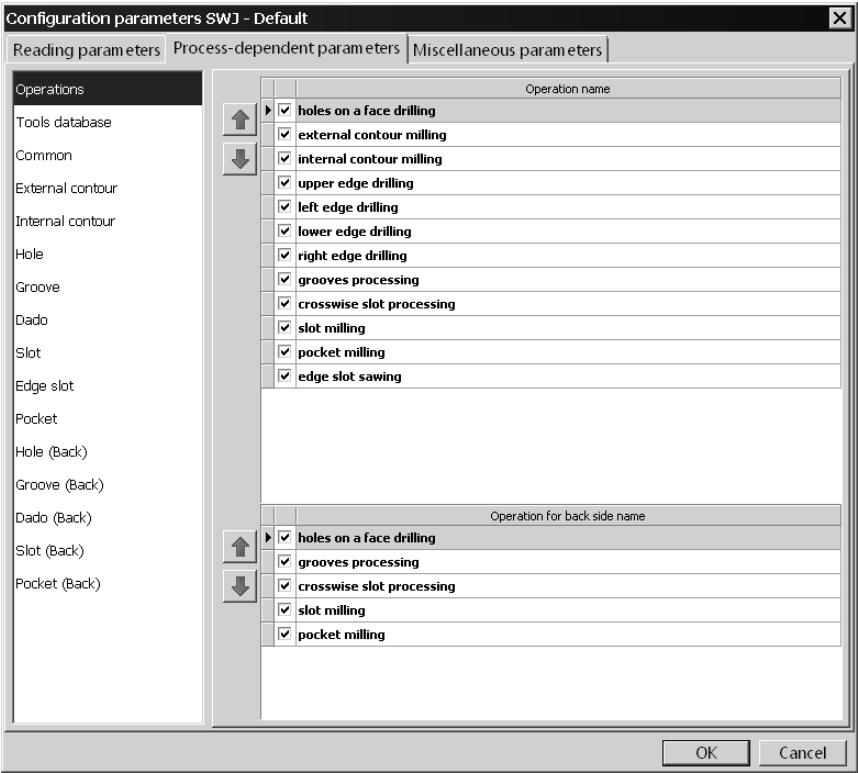


Fig. 10.98.

Usable tools

Tools database section control elements (fig. 10.99) allow to create a list of tools available for use in control programs.

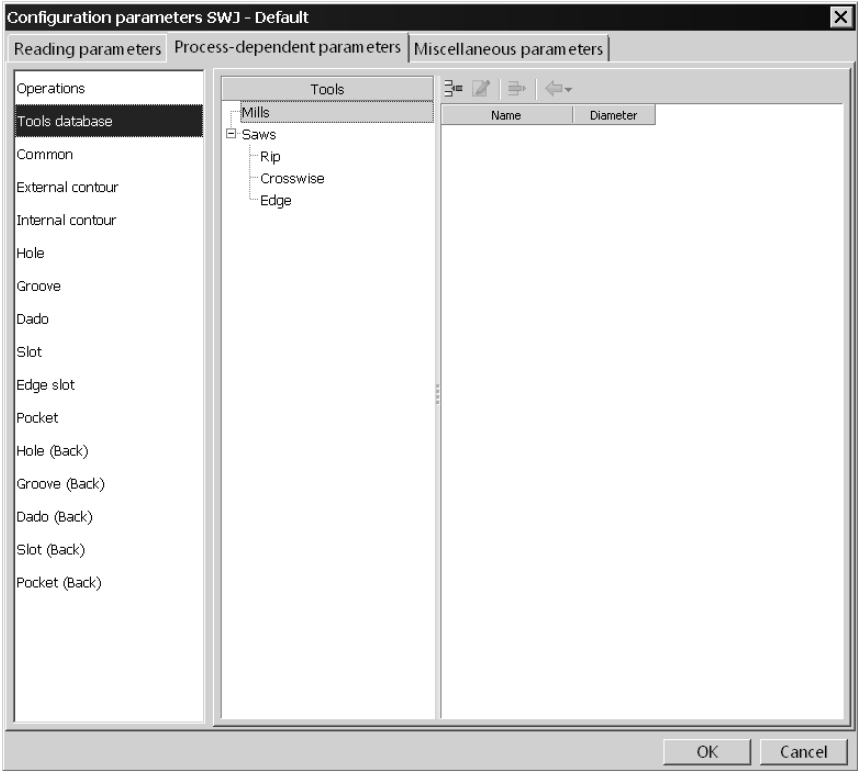


Fig. 10.99.



To add a tool to the list push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.100).

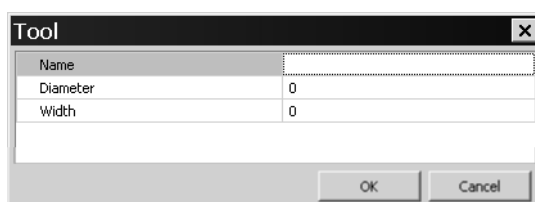


Fig. 10.100.



In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list.



Delete button allows to delete selected tool from a list.

Common parameters

Common section control elements (fig. 10.101) allow to set control programs common parameters.

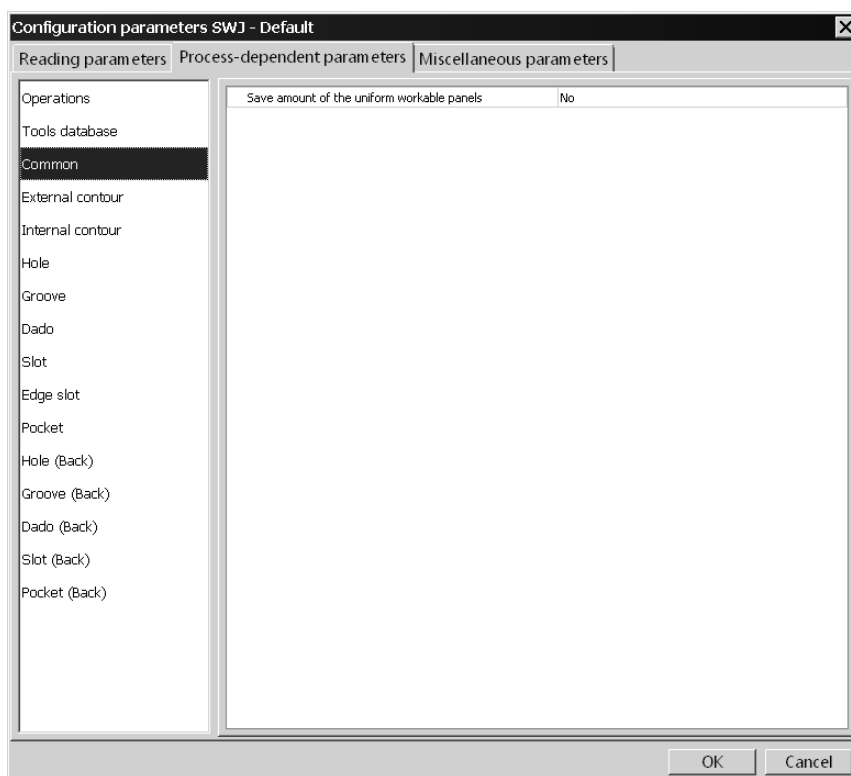


Fig. 10.101.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** and **Internal contour** sections (fig. 10.102).

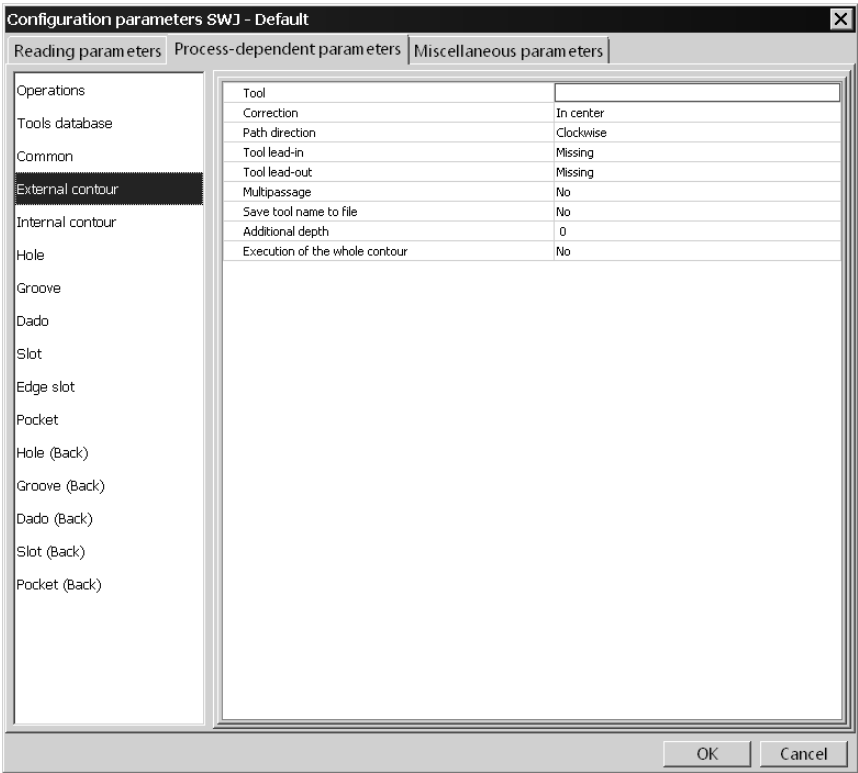


Fig. 10.102.

Section control elements allow to choose the tool, correction types, tool lead-in and lead-out, etc.

Hole parameters

Hole processing from front side is set in **Hole** section (fig. 10.103), holes from back side – in **Hole (Back)** section in the same way.

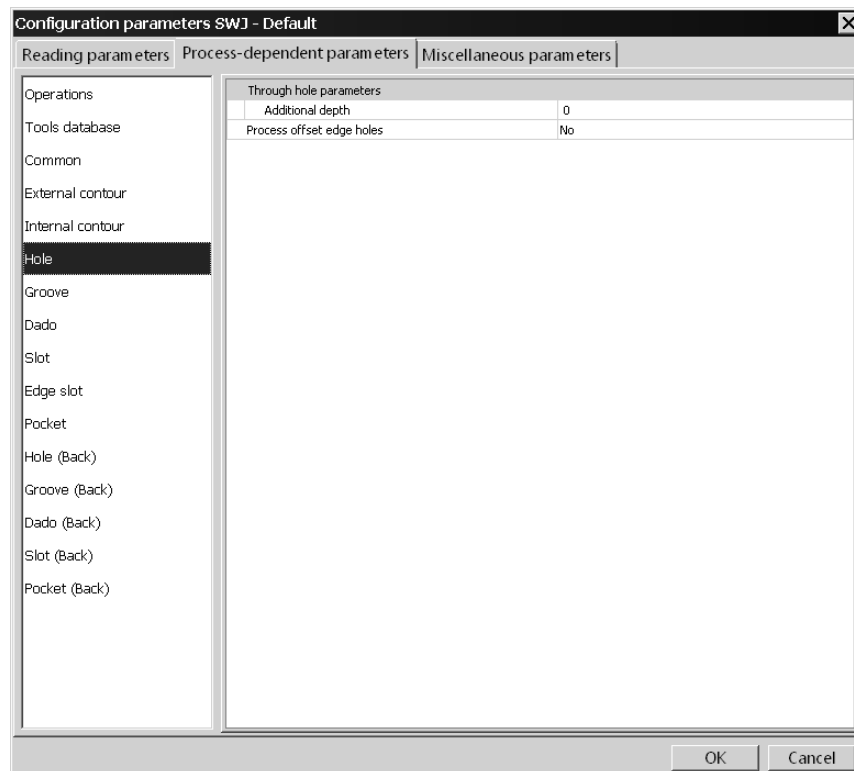


Fig. 10.103.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Grooves processing from front side is set in the following sections: **Groove** (fig. 10.104) and **Dado**. Grooves processing from back side of the panel is set in **Groove (Back)** and **Dado (Back)** sections.

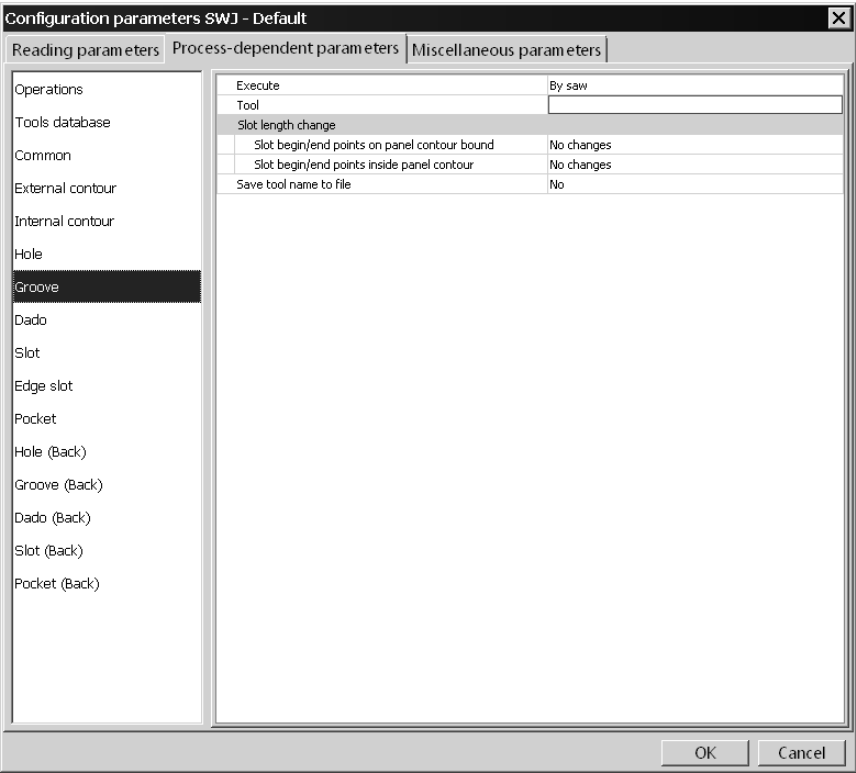


Fig. 10.104.

The section control elements allow to select a tool type for groove creation, data-base tool and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory .

Slot parameters of custom trajectory

Slot processing from front side is set in section **Slot** (fig. 10.105), from panel back side – in **Slot (Back)** section.

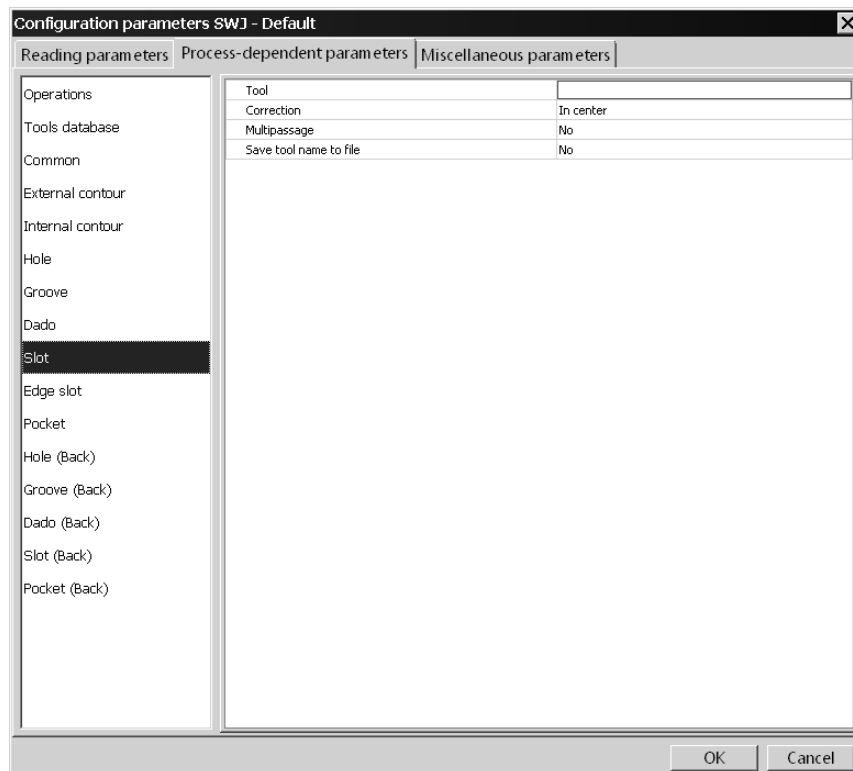


Fig. 10.105.

Section control elements allow to choose database tool, correction type and multipassage use.

Pocket parameters

Pocket processing parameters from front side are set in **Pocket** section (fig. 10.106), from panel back side – in **Pocket (Back)** section.

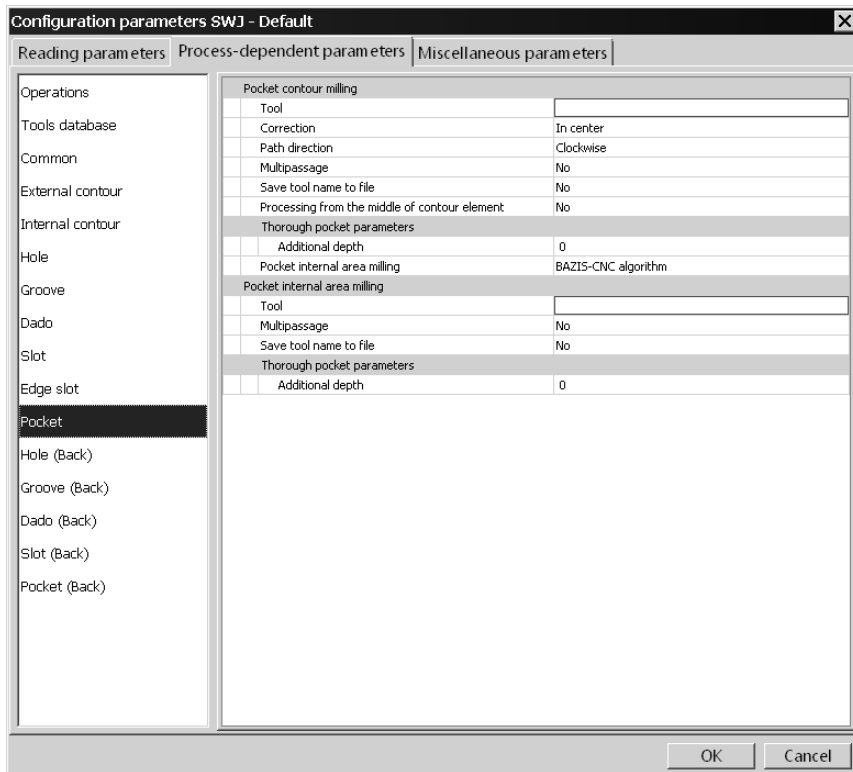


Fig. 10.106.

Tool combo box allows to select a milling tool name from a database. For pocket contour select milling path direction, correction type and multipassage use.

Setup completion

To complete control programs setup for Swift Turbo, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.9. Syntech



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.107) allow to specify operations that can be used in control program.

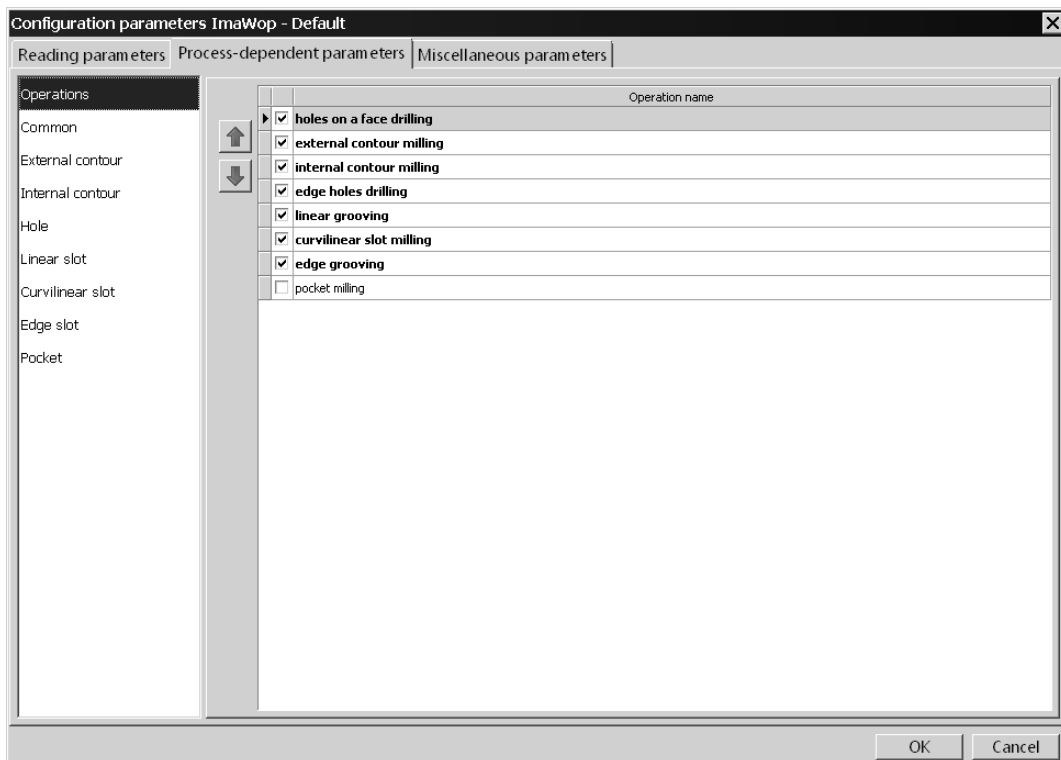


Fig. 10.107.

Common parameters

Common section control elements (fig. 10.108) allow to set common parameters of control programs creation.

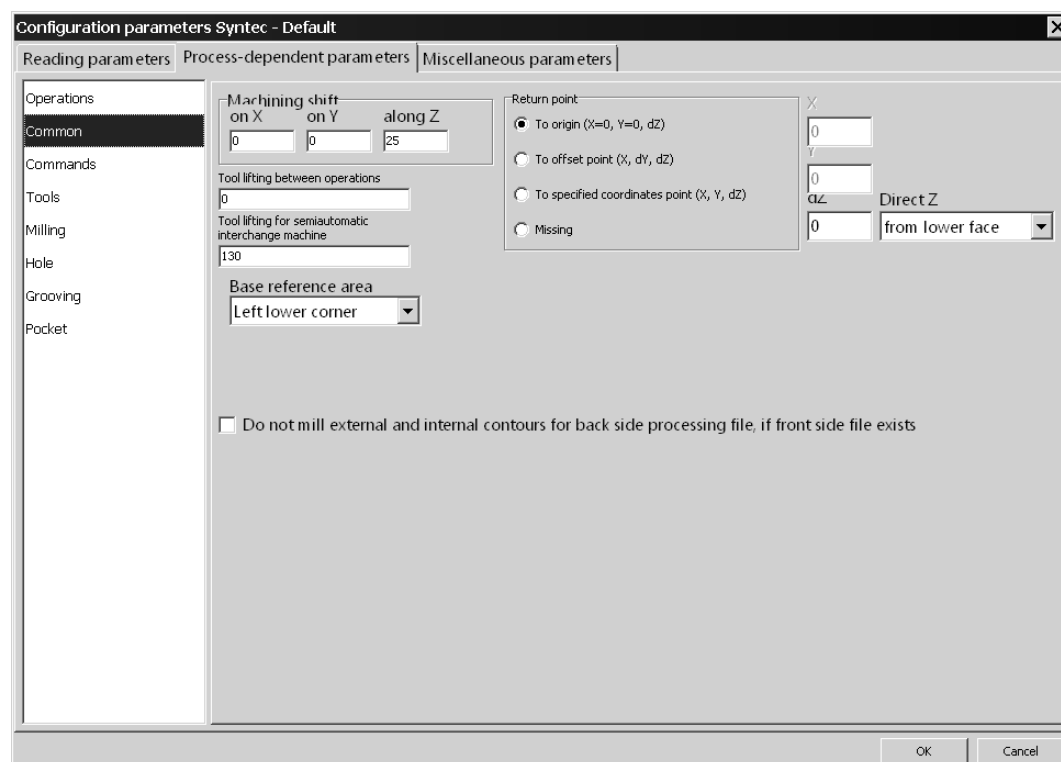


Fig. 10.108.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height.

Usable commands

Commands section control elements (fig. 10.109) allow to control separate commands to control program inclusion.

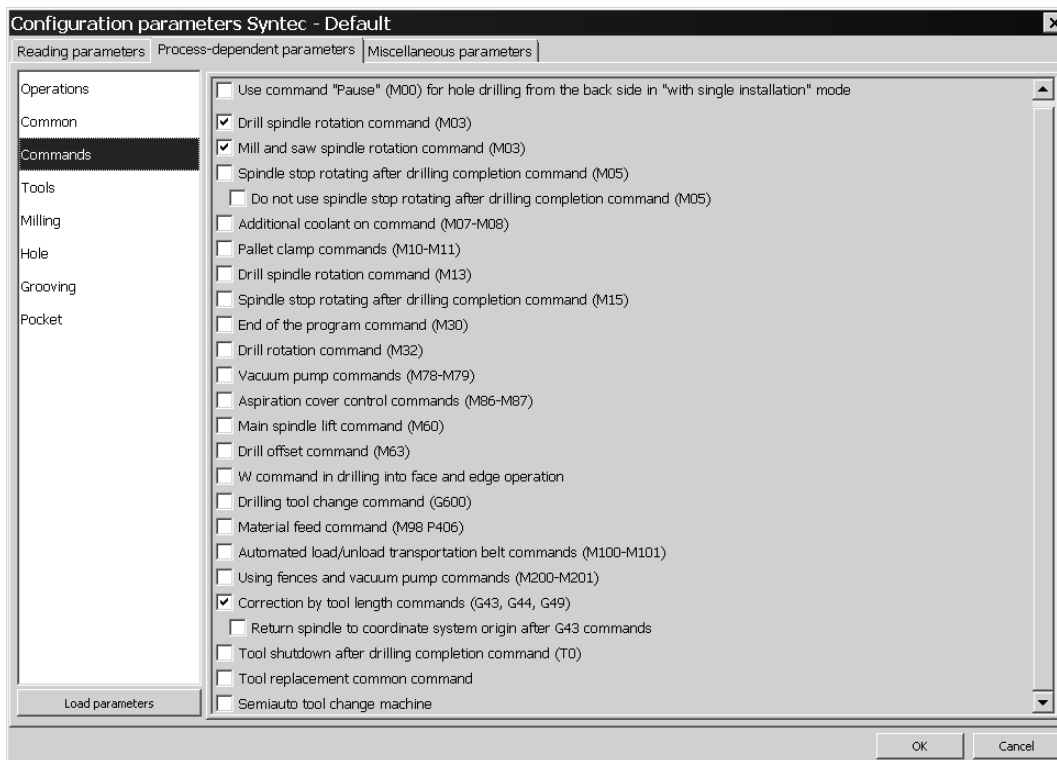


Fig. 10.109.

Usable tools

Tools section control elements (fig. 10.110) allow to create a list of tools available for use in control programs.

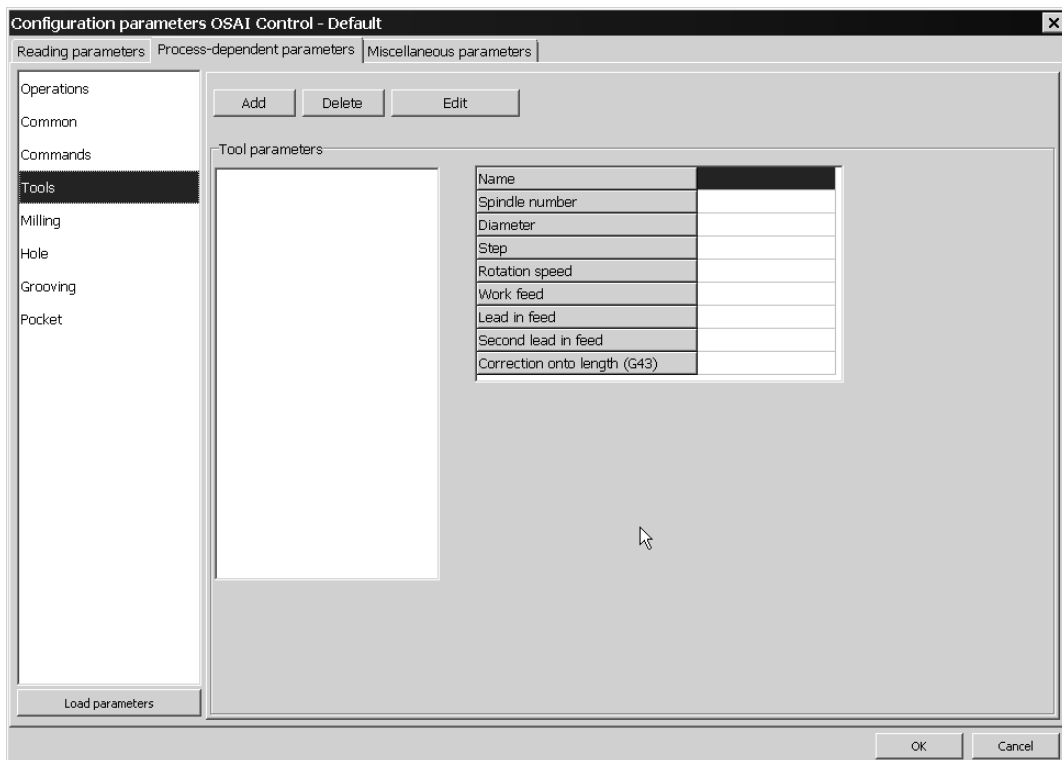


Fig. 10.110.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.111).

Fig. 10.111.

In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.112) allow to set contour milling parameters.

Fig. 10.112.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.113).

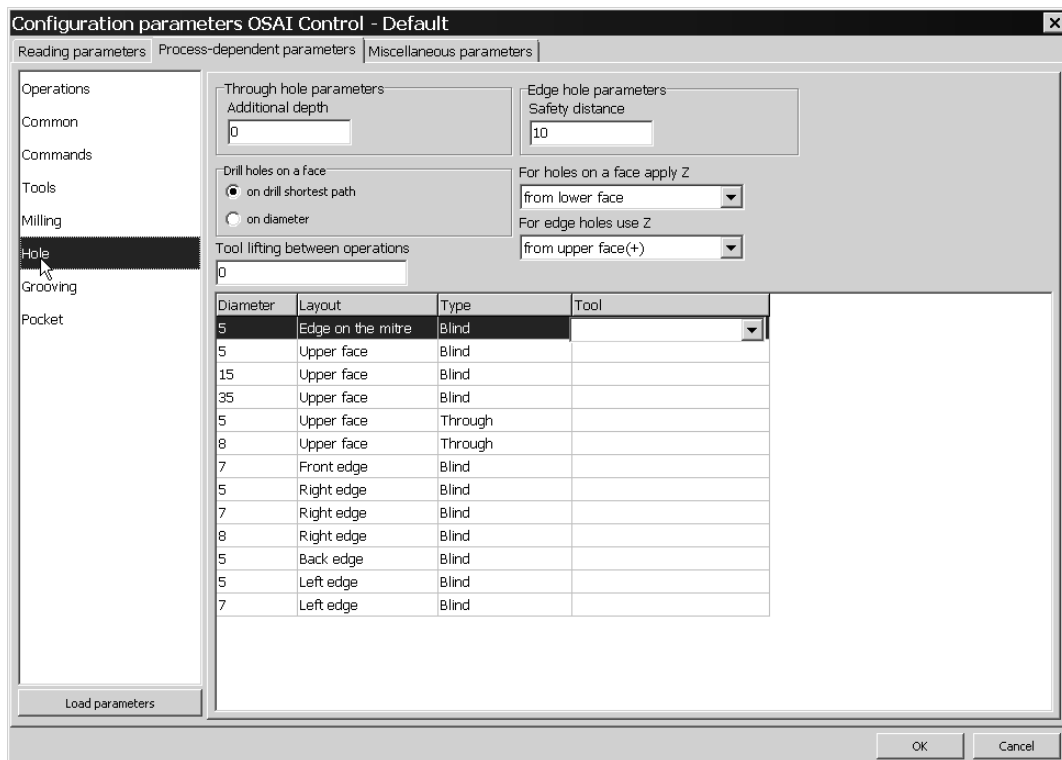


Fig. 10.113.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **For edge holes use Z** and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height.

Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the center of the hole, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slot parameters

Grooving section control elements (fig. 10.114) allow to set slots processing parameters.

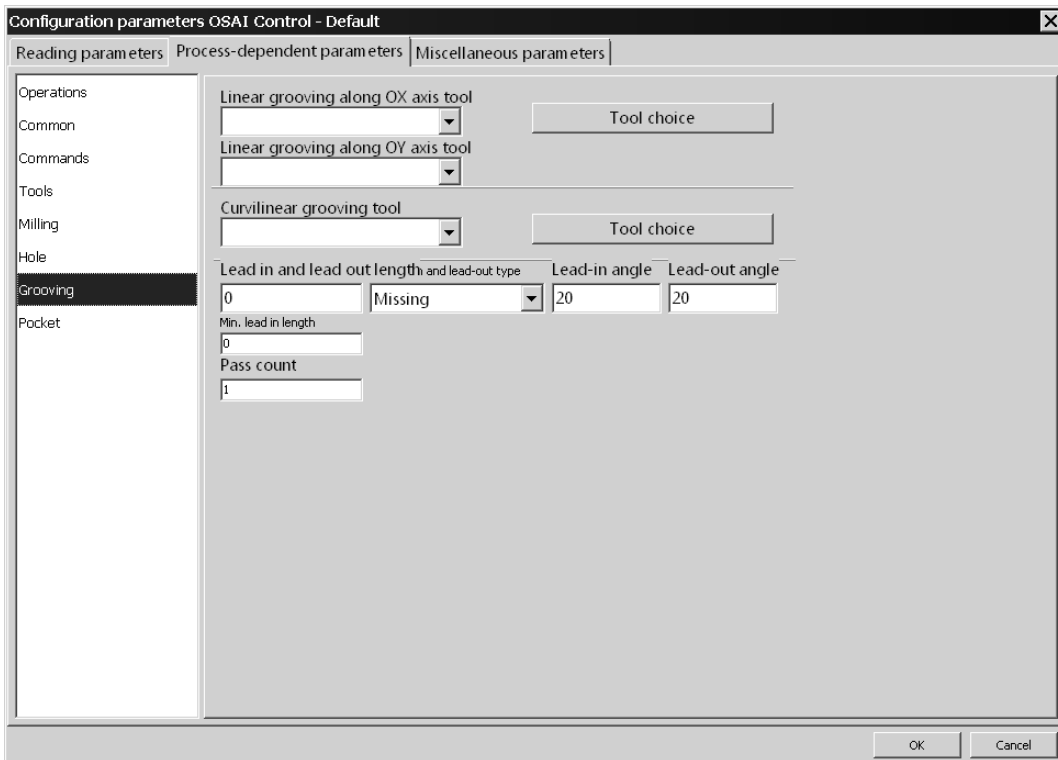


Fig. 10.114.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set a tool for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.115).

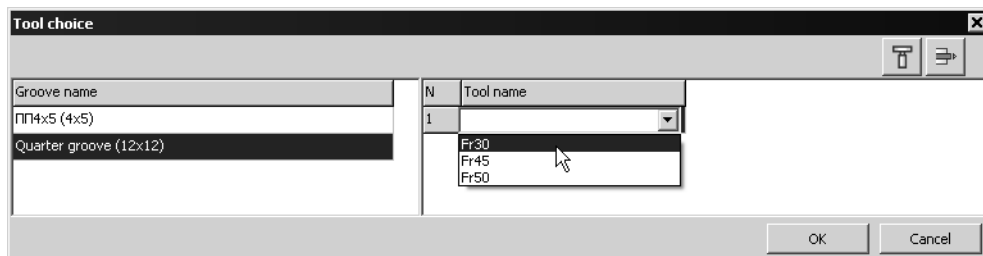


Fig. 10.115.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.116) allow to set pocket processing parameters.

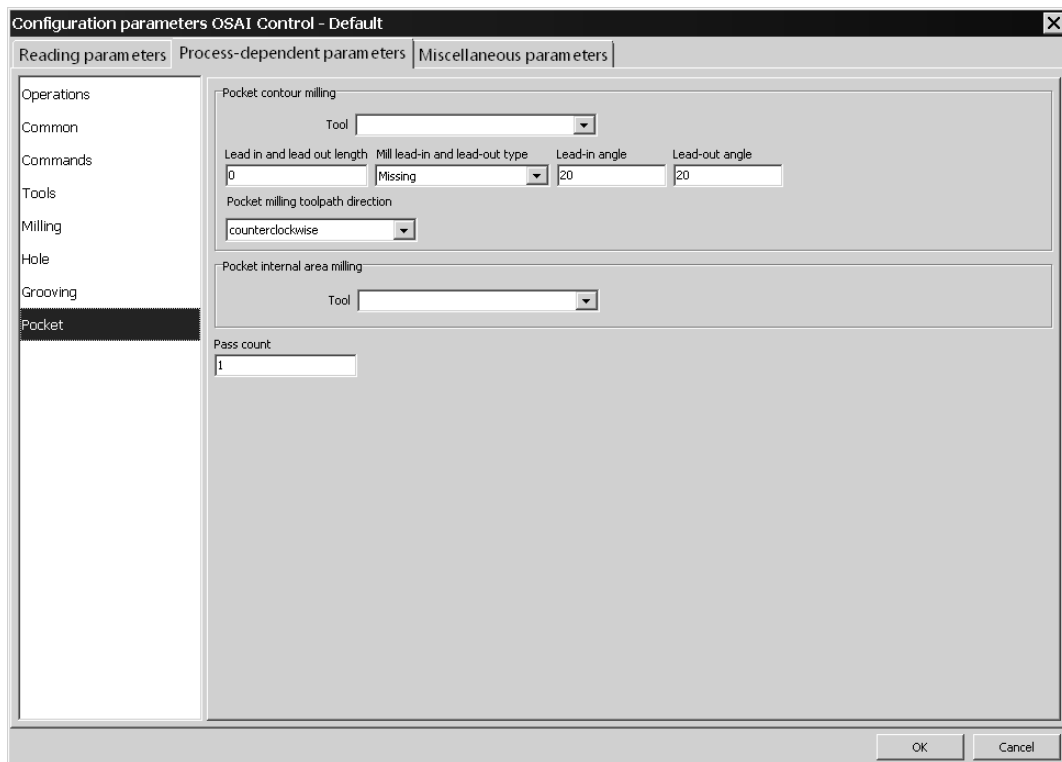


Fig. 10.116.

Section control elements allow to set control programs creation parameters for notch contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.10.TPA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.117) allow to specify operations that can be used in control program

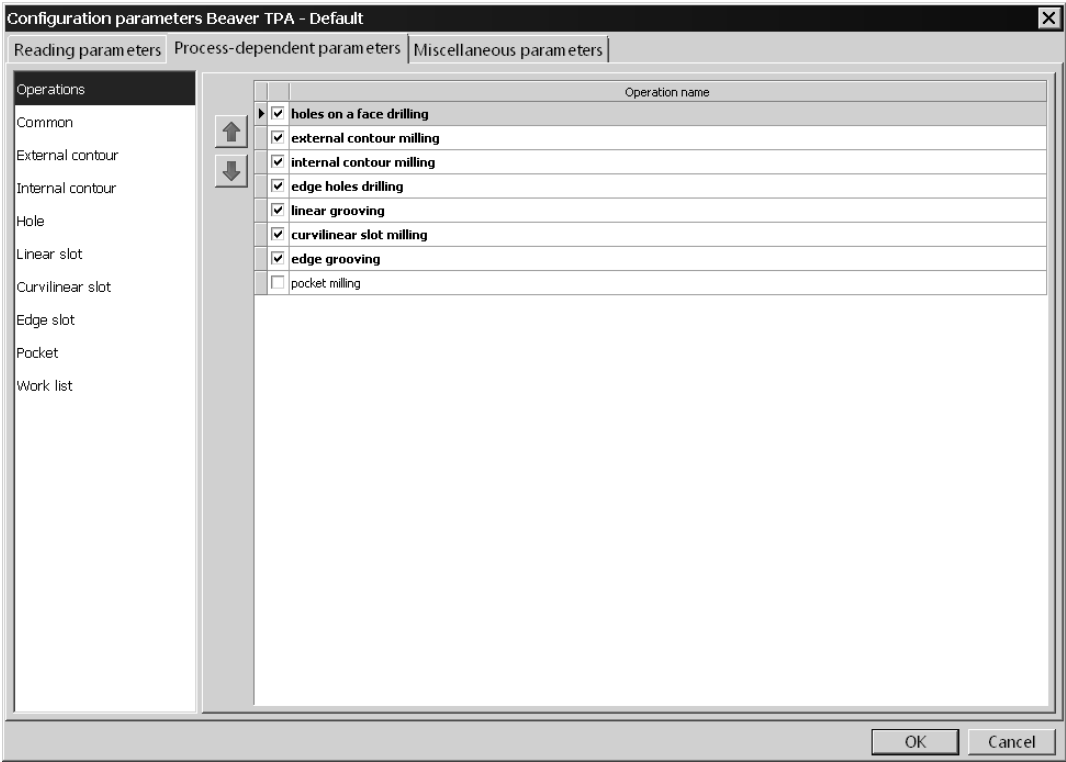


Fig. 10.117.

Common parameters

Common section control elements (fig. 10.118) allow to set common parameters.

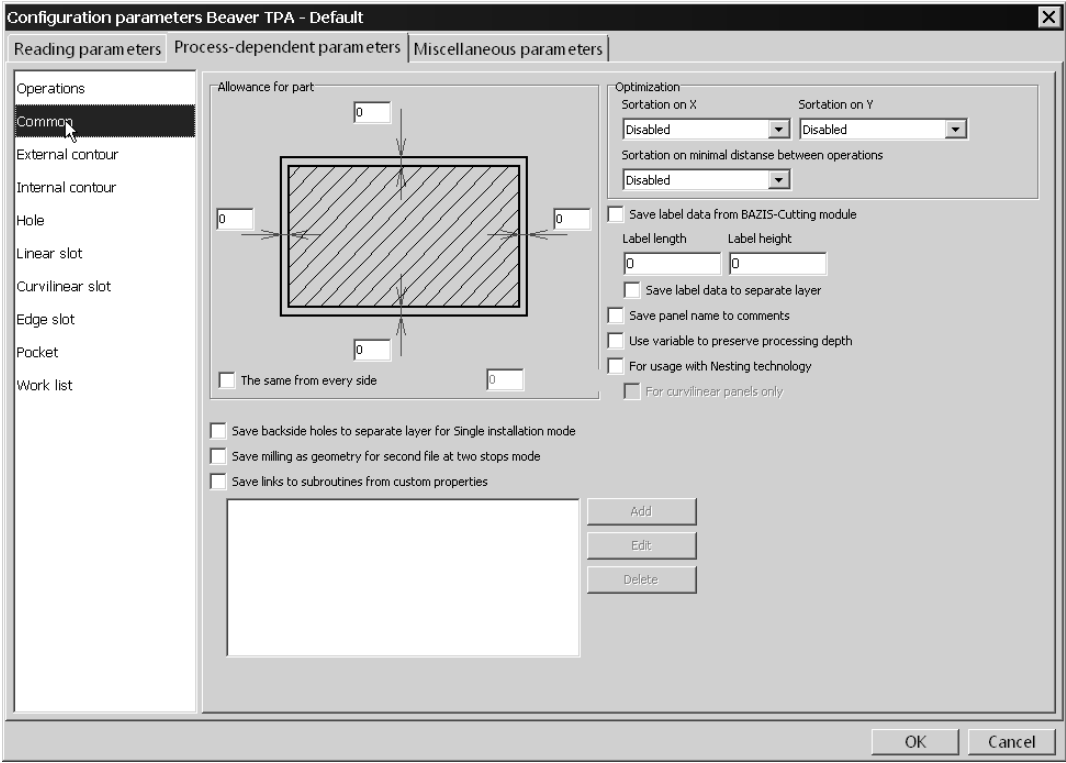


Fig. 10.118.

Allowance for part section control elements allow to set allowance values. Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, the same from every panel edge allowance field will become available. **Append operations to CNC program** group control elements allow to control operations of options names appending to control program. To add the program to the program, enable the option next to its name.

Contours parameters

Contours milling parameters are set in **Outer contour** (fig. 10.119) and **Internal contour** sections of **Process-dependent parameters** tab.

The screenshot shows a software window titled "Configuration parameters Beaver TPA - Default". It has three tabs: "Reading parameters", "Process-dependent parameters", and "Miscellaneous parameters". The "Process-dependent parameters" tab is selected. On the left, there is a tree view with categories: "Operations", "Common", "External contour" (selected), "Internal contour", "Hole", "Linear slot", "Curvilinear slot", "Edge slot", "Pocket", and "Work list". The main area displays parameters for "External contour milling".

Tool		Processing from the middle of contour element	No
Correction	In center	Add. depth	0
Lead-in type	Missing	Lead-out type	Missing
Lead-in Length/Radius	0	Lead-out Length/Radius	0
Lead-in Z start	0	Final Z	0
Lead in/Lead out point trajectory overlapping		0	
Pass count	1	Execution of the whole contour	Yes
First pass depth	0		
Milling path direction:			
☐ clockwise ☒ counterclockwise			

At the bottom right, there are "OK" and "Cancel" buttons.

Fig. 10.119.

Finish machining and **Rough machining** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Rough machining** option allows to control this operation. Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Hole section control elements allow to set holes parameters (fig. 10.120).

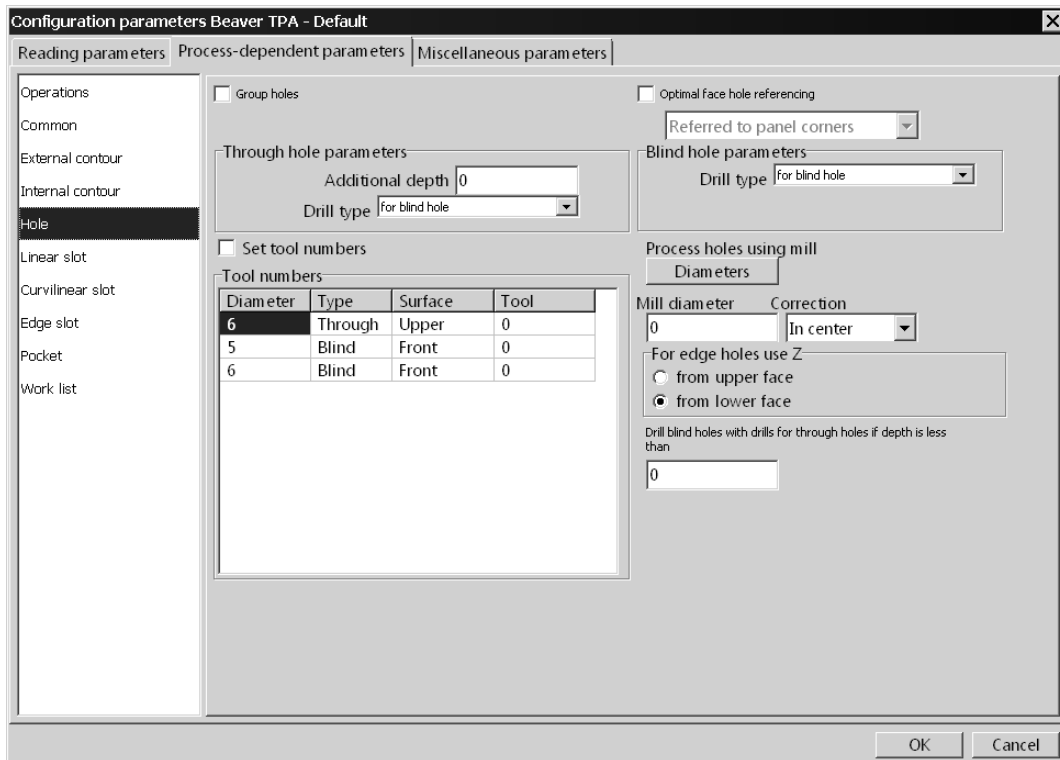


Fig. 10.120.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Through holes can be bored with drills for through and blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **Tool numbers** table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.121).

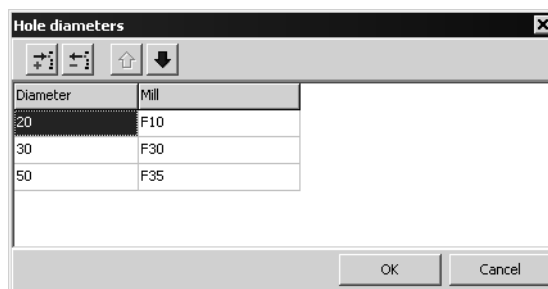


Fig. 10.121.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To finish setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter application.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.122).

Fig. 10.122.

Through slot always cut with saw and **Grooving by saw** options allow to set through slots direction that will be processed with a saw. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.123).

N	Tool name	Correction	Diameter
1	D15	At a center	0

Fig. 10.123.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be se-



lected from **Correction** combo box. **Add saw** button allows to choose a saw for slot processing.

The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.124).

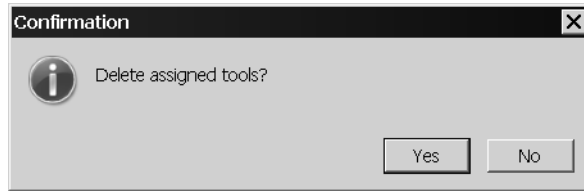


Fig. 10.124.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. You can map slot width and mill parameters that will be automatically selected for its processing. For this purpose push **Mill parameters** button. **Mill parameters** dialogue will appear on the screen (fig. 10.125).

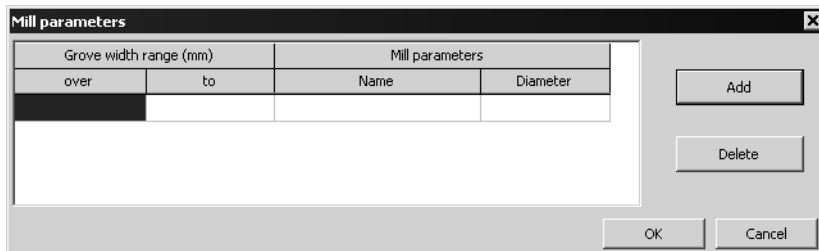


Fig. 10.125.

Dialogue table contains slot width ranges and corresponding mill parameters. The fields should be filled via a keyboard. **Add** button allows to add a blank row to the table. To delete the current row push **Delete** button. To finish parameters setup push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.126).

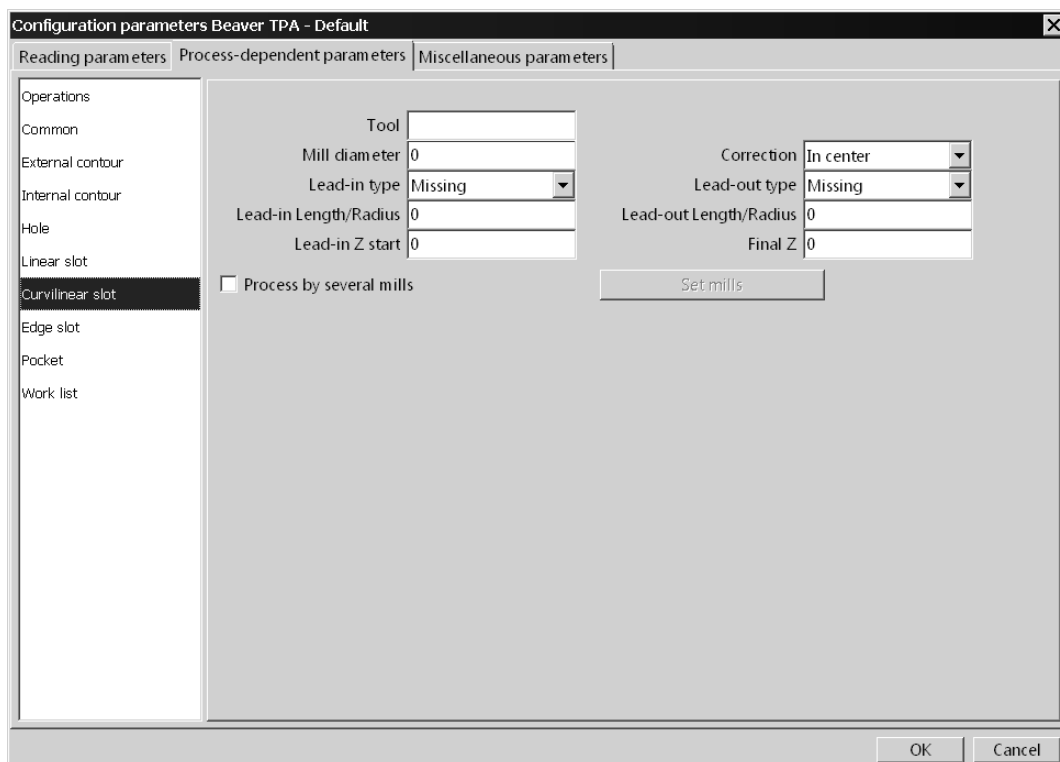


Fig. 10.126.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.127).

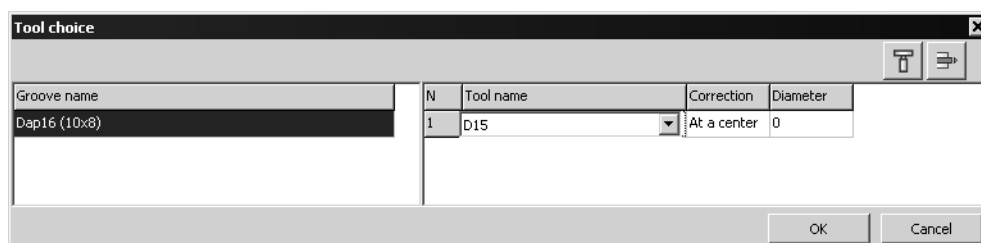


Fig. 10.127.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box. **Delete** button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.128).

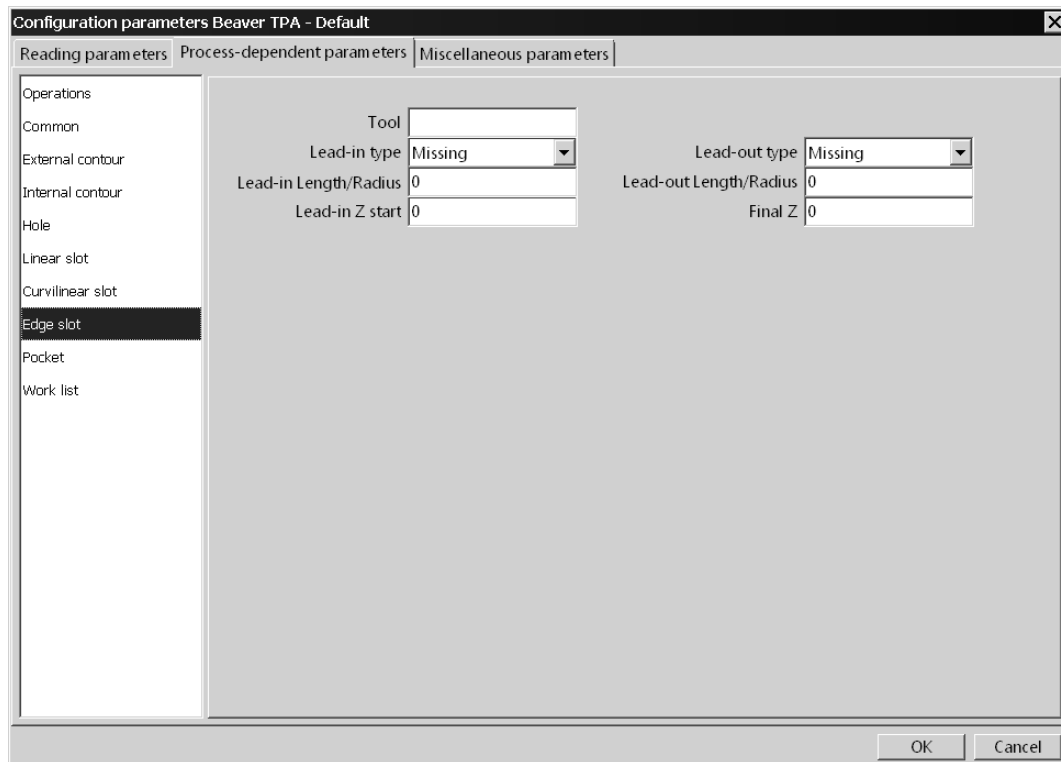


Fig. 10.128.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.129) allow to set pocket processing parameters.

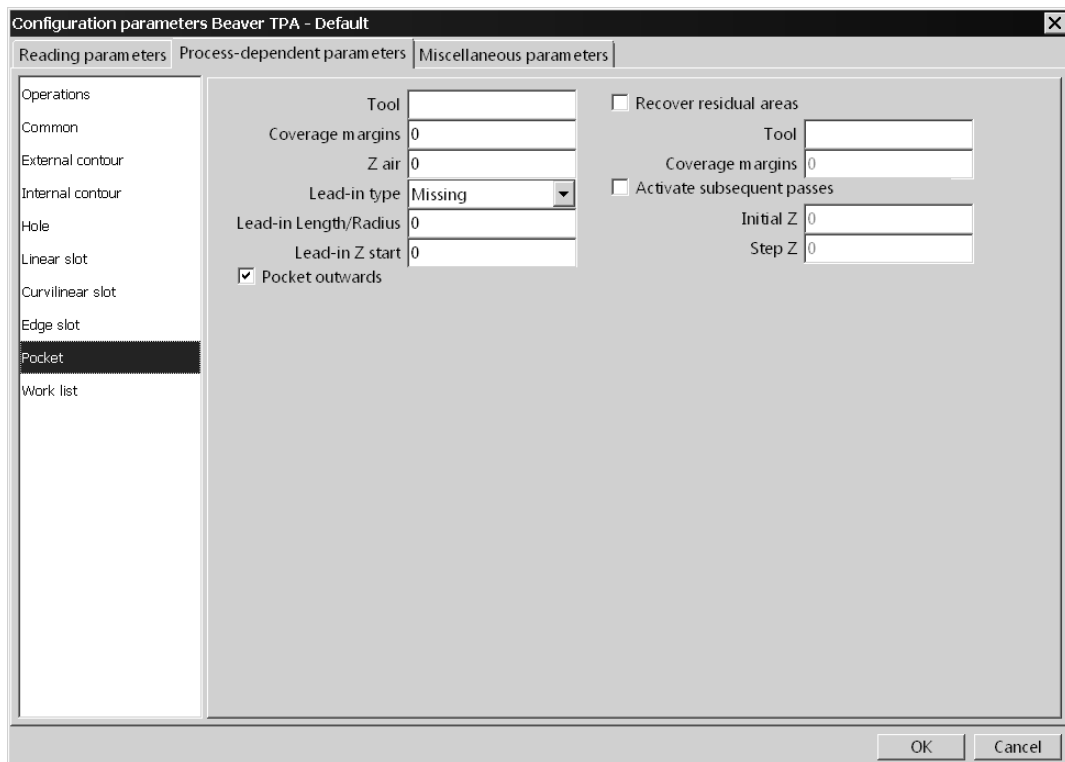


Fig. 10.129.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Work list

Work list can be set in **Work list** section (fig. 10.130).

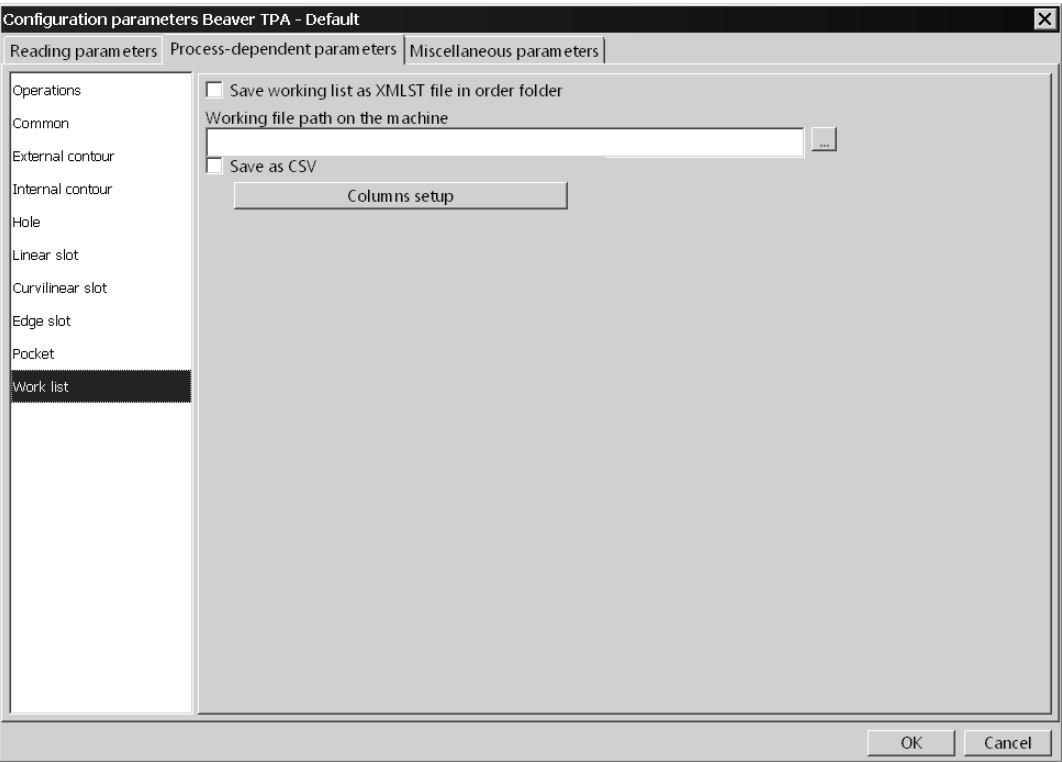


Fig. 10.130.



Save working list as file in order folder option allows to control the file saving. To specify the folder push **Browse** button and choose it in a standard Windows dialogue. **Save as CSV** option allows to control file saving. **Columns setup** button allows to setup file table columns. After pushing the button, **Table columns setup** dialogue (fig. 10.131).

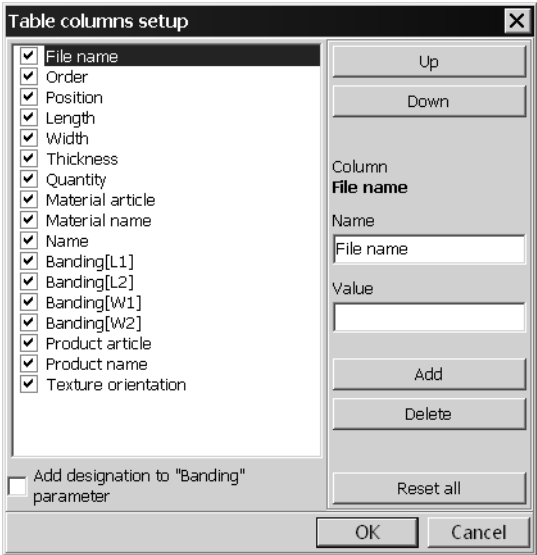


Fig. 10.131.

To add data from a column to a file, enable the option for it. **Up** and **Down** buttons allow to move the selected column through the list. **Add** button allows to add a column to the list. In **Name** field, you can change the name of the current column. For columns added by the user, you can set a value in the appropriate field. **Delete** but-

ton allows to delete a column added by the user. After pushing the button, a warning will appear on the screen (fig. 10.132).

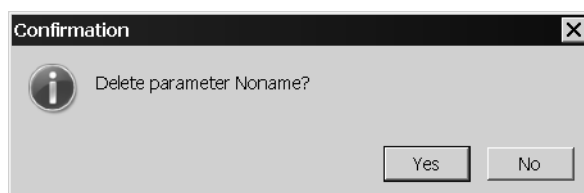


Fig. 10.132.

Message buttons allow to confirm or reject the deletion.

Reset all button allows to return the table settings to their default values. After pushing the button, a warning will appear on the screen (fig. 10.133).

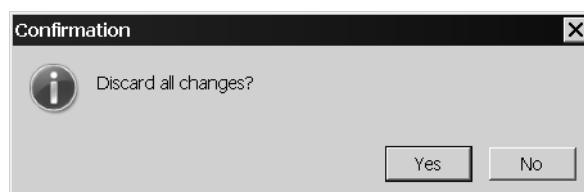


Fig. 10.133.

The message buttons allow to confirm or cancel the reset of parameters. To complete the table setup, push **OK** button.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.11.VIC Control



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.134) allow to set common parameters of control programs creation.

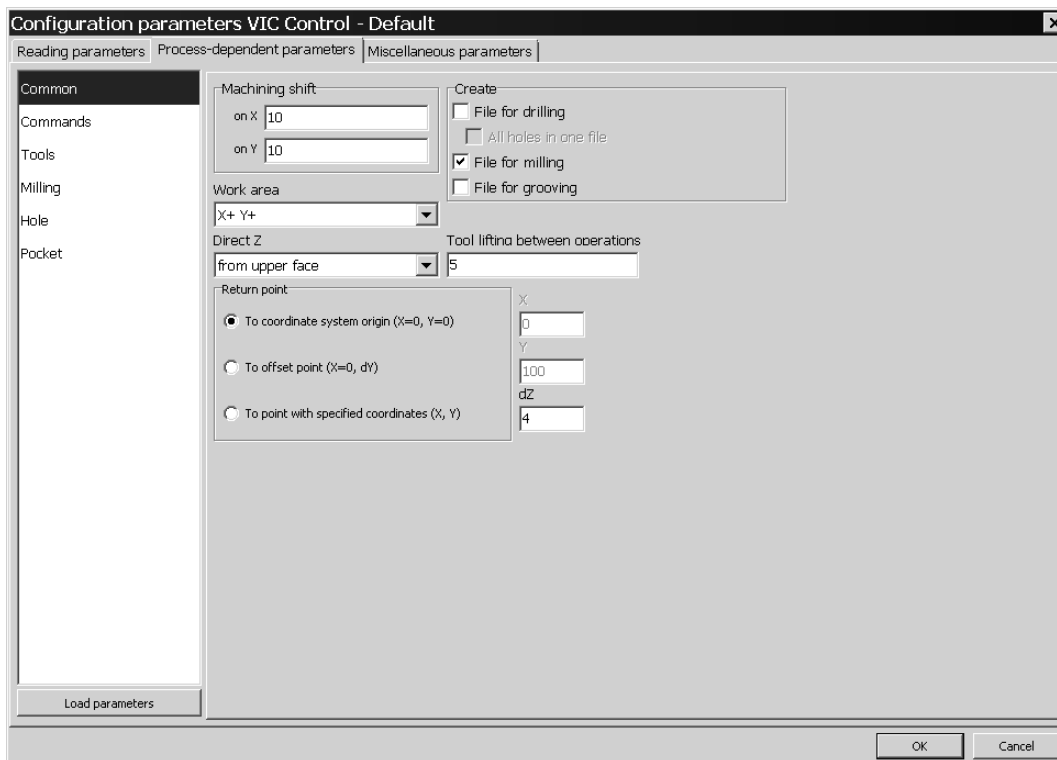


Fig. 10.134.

On X and **on Y** fields of **Machining shift** group allow to set work area initial point moving. **Create** group options allow to control programs creation for separate operations. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Work area** combo box variants allow to set work area coordinate axes direction.

Usable commands

Commands section control elements (fig. 10.135) allow to control separate commands to control program inclusion.

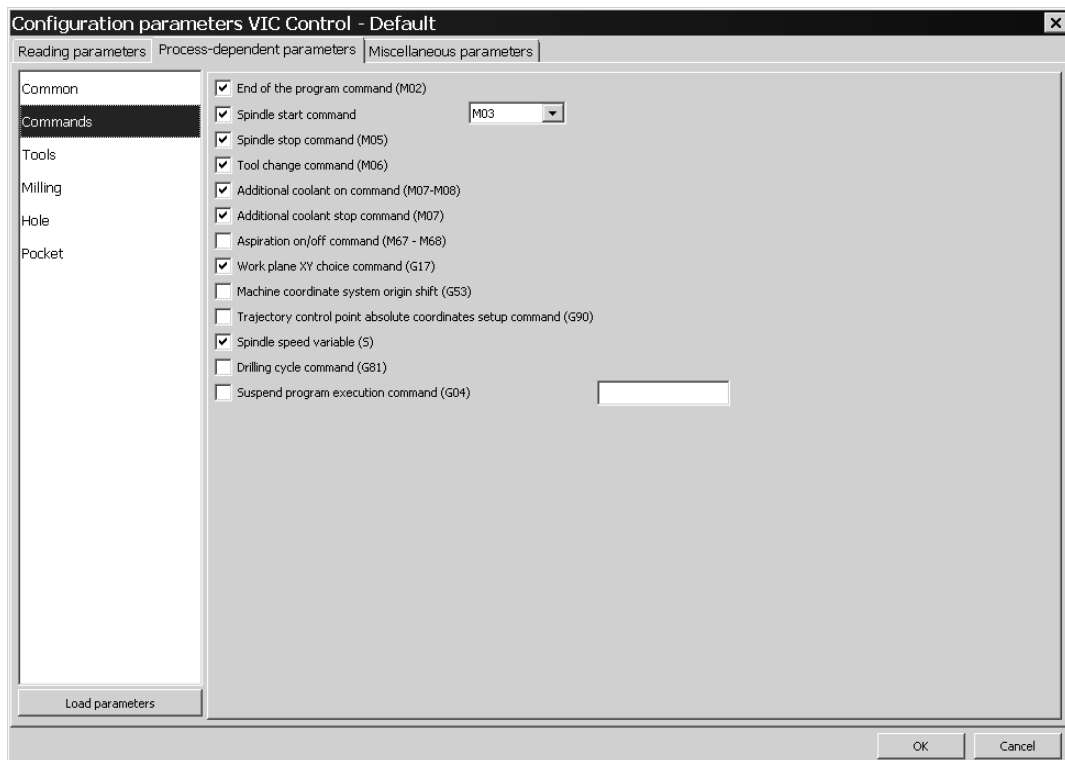


Fig. 10.135.

Spindle start command can be selected from variants in a combo box. If program execution delay command is activated (G04) a delay setup field will become available.

Usable tools

Tools section control elements (fig. 10.136) allow to create a list of tools available for use in control programs.

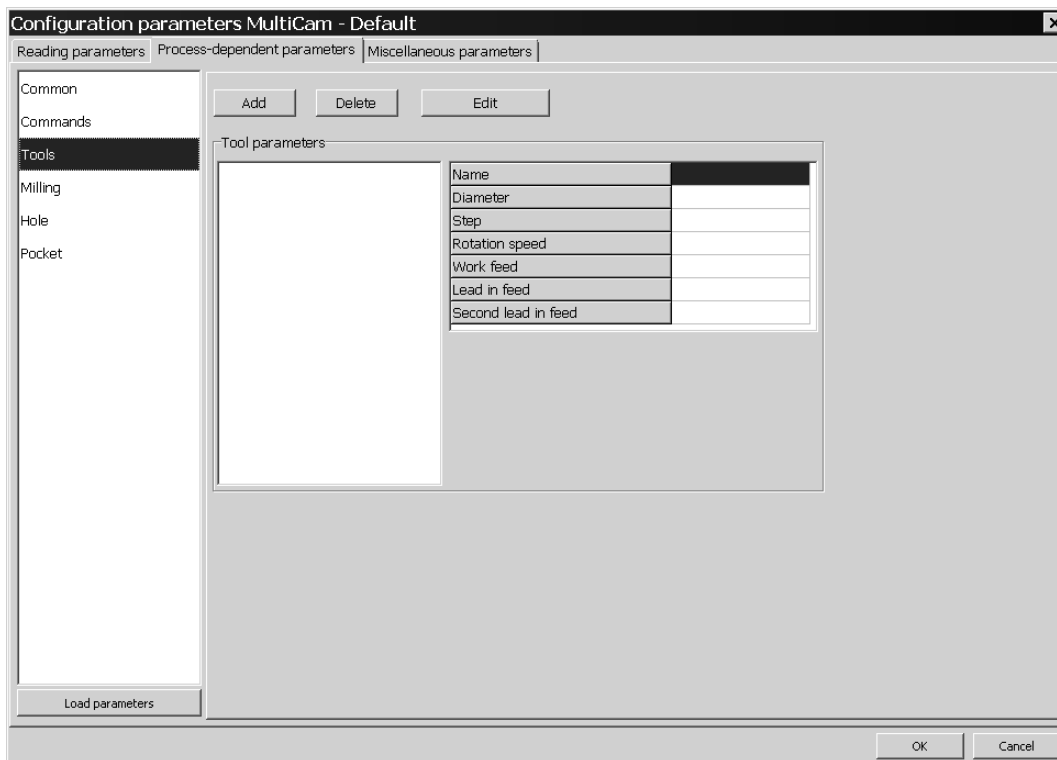


Fig. 10.136.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.137).

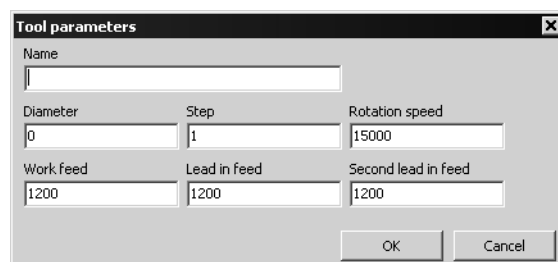


Fig. 10.137.

In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.138) allow to set milling parameters.

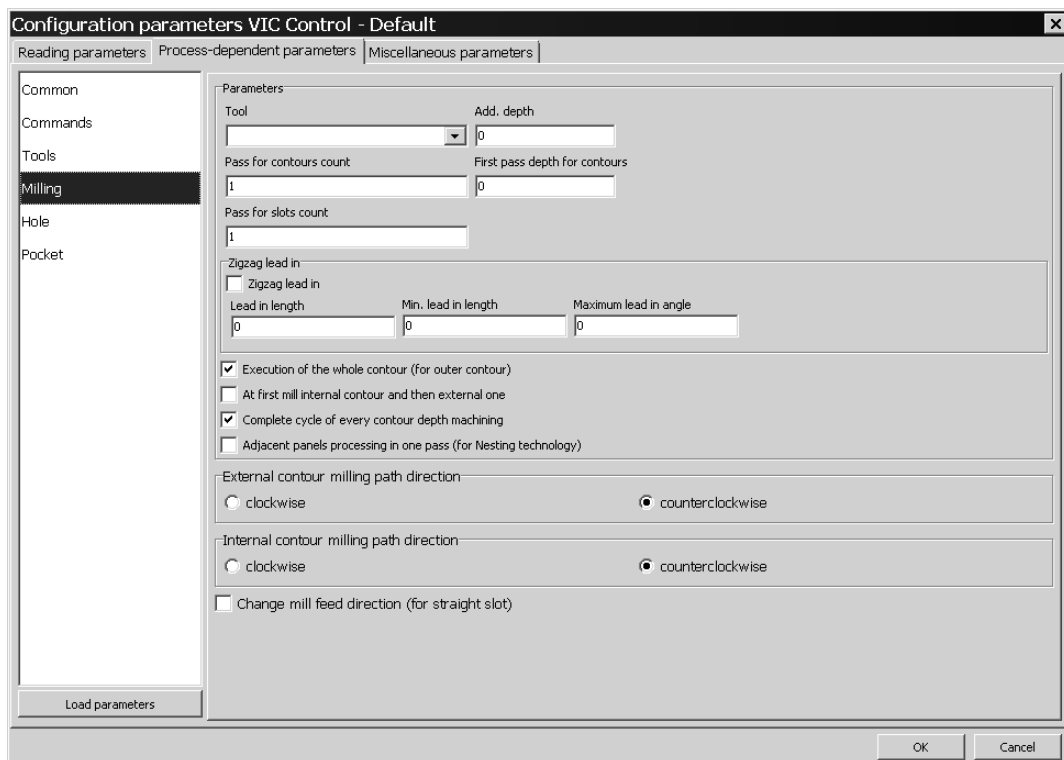


Fig. 10.138.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Zigzag lead in** option allows to activate this operation. Group fields allow to set option parameters. **Milling path direction** group variants allow to select tool moving direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.139).

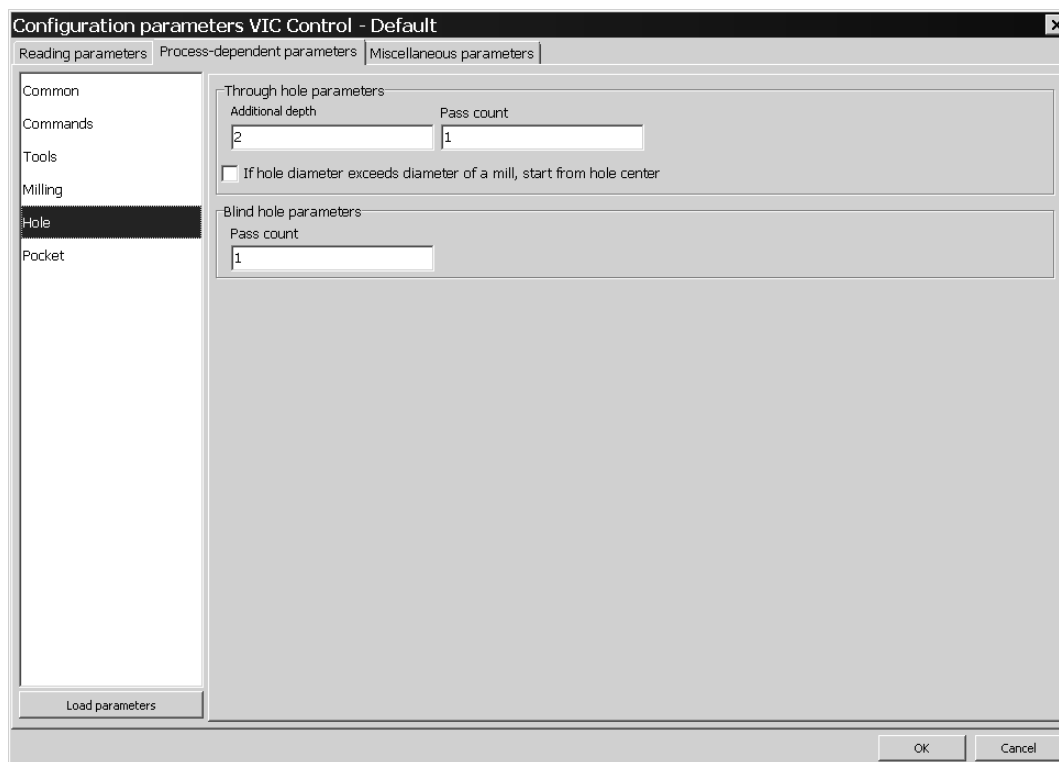


Fig. 10.139.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **If hole diameter exceeds diameter of a mill, start from hole center** option allows to control holes with a mill processing order, from center or from perimeter.

Pocket parameters

Pocket section control elements (fig. 10.140) allow to set pocket processing parameters.

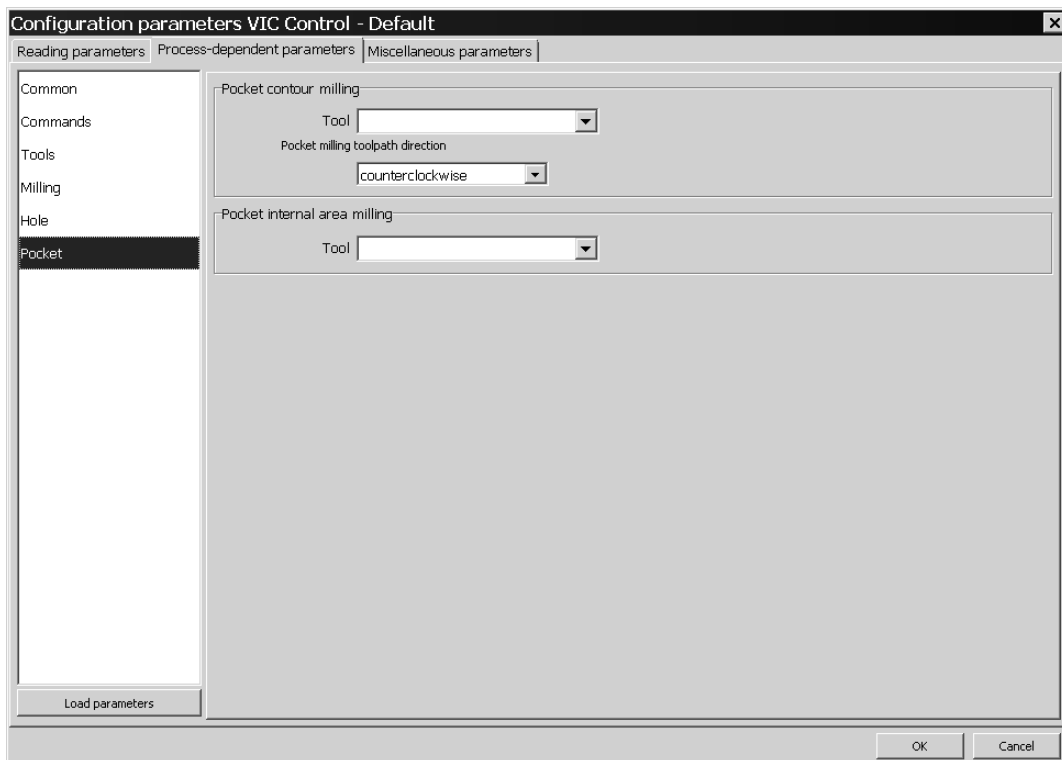


Fig. 10.140.

Tool combo boxes in **Pocket contour milling** and **Pocket internal area milling** groups allow to select tool name for corresponding objects milling. Tools should be created beforehand in **Tools** section. It is necessary to select milling execution direction for notch contour in corresponding combo box.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.5.12.Weihong – i-Nest, NestConcept



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.141) allow to specify operations that can be used in control program.

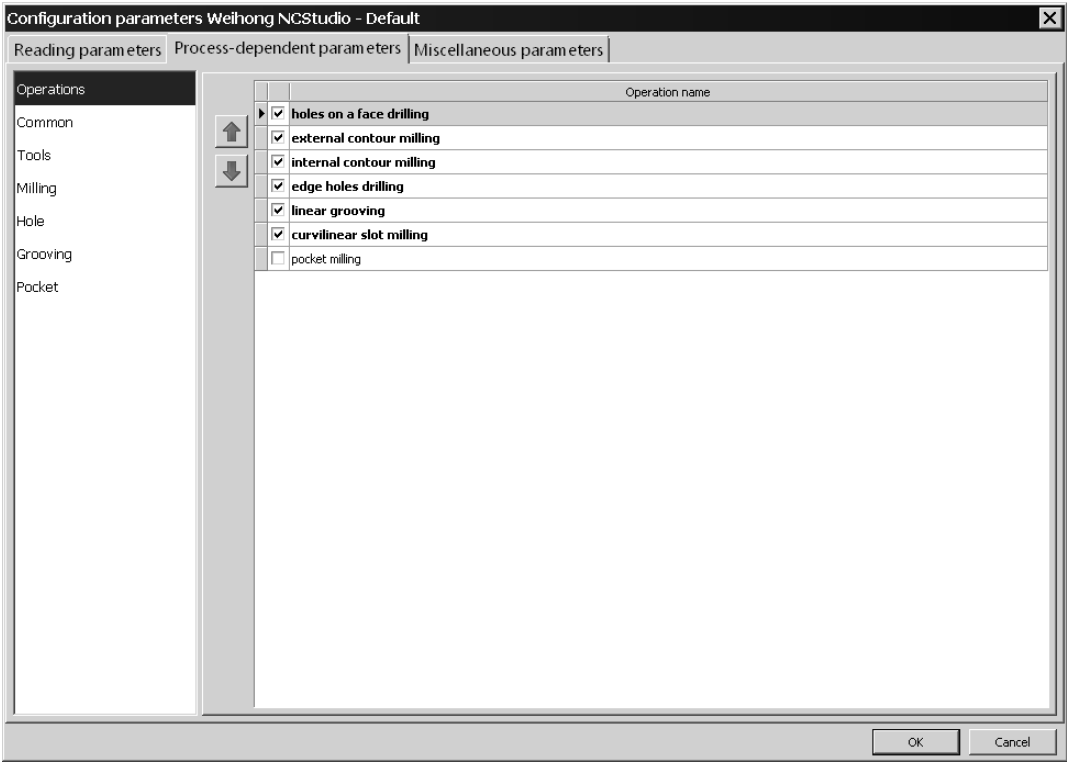


Fig. 10.141.

Common parameters

Common section control elements (fig. 10.142) allow to setup common parameters of control programs creation.

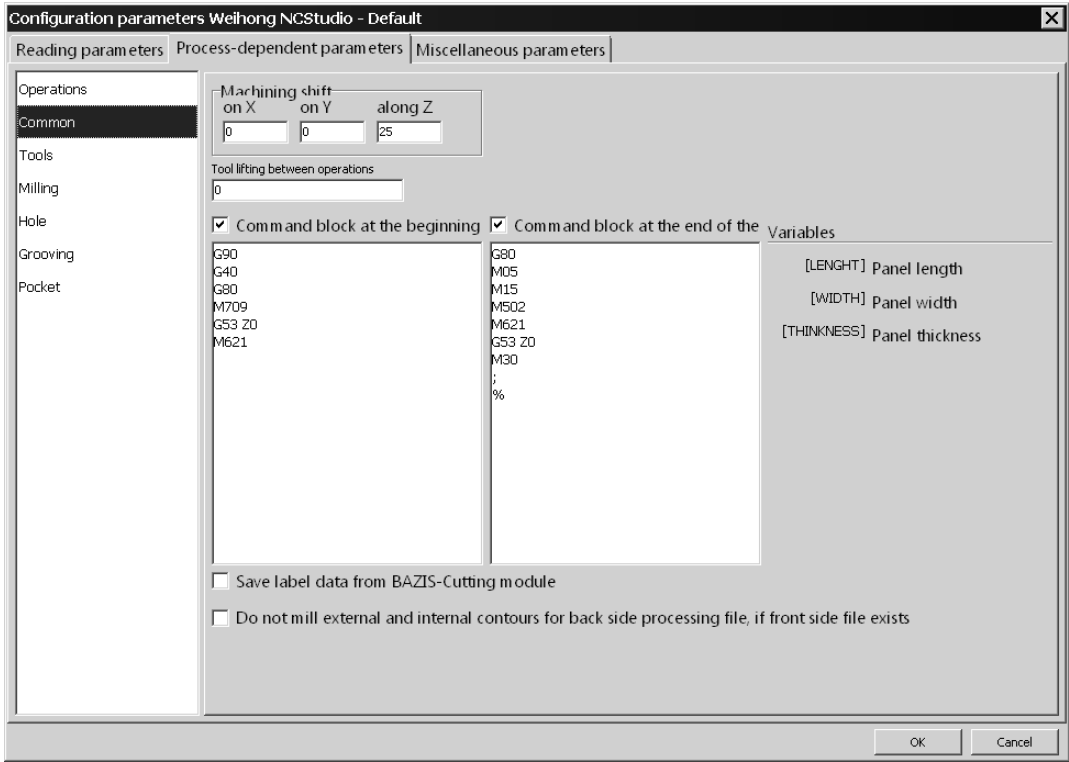


Fig. 10.142.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas. To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable tools

Tools section control elements (fig. 10.143) allow to create a list of tools available for use in control programs.

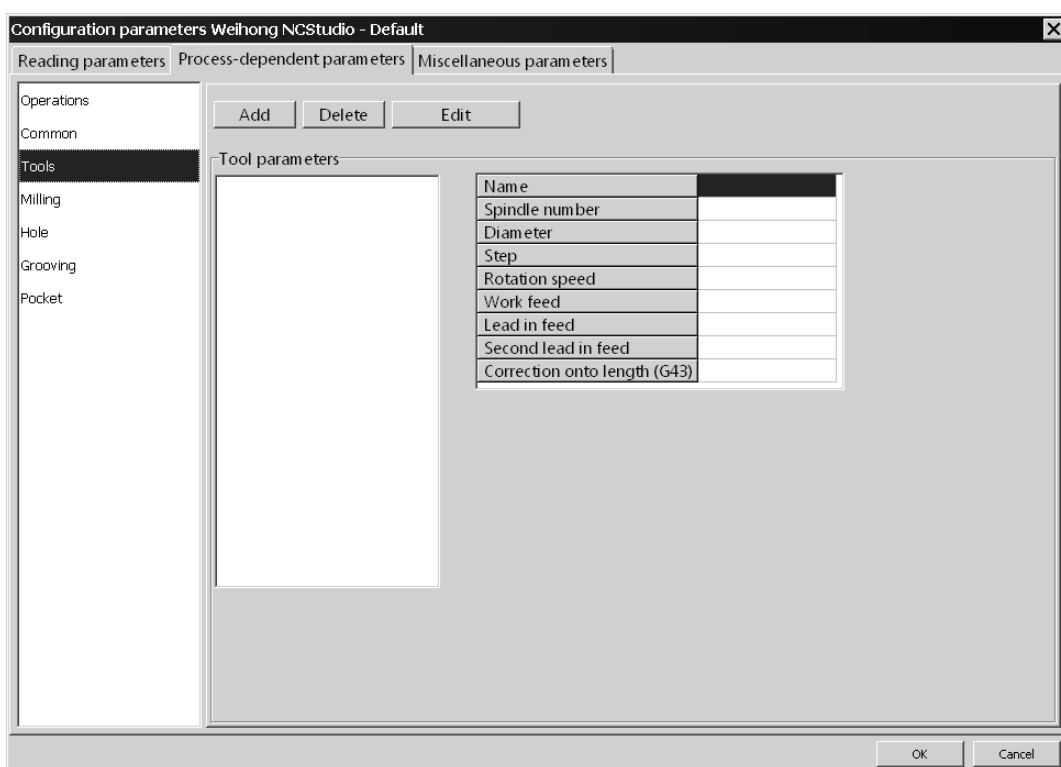
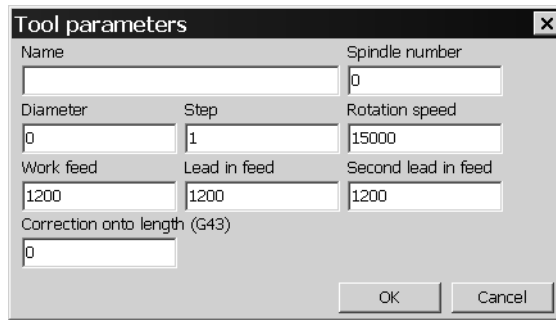


Fig. 10.143.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.144).



The 'Tool parameters' dialog box contains the following fields and controls:

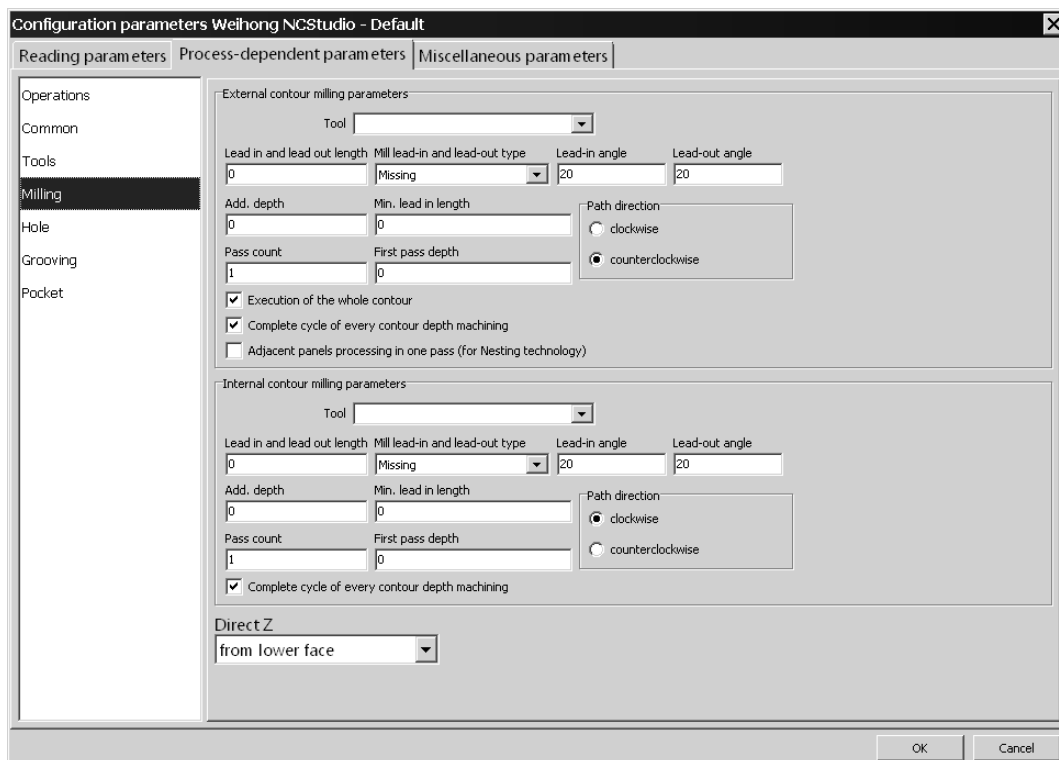
Name		Spindle number
		0
Diameter	Step	Rotation speed
0	1	15000
Work feed	Lead in feed	Second lead in feed
1200	1200	1200
Correction onto length (G43)		
0		
OK		Cancel

Fig. 10.144.

In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.145) allow to set contour milling parameters.



The 'Configuration parameters WeiHong NCStudio - Default' dialog box has three tabs: 'Reading parameters', 'Process-dependent parameters', and 'Miscellaneous parameters'. The 'Process-dependent parameters' tab is active.

Operations (left sidebar):

- Common
- Tools
- Milling**
- Hole
- Grooving
- Pocket

External contour milling parameters:

- Tool:
- Lead in and lead out length: Mill lead-in and lead-out type: Lead-in angle: Lead-out angle:
- Add. depth: Min. lead in length:
- Pass count: First pass depth:
- Path direction: ☒ clockwise ☐ counterclockwise
- ☒ Execution of the whole contour
- ☒ Complete cycle of every contour depth machining
- ☐ Adjacent panels processing in one pass (for Nesting technology)

Internal contour milling parameters:

- Tool:
- Lead in and lead out length: Mill lead-in and lead-out type: Lead-in angle: Lead-out angle:
- Add. depth: Min. lead in length:
- Pass count: First pass depth:
- Path direction: ☒ clockwise ☐ counterclockwise
- ☒ Complete cycle of every contour depth machining

Direct Z:

OK Cancel

Fig. 10.145.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rect-

angular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.146).

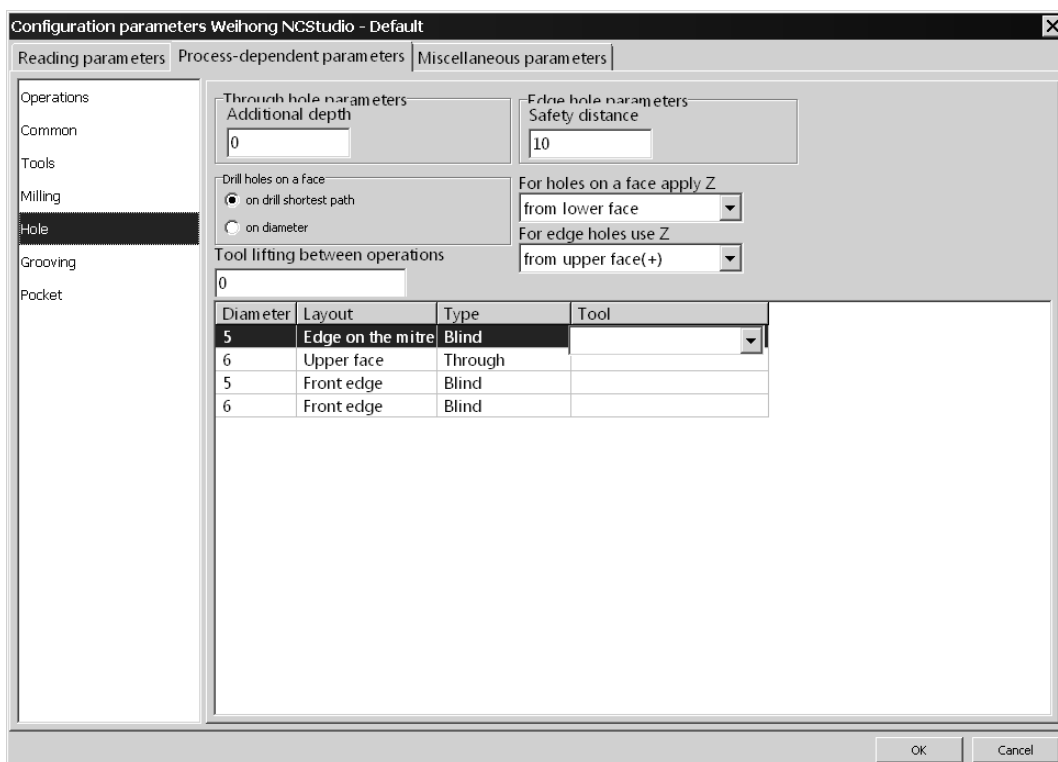


Fig. 10.146.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **For edge holes use Z** and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height.

Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

Slot parameters

Grooving section control elements (fig. 10.147) allow to set slots processing parameters.

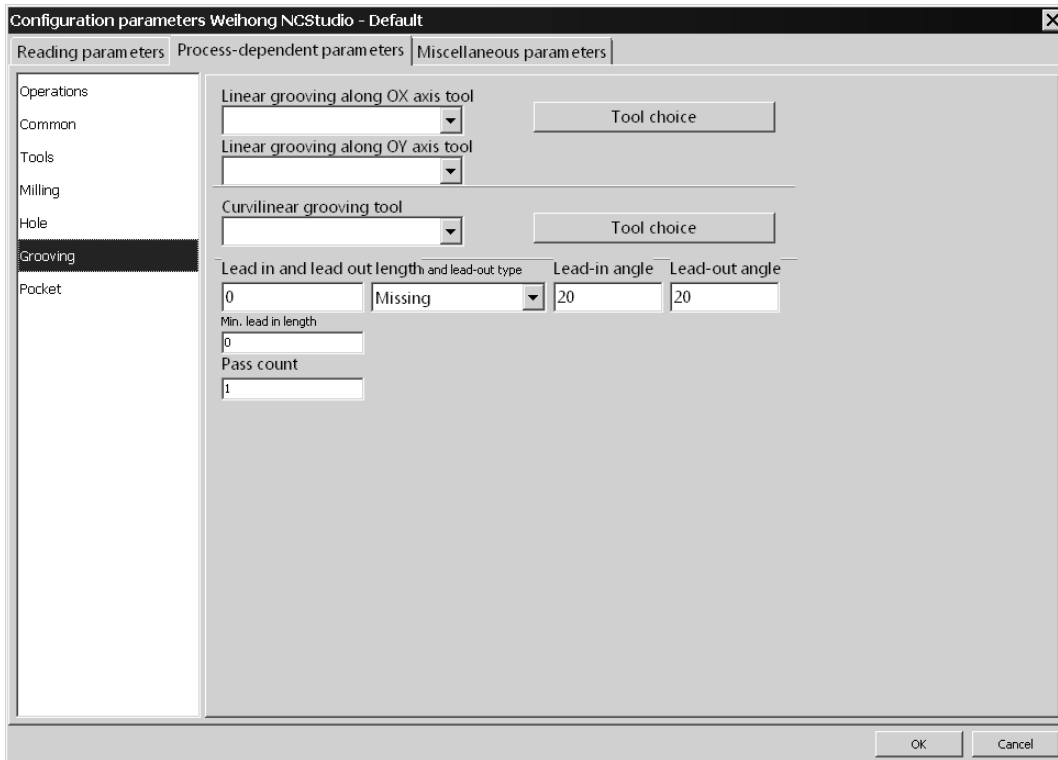


Fig. 10.147.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set a tool for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.148).

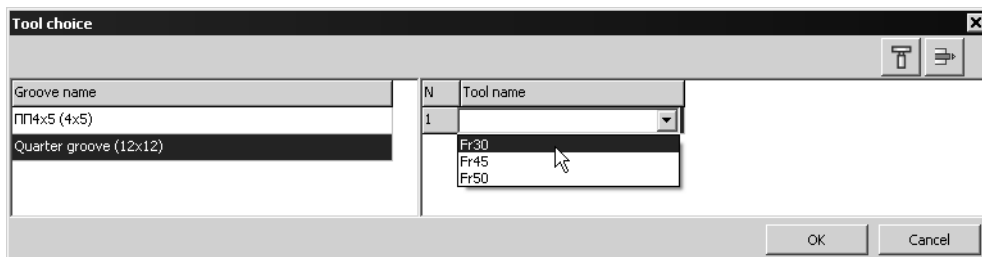


Fig. 10.148.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.149) allow to set pocket processing parameters.

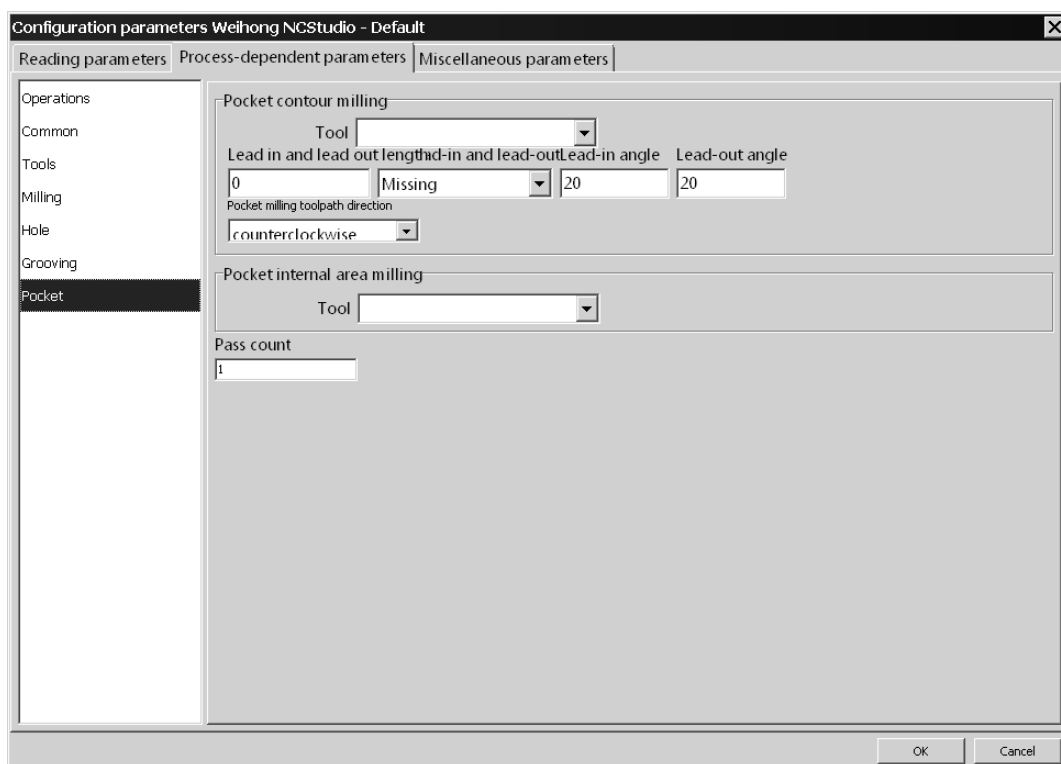


Fig. 10.149.

Section control elements allow to set control programs creation parameters for notch contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.6. Biesse Group

10.6.1. BiesseWorks (*.bpp)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.150) allow to specify operations that can be used in control program.

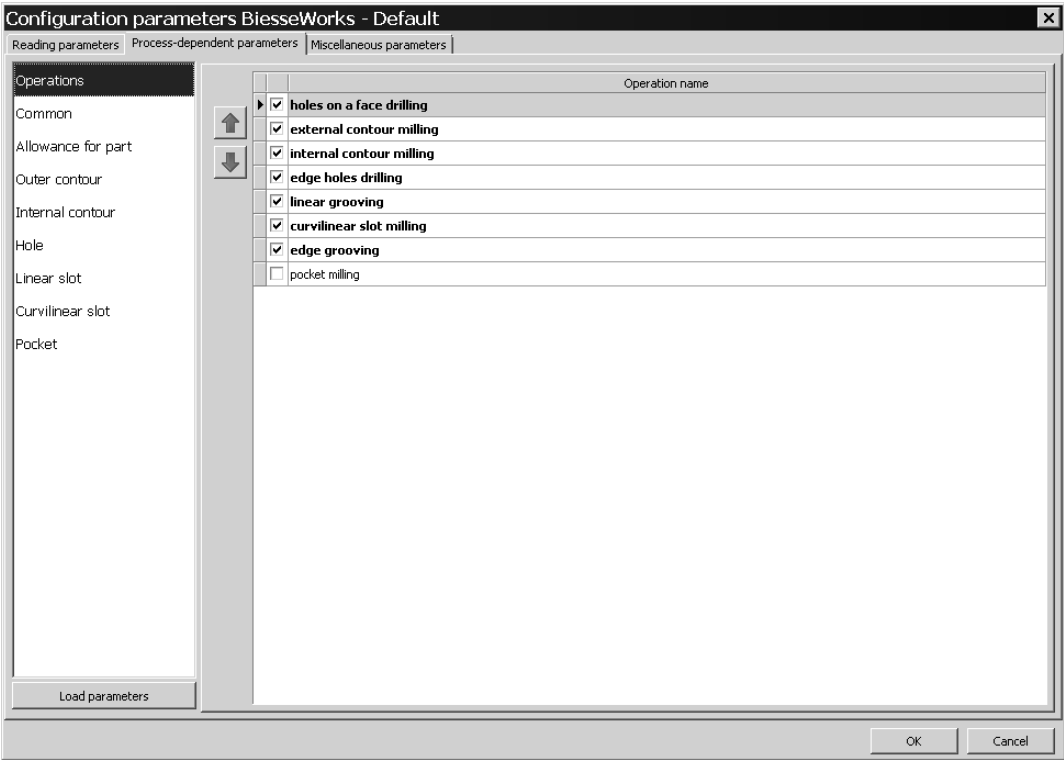


Fig. 10.150.

Common parameters

Common parameters are set in **Common** section of **Process-dependent parameters** tab (fig. 10.151).

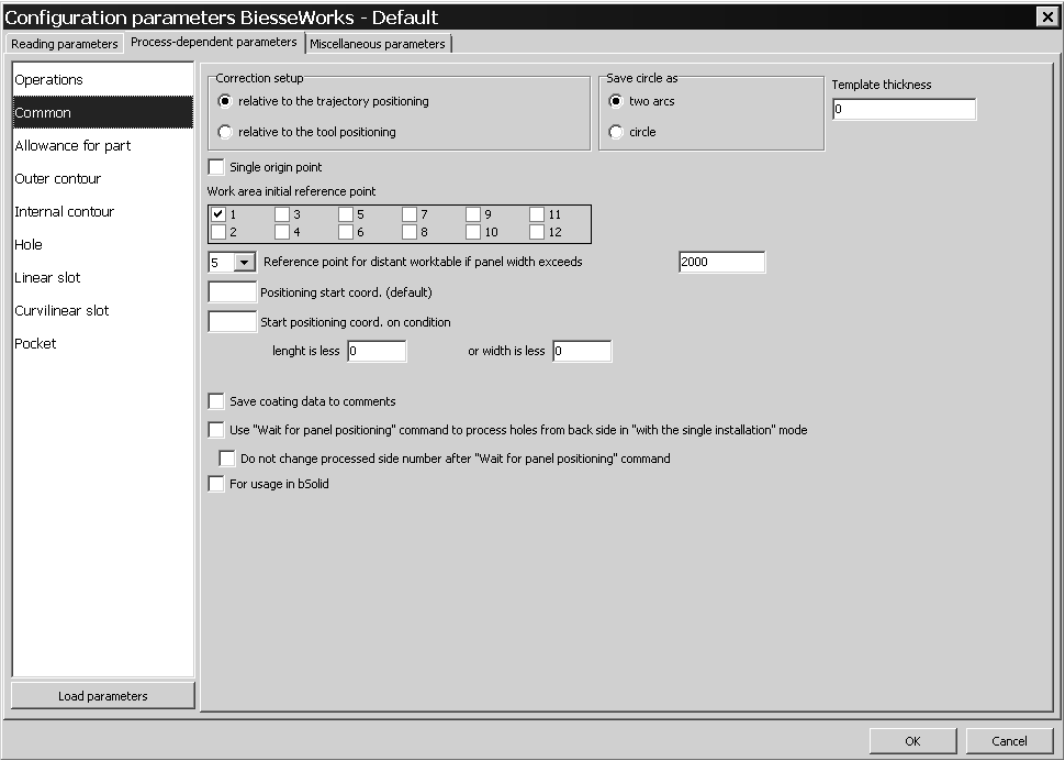


Fig. 10.151.

Tab control elements allow to set tool correction, circles saving mode, work area initial reference point, etc.

Allowance for part

Allowance for part section control elements (fig. 10.152) allow to set allowances values.

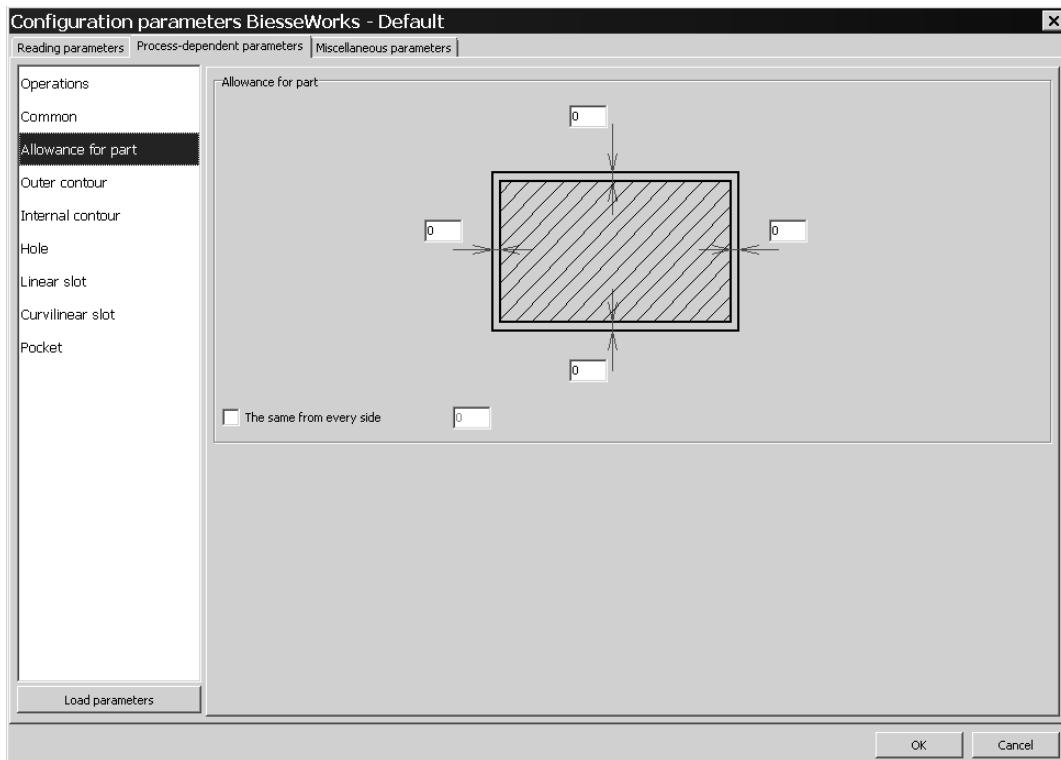


Fig. 10.152.

Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, field of allowance for all panel edges will become available.

Contour parameters

Contours milling parameters are set in **Outer contour** (fig. 10.153) and **Internal contour** sections.

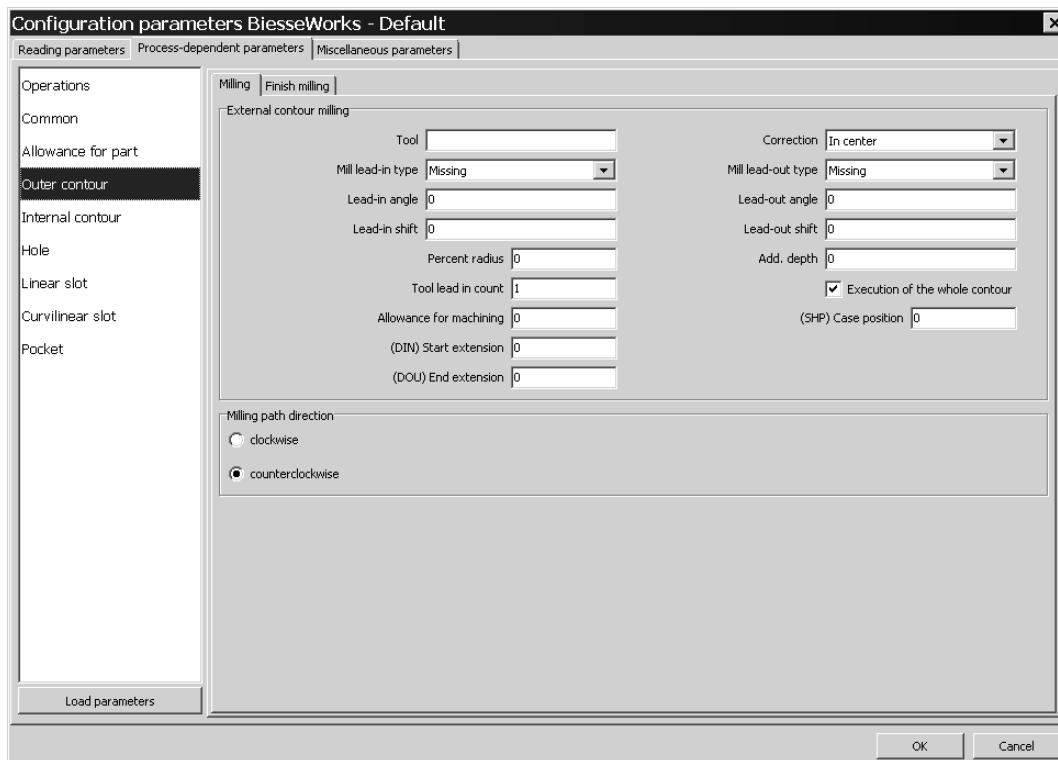


Fig. 10.153.

Milling and **Finish milling** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Finish milling** option allows to control this operation.

These tabs control elements are the same. Additional control elements are on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Holes section control elements (fig. 10.154) allow to set drilling parameters.

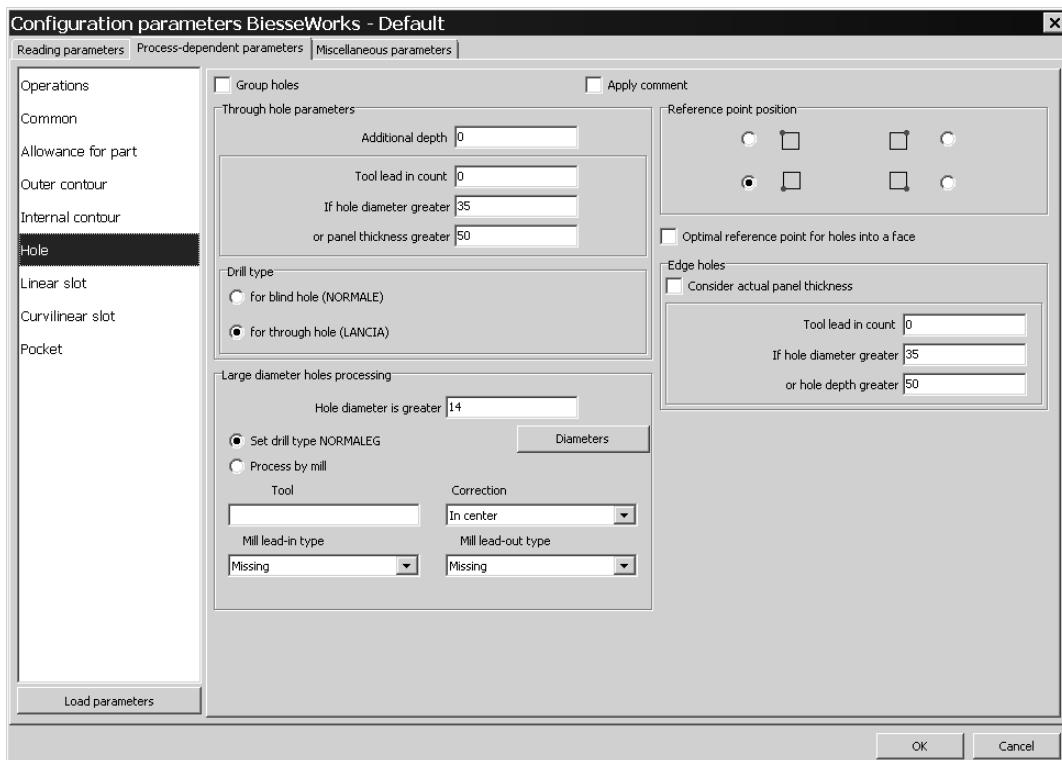


Fig. 10.154.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Through holes can be bored with drills for through and blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Holes can be bored with a drill as well as with a mill. Hole diameter usually specifies tool selection. **Hole diameter is greater** field allows to set hole diameter that specifies automatic tool selection in program. **Set drill type NORMALEG** and **Process by mill** variants allow to select a tool type when set diameter value is exceeded. If **Process by mill** variant is selected, input a tool name and correction types, mill lead-in and lead-out in corresponding fields. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameter** button. **Hole diameters** dialogue will appear on the screen (fig. 10.155).

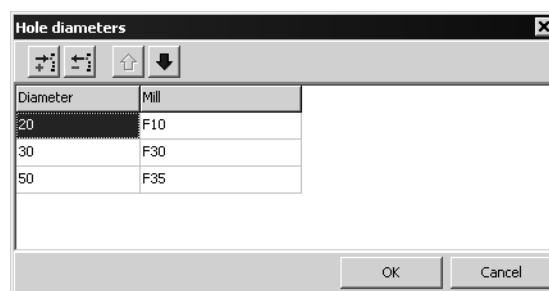


Fig. 10.155.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To complete setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Reference point position** group variants allow to select this point position for holes referencing. If **Optimal reference point for holes into a face** option is enabled, referencing will be set for each hole individually. **Edge holes** group control elements allow to control panel actual thickness accounting. It is possible to set tool lead-in count depending on holes diameter or depth.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.156).

Fig. 10.156.

Create slot group variants allow to select a tool type for slot processing. If **Through slot - by saw, else - by mill** variant is selected, pass-through slots will be processed with a saw, other slots - with a mill. **Through slot along Y axis - by mill** option allows to set pass-through grooves direction for saw processing. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.157).

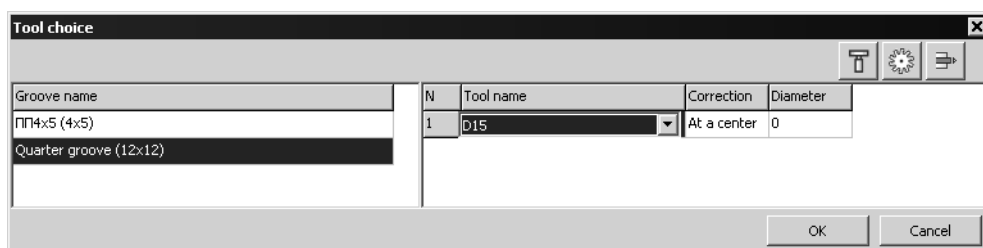


Fig. 10.157.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to select a saw for slot processing.

The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.158).

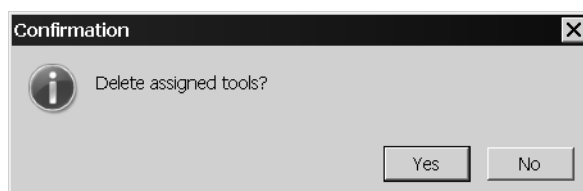


Fig. 10.158.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot - by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot - by mill group control elements allow to set control programs creation parameters for grooving by mill – a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. You can map slot width and mill parameters that will be automatically selected for its processing. For this purpose push **Mill parameters** button. **Mill parameters** dialogue will appear on the screen (fig. 10.159).

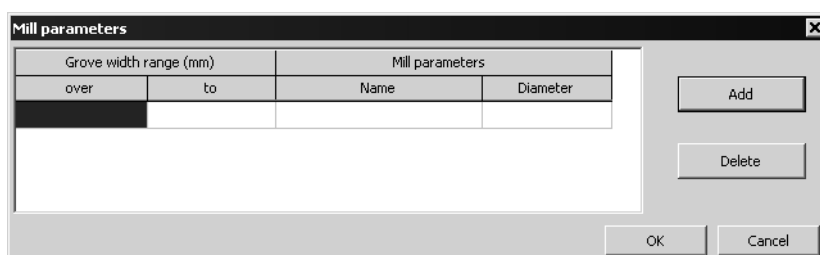


Fig. 10.159.

Dialogue table contains slot width ranges and corresponding mill parameters. The fields should be filled via a keyboard. **Add** button allows to add a blank row to the table. To delete the current row push **Delete** button. To complete parameters setup push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.160).

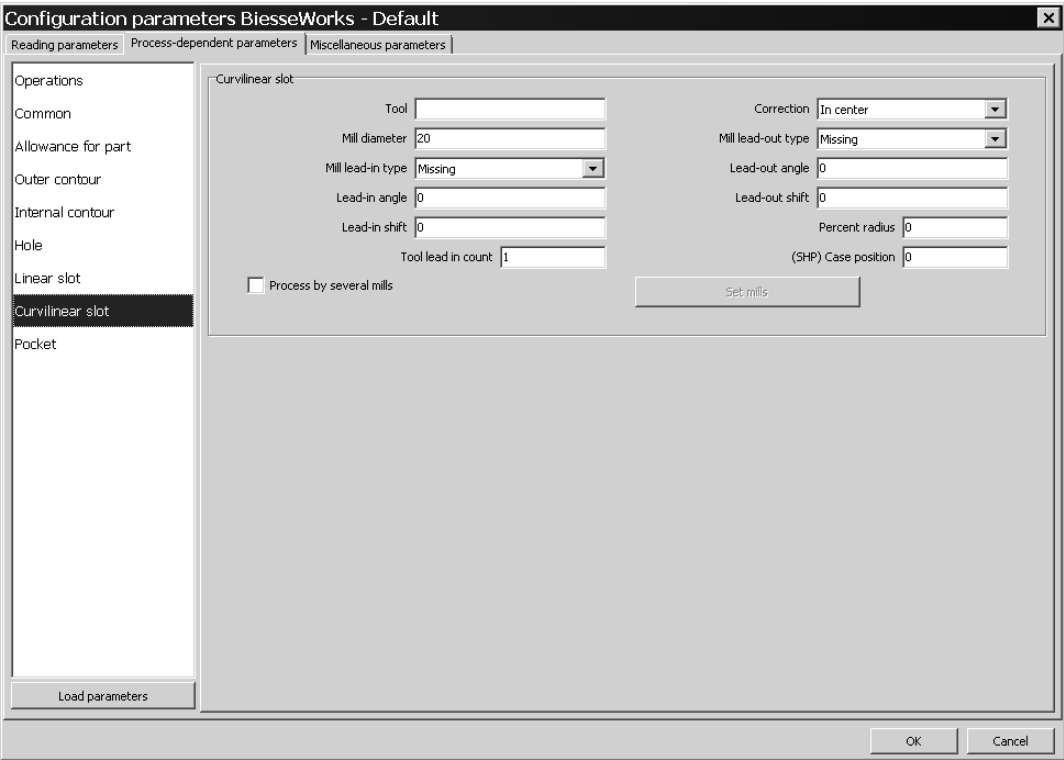


Fig. 10.160.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.161).

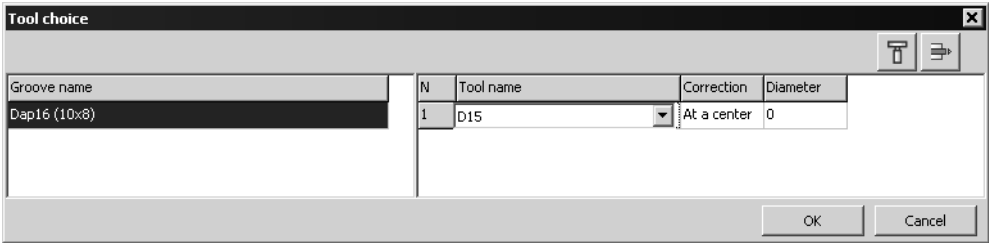


Fig. 10.161.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Delete button allows to delete selected tool from a list. To complete tools assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.162) allow to set pocket processing parameters.

Fig. 10.162.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs for Biesse Works setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.6.2. NC500



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.163) allow to set control programs common parameters.

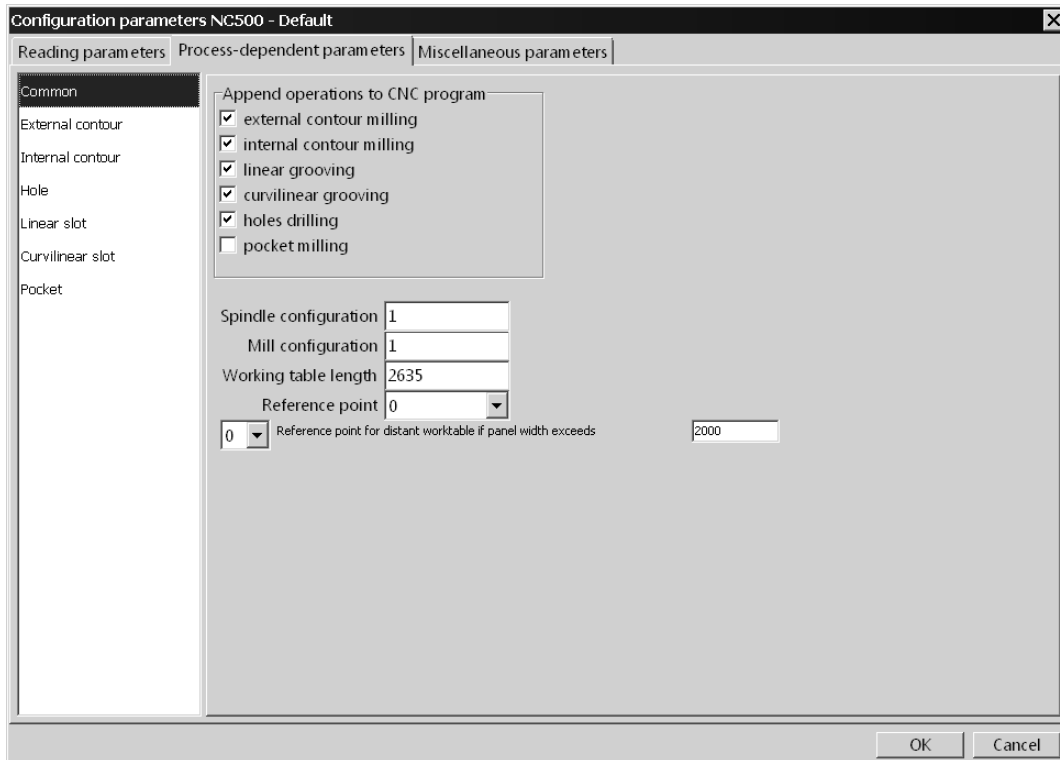


Fig. 10.163.

Spindle configuration, **Mill configuration**, **Working table length** fields allow to set corresponding parameters values for control program. **Append operations to CNC program** group options allow to control the inclusion in control program of operations corresponding to the options names.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** and **Internal contour** sections (fig. 10.164). Rough and finish milling setup is executed separately on corresponding tabs. Tabs control elements content is the same and described for finish milling.

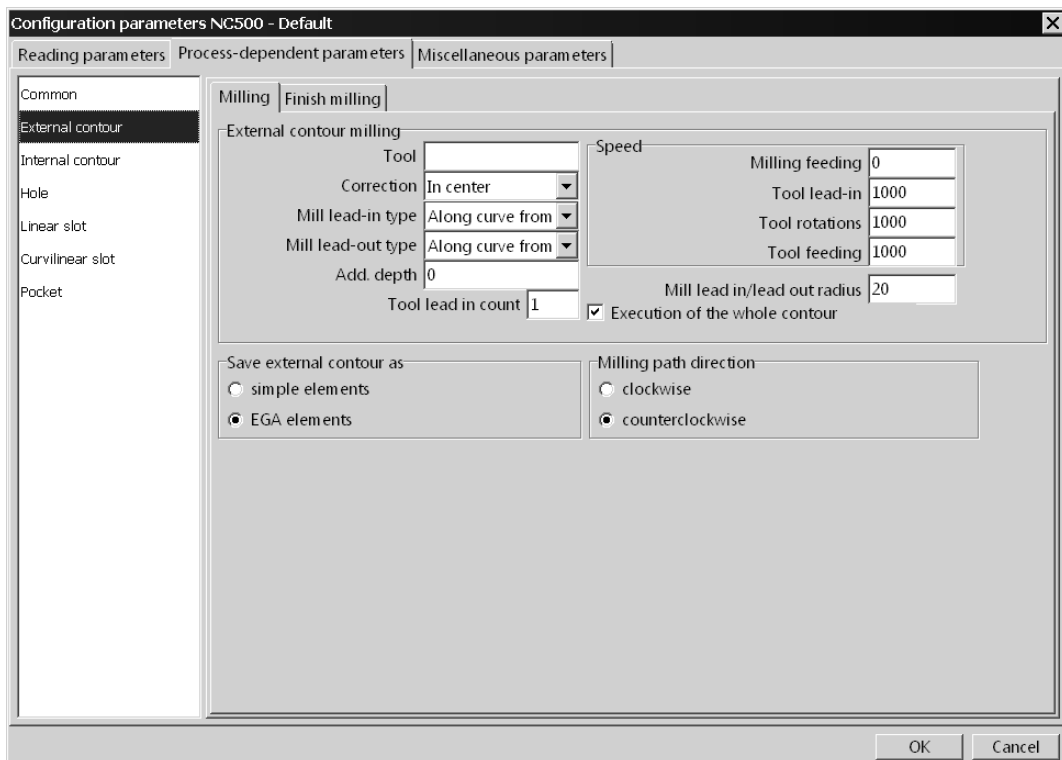


Fig. 10.164.

Finish milling option allows to control this operation. Tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction. **Save contour as** group variants allow to select milling contour referencing mode: as arcs and line segments (**simple elements**) or in internal format of machine program system (**EGA elements**).

Hole parameters

Holes processing is set in **Hole** section (fig. 10.165).

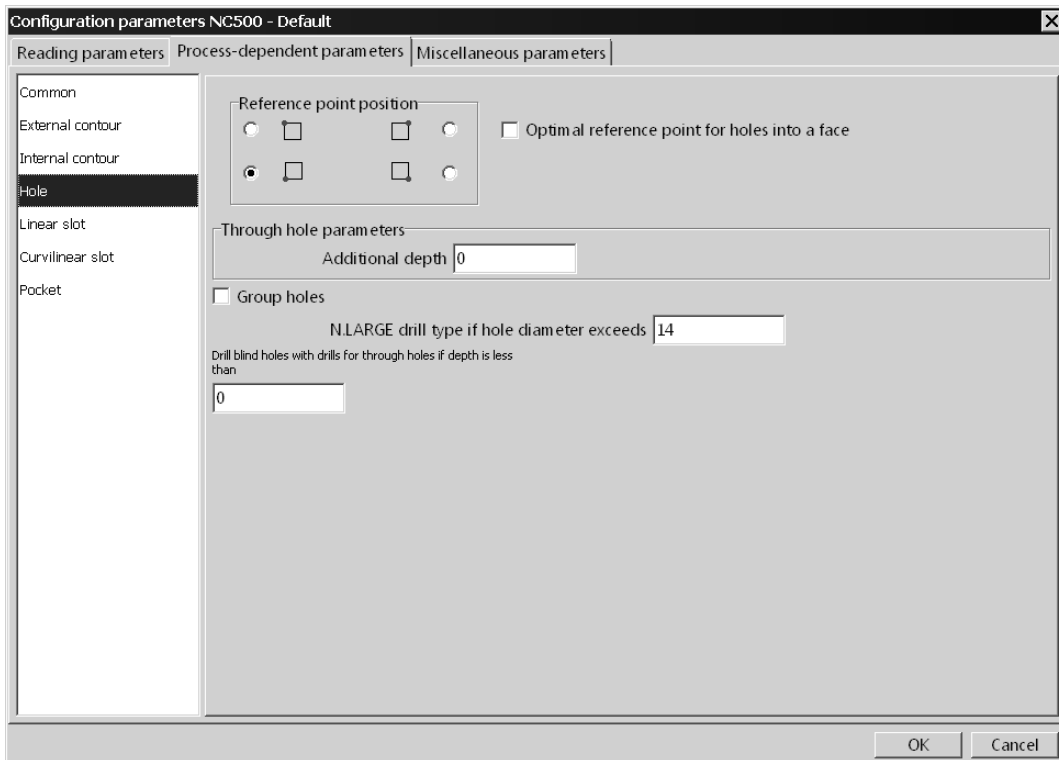


Fig. 10.165.

Reference point position group variants allow to select panel angle as origin of coordinates. If **Optimal reference point for holes into a face** option is enabled, referencing will be set for each hole individually. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Holes can be bored with a common drill as well as with N.LARGE type drill. Hole diameter specifies tool selection. **N.LARGE drill type if hole diameter exceeds** field allows to set hole diameter that will specify automatic tool selection in the program.

Linear slot parameters

Linear slots parameters are set on **Linear slot** tab (fig. 10.166).

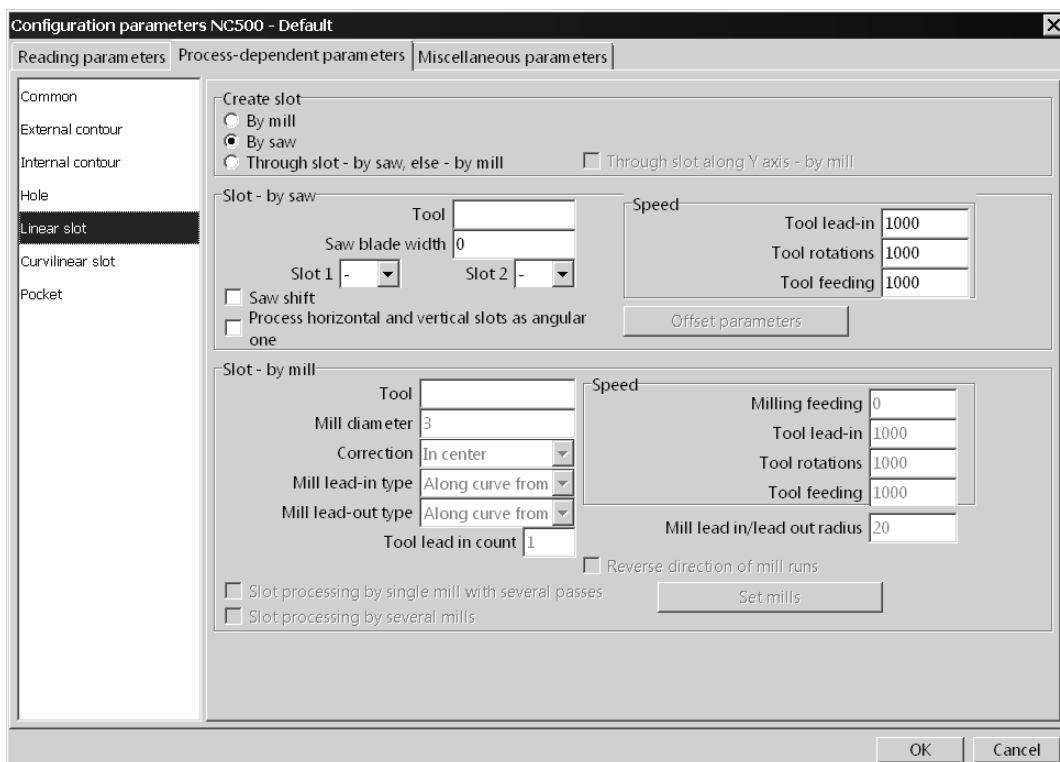


Fig. 10.166.

Create slot group control elements allow to set tool type for slot processing.

Grooving by saw parameters

Slot - by saw group control elements allow to set control programs creation parameters for grooving by saw. If **Saw shift** option is enabled, **Offset parameters** button will become available. After pushing the button **Saw offset parameters** dialogue will appear on the screen (fig. 10.167).

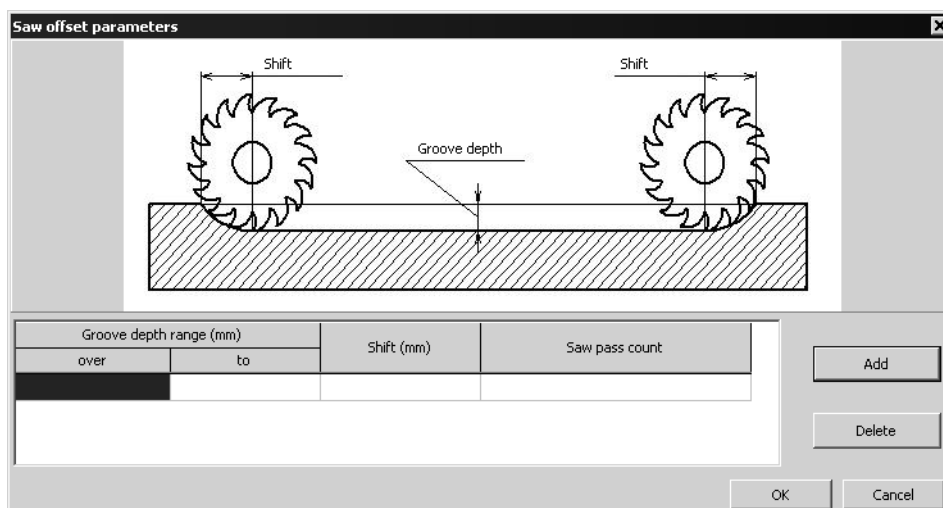


Fig. 10.167.

Table each row allows to set offset value and saw pass count for set slot depth range. To set this parameter click with left mouse button in table field and then enter the value. **Add** button allows to add offset parameters new row to the table for depth

interval. To delete the current row push **Delete** button. To complete offset setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill. If **Slot processing by several mills** or **Slot processing by single mill with several passes** option is enabled, **Set mills** button will become available. If **Process by several mills** option is enabled, after pushing the button, **Tool choice** dialogue will appear on the screen (fig. 10.168).

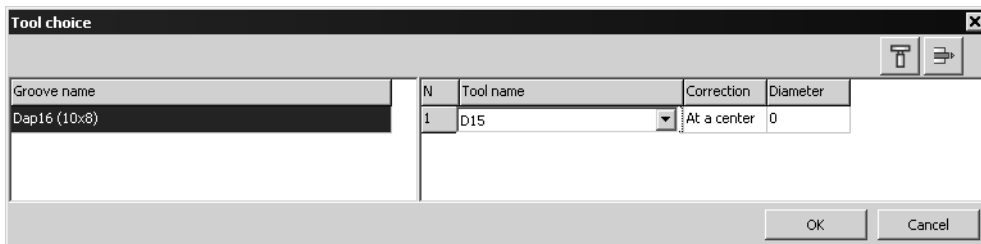


Fig. 10.168.



Slot name table contains all slots names in the model. You can assign particular mills for each slot processing. **Add mill** button allows to add a tool to the list. Correction type should be selected from combo box. Mill name and diameter can be input in a list field.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. If **Slot processing by single mill with several passes** option is enabled, after pushing **Set mills** button, **Mill parameters** dialogue will appear on the screen (fig. 10.169).

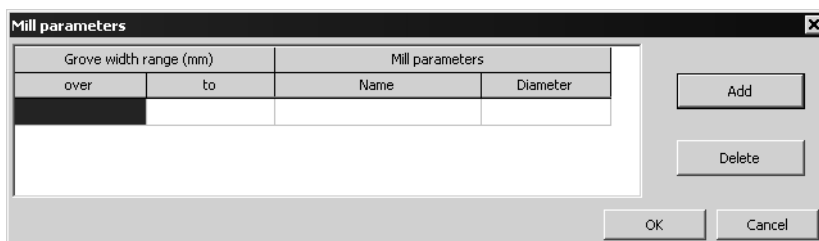


Fig. 10.169.

Each table row allows to set mill diameter and name for set slot width range. To set parameter value, click with mouse left button in a table field and then input the value. **Add** button allows to add a new row of slot parameters to the table. To delete the current row push **Delete** button. To complete tools setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear slots parameters are set in **Curvilinear slot** section (fig. 10.170).

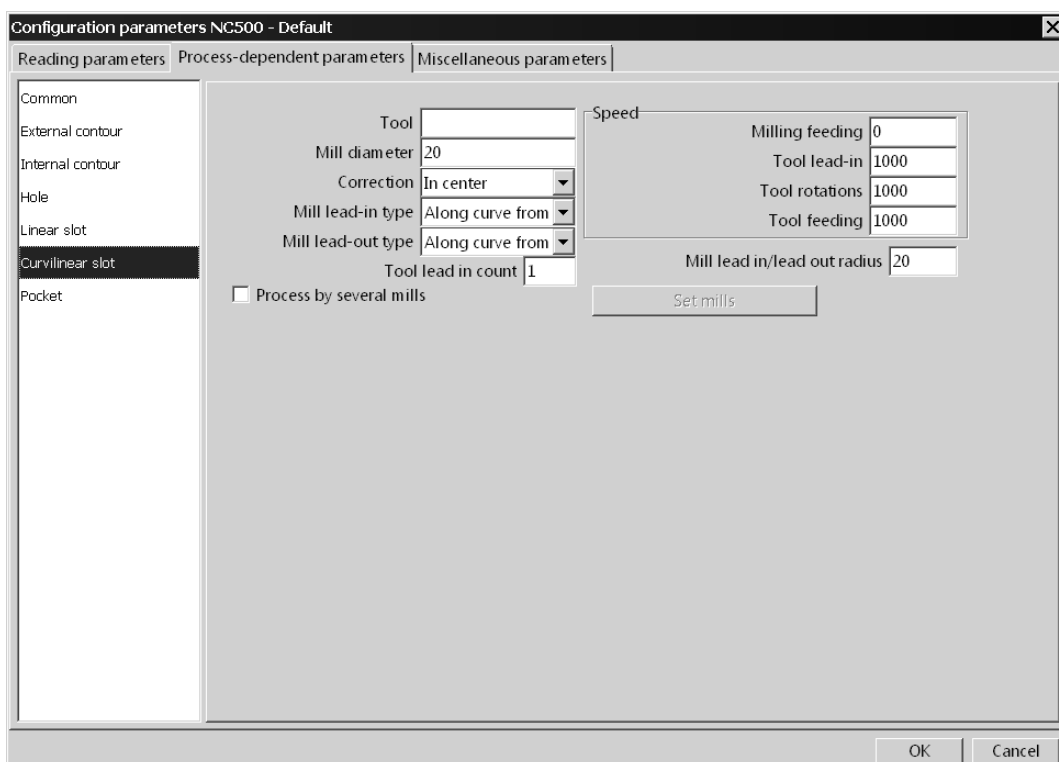


Fig. 10.170.

Control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.171).

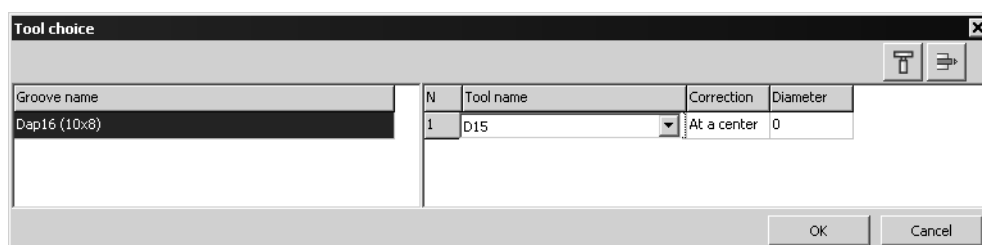


Fig. 10.171.



Slot name table contains all slots names in the model. You can assign particular mills for each slot processing. **Add mill** button allows to add a tool to the list. Correction type should be selected from combo box. Mill name and its diameter can be changed in list fields.



Delete button allows to delete a tool from the list.

To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.172) allow to set pocket processing parameters.

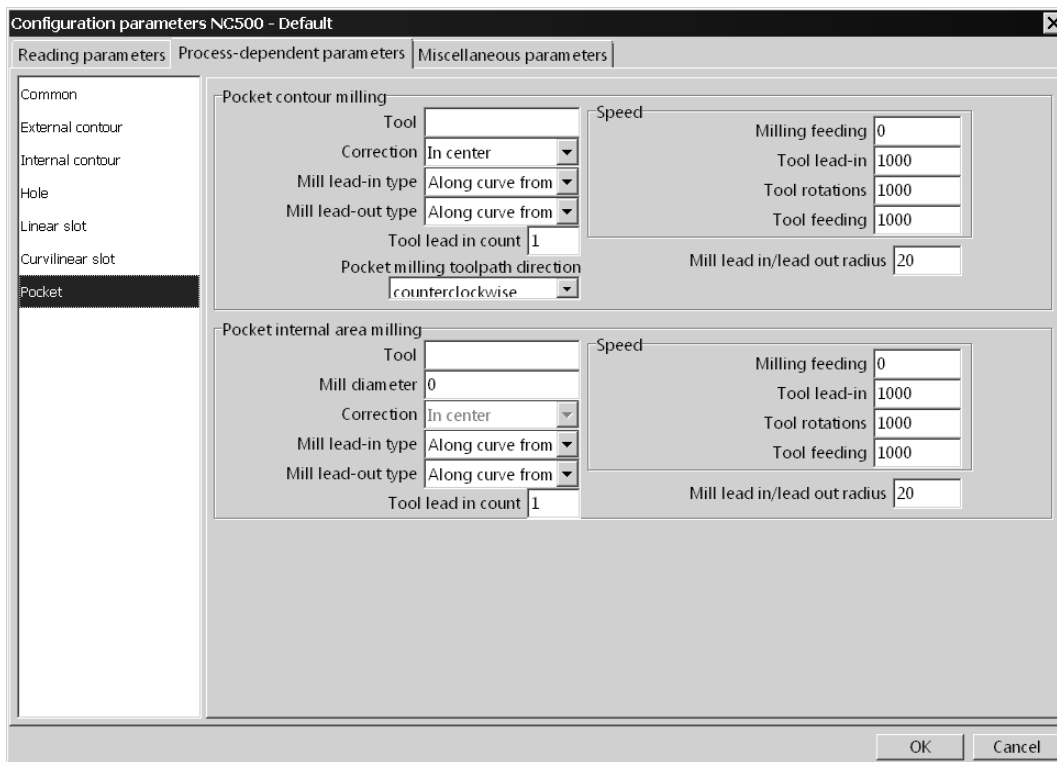


Fig. 10.172.

Section control elements allow to set control programs creation parameters for notch contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.6.3. Biesse XNC



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common parameters are set in **Common** section of **Process-dependent parameters** tab (fig. 10.173).

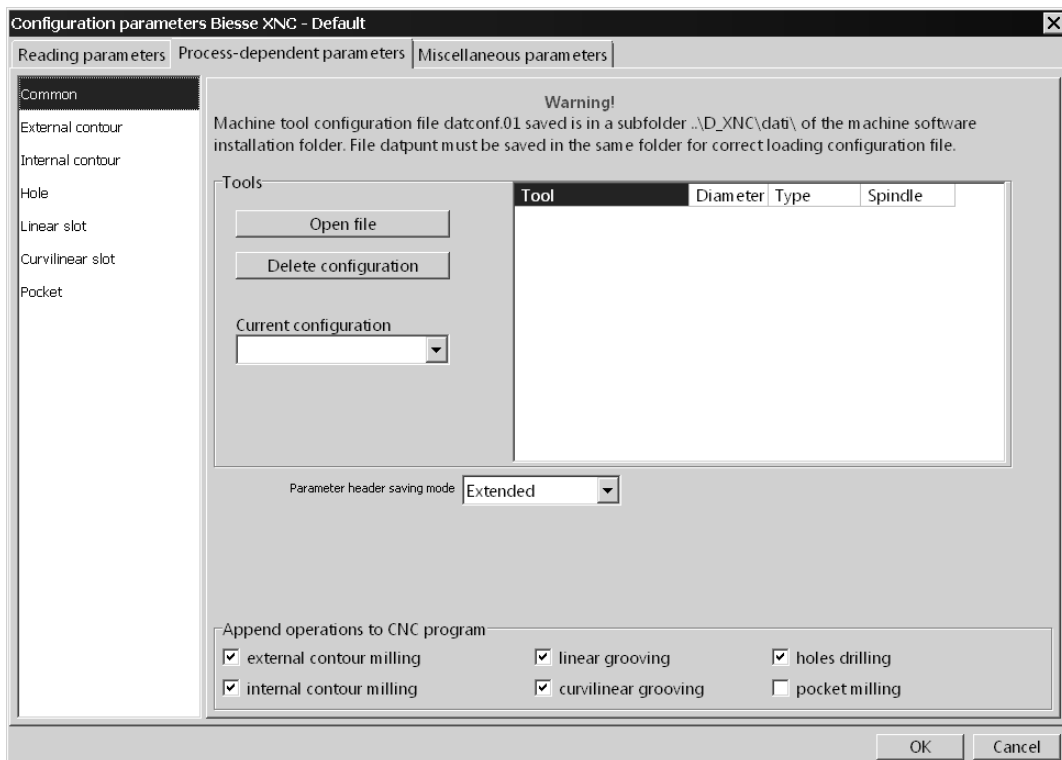


Fig. 10.173.

Load file button allows to load tools configuration file. After pushing the button, a standard Windows dialogue will appear on the screen. In this dialogue open a file of *.01* extension. Tool parameters list will be automatically filled with file data. **Delete configuration** button allows to delete the current configuration. After pushing the button, a warning will appear on the screen (fig. 10.174).

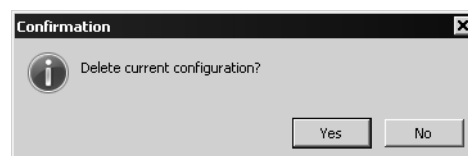


Fig. 10.174.

If you confirm, the tool parameters list will be cleaned up. **Append operations to CNC program** group control elements allow to control operations automatic appending to control program. To append the operation to the program, enable option next to its name.

Contours parameters

Contours milling parameters are set in **Outer contour** (fig. 10.175) and **Internal contour** sections of **Process-dependent parameters** tab.

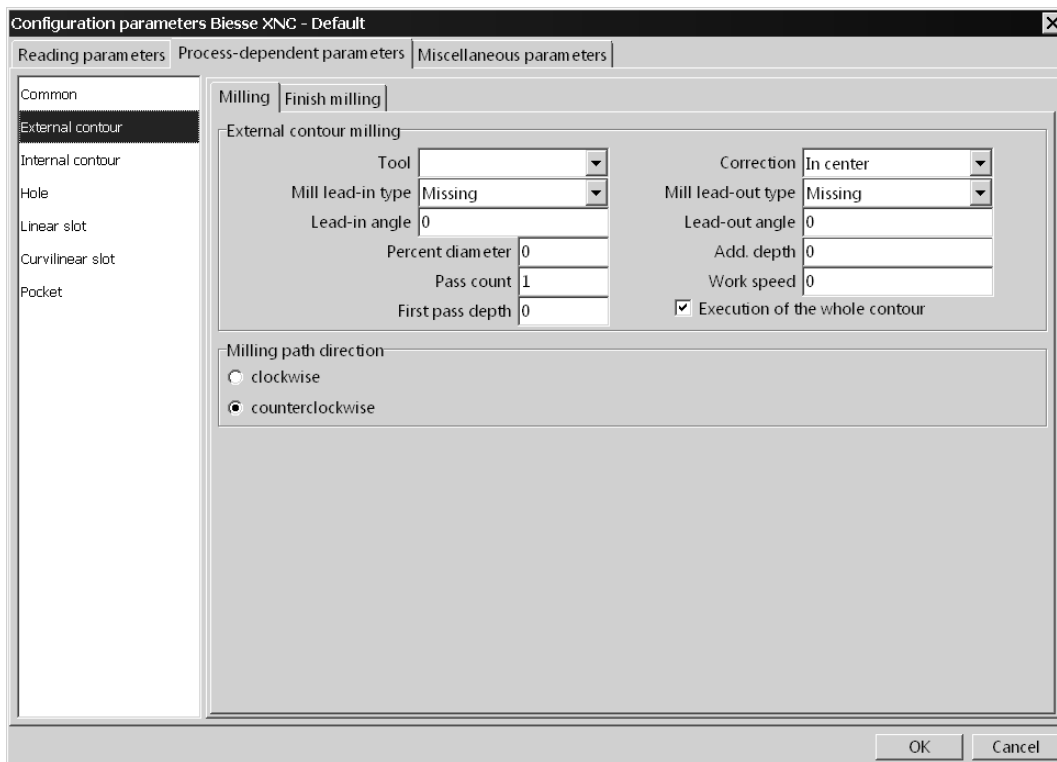


Fig. 10.175.

Milling and **Finish milling** tabs control elements allow to choose a tool, correction type, tool lead-in and lead-out, etc. **Finish milling** option allows to control this operation. **Tools** combo box contains tools designations of configuration file. These tabs control elements are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Holes section control elements (fig. 10.176) allow to set boring parameters.

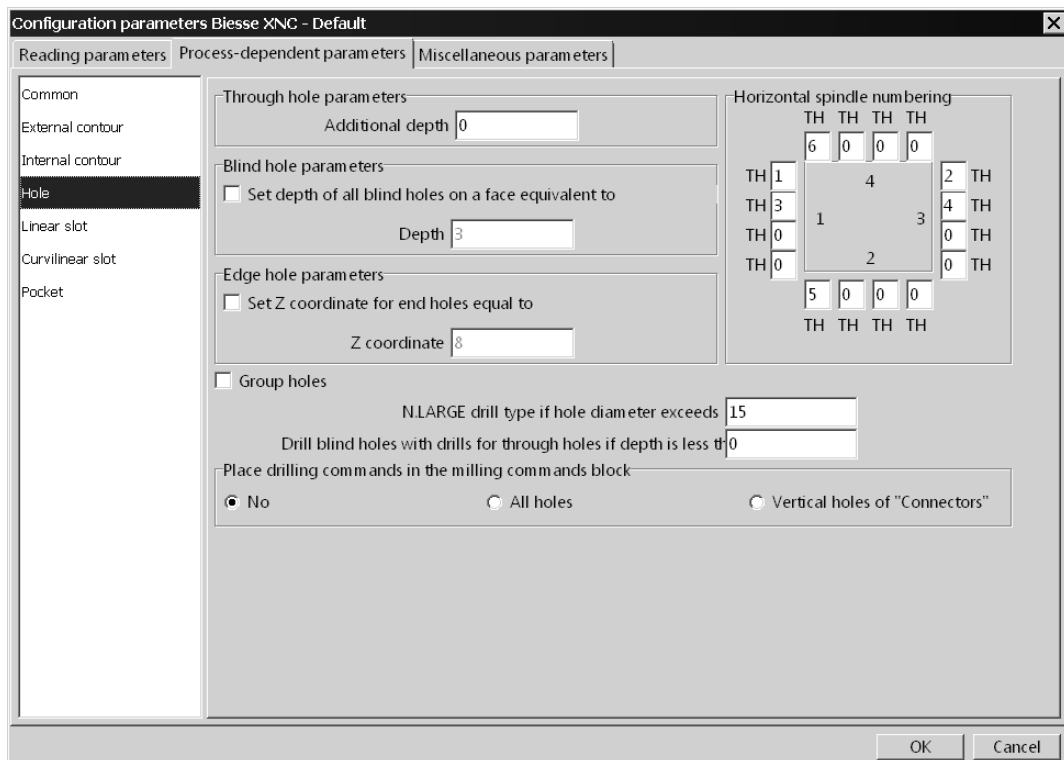


Fig. 10.176.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. You can set the same depth for blind holes. For this purpose enable the option in **Blind hole parameters** section and set depth value. It is possible to set the same Z parameter values for edge holes. For this purpose enable the option and set parameter in **Edge hole parameters** group. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option.

N.LARGE drill type if hole diameter exceeds field allows to set hole diameter that will specify automatic tool selection in the program. **Place drilling commands in the milling commands block** group variants allow to specify holes processing commands placing in milling commands block.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.177).

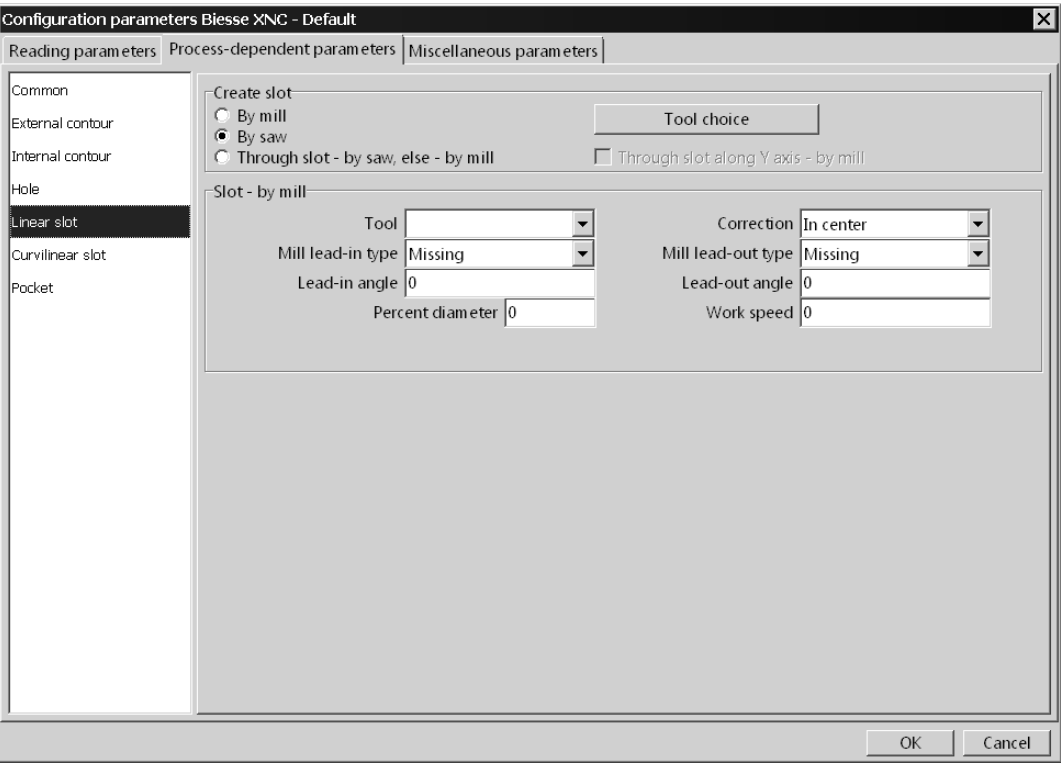


Fig. 10.177.

Create slot group variants allow to select a tool type for slot processing. If **Through slot - by saw, else - by mill** variant is selected, pass-through grooves will be processed with a saw, other slots - with a mill. **Through slot along Y axis - by mill** option allows to set pass-through grooves direction for saw processing. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.178).

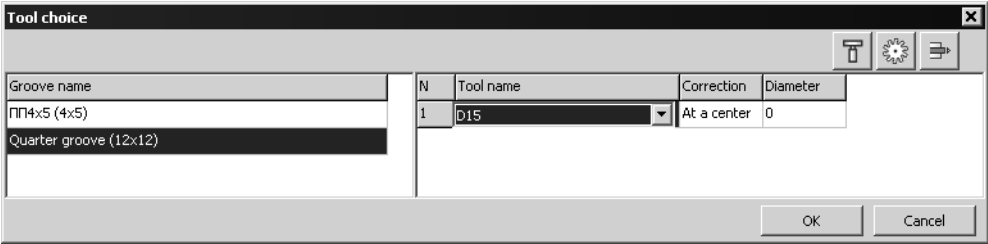


Fig. 10.178.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.179).

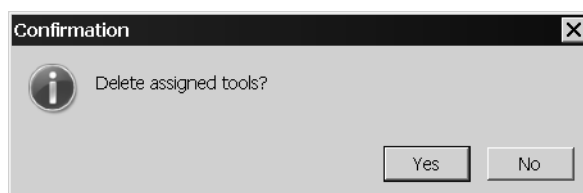


Fig. 10.179.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Slot – by mill** group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. **Tools** combo box contains tools designations of configuration file.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.180).

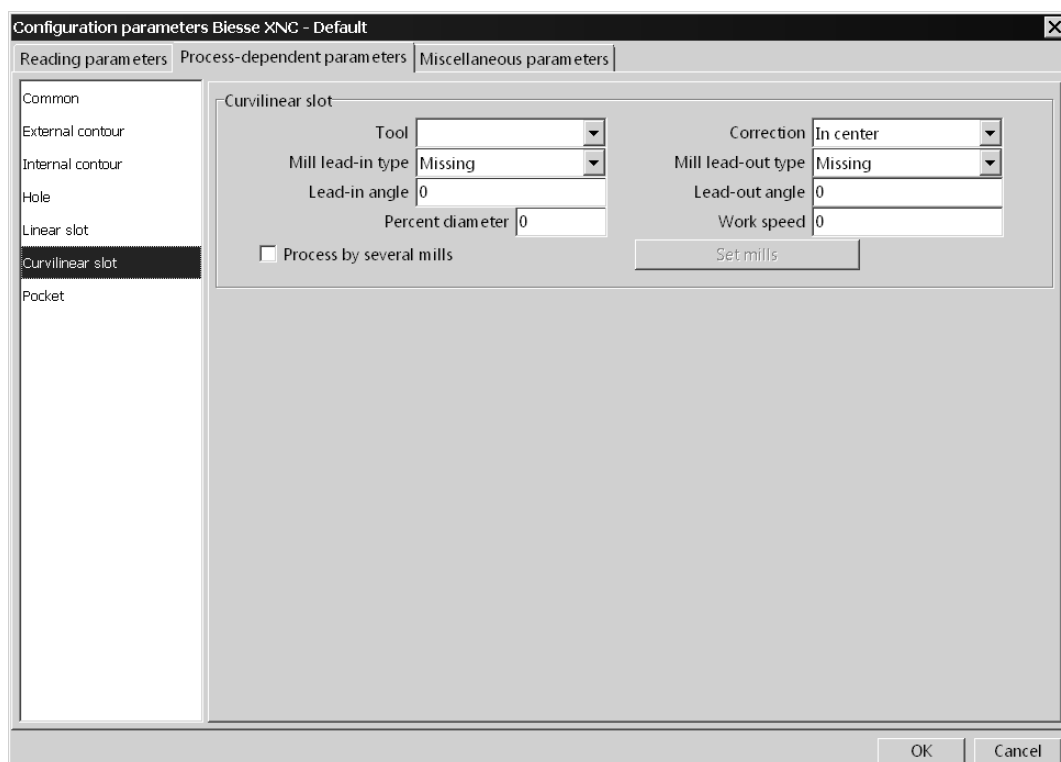


Fig. 10.180.

Section control elements allow to set control programs creation parameters for slots processing. **Tools** combo box contains tools designations of configuration file. If **Process by several mills** option is enabled, **Set mills** button will become available.

After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.181).

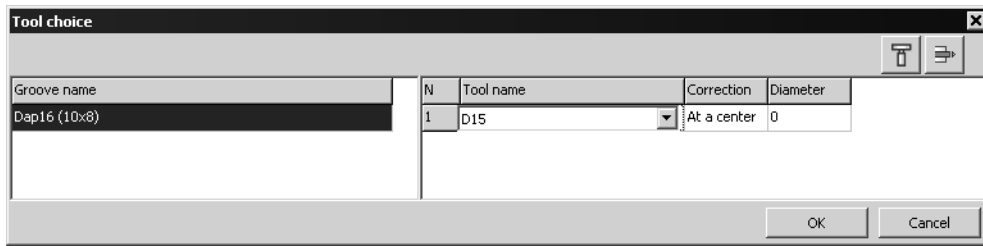


Fig. 10.181.

Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.

Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.182) allow to set pocket processing parameters.

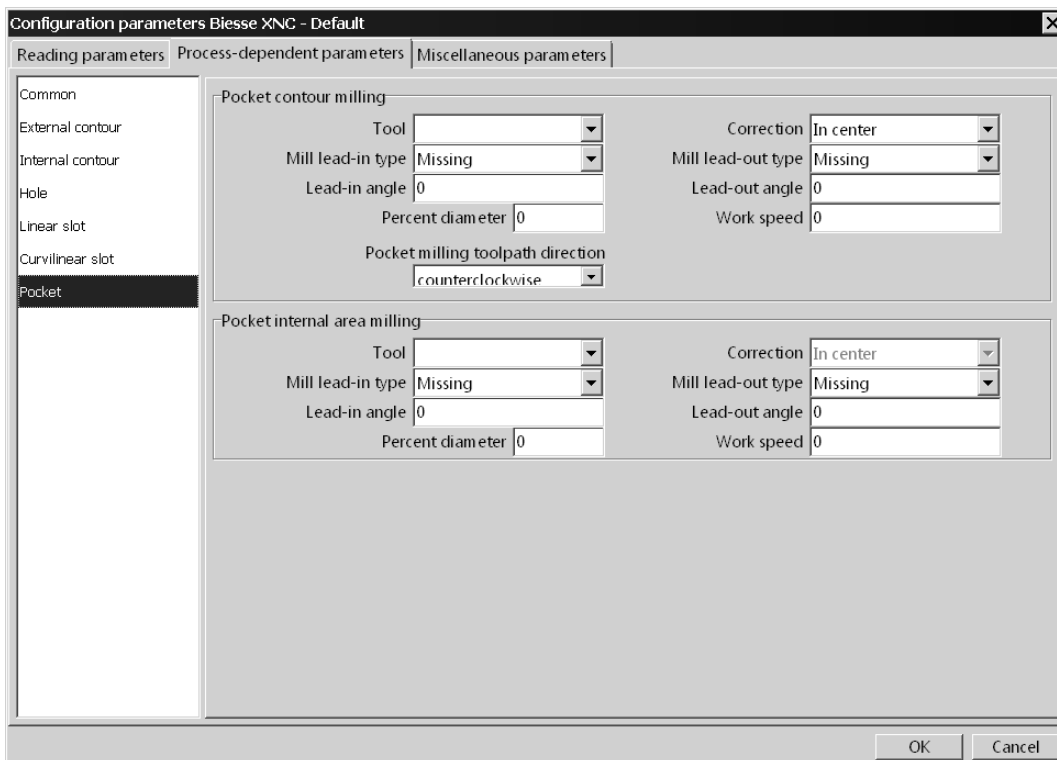


Fig. 10.182.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs for Biesse Works XNC setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.6.4. bSolid CIX

Operations

Operations section control elements (fig. 10.183) allow to specify operations that can be used in control program.

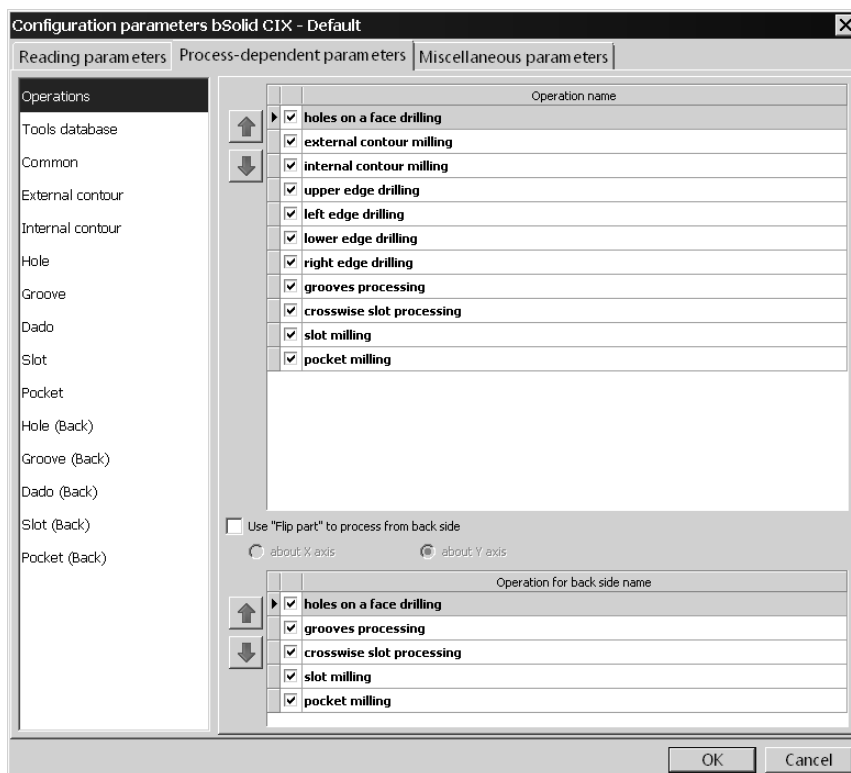


Fig. 10.183.

Usable tools

Tools database section control elements (fig. 10.184) allow to create a tools list available for use in control programs.

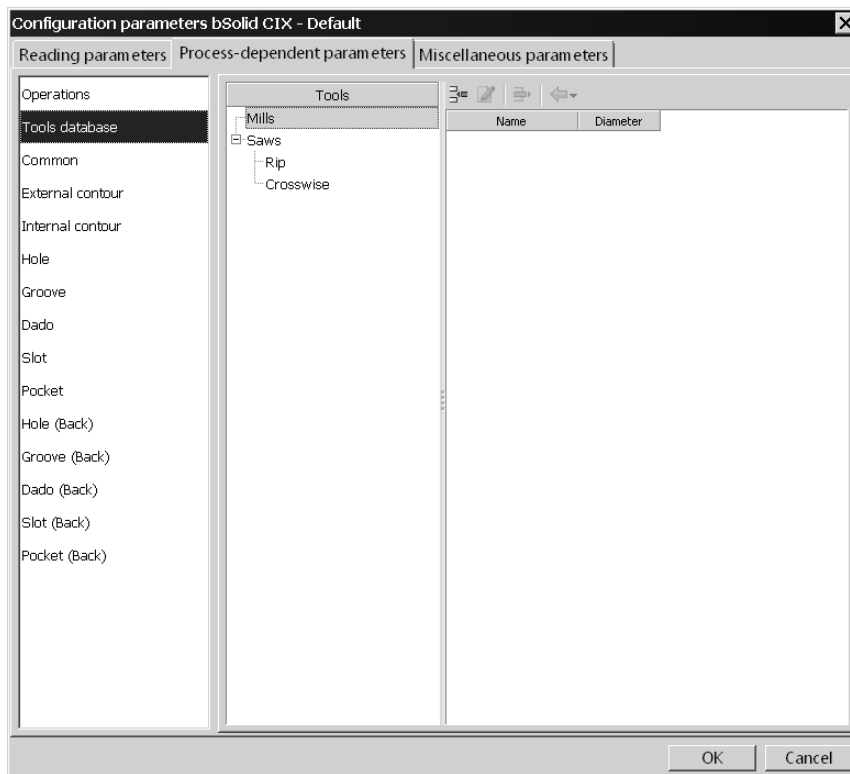


Fig. 10.184.



To add a tool to the list push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.185).

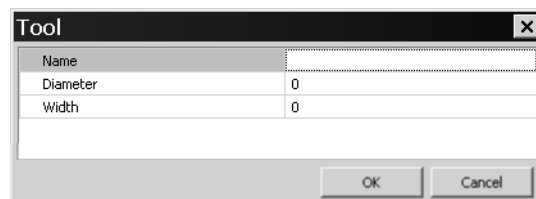


Fig. 10.185.



In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes.



Edit button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Common parameters

Common section control elements (fig. 10.186) allow to set common parameters of control programs creation.

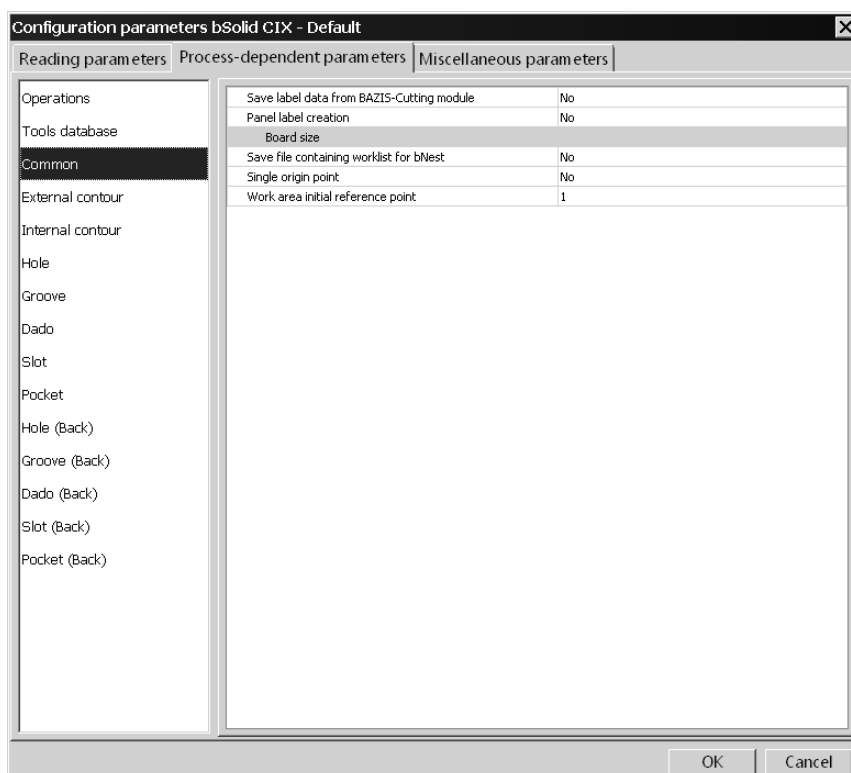


Fig. 10.186.

Contour processing parameters

Outer and internal contour processing parameters are set in **Outer contour** and **Internal contour** sections (fig. 10.187).

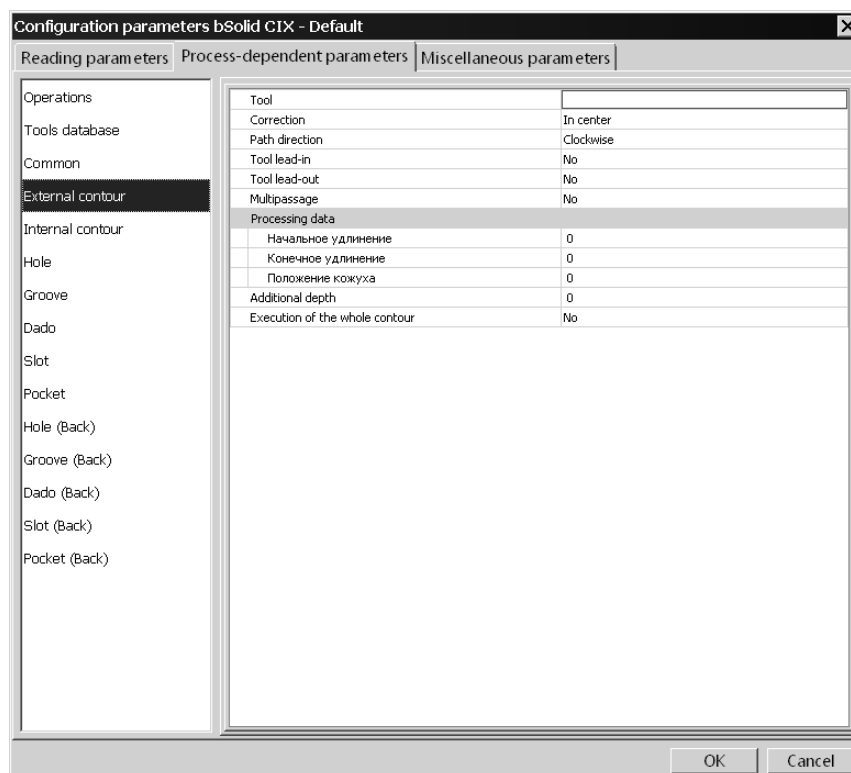


Fig. 10.187.

Sections control elements allow to choose the tool, correction types, lead-in, etc.

Hole parameters

Hole processing from front side is set in **Hole** section (fig. 10.188), holes from back side – in **Hole (Back)** section in the same way.

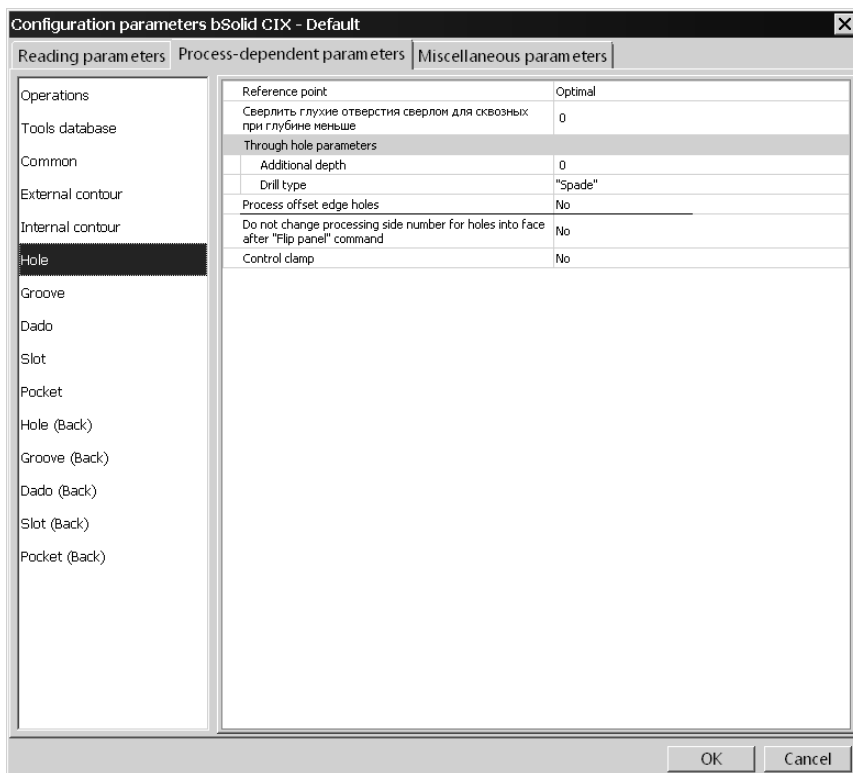


Fig. 10.188.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Grooves processing from front side is set in the following sections: **Groove** (fig. 10.189) and **Dado**. Grooves processing from back side of the panel is set in **Groove (Back)** and **Dado (Back)** sections.

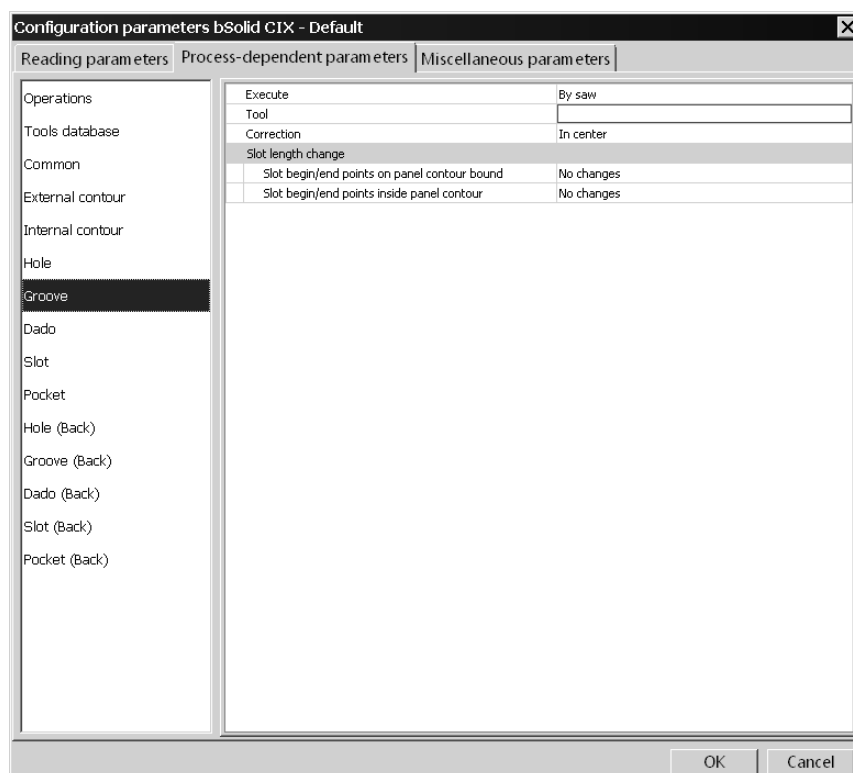


Fig. 10.189.

The section control elements allow to select a tool type for groove creation, database tool and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory .

Slot parameters of custom trajectory

Slot processing from front side is set in section **Slot** (fig. 10.190), from panel back side – in **Slot (Back)** section.

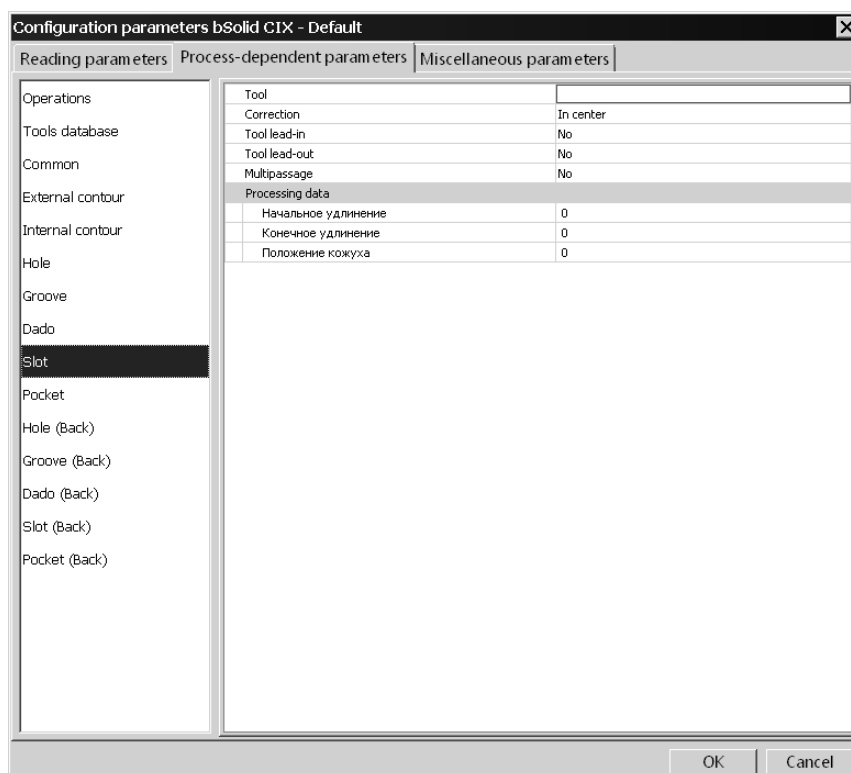


Fig. 10.190.

Section control elements allow to choose database tool, correction type and multipassage use.

Pocket parameters

Pocket processing parameters from front side are set in **Pocket** section (fig. 10.191), from panel back side – in **Pocket (Back)** section.

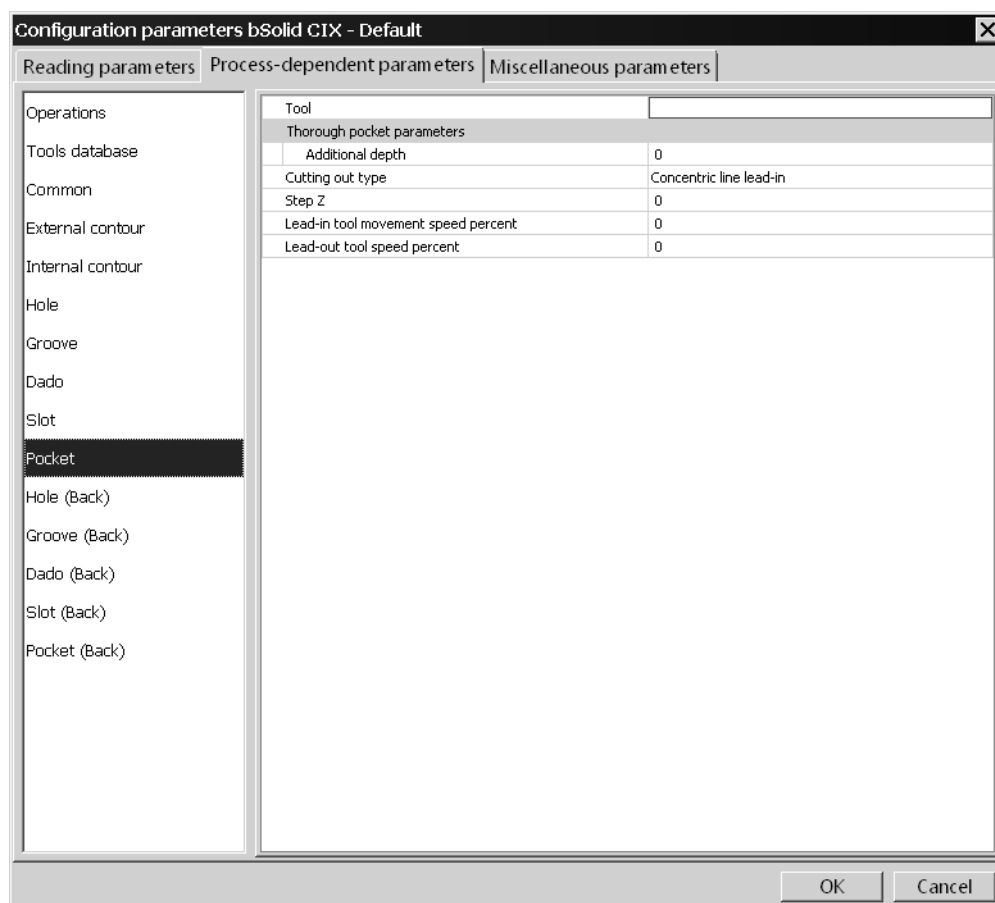


Fig. 10.191.

Tool combo box allows to select a milling tool name from a database. For pocket contour select milling path direction, correction type and multipassage use.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.7. BJD

10.7.1. HiWCAM



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.192) allow to specify operations that can be used in control program.

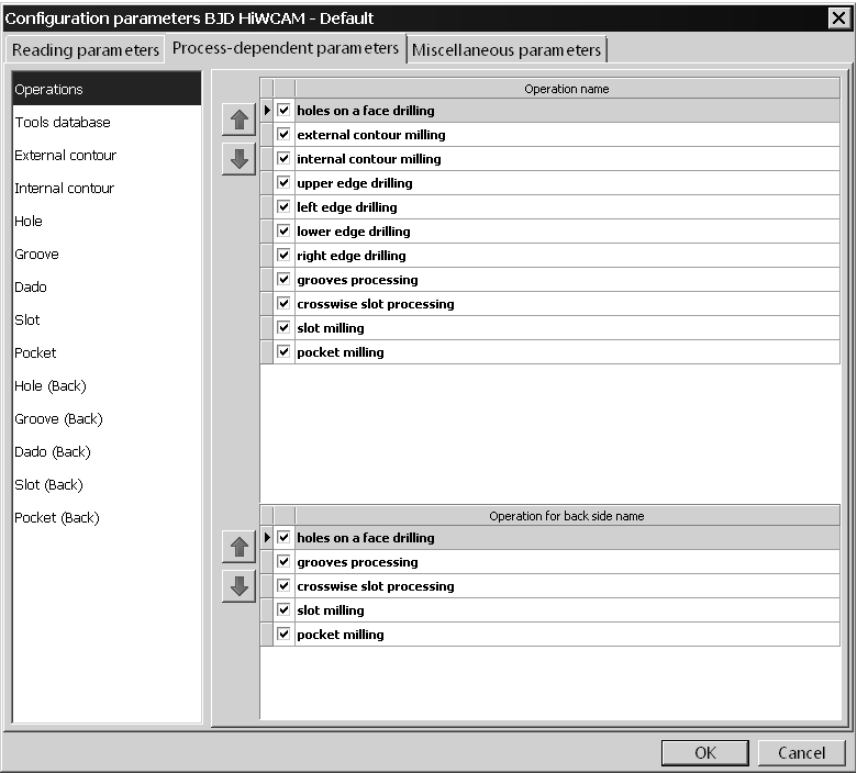


Fig. 10.192.

Usable tools

Tools database section control elements (fig. 10.193) allow to create a list of tools available for use in control programs.

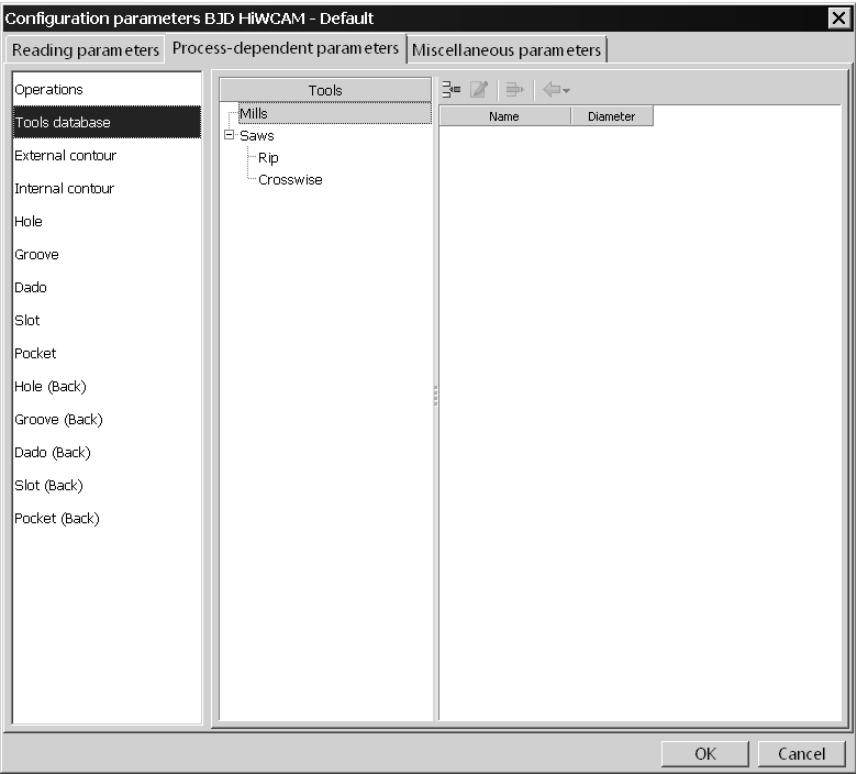


Fig. 10.193.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.194).

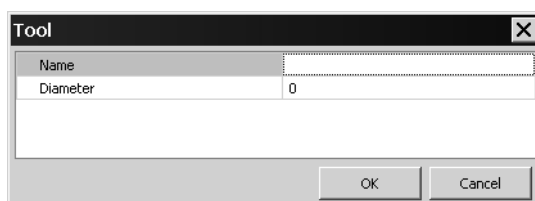


Fig. 10.194.



In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Contours parameters

Contours milling parameters are set in **Outer contour** and **Internal contour** sections (fig. 10.195).

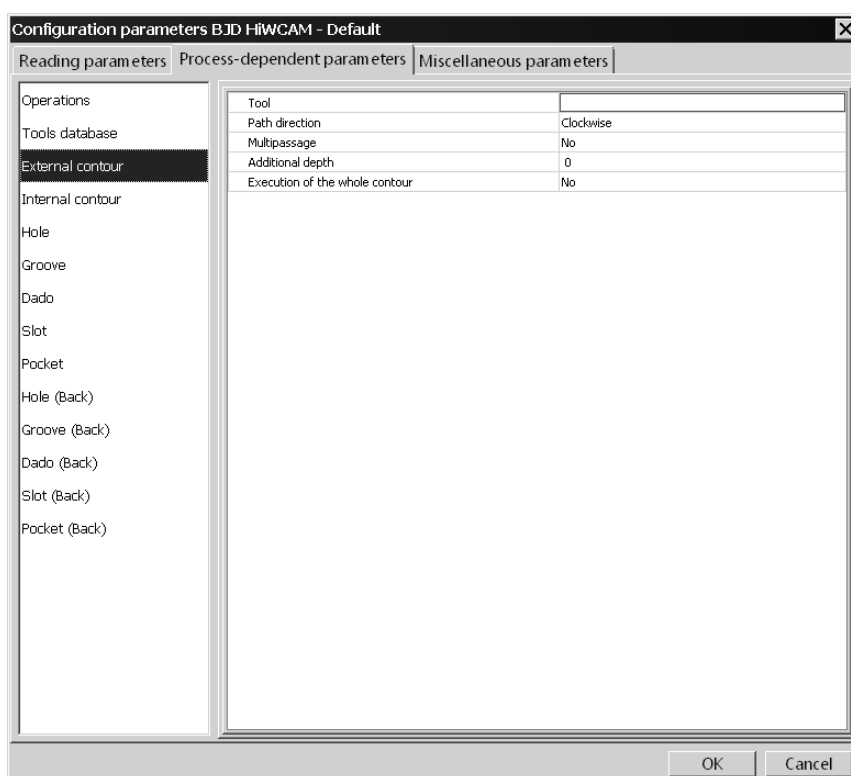


Fig. 10.195.

Section control elements allow to choose a tool, lead-in type, correction type, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.196), holes from back side – in **Hole (Back)** section in the same way.

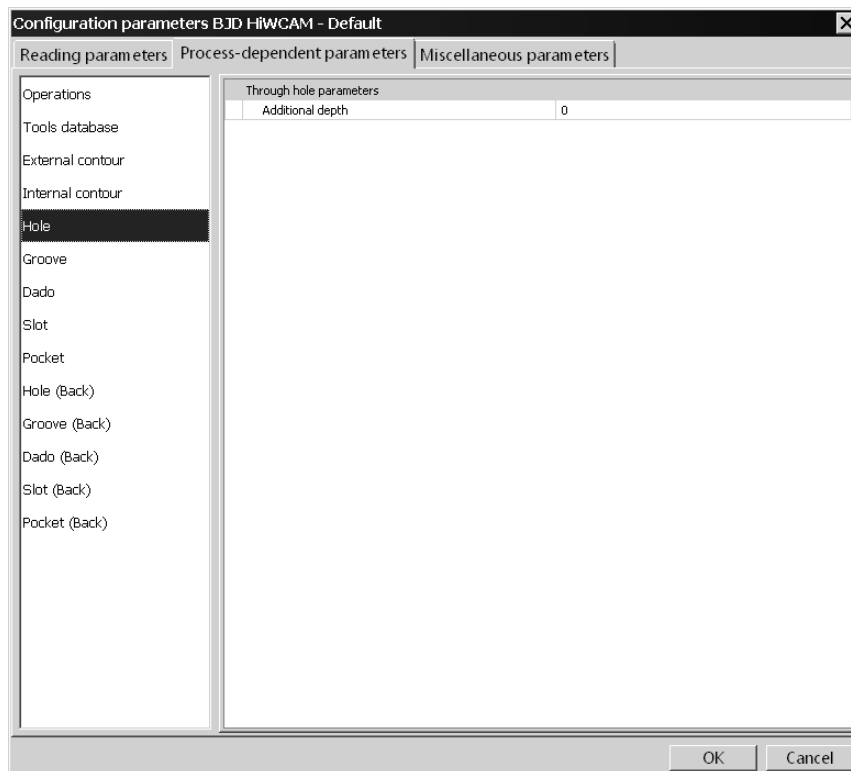


Fig. 10.196.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.197), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

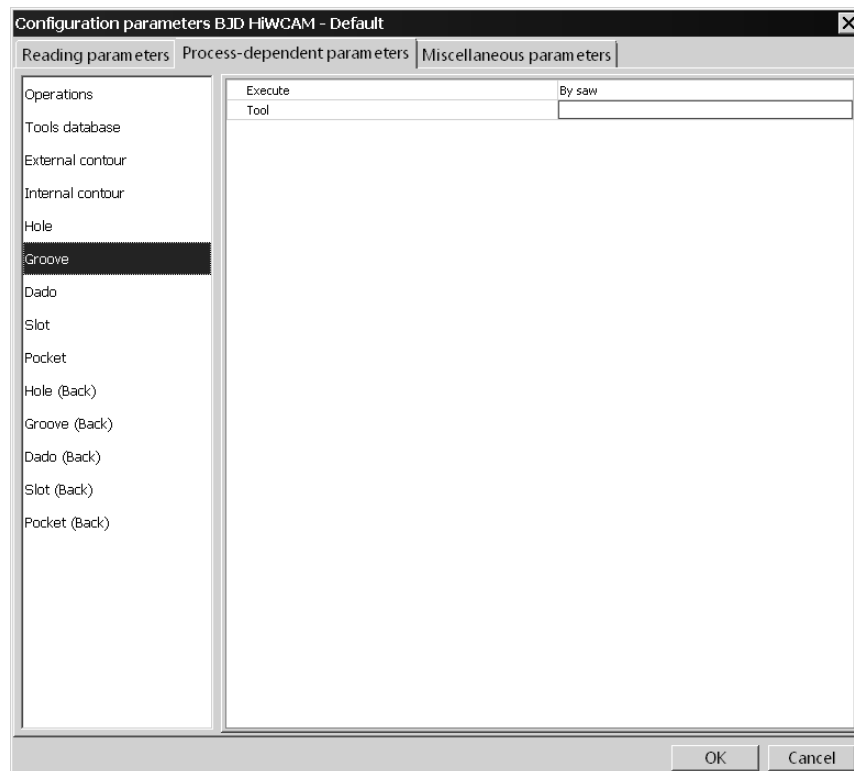


Fig. 10.197.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.198), from the back panel side – in **Slot (Back)** section.

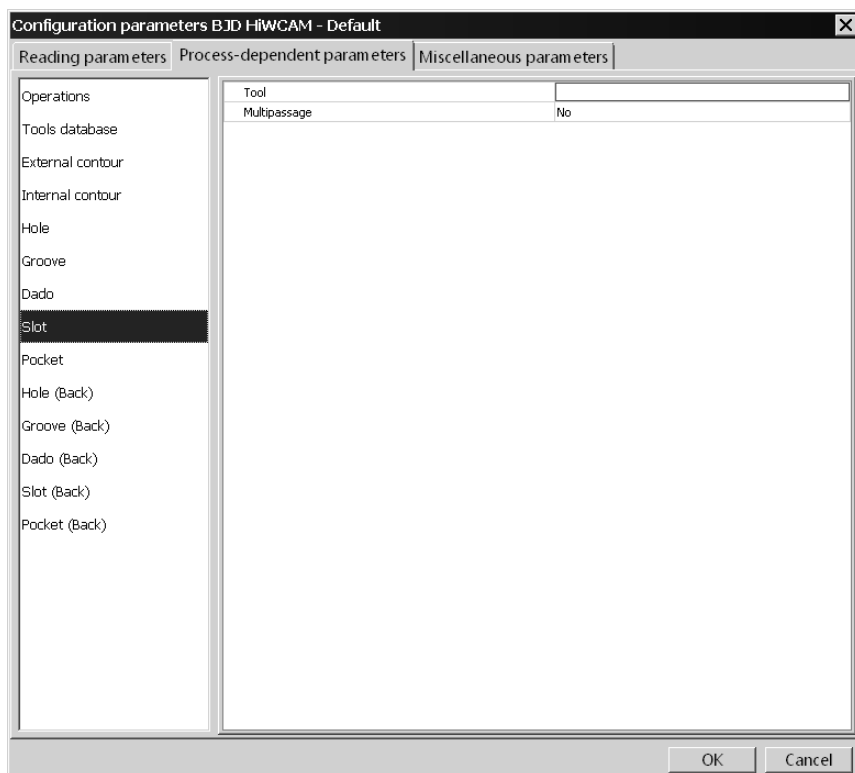


Fig. 10.198.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.199), on the back panel side – in **Pocket (Back)** section.

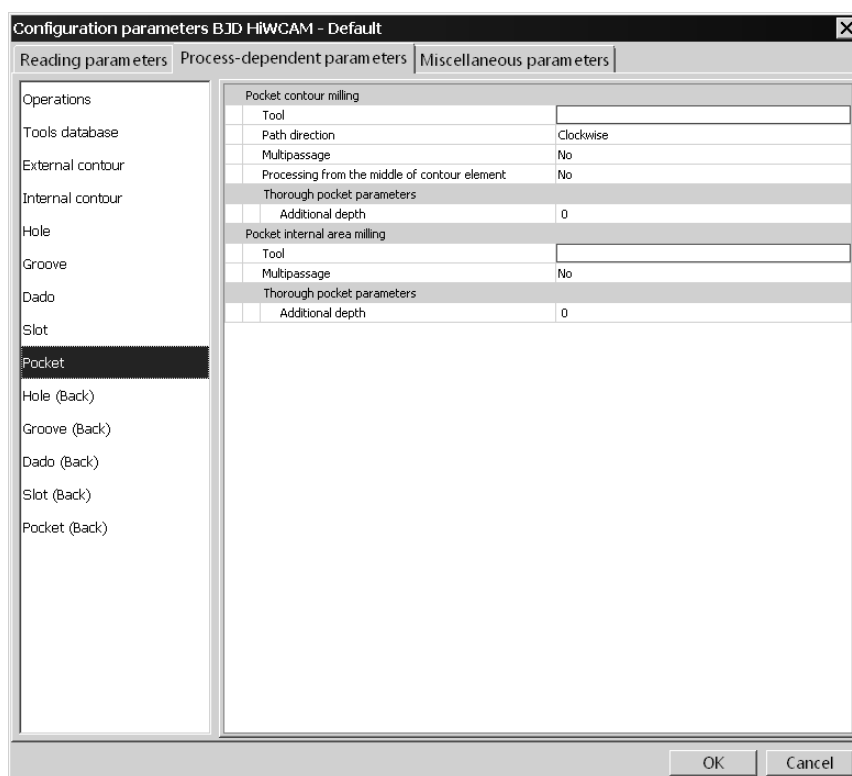


Fig. 10.199.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects from database. For pocket contour select milling execution direction, correction type and multipassage use. For internal pocket part – correction type and multipassage use.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.7.2. BJD_Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide

Operations

Operations section control elements (fig. 10.200) allow to specify operations that can be used in control program.

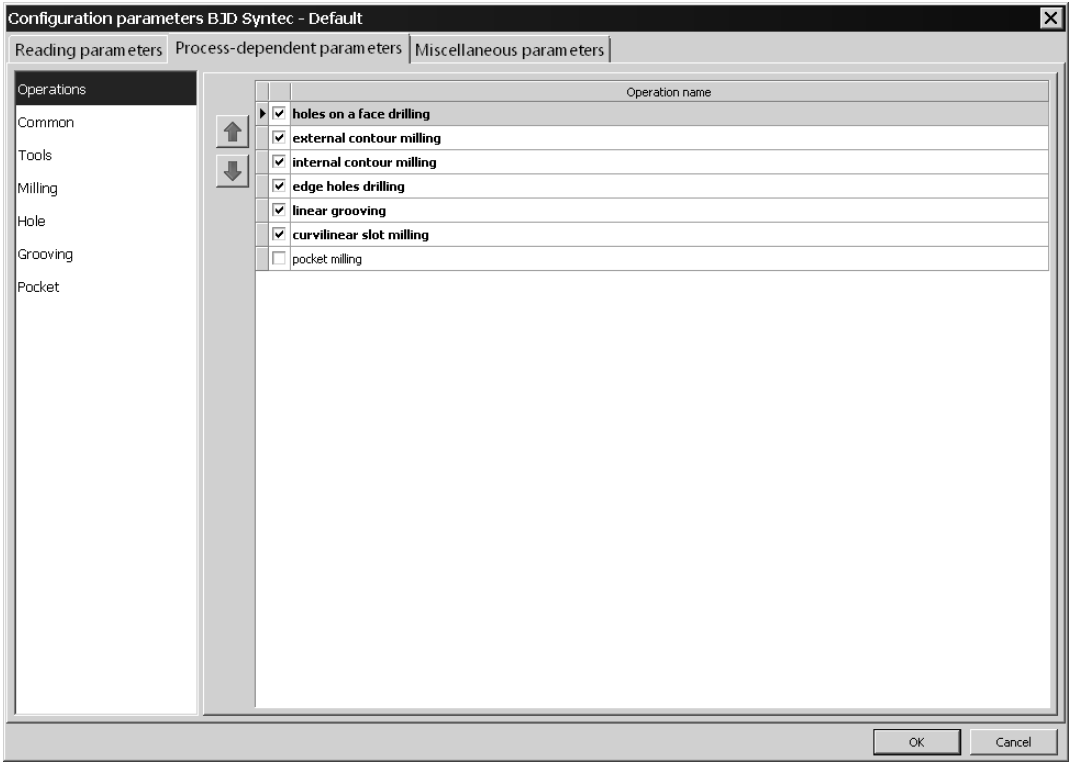


Fig. 10.200.

Common parameters

Common section control elements allow to setup common parameters of control programs creation (fig. 10.201).

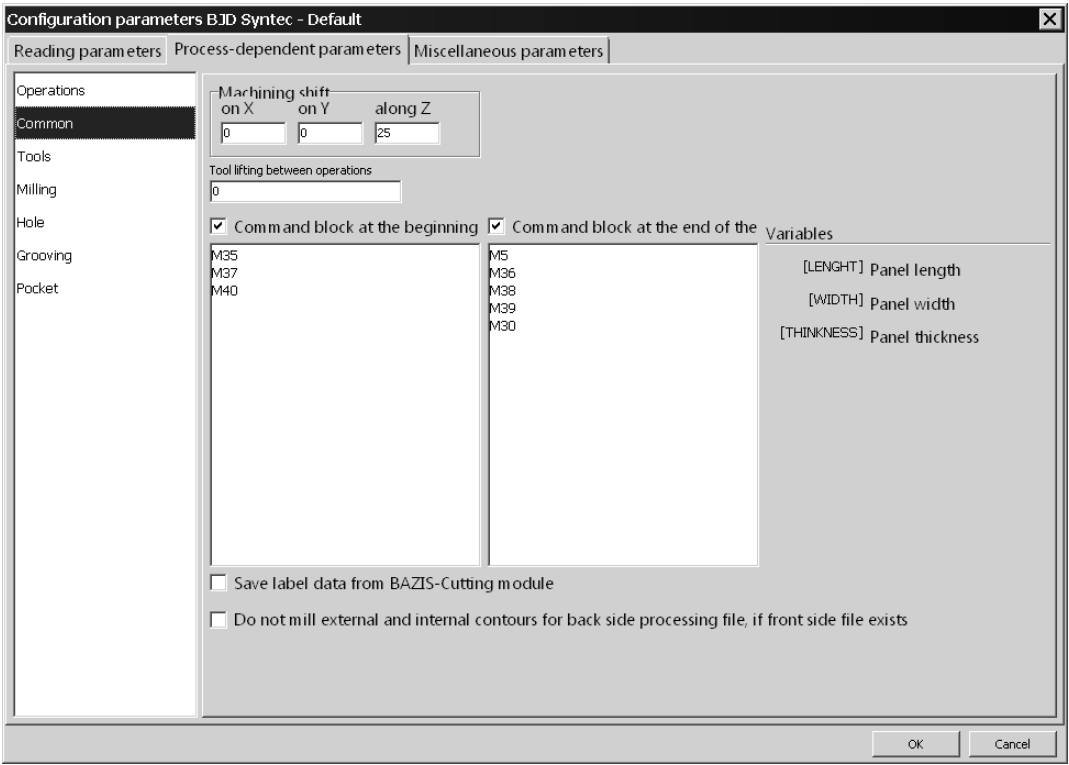


Fig. 10.201.

Machining shift group fields allow to set work area initial point moving. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.202).

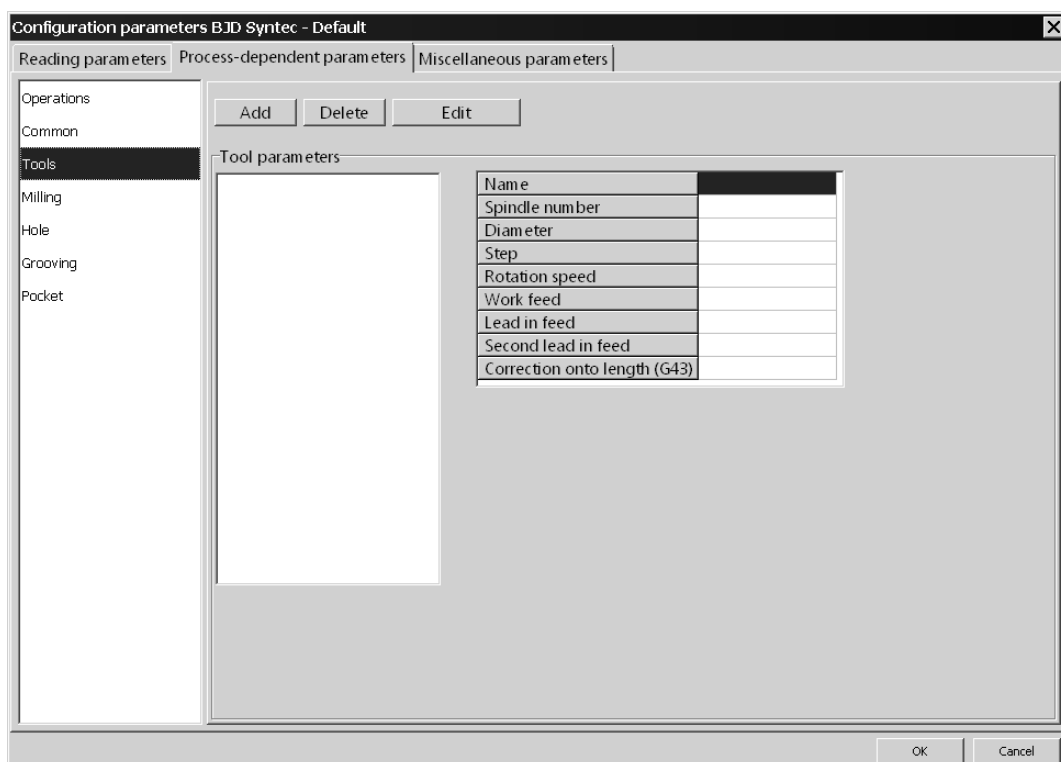


Fig. 10.202.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.203).

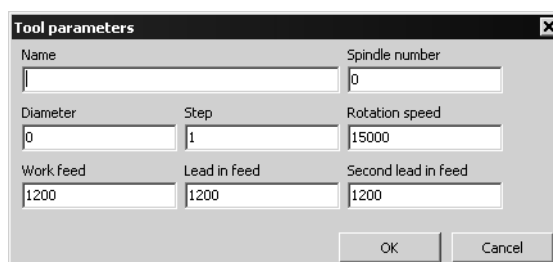


Fig. 10.203.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is in fig. 10.204.

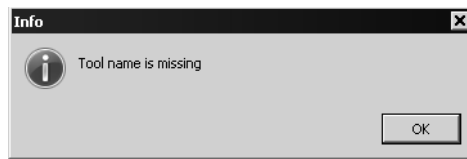


Fig. 10.204.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.203). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Milling** tab (fig. 10.205).

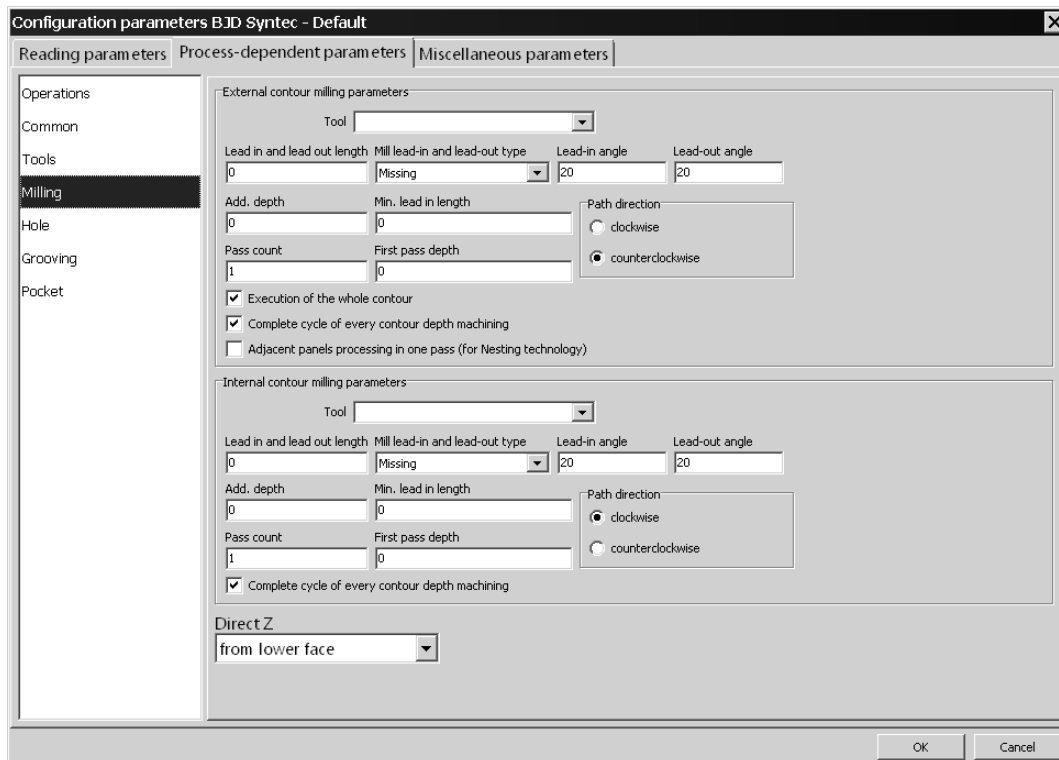


Fig. 10.205.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.206).

The screenshot shows a software interface for configuring hole parameters. The 'Hole' section is selected in the left sidebar. The main area contains two groups of parameters: 'Through hole parameters' and 'Edge hole parameters'. The 'Through hole parameters' group includes 'Additional depth' (set to 0) and 'Drill holes on a face' (with options 'on drill shortest path' and 'on diameter'). The 'Edge hole parameters' group includes 'Safety distance' (set to 10), 'For holes on a face apply Z' (set to 'from lower face'), and 'For edge holes use Z' (set to 'from upper face(+)'). Below these is a 'Tool lifting between operations' field set to 0. At the bottom is a table with columns 'Diameter', 'Layout', 'Type', and 'Tool'.

Diameter	Layout	Type	Tool
5	Edge on the mitre	Blind	
6	Upper face	Through	
5	Front edge	Blind	
6	Front edge	Blind	

Fig. 10.206.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Slot parameters

Grooving section control elements (fig. 10.207) allow to set slots processing parameters.

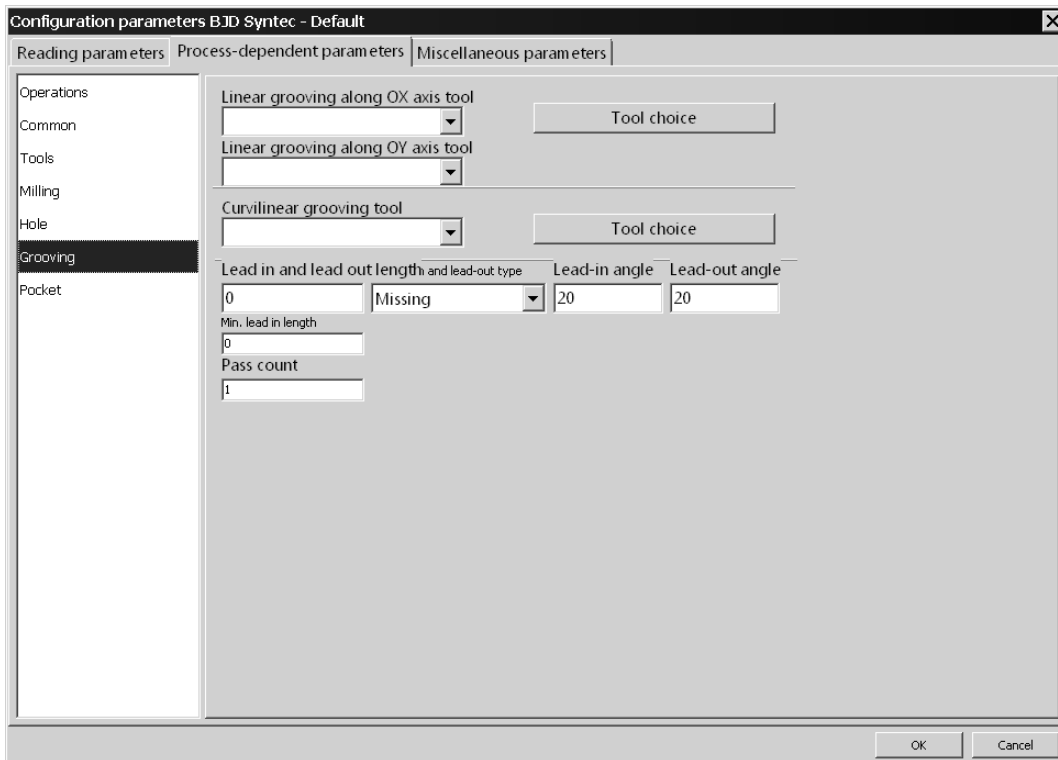


Fig. 10.207.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.208).

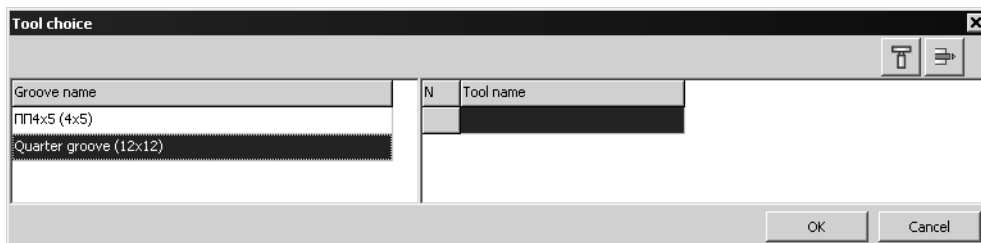


Fig. 10.208.



Slot name table contains all slots names of the model. You can set particular tools for each slot processing. **Add mill** button allows to assign a tool to the slot from the list, created in **Tools** section.



Delete button allows to delete selected tool from the list.

To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.209) allow to set pocket processing parameters.

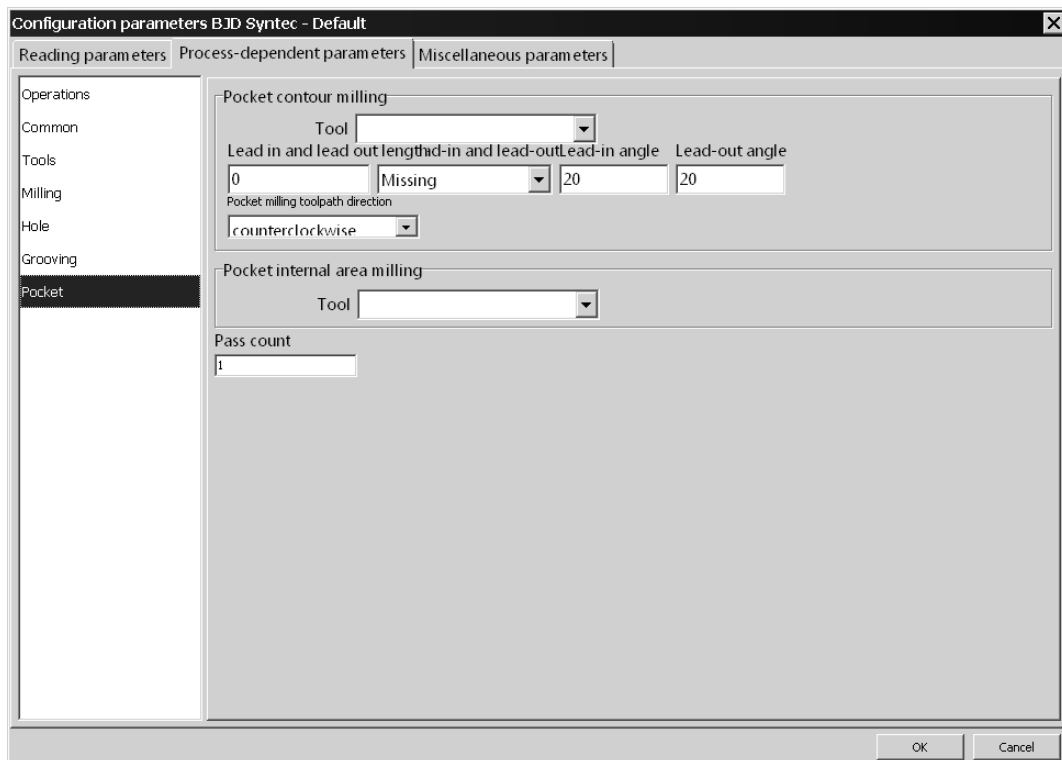


Fig. 10.209.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.8. Blue Elephant

10.8.1. Blue Elephant



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.210) allow to specify operations that can be used in control program.

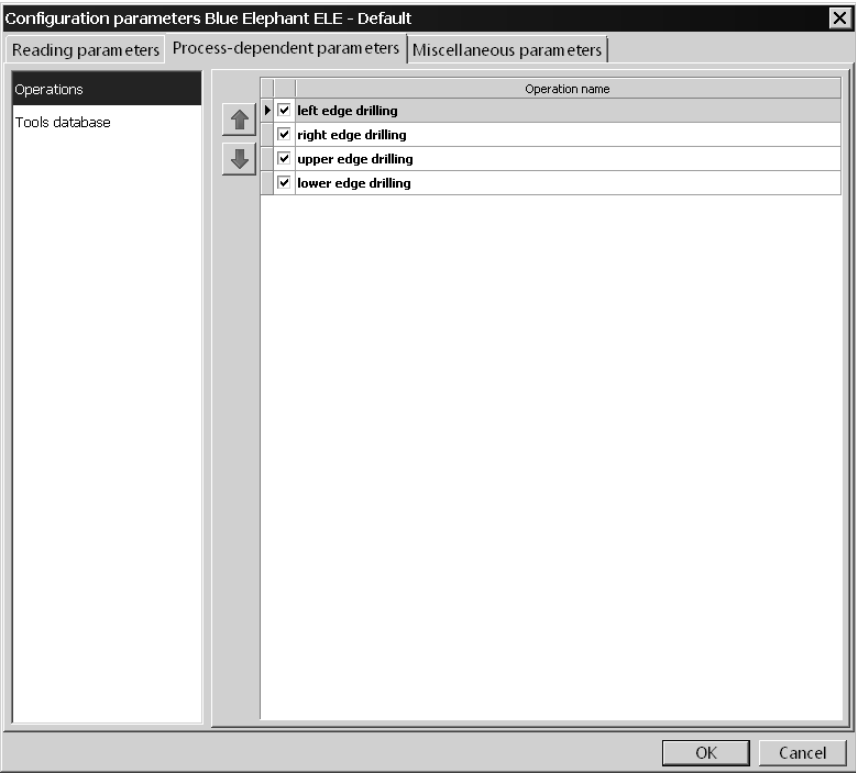


Fig. 10.210.

Usable tools

Tools database section control elements (fig. 10.211) allow to create a list of tools available for use in control programs.

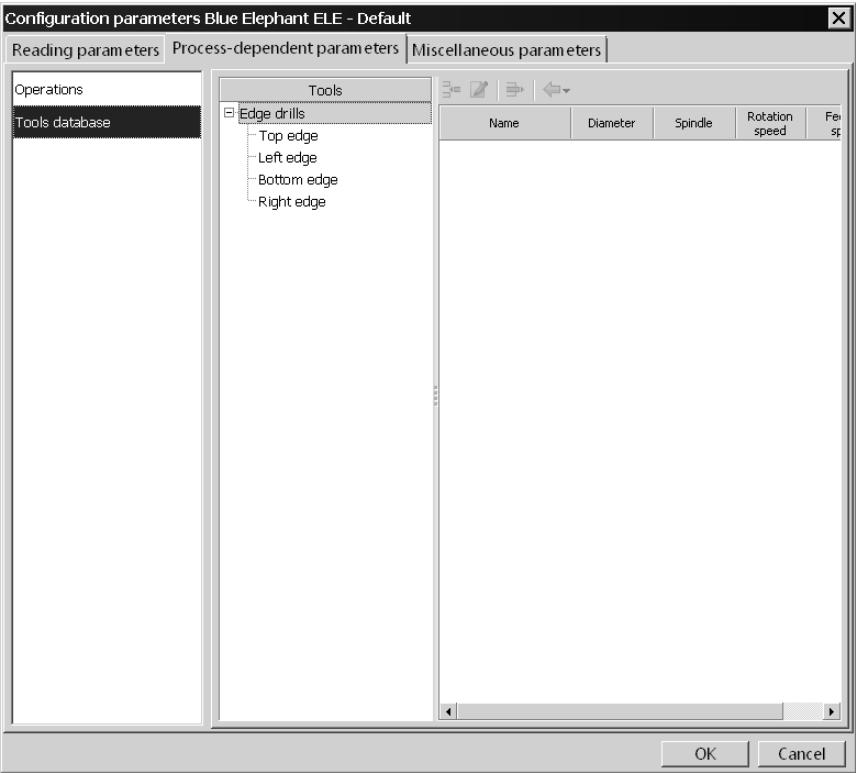


Fig. 10.211.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.212).

Tool	
Name	
Diameter	0
Spindle	0
Rotation speed	0
Feeding speed	0

Fig. 10.212.



In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

10.9. Busellato

10.9.1. Genesis Evolution



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.213) allow to specify operations that can be used in control program.

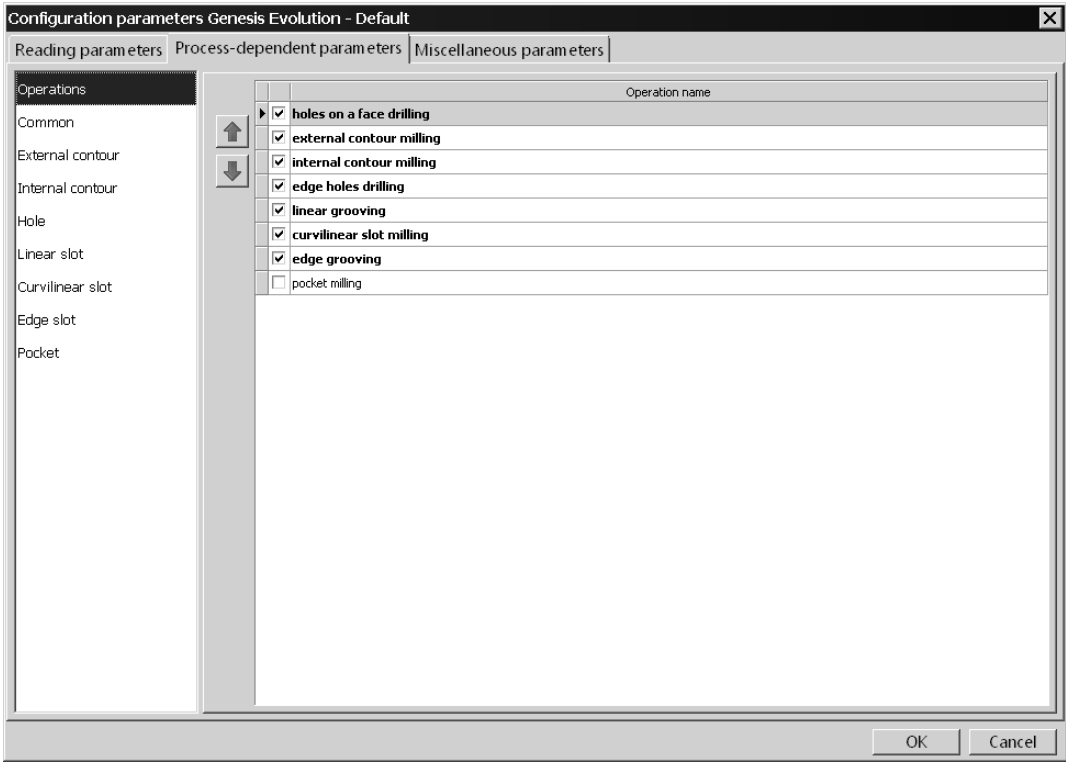


Fig. 10.213.

Common parameters

Common section control elements (fig. 10.214) allow to set control programs common parameters.

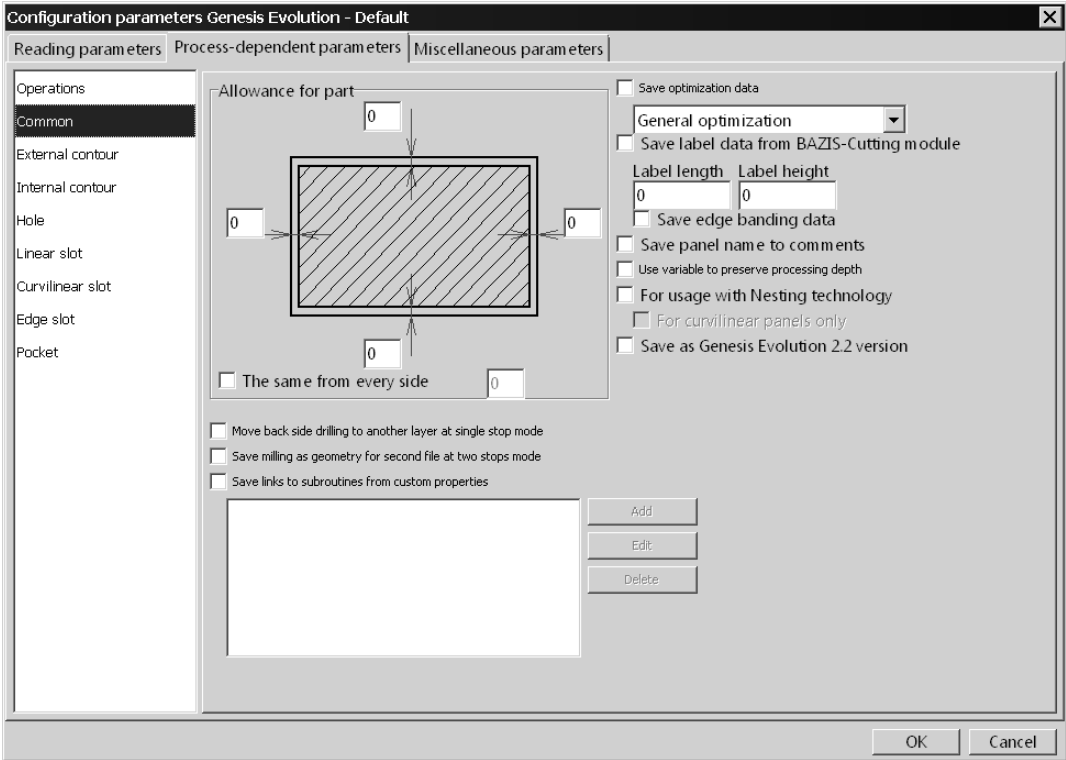


Fig. 10.214.

Allowance for machining group fields of panel image sides allow to set allowance values for each edge. To set the same allowance values for all edges enable **The same from every side** option and enter this allowance value. **Append operations to CNC program** group options allow to control operations appending to control program.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** and **Internal contour** sections of **Process-dependent parameters** tab (fig. 10.215). Rough and finish milling setup is executed separately on corresponding tabs. Tabs control elements content is the same and described for rough milling.

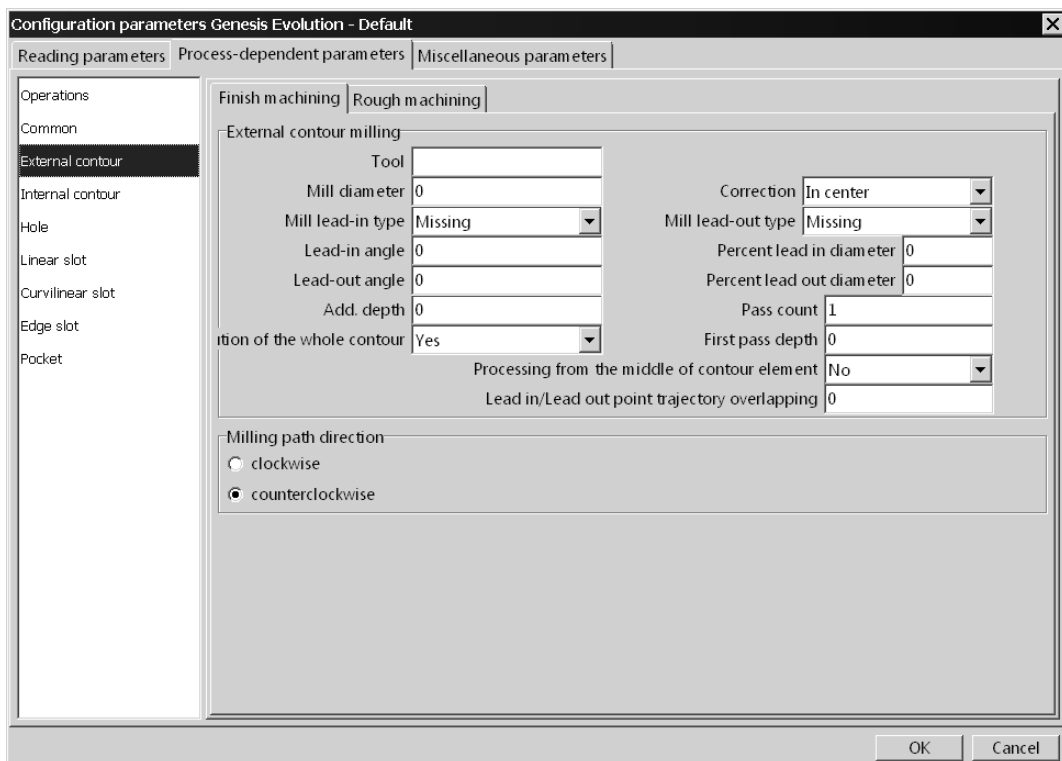


Fig. 10.215.

Rough machining option allows to control this operation. Tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.216).

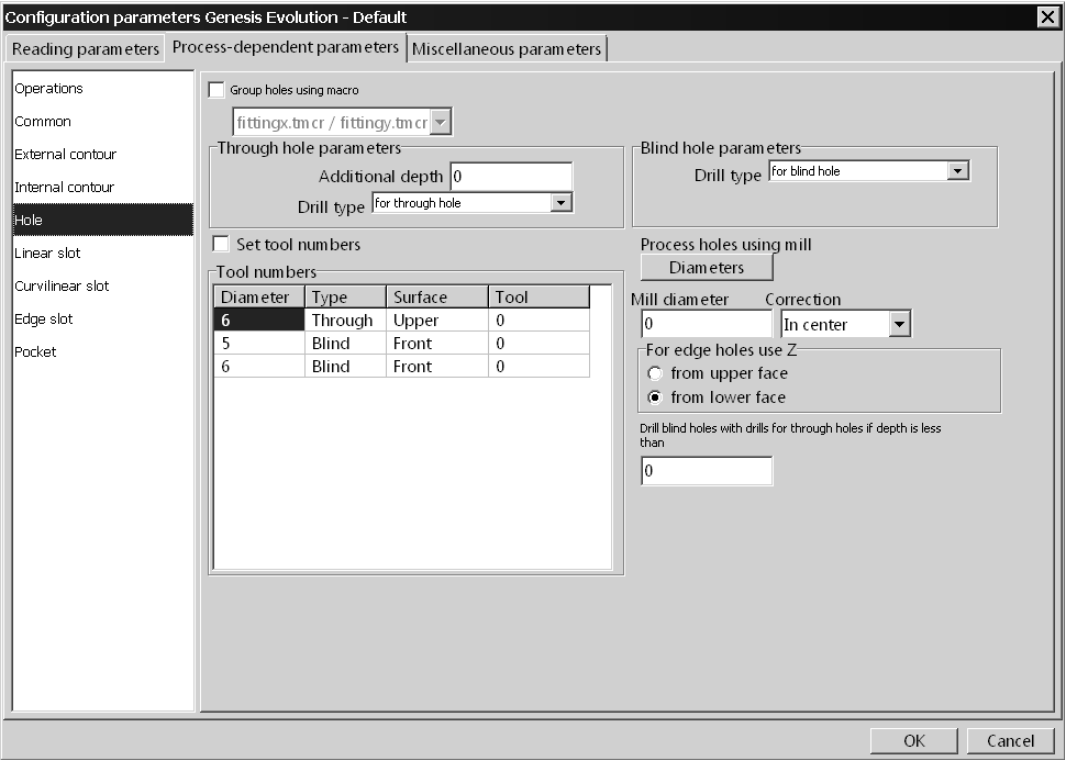


Fig. 10.216.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. Holes can be bored with a drill as well as with a mill. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameter** button. **Hole diameters** dialogue will appear on the screen (fig. 10.217).

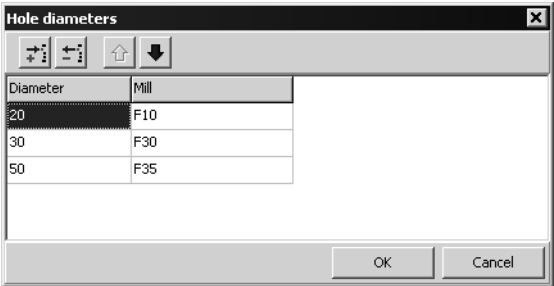


Fig. 10.217.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To complete setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.218).

Fig. 10.218.

Tool choice button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.219).

N	Tool name	Correction	Diameter
1	D15	At a center	0
	Quarter groove (12x12)		

Fig. 10.219.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing.

The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.220).

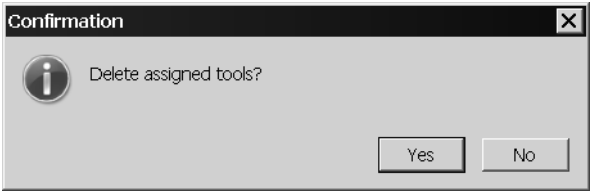


Fig. 10.220.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Through slot cut with saw** and **Grooving by saw** options allow to choose tool type for grooving. If they are disabled, slot will be milled.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill.

Curvilinear slot parameters

Curvilinear slots parameters are set in **Curvilinear slot** section (fig. 10.221).

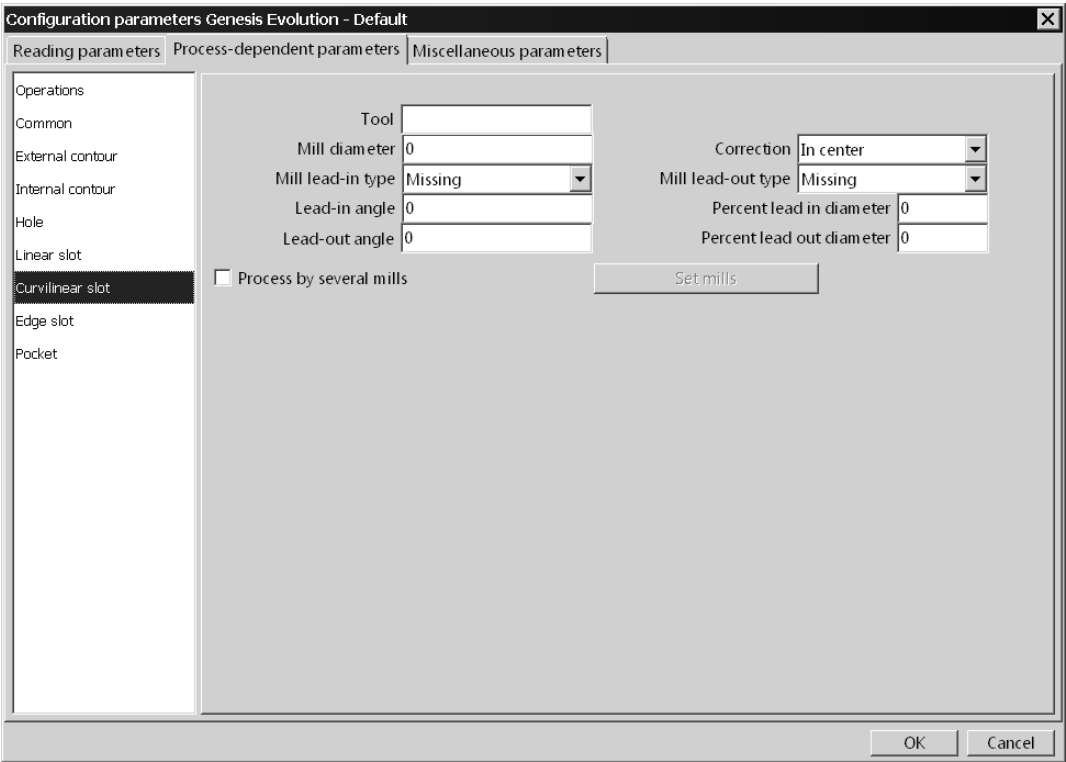


Fig. 10.221.

Control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.222).

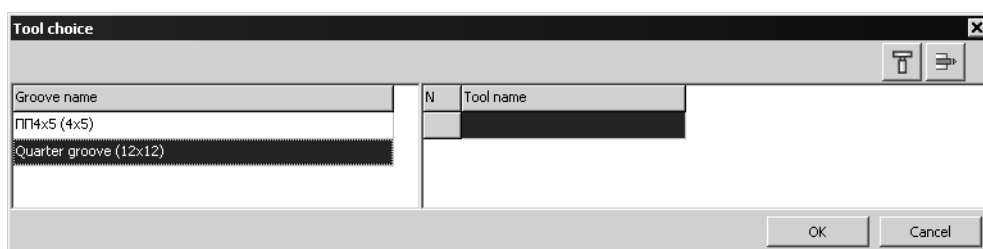


Fig. 10.222.



Slot name table contains all slots names in the model. You can assign particular tools for each slot processing. **Add mill** button allows to assign a tool for slot processing.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes

Edge slot parameters

Edge slot parameters are set in **Edge slot** section (fig. 10.223).

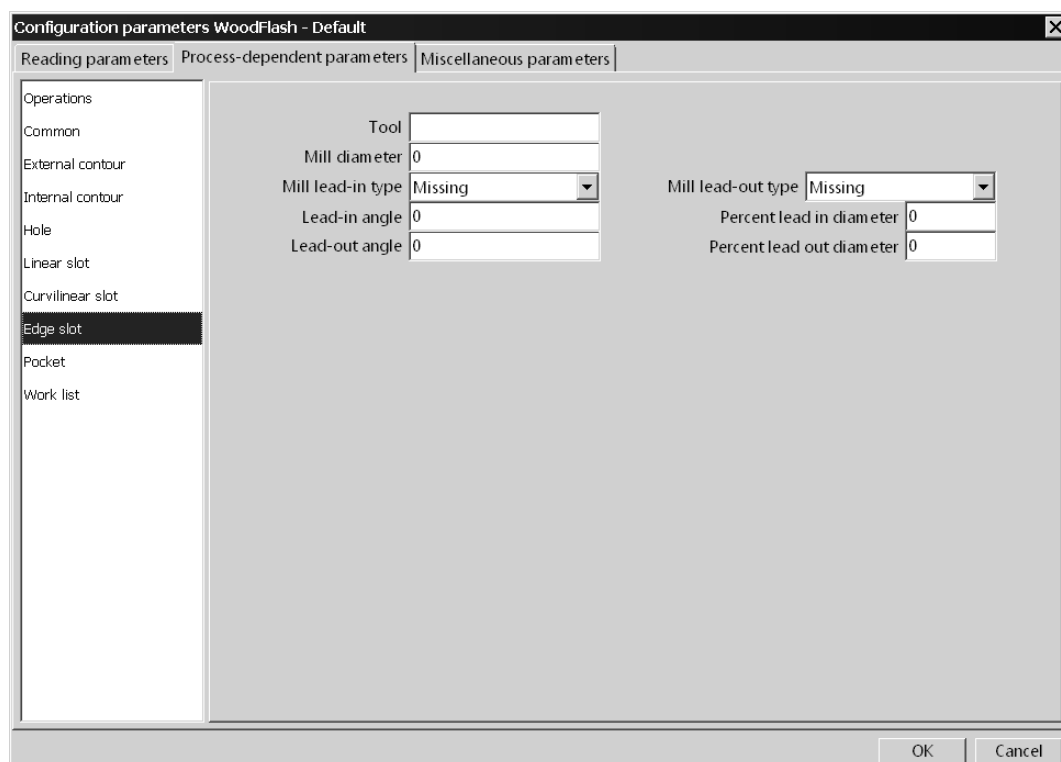


Fig. 10.223.

Section control elements allow to set control programs creation parameters for grooving.

Pocket parameters

Pocket section control elements (fig. 10.224) allow to set pocket processing parameters.

Fig. 10.224.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.9.2. Wave (*.)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.225) allow to specify the operations that can be used in control program.

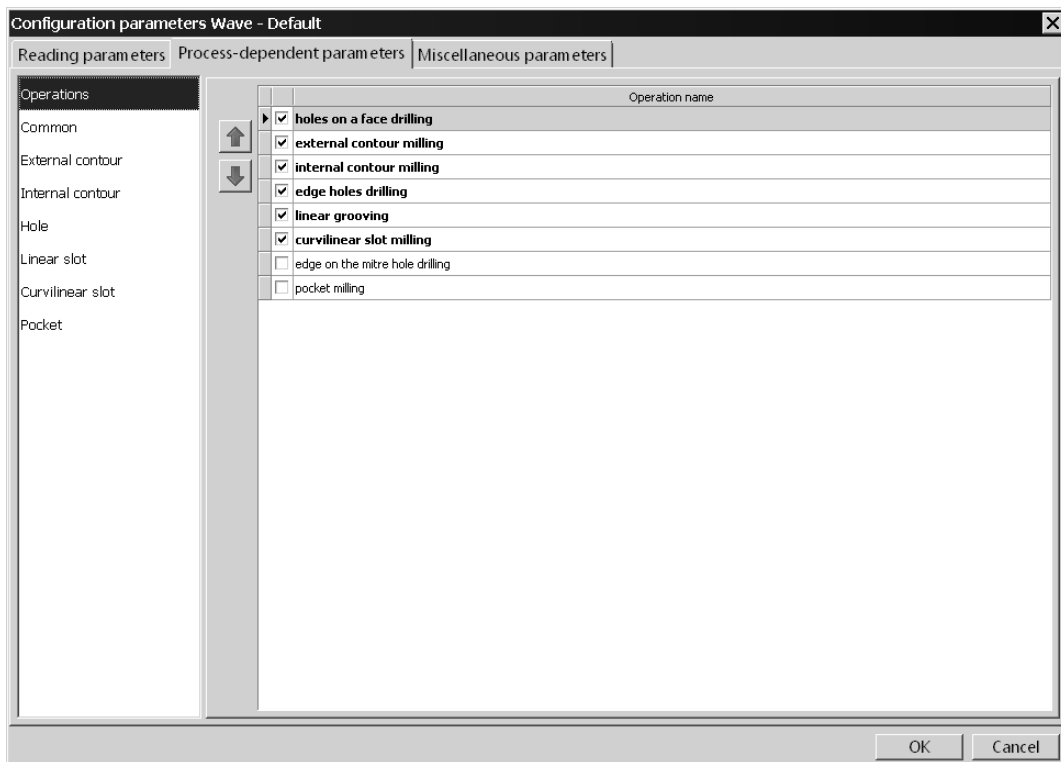


Fig. 10.225.

Common parameters

Common section control elements (fig. 10.226) allow to set control programs creation common parameters.

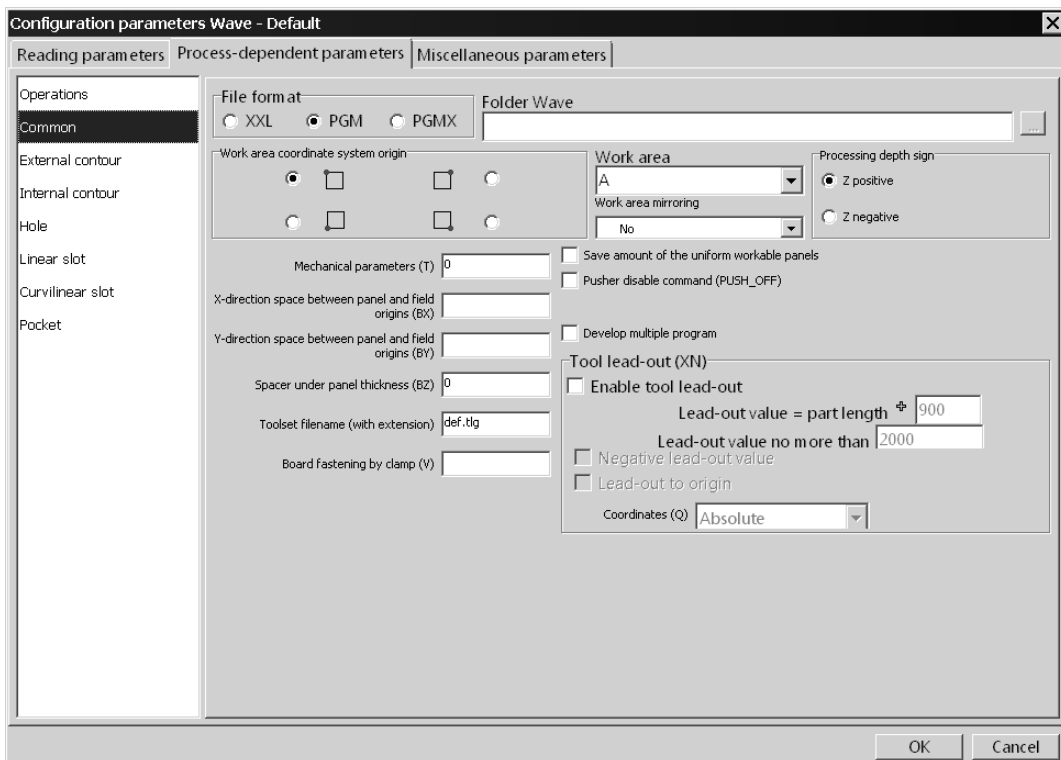


Fig. 10.226.



File format group variants allow to select control program saving format. To select a folder with saved Wave program, push **Browse** button and choose a folder in a standard **Browse** dialogue. **Work area coordinate system origin** group variants allow to select panel angle as origin of coordinates. **Work area** group variants allow to select machine work area.

Contours processing parameters

Outer and internal contours processing are set in **Outer contour** and **Internal contour** sections (fig. 10.227).

Rough and finish milling setup is executed separately on corresponding tabs. Tabs control elements content is the same and described for finish milling.

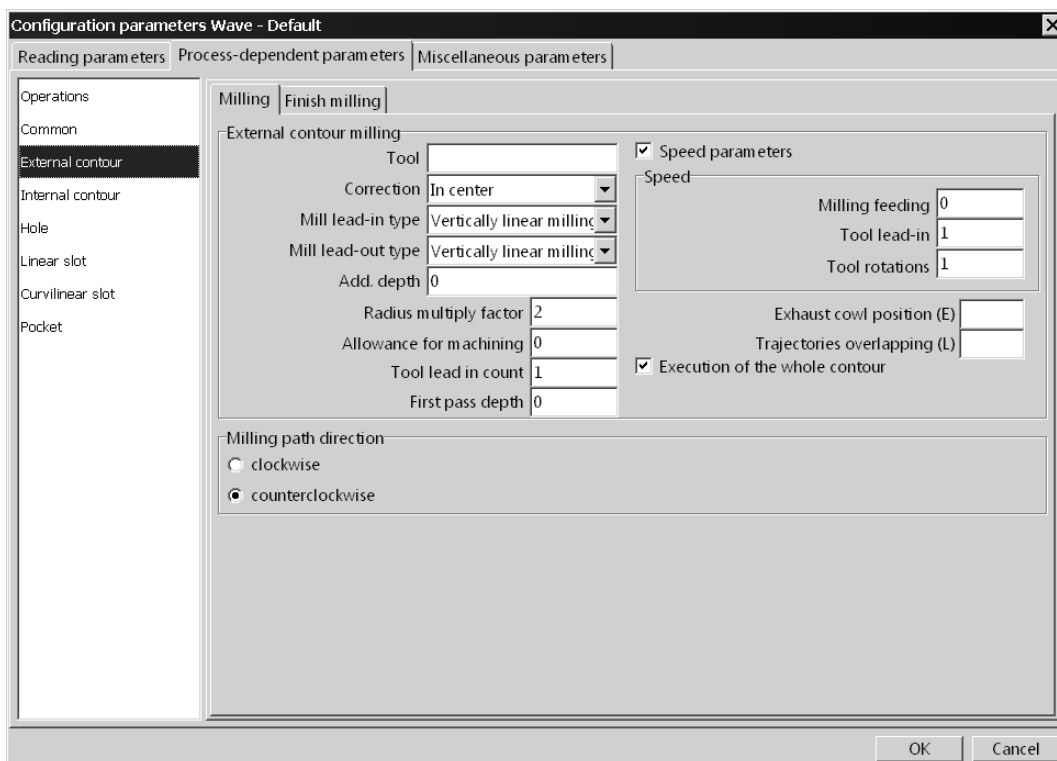


Fig. 10.227.

Finish milling option allows to control this operation. Tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section.

Processing using drill bit

Processing using drill bit (fig. 10.228) tab control elements allow to set holes boring parameters.

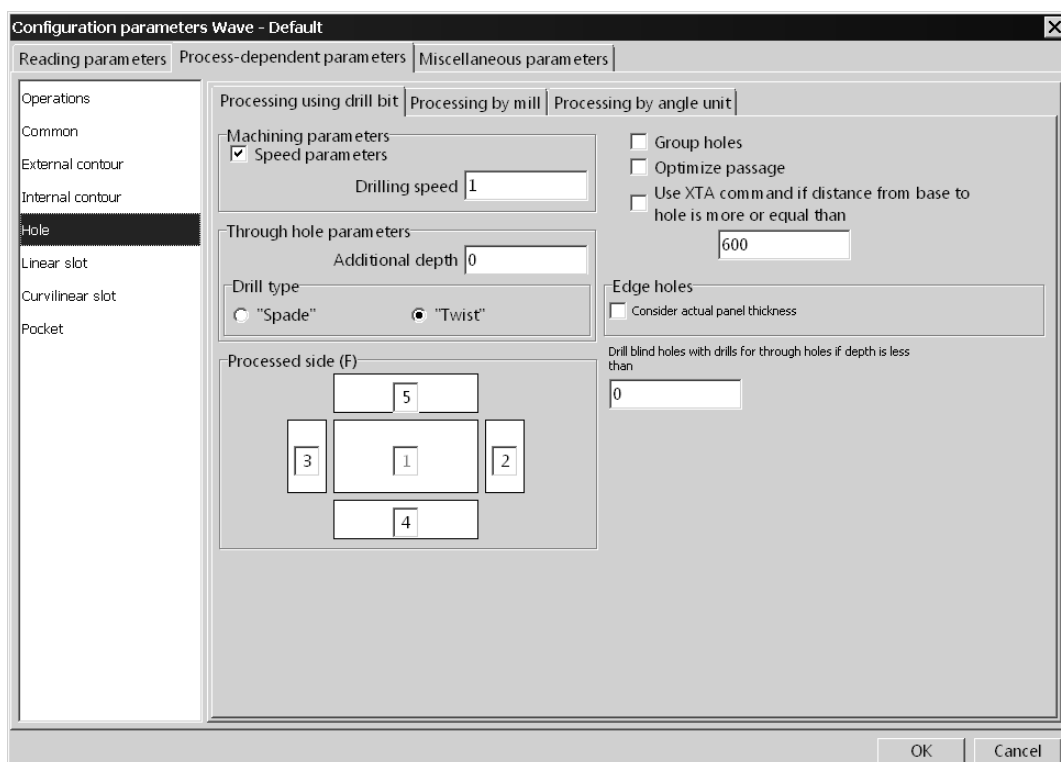


Fig. 10.228.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Control elements allow to set control programs creation parameters: speed parameters, borer type, processed side, etc.

Processing by mill

Processing by mill (fig. 10.229) tab control elements allow to set holes milling parameters.

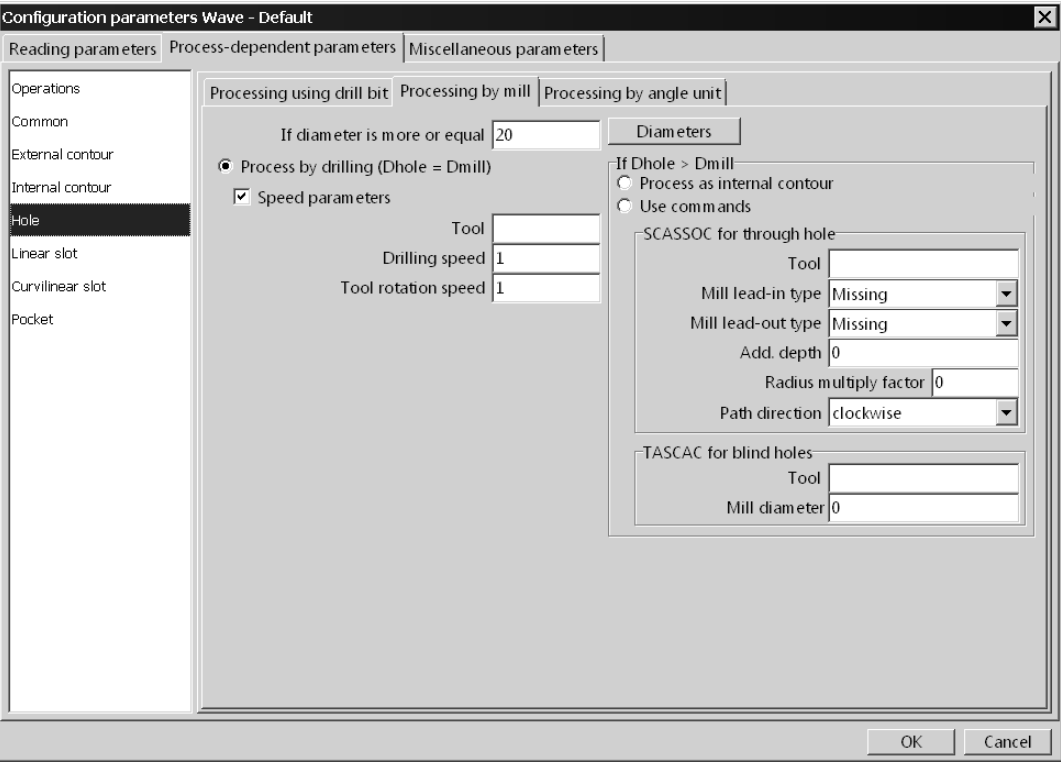


Fig. 10.229.

If **diameter is more or equal than** field allows to set hole diameter, if it exceeds, hole will be milled. To map hole diameter to a mill for its processing, push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.230).

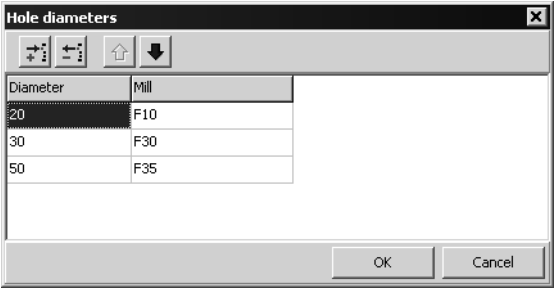


Fig. 10.230.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To finish setup push **OK** button. **Cancel** button allows to close the dialogue without changes. «SCASSOC» command can be used for through holes of great diameter milling. «TASCAC» program can be used for blind holes of great diameter milling. **SCASSOC for through hole** and **TASCAC for blind hole** groups control elements allow to set these commands use.

Processing by angle unit

Processing by angle unit (fig. 10.231) tab control elements allow to set angle unit parameters for edge holes processing.

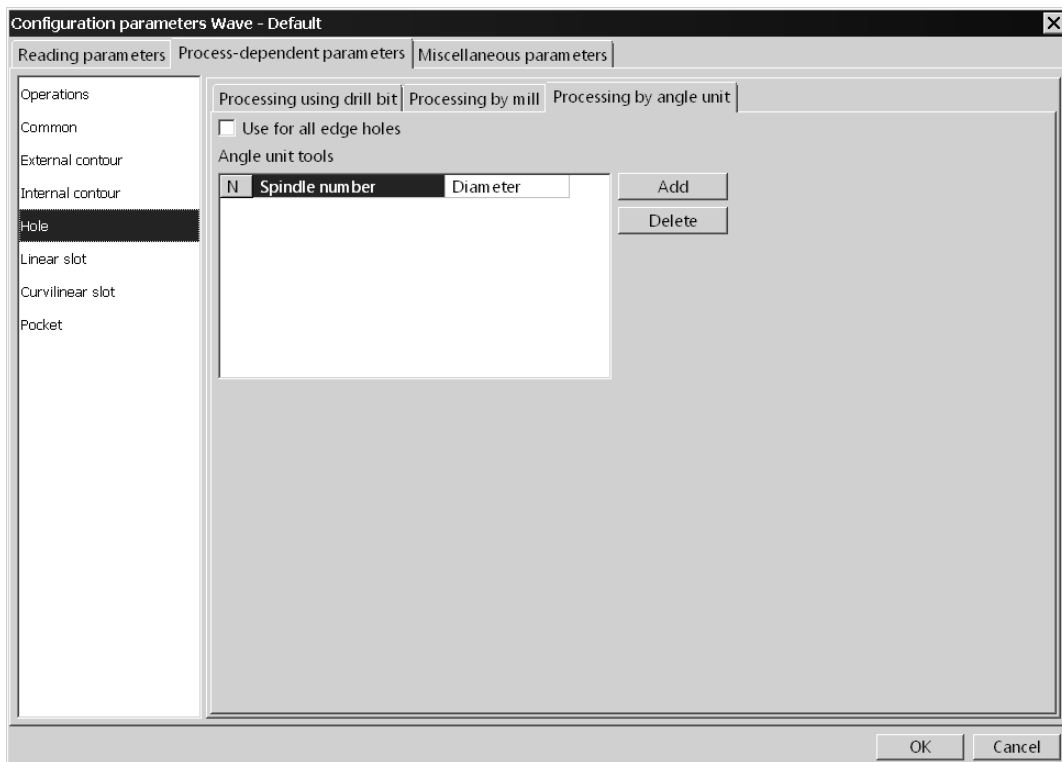


Fig. 10.231.

Use for all edge holes option allows to control unit use. **Angle unit tools** table fields allow to map holes diameter to spindle numbers. **Add** button allows to add a blank row to the table. To delete a row, select it in the table and push **Delete** button.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.232).

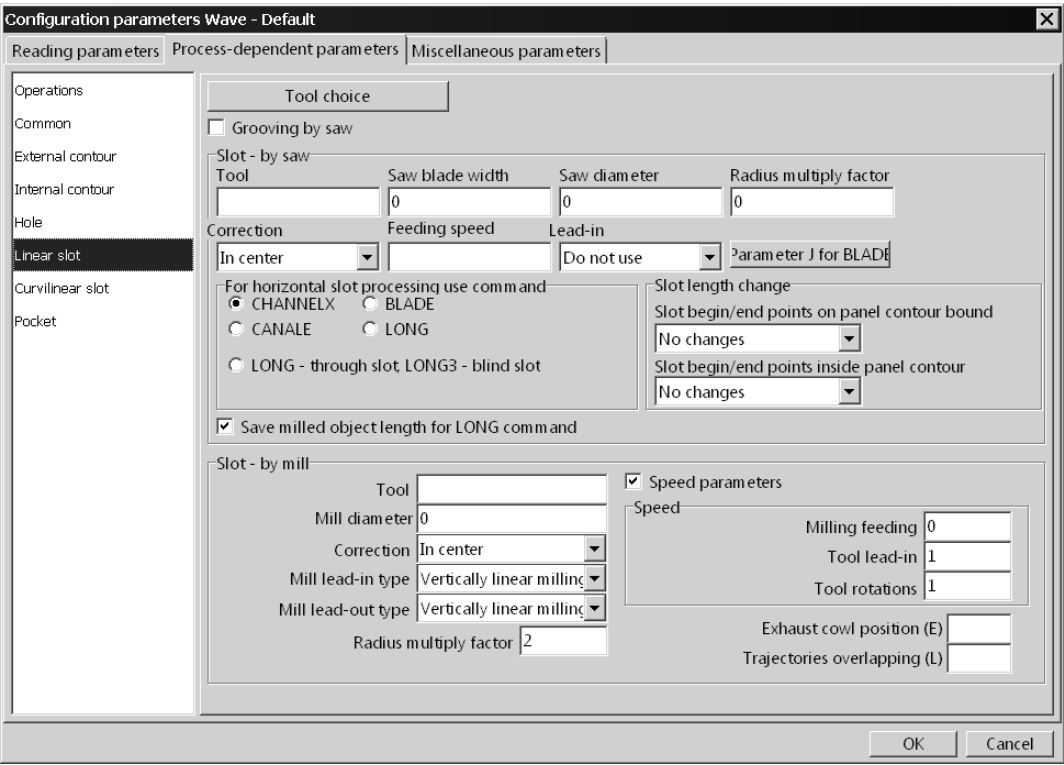


Fig. 10.232.

Tool choice button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.233).

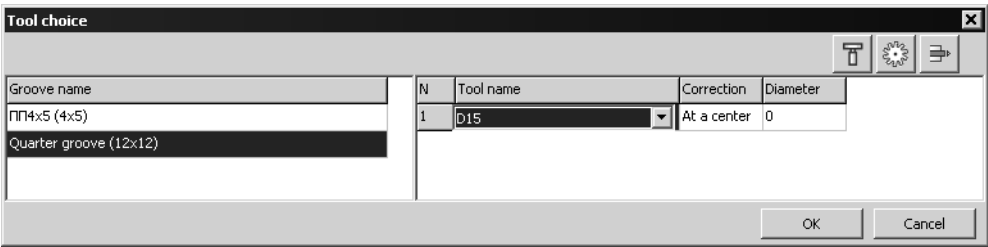


Fig. 10.233.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.234).

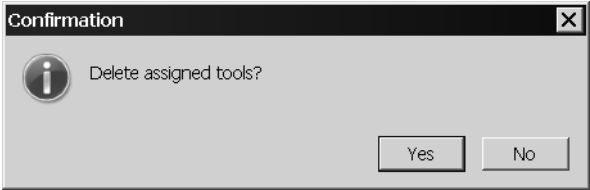


Fig. 10.234.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Parameter J for BLADE** button allows to set J parameter depending on slots length. After pushing the button **J parameter setup** dialogue will appear on the screen (fig. 10.235).

Groove length range (mm)		J value
from	to	

Fig. 10.235.

From and **to** table fields allow to set slot length bounds. Input range value parameter value in **J value** field. **Add** and **Delete** buttons allow to operate with table rows. To complete parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Grooving by saw group control elements allow to set control programs creation parameters for grooving by saw. **For horizontal slot processing use command** group variants allow to select corresponding commands for use in the program.

Grooving by mill parameters

Slot - by saw group control elements allow to set control programs creation parameters for grooving by saw.

Curvilinear slot parameters

Curvilinear slots parameters are set in **Curvilinear slot** section (fig. 10.236).

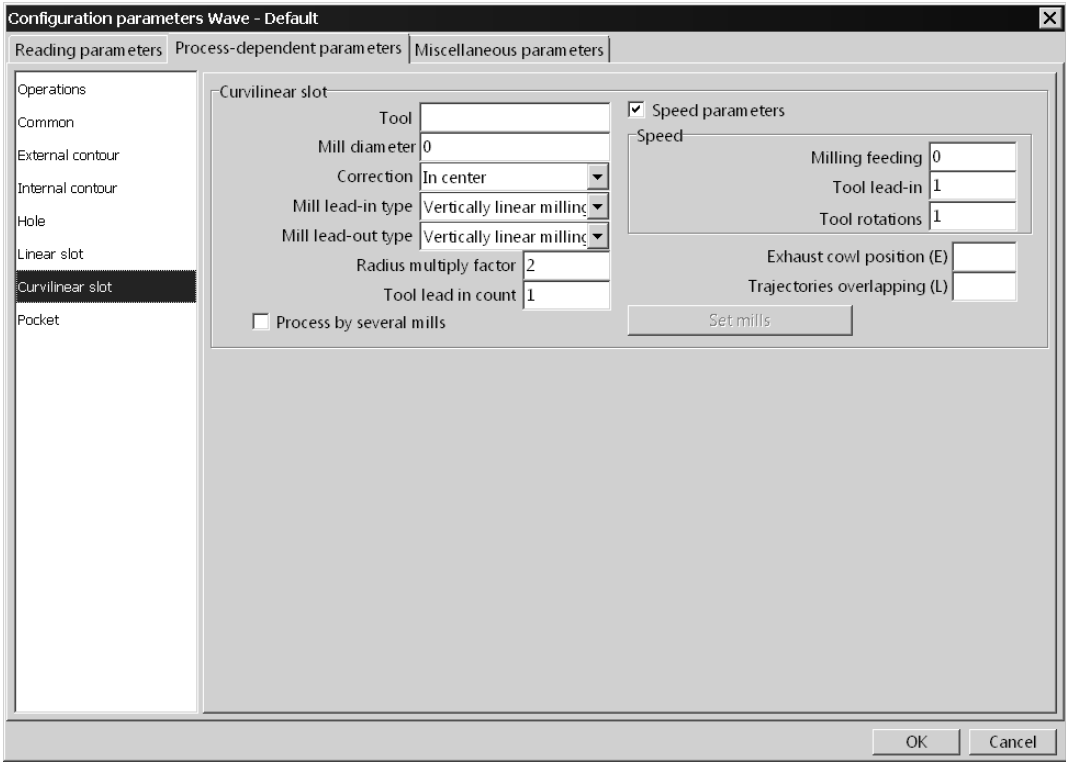


Fig. 10.236.

Control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.237).

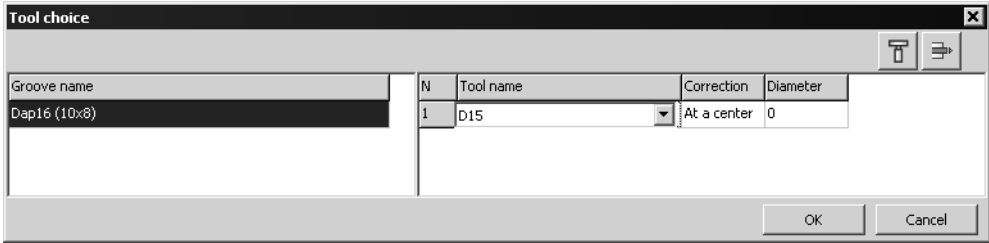


Fig. 10.237.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Correction type should be selected from a combo box. Mill name and its diameter can be input in list fields.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.238) allow to set pocket processing parameters.

Fig. 10.238.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.10. CAD System

10.10.1.CAD System



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.239) allow to specify operations that can be used in control program.

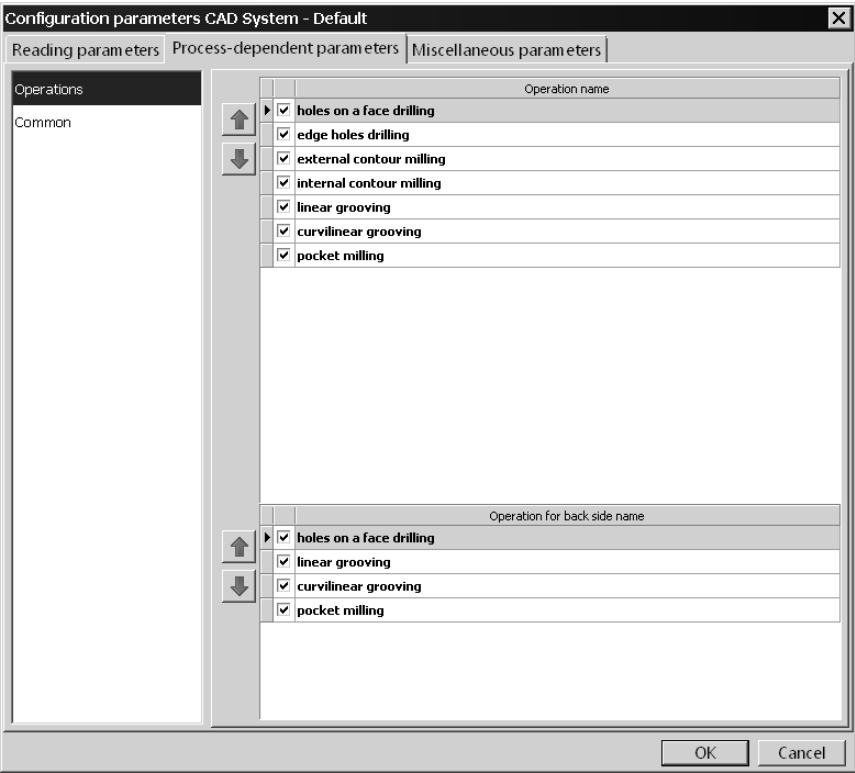


Fig. 10.239.

Common parameters

Common section control elements (fig. 10.240) allow to setup common parameters of control programs creation.

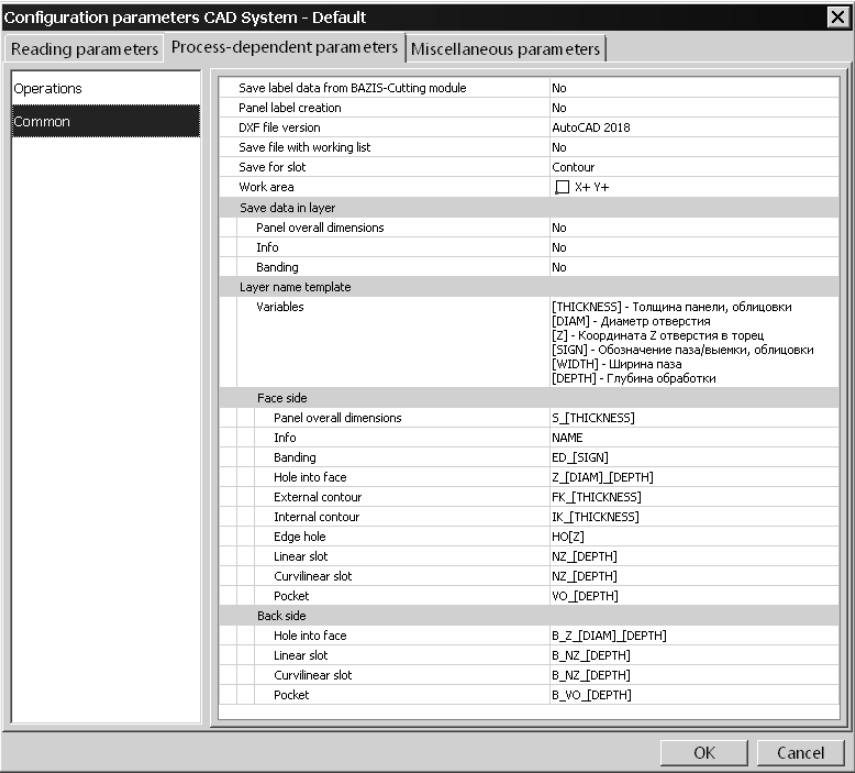


Fig. 10.240.

Setup completion

To complete control program parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.11. Felder Group

10.11.1.F4Integrate



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.241) allow to specify operations that can be used in control program.

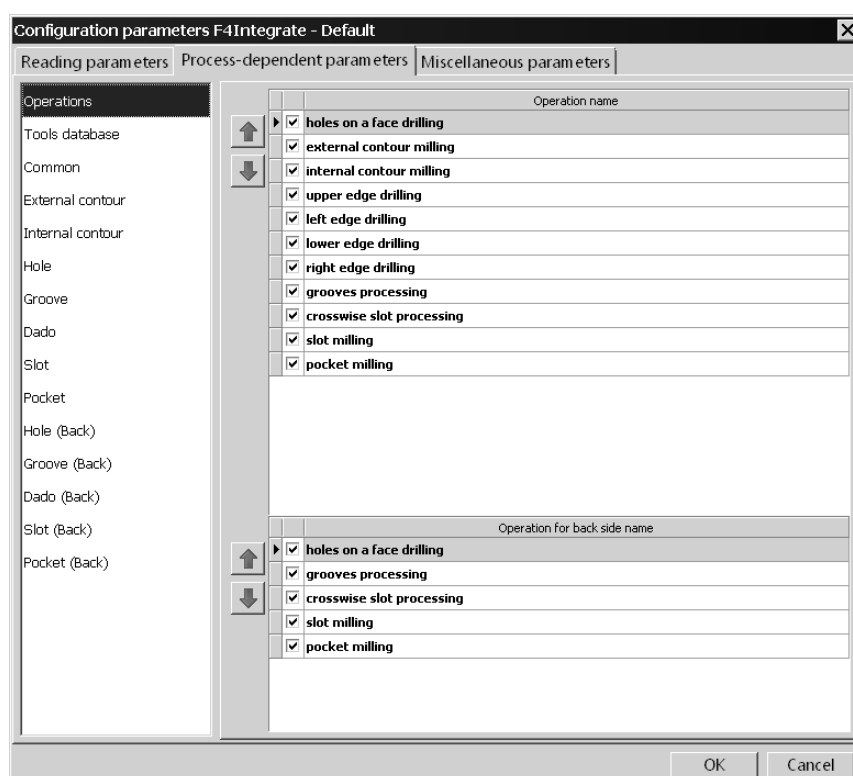


Fig. 10.241.

Usable tools

Tools database section control elements (fig. 10.242) allow to create a list of tools available for use in control programs.

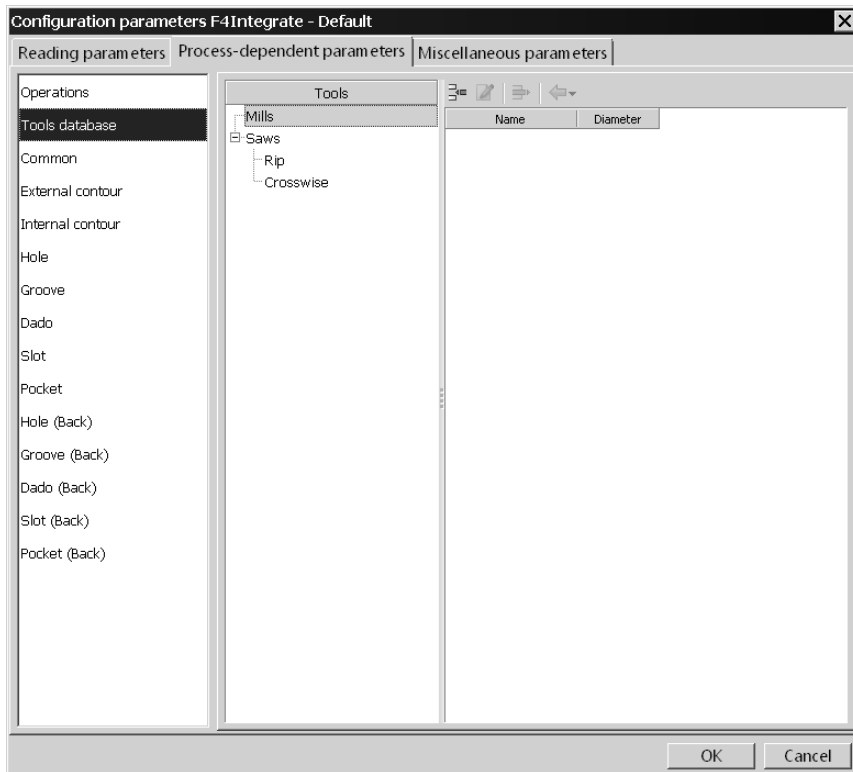


Fig. 10.242.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.243).

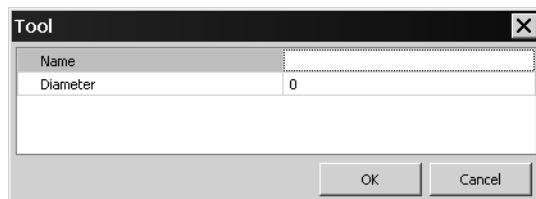


Fig. 10.243.



In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common parameters are set in **Common** section (fig. 10.244).

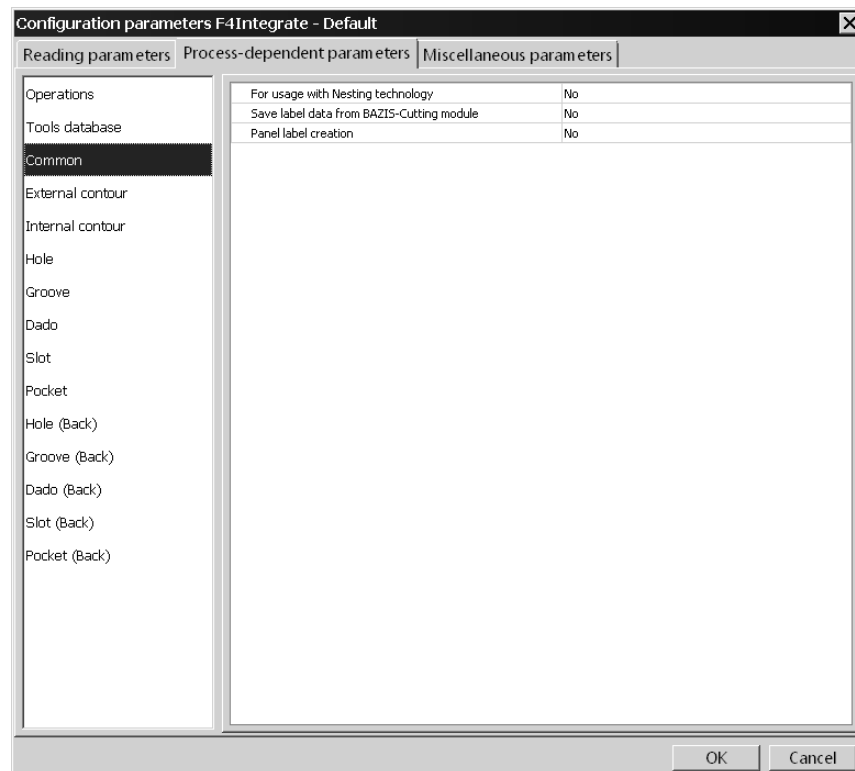


Fig. 10.244.

For usage with Nesting technology, Save label data from BAZIS-Cutting, Panel label creation combo boxes variants allow to control the corresponding parameters.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.245).

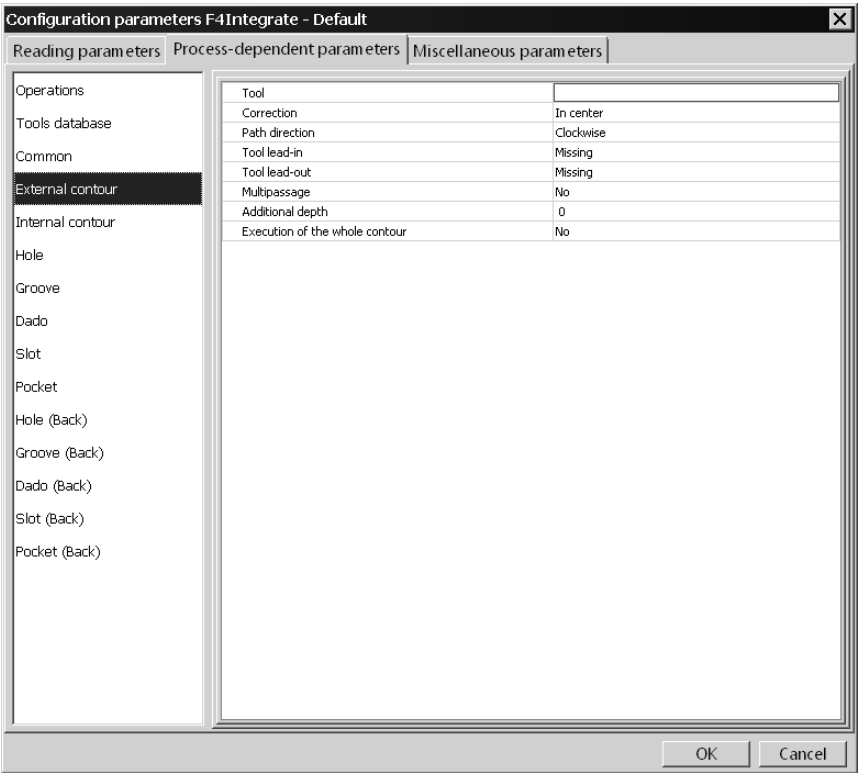


Fig. 10.245.

Control elements in sections allow to select the tool, correction types, lead-in, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.246), holes from back side – in **Hole (Back)** section in the same way.

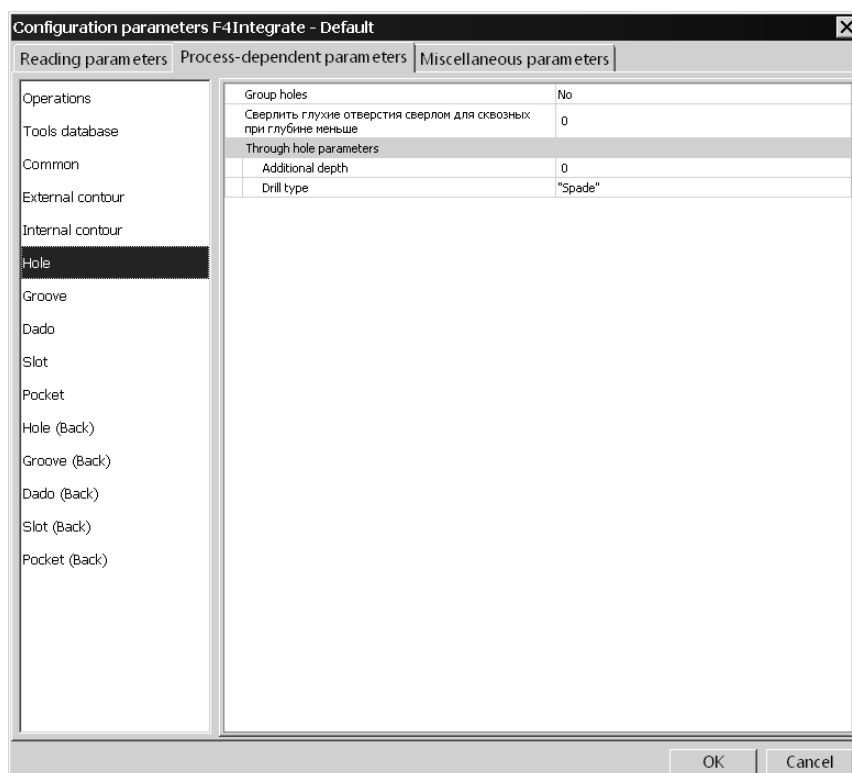


Fig. 10.246.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.247), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

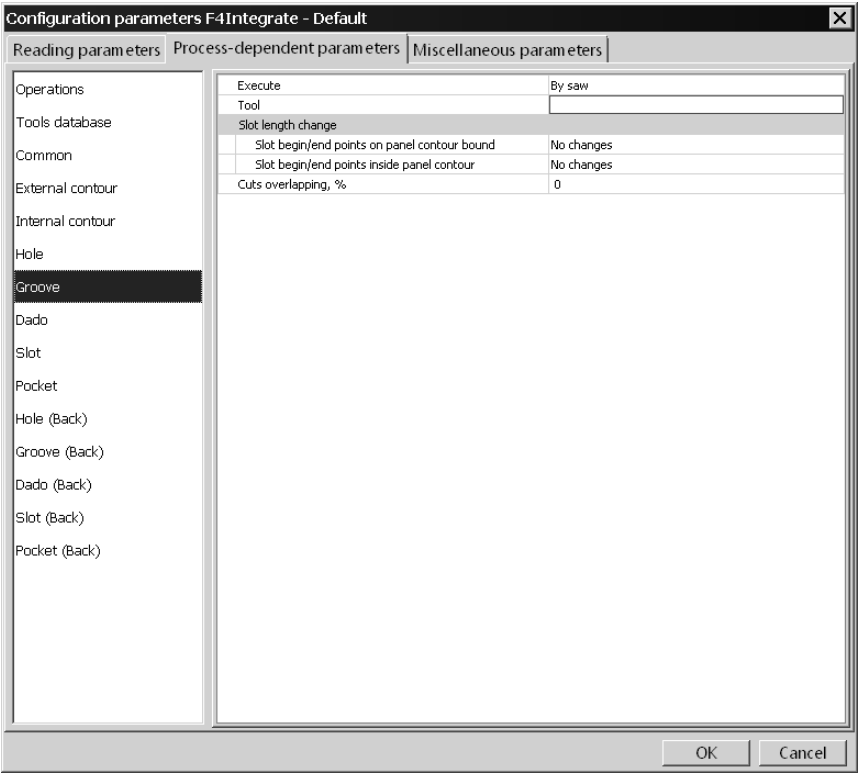


Fig. 10.247.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.248), from the back panel side – in **Slot (Back)** section.

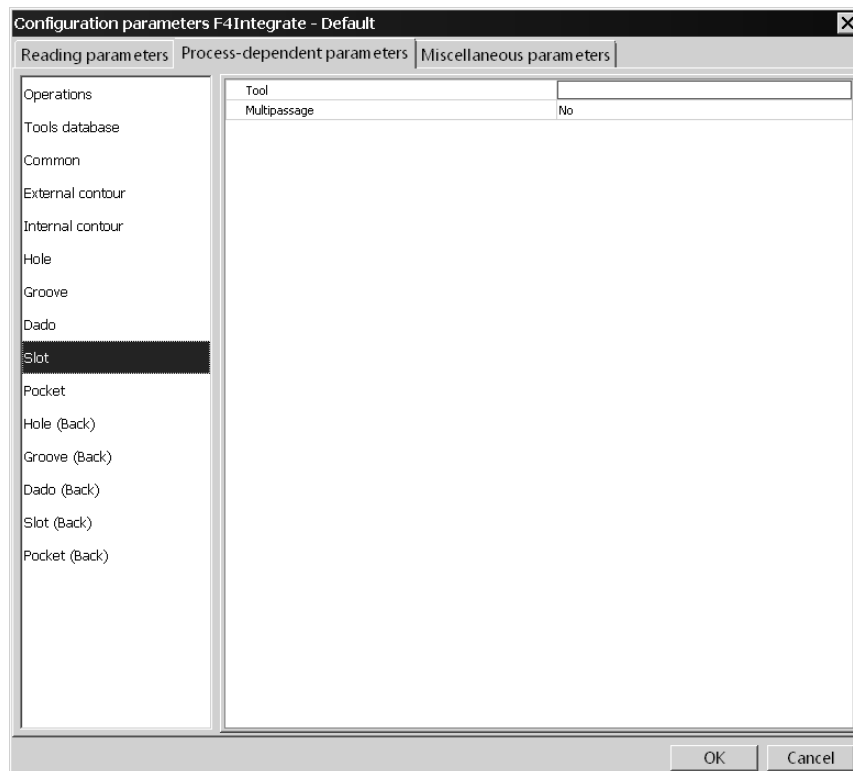


Fig. 10.248.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.249), on the back panel side – in **Pocket (Back)** section.

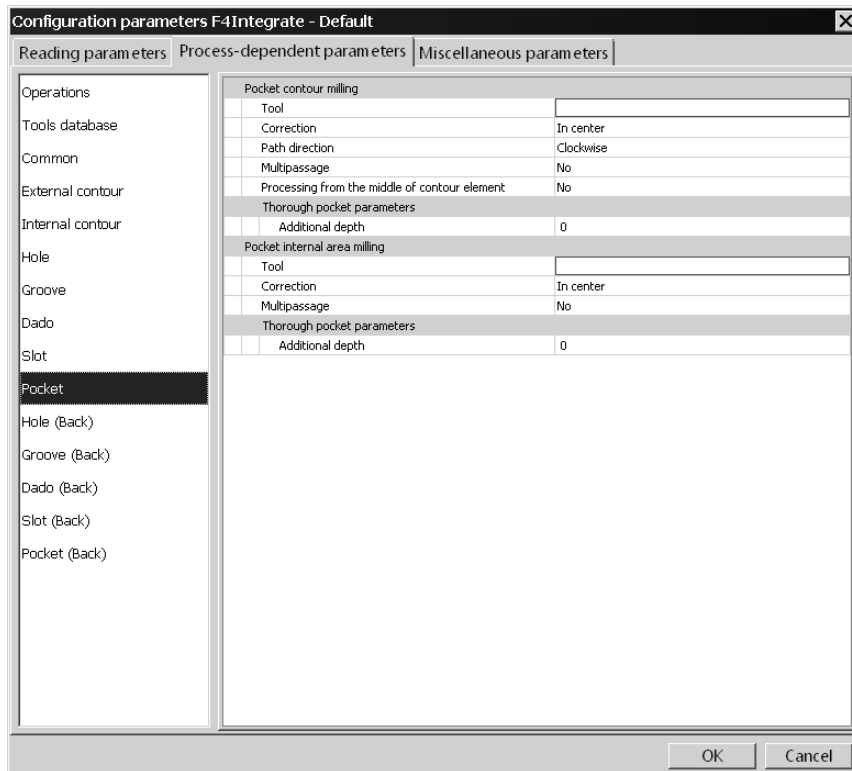


Fig. 10.249.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects from database. For pocket contour select milling execution direction, correction type and multipassage use. For internal pocket part – correction type and multipassage use.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.11.2.WoodFlash



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.250) allow to specify operations that can be used in control program.

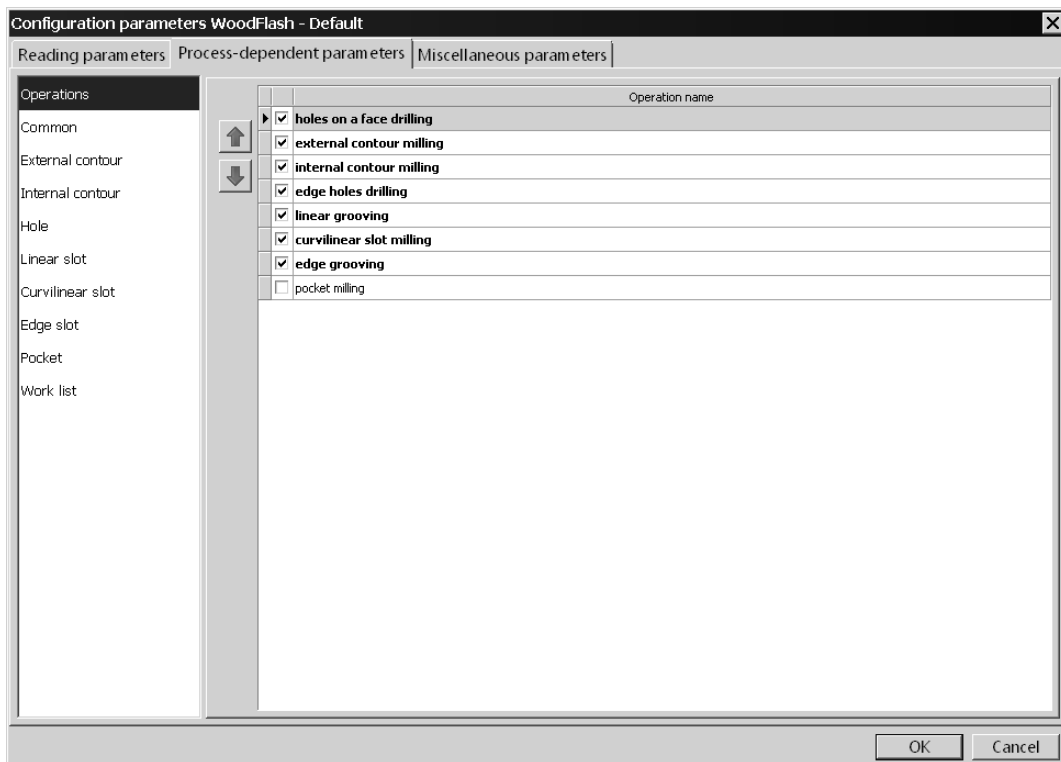


Fig. 10.250.

Common parameters

Common parameters are set in **Common** section (fig. 10.251).

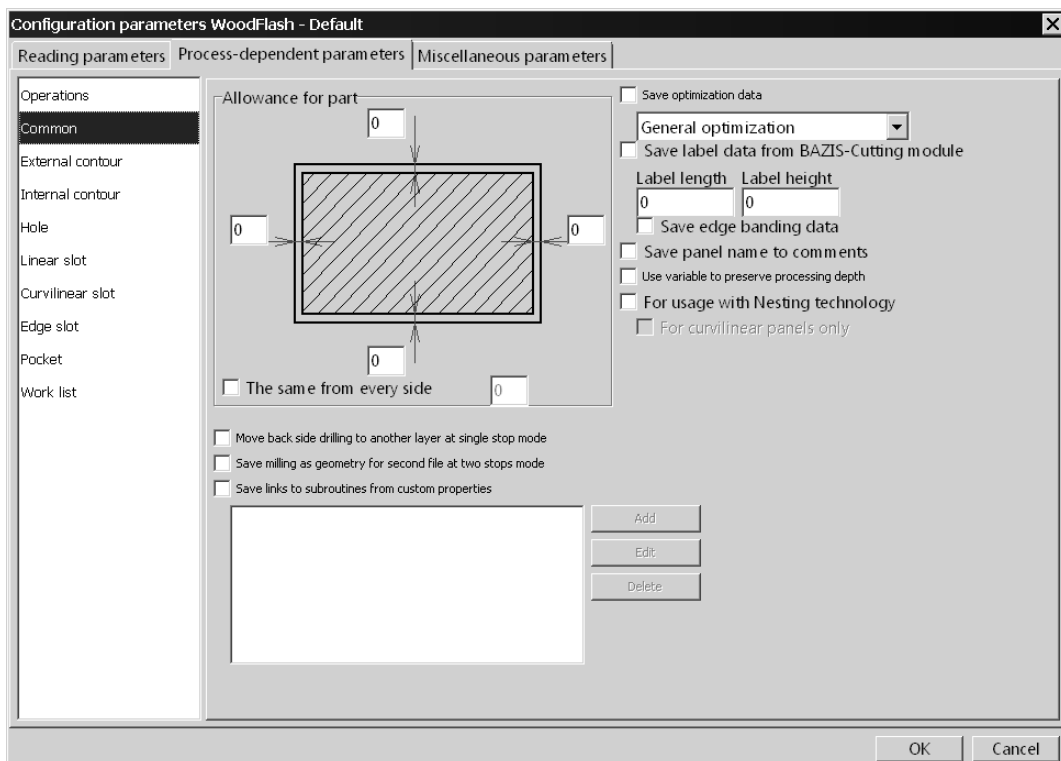


Fig. 10.251.

Allowance for part section control elements allow to set allowance values. Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, the same from every panel edge allowance field will become available. **Append operations to CNC program** group options allow to control the inclusion in control program of operations. To include the operation to the program, enable an option next to its name.

Contours parameters

Contours milling parameters are set in **Outer contour** (fig. 10.252) and **Internal contour** sections of **Process-dependent parameters** tab.

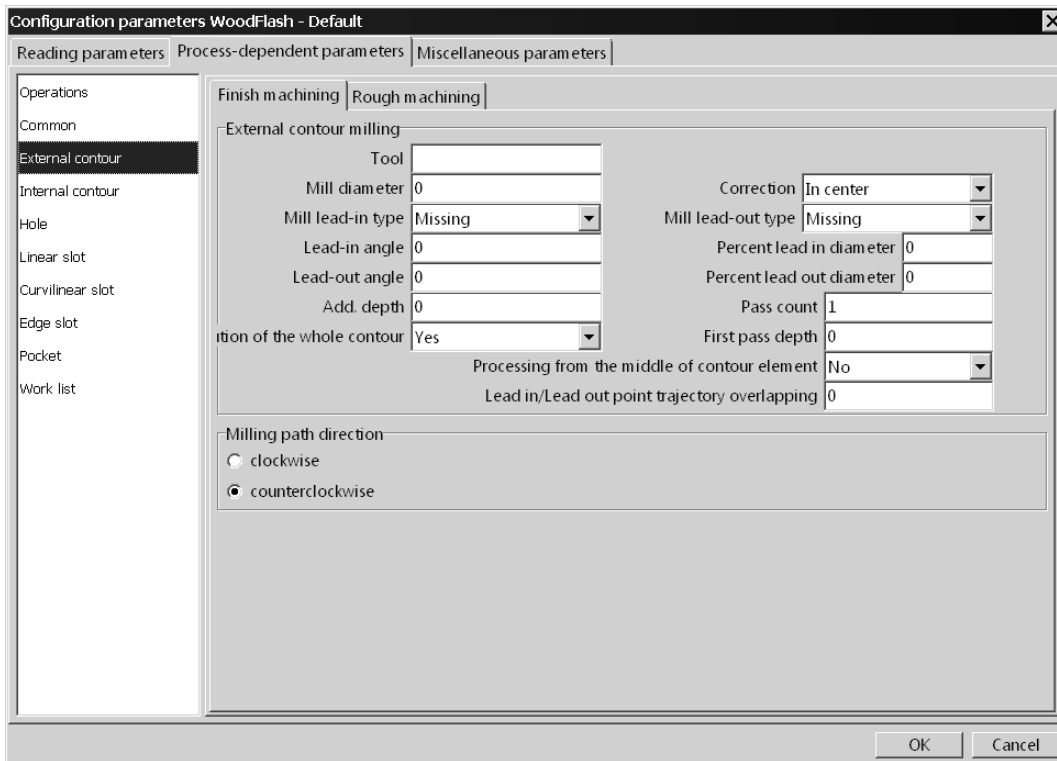


Fig. 10.252.

Finish machining and **Rough machining** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Rough machining** option allows to control this operation. Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Hole section control elements allow to set holes parameters (fig. 10.253).

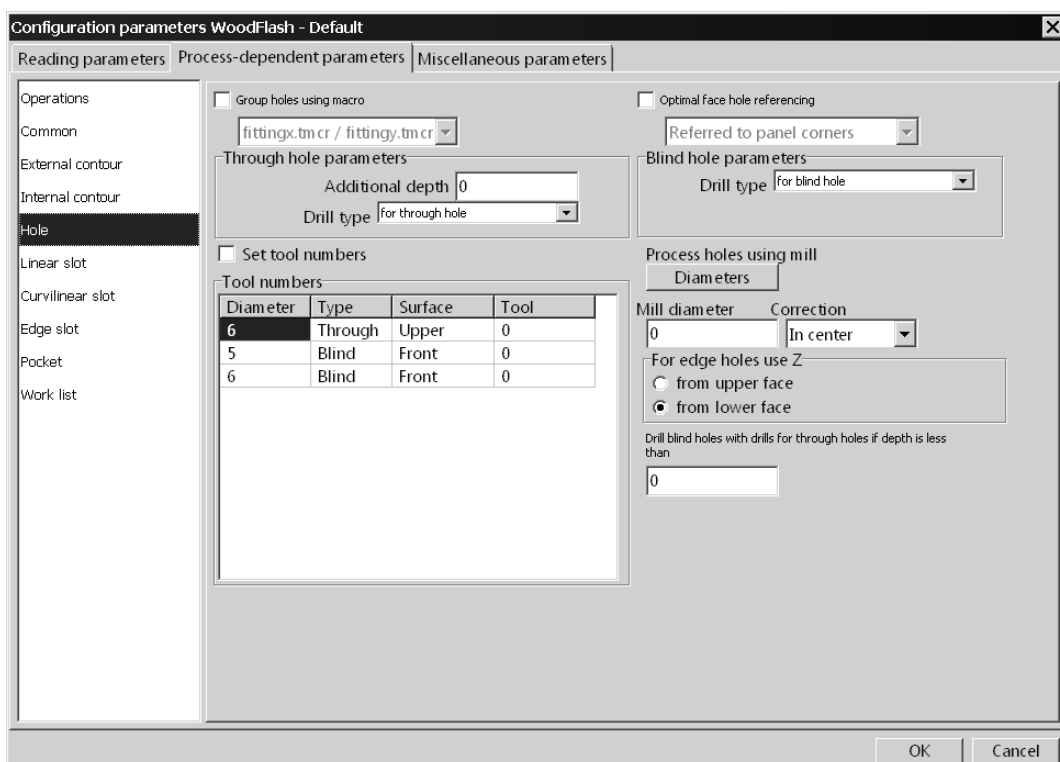


Fig. 10.253.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Through holes can be bored with drills for through and blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.254).

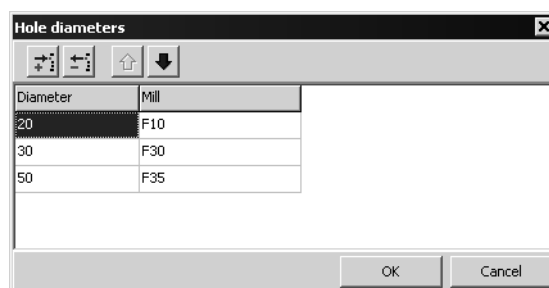


Fig. 10.254.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To finish setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter application.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.255).

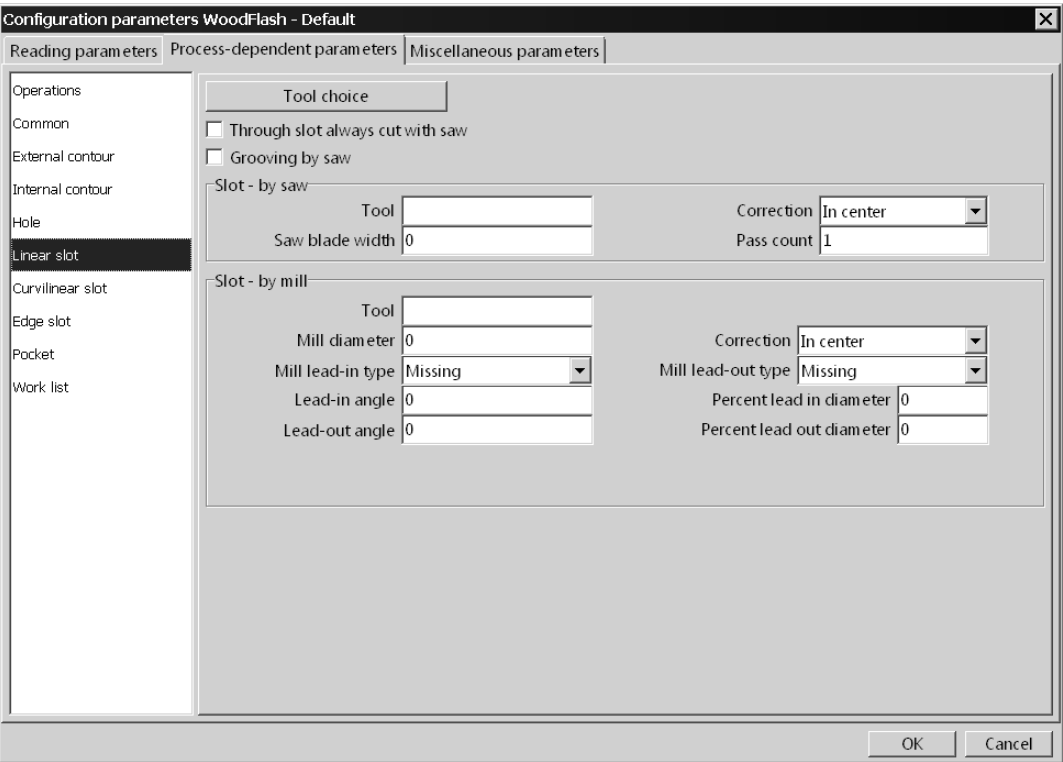


Fig. 10.255.

Through slot always cut with saw and **Grooving by saw** options allow to set through slots direction that will be processed with a saw. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.256).

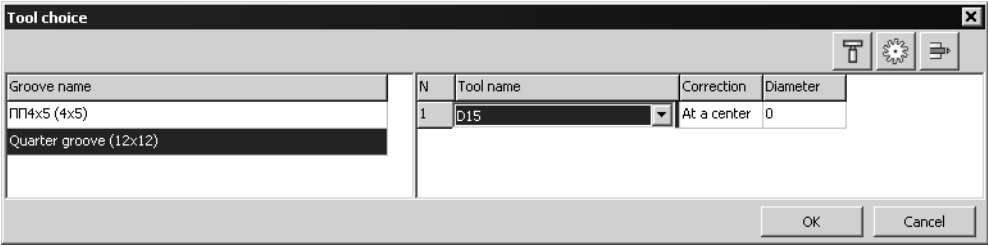


Fig. 10.256.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing.

The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.257).

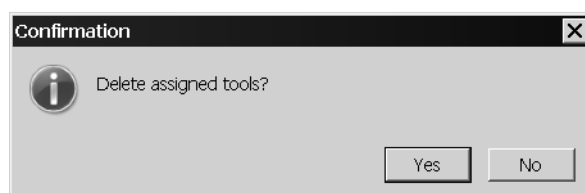


Fig. 10.257.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. You can map slot width and mill parameters that will be automatically selected for its processing. For this purpose push **Mill parameters** button. **Mill parameters** dialogue will appear on the screen (fig. 10.258).

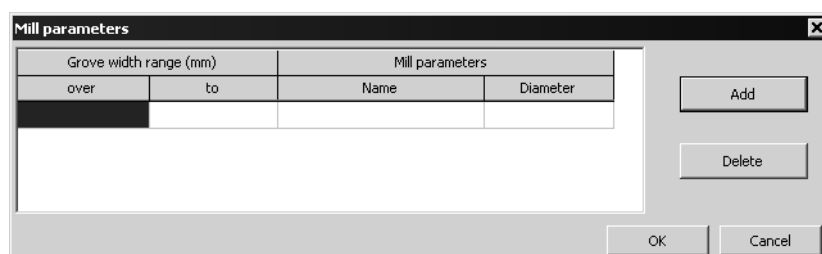


Fig. 10.258.

Dialogue table contains slot width ranges and corresponding mill parameters. The fields should be filled via a keyboard. **Add** button allows to add a blank row to the table. To delete the current row push **Delete** button. To finish parameters setup push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.259).

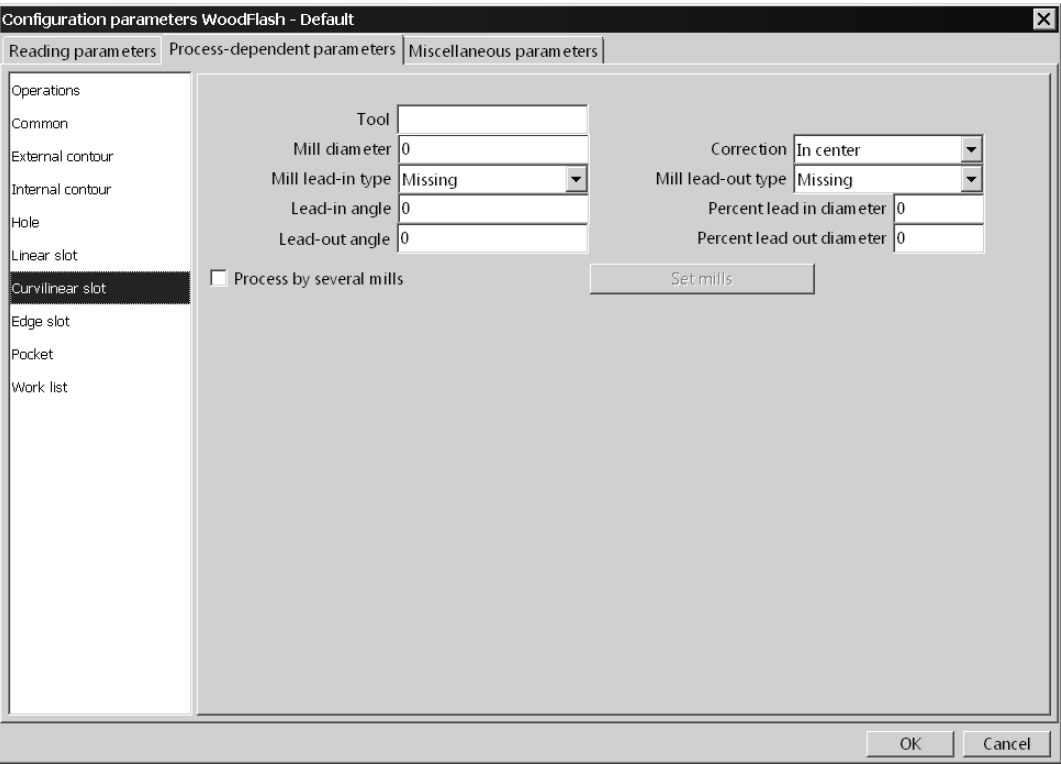


Fig. 10.259.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.260).

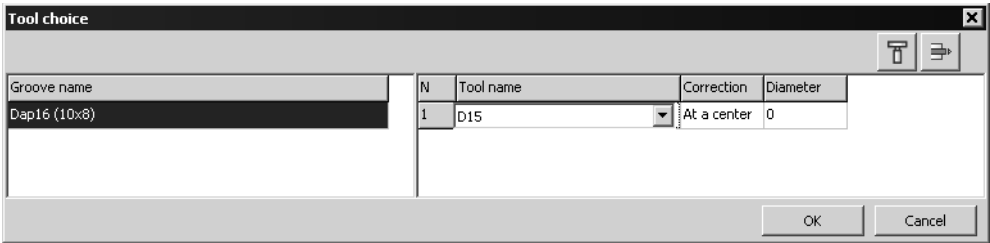


Fig. 10.260.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.261).

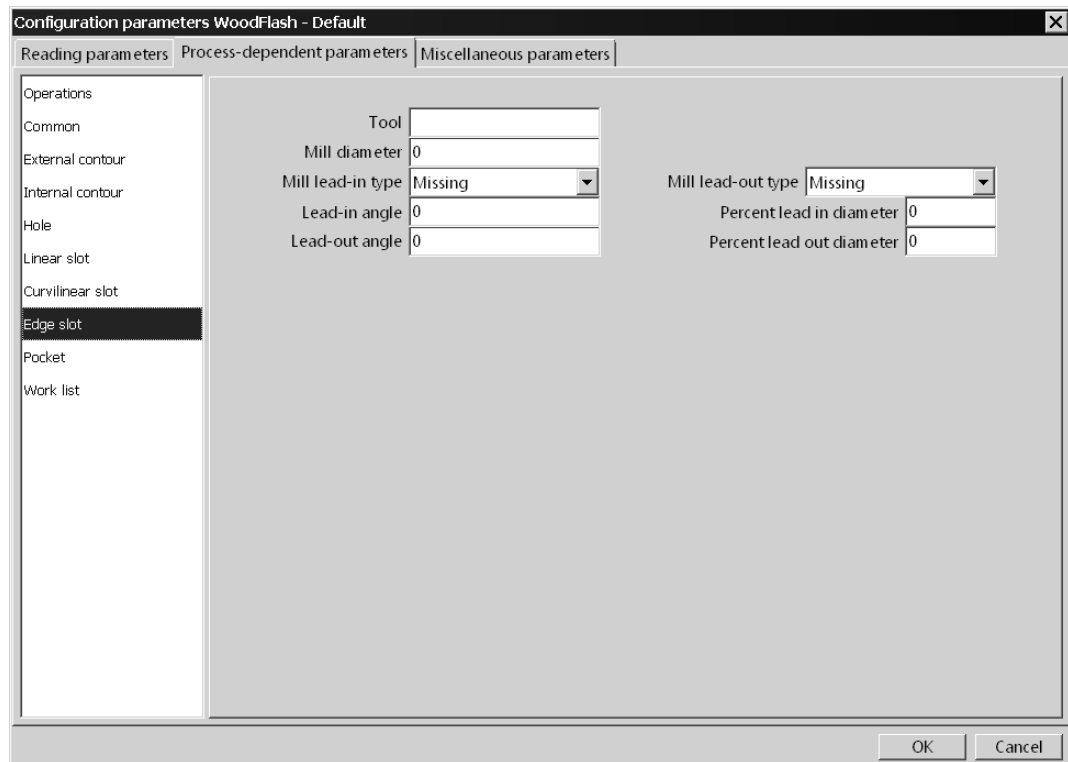


Fig. 10.261.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.262) allow to set pocket processing parameters.

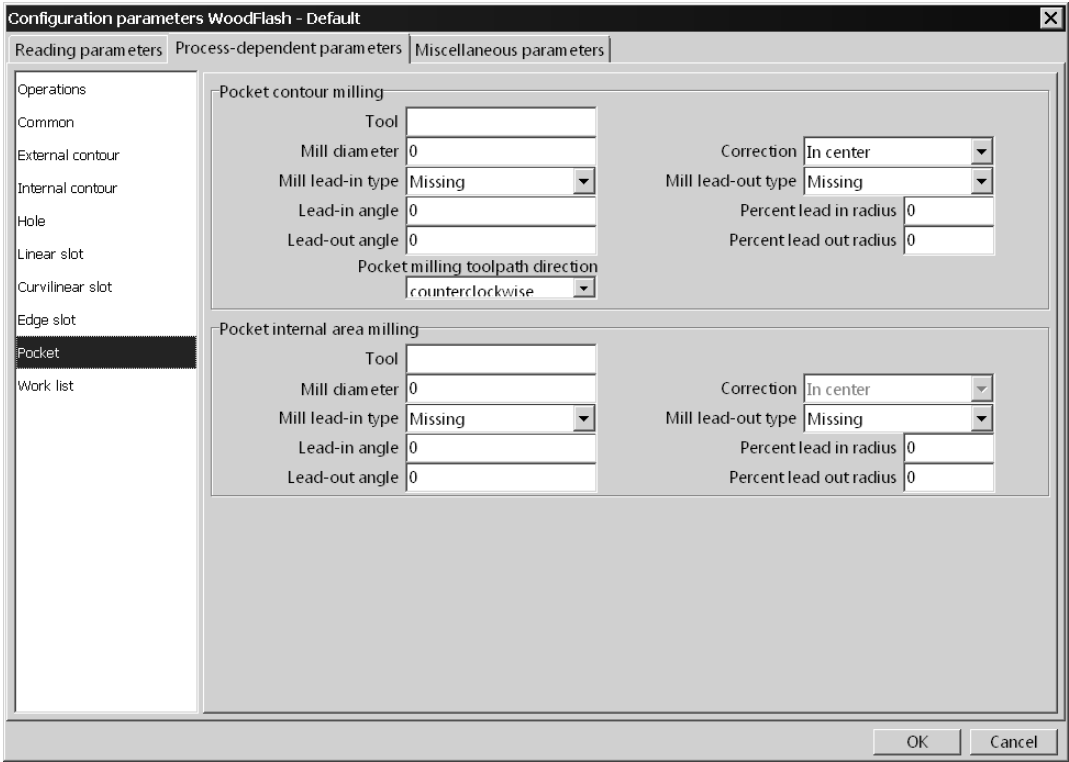


Fig. 10.262.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Work list

Work list can be set in **Work list** section (fig. 10.263).

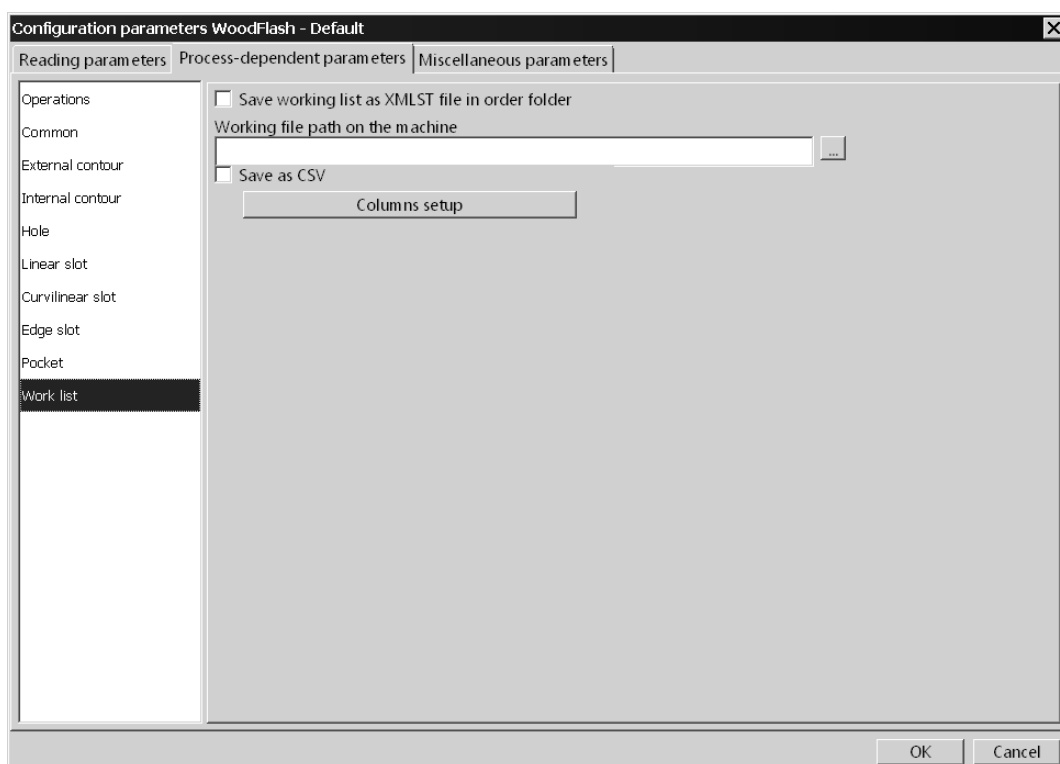


Fig. 10.263.



Save work list file as DST format option allows to control this file saving. To specify the saving folder, push **Browse** button and select it in a standard Windows dialogue. **Save as CSV** option allows to control file saving. **Columns setup** button allows to setup file table columns. After pushing the button, **Table columns setup** dialogue (fig. 10.264).

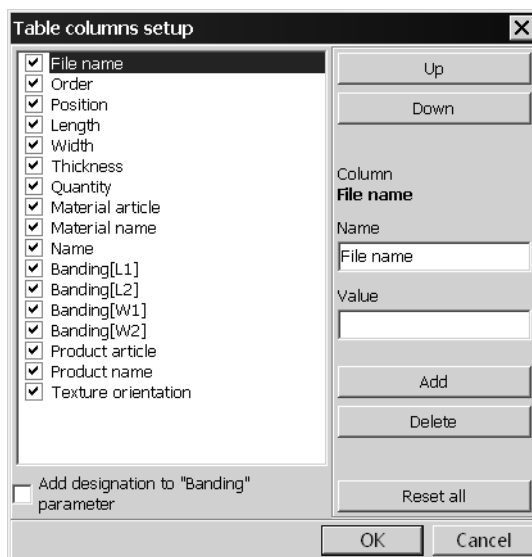


Fig. 10.264.

To add data from a column to a file, enable the option for it. **Up** and **Down** buttons allow to move the selected column through the list. **Add** button allows to add a column to the list. In **Name** field, you can change the name of the current column. For columns added by the user, you can set a value in the appropriate field. **Delete** but-

ton allows to delete a column added by the user. After pushing the button, a warning will appear on the screen (fig. 10.265).

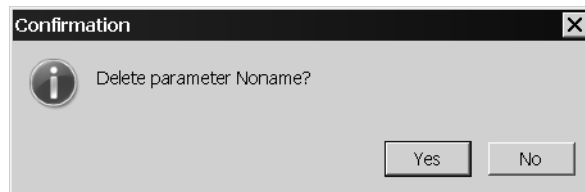


Fig. 10.265.

Message buttons allow to confirm or reject the deletion.

Reset all button allows to return the table settings to their default values. After pushing the button, a warning will appear on the screen (fig. 10.266).

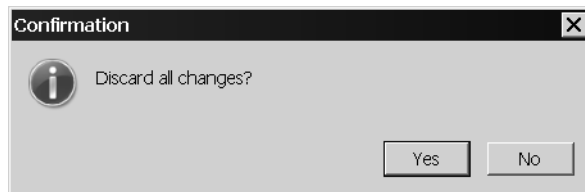


Fig. 10.266.

The message buttons allow to confirm or cancel the reset of parameters. To complete the table setup, push **OK** button.

Setup completion

To complete control programs for WoodFlash setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.12. Filato

10.12.1.ARIX-OpenCNC (*.tnc, *.tn2)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.267) allow to set common parameters of control programs creation.

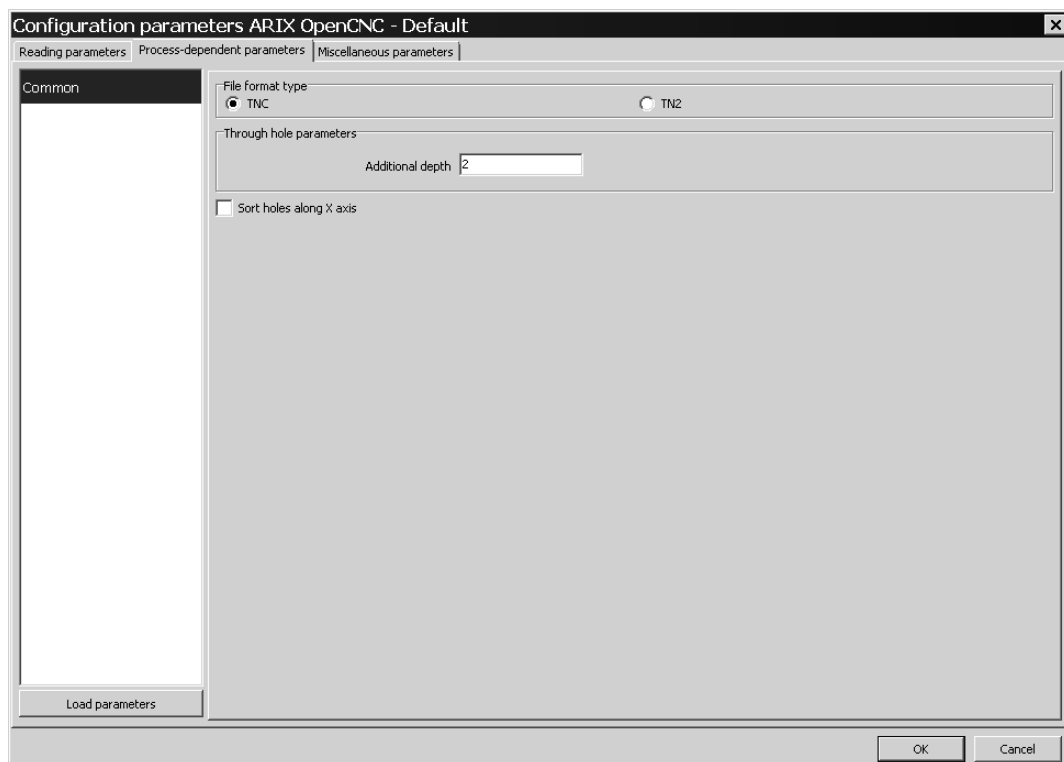


Fig. 10.267.

TN2 and **TNC** variants of **File format type** group allow to select output file format. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **Sort holes along X axis** option allows to sort holes drilling. If it is enabled, the holes will be drilled one by one depending on OX axis coordinate.

Setup completion

To complete ARIX-OpenCNC control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.12.2.Filato DS3



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.268) allow to specify operations that can be used in control program.

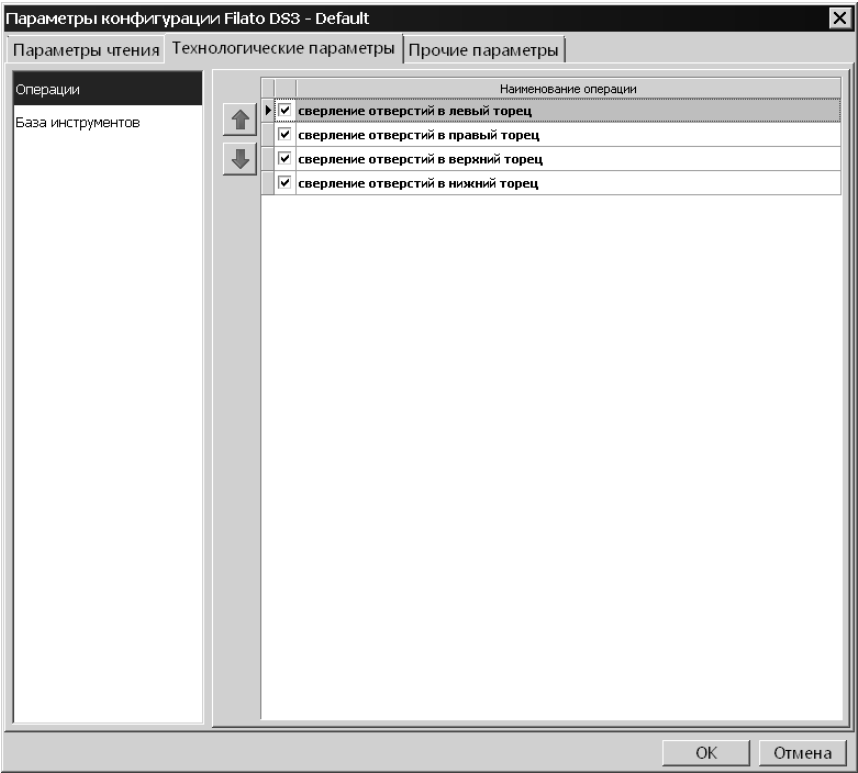


Fig. 10.268.

Usable tools

Tools database section control elements (fig. 10.269) allow to create a list of tools available for use in control programs.

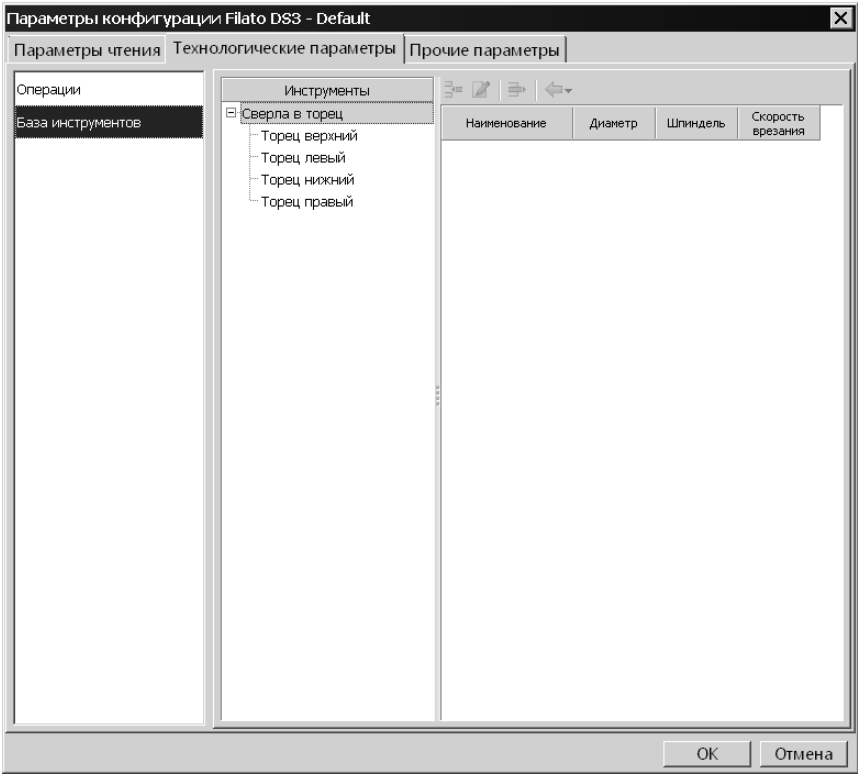


Fig. 10.269.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.270).

Tool	
Name	
Diameter	0
Spindle	0

OK Cancel

Fig. 10.270.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.



Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Create** group options allow to control the creation of control programs files that correspond to options names.

10.12.3.Filato Next



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Hole parameters

Holes processing is set in **Common** section (fig. 10.271).

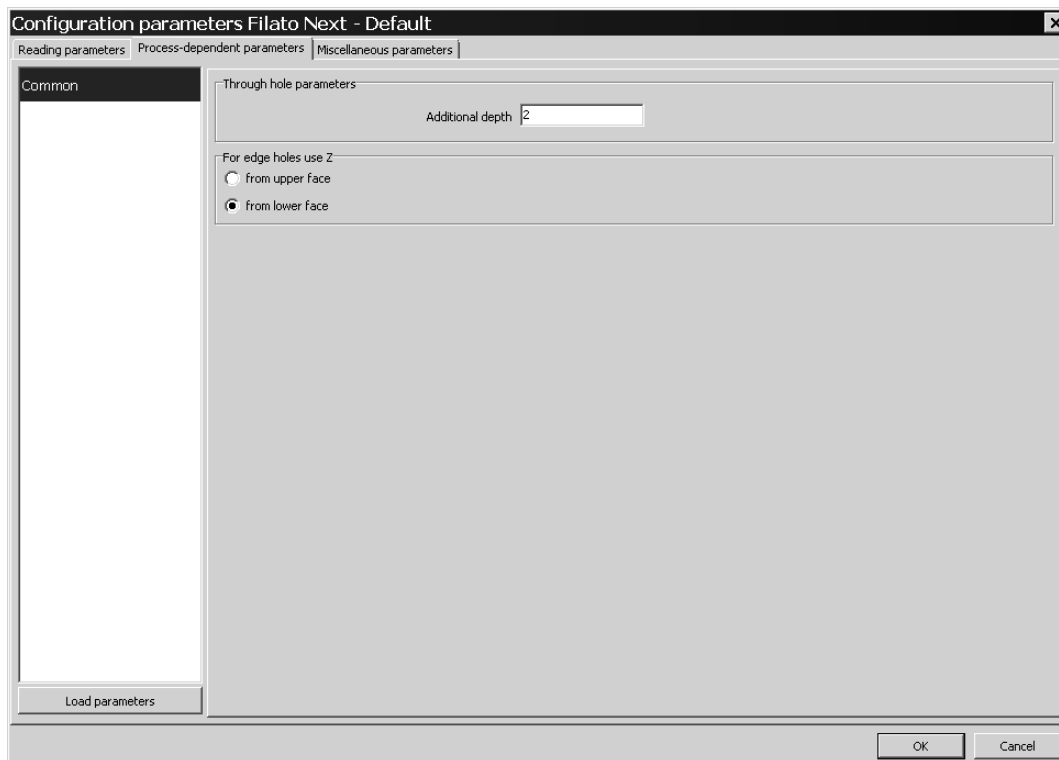


Fig. 10.271.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **For edge holes use Z** group variants allow to select the face for Z value setting.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.12.4.Higerman



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.272) allow to specify operations that can be used in control program.

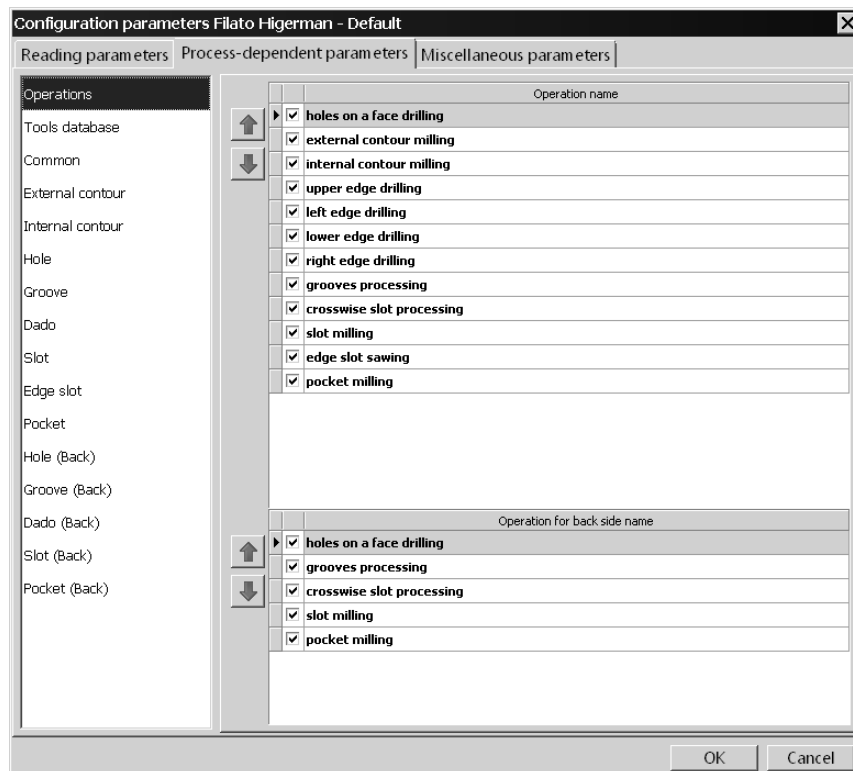


Fig. 10.272.

Usable tools

Tools database section control elements (fig. 10.273) allow to create a list of tools available for use in control programs.

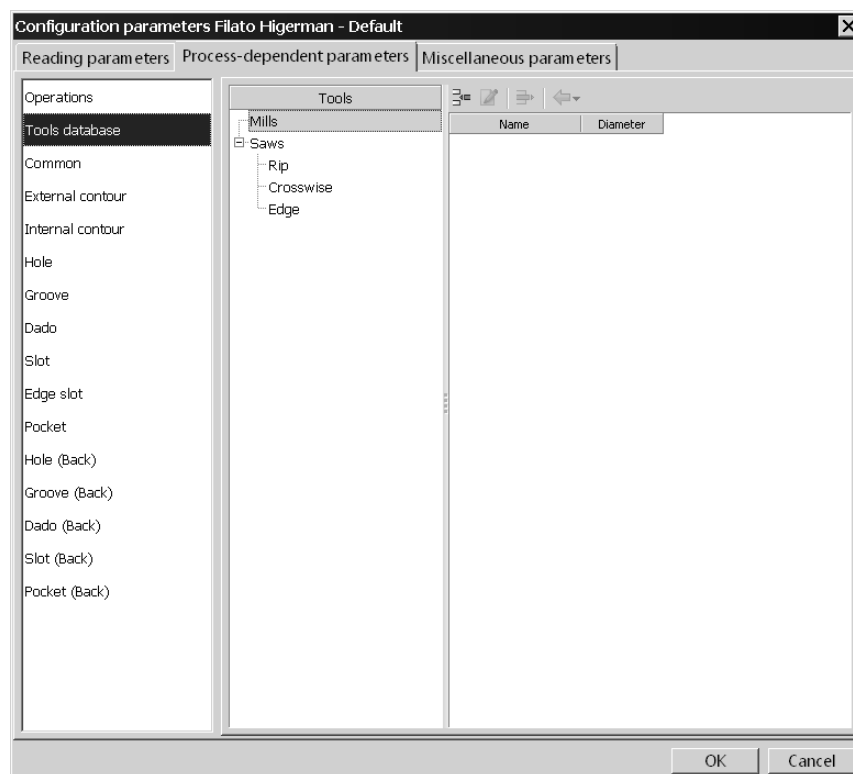


Fig. 10.273.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.274).



Fig. 10.274.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common parameters are set in **Common** section (fig. 10.275).

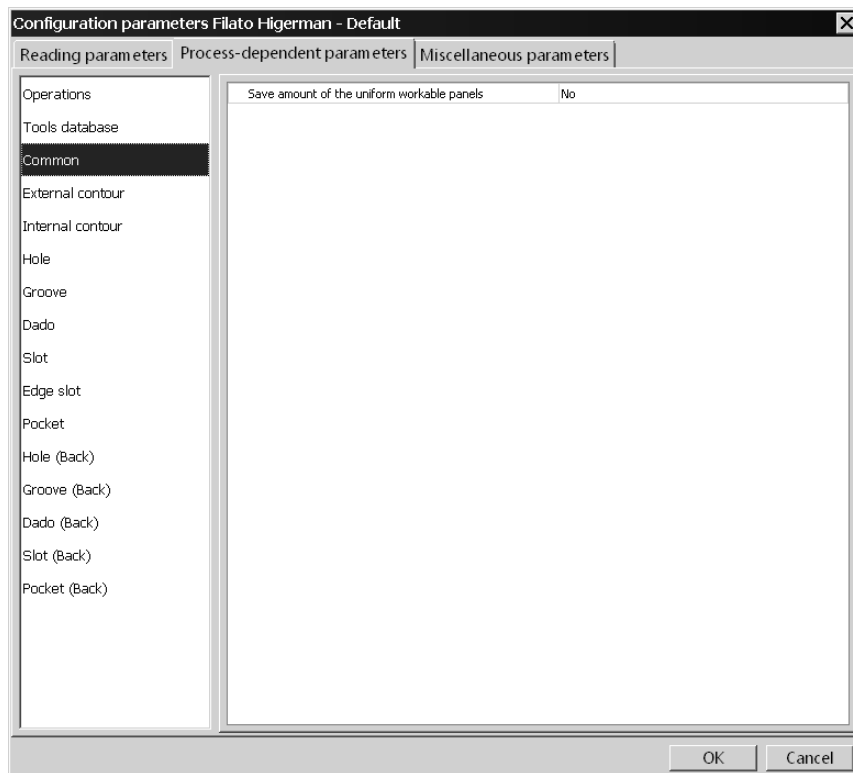


Fig. 10.275.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.276).

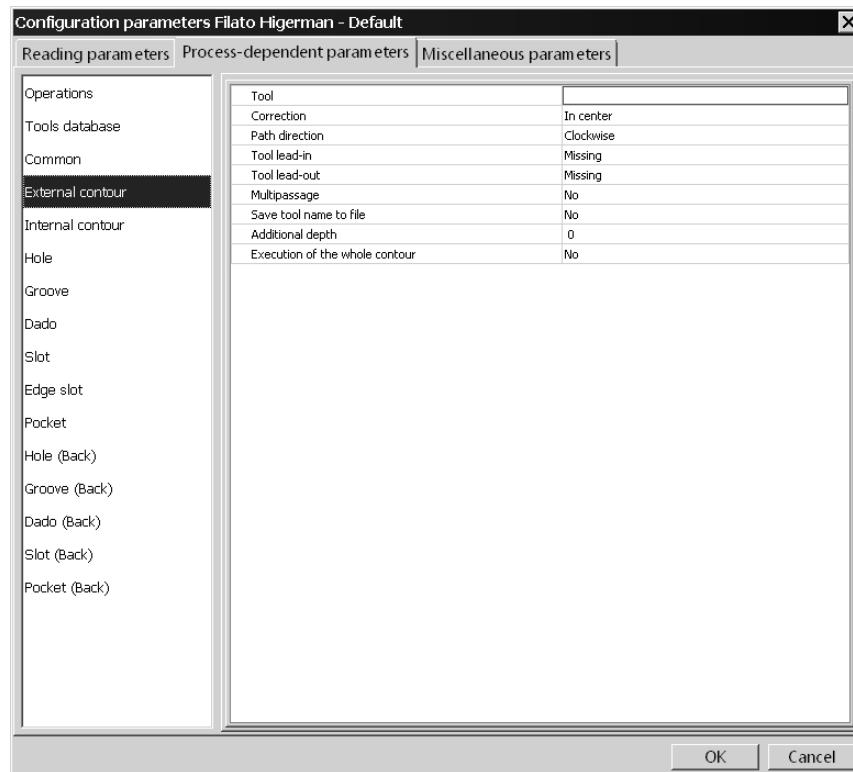


Fig. 10.276.

Control elements in sections allow to select the tool, correction types, lead-in, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.277), holes from back side – in **Hole (Back)** section in the same way.

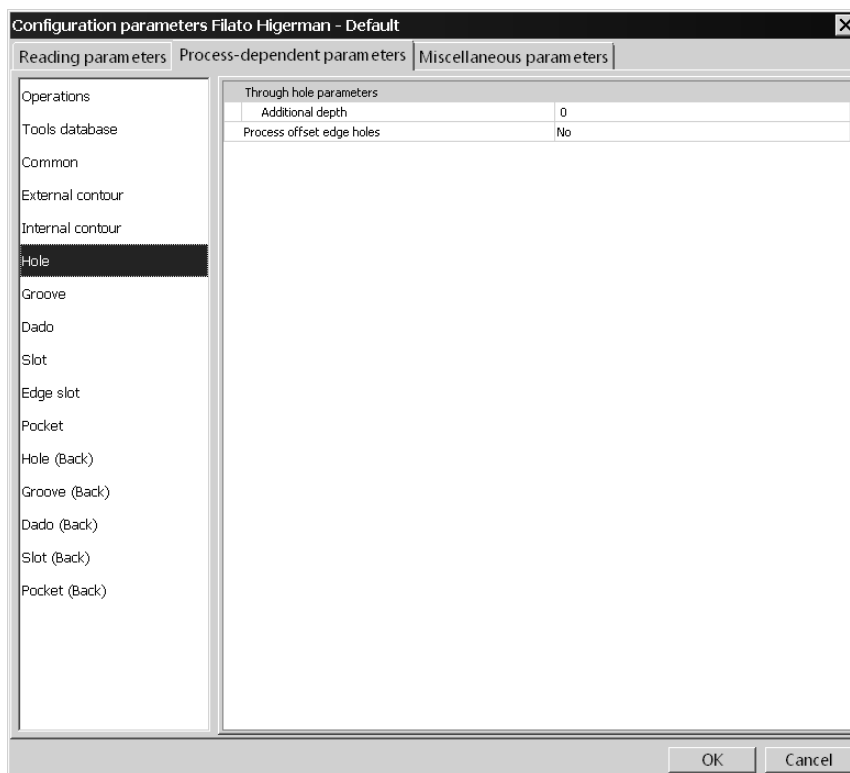


Fig. 10.277.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.278), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

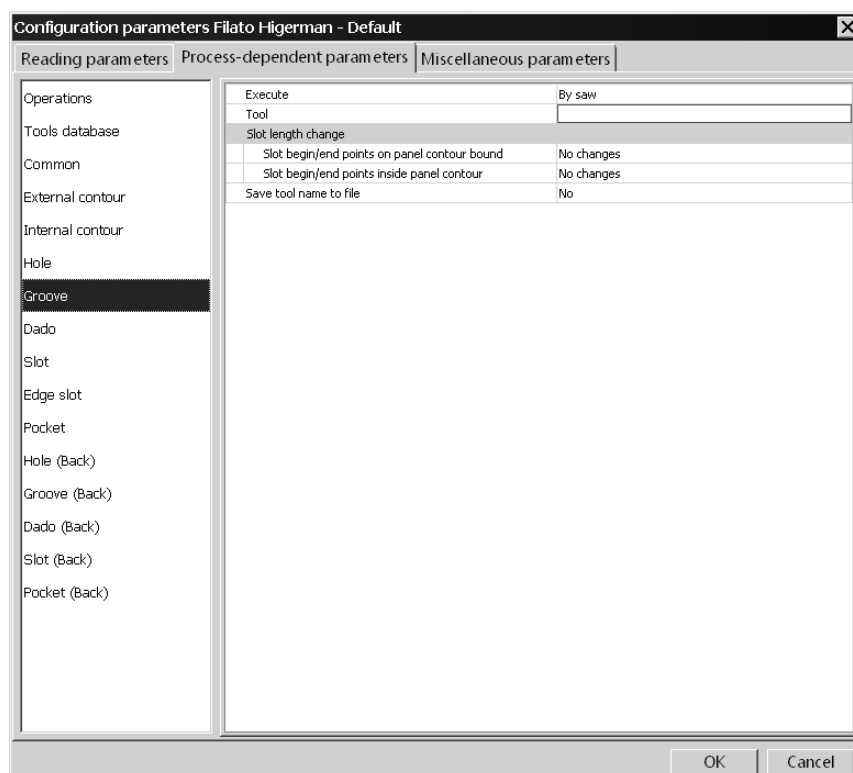


Fig. 10.278.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.279), from the back panel side – in **Slot (Back)** section.

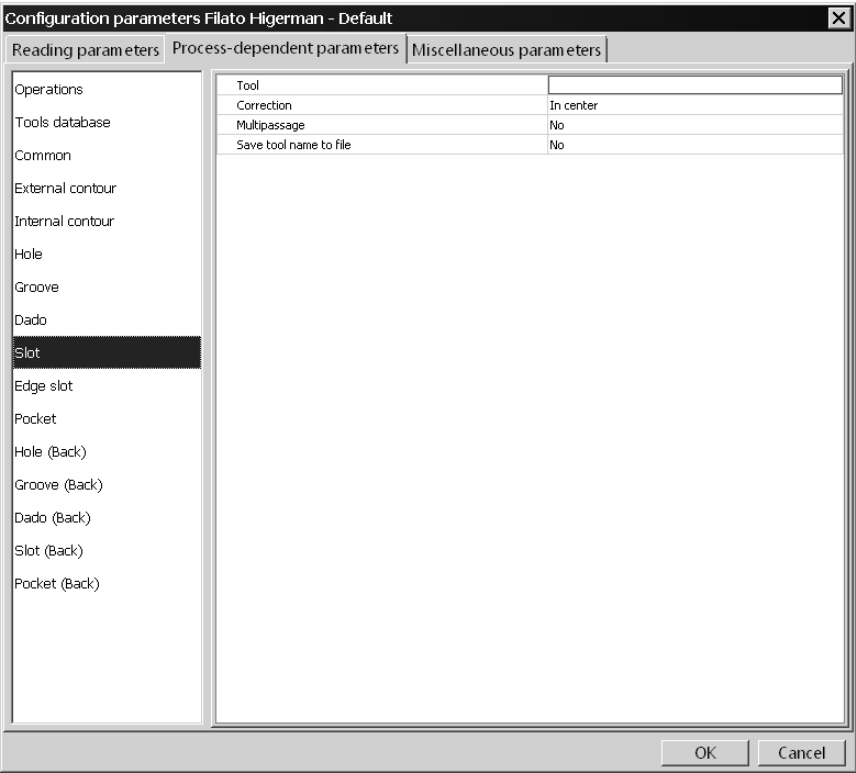


Fig. 10.279.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.280), on the back panel side – in **Pocket (Back)** section.

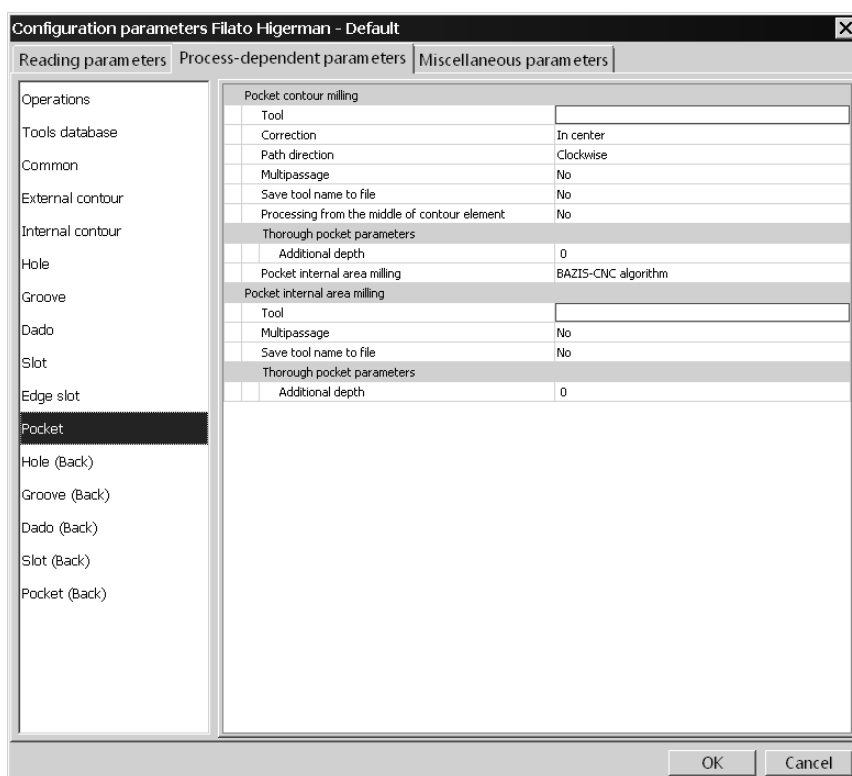


Fig. 10.280.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects from database. For pocket contour select milling execution direction, correction type and multipassage use. For internal pocket part – correction type and multipassage use.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.12.5.Filato SV



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.281) allow to specify operations that can be used in control program.

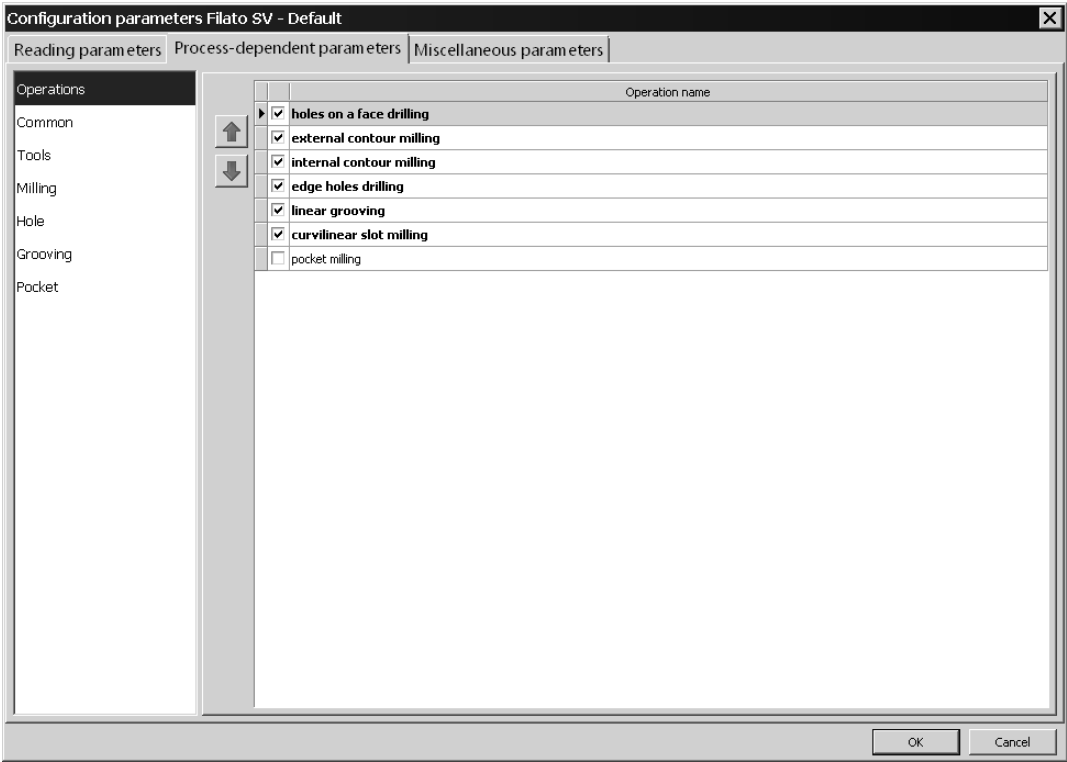


Fig. 10.281.

Common parameters

Common section control elements (fig. 10.282) allow to set common parameters of control programs creation.

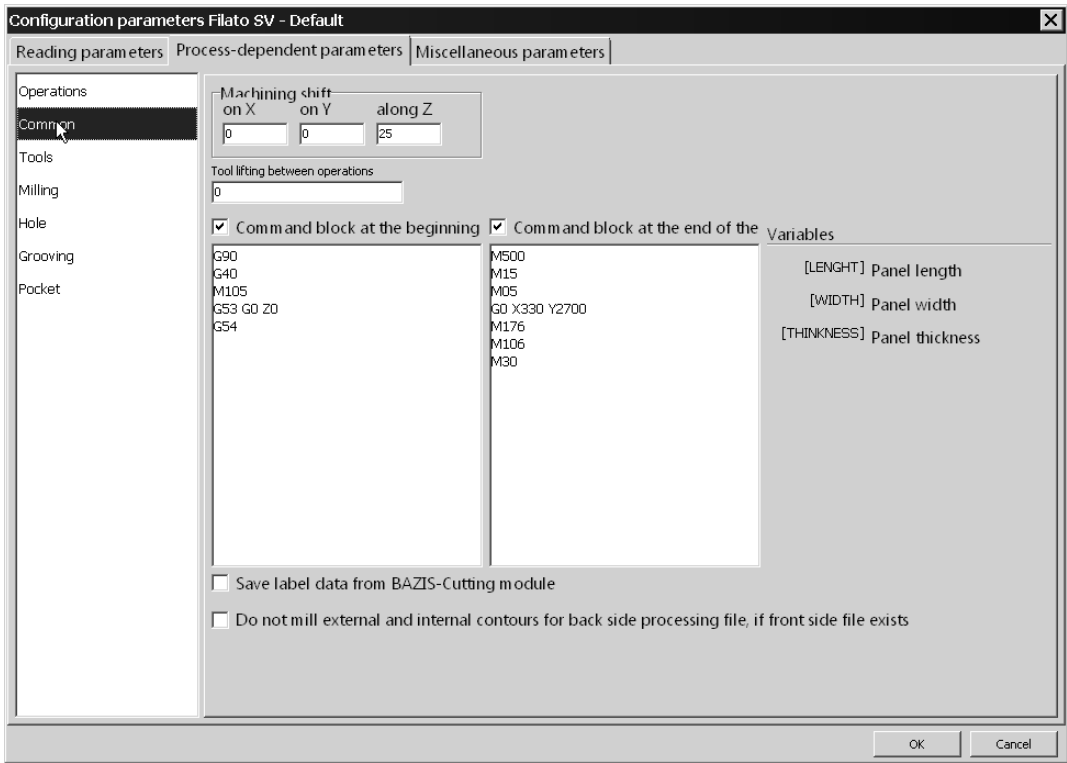


Fig. 10.282.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas. To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.283).

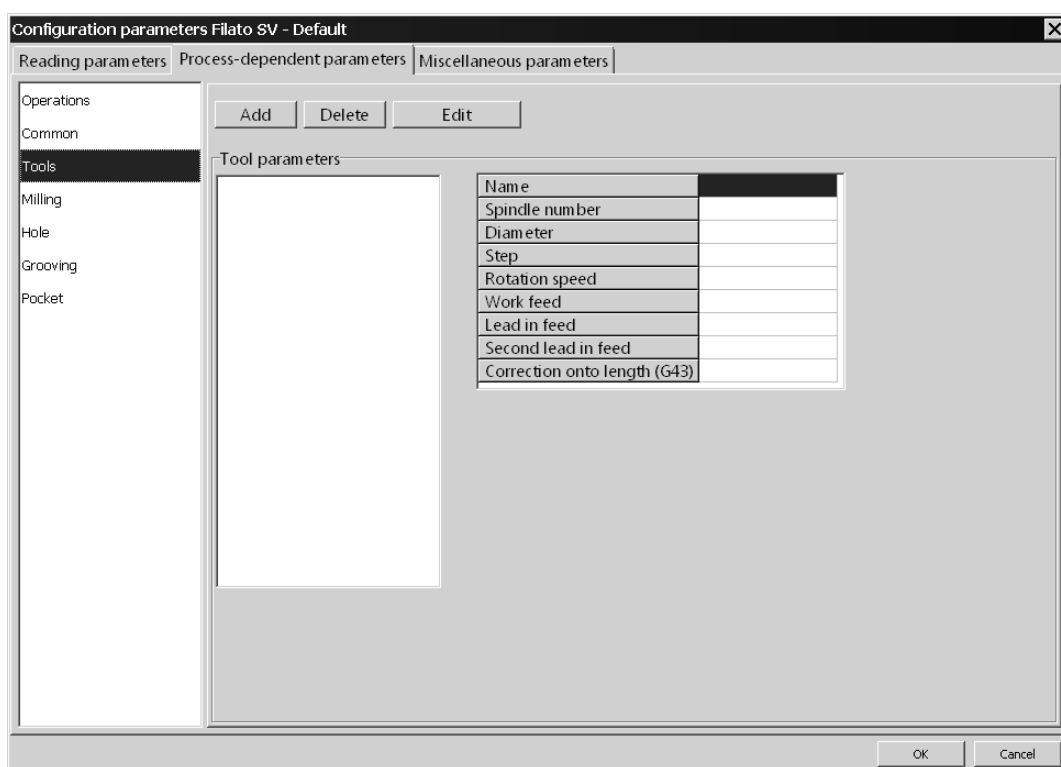


Fig. 10.283.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.284).

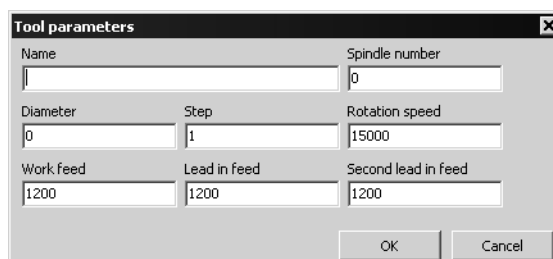


Fig. 10.284.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is fig. 10.285.

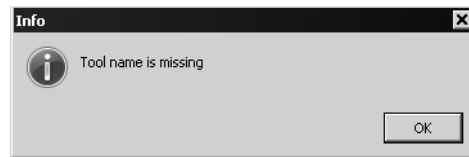


Fig. 10.285.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.284). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.286).

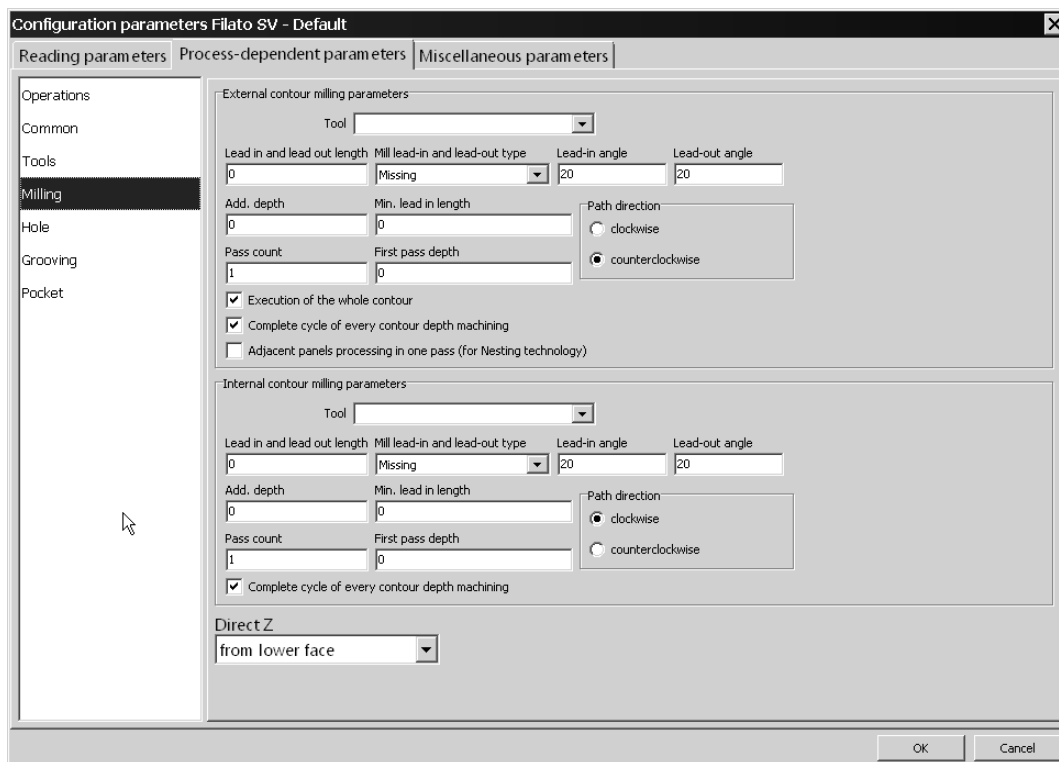


Fig. 10.286.

Tool combo box allows to select a milling tool name created in **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.287).

The screenshot shows a software interface for configuring hole parameters. The 'Hole' section is selected in the left sidebar. The main area contains several input fields and a table.

Through hole parameters

Additional depth: 0

Drill holes on a face

☒ on drill shortest path
☐ on diameter

Edge hole parameters

Safety distance: 10

For holes on a face apply Z: from lower face (dropdown)
 For edge holes use Z: from upper face(+) (dropdown)

Tool lifting between operations: 0

Diameter	Layout	Type	Tool
5	Edge on the mitre	Blind	
6	Upper face	Through	
5	Front edge	Blind	
6	Front edge	Blind	

Fig. 10.287.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Slot parameter

Grooving section control elements (fig. 10.288) allow to set slots processing parameters.

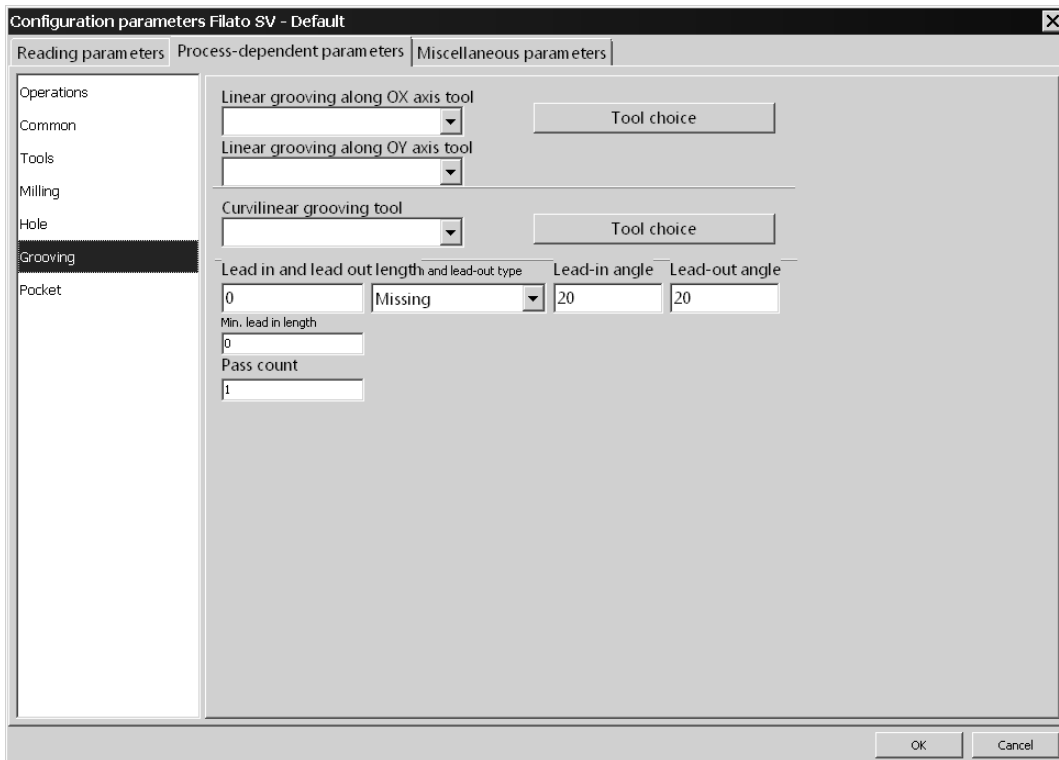


Fig. 10.288.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.289).

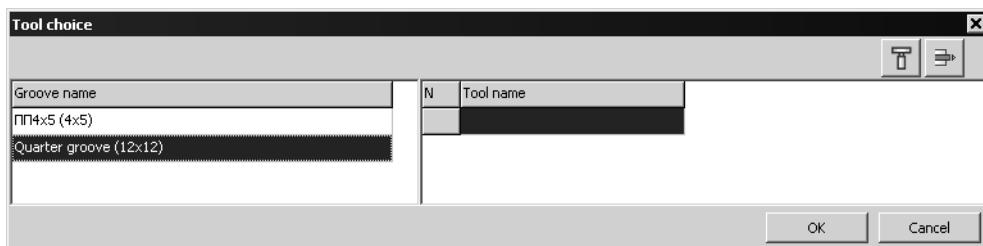


Fig. 10.289.



Slot name list contains slot names of the model. Several tools can be assigned for one slot. **Add mill** button allows to map selected slot with a tool from the list created in **Tools** section.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.290) allow to set pocket processing parameters.

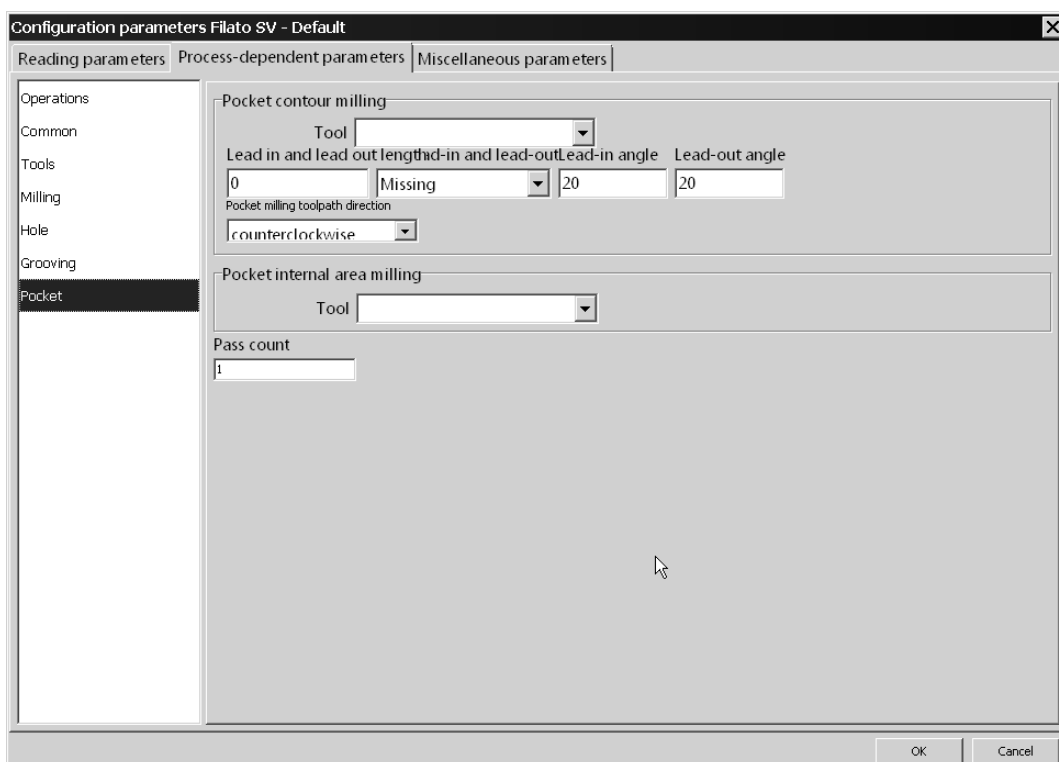


Fig. 10.290.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.12.6.Filato Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.291) allow to specify operations that can be used in control program.

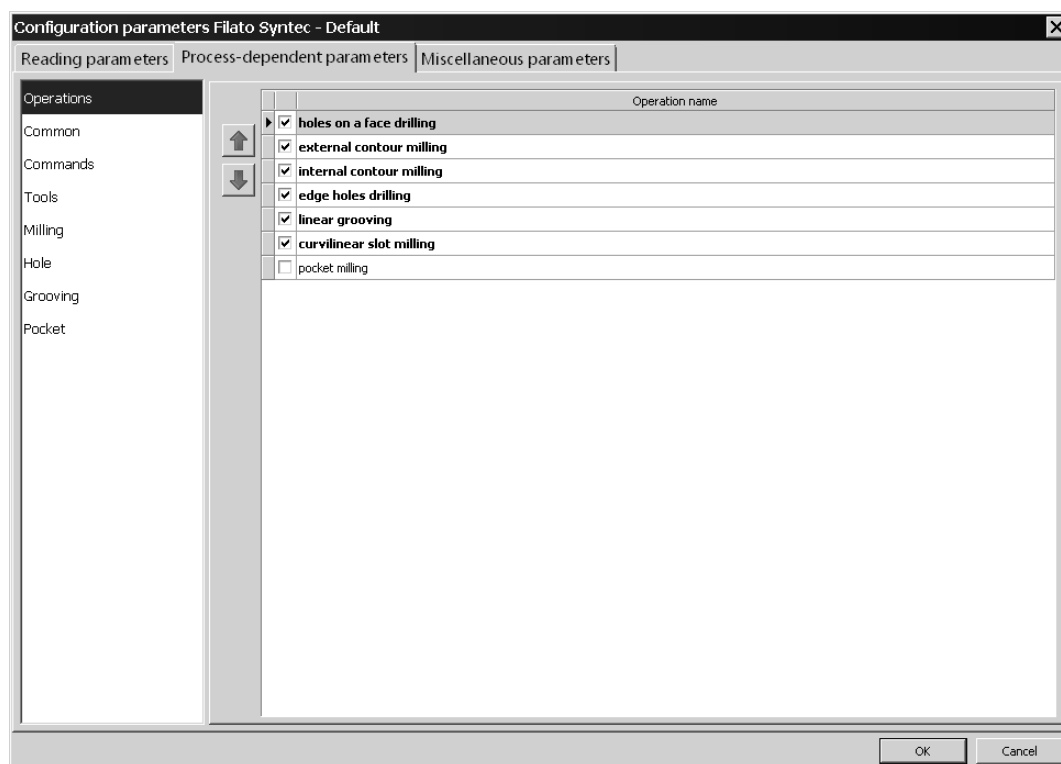


Fig. 10.291.

Common parameters

Common section control elements (fig. 10.292) allow to set common parameters of control programs creation.

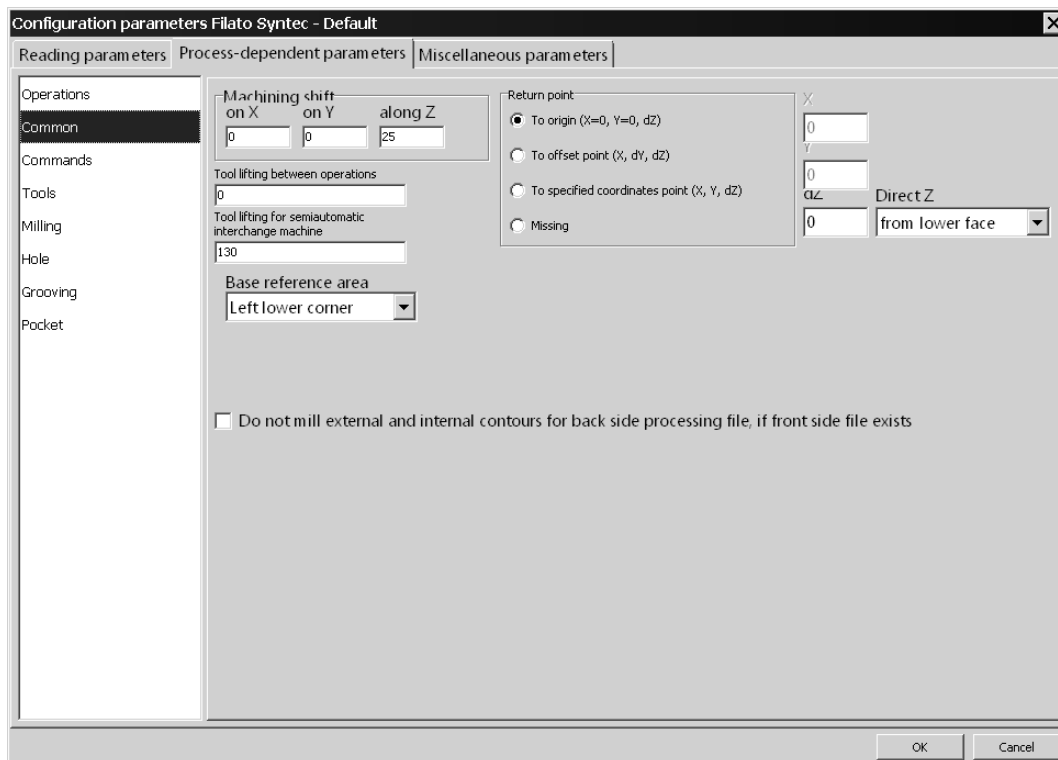


Fig. 10.292.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.293) allow to control separate commands to control program inclusion.

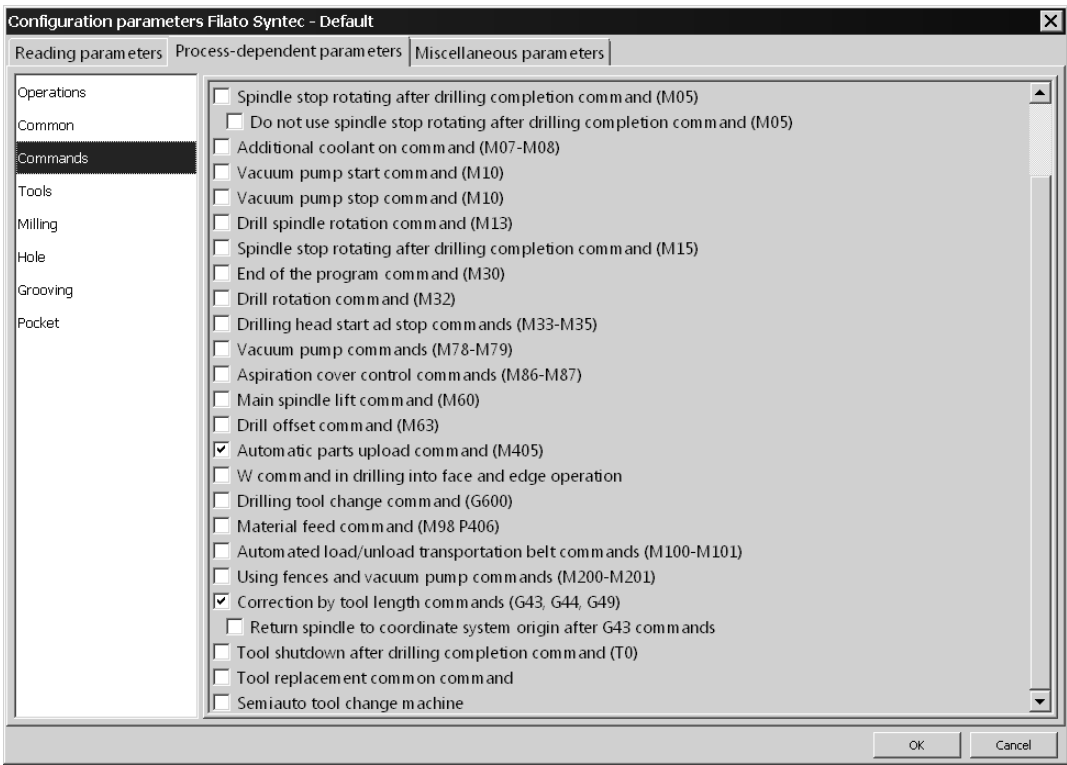


Fig. 10.293.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.294).

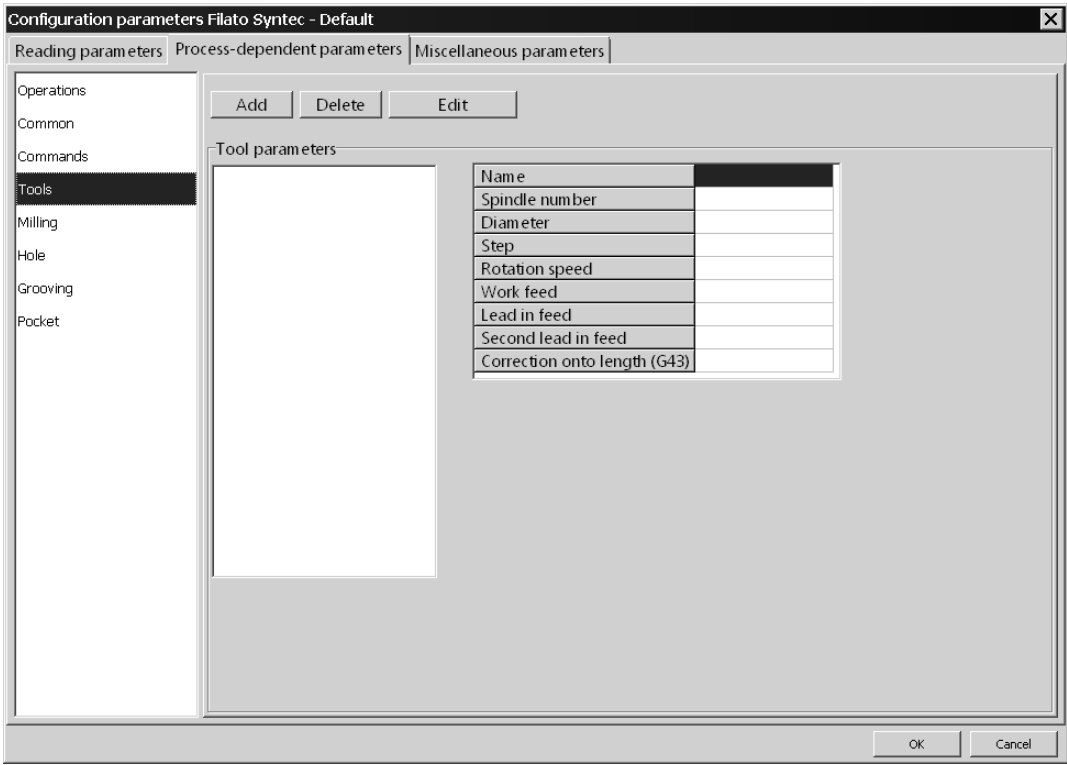
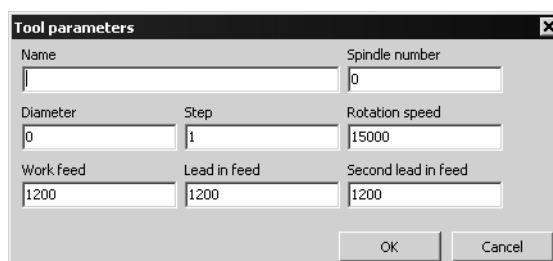


Fig. 10.294.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.295).



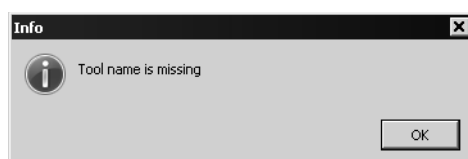
Tool parameters

Name		Spindle number
		0
Diameter	Step	Rotation speed
0	1	15000
Work feed	Lead in feed	Second lead in feed
1200	1200	1200

OK Cancel

Fig. 10.295.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is (fig. 10.296).



Info

 Tool name is missing

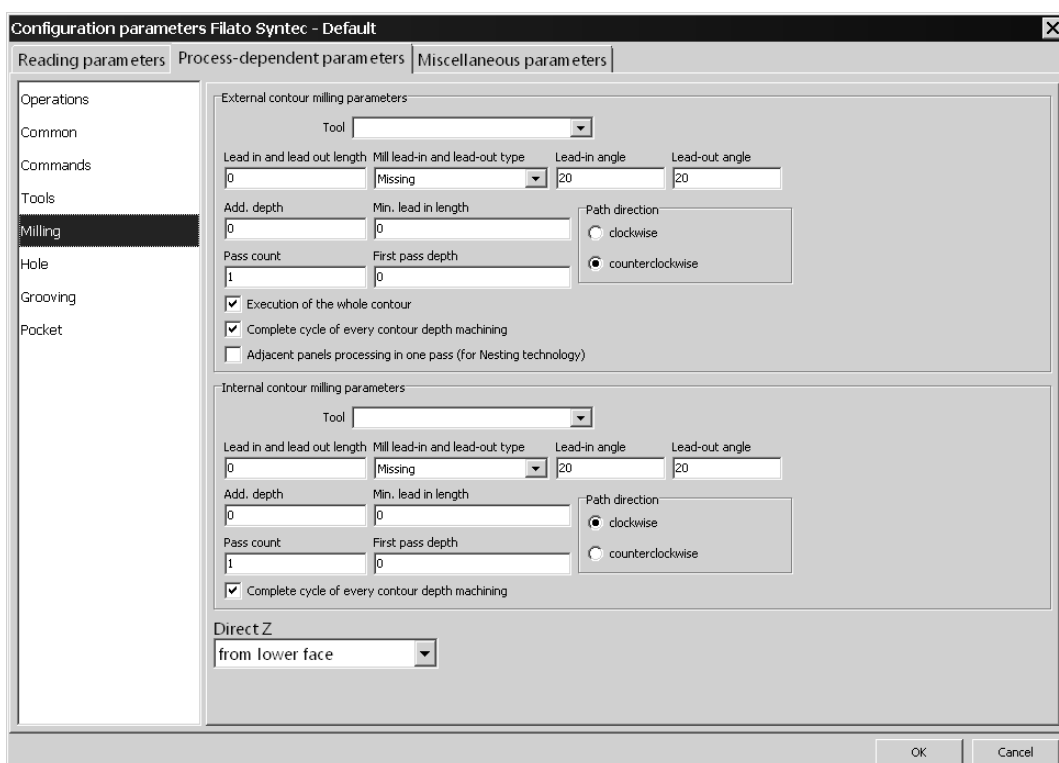
OK

Fig. 10.296.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.295). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.297).



Configuration parameters Filato Syntec - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Operations
Common
Commands
Tools
Milling
Hole
Grooving
Pocket

External contour milling parameters

Tool

Lead in and lead out length	Mill lead-in and lead-out type	Lead-in angle	Lead-out angle
0	Missing	20	20

Add. depth	Min. lead in length	Path direction ☐ clockwise ☒ counterclockwise
0	0	

Pass count	First pass depth
1	0

☒ Execution of the whole contour
☒ Complete cycle of every contour depth machining
☐ Adjacent panels processing in one pass (for Nesting technology)

Internal contour milling parameters

Tool

Lead in and lead out length	Mill lead-in and lead-out type	Lead-in angle	Lead-out angle
0	Missing	20	20

Add. depth	Min. lead in length	Path direction ☒ clockwise ☐ counterclockwise
0	0	

Pass count	First pass depth
1	0

☒ Complete cycle of every contour depth machining

Direct Z
from lower face

OK Cancel

Fig. 10.297.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.298).

Diameter	Layout	Type	Tool
5	Edge on the mitre	Blind	
6	Upper face	Through	
5	Front edge	Blind	
6	Front edge	Blind	

Fig. 10.298.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from**

the center option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slot parameters

Grooving section control elements (fig. 10.299) allow to set slots processing parameters.

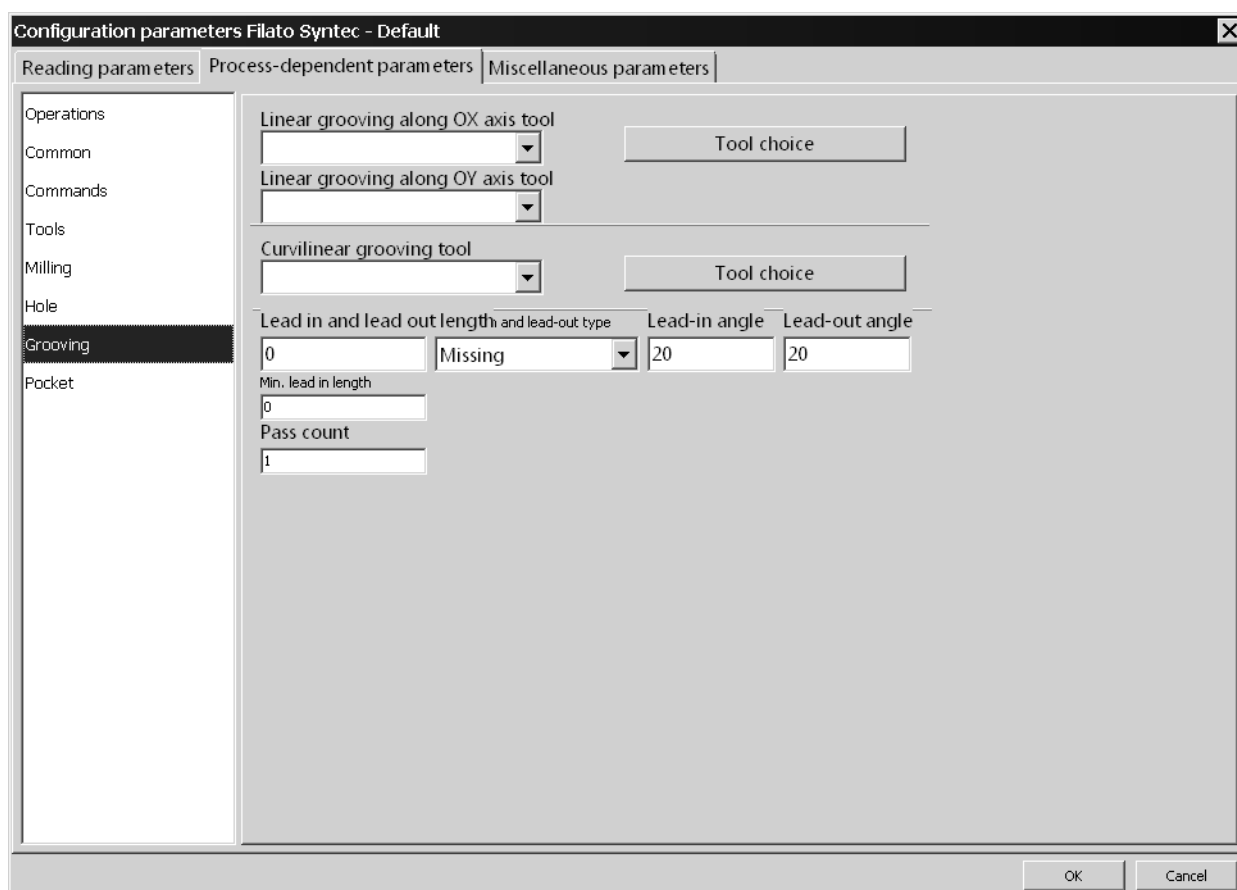


Fig. 10.299.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.300).

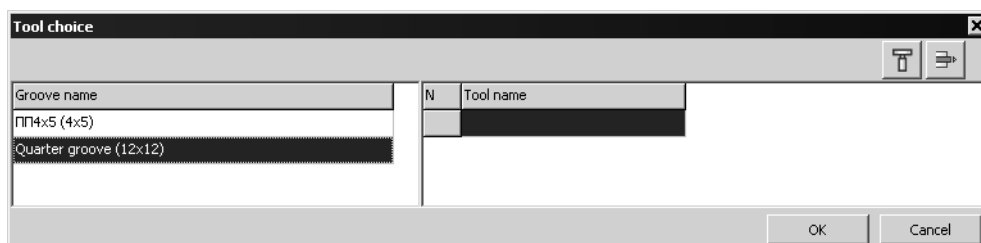


Fig. 10.300.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list created in **Tools** section.



Delete button allows to delete selected tool from the list.

To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.301) allow to set pocket processing parameters.

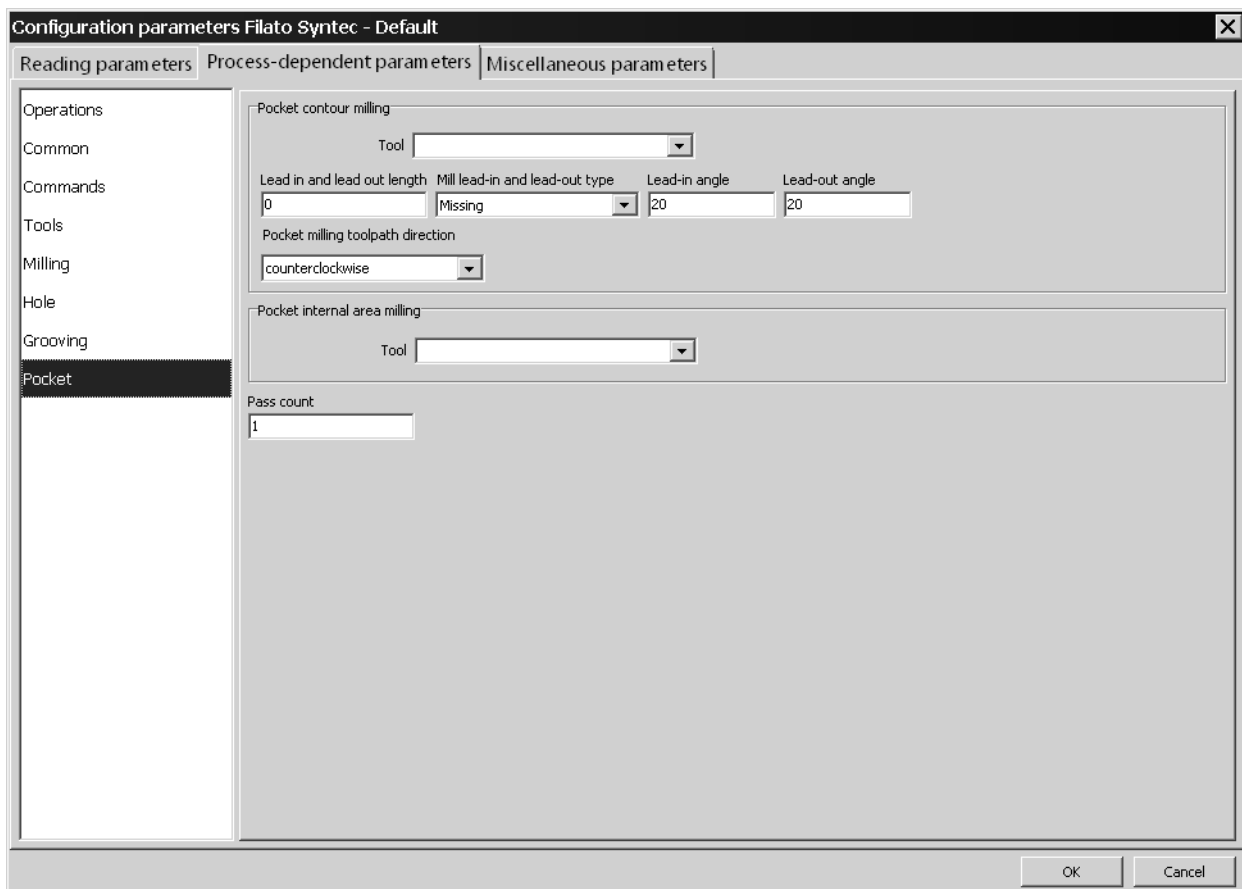


Fig. 10.301.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.12.7.Filato NC Studio



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.302) allow to set common parameters of control programs creation.

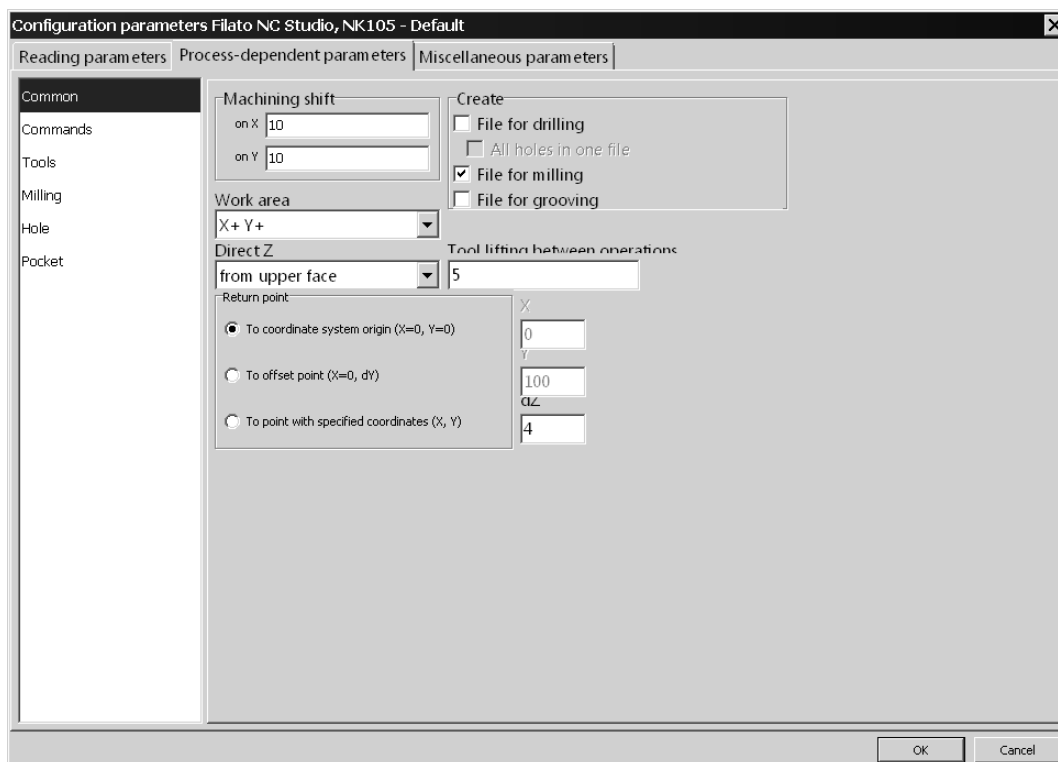


Fig. 10.302.

On X and **on Y** fields of **Machining shift** group allow to set work area initial point moving. **Create** group options allow to control programs creation for separate operations. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **dY** and **dZ** fields values.

Tool lifting between operations field allows to set tool height. **Work area** combo box variants allow to set work area coordinate axes direction.

Usable commands

Commands section control elements (fig. 10.303) allow to control separate commands to control program inclusion.

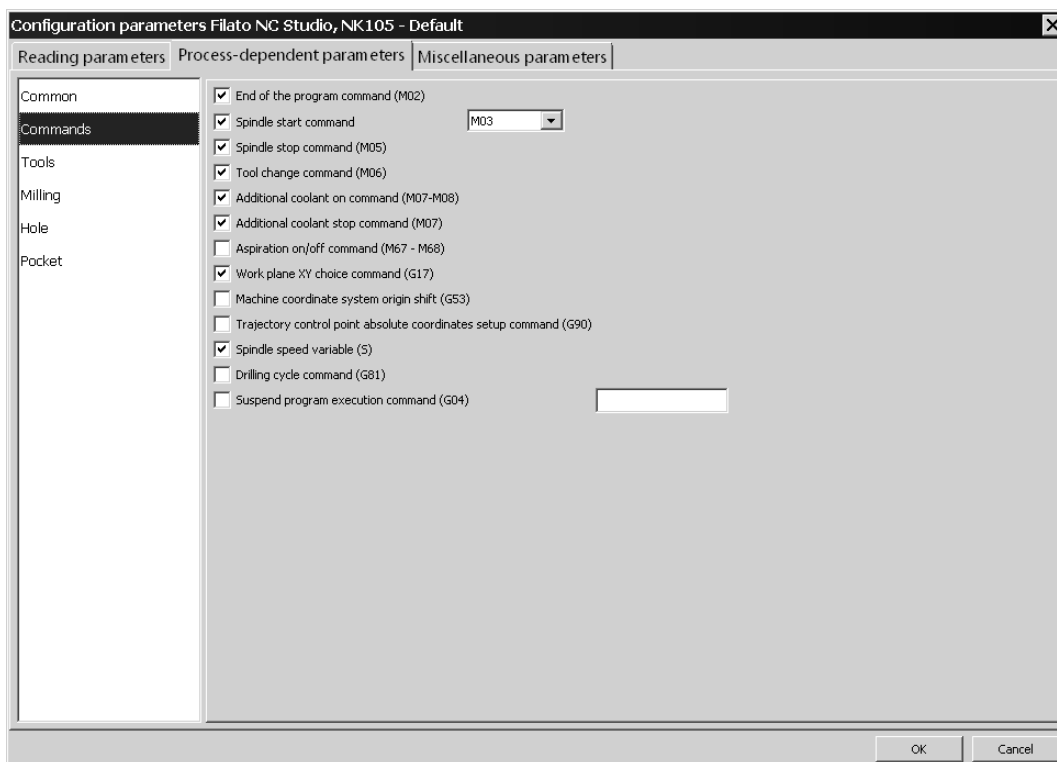


Fig. 10.303.

Spindle start command can be selected from two variants in a combo box.

Usable tools

Tools section control elements (fig. 10.304) allow to create a list of tools available for use in control programs.

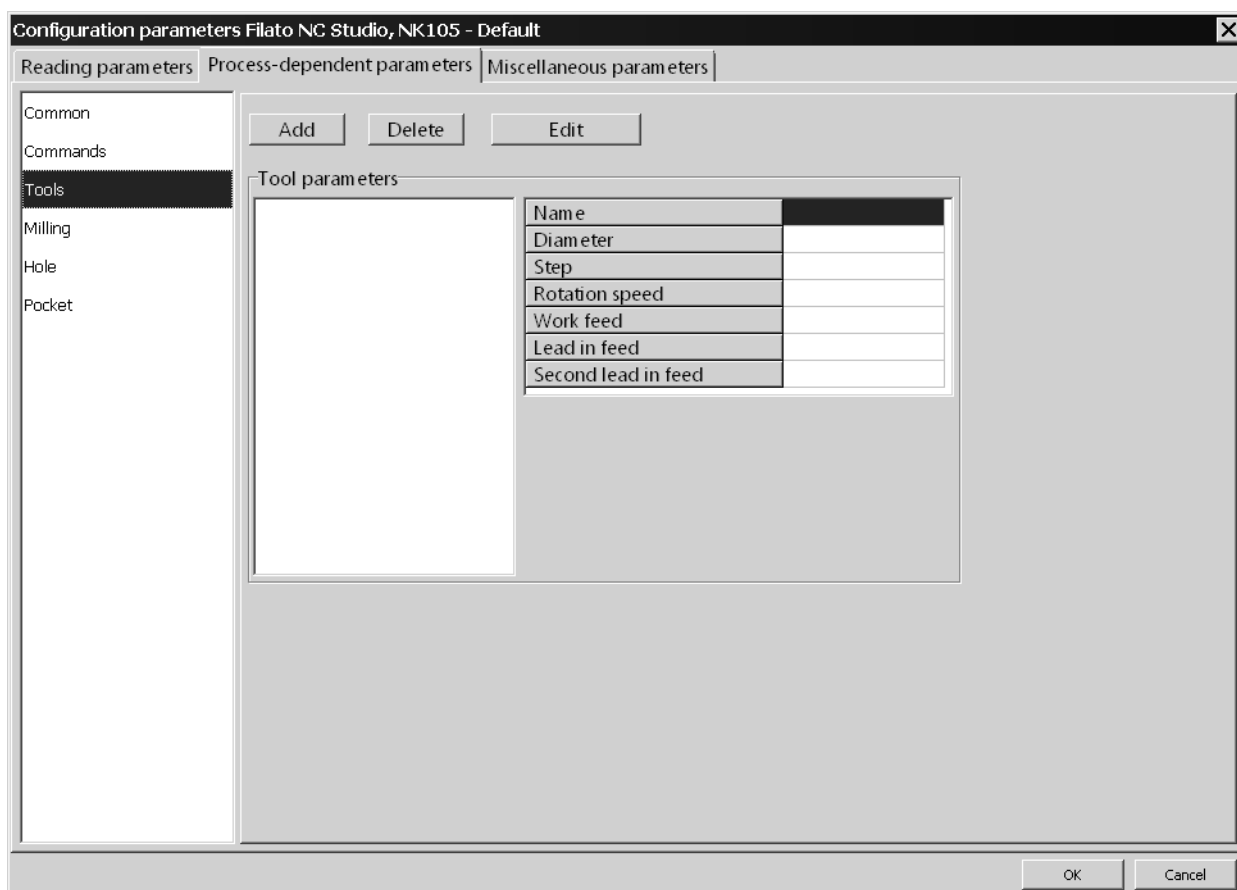


Fig. 10.304.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.305).

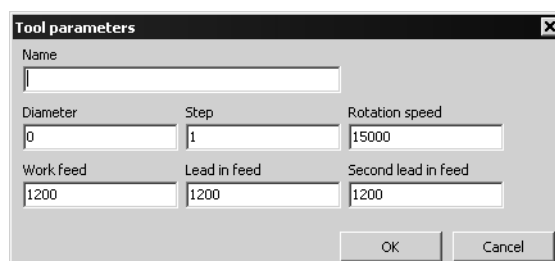


Fig. 10.305.

In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.306) allow to set contour milling parameters.

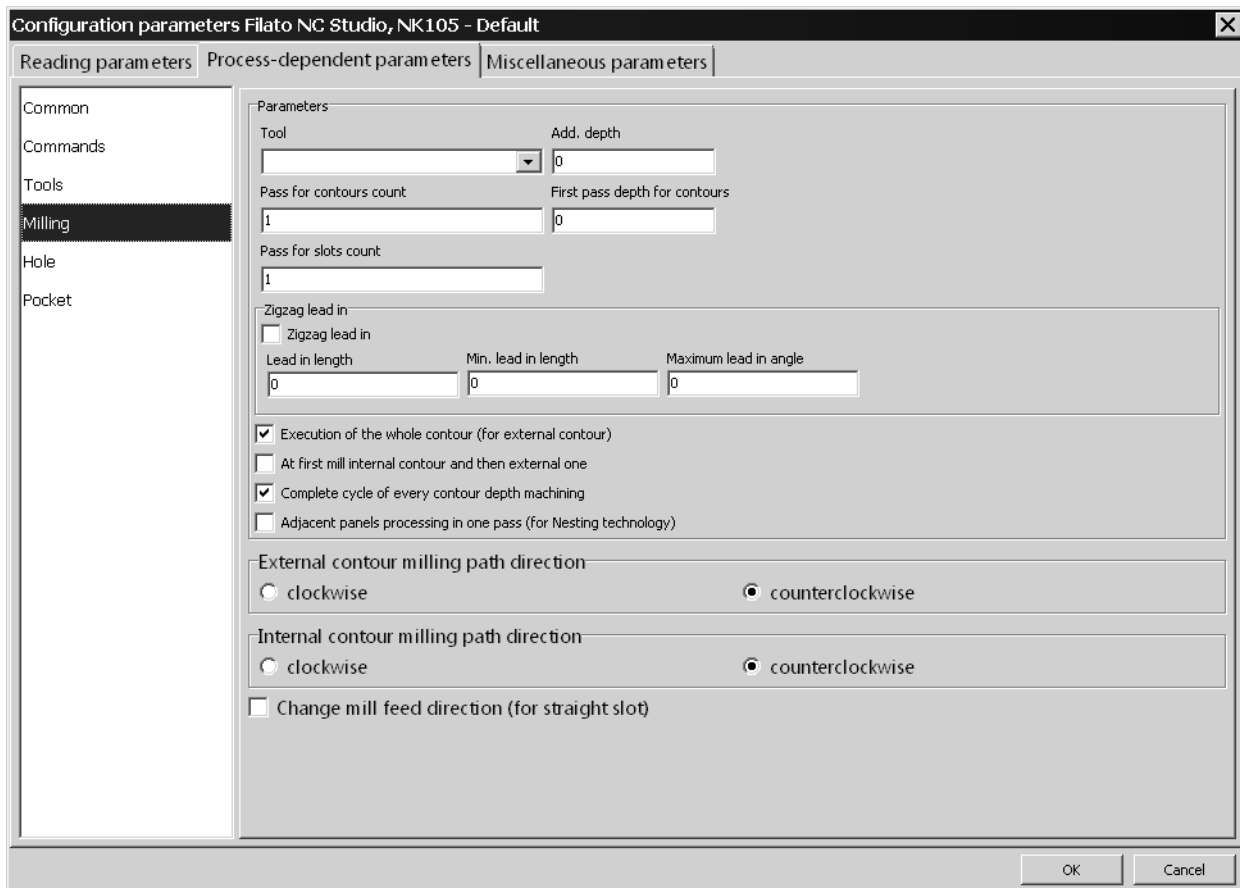


Fig. 10.306.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc.

Execution of the whole contour option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Zigzag lead in** option allows to activate this operation. Group fields allow to set option parameters. **Milling path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.307).

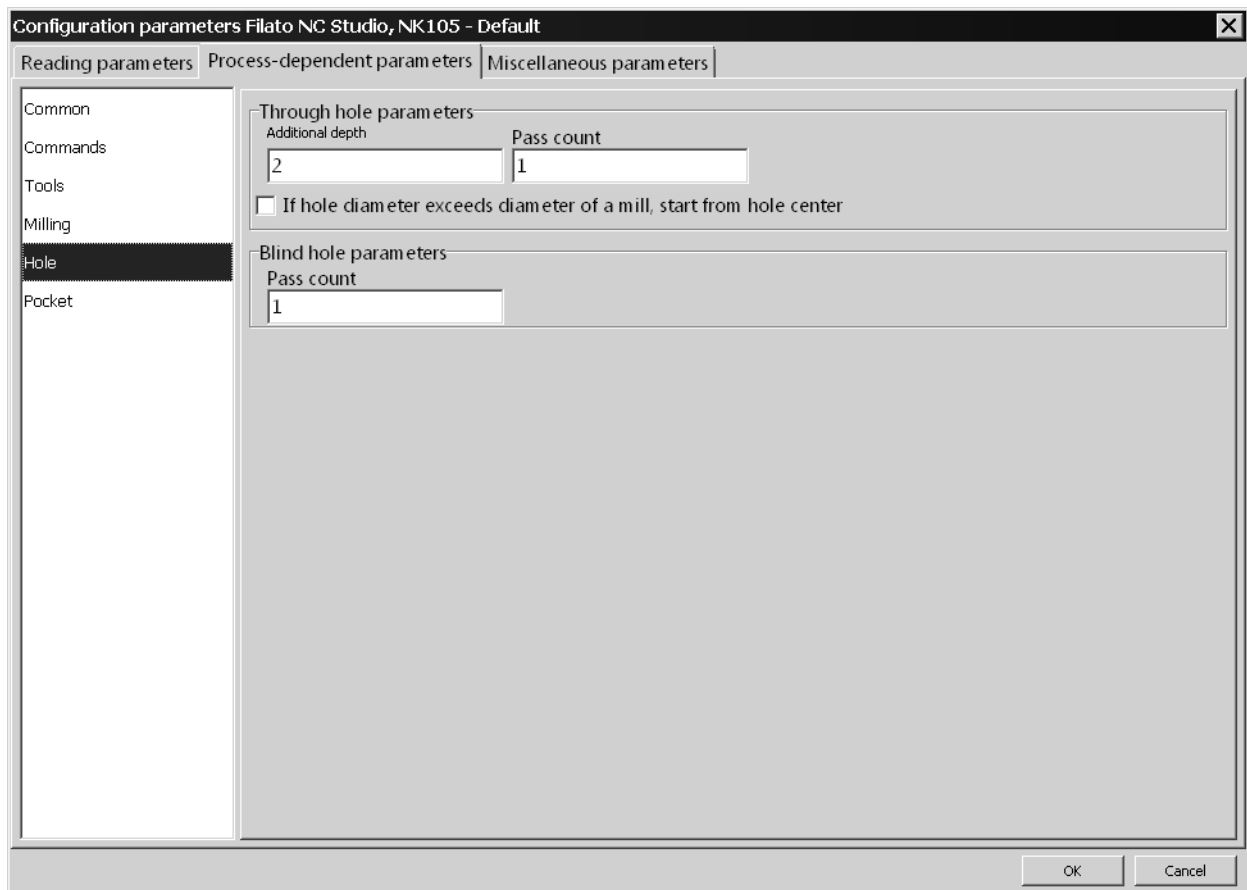


Fig. 10.307.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **If hole diameter exceeds diameter of a mill, start from hole center** option allows to control holes with a mill processing order, from center or from perimeter.

Pocket parameters

Pocket section control elements (fig. 10.308) allow to set pocket processing parameters.

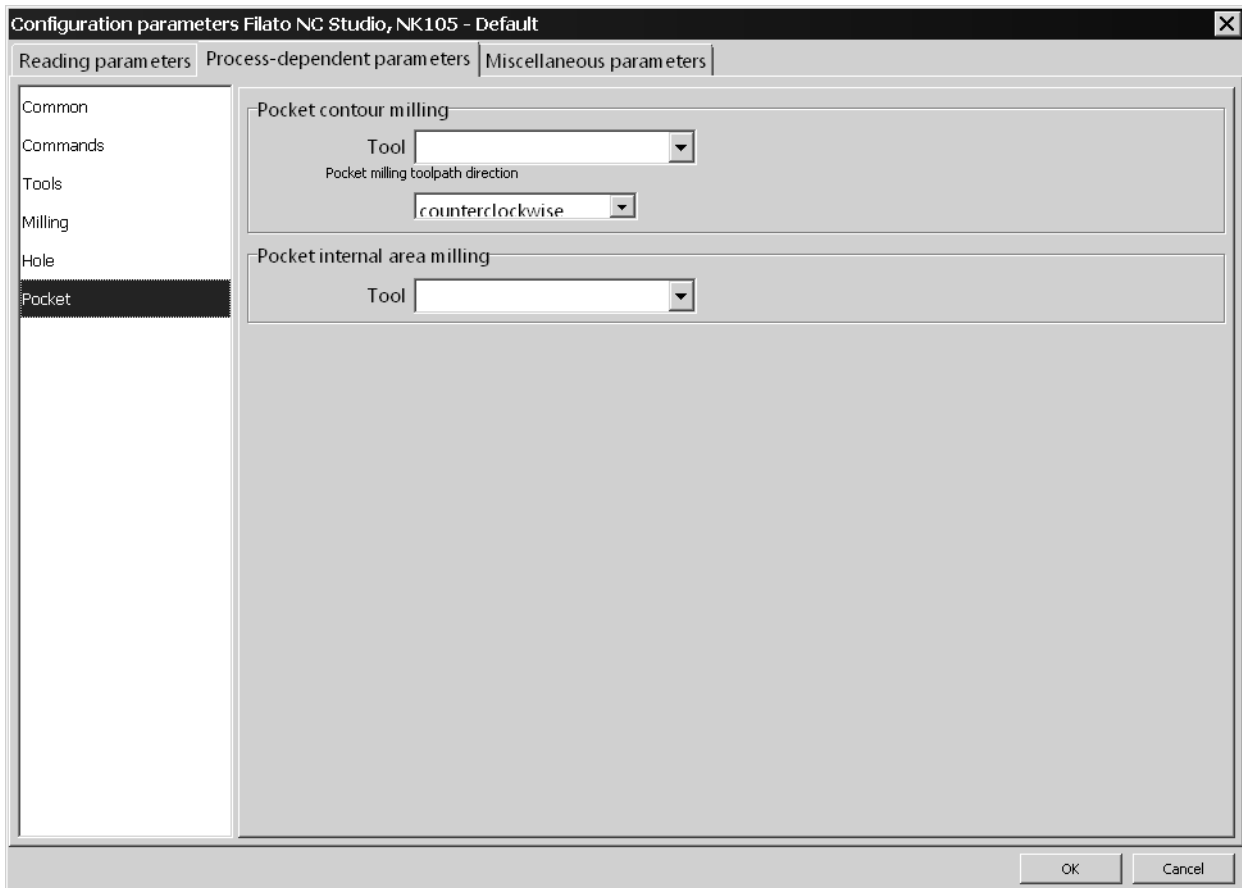


Fig. 10.308.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects. Tools should be created beforehand in **Tools** section. For pocket contour select milling execution direction in a combo box.

Setup completion

To complete control programs parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes

Create group control elements allow to control the creation of control programs files corresponding to option names.

10.12.8.Filato RS Lab



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.309) allow to specify operations that can be used in control program.

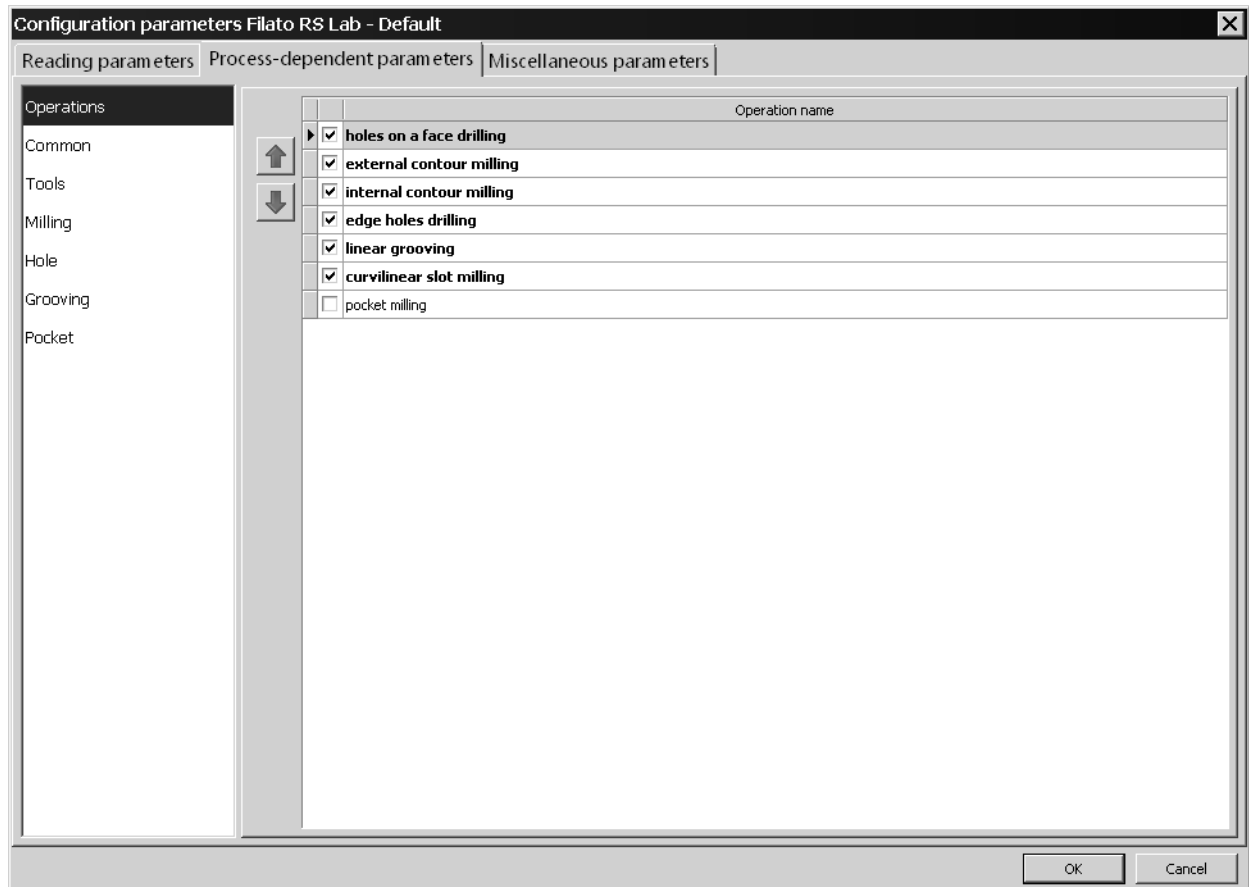


Fig. 10.309.

Common parameters

Common section control elements (fig. 10.310) allow to set common parameters of control programs creation.

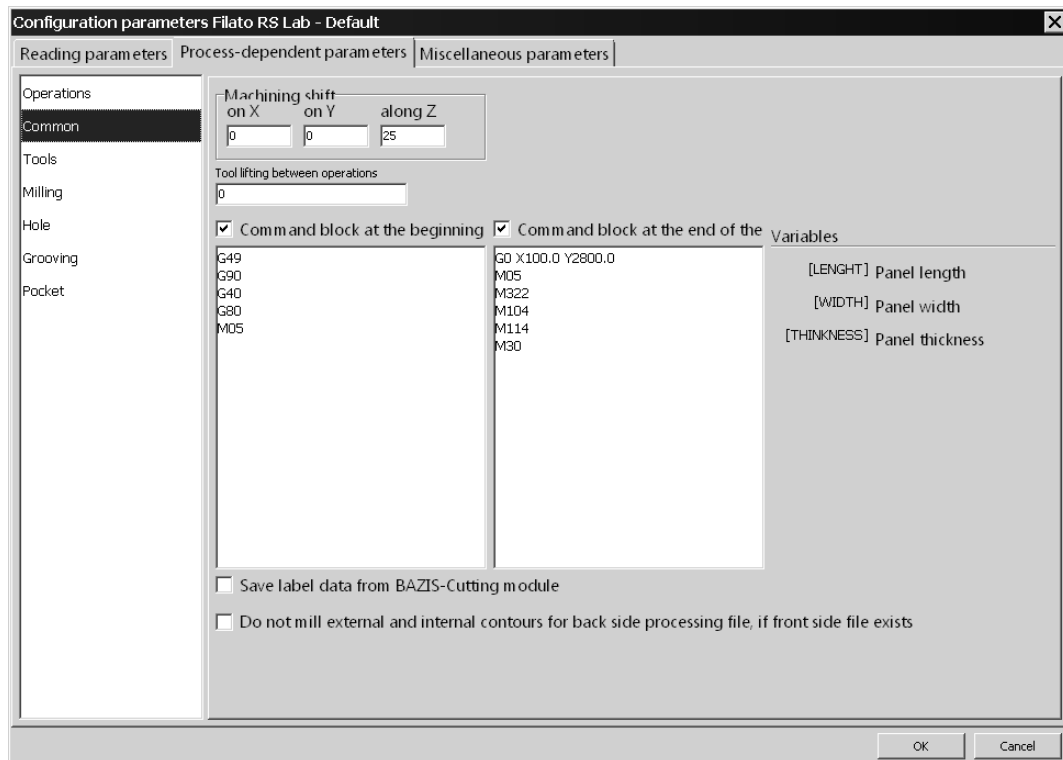


Fig. 10.310.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas. To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable tools

Tools section control elements (fig. 10.311) allow to create a list of tools available for use in control programs.

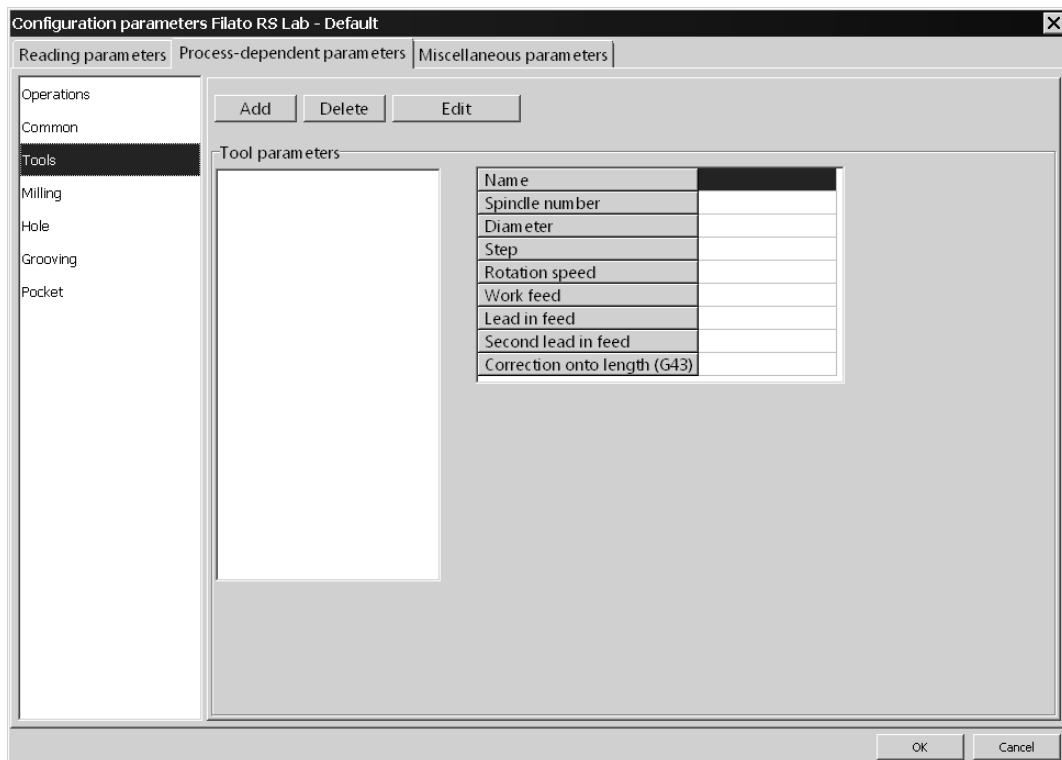


Fig. 10.311.

To add a tool to the list push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.312).

Name		Spindle number
		0
Diameter	Step	Rotation speed
0	1	15000
Work feed	Lead in feed	Second lead in feed
1200	1200	1200
Correction onto length (G43)		
0		

Fig. 10.312.

In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.313) allow to set contour milling parameters.

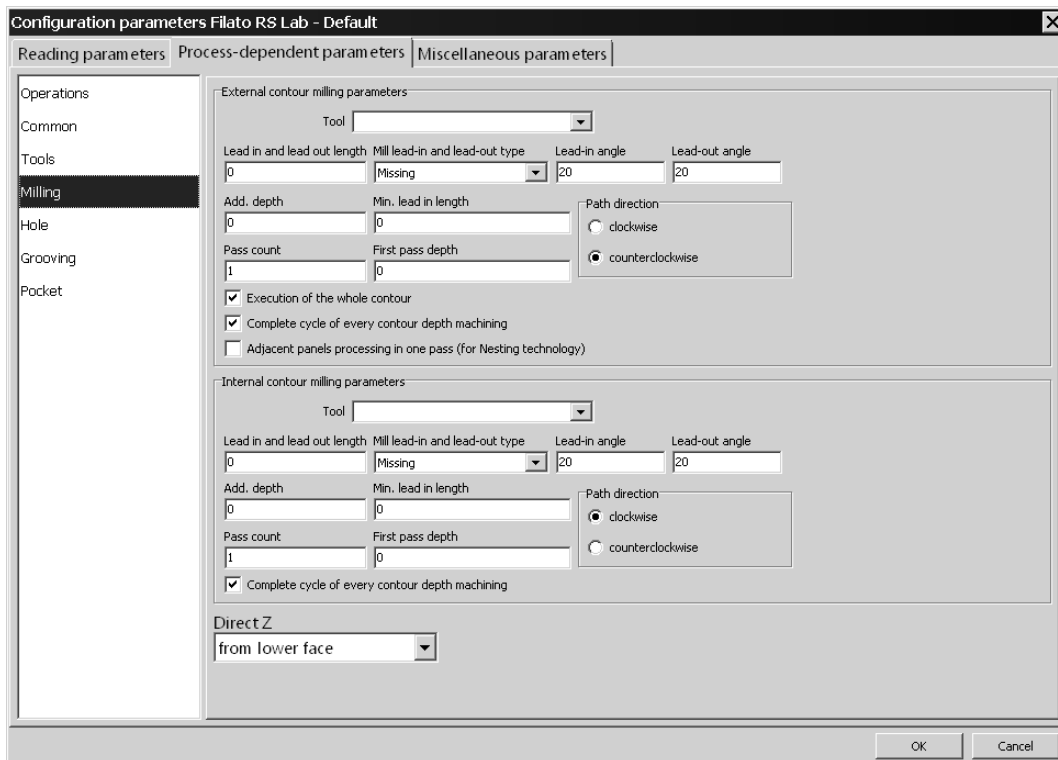


Fig. 10.313.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.314).

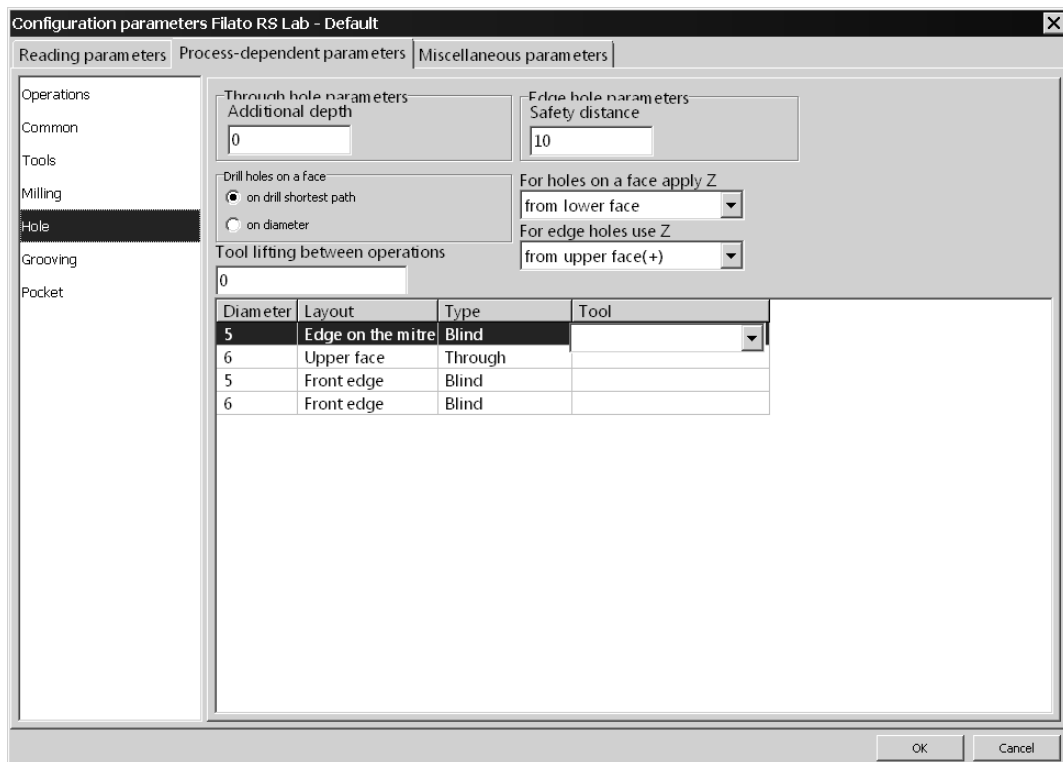


Fig. 10.314.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **For edge holes use Z** and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height. Section table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants of **Drill holes on a face** group allow to set holes boring order.

Slot parameters

Grooving section control elements (fig. 10.315) allow to set slots processing parameters.

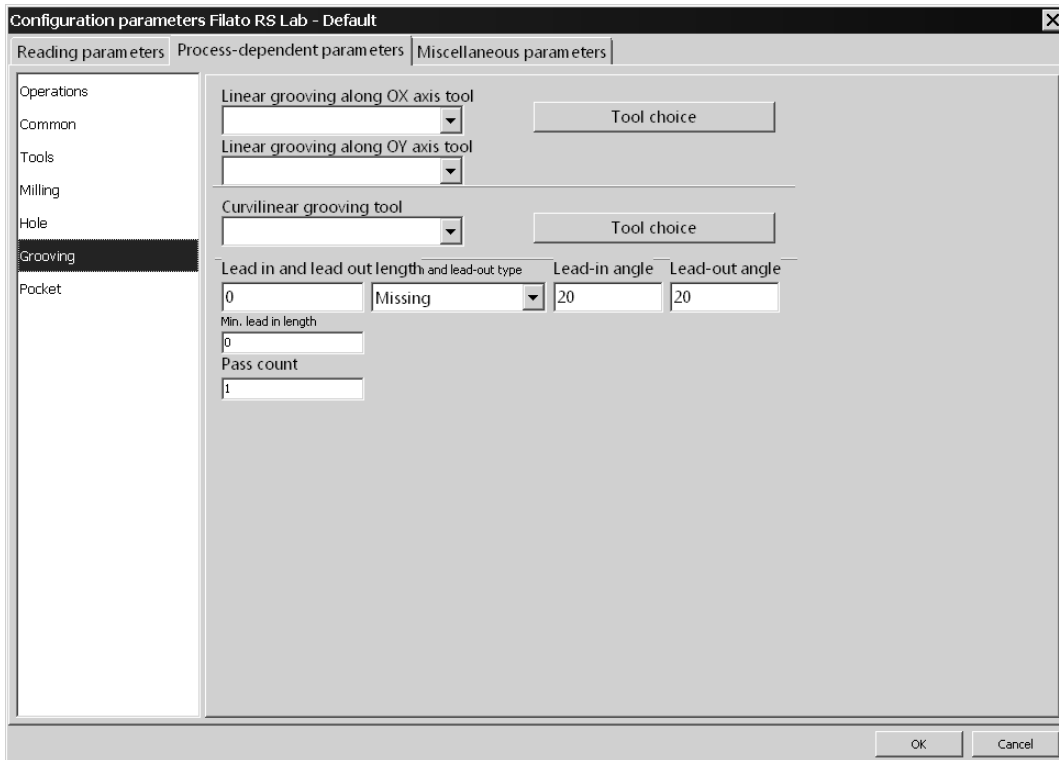


Fig. 10.315.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set a tool for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.316).

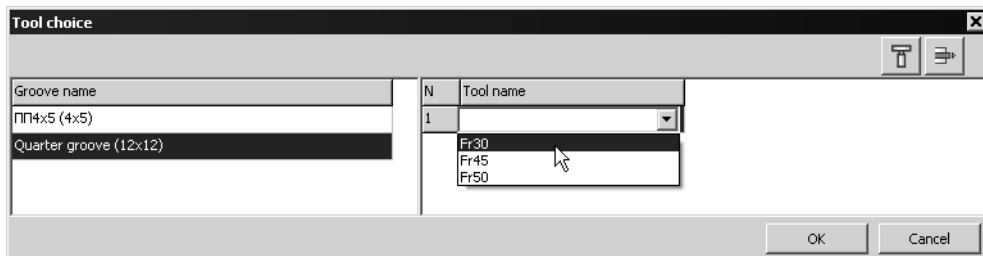


Fig. 10.316.



Slot name list contains slot names of the model. **Add mill** button allows to map selected slot with a tool from the list. Several tools can be assigned for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.317) allow to set pocket processing parameters.

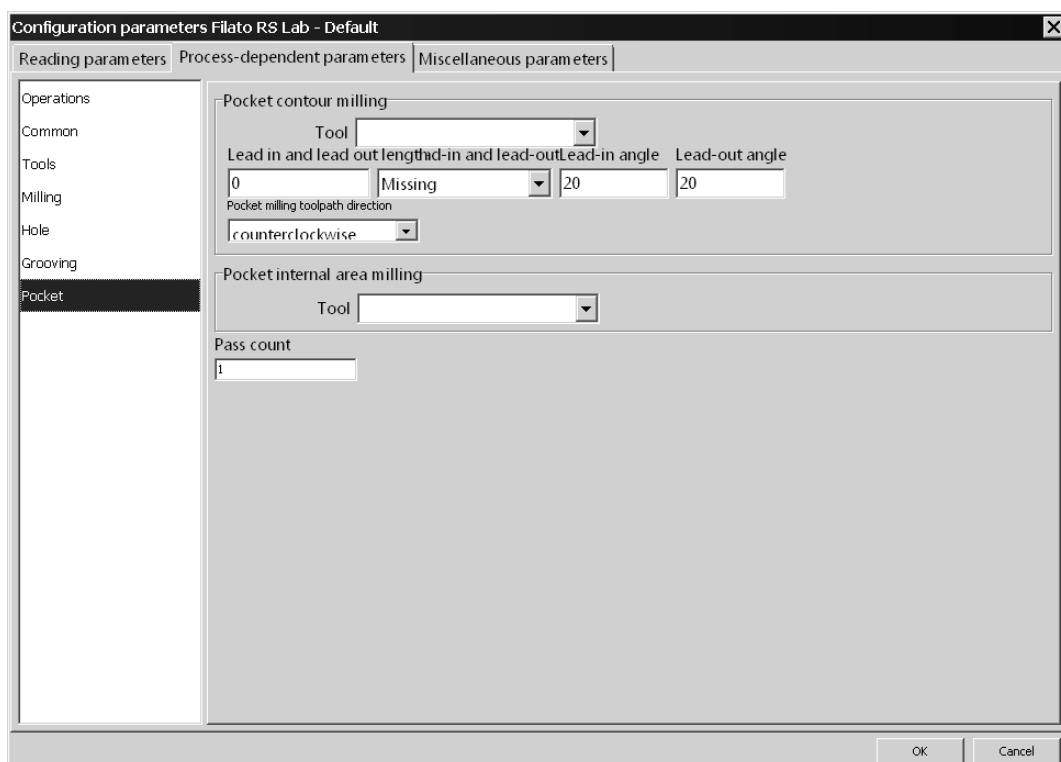


Fig. 10.317.

Section control elements allow to set control programs creation parameters for notch contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.13. Gannomat

10.13.1. GannoMat Editor (*.ascii)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.318) allow to set common parameters of control programs creation.

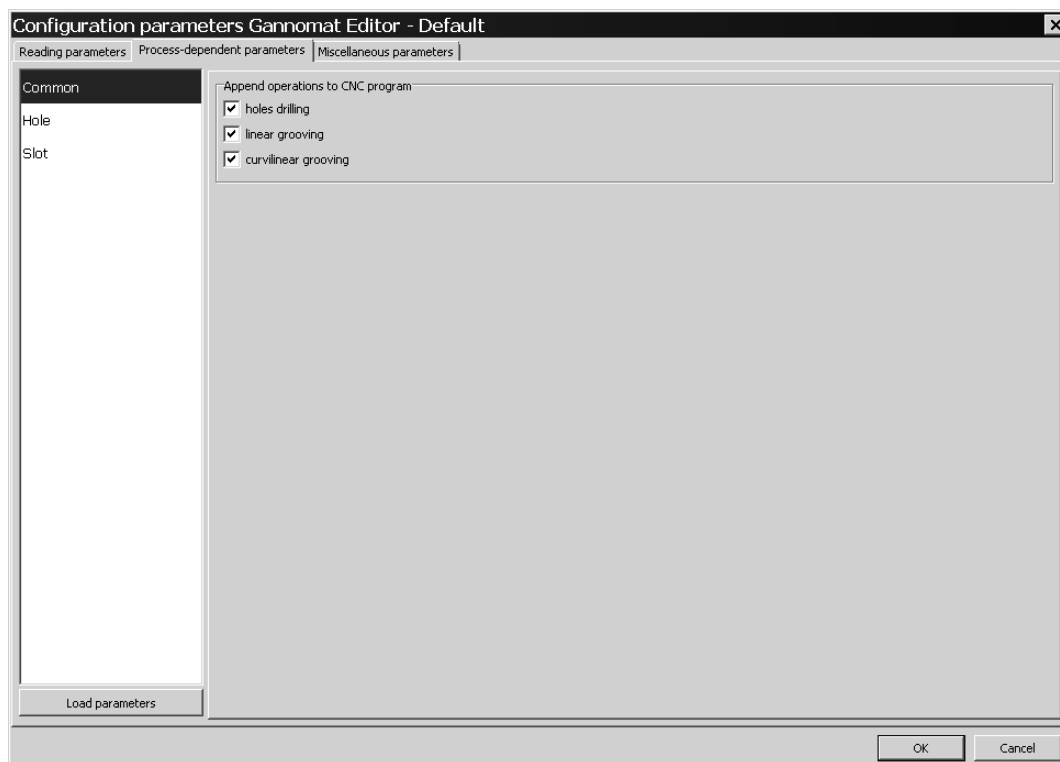


Fig. 10.318.

Append operations to CNC program group options allow to control the inclusion in control program of operations corresponding to the options names.

Hole parameters

Hole section control elements allow to set holes parameters (fig. 10.319).

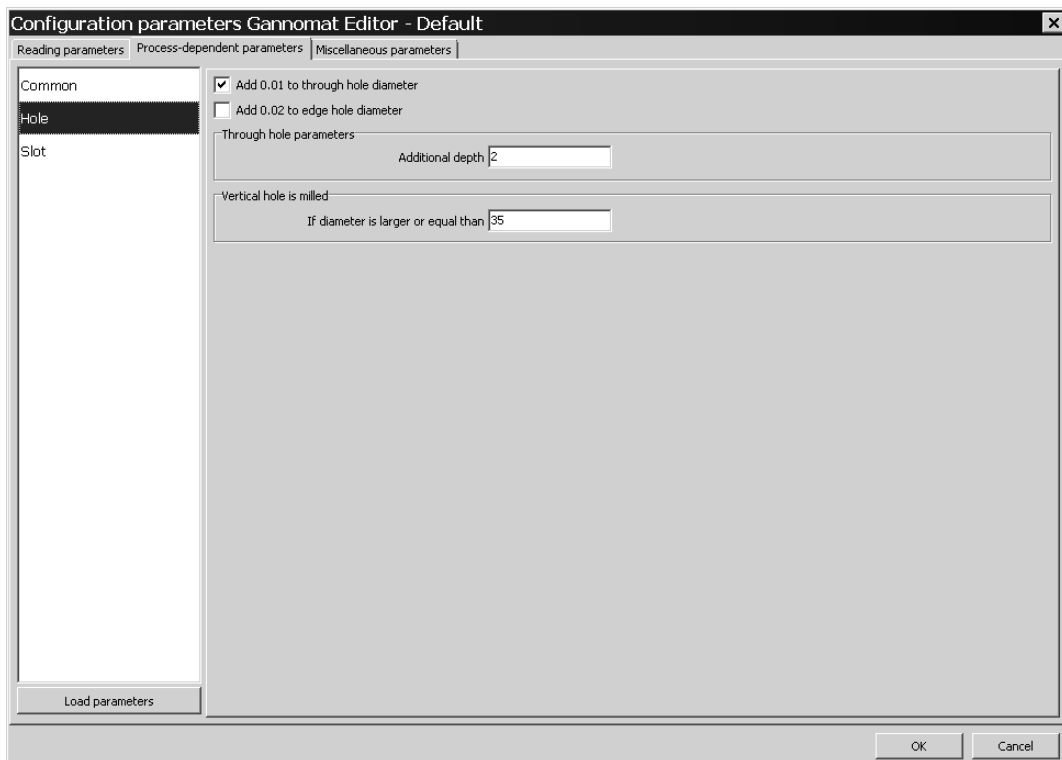


Fig. 10.319.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Depending on hole diameter it may be processed with a drill or a mill. **If diameter is more or equal** field in **Vertical hole is milled** group allows to set hole diameter that specifies tool selection.

Slot parameters

Slot section control elements (fig. 10.320) allow to set slots parameters.

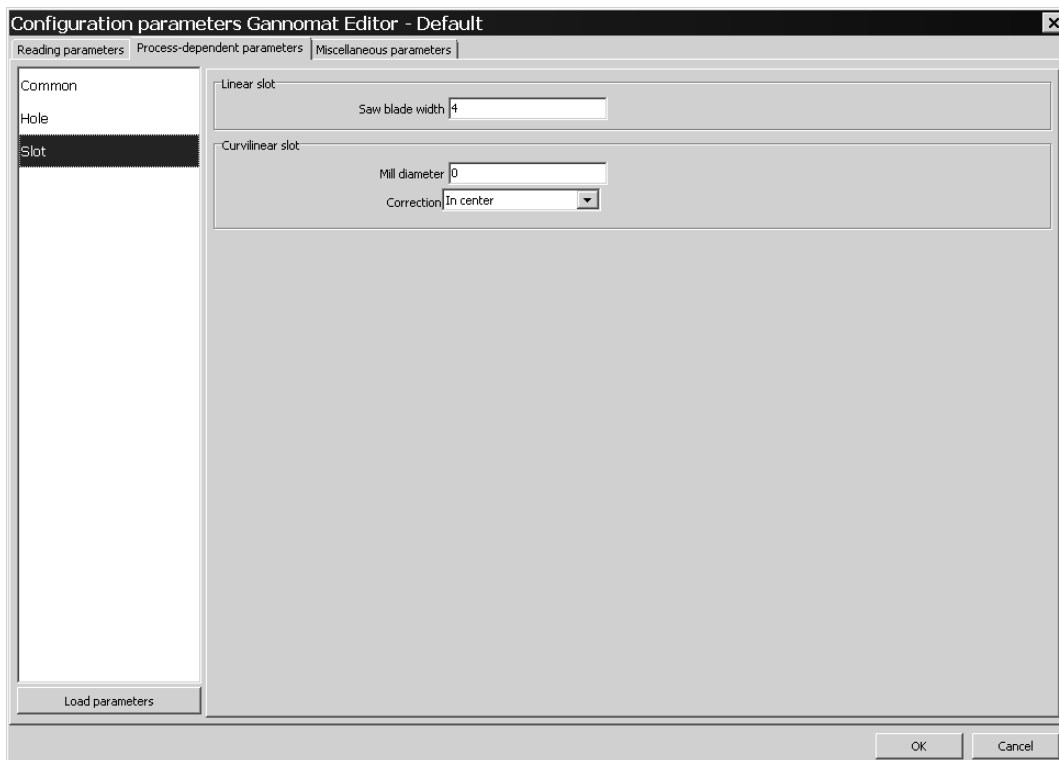


Fig. 10.320.

Saw blade width field allows to set saw blade width at linear slots processing. Mill diameter and correction type can be set for curvilinear slots.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.14. Giben

10.14.1. WoodCam



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common parameters are set in **Common** section (fig. 10.321) of **Process-dependent parameters** tab.

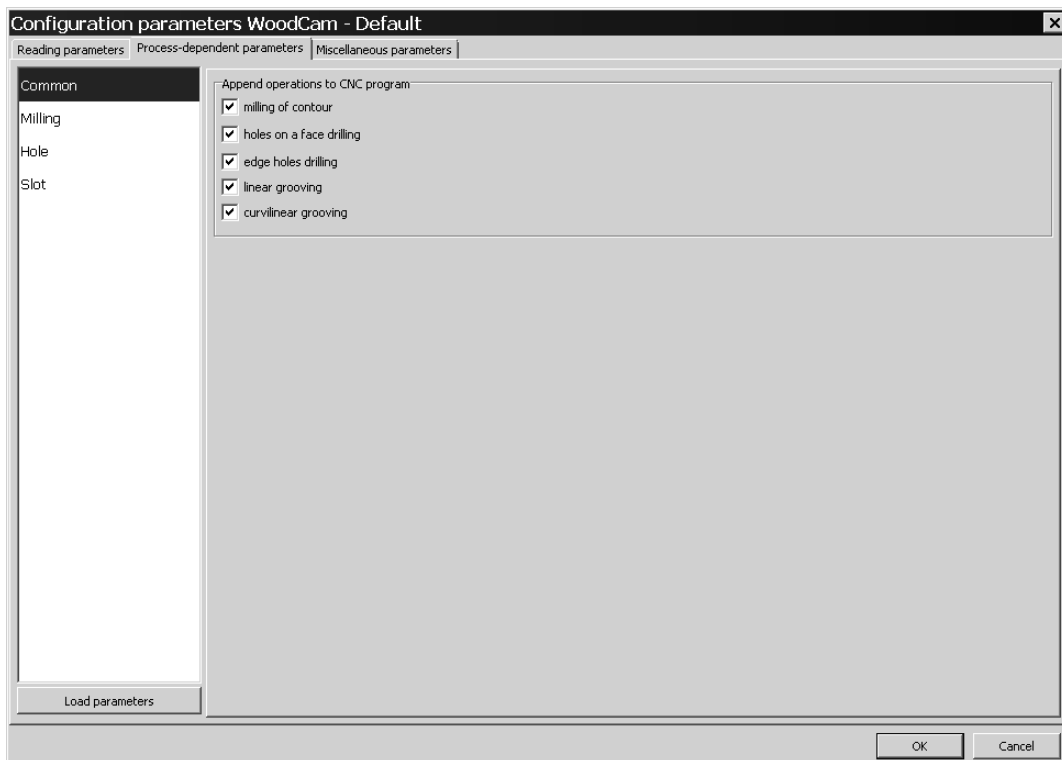


Fig. 10.321.

Tab control elements allow to choose operations of control program.

Milling parameters

Milling section control elements (fig. 10.322) allow to set contour milling parameters.

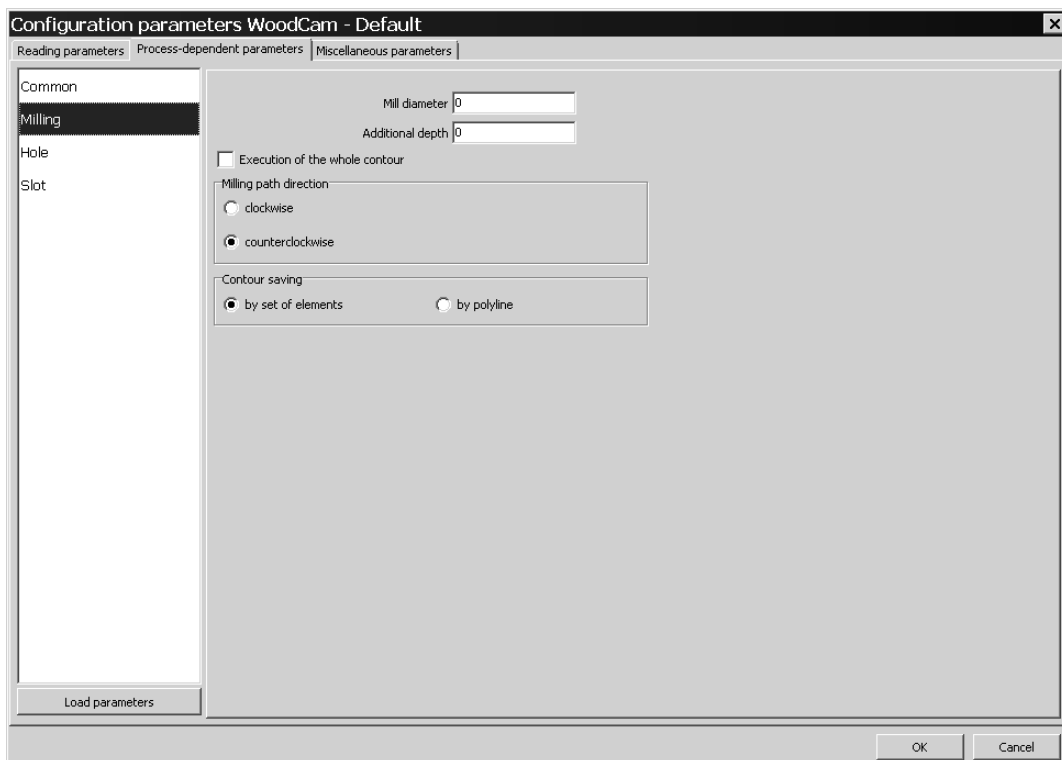


Fig. 10.322.

Section control elements allow to set milling parameters – mill diameter, milling direction, trajectory saving method, etc.

Hole parameters

Hole section control elements allow to set holes parameters (fig. 10.323).

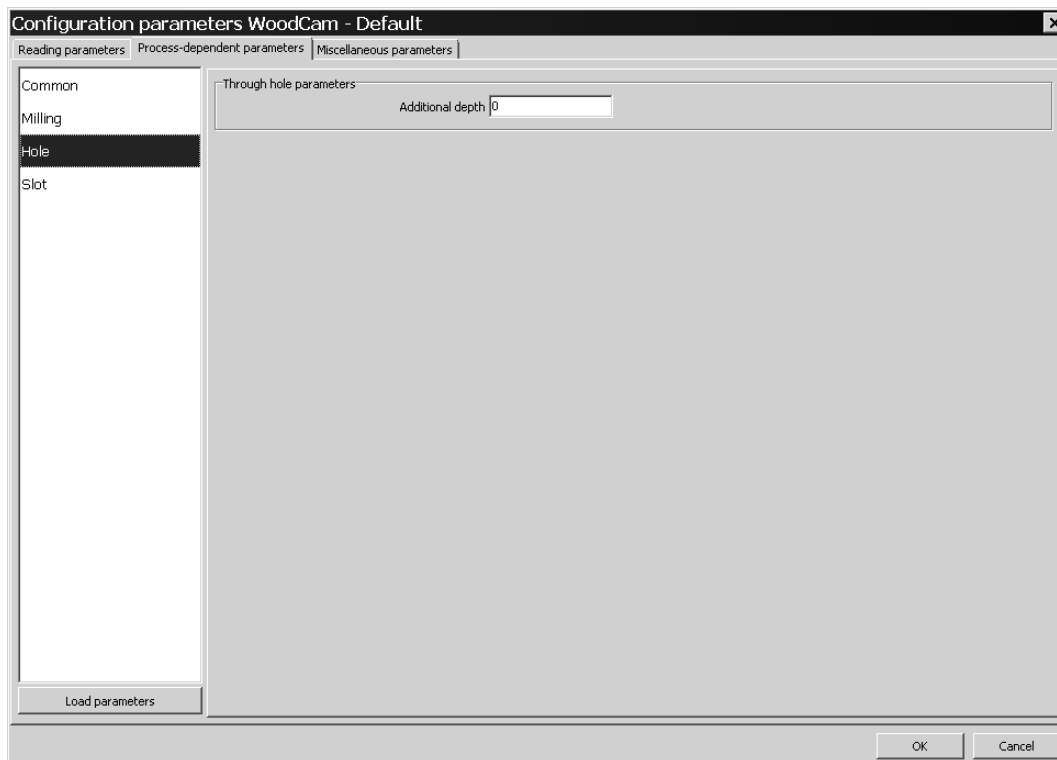


Fig. 10.323.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Slot parameters

Slot section control elements (fig. 10.324) allow to set slots processing parameters.

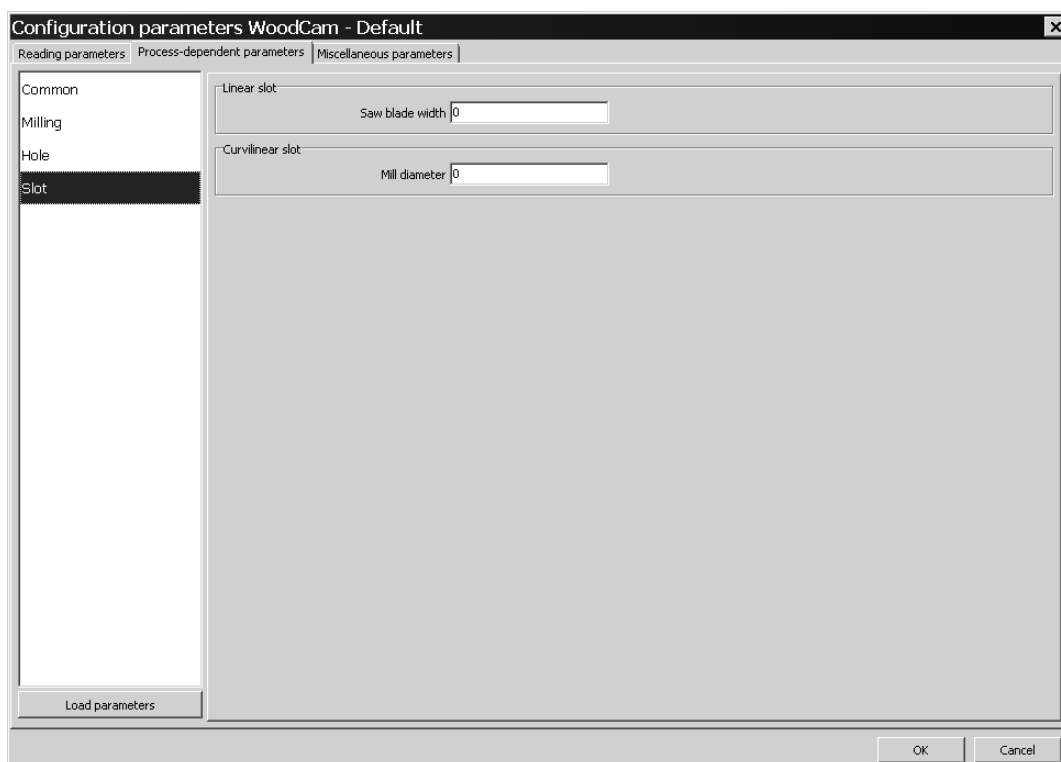


Fig. 10.324.

Saw blade width and **Mill diameter** fields allow to set the corresponding tools size for linear and curvilinear slots processing.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.15. Gorelovsky System



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.325) allow to specify operations that can be used in control program.

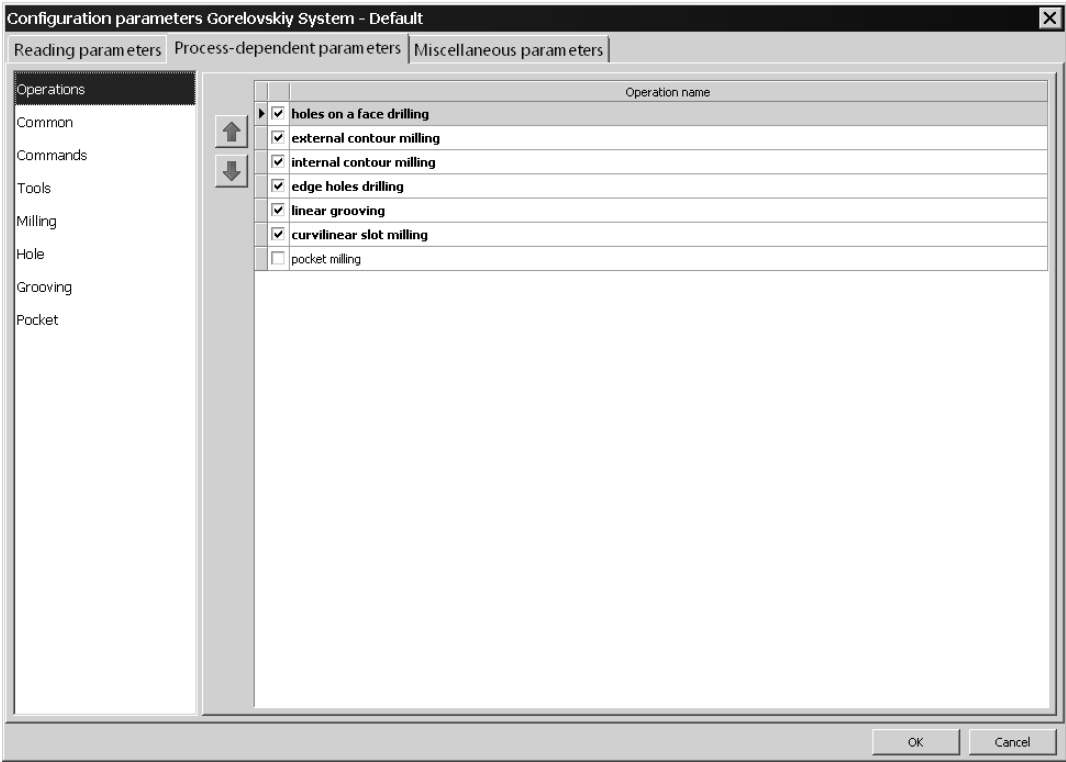


Fig. 10.325.

Common parameters

Common section control elements (fig. 10.326) allow to set common parameters of control programs creation.

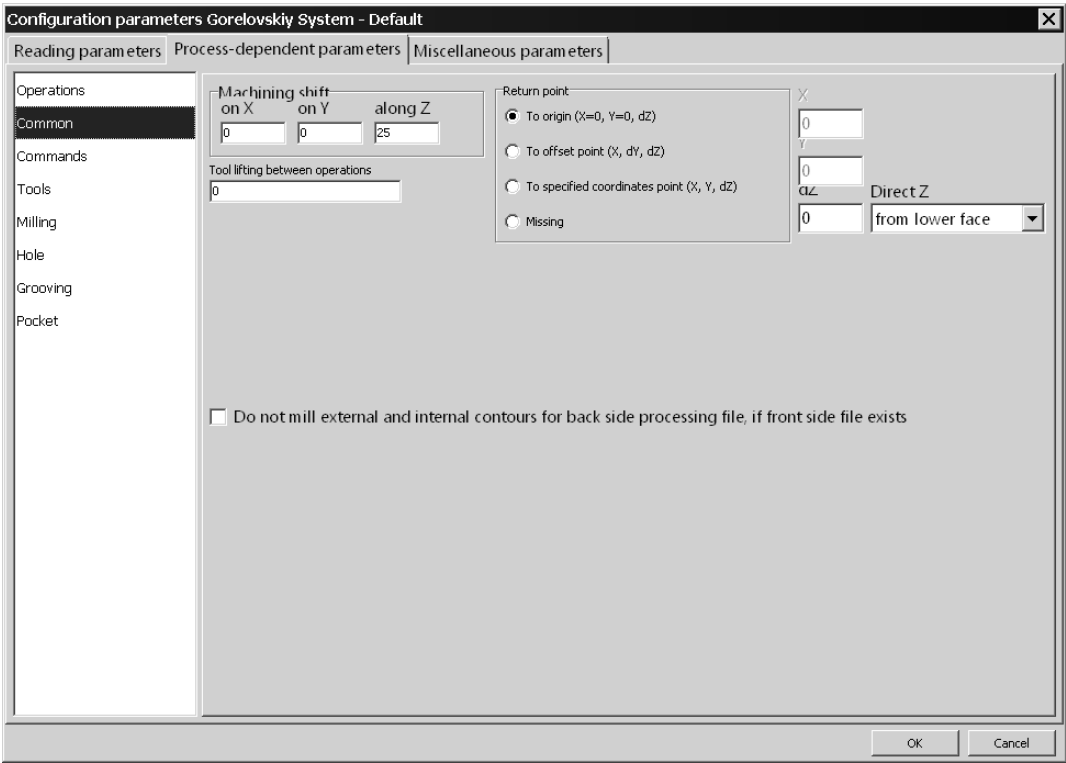


Fig. 10.326.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** variant is selected, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height.

Usable commands

Commands section control elements (fig. 10.327) allow to control separate commands to control program inclusion.

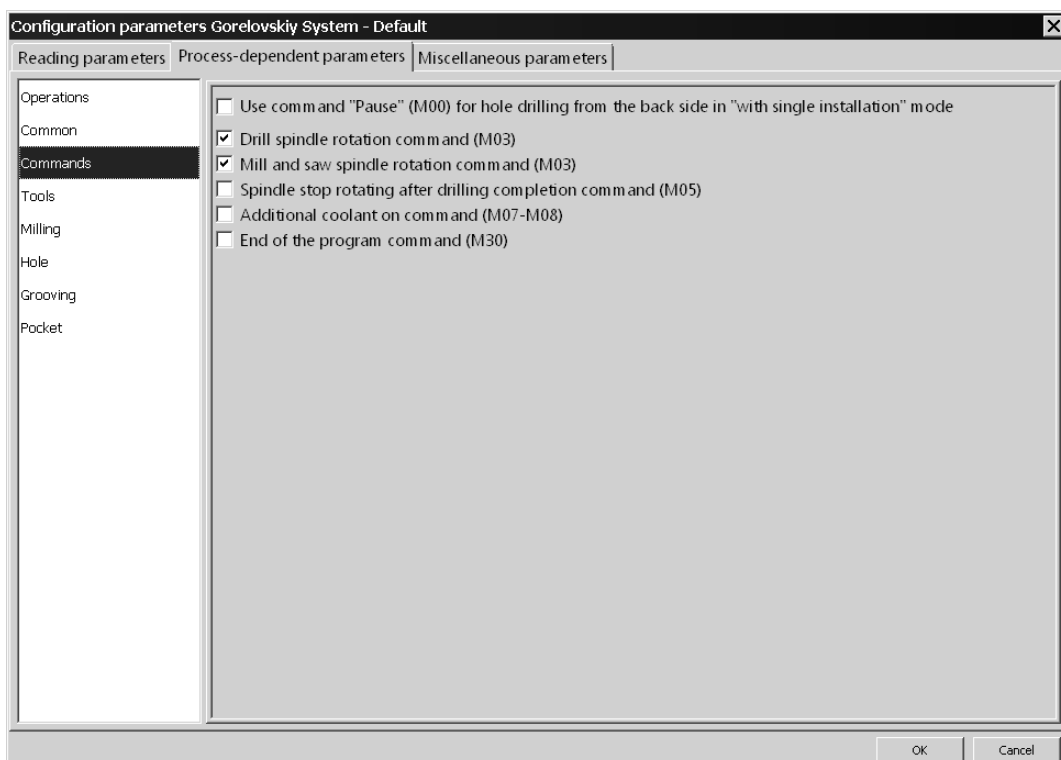


Fig. 10.327.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.328).

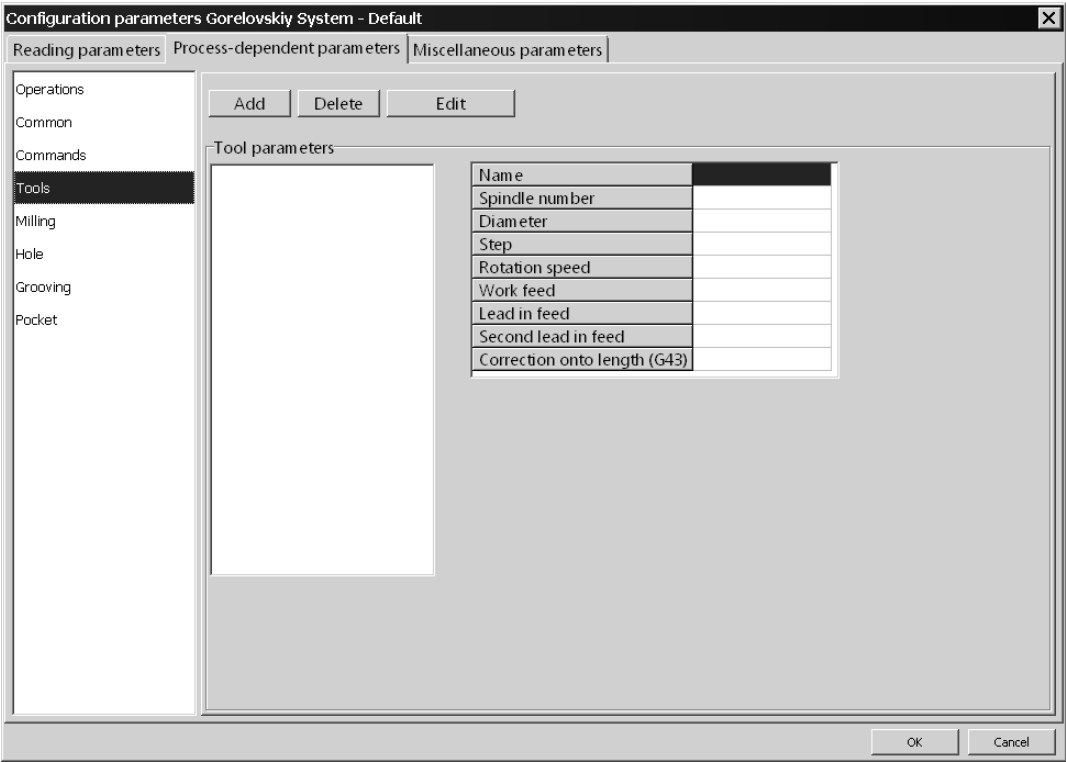


Fig. 10.328.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.329).

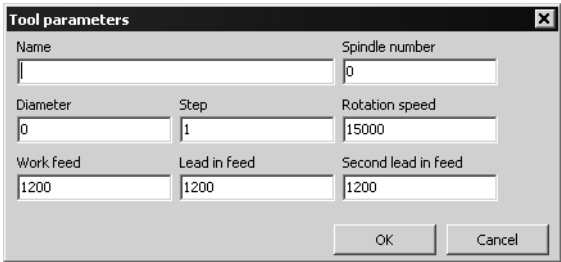


Fig. 10.329.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is in fig. 10.330.

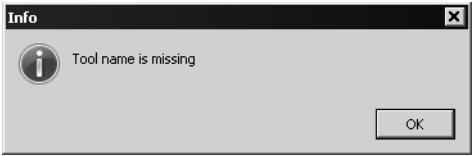


Fig. 10.330.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.329). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Milling** tab (fig. 10.331).

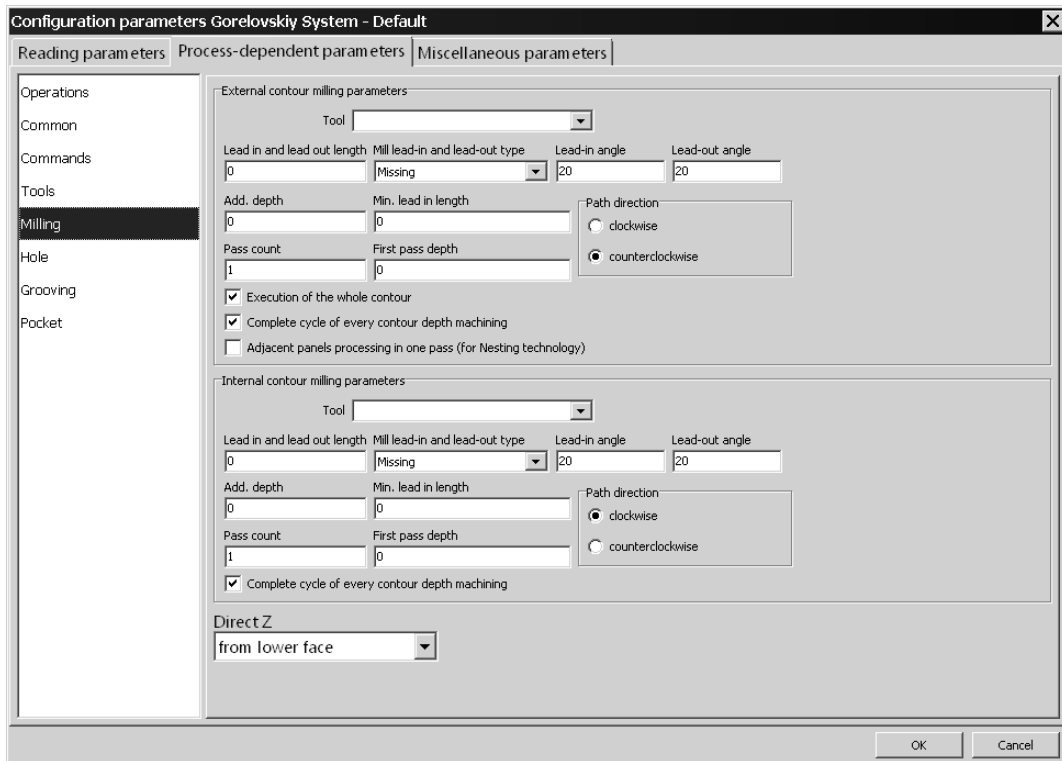


Fig. 10.331.

Tool combo box allows to select a milling tool from the list created in **Tools** tab. Tool lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Holes processing setup is performed in **Hole** section (fig. 10.332).

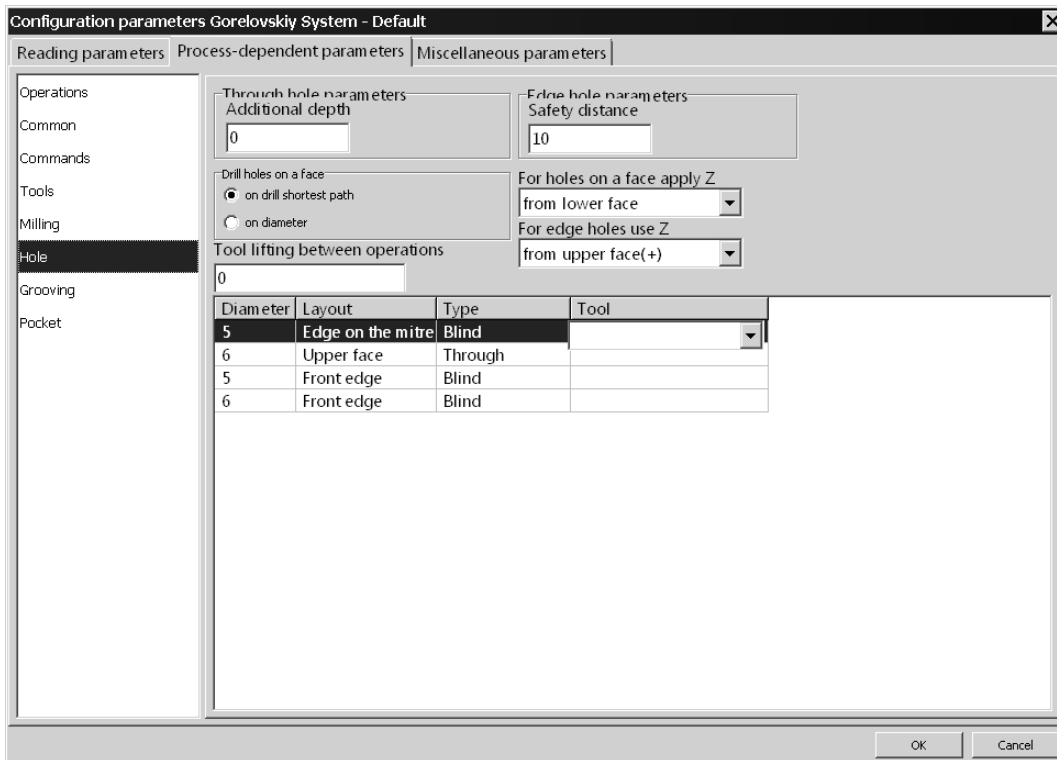


Fig. 10.332.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

Drill holes on a face group table allows to set automatic spindle number selection depending on hole parameters.

On drill shortest path and **on diameter** variants allow to set holes boring order. If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slot parameters

Grooving section control elements (fig. 10.333) allow to set slots processing parameters.

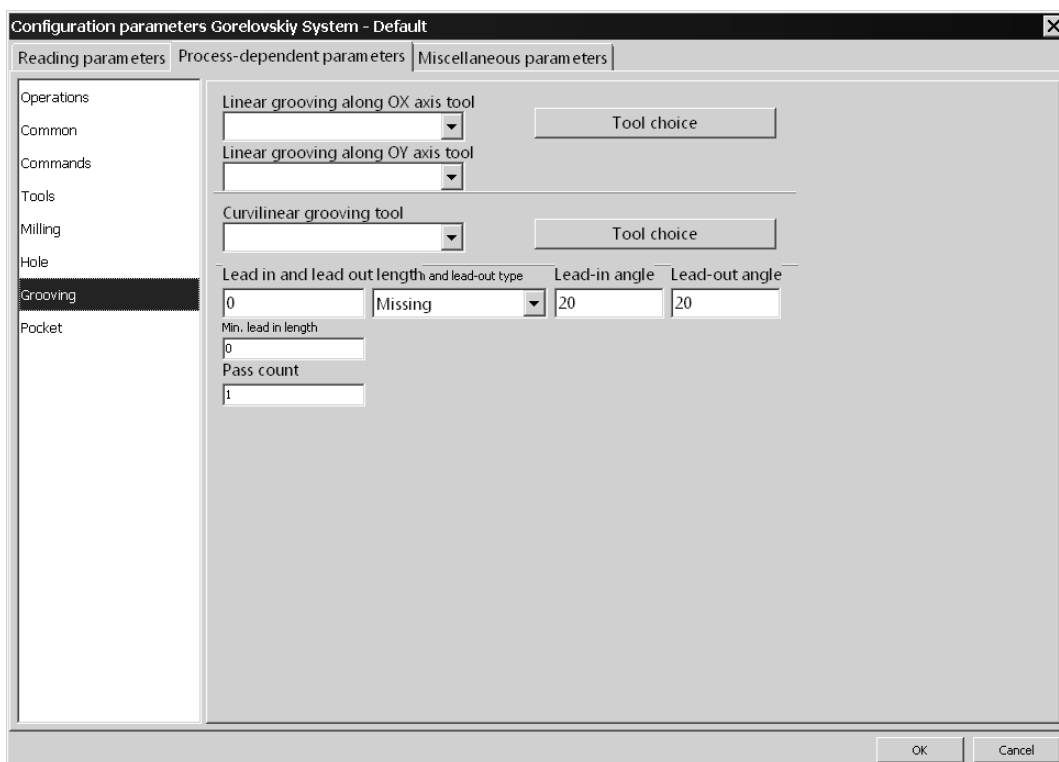


Fig. 10.333.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button, **Tool choice** dialogue will appear on the screen (fig. 10.334).

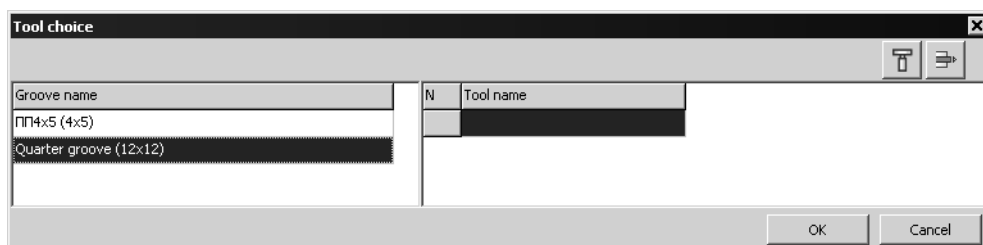


Fig. 10.334.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to map selected slot with a tool from the list, created in **Tools** section.



Delete button allows to delete selected tool from the list.

To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.335) allow to set pocket processing parameters.

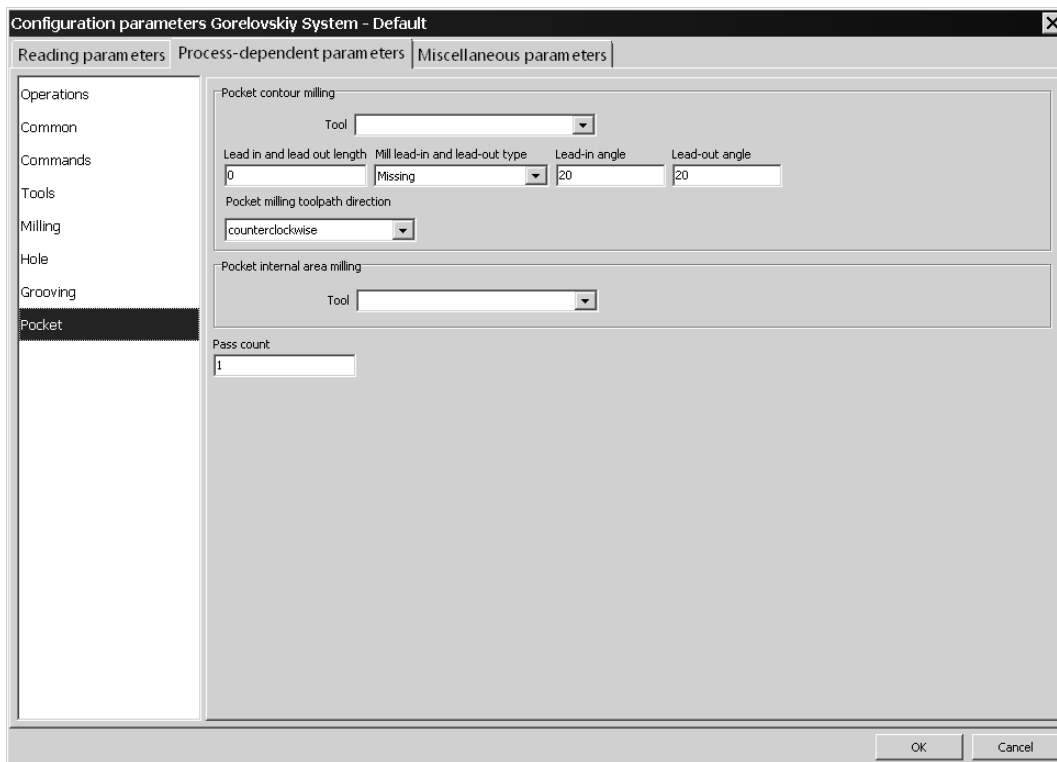


Fig. 10.335.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.16. Griggio

10.16.1.G-Cad Lepton



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.336) allow to set slots and through holes parameters.

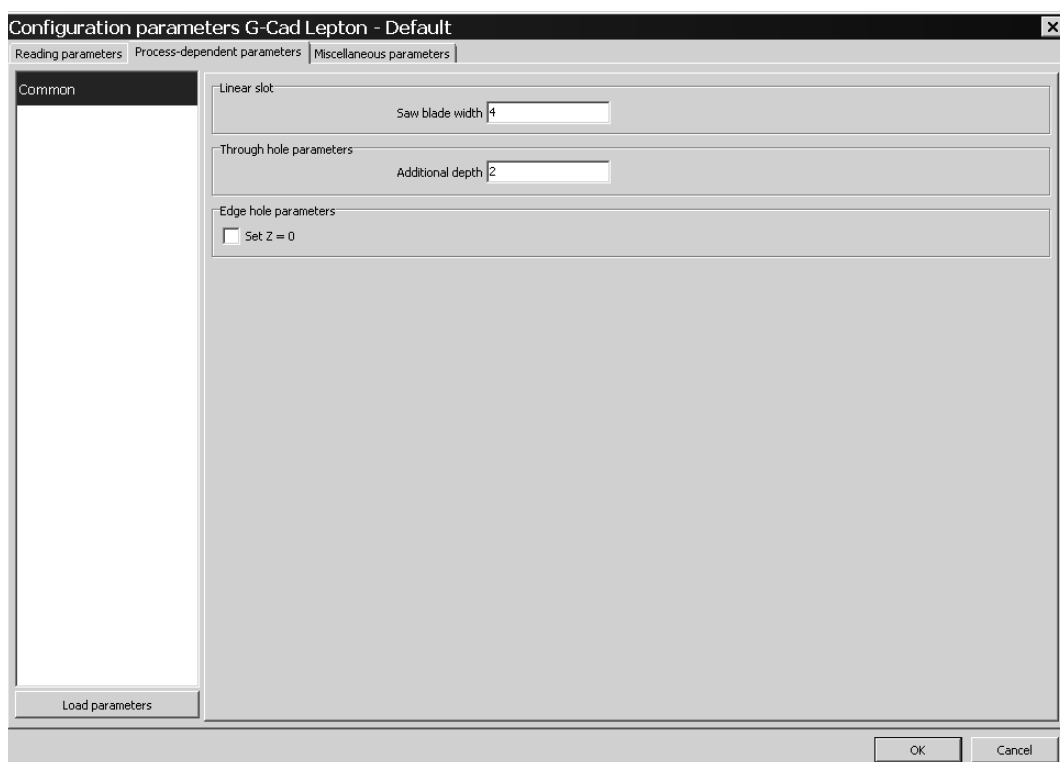


Fig. 10.336.

Saw blade width field allows to set saw blade width at linear slots processing. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.16.2.Griggio TPA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.337) allow to specify operations that can be used in control program.

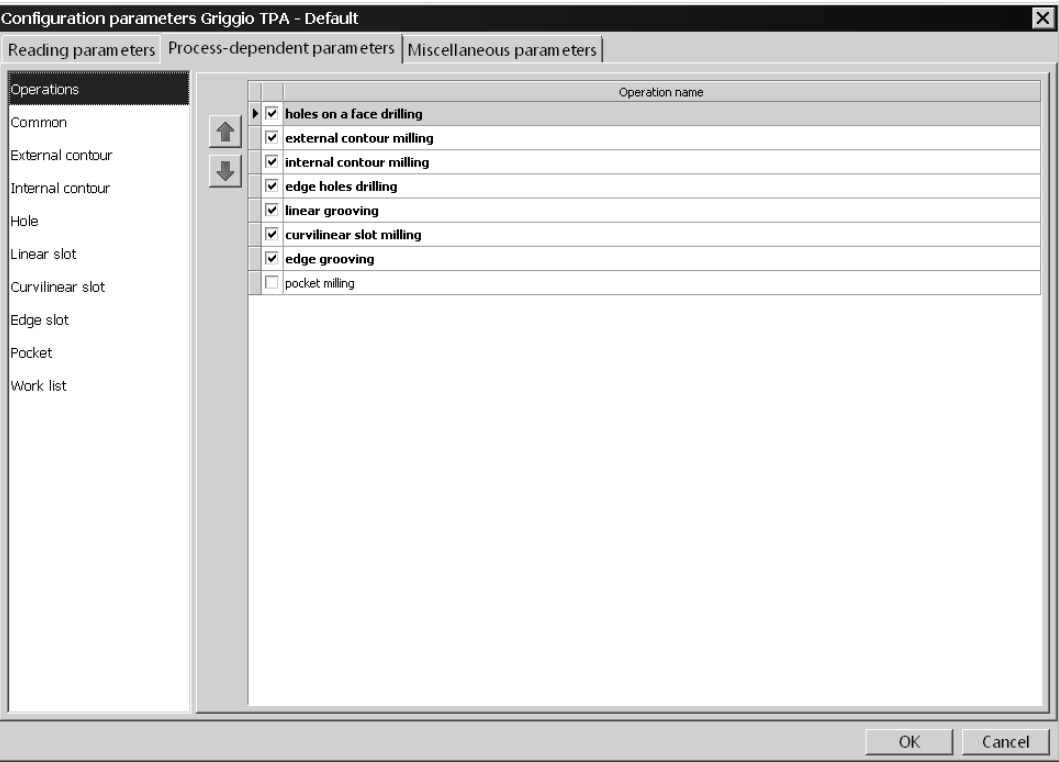


Fig. 10.337.

Common parameters

Common section control elements (fig. 10.338) allow to set common parameters.

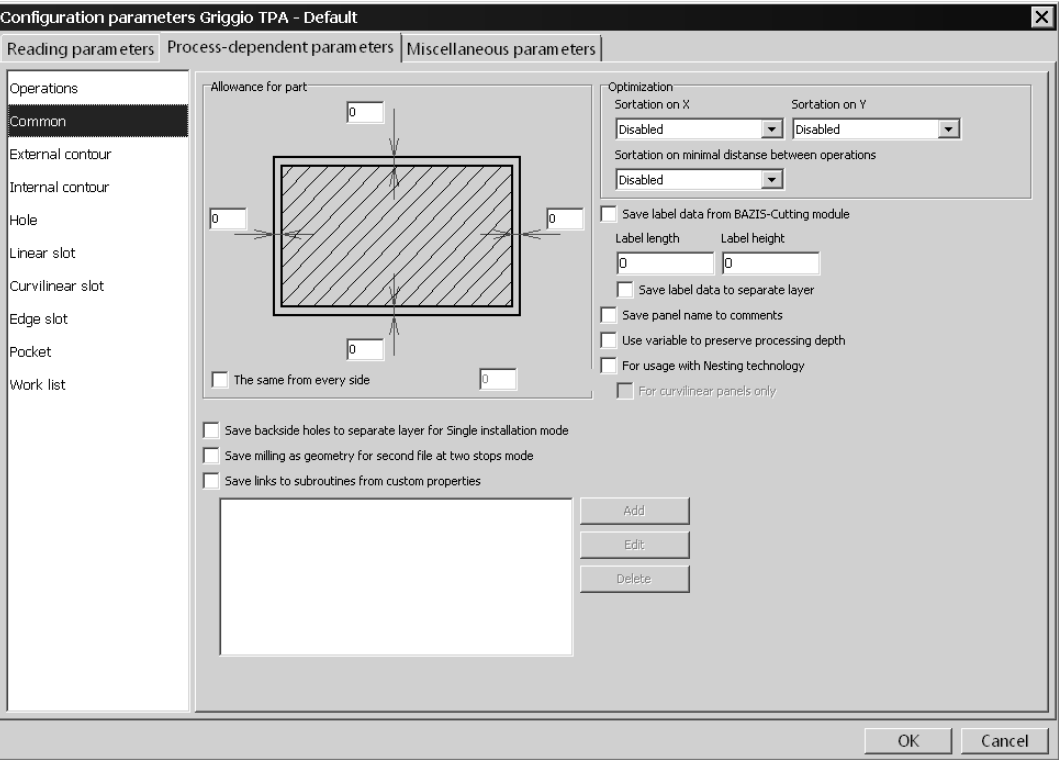


Fig. 10.338.

Tab control elements allow to control saving of optimization parameters, label data, etc. **Allowance for part** section control elements allow to set allowance values.

Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, the same from every panel edge allowance field will become available.

Contours parameters

Contours milling parameters are set in **Outer contour** (fig. 10.339) and **Internal contour** sections.

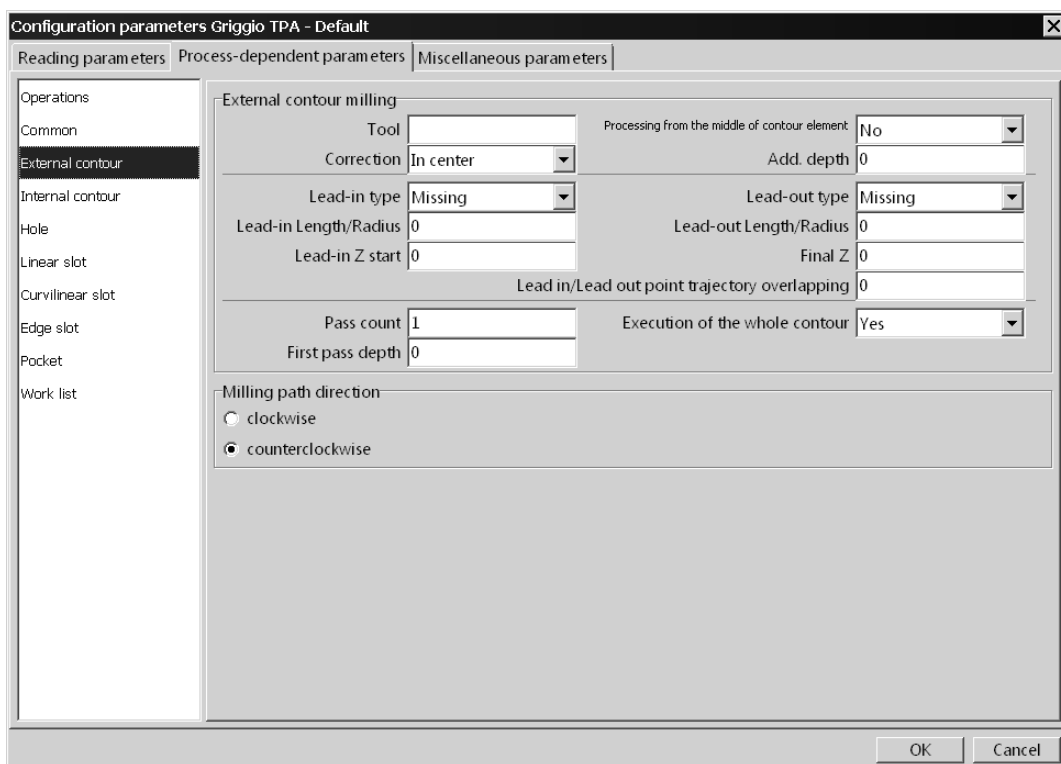


Fig. 10.339.

Finish machining and **Rough machining** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Rough machining** option allows to control this operation.

Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Holes section control elements (fig. 10.340) allow to setup parameters of drilling operations.

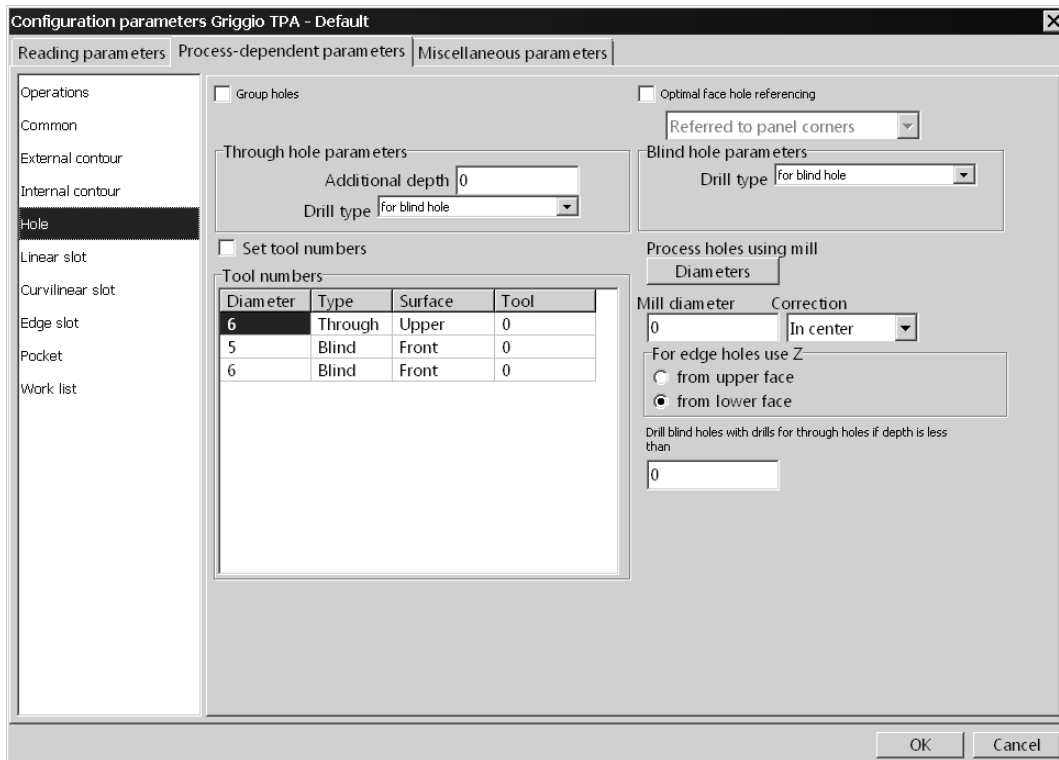


Fig. 10.340.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Macros files names should be selected from a combo box.

Through holes can be bored with drills for through and blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.341).

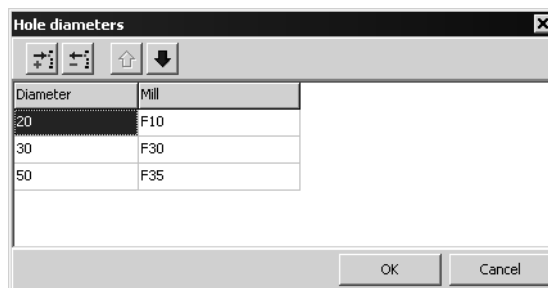


Fig. 10.341.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To finish setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter application.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.342).

Fig. 10.342.

Through slot always cut with saw and **Grooving by saw** options allow to set through slots direction that will be processed with a saw. **Tool choice** button allows to set tools for particular slots processing. After pushing the button, **Tool choice** dialogue will appear on the screen (fig. 10.343).

N	Tool name	Correction	Diameter
1	D15	At a center	0

Fig. 10.343.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.344).

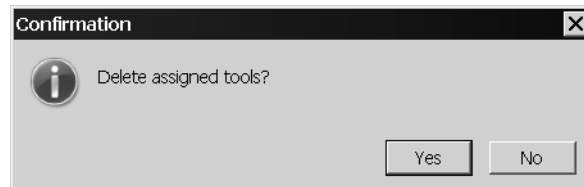


Fig. 10.344.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. To finish parameters setup push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.345).

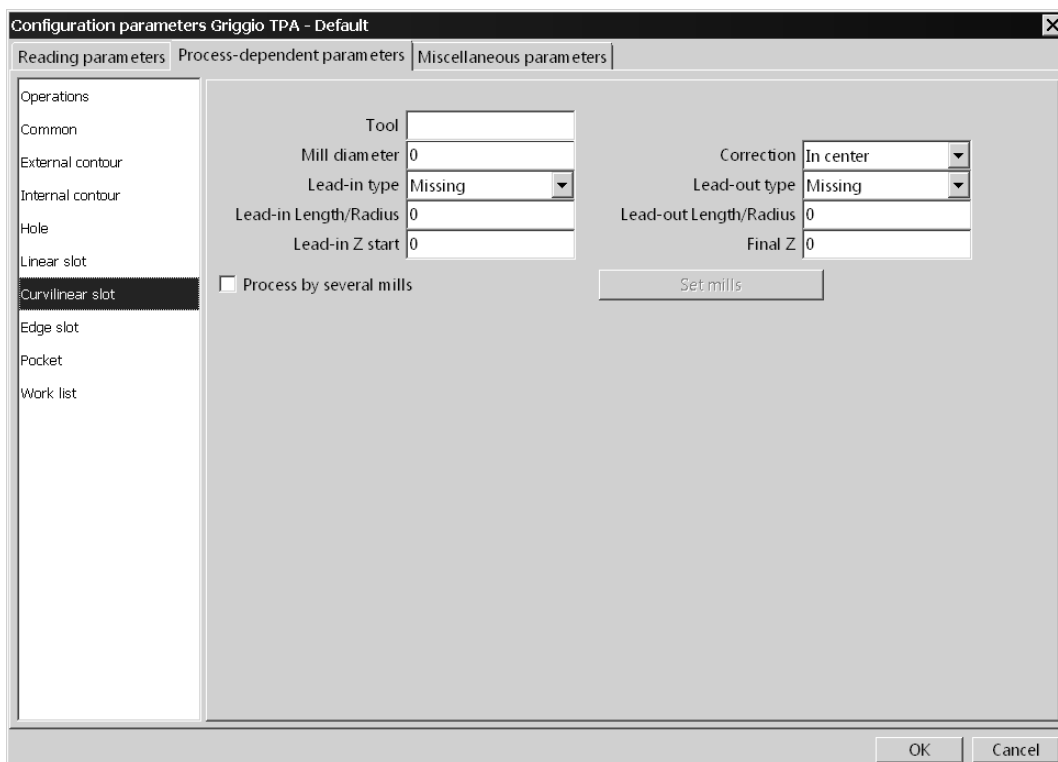


Fig. 10.345.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button, **Tool choice** dialogue will appear on the screen (fig. 10.346).

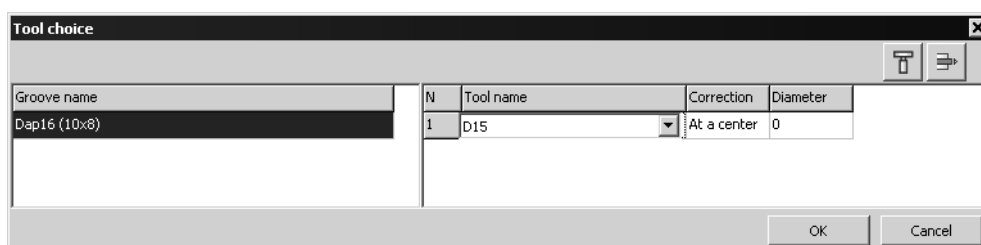


Fig. 10.346.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.347).

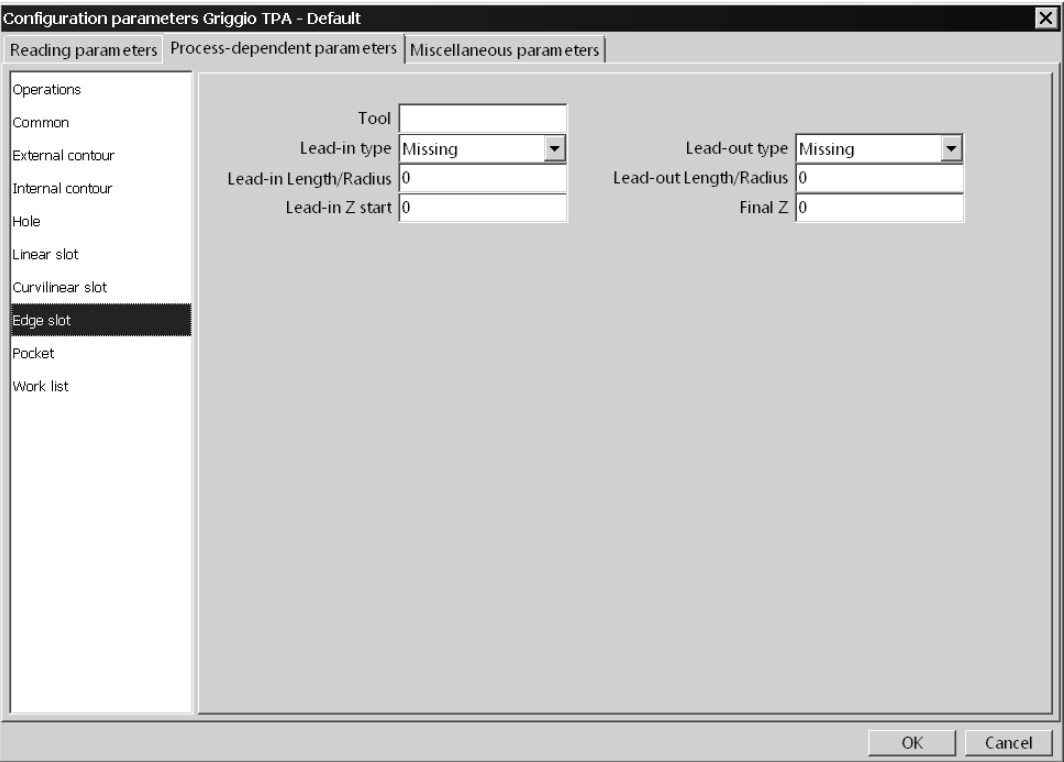


Fig. 10.347.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.348) allow to set pocket processing parameters.

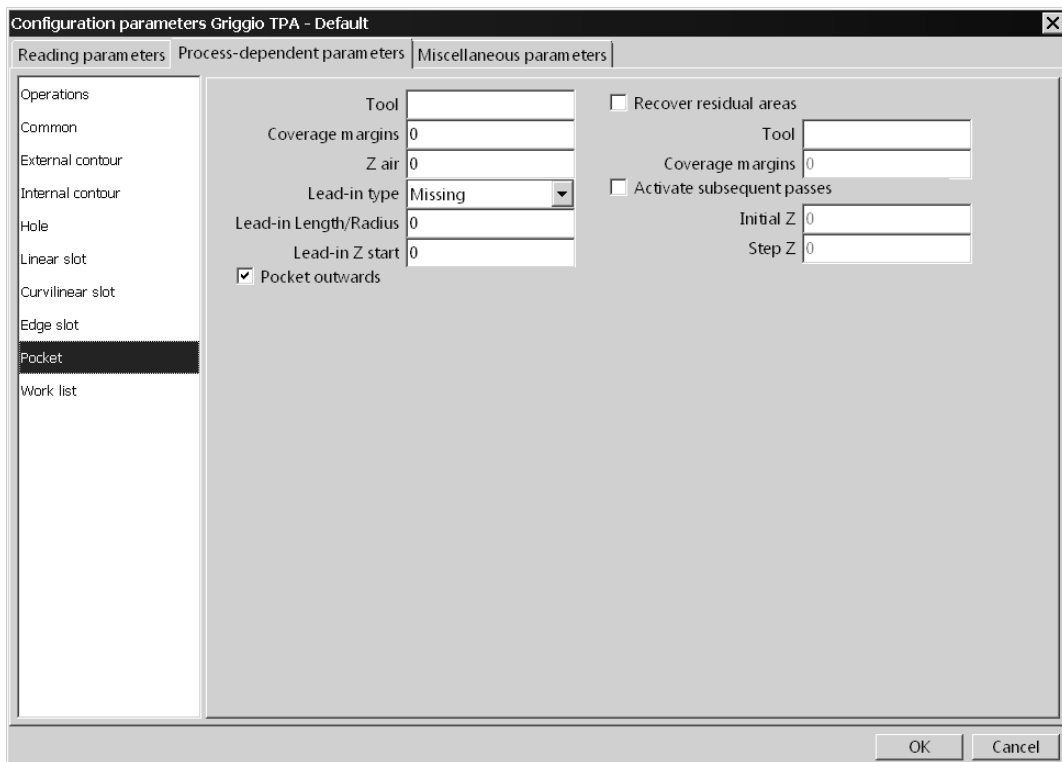


Fig. 10.348.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Work lists

Work list section control elements allow to set parameters of work lists use (fig. 10.349).

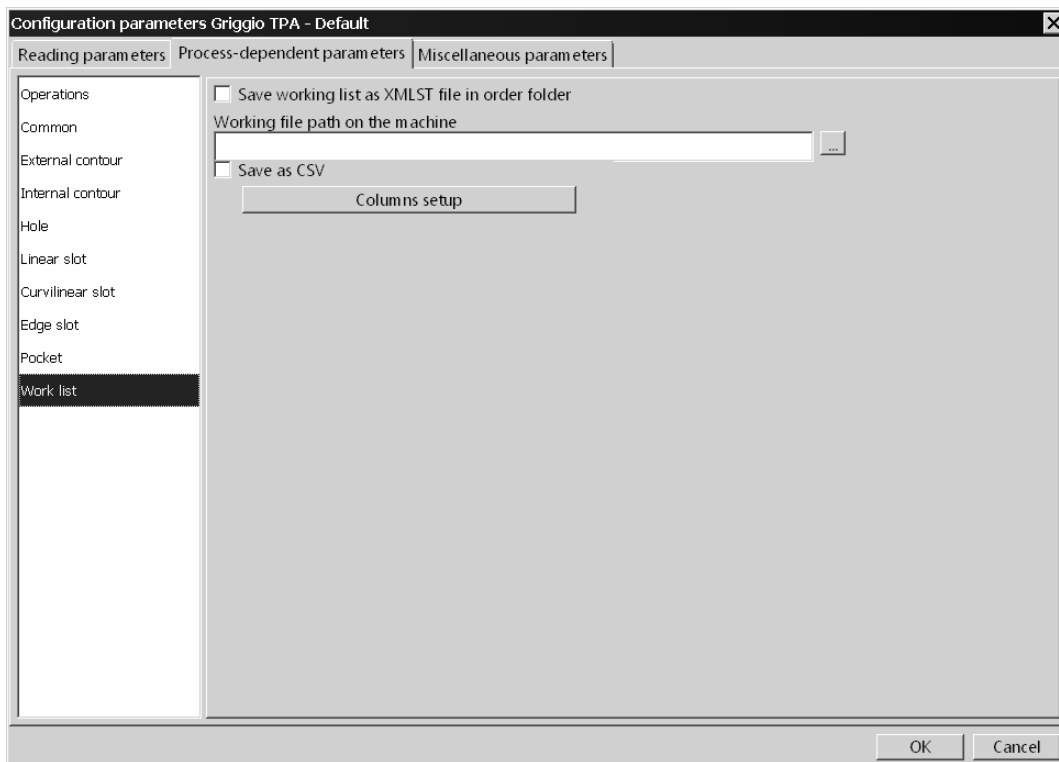


Fig. 10.349.



Save working list as XMLST file in order folder option allows to control the corresponding document creation. **Browse** button allows to set a path to working files. After pushing the button **Browse** dialogue will appear on the screen.

Setup completion

To complete control programs for Griggio TPA setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.17. Hirzt

10.17.1. Hirzt TPA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.350) allow to specify the operations that can be used in control program.

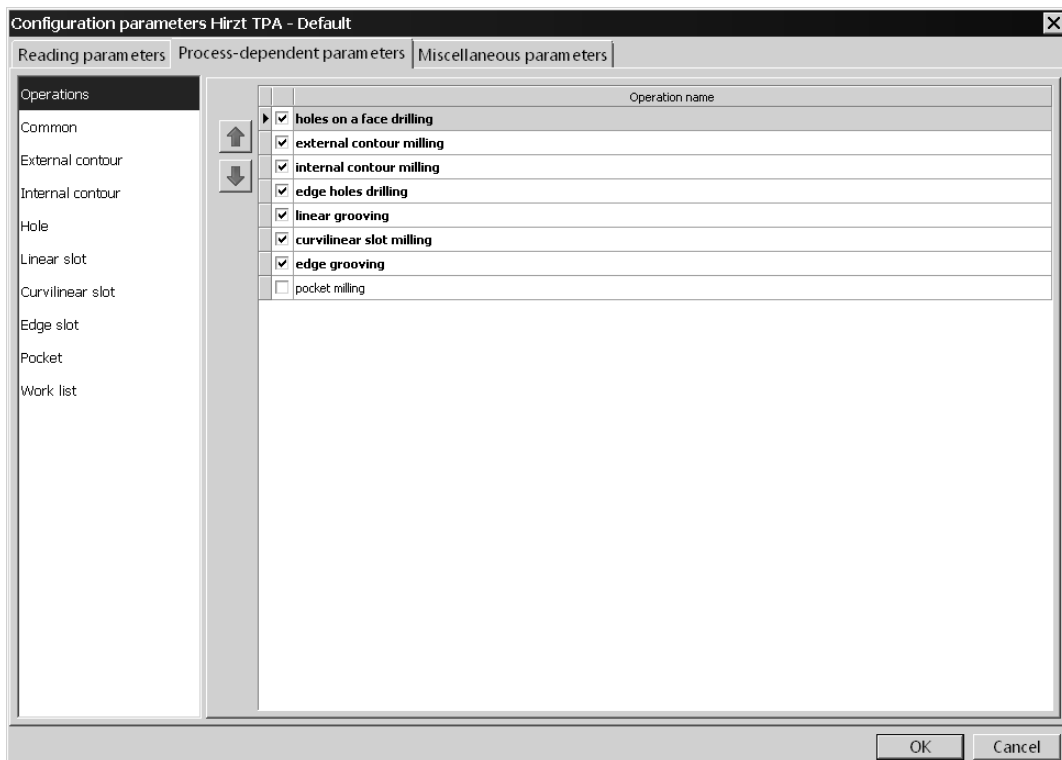


Fig. 10.350.

Common parameters

Common parameters are set in **Common** section (fig. 10.351).

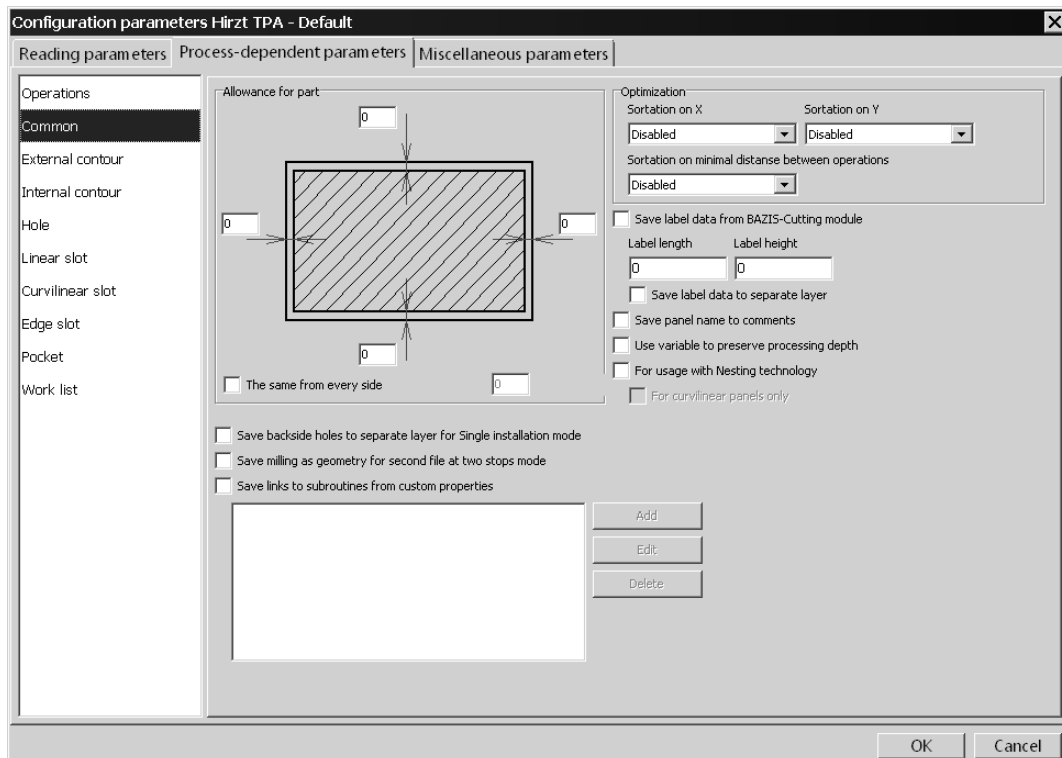


Fig. 10.351.

Tab control elements allow to control saving of optimization parameters, label data, etc. **Allowance for part** section control elements allow to set allowance values. Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, the same from every panel edge allowance field will become available.

Contour parameters

Contour milling parameters setup is performed in **Outer contour** (fig. 10.352) and **Internal contour** sections.

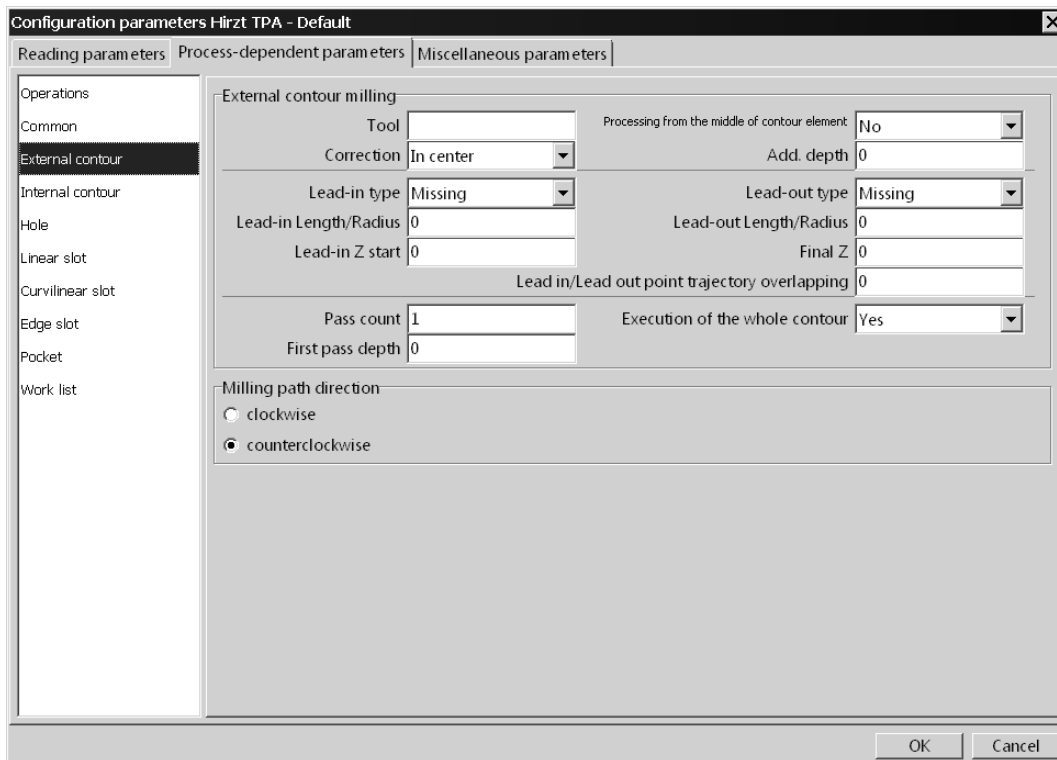


Fig. 10.352.

Finish machining and **Rough machining** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Rough machining** option allows to control this operation.

Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Holes section control elements (fig. 10.353) allow to setup parameters of drilling operations.

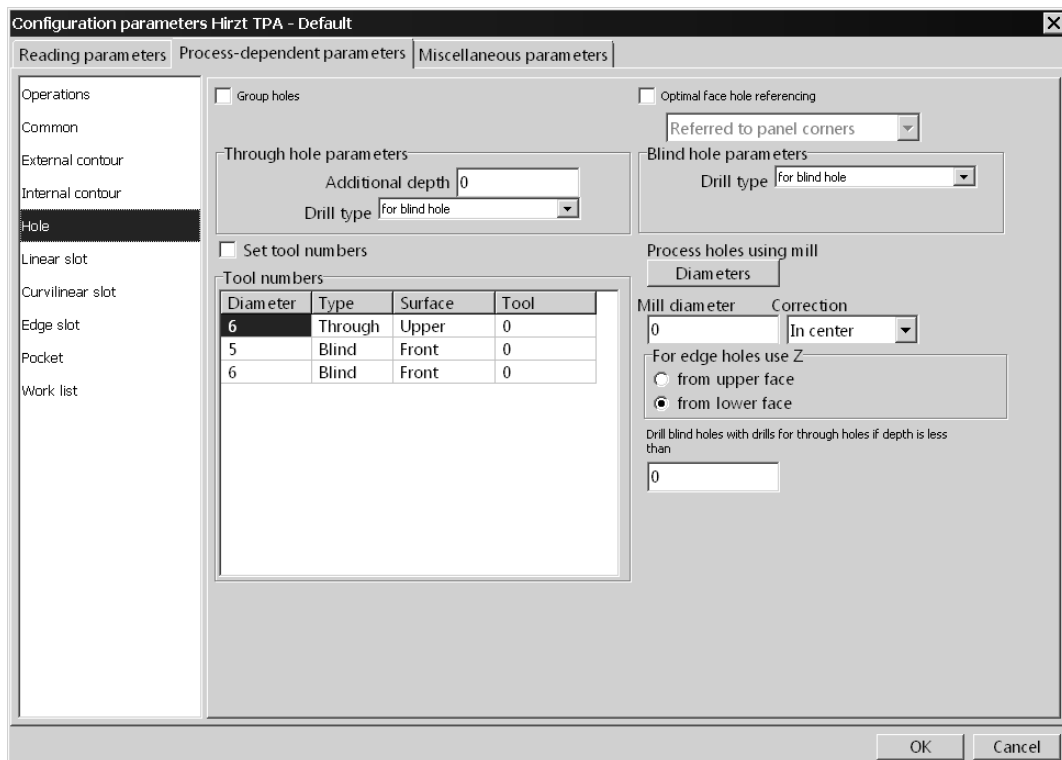


Fig. 10.353.

If several holes are located on the same straight line and the distance between them (step) is the same, it is possible to combine them and reduce the program volume. For this purpose enable **Group holes using macro** option. Macros files names should be selected from a combo box.

Through holes can be drilled with drills for through as well as for blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameter** button. **Hole diameters** dialogue will appear on the screen (fig. 10.354).

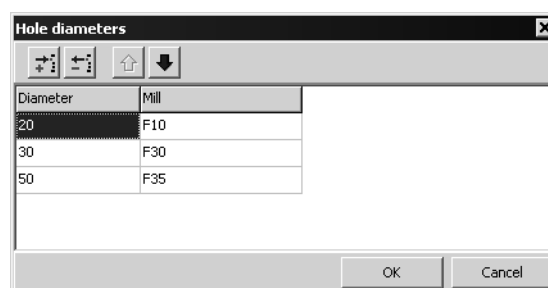


Fig. 10.354.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to edit the current row position in the table. To complete setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter.

Linear slot parameters

Linear slot parameters setup is performed in **Linear slot** section (fig. 10.355).

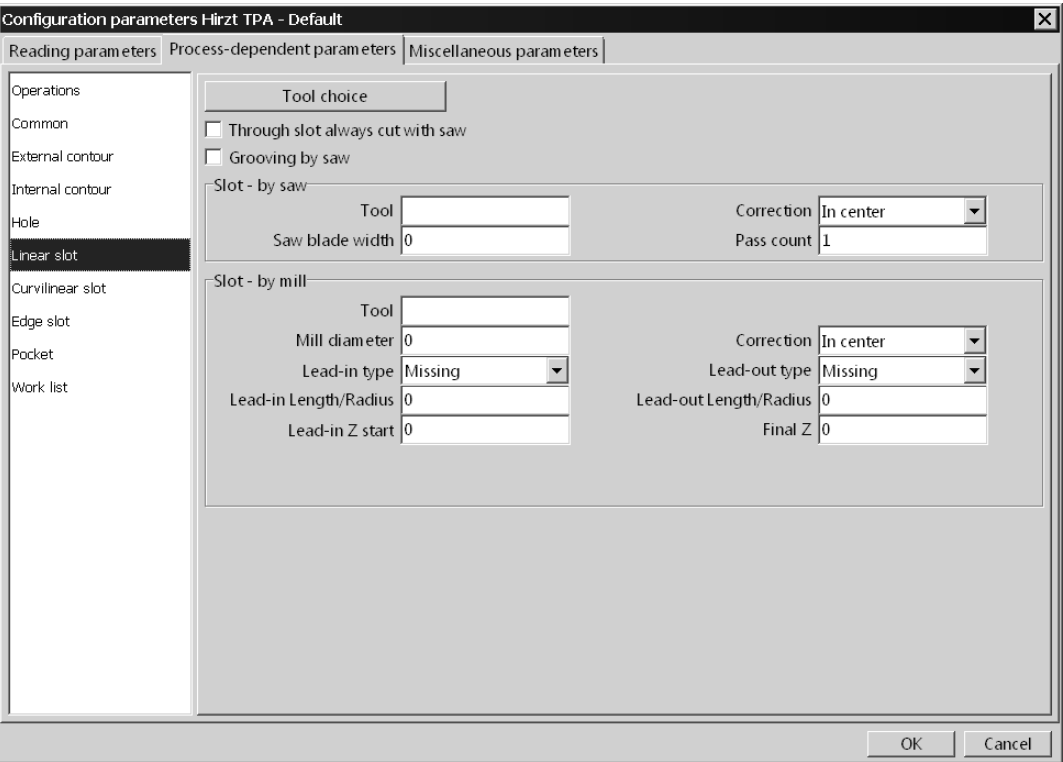


Fig. 10.355.

Through slot always cut with saw and **Grooving by saw** options allow to set through slots direction that will be processed with a saw. **Tool choice** button allows to set tools for particular slots processing. After pushing the button, **Tool choice** dialogue will appear on the screen (fig. 10.356).

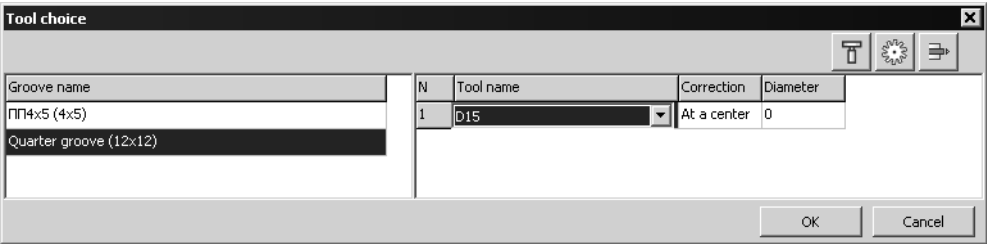


Fig. 10.356.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.357).

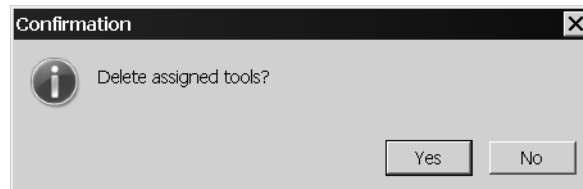


Fig. 10.357.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. To complete parameters setup, push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear slots parameters setup is performed in **Curvilinear slot** section (fig. 10.358).

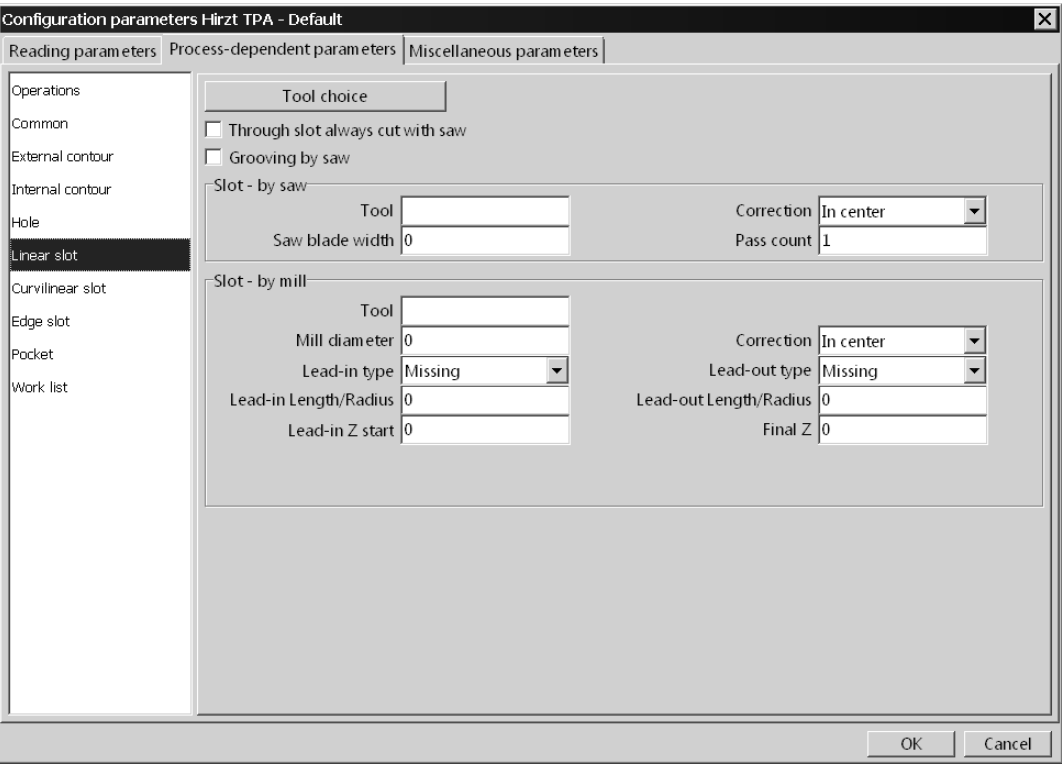


Fig. 10.358.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.359).

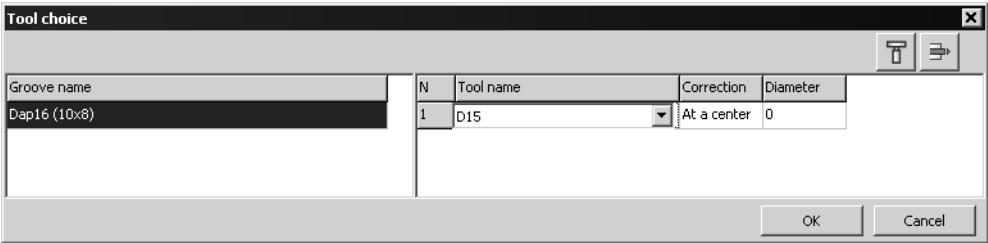


Fig. 10.359.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Cancel button allows to delete selected tool from the list. To complete tool assignment, push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slot parameters setup is performed in **Edge slot** section (fig. 10.360).

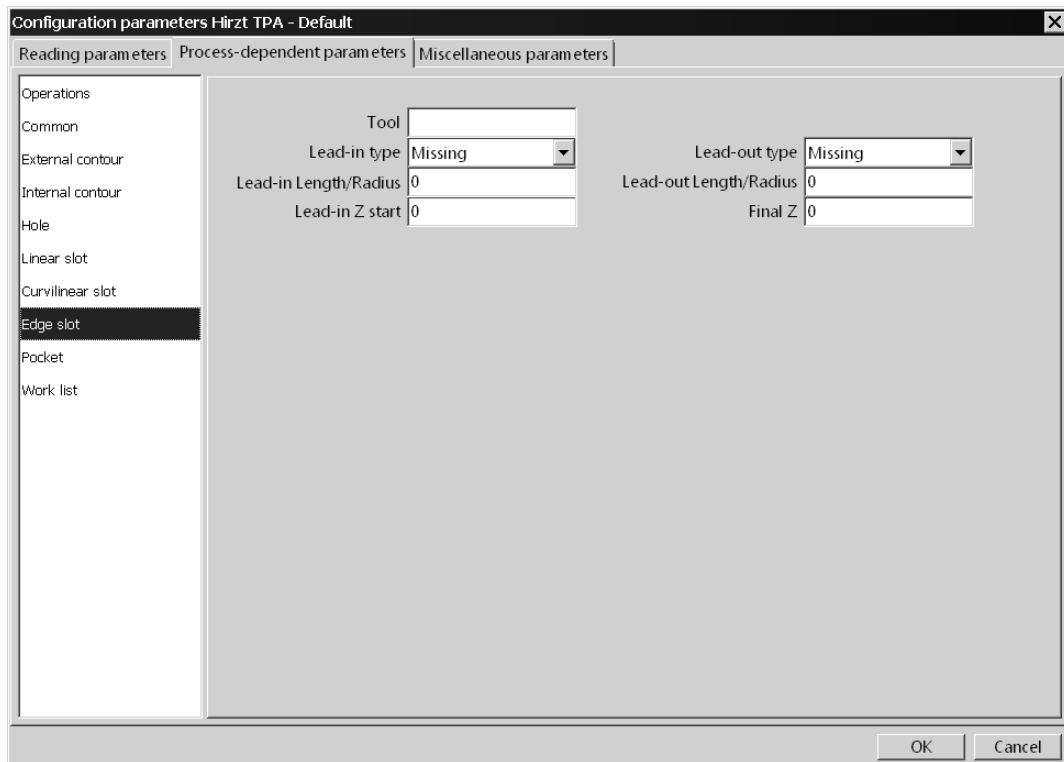


Fig. 10.360.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.361) allow to setup pocket processing parameters.

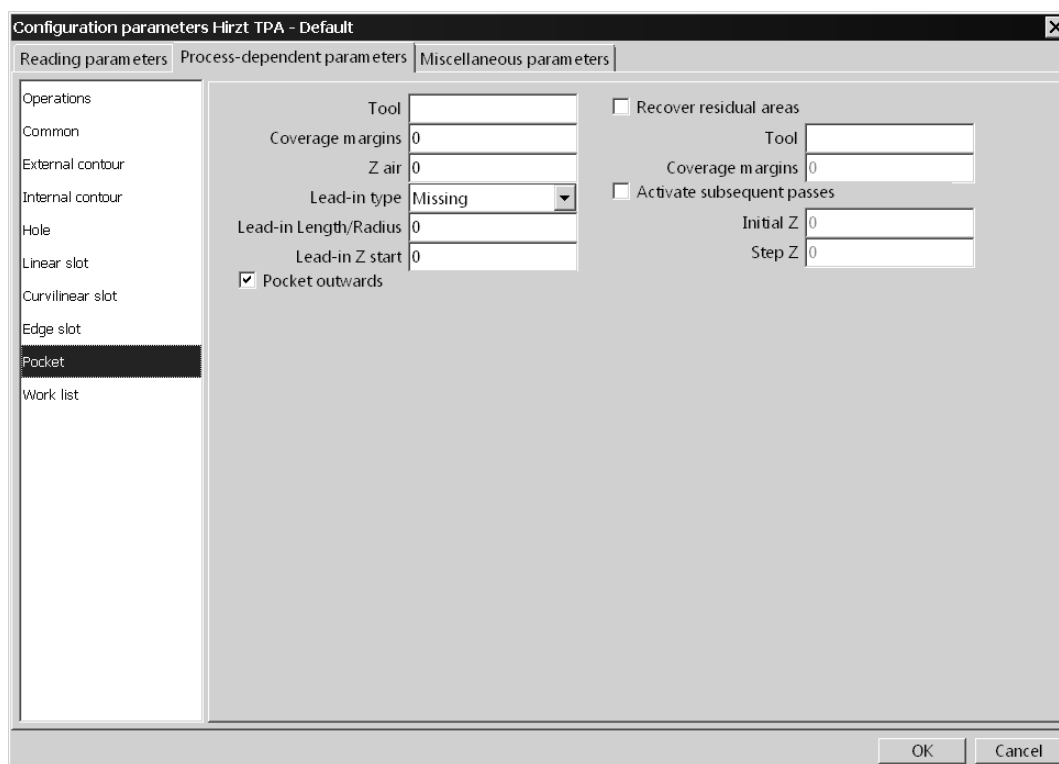


Fig. 10.361.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Work lists

Work list section control elements (fig. 10.362) allow to setup parameters of lists use.

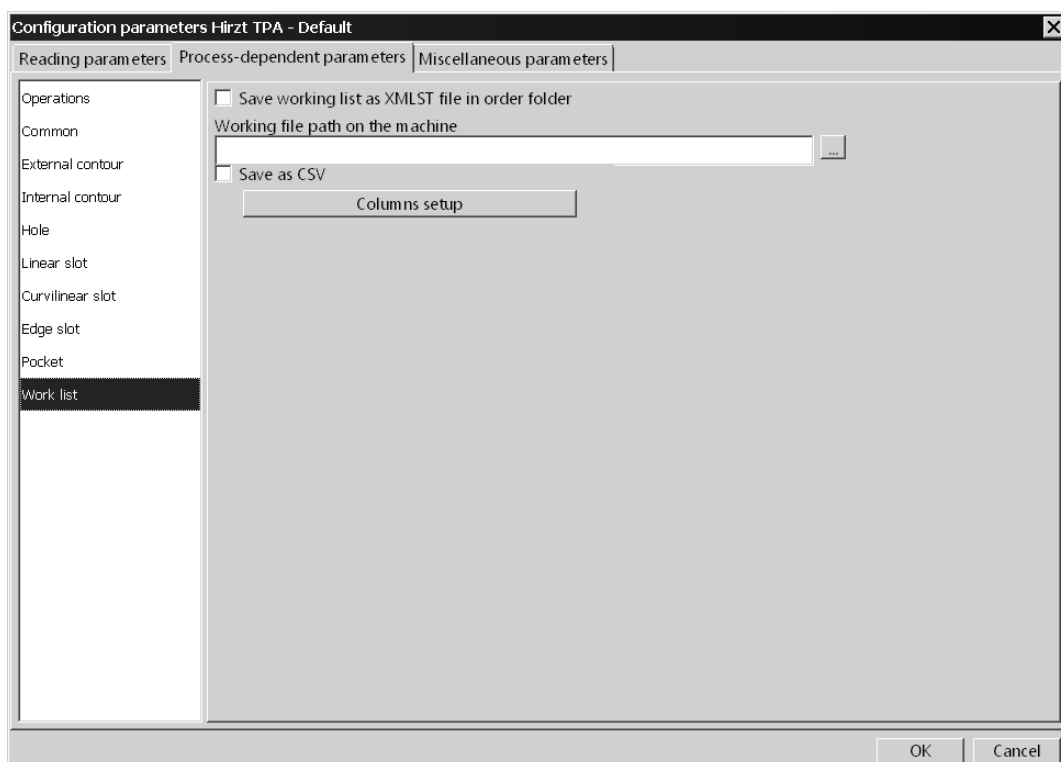


Fig. 10.362.

Save working list as XMLST file in order folder option allows to control the corresponding document creation. **Browse** button allows to set a path to working files. After pushing the button **Browse** dialogue will appear on the screen.

Save as CSV option allows to control export file saving. **Columns setup** button allows to setup file table columns. **Table columns setup** dialogue will appear on the screen (fig. 10.363).

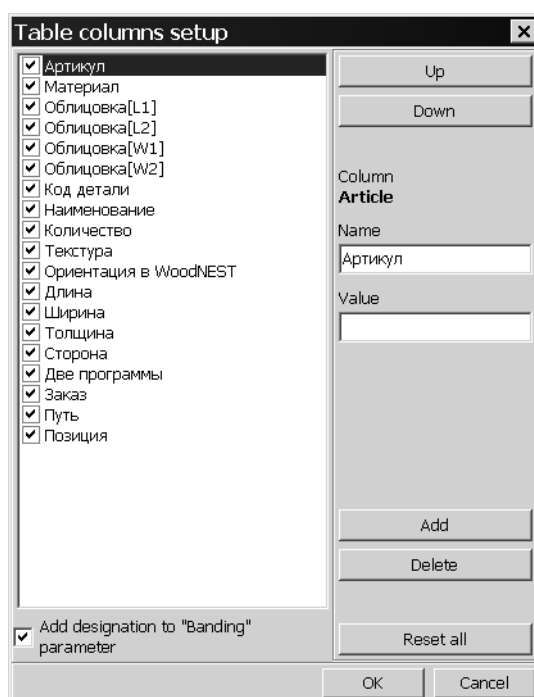


Fig. 10.363.

Enable the columns options you need to include to the table. **Up** and **Down** buttons allow to change columns position. Selected column name can be edited in **Name** field. To add the column push **Add** button. New column name will appear in the list. It can be changed as well. Added columns values can be set in corresponding field. **Delete** button allows to delete selected column. **Reset all** button allows to cancel all custom changes. To finish table setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

Setup completion

To complete control programs for Hirzt TPA setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.17.2.iLENIA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

iLENIA control programs are saved in files of *pd4* extension.

Common parameters

Common section control elements (fig. 10.364) allow to set common parameters of control programs creation.

Configuration parameters iLENIA - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Common

Slot

Work list

Reference point position

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

☐ Optimal reference point for holes into a face

☐ Group holes

Addition through hole depth

Hinge sinker "Cup" drill type if hole diameter exceeds

Machine control system

☒ Hirzt ☐ Maggi

☐ Set drill types

Diameter	Hole type	Drill type	Drilling mode
5	Blind	Blind	Automatically
15	Blind	Blind	Automatically
35	Blind	Cup	Automatically
5	Through	Through	Automatically
8	Through	Through	Automatically

Load parameters

OK Cancel

Fig. 10.364.

Reference point position group variants allow to select panel angle as origin of coordinates. If **Optimal reference point for holes into a face** option is enabled, referencing will be set for each hole individually. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional through hole depth** field allows to take drill center point into account. Holes can be bored with a common drill as well as with a cup borer type. Tool selection is specified with a hole diameter. **Hinge sinker "Cup" drill type if hole diameter exceeds** field allows to set hole diameter that will specify automatic tool selection in the program. **Machine control system** group variants allow to select control system type. **Set drill types** option allows to set automatic borer type and boring mode selection depending on hole diameter and type. If it is enabled, drill table control elements will become available.

Slot parameters

Slots parameters are set in **Slot** section (fig. 10.365).

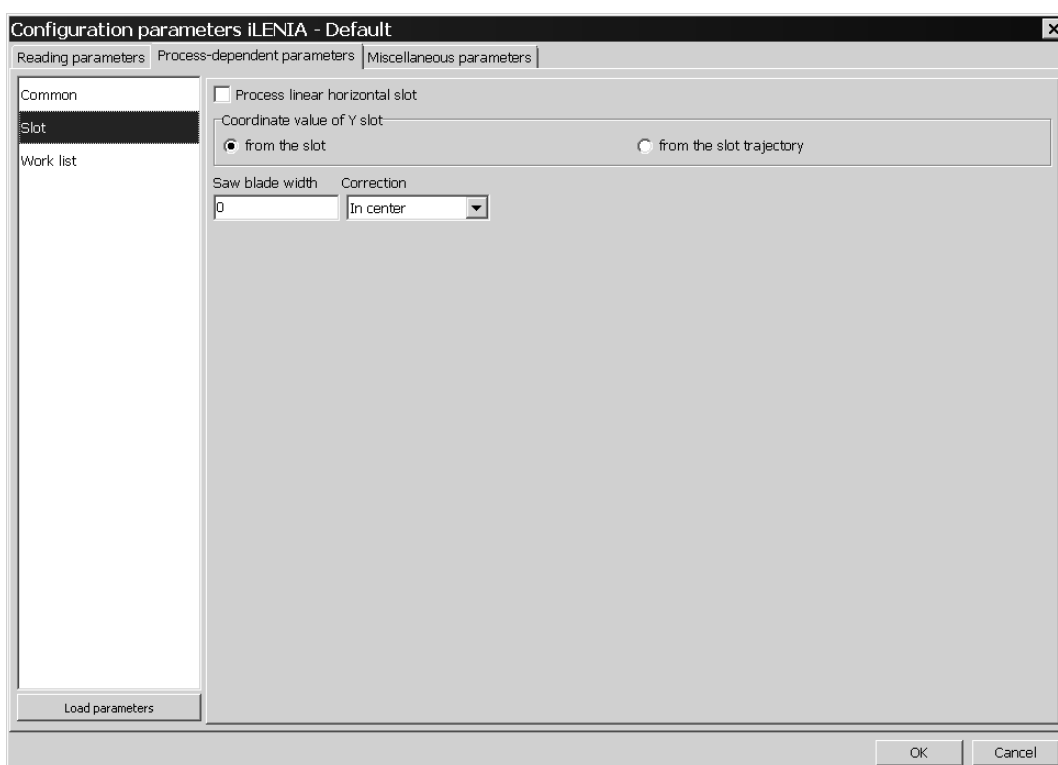


Fig. 10.365.

Process linear horizontal slot button allows to append slots processing to the program. **Coordinate value of Y slot** group variants allow to select reference line of corresponding coordinate. **Correction** combo box allows to select correction type for set saw blade width.

Work file list

Work file list can be saved to a file. Open **Work list** section (fig. 10.366) to setup saving.

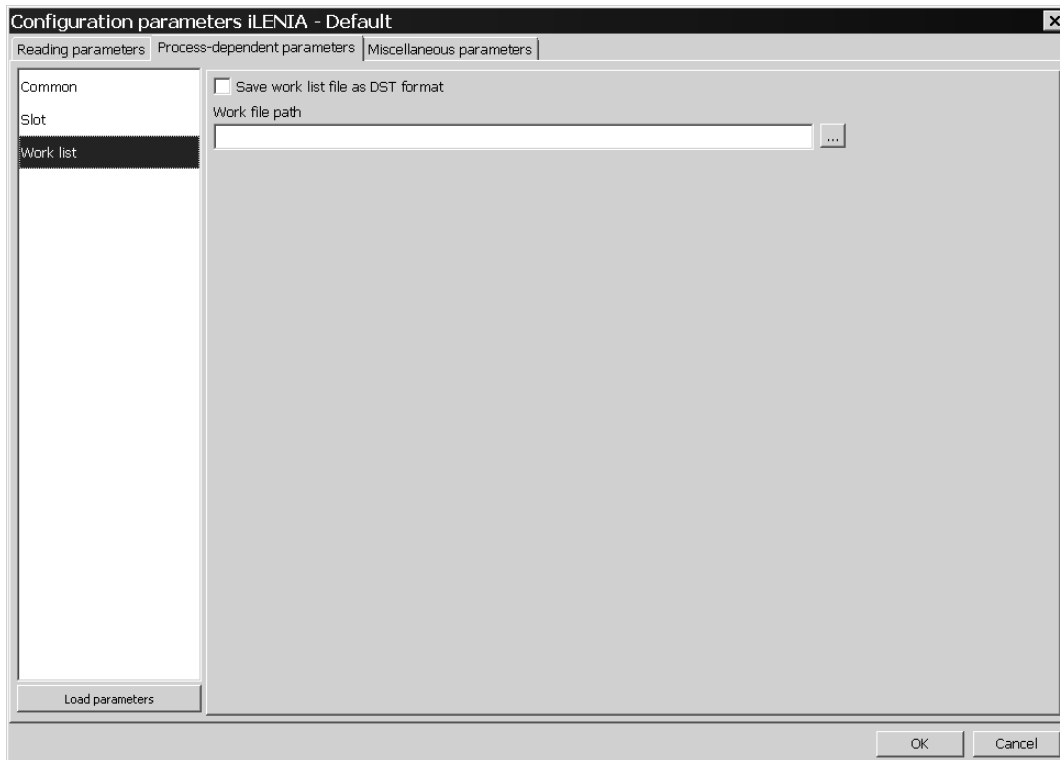


Fig. 10.366.



Save work list file as DST format option allows to enable saving. **Browse** button allows to specify file saving folder.

Setup completion

To complete control program for iLENIA Cad (*.pd4) parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.18. HOLD

10.18.1.CNC Drill



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.367) allow to specify operations that can be used in control program.

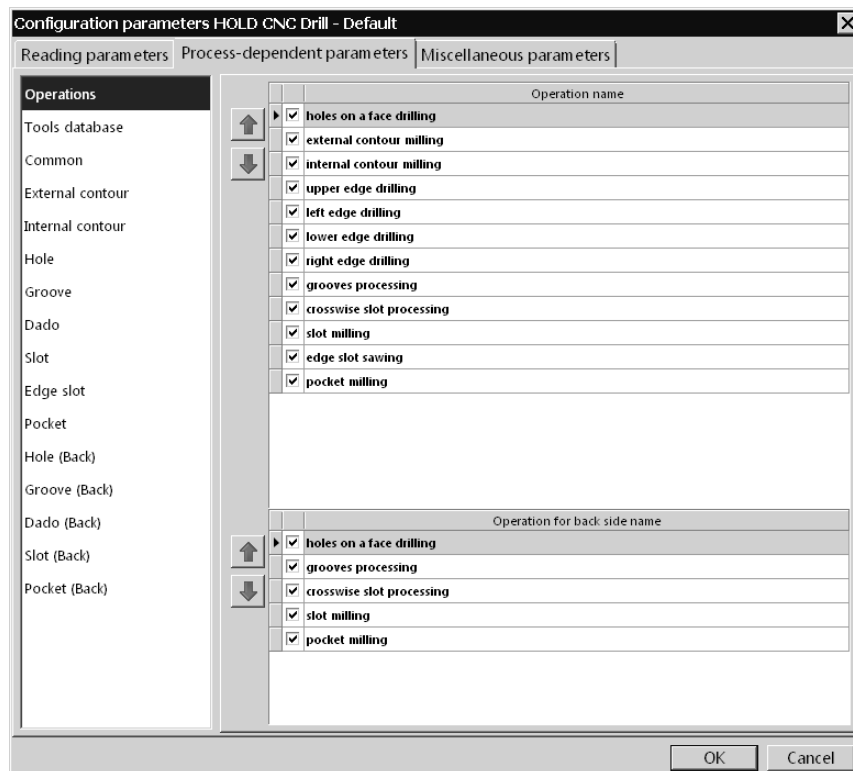


Fig. 10.367.

Usable tools

Tools database section control elements (fig. 10.368) allow to create a list of tools available for use in control programs.

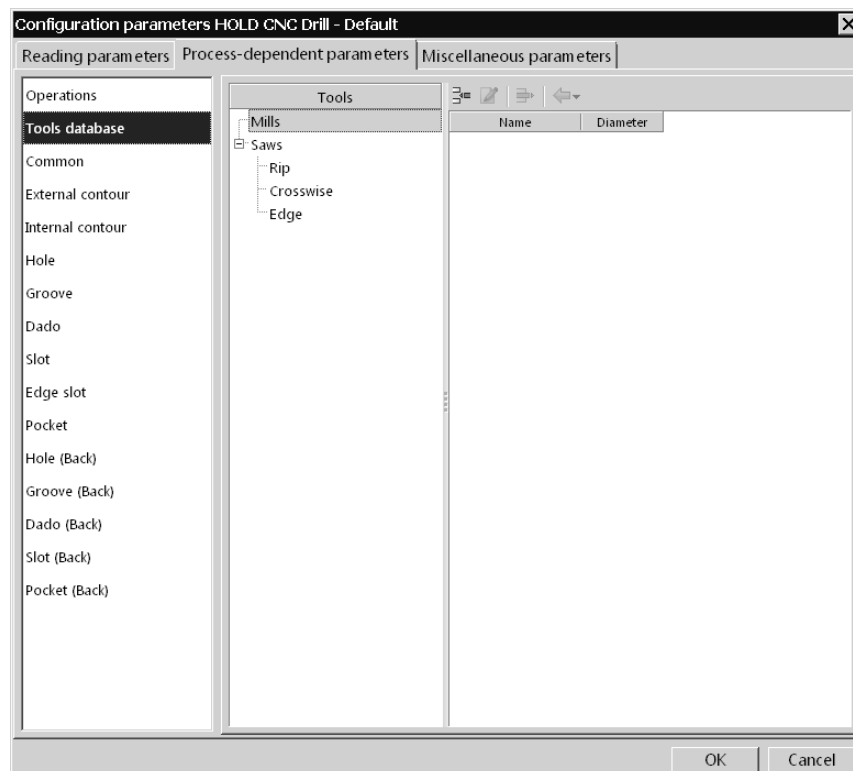


Fig. 10.368.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.369).

Fig. 10.369.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes.



Edit button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common section control elements (fig. 10.370) allow to set common parameters of control programs creation.

Fig. 10.370.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.371).

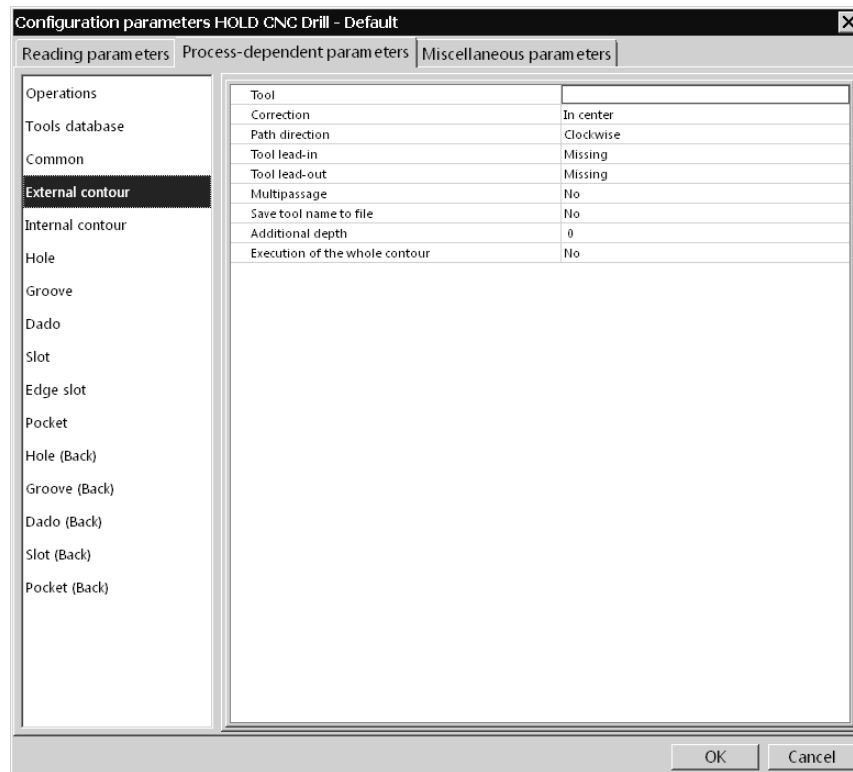


Fig. 10.371.

Section control elements allow to choose a tool, lead-in type, correction type, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.372), holes from back side – in **Hole (Back)** section in the same way.

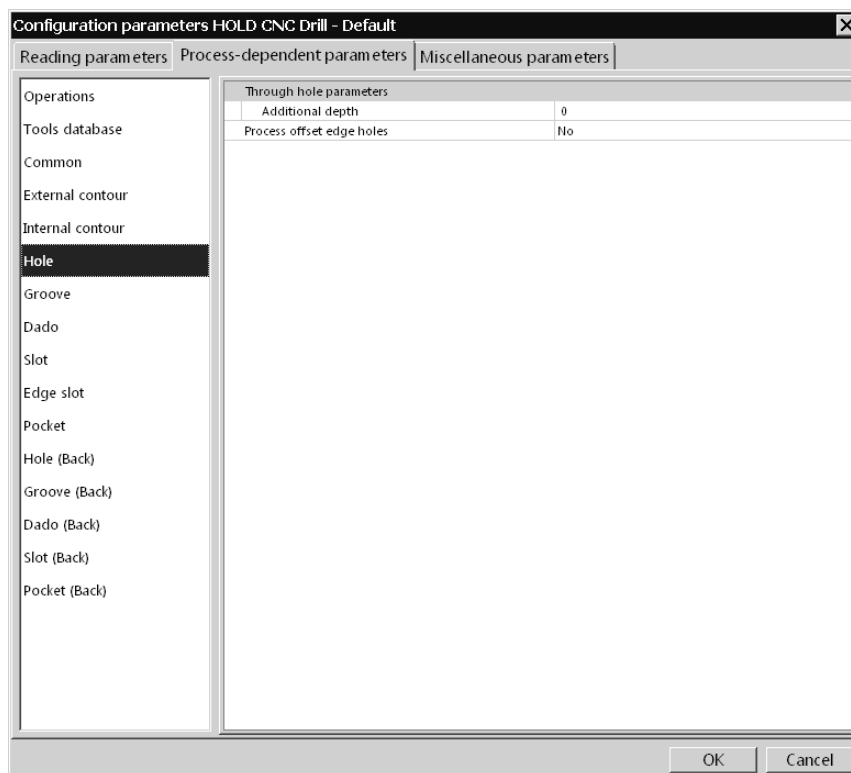


Fig. 10.372.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.373), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

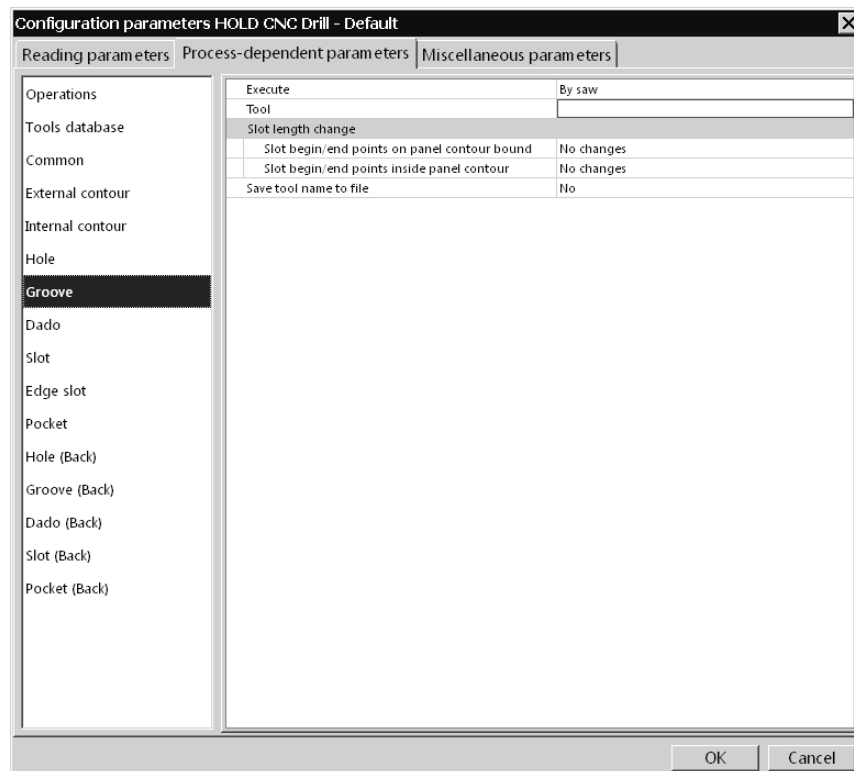


Fig. 10.373.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.374), from the back panel side – in **Slot (Back)** section.

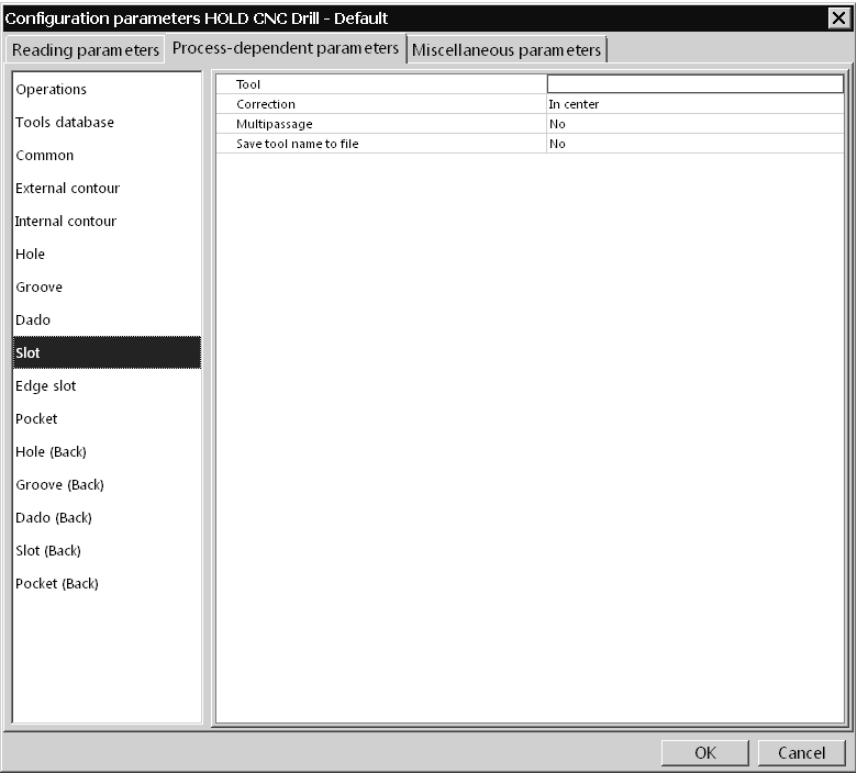


Fig. 10.374.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.375), on the back panel side – in **Pocket (Back)** section.

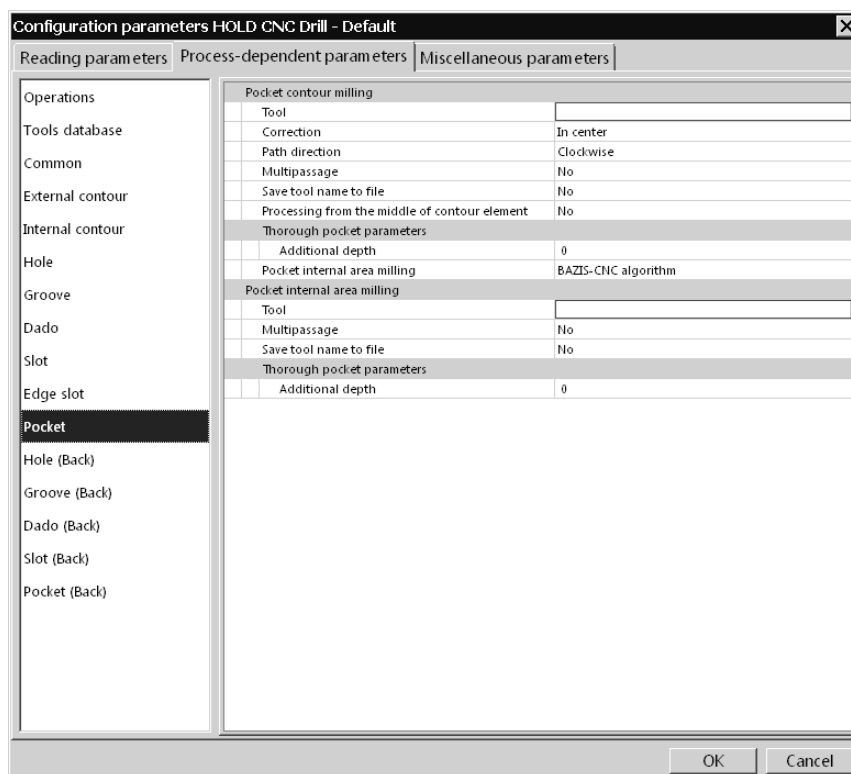


Fig. 10.375.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects from database. For pocket contour select milling execution direction, correction type and multipassage use. For internal pocket part – correction type and multipassage use.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.18.2.HOLD LNC



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.376) allow to specify operations that can be used in control program.

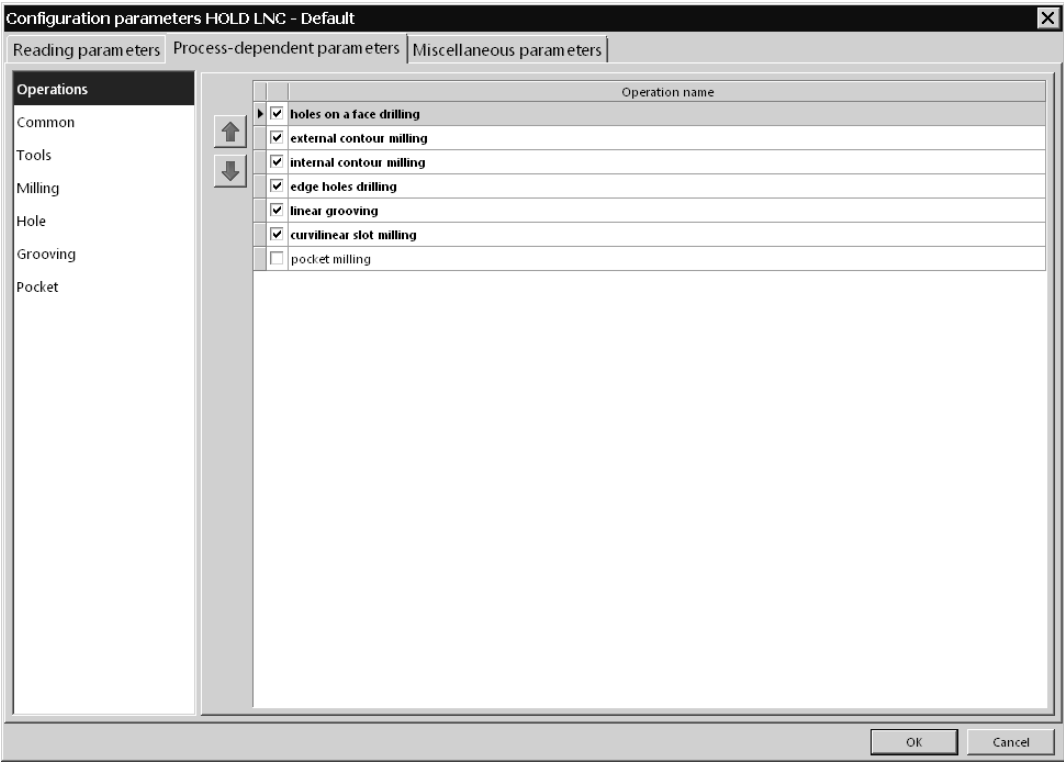


Fig. 10.376.

Common parameters

Common section control elements (fig. 10.377) allow to set common parameters of control programs creation.

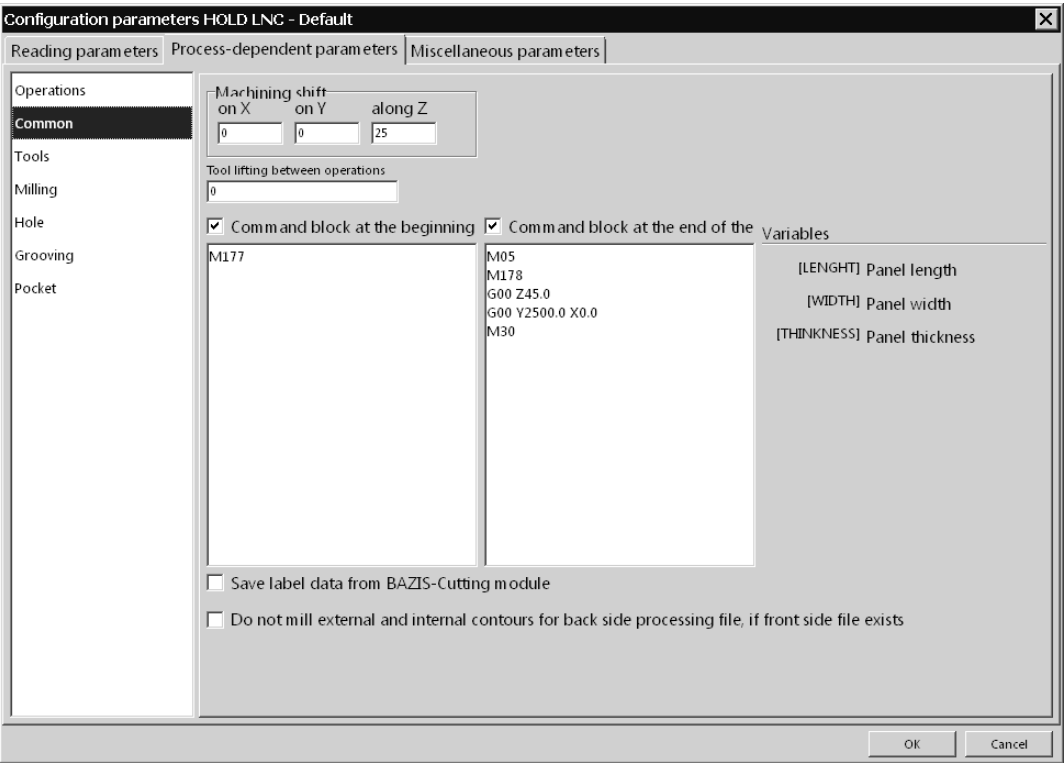


Fig. 10.377.

Machining shift group fields allow to set work area initial point moving. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas.

Save label data option allows to save the corresponding data. To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Tool parameters

Applied tools parameters are setup is **Tools** section (fig. 10.378).

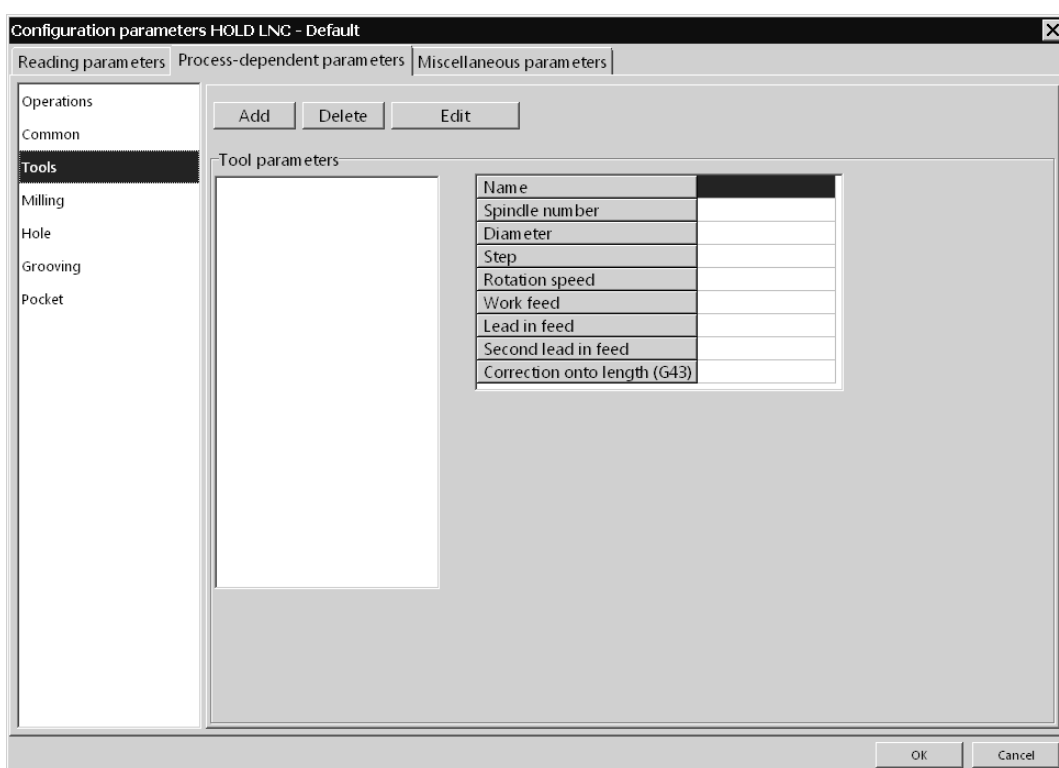


Fig. 10.378.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.379).

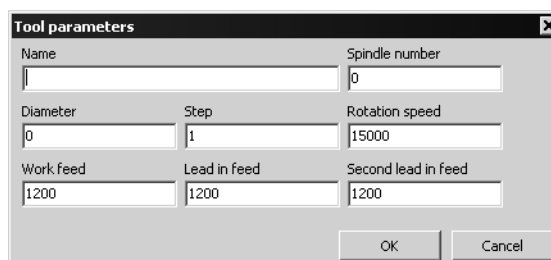


Fig. 10.379.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is shown in fig. 10.380.

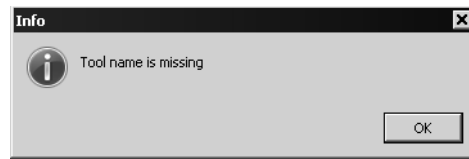


Fig. 10.380.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.379). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.381).

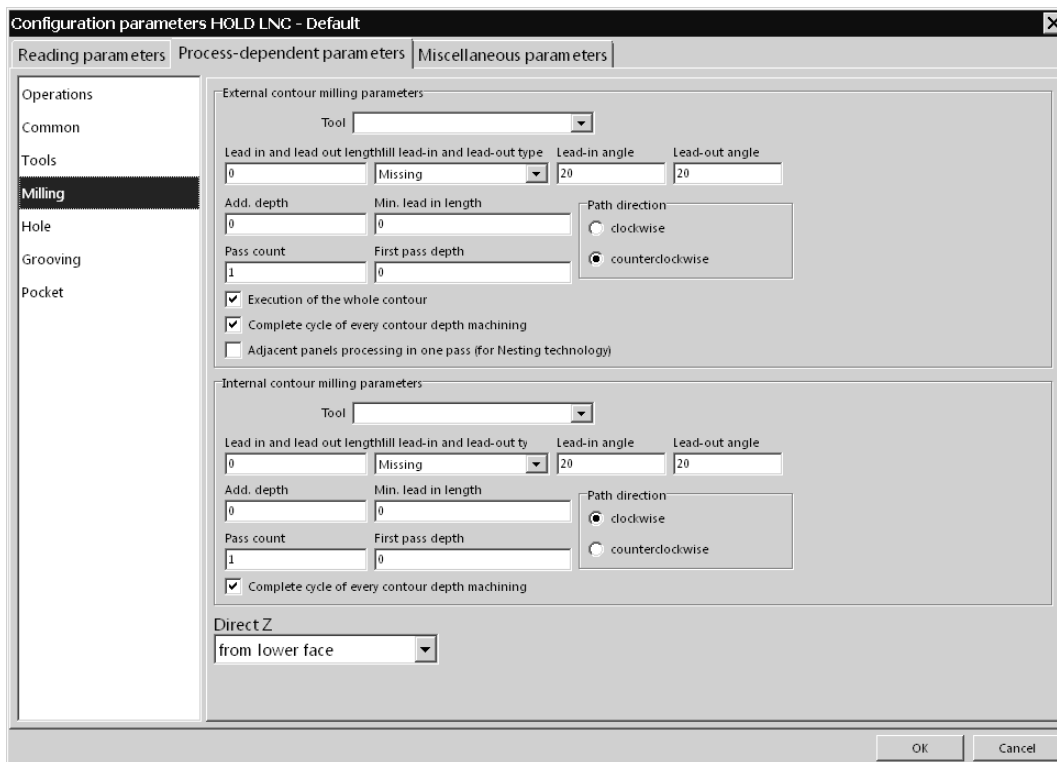


Fig. 10.381.

Tool combo box allows to select a milling tool name from the list, created on **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.382).

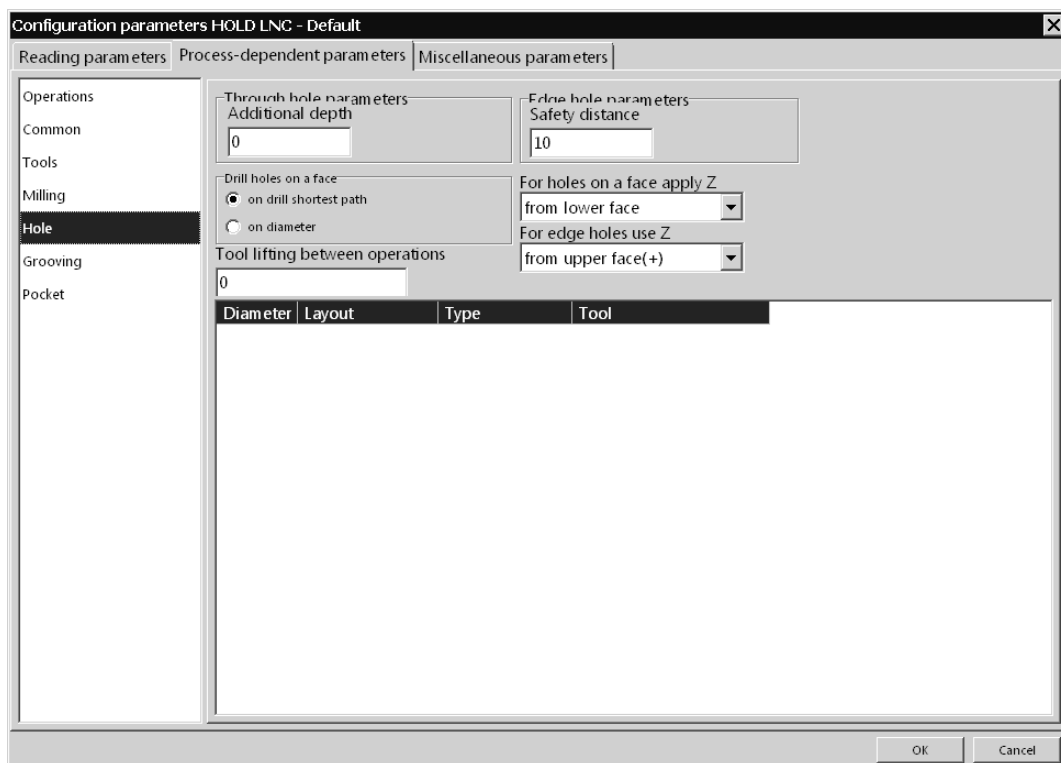


Fig. 10.382.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Groove parameters

Groove processing setup is performed in **Grooving** section (fig. 10.383).

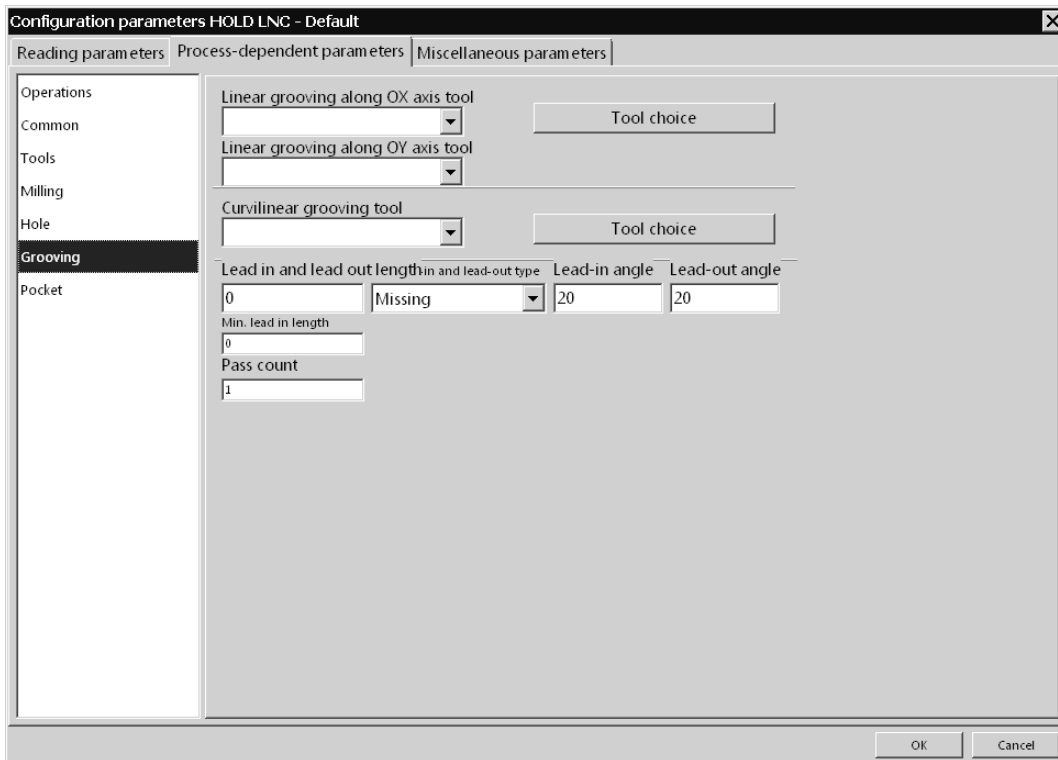


Fig. 10.383.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.384).

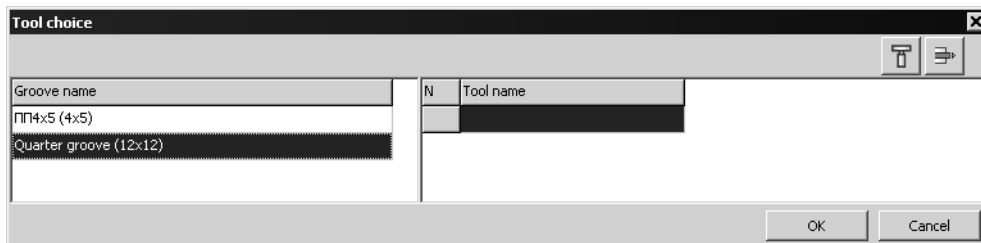


Fig. 10.384.



Slot name table contains all grooves names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool from the list created in **Tools** section.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.385) allow to set pocket processing parameters.

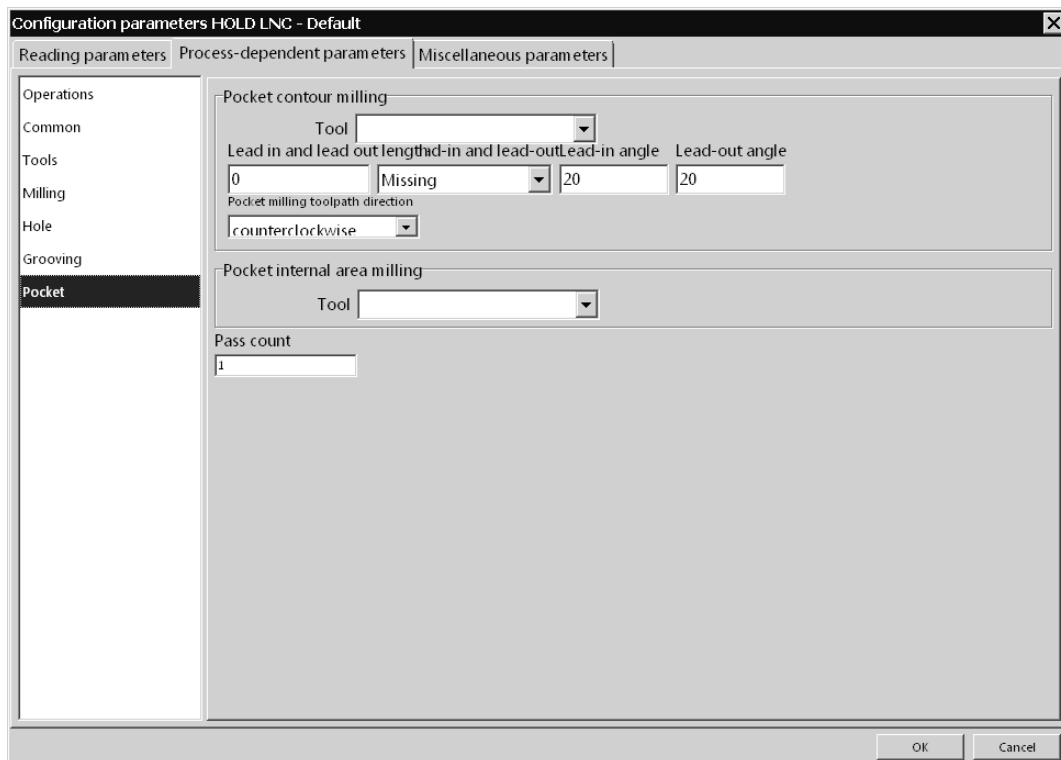


Fig. 10.385.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.19. HOLZ-HER

10.19.1.Campus Hops (*.hop)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Campus Hops control programs are saved in files of *hop* extension.

Operations

Operations section control elements (fig. 10.386) allow to specify operations that can be used in control program.

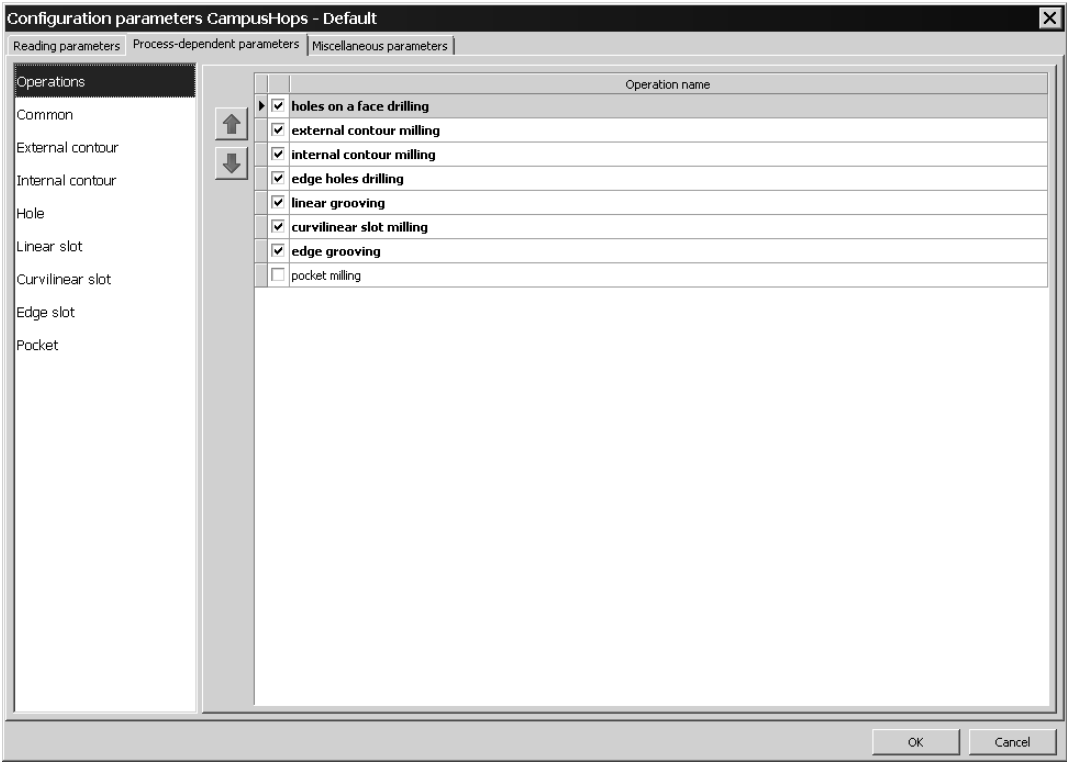


Fig. 10.386.

Common parameters

Common section control elements (fig. 10.387) allow to set common parameters of control programs creation.

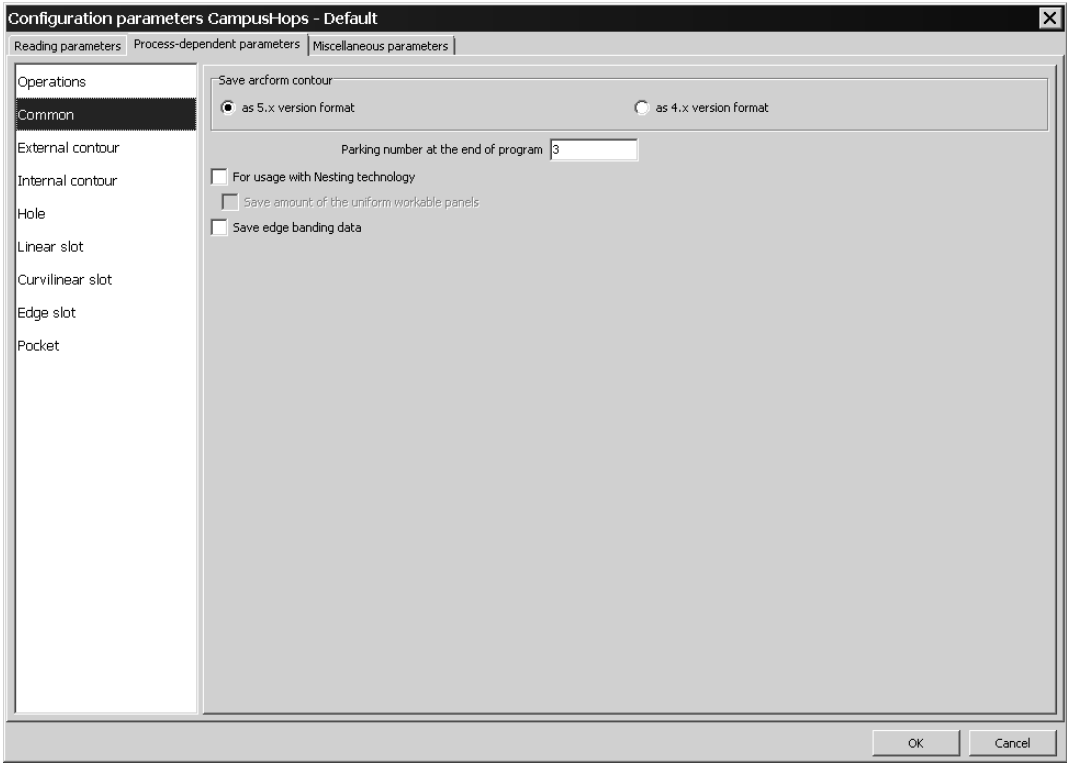


Fig. 10.387.

Save arcform contour group variants allow to select contour saving format. **For usage with Nesting technology** option allows to add automatically data to control program for Nesting technology use.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** and **Internal contour** sections of **Process-dependent parameters** dialogue (fig. 10.388). Rough and finish milling setup is executed separately on corresponding tabs. Tabs control elements content is the same and described for finish milling.

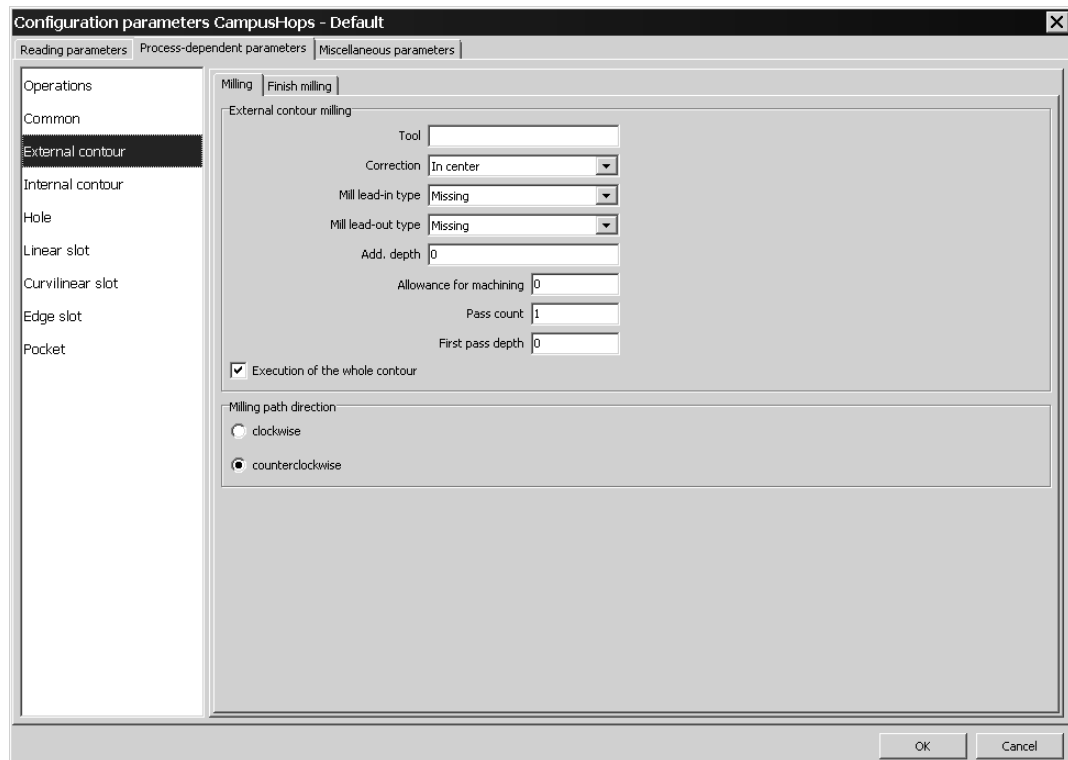


Fig. 10.388.

Finish milling option allows to control this operation. Tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.389).

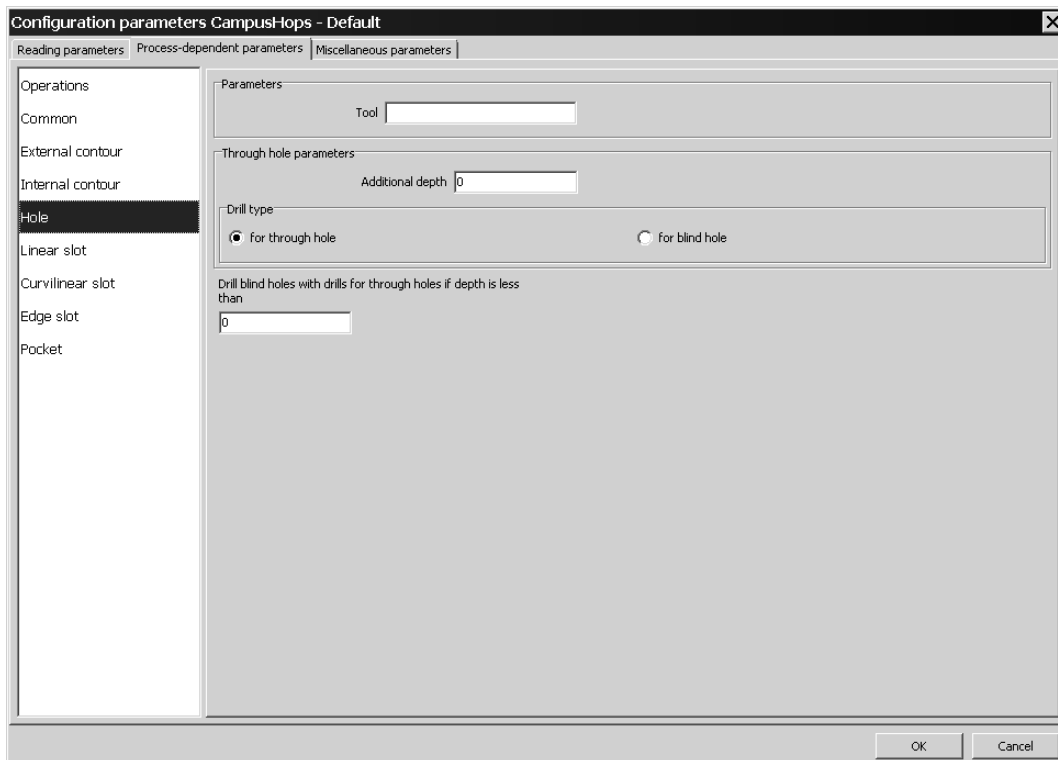


Fig. 10.389.

Tool field allows to set the corresponding tool name. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **Drill type** group variants allow to select borer type.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.390).

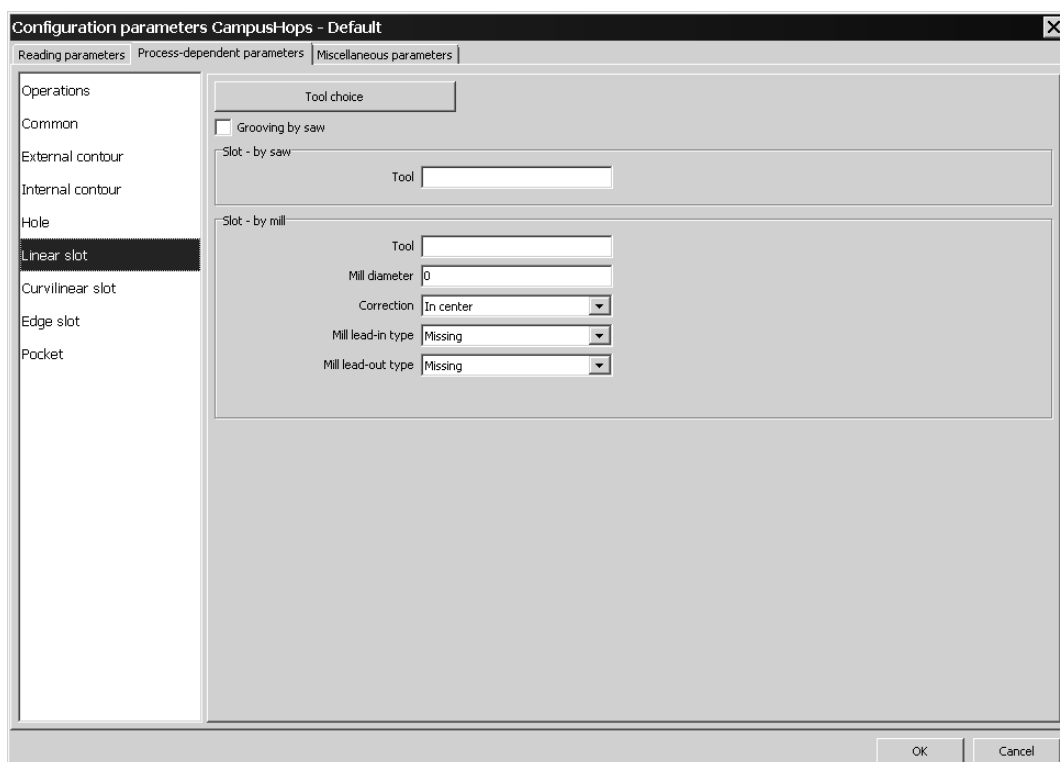


Fig. 10.390.

Grooving by saw option allows to select a tool type for slots processing. If it is disabled, slot will be milled. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.391).

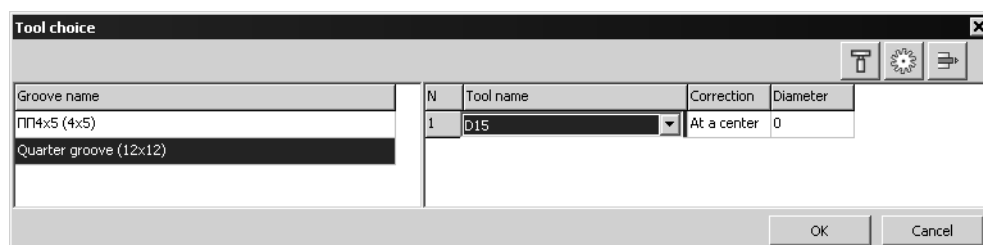


Fig. 10.391.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to select a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.392).

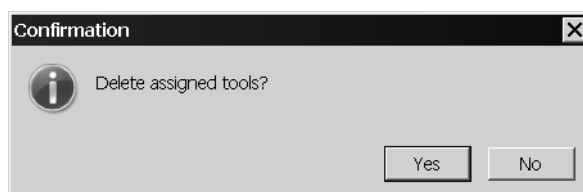


Fig. 10.392.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Tool field allows to input tool designation.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill.

Curvilinear and edge slots parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** (fig. 10.393) and **Edge slot** sections. These slot types parameters are almost the same, some parameters are specific for curvilinear slots.

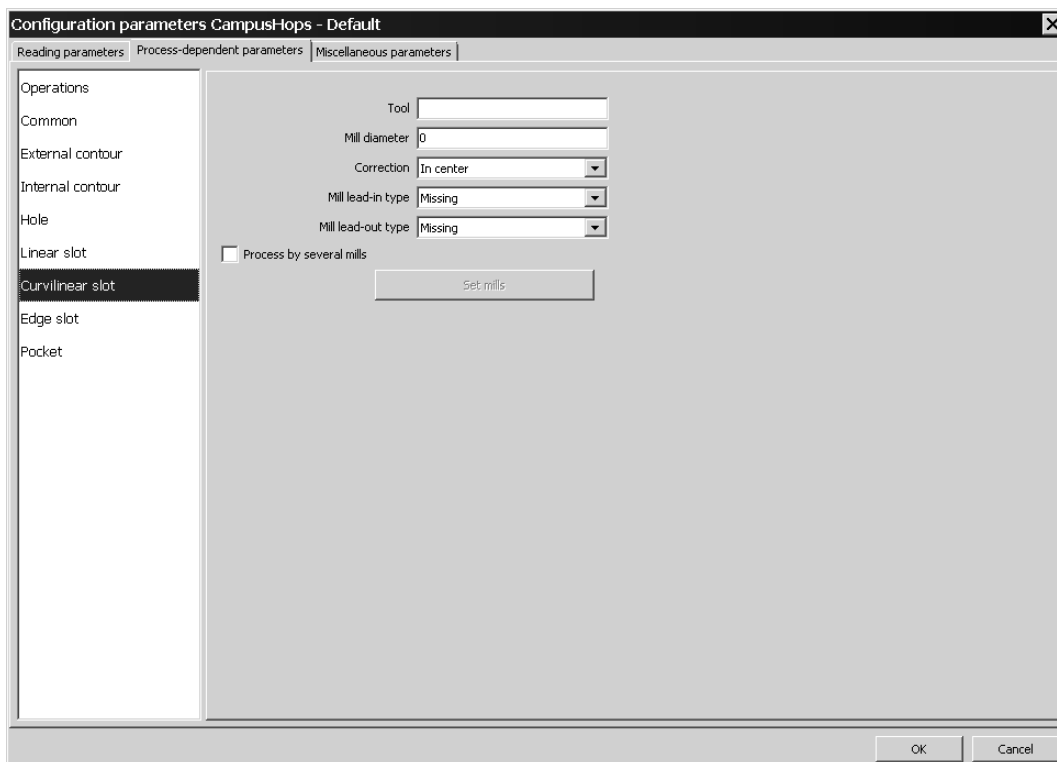


Fig. 10.393.

If **Process by several mills** option is enabled in **Curvilinear slot** section, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.394).

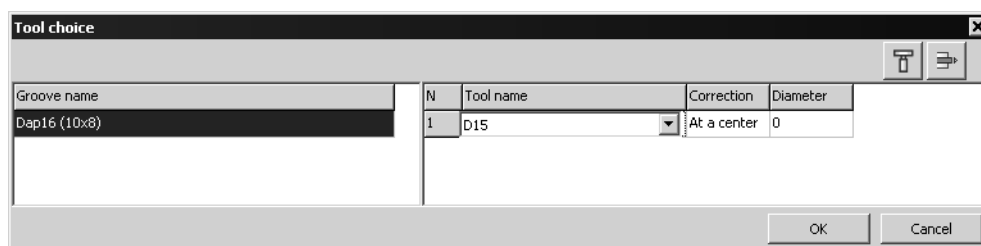


Fig. 10.394.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.

Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements allow to set pocket processing parameters (fig. 10.395).

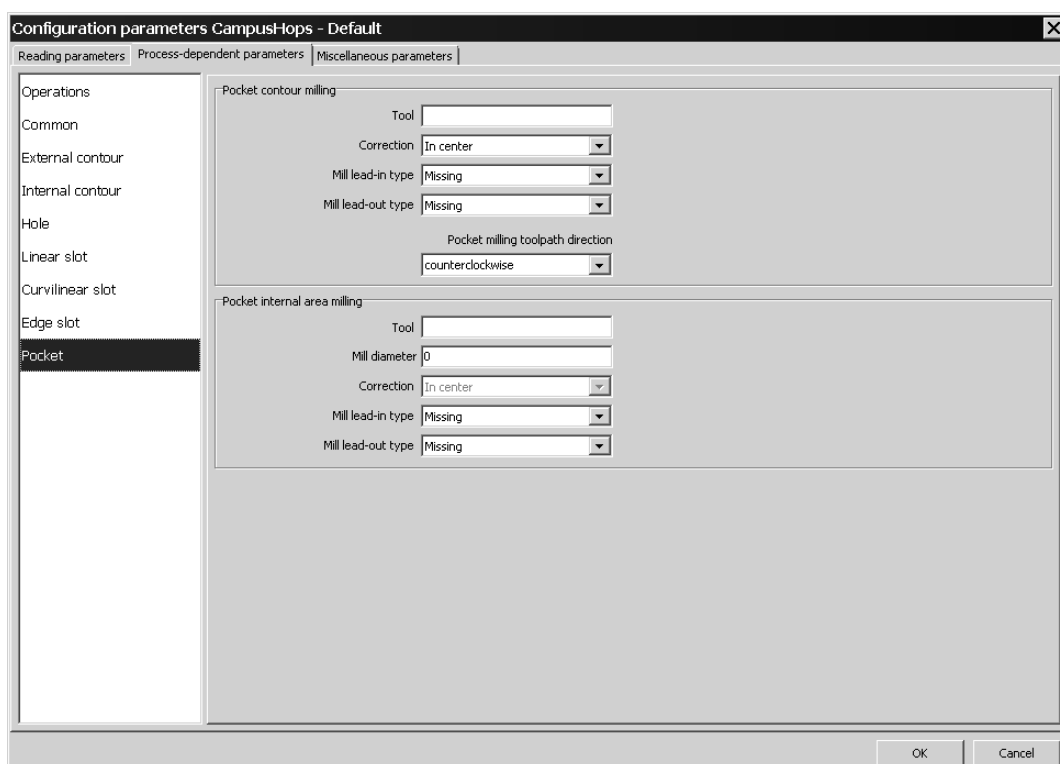


Fig. 10.395.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs for Campus Hops setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.19.2.TwinCam



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.396) allow to set common parameters of control programs creation.

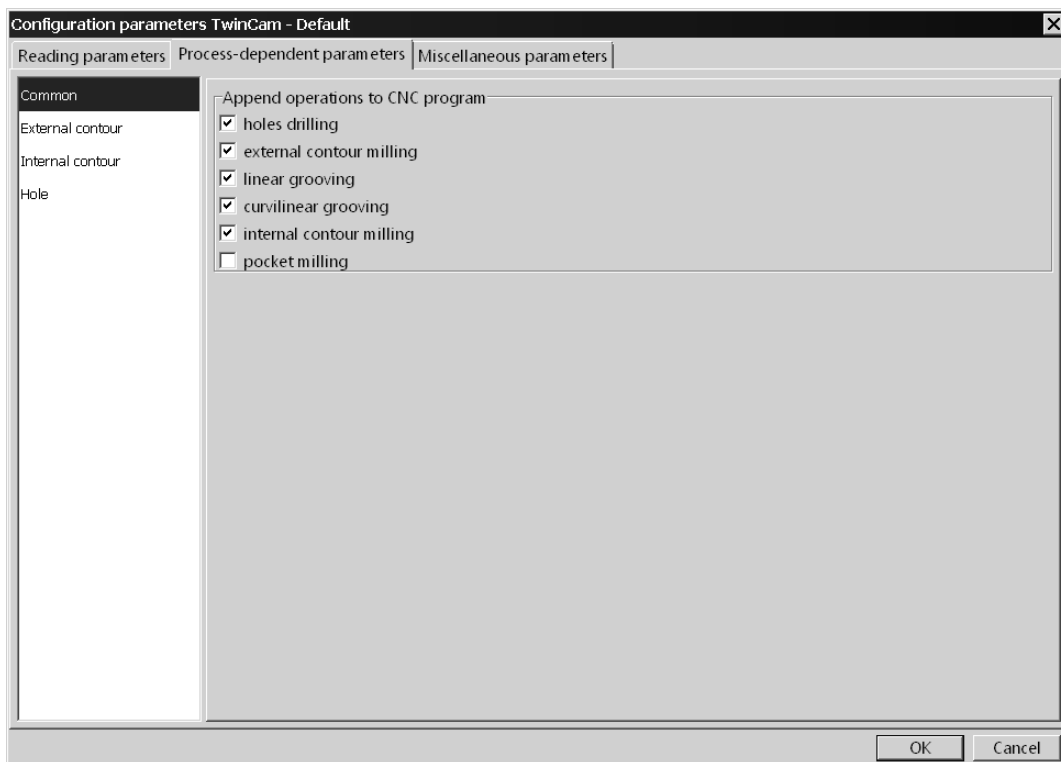


Fig. 10.396.

Append operations to CNC program group options allow to control operations automatic appending to control program.

Contours processing parameters

Outer and internal contours processing parameters are set on **Outer contour** and **Internal contour** tabs (fig. 10.397).

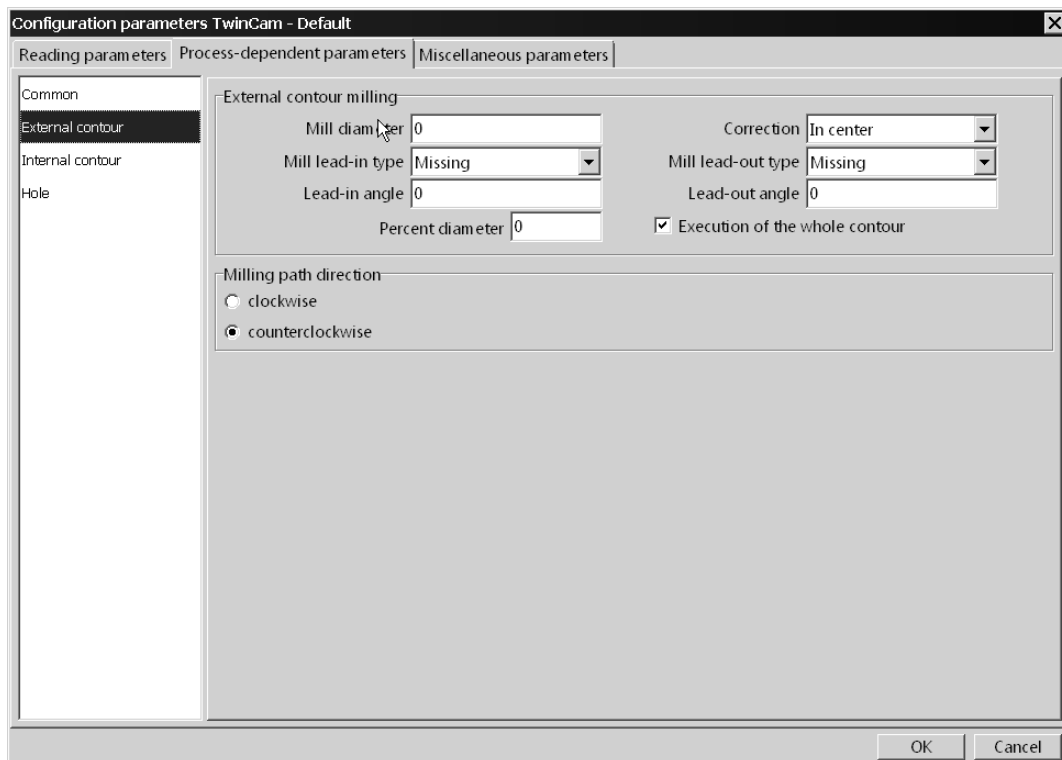


Fig. 10.397.

Contours processing setup is executed in the same way and is described in outer contour processing example. Tabs control elements allow to select mill diameter, correction type, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.398).

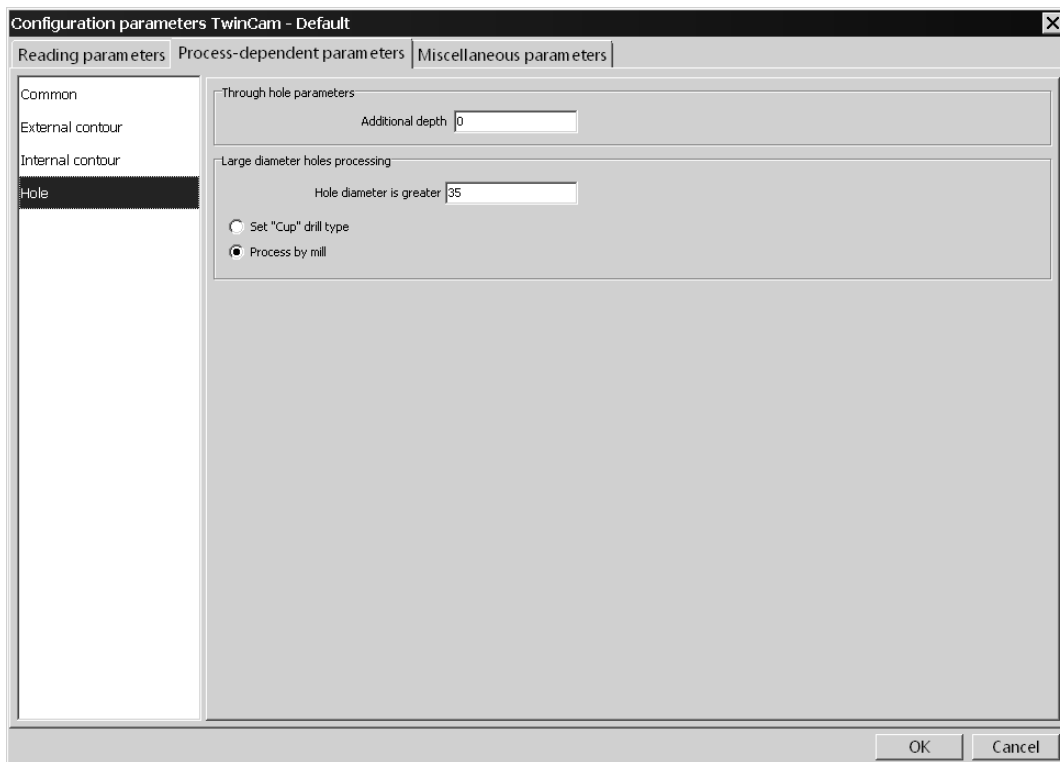


Fig. 10.398.

Hole can be milled or bored depending on its diameter. **Vertical hole is milled if its diameter is more or equal** field allows to set hole diameter that will specify tool choice.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.20. Holztek

10.20.1.Horizontal Drill



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.399) allow to specify operations that can be used in control program.

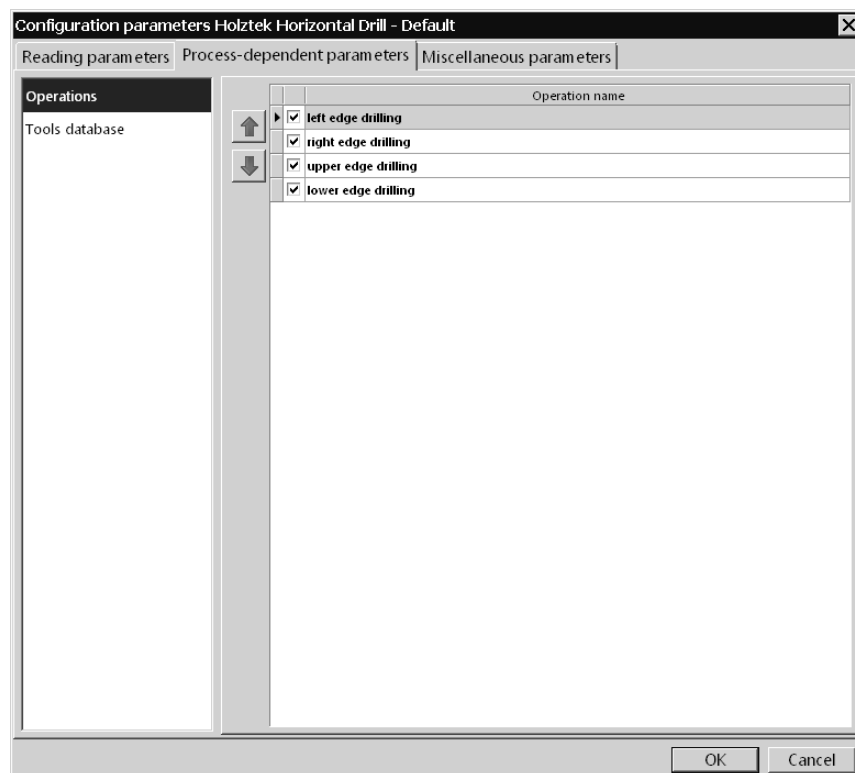


Fig. 10.399.

Usable tools

Tools database section control elements (fig. 10.400) allow to create a list of tools available for use in control programs.

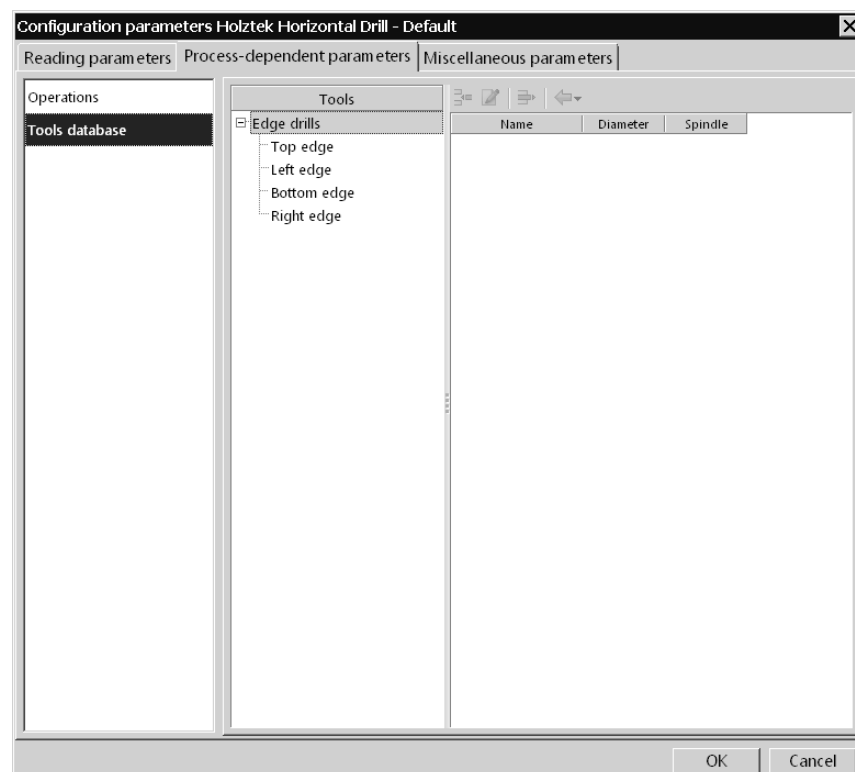


Fig. 10.400.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.401).

Tool	
Name	
Diameter	0
Spindle	0
OK Cancel	

Fig. 10.401.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Create** group options allow to control the creation of control programs files that correspond to options names.

10.21. HOMAG Group

10.21.1. WoodWop



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.402) allow to specify operations that can be used in control program.

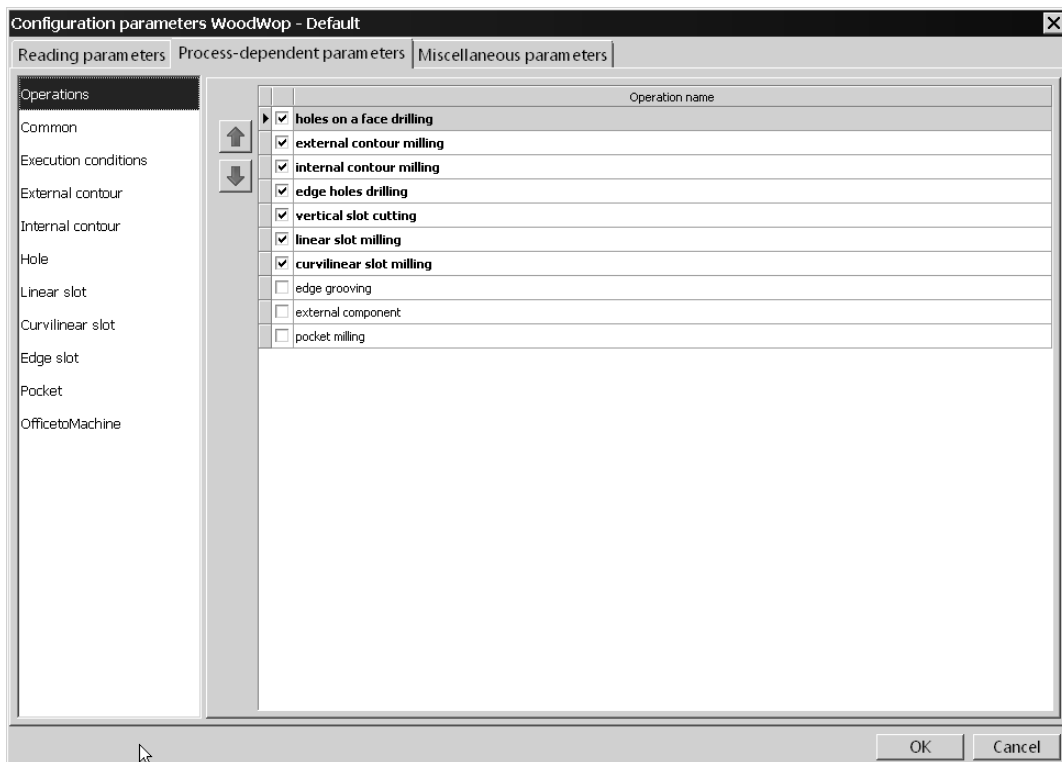


Fig. 10.402.

Common parameters

Common parameters are set in **Common** section (fig. 10.403).

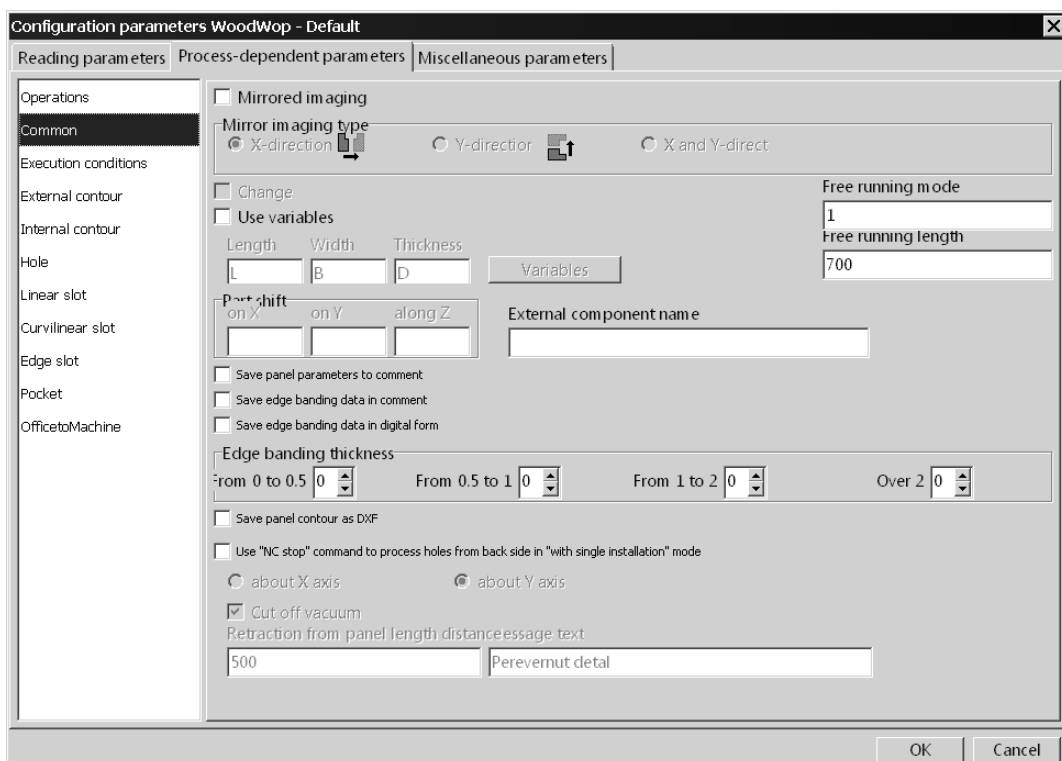


Fig. 10.403.

Mirrored imaging option allows to control panel image creation, that is mirrored relatively to selected axis. To set an axis, select a necessary variant in **Mirror imaging type** group. **Use variables** option allows to control variables use in control programs. If it is enabled, **Length**, **Width**, **Thickness** fields will become available. Set variable names in these fields. **Variables** button allows to create arbitrary variables. After pushing the button, **Variables** dialogue will appear on the screen (fig. 10.404).

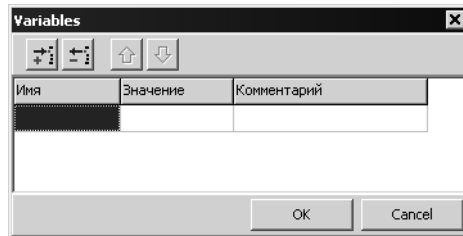


Fig. 10.404.



Dialogue table contains variables. Variable name, its value and comment should be input in table fields. **Add row** button allows to add a blank row to the table for new variable. To delete the current row push **Delete row** button. To finish work with variables push **OK** button. **Cancel** button allows to close the dialogue without changes.

Save panel parameters to comment, **Save banding data to comments**, **Save panel contour as DXF**, **Use "NC stop" command to process holes from back side in "with single installation" mode**, **Create file for each identical panel (for Nesting technology)** options designations correspond to their names. **Append operations to CNC program** group options allow to control the inclusion in control program of operations corresponding to the options names. To append operation to the program, enable option next to its name.

Operations execution conditions

Execution conditions section fields (fig. 10.405) allow to create program operations execution conditions.

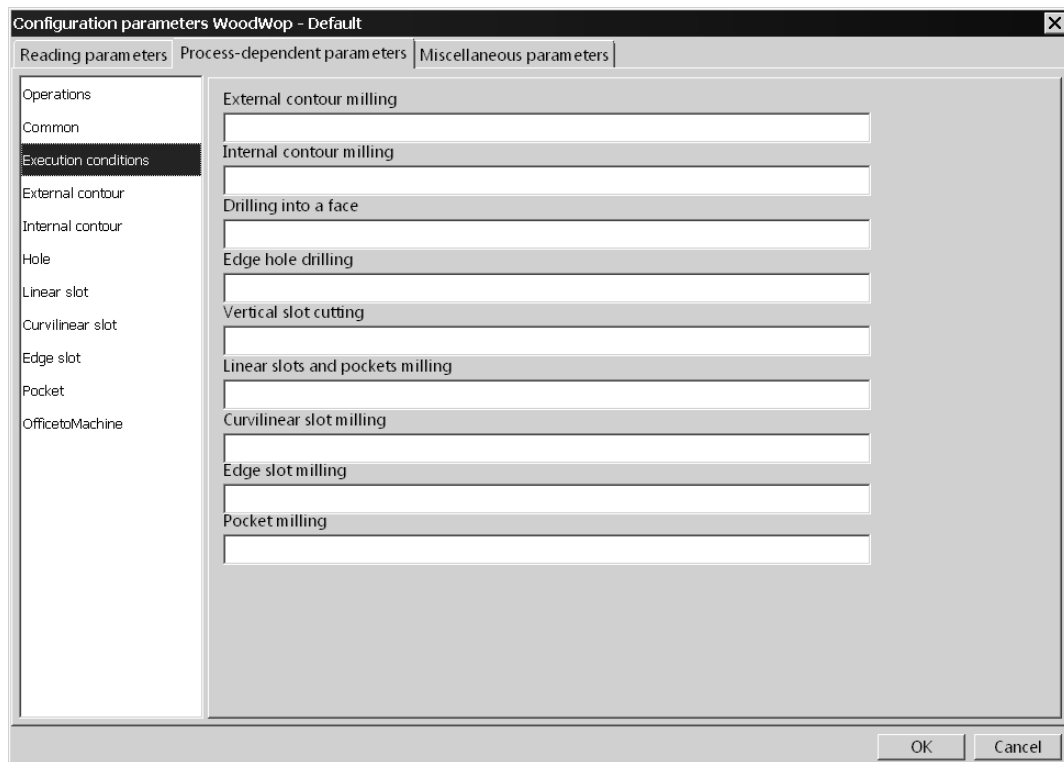


Fig. 10.405.

Conditions are created in the following way. Variables are created for CNC program. These variables values are conditions for particular operations execution or their rejection in particular program. Conditions can be created for operations listed in section.

Contours parameters

Contours milling parameters are set in **Outer contour** (fig. 10.406) and **Internal contour** sections of **Process-dependent parameters** dialogue.

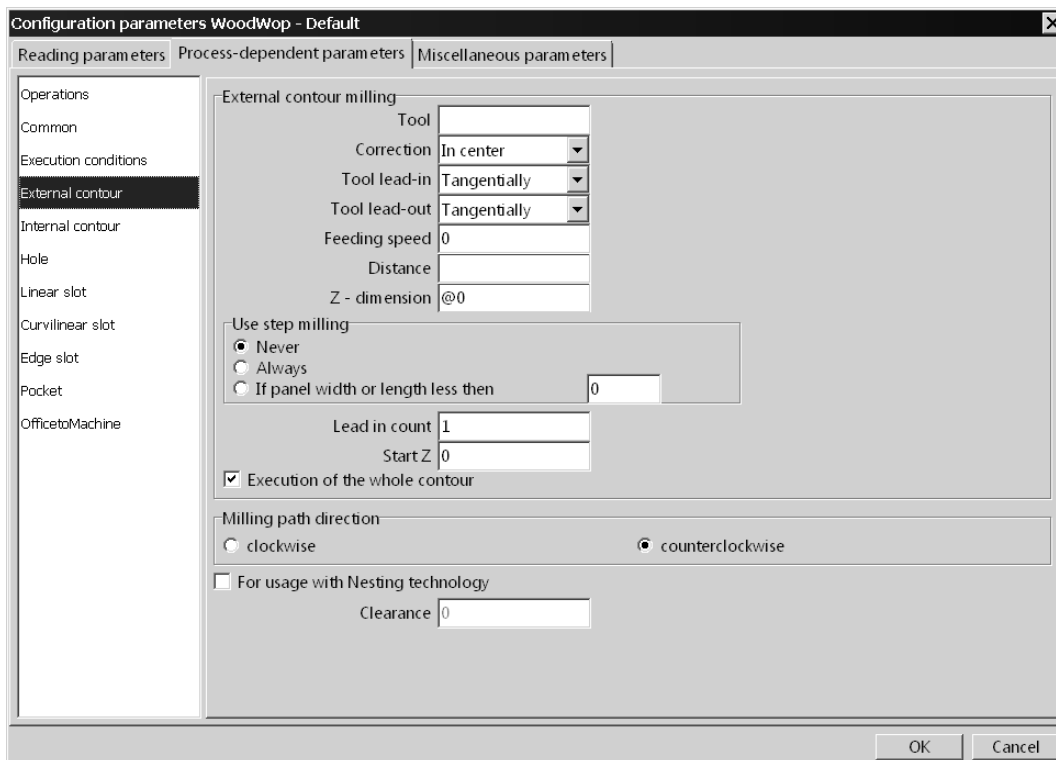


Fig. 10.406.

Section control elements allow to select a tool, correction type, tool lead-in and lead-out, etc. These tabs control elements are the same. Additional control elements are on **Outer contour** tab.

Execution of the whole contour option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **For usage with Nesting technology** option on **Outer contour** tab allows to add automatically data to control program for its processing via WoodNest program. If it is enabled, **Clearance** field will become available, it allows to set corresponding parameter.



BAZIS-Cutting module allows to create Nesting technology control program for panel processing.

Hole parameters

Hole section control elements allow to set holes parameters (fig. 10.407).

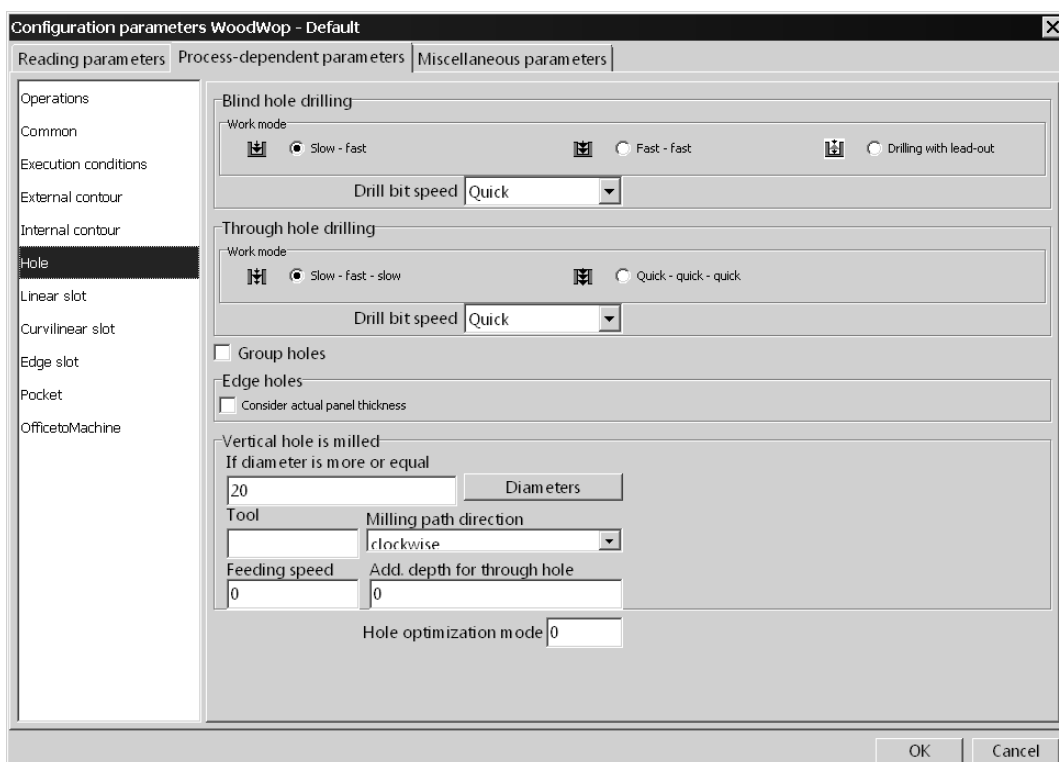


Fig. 10.407.

Blind hole work mode and **Through hole work mode** groups variants allow to select boring stages operating speed. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option.

Holes can be bored with a drill as well as with a mill. Hole diameter usually specifies tool selection. **If diameter is more or equal** field allows to set hole diameter, if it exceeds, hole will be milled, mill name is in **Tool** field. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameter** button. **Hole diameters** dialogue will appear on the screen (fig. 10.408).

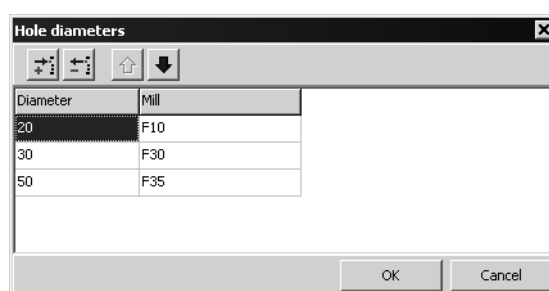


Fig. 10.408.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To finish setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

Linear slot

Linear slot section (fig. 10.409) control elements allow to set linear slots processing parameters.

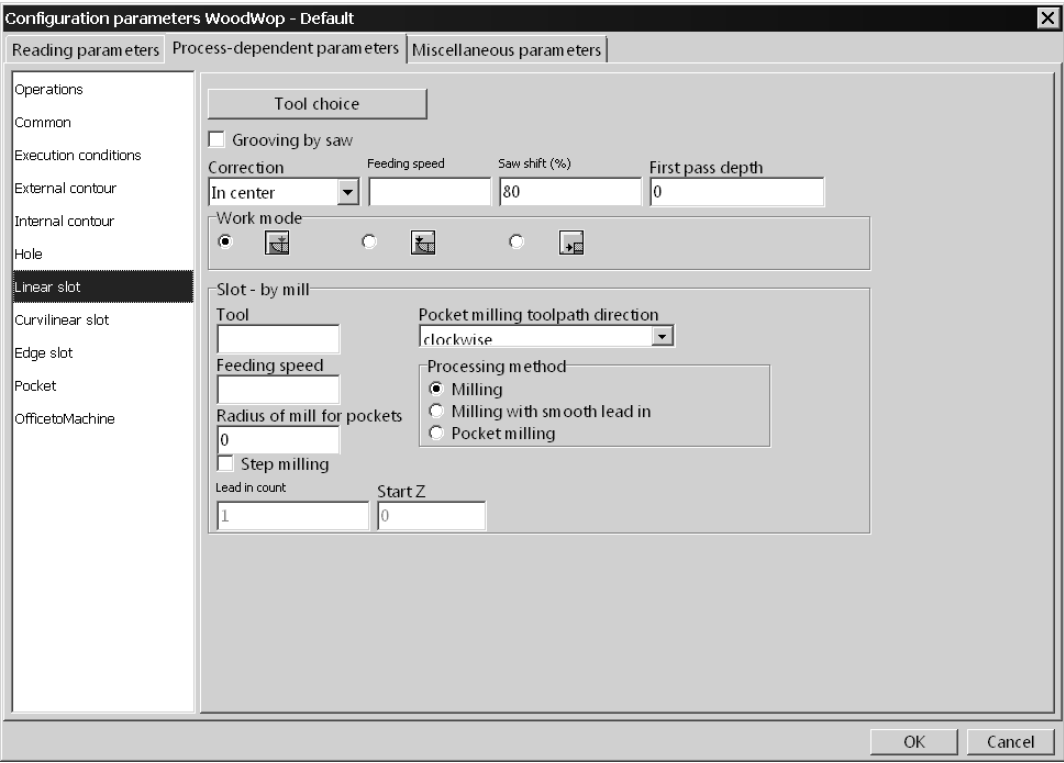


Fig. 10.409.

Tool choice button allows to set tools for particular slots processing. After pushing the button, **Tool choice** dialogue will appear on the screen (fig. 10.410).

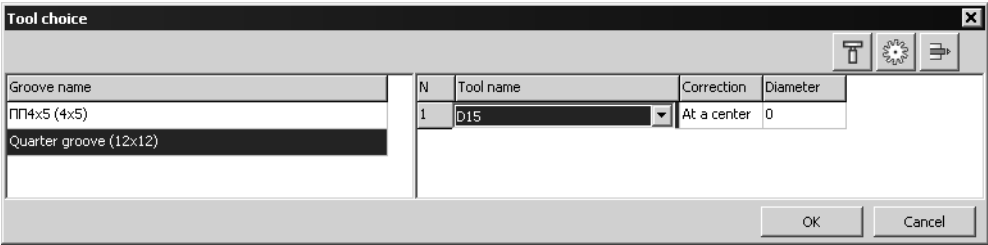


Fig. 10.410.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.411).

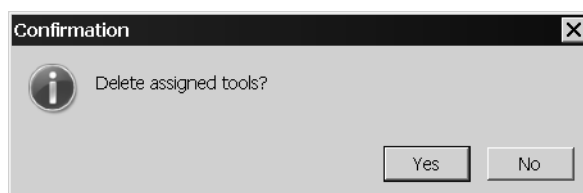


Fig. 10.411.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. Slots can be processed with a drill or a mill. Correction type should be selected from **Correction** combo box if you use a saw.

Curvilinear slot

Curvilinear slot (fig. 10.412) section control elements allow to set curvilinear slots processing parameters.

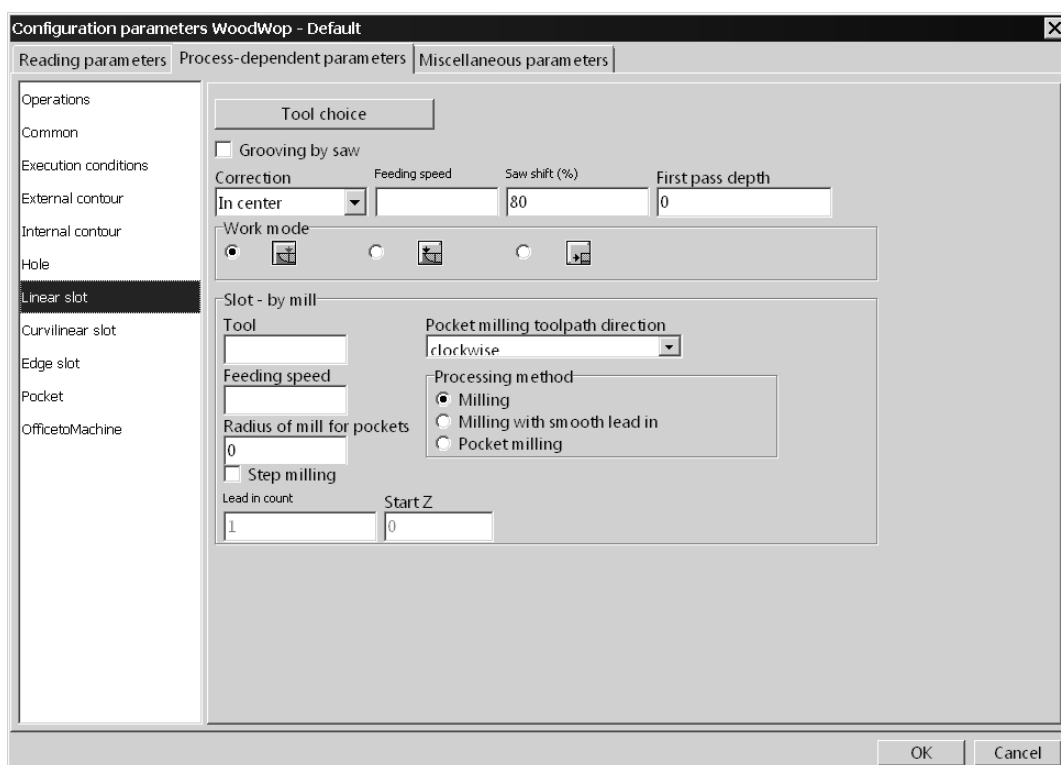


Fig. 10.412.

Tab control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.413).

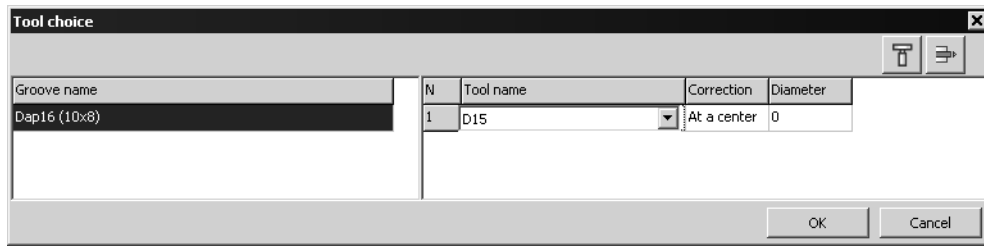


Fig. 10.413.

Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.

Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot

Edge slot section control elements (fig. 10.414) allow to set edge slots processing parameters.

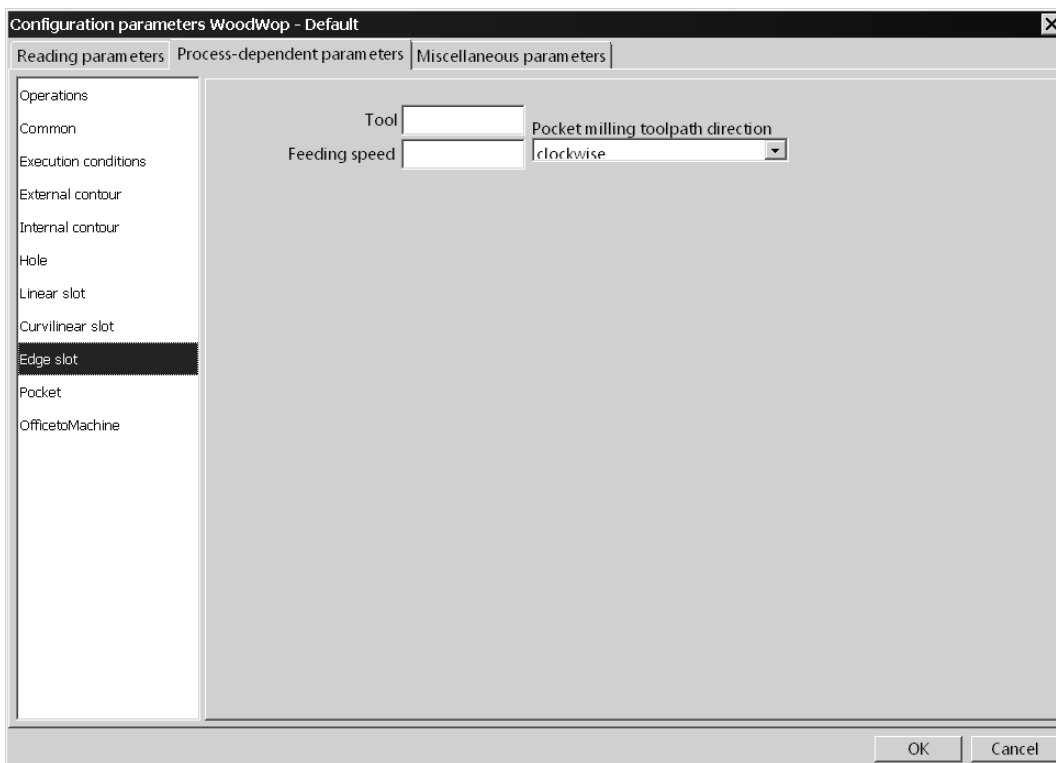


Fig. 10.414.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.415) allow to set pocket processing parameters.

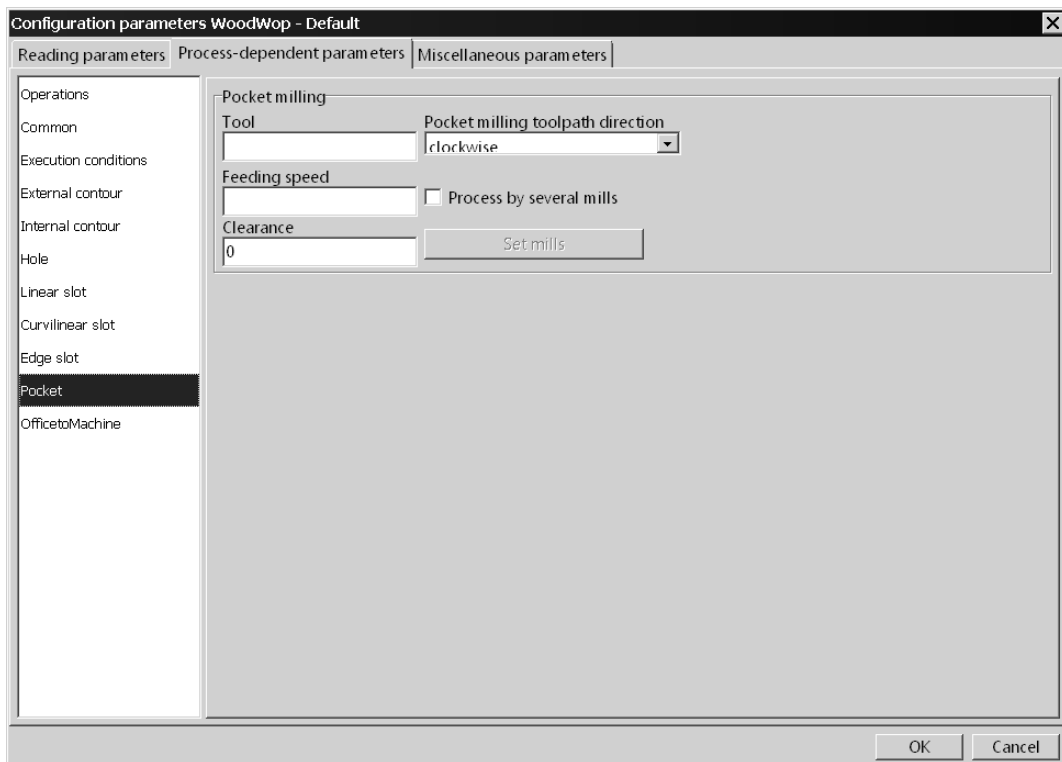


Fig. 10.415.

Section control elements allow to set control programs creation parameters for pocket processing: tool name, path direction, feeding speed, etc.

OfficetoMachine

OfficetoMachine section control elements (fig. 10.416) allow to set export parameters.

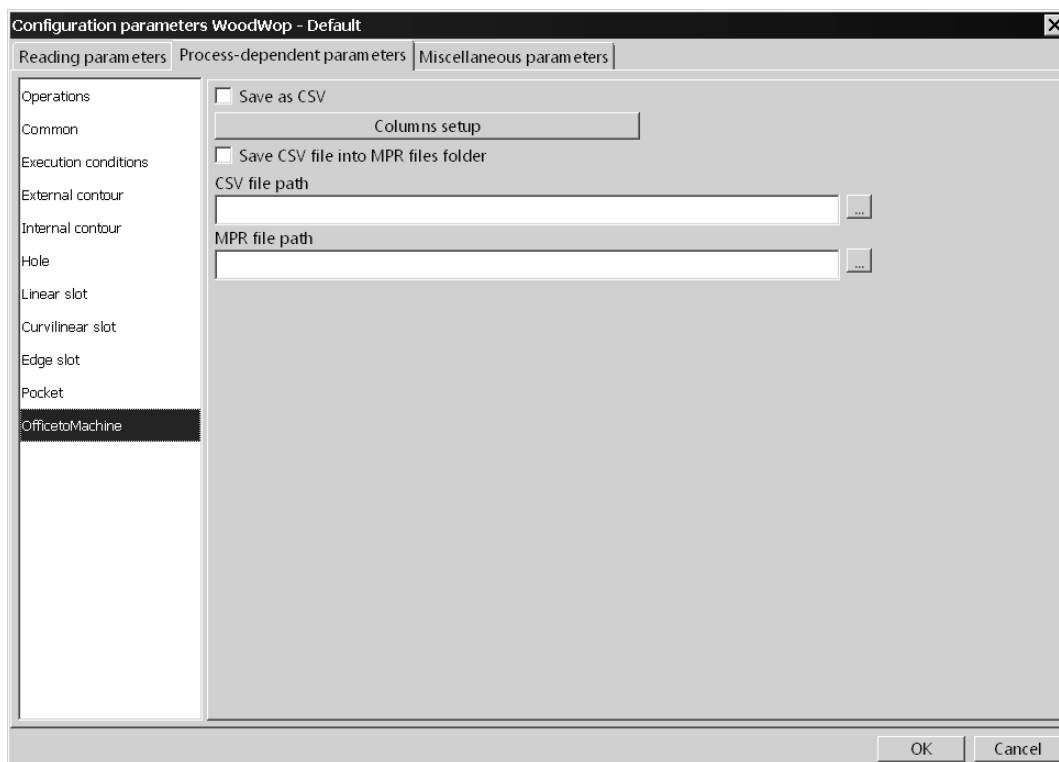


Fig. 10.416.

Save as CSV option allows to control export file saving. To set table columns push **Columns setup** button. **Columns setup** dialogue will appear on the screen (fig. 10.417).

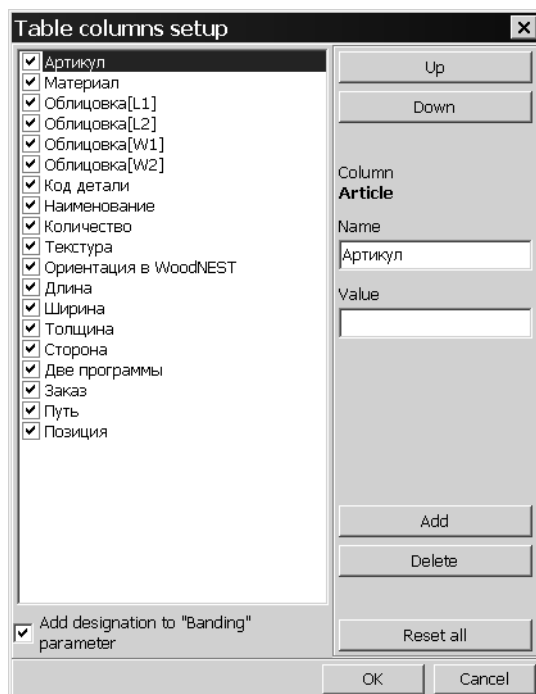


Fig. 10.417.

Enable the columns options you need to include to the table. **Up** and **Down** buttons allow to change columns position. Selected column name can be edited in **Name** field. To add the column push **Add** button. New column name will appear in the list.



It can be changed as well. Added columns values can be set in corresponding field. **Delete** button allows to delete selected column. **Reset all** button allows to cancel all custom changes. To finish table setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Browse** button of **CSV file path** and **MPR file path** fields allows to set paths for corresponding files saving in **Browse** dialogue.

Setup completion

To complete control programs for WoodWop setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.22. Huhua

10.22.1.PCDrillCAM



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.418) allow to specify operations that can be used in control program.

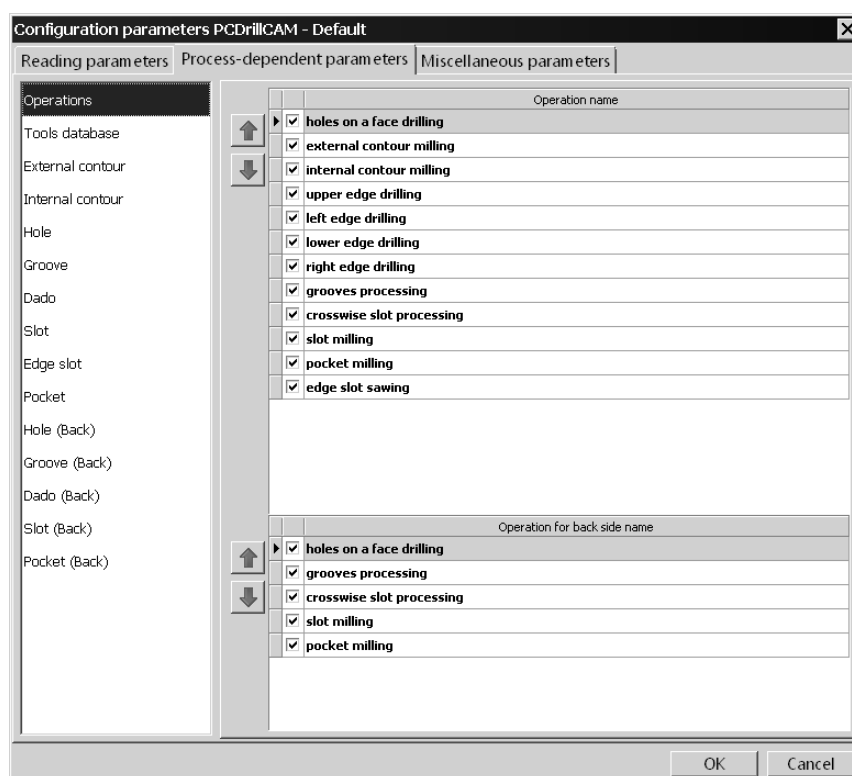


Fig. 10.418.

Usable tools

Tools database section control elements (fig. 10.419) allow to create a list of tools available for use in control programs.

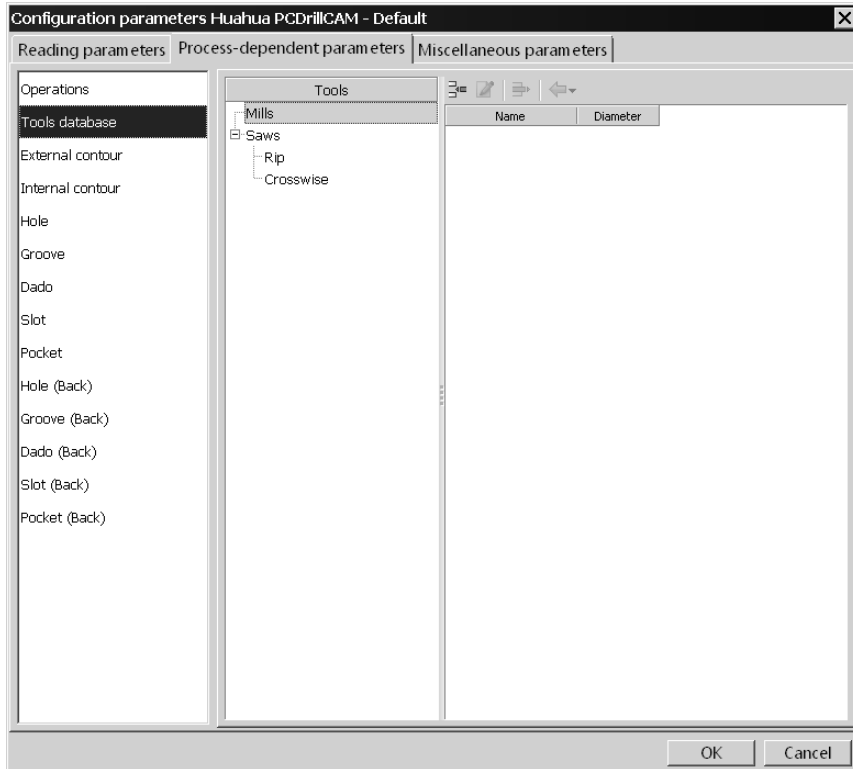


Fig. 10.419.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.420).

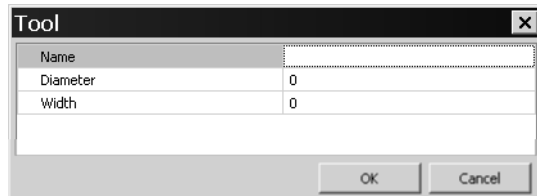


Fig. 10.420.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.421).

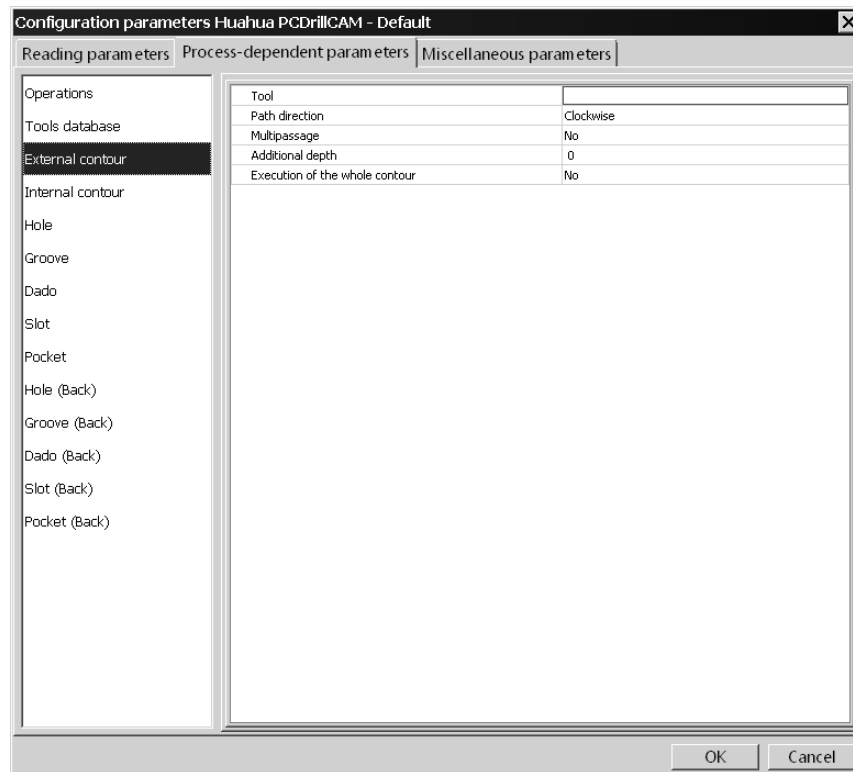


Fig. 10.421.

Section control elements allow to choose a tool, lead-in type, correction type, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.422), holes from back side – in **Hole (Back)** section in the same way.

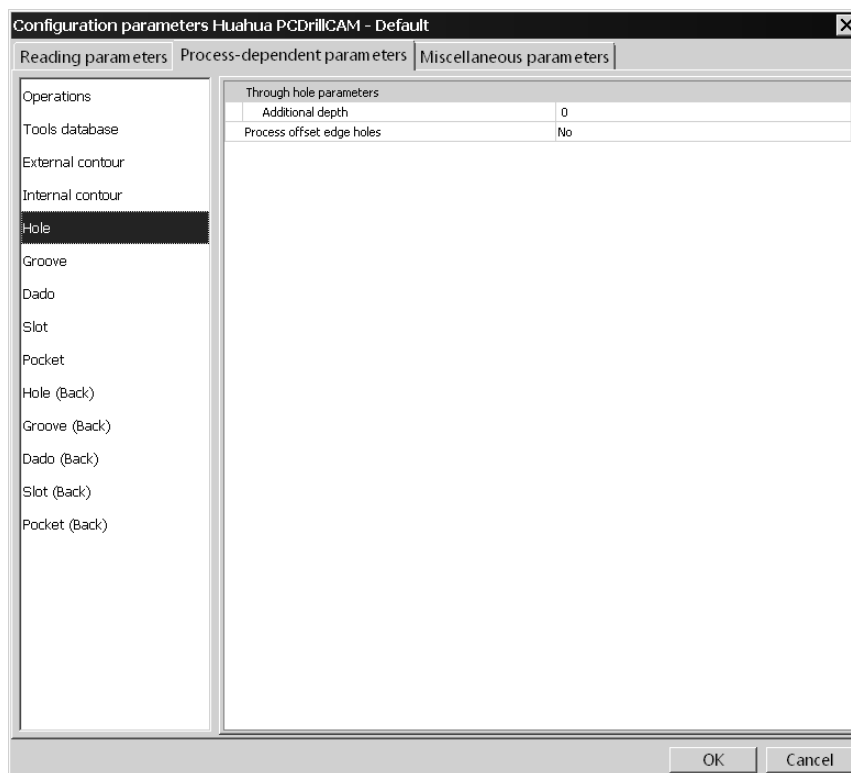


Fig. 10.422.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.423), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

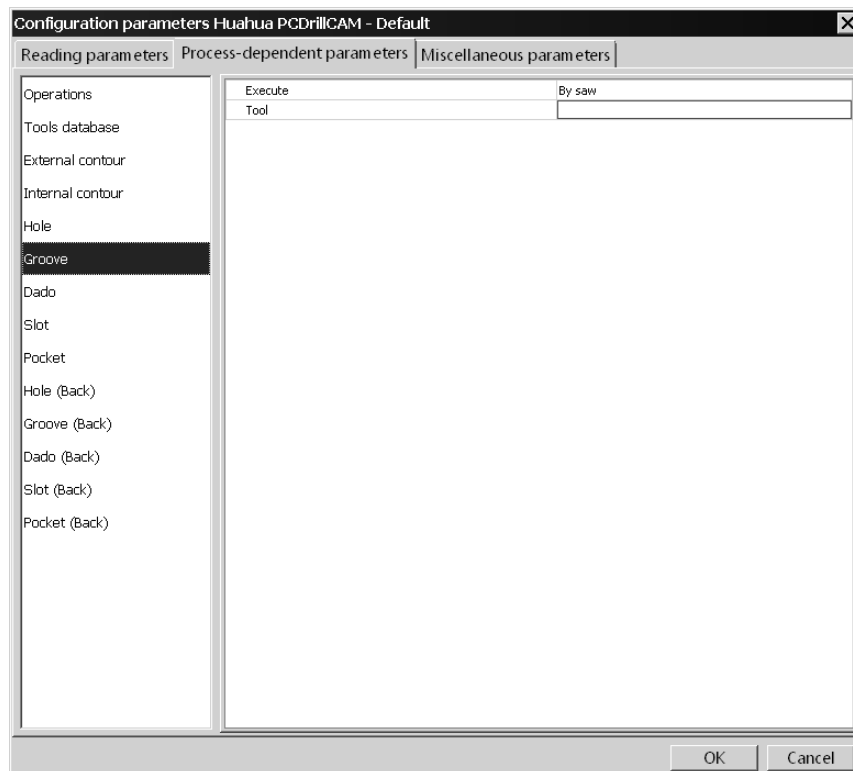


Fig. 10.423.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.424), from the back panel side – in **Slot (Back)** section.

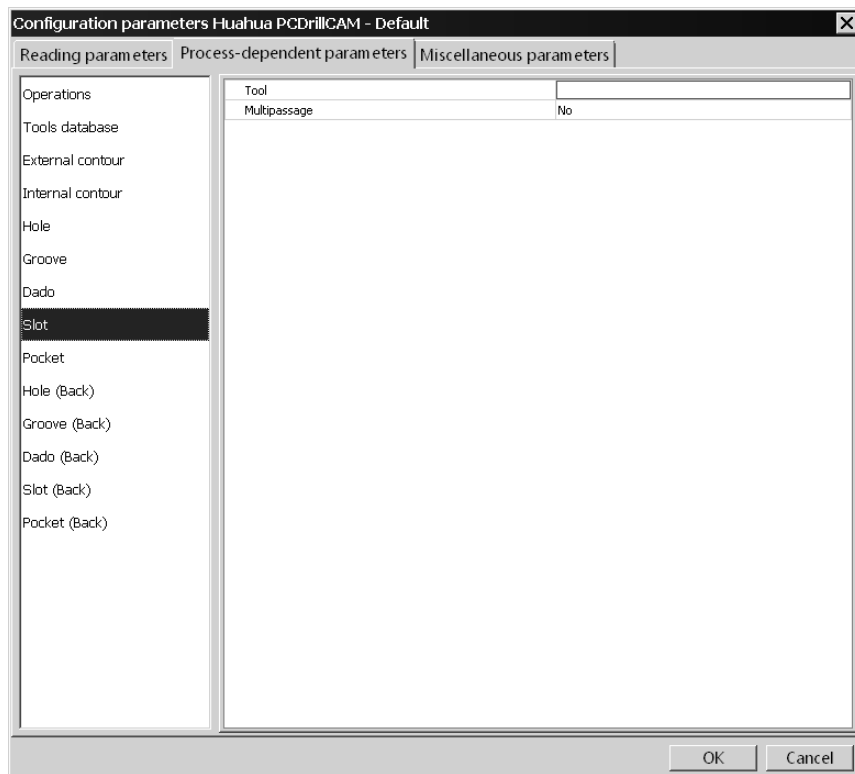


Fig. 10.424.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.425), on the back panel side – in **Pocket (Back)** section.

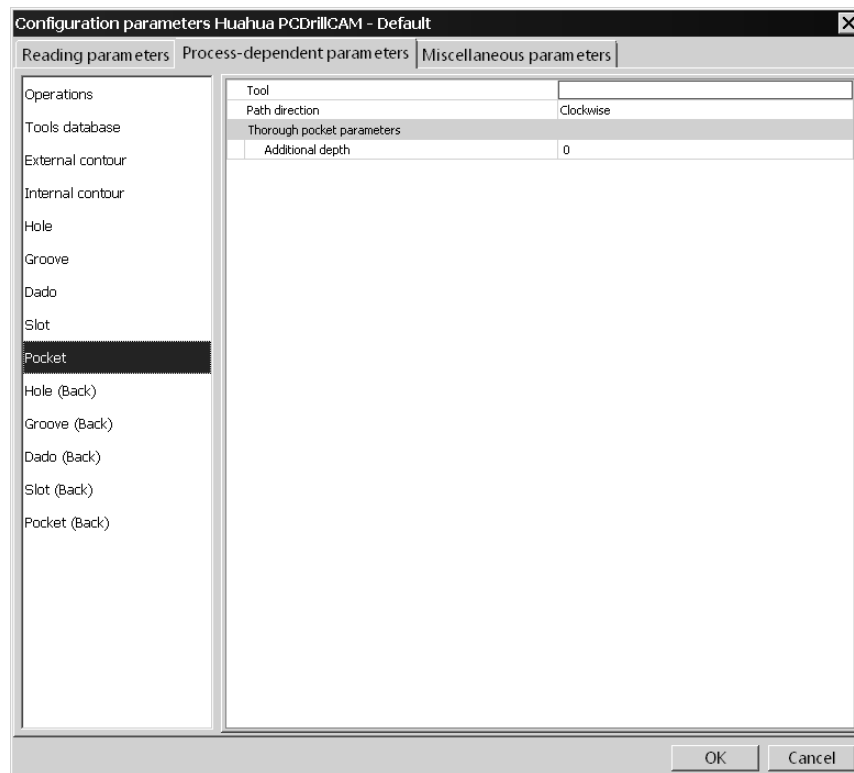


Fig. 10.425.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects from database. For pocket contour select milling execution direction, correction type and multipassage use. For internal pocket part – correction type and multipassage use.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.23. IMA

10.23.1. ImaWop 4.0 (*.fmc)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

ImaWop control programs are saved in files of *fmc* extension.

Operations

Operations section control elements (fig. 10.426) allow to specify the operations that can be used in control program.

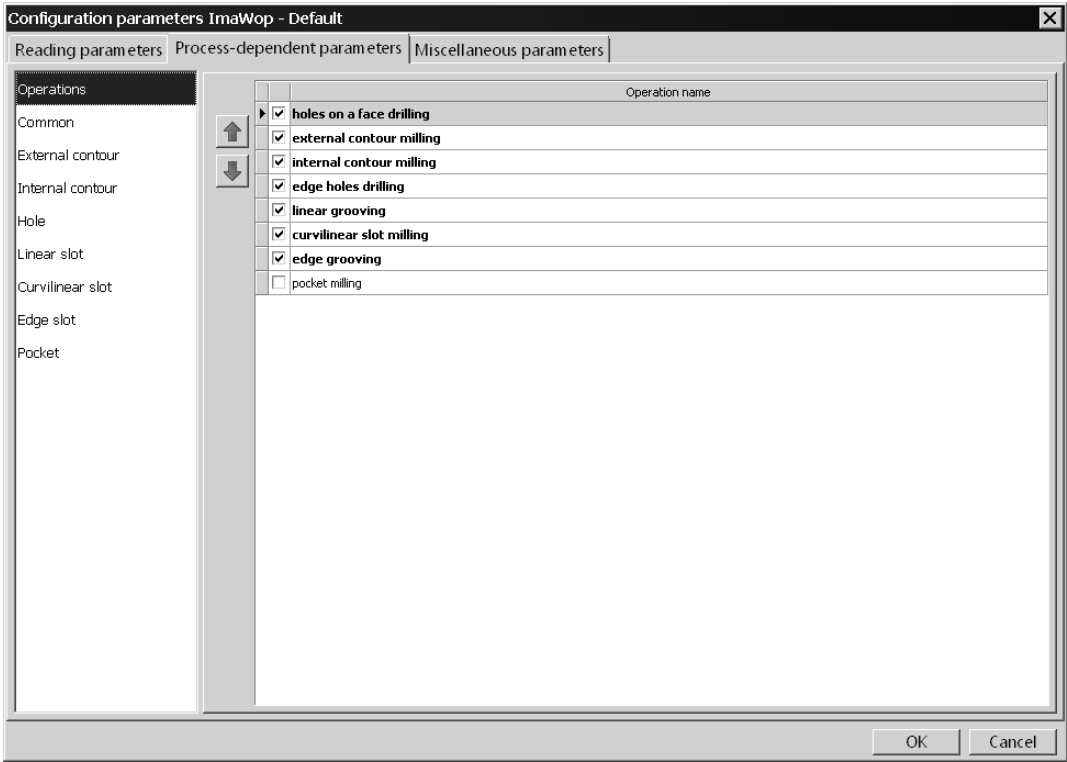


Fig. 10.426.

Common parameters

Common section control elements allow to set common parameters of control programs creation (fig. 10.427).

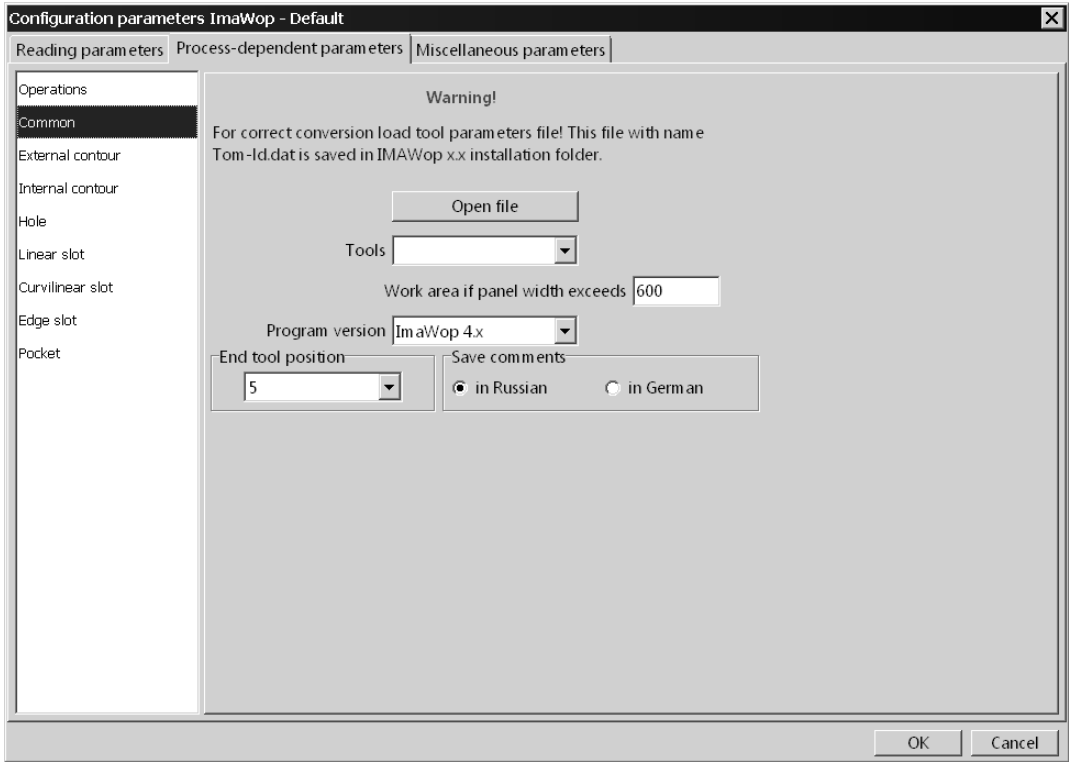


Fig. 10.427.

To set control programs creation parameters, load *Tom-Id.dat* file where tool parameters are saved. By default a file is saved in a folder, where ImaWop 4.0 is installed. Tool parameters file will be loaded automatically from the folder. To load tool parameters push **Load file** button and select tool parameters file in a standard Windows dialogue. Tool name can be chosen in **Tools** combo box.

Work area if panel width exceeds field allows to set panel width, and if it exceeds the outermost work area will be selected automatically. **Program version** combo box allows to select usable program version. **End tool position** combo box allows to set tool position after operation completion. Comments are automatically added to control program. **Save comments** group variants allow to select the language. **Append operations to CNC program** group options allow to control the inclusion in control program of operations corresponding to the options names.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** (fig. 10.428) and **Internal contour** sections of **Process-dependent parameters** dialogue. Rough and finish milling setup is executed separately on corresponding tabs. Tabs control elements content is the same and described for finish milling.

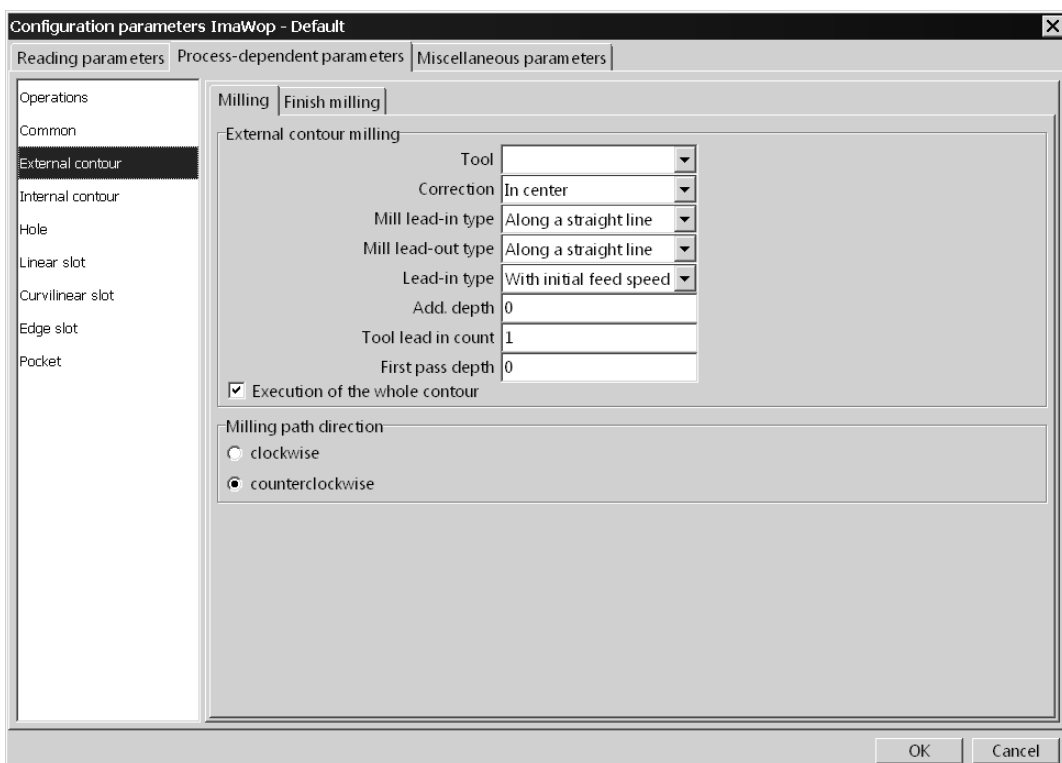


Fig. 10.428.

Finish milling option allows to control this operation. Tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.429).

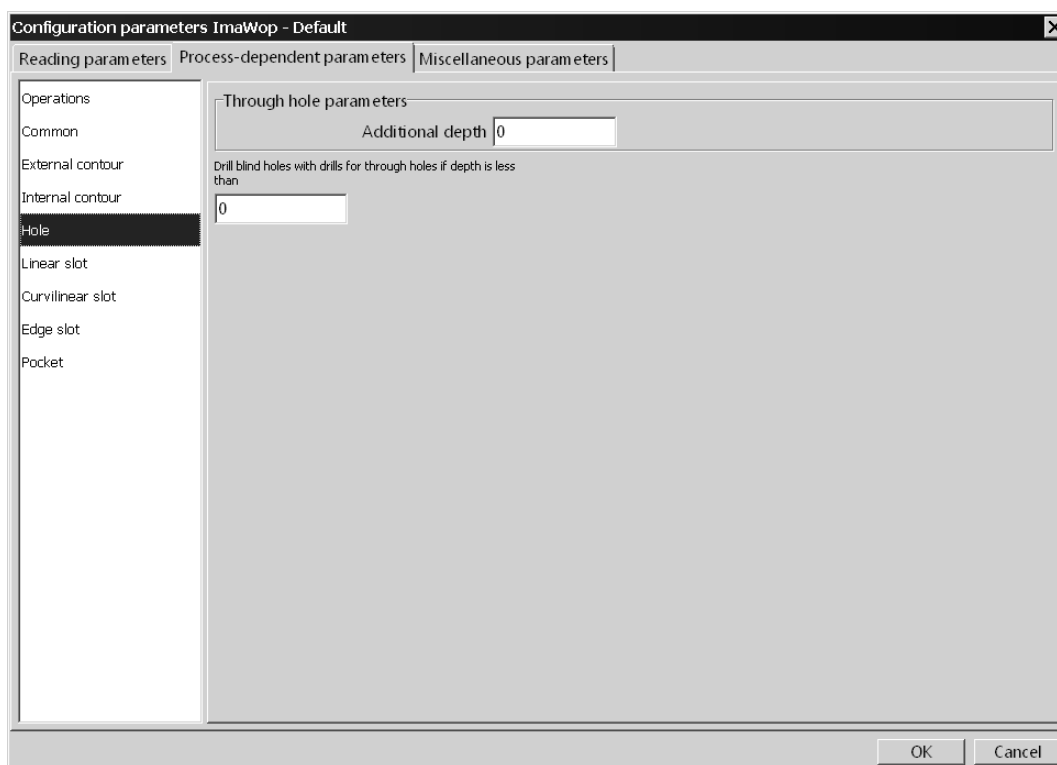


Fig. 10.429.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.430).

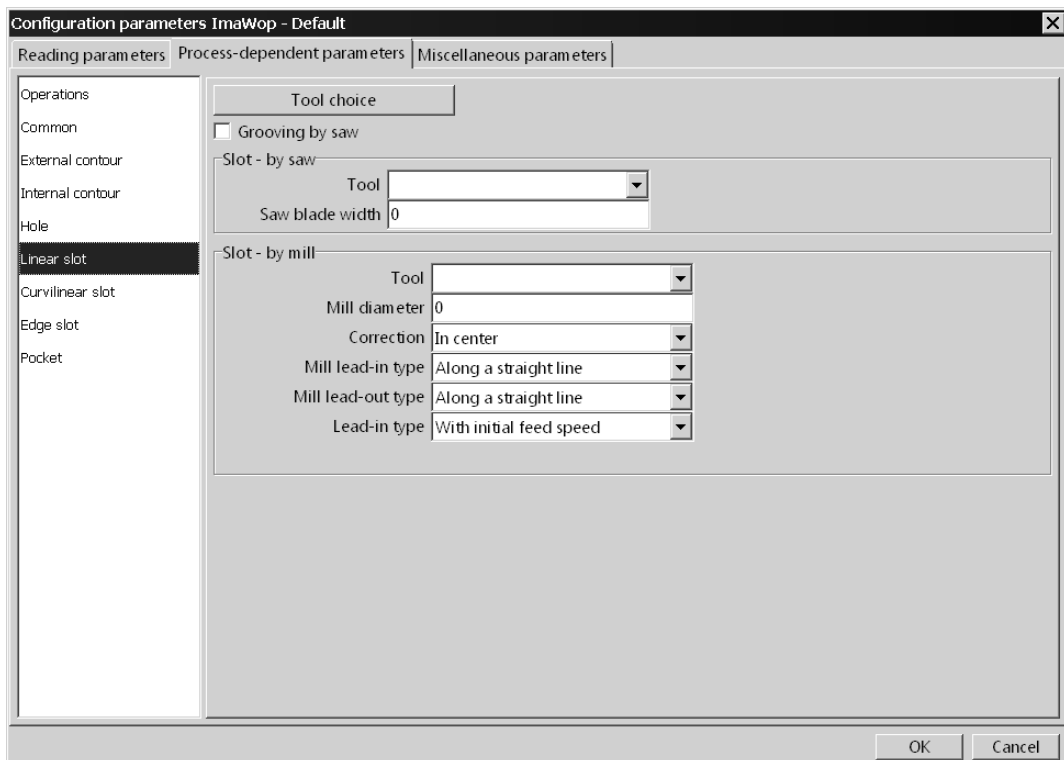


Fig. 10.430.

Tool choice button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.431).

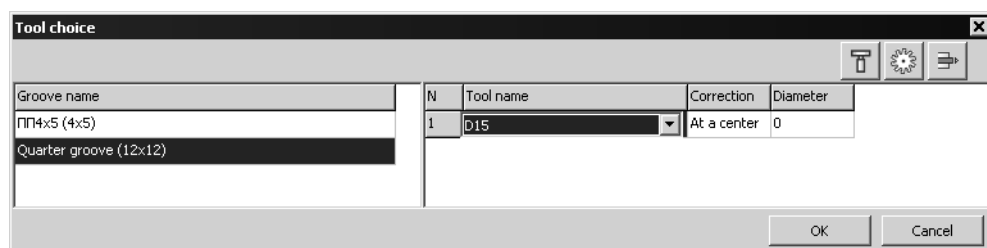


Fig. 10.431.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill diameter can be edited in list fields. Tool name and correction type should be selected from combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.432).

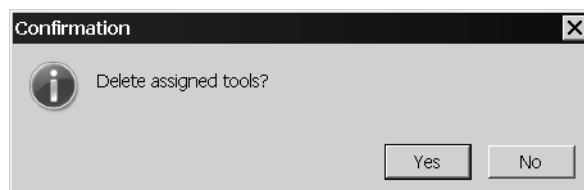


Fig. 10.432.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Grooving by saw** option allows to select a tool type for slots processing. If it is disabled, slot will be milled.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool and saw blade width.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill.

Curvilinear slot parameters

Curvilinear slots parameters are set in **Curvilinear slot** section (fig. 10.433).

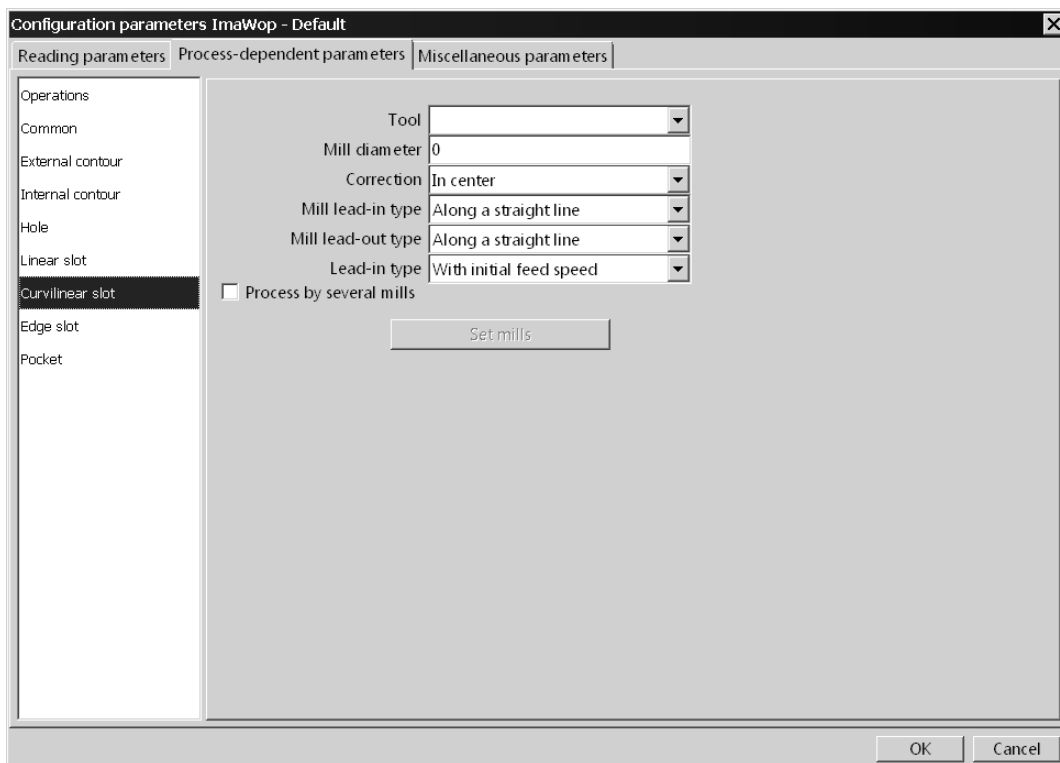


Fig. 10.433.

Section control elements allow to set control programs creation parameters for slots processing: tool selection, correction type, etc.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.434).

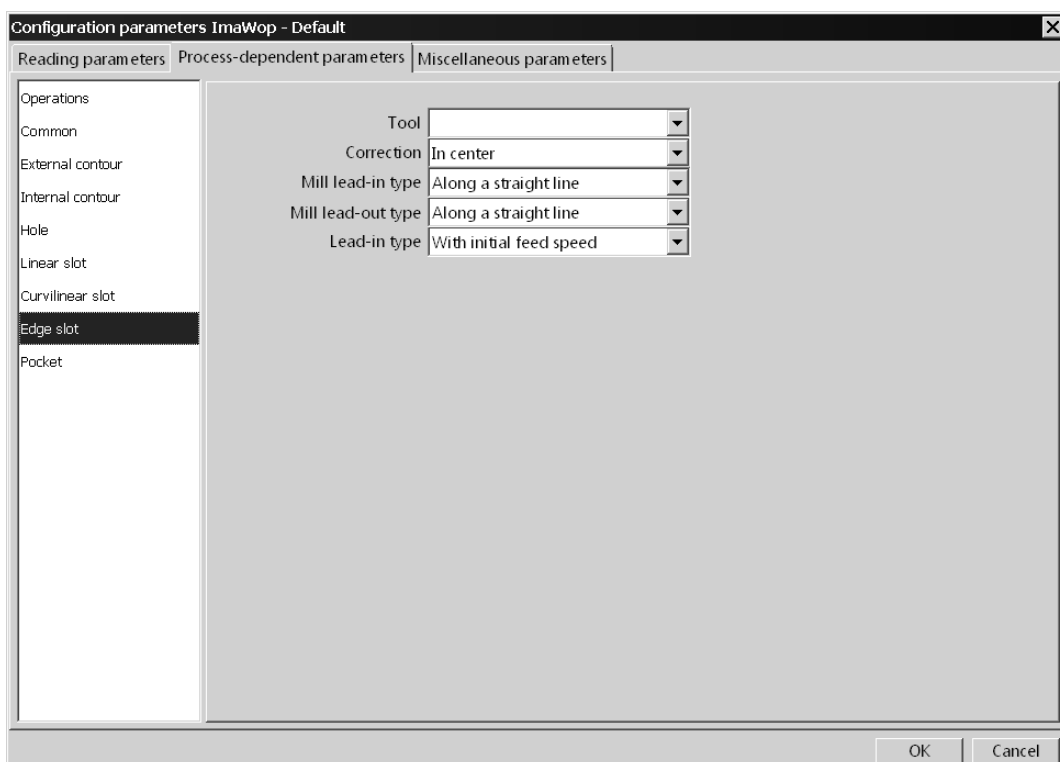


Fig. 10.434.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.435).

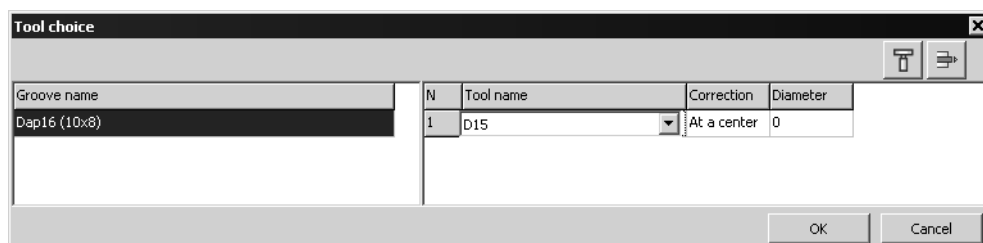


Fig. 10.435.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and correction type should be selected from combo box. Its diameter can be changed in list field.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.436) allow to set pocket processing parameters.

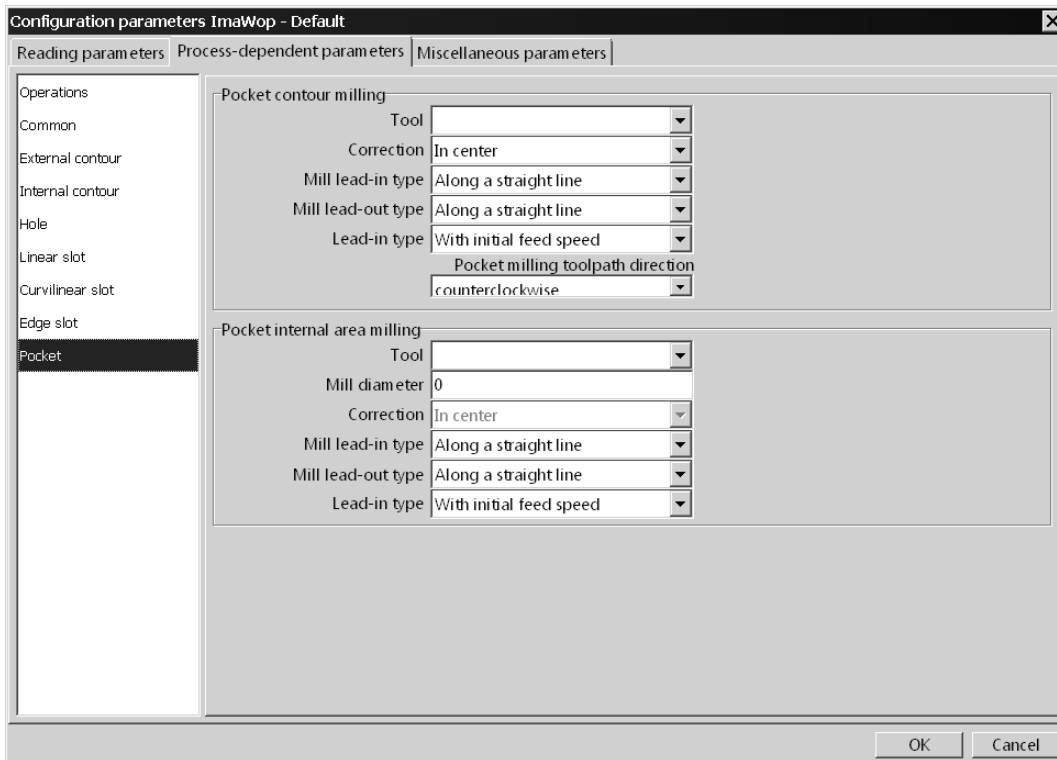


Fig. 10.436.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.24. KDT

10.24.1.KDT Delta (*.tnc, *.tn2)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation. These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.437) allow to specify the operations that can be used in control program.

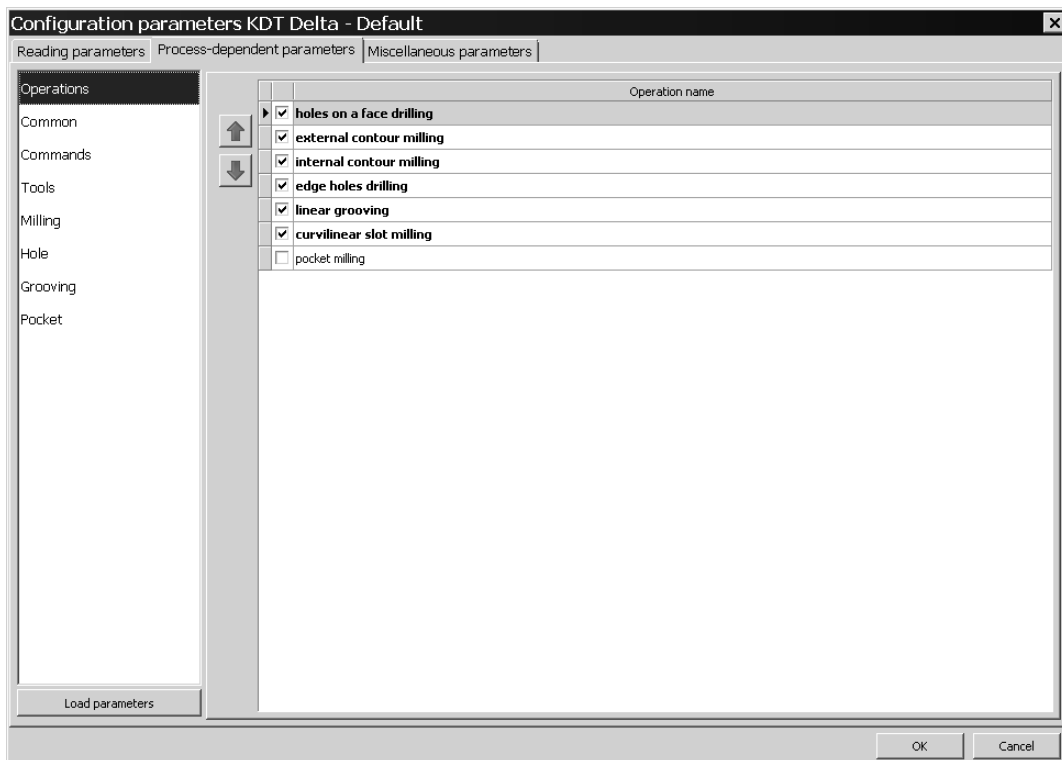


Fig. 10.437.

Common parameters

Common (fig. 10.438) section control elements allow to setup common parameters of control programs generation.

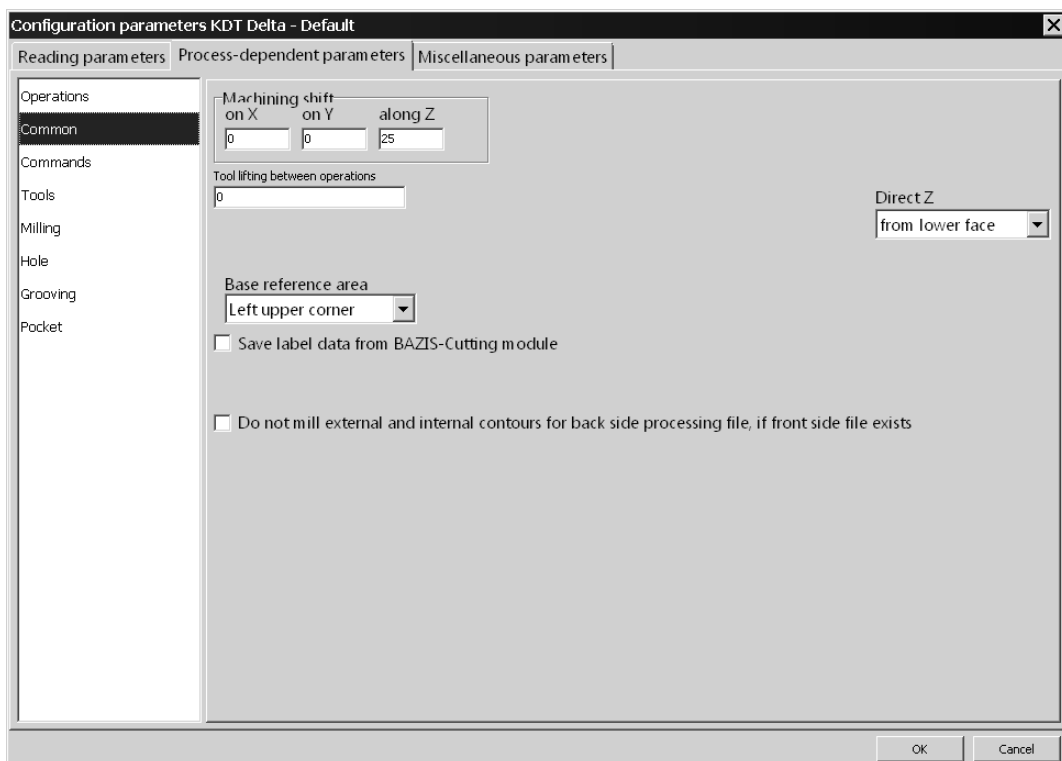


Fig. 10.438.

Machining shift group fields allow to set the shift of start point work area. **Create** group options allow to control files generation of control programs for separate operations. **Tool lifting between operations** field allows to set tool height value during moving between operations. **Base reference area** combo box variants allow to set base area position. **Save label data** option allows to save the corresponding data.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.439) allow to control separate commands to control program inclusion.

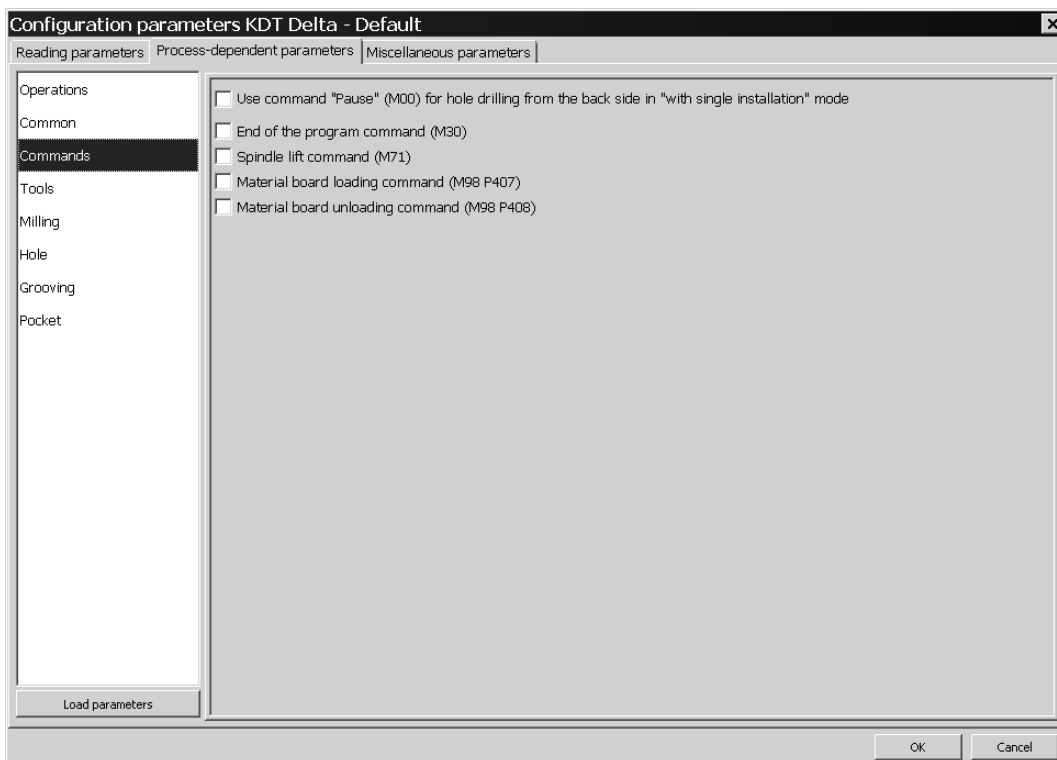


Fig. 10.439.

Usable tools

Tools section control elements (fig. 10.440) allow to create a list of tools available for use in control programs.

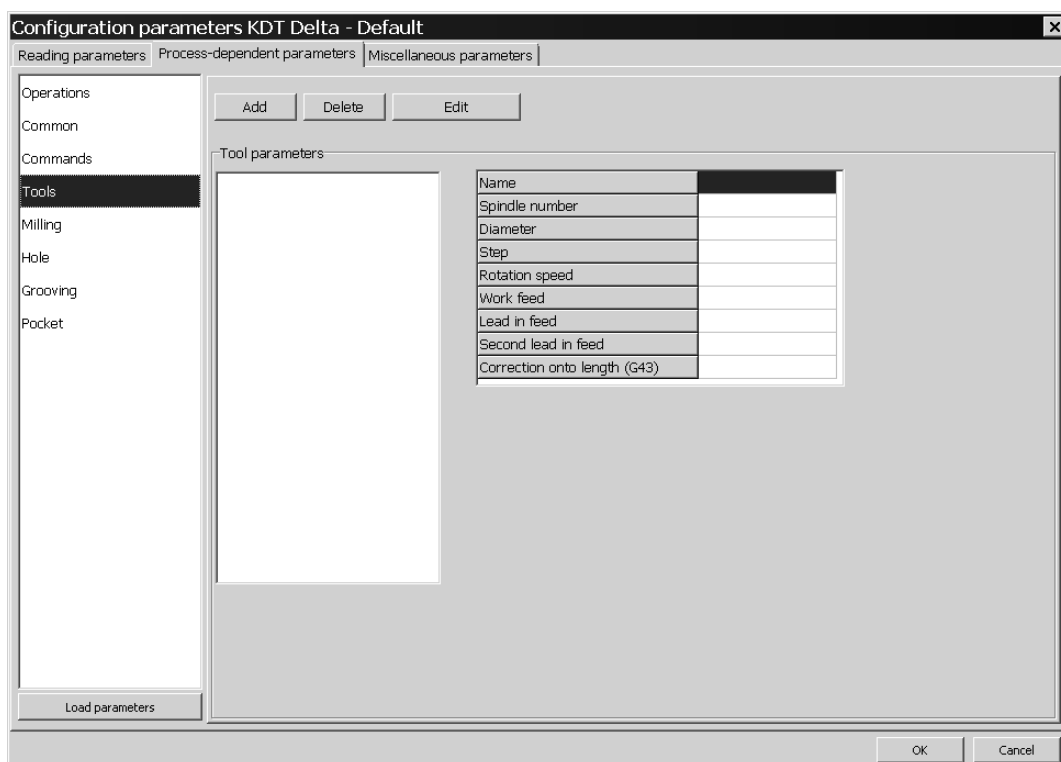


Fig. 10.440.

To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.441).

Fig. 10.441.

In this dialogue enter a tool name and set its parameters values. To complete tool adding, push **OK** button. The dialogue will be closed, a tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change selected tool parameters. **Delete** button allows to delete selected tool from the list.

Milling parameters

Milling section control elements (fig. 10.442) allow to setup milling parameters of outer and internal contour.

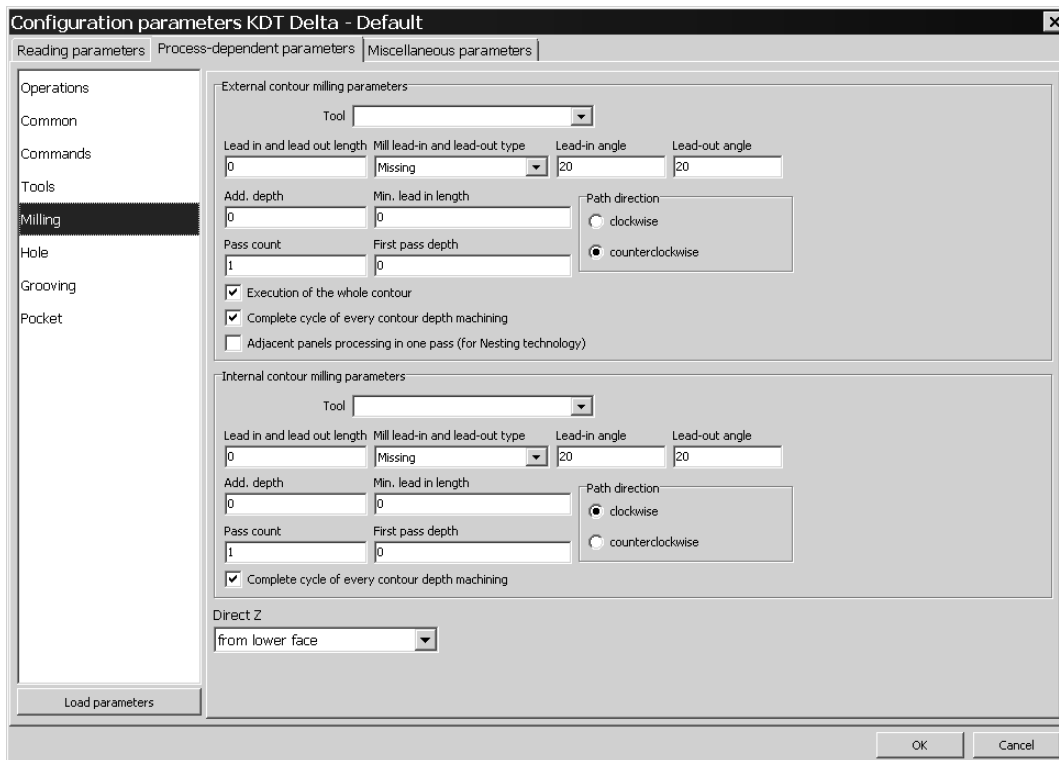


Fig. 10.442.

Tool combo box allows to select a tool name for milling. Tools should be created beforehand in **Tools** section. Section control elements allow to set processing depth, pass count, etc.

Execution of the whole contour option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Path direction** of internal and external contour milling group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.443).

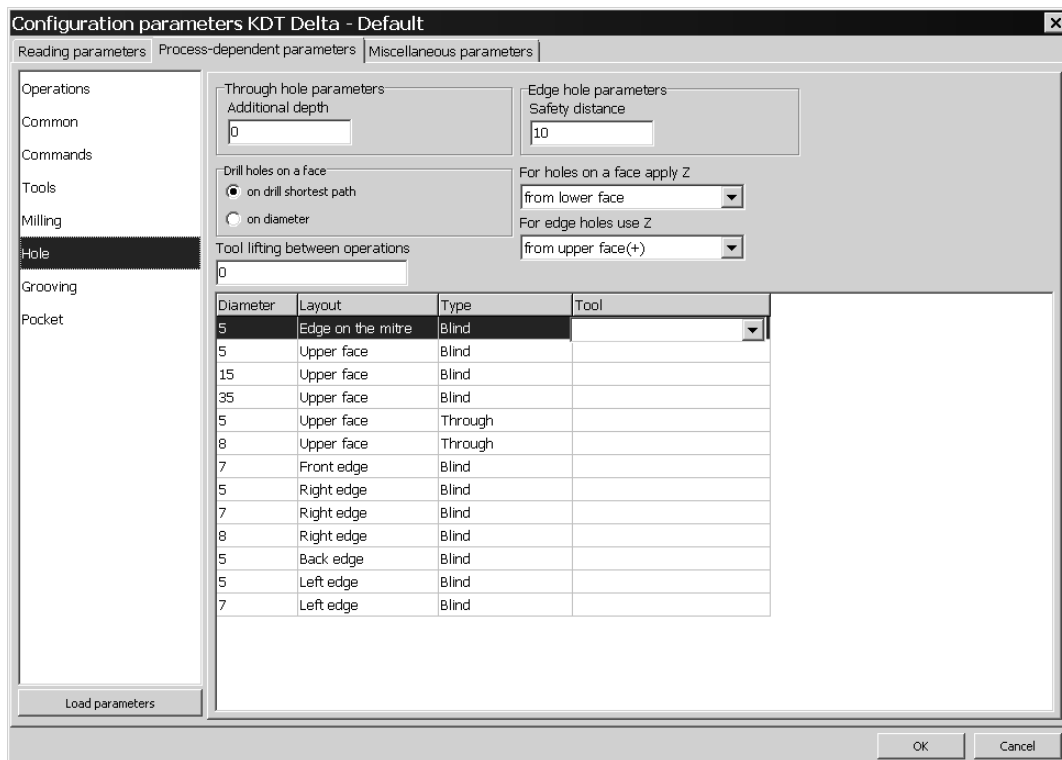


Fig. 10.443.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slots processing parameters

Grooving section control elements (fig. 10.444) allow to set slots processing parameters.

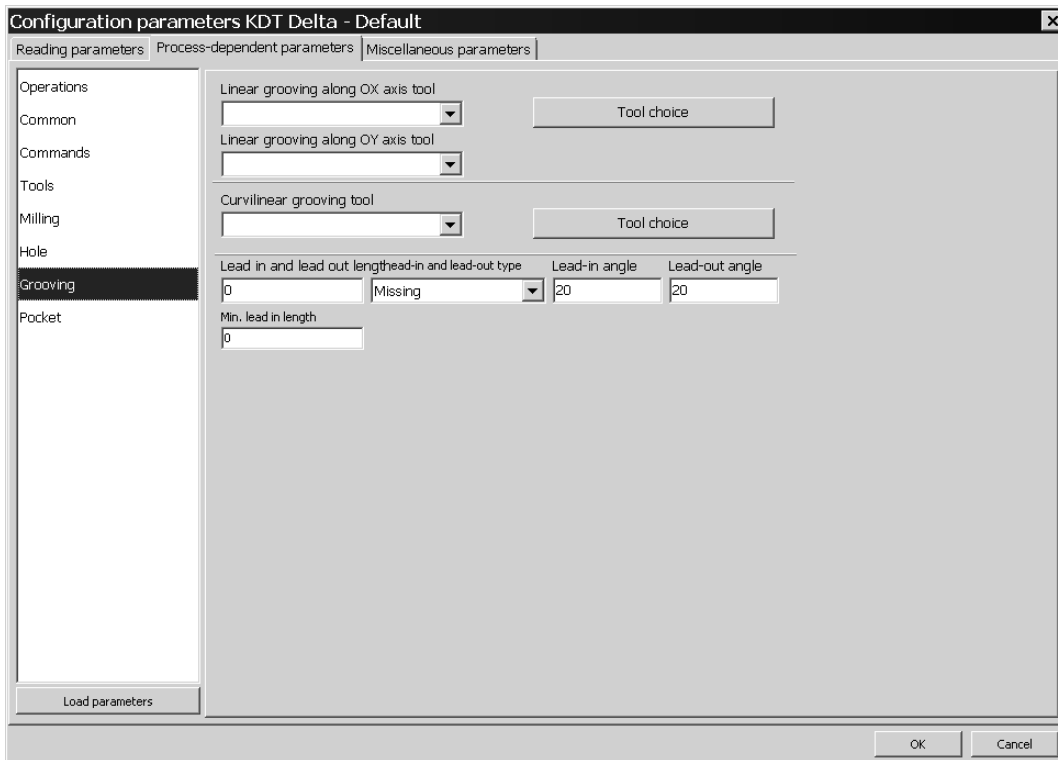


Fig. 10.444.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.445).

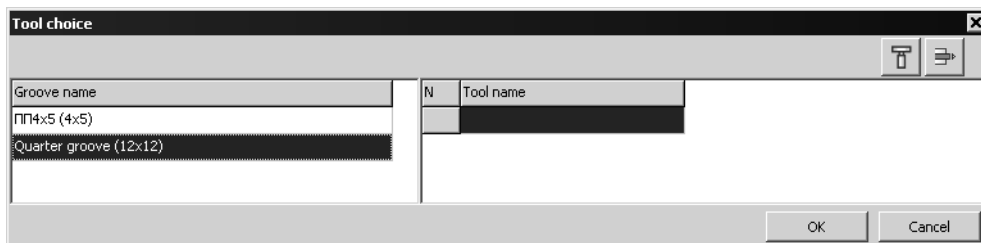


Fig. 10.445.



Slot name table contains all slots names of the model. You can set particular tools for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in section Usable tools on p. 362.



Delete button allows to delete selected tool from the list. To complete tools assignment, push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.446) allow to set slots processing parameters.

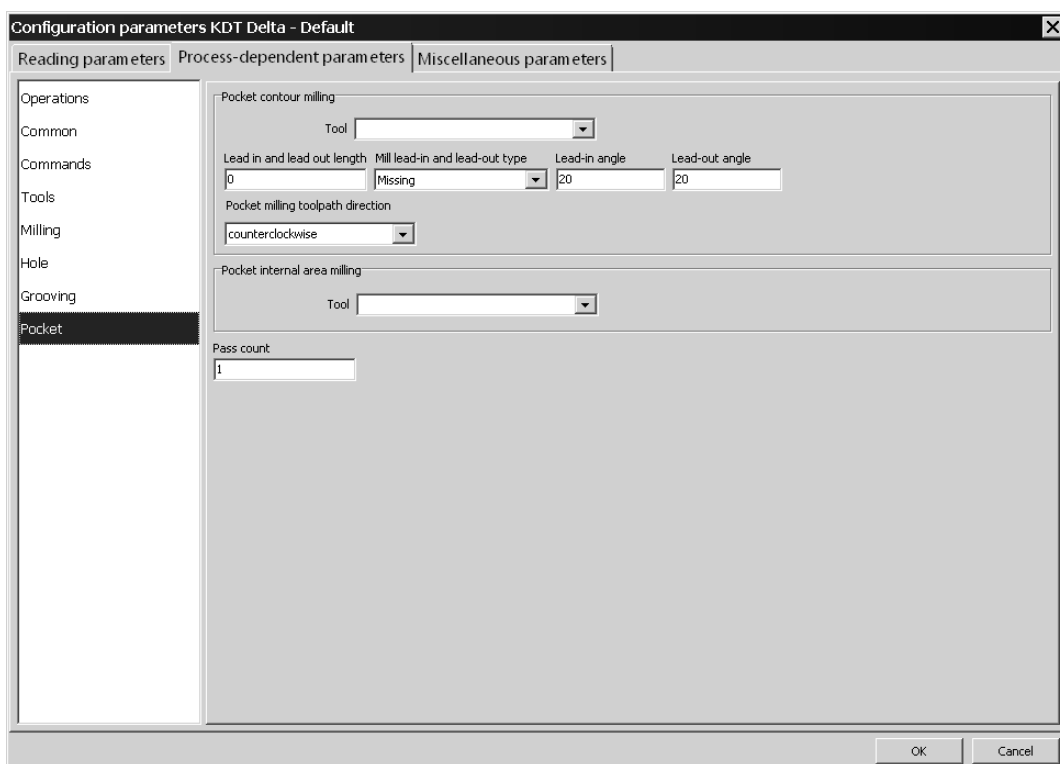


Fig. 10.446.

Tool combo boxes in **Pocket contour milling** and **Pocket internal area milling** groups allow to select a tool name for corresponding objects milling. The tools should be created beforehand in **Tools** section. It is necessary to select milling execution direction for pocket contour in corresponding combo box.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.24.2.KDT Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.447) allow to specify the operations that can be used in control program.

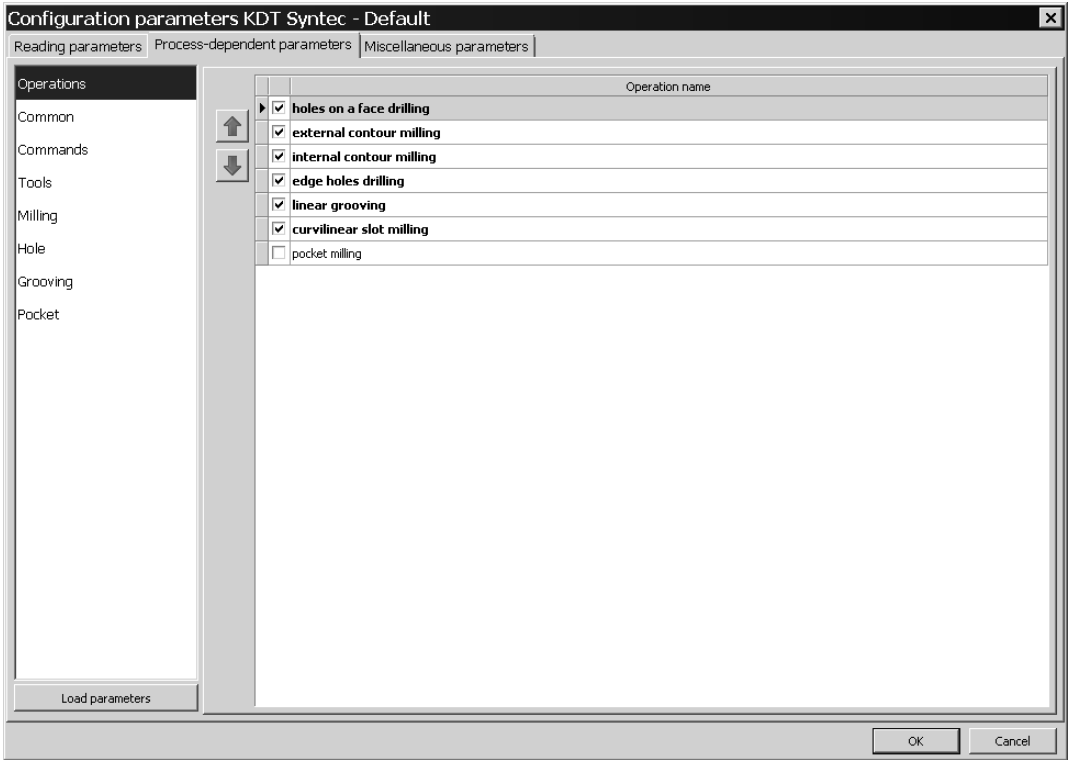


Fig. 10.447.

Common parameters

Common section control elements (fig. 10.448) allow to set common parameters of control programs.

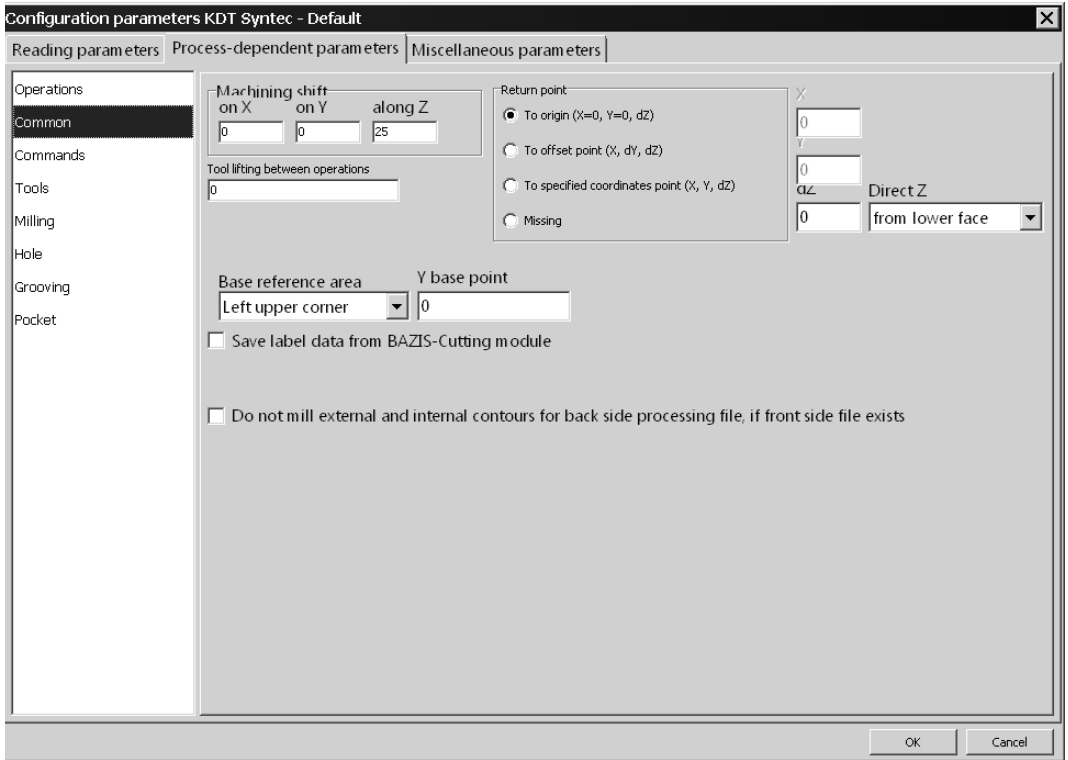


Fig. 10.448.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If it is disabled, return point coordinates are set by **X**, **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Base reference area** combo box variants allow to set base reference area position. **Y base point** field allows to set base point position.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.449) allow to control command to control program inclusion.

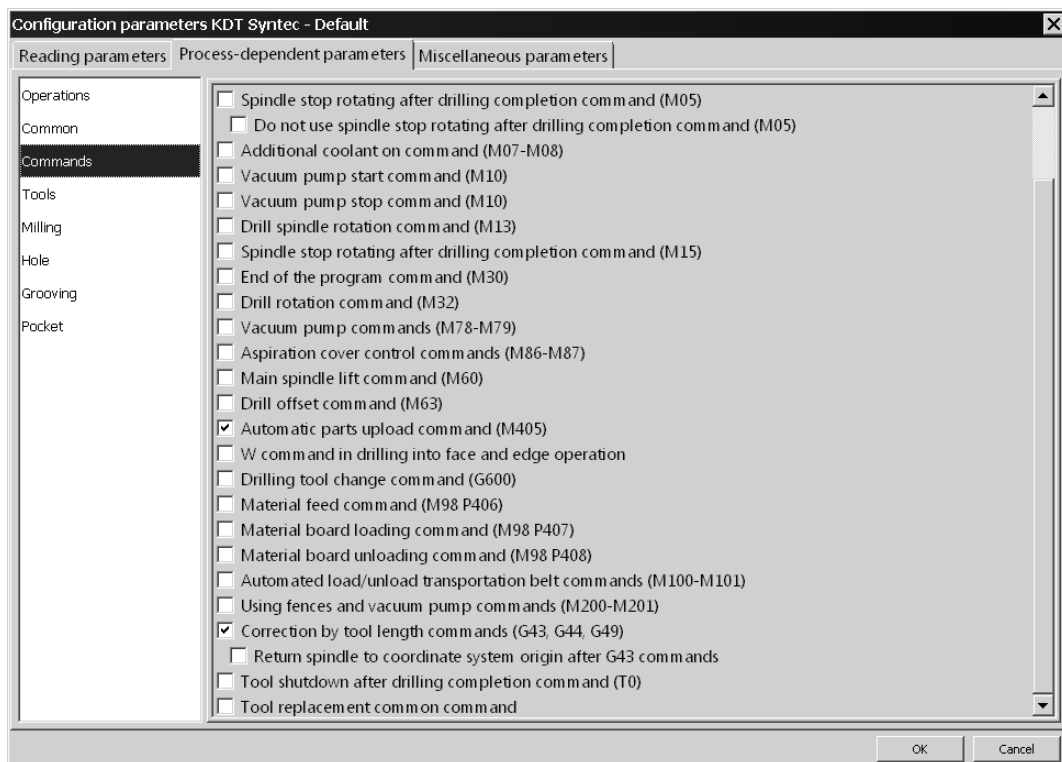


Fig. 10.449.

Usable tools

Tools section control elements (fig. 10.450) allow to create a list of tools available for use in control programs.

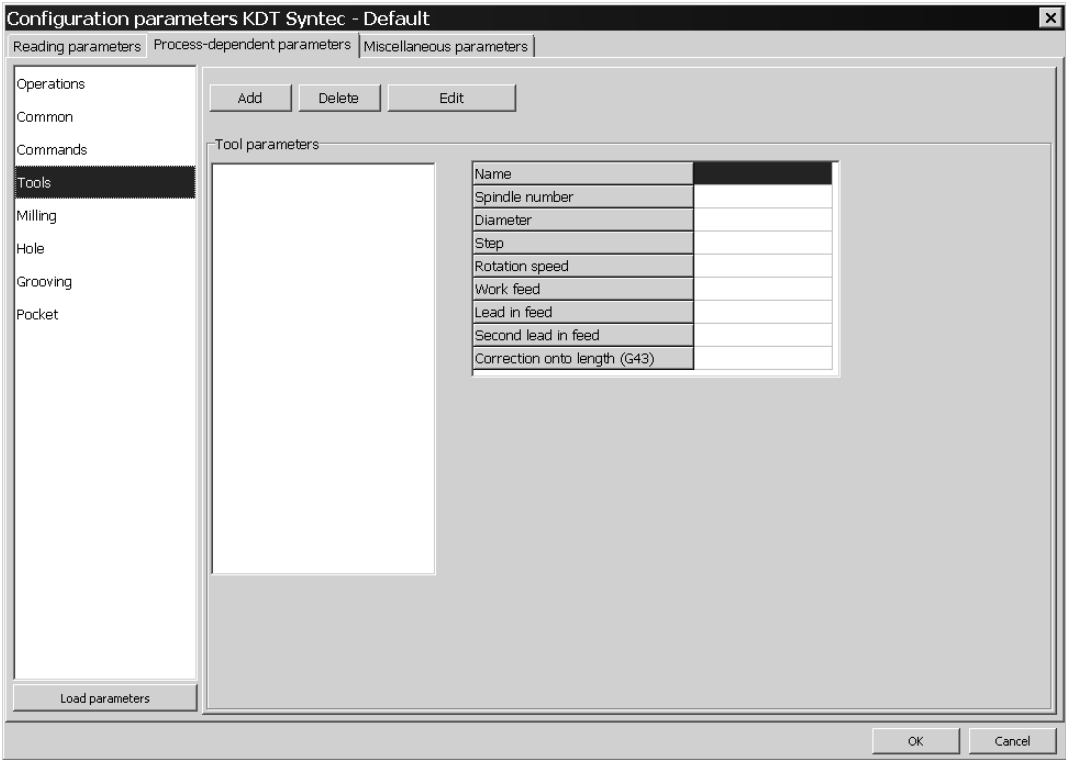


Fig. 10.450.

To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.451).

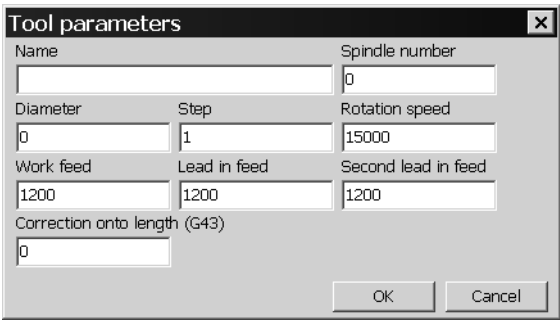


Fig. 10.451.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.452) allow to set milling parameters of outer and internal contours.

Configuration parameters KDT Syntec - Default

Reading parameters | Process-dependent parameters | **Miscellaneous parameters**

Operations
Common
Commands
Tools
Milling
Hole
Grooving
Pocket

External contour milling parameters

Tool: []

Lead in and lead out length: [0] Mill lead-in and lead-out type: [Missing] Lead-in angle: [20] Lead-out angle: [20]

Add. depth: [0] Min. lead in length: [0] Path direction: ☐ clockwise ☒ counterclockwise

Pass count: [1] First pass depth: [0]

☒ Execution of the whole contour
☒ Complete cycle of every contour depth machining
☐ Adjacent panels processing in one pass (for Nesting technology)

Internal contour milling parameters

Tool: []

Lead in and lead out length: [0] Mill lead-in and lead-out type: [Missing] Lead-in angle: [20] Lead-out angle: [20]

Add. depth: [0] Min. lead in length: [0] Path direction: ☒ clockwise ☐ counterclockwise

Pass count: [1] First pass depth: [0]

☒ Complete cycle of every contour depth machining

Direct Z: [from lower face]

Load parameters

OK Cancel

Fig. 10.452.

Tool combo box allows to select a milling tool name. Tools should be created in **Tools** section. Section control elements allow to set pass depth, pass count, etc.

Execution of the whole contour option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool moving direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.453).

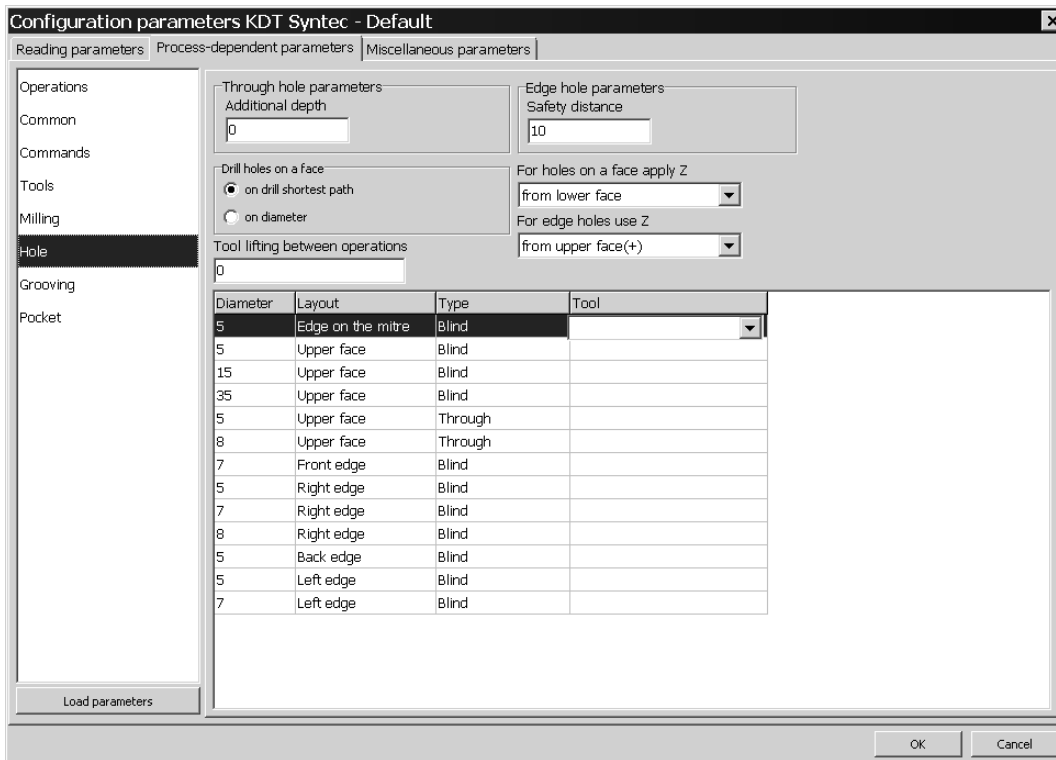


Fig. 10.453.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. Hole table allows to set tool automatic selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order. **For edge holes use Z** and **For holes on a face apply Z** combo boxes variants allow to choose a face for this parameter application. **Tool lifting between operations** field allows to set tool height between operations.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Groove parameters

Groove processing setup is performed in **Grooving** section (fig. 10.454).

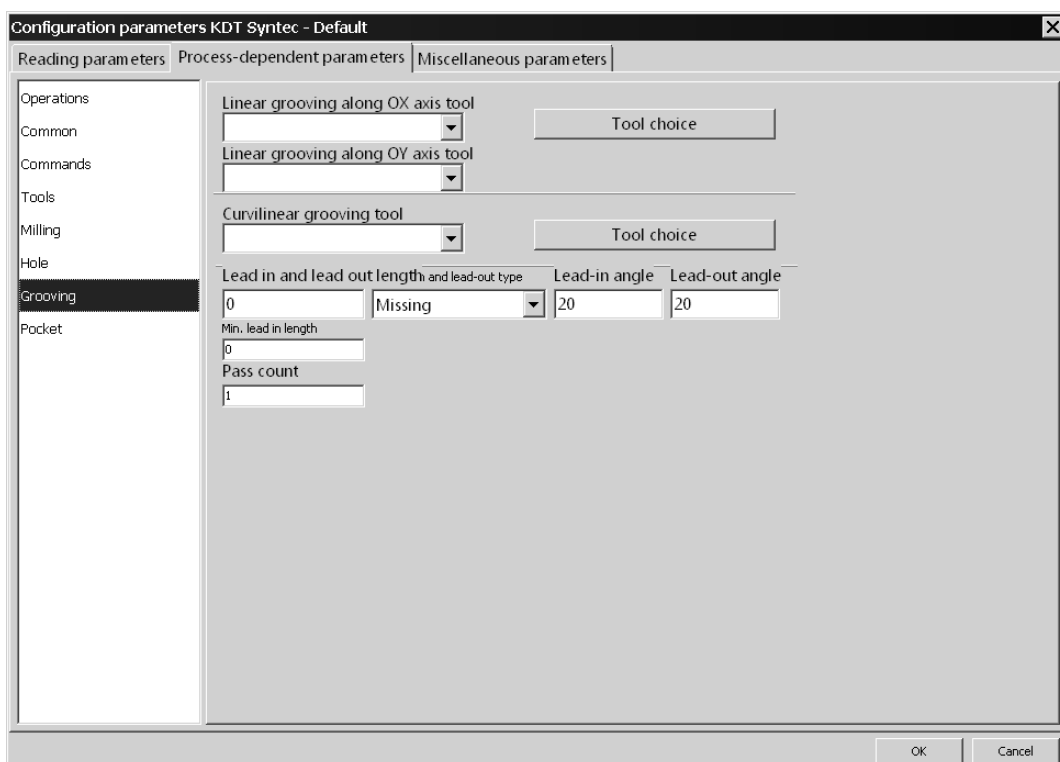


Fig. 10.454.

Linear grooving along OX axis tool, Linear grooving OY axis tool, Curvilinear grooving tool combo boxes allow to choose a tool for corresponding operations. Tool list should be created beforehand. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.455).

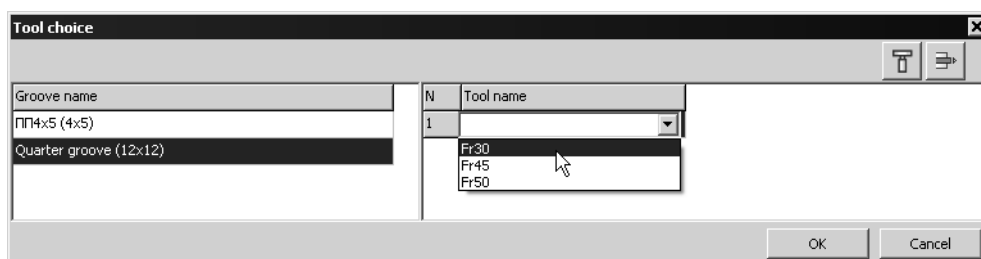


Fig. 10.455.



Slot name list contains slots names in the model. **Add mill** button allows to map a tool from prepared list to selected slot. Several tools can be set for one slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes. **Lead in and lead out length, Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Pocket parameters

Pocket section control elements (fig. 10.456) allow to set pocket processing parameters.

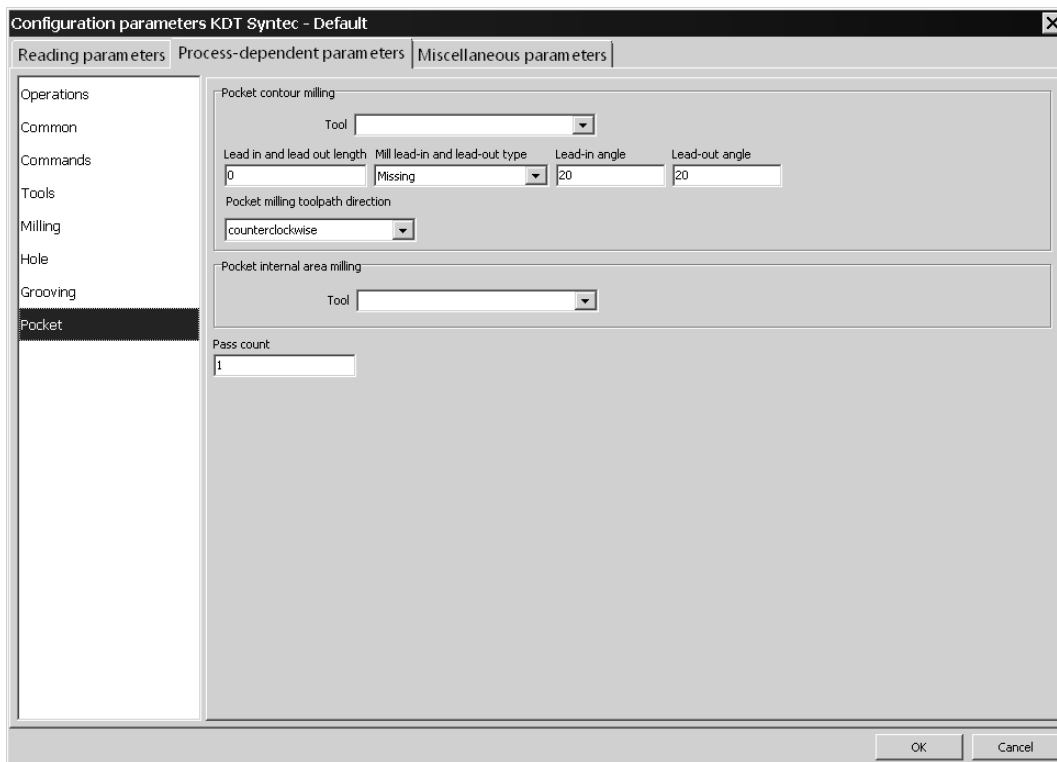


Fig. 10.456.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select tool name for corresponding objects from database milling. **Lead in and lead out length**, **Lead-in angle** section fields allow to set supplementary parameters of slot processing.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.24.3.KDT TPA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.457) allow to specify the operations that can be used in control program.

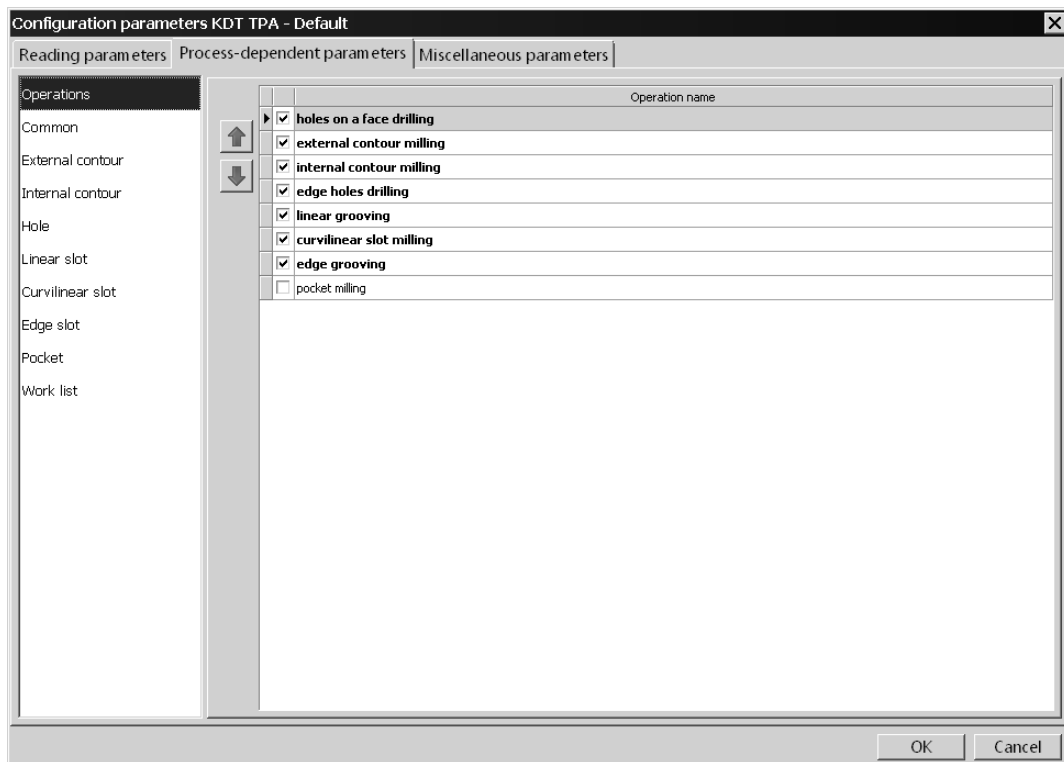


Fig. 10.457.

Common parameters

Common parameters are set in **Common** section (fig. 10.458).

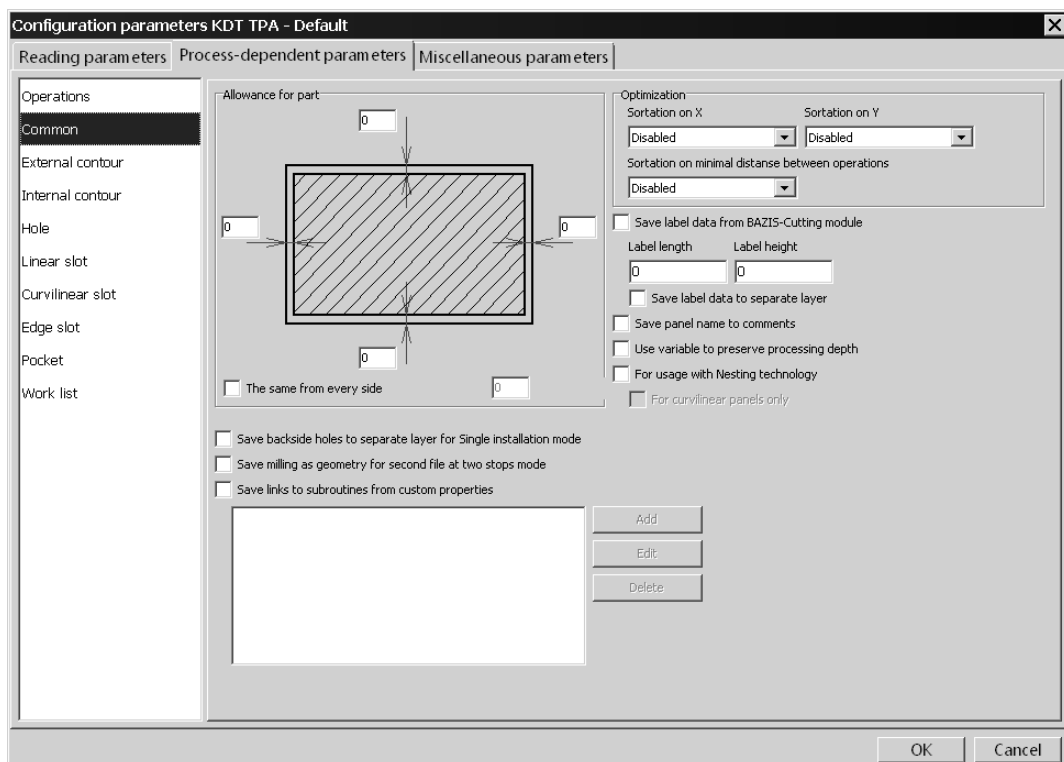


Fig. 10.458.

Tab control elements allow to control the saving of optimization parameters, label data, etc. **Allowance for part** section control elements allow to set allowance values. Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, input field for the same allowance for every panel edge will become available.

Contours parameters

Contour milling parameters setup is performed in **Outer contour** (fig. 10.459) and **Internal contour** sections.

Configuration parameters KDT TPA - Default	
Reading parameters Process-dependent parameters Miscellaneous parameters Operations Common External contour Internal contour Hole Linear slot Curvilinear slot Edge slot Pocket Work list External contour milling Tool Correction: In center Processing from the middle of contour element: No Add. depth: 0 Lead-in type: Missing Lead-in Length/Radius: 0 Lead-in Z start: 0 Lead-out type: Missing Lead-out Length/Radius: 0 Final Z: 0 Lead in/Lead out point trajectory overlapping: 0 Pass count: 1 First pass depth: 0 Execution of the whole contour: Yes Milling path direction ☐ clockwise ☒ counterclockwise OK Cancel	

Fig. 10.459.

Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Holes section control elements (fig. 10.460) allow to setup drilling operations parameters.

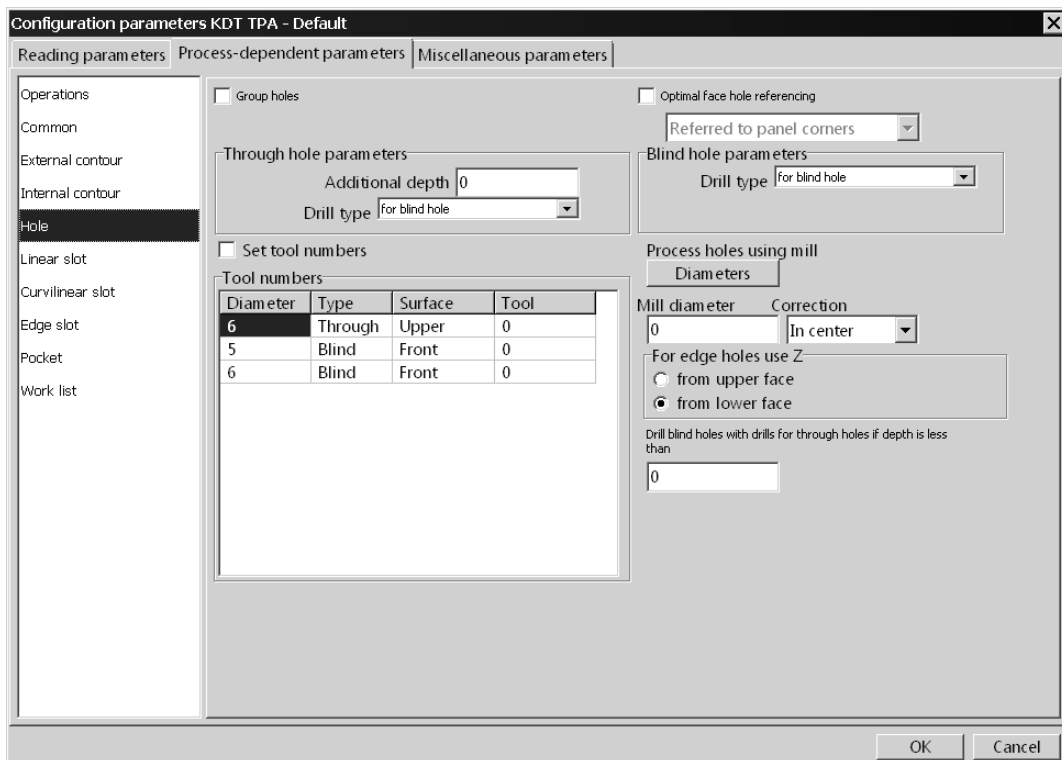


Fig. 10.460.

If several holes are located on the same straight line and the distance between them (step) is the same, it is possible to combine them and reduce the program volume. For this purpose enable **Group holes using macro** option. Macros files names should be selected from a combo box.

Through holes can be drilled with drills for through as well as for blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameter** button. **Hole diameters** dialogue will appear on the screen (fig. 10.461).

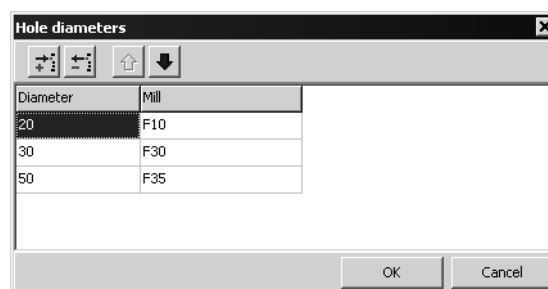


Fig. 10.461.



The dialogue table contains holes diameter values and corresponding mill designations. Fill in the fields using a keyboard. **Add row** button allows to add an empty row to the table.



To delete the current row, push **Delete row** button. The arrow buttons allow to change the current row position in the table. To complete the setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.462).

Fig. 10.462.

Through slot always cut with saw and **Grooving by saw** options allow to specify through slots direction that will be processed with saw. **Tool choice** button allows to set the tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.463).

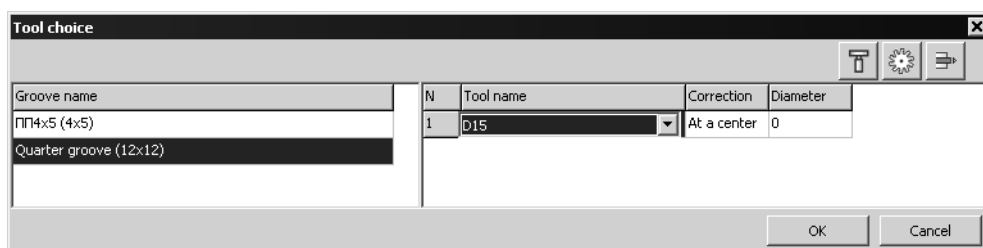


Fig. 10.463.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add the corresponding tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to select a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.464).

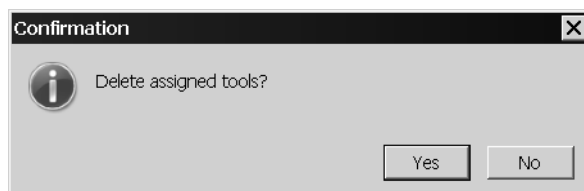


Fig. 10.464.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. To complete parameters setup, push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear slots parameters setup is performed in **Curvilinear slot** section (fig. 10.465).

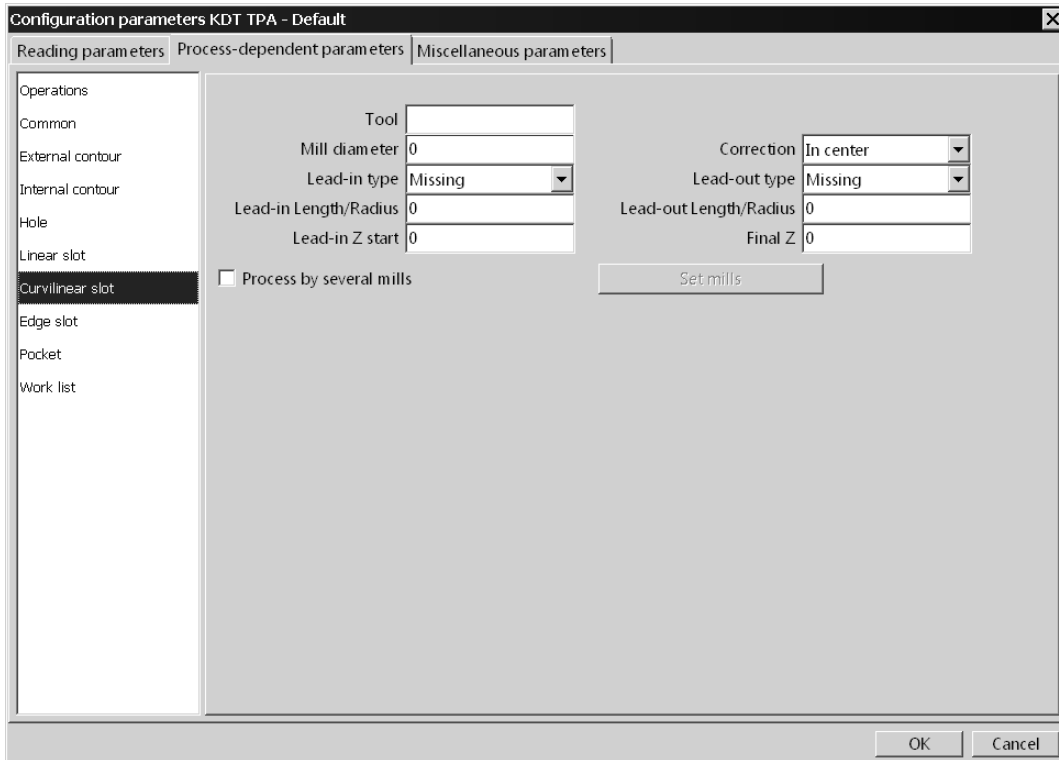


Fig. 10.465.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.466).

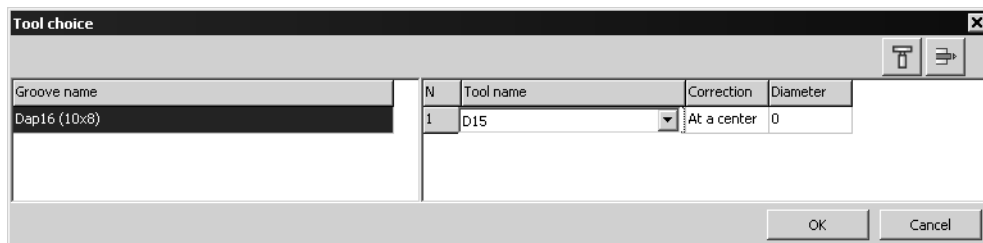


Fig. 10.466.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Delete button allows to delete selected tool from the list. To complete tools assignment, push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slots parameters setup is performed in **Edge slot** section (fig. 10.467).

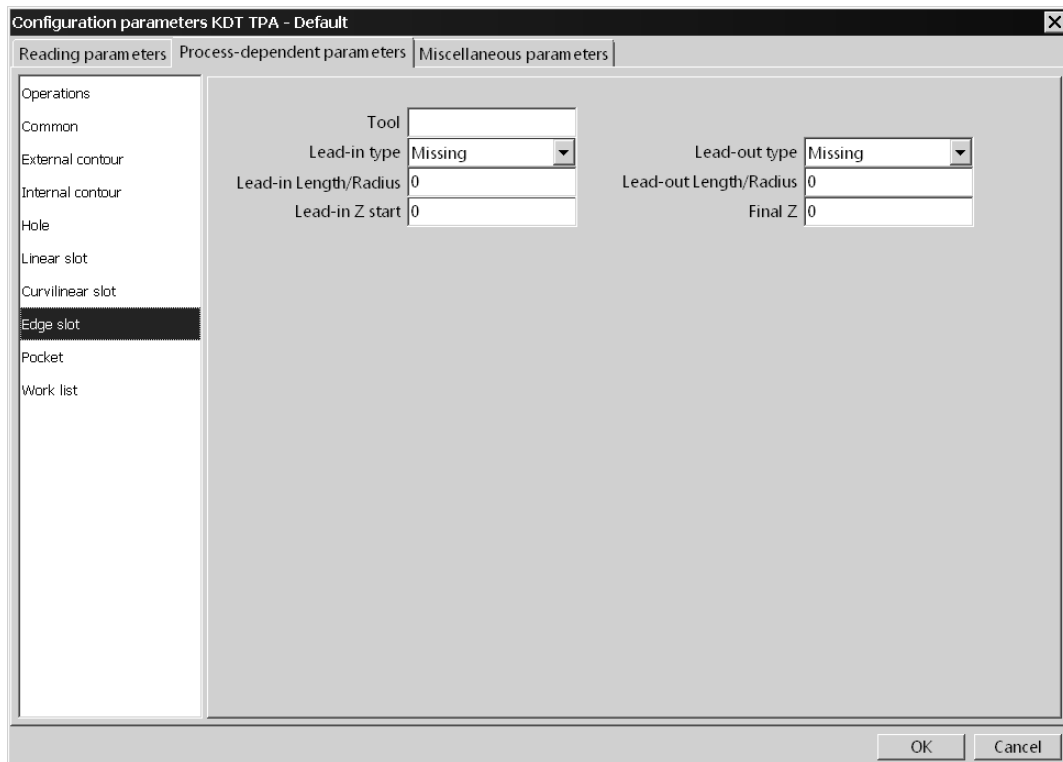


Fig. 10.467.

Section control elements allow to setup creation parameters of control programs for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.468) allow to setup pocket processing parameters.

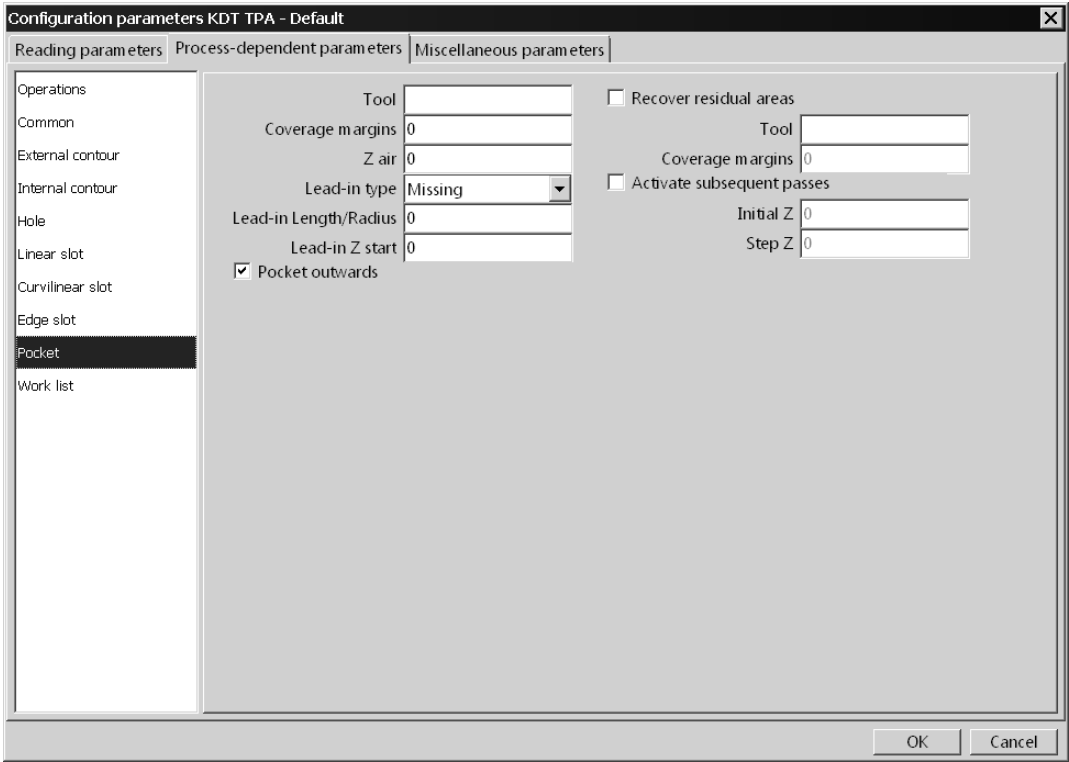


Fig. 10.468.

Section control elements allow to setup creation parameters of control programs for contour and pocket internal area processing: tool name, mill diameter, lead-in type, correction type, etc.

Work lists

Work list section control elements (fig. 10.469) allow to setup parameters of lists use.

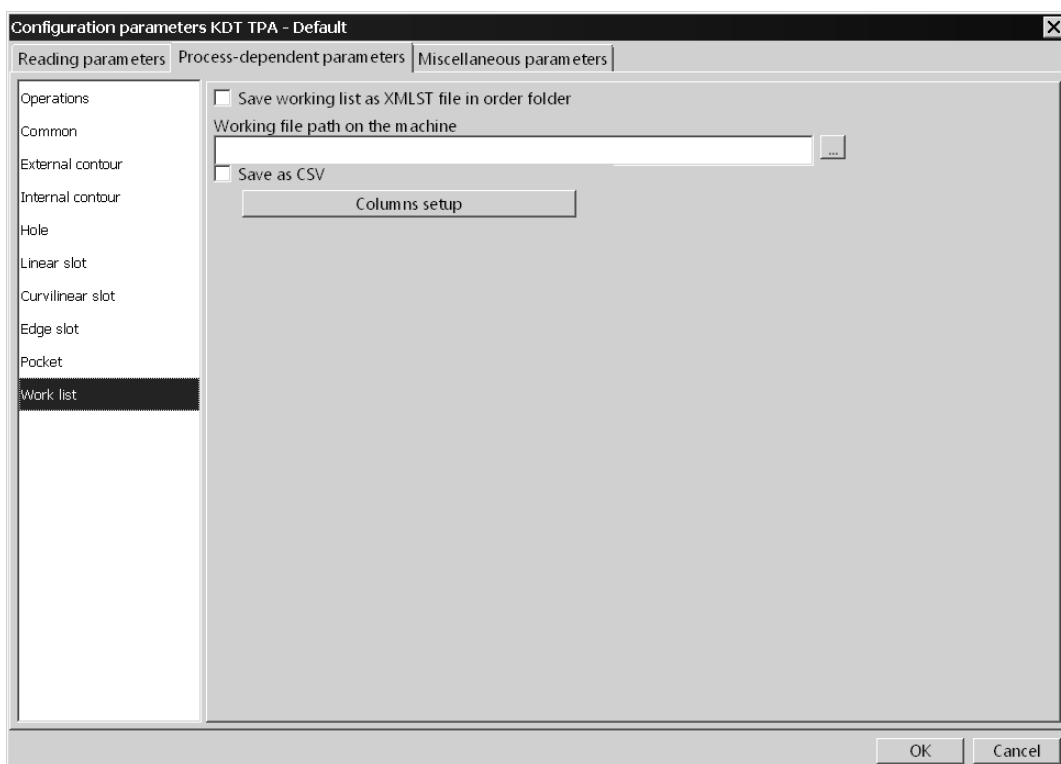


Fig. 10.469.



Save working list as XMLST file in order folder option allows to control the corresponding document creation. **Browse** button allows to set a path to working files. After pushing the button **Browse** dialogue will appear on the screen.

Setup completion

To complete control programs for KDT TPA setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.24.4. CNC Drill



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.470) allow to specify operations that can be used in control program.

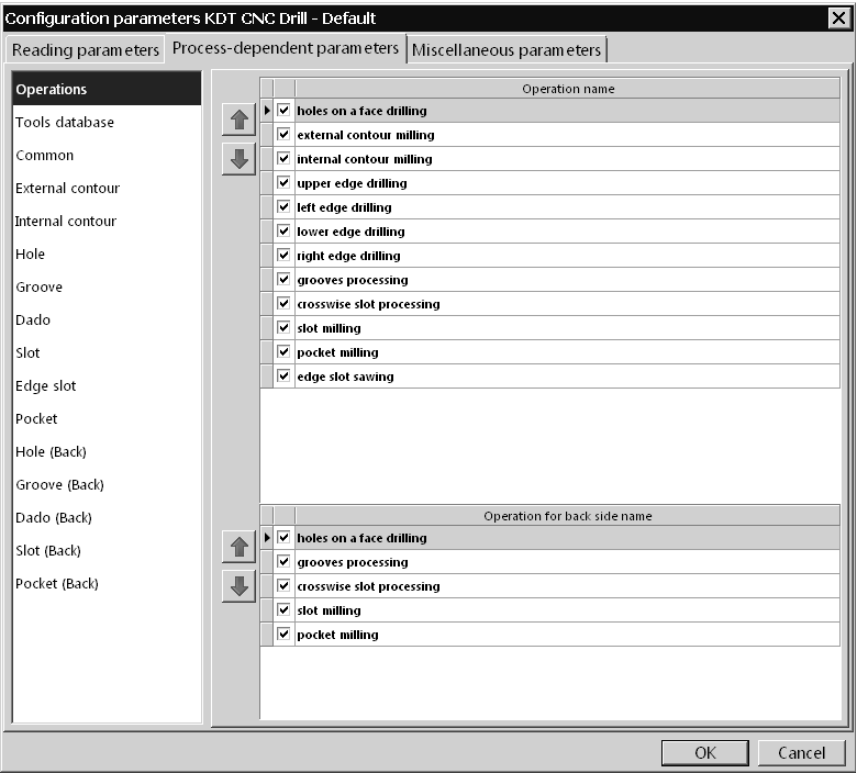


Fig. 10.470.

Usable tools

Tools database section control elements (fig. 10.471) allow to create a list of tools available for use in control programs.

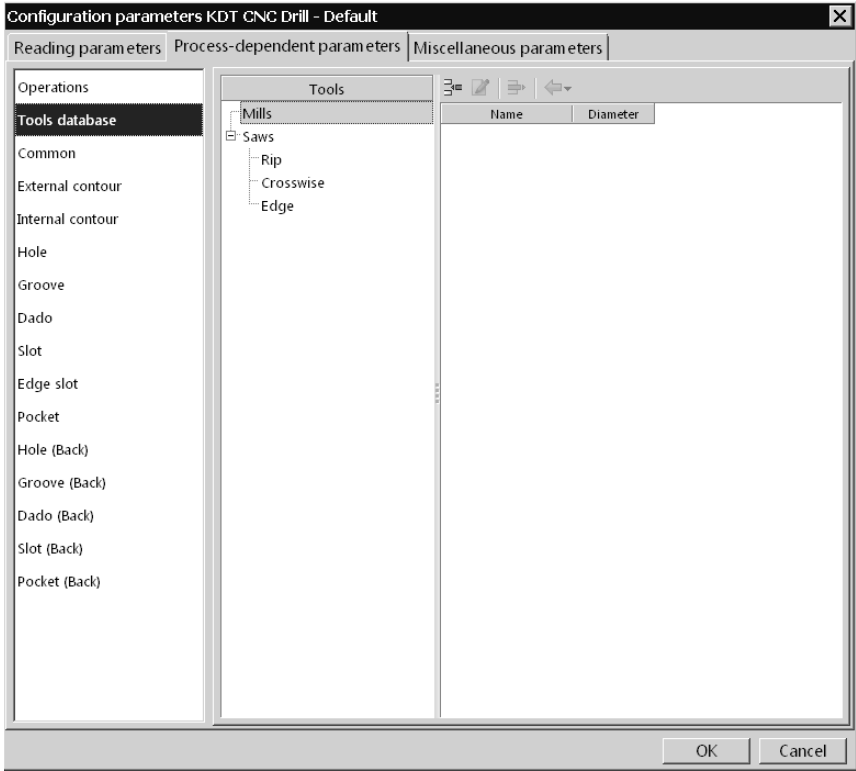


Fig. 10.471.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.472).

Name	
Diameter	0
Width	0

Fig. 10.472.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common section control elements (fig. 10.473) allow to setup common parameters of control programs.

Save amount of the uniform workable panels	No
Save edge banding data	No
Comment to Latin transliteration	No

Fig. 10.473.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.474).

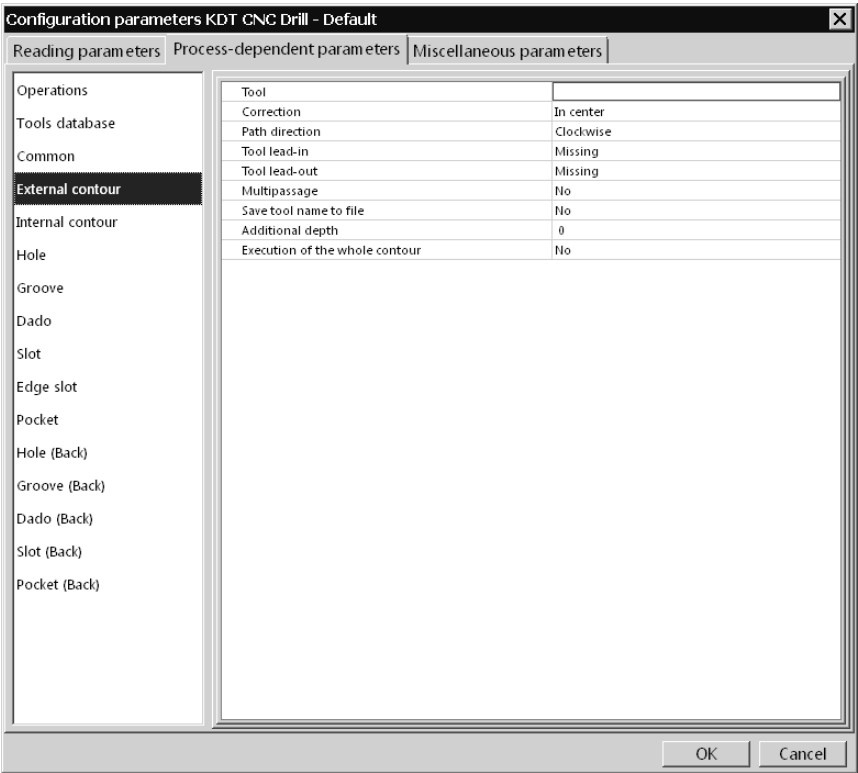


Fig. 10.474.

Control elements in sections allow to select the tool, correction types, lead-in, etc.

Hole parameters

Hole processing from face setup is performed in **Hole** section (fig. 10.475), holes from back side – in **Hole (Back)** section in the same way.

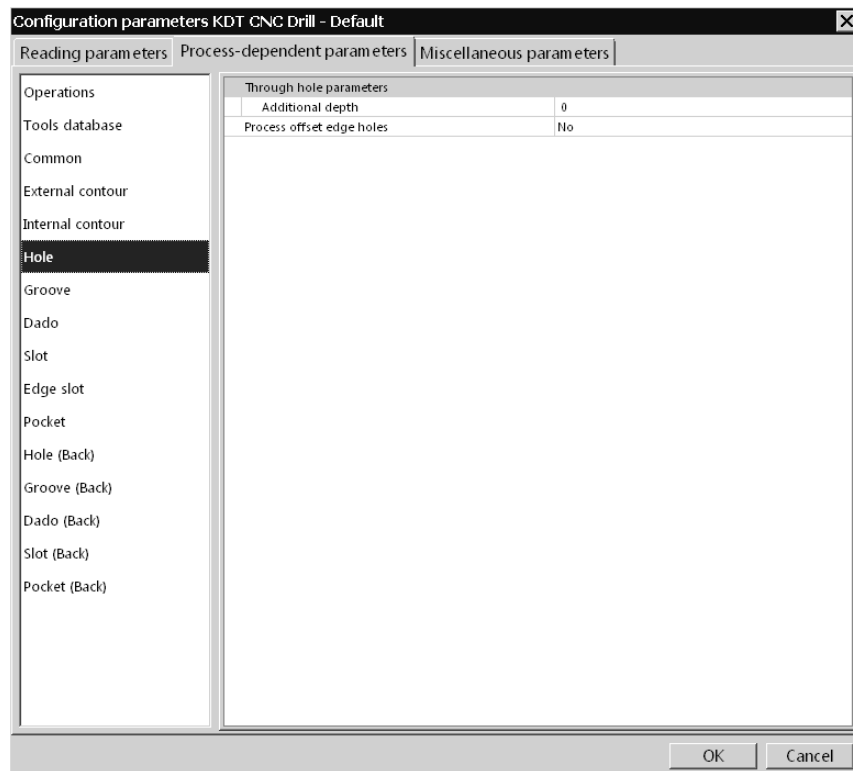


Fig. 10.475.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.476), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

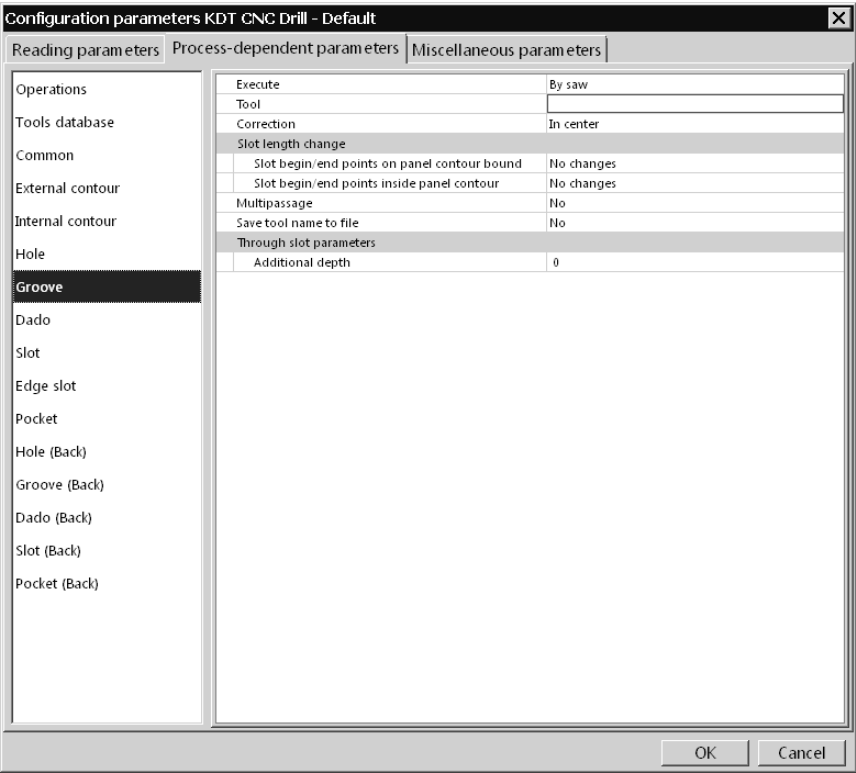


Fig. 10.476.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.477), from the back panel side – in **Slot (Back)** section.

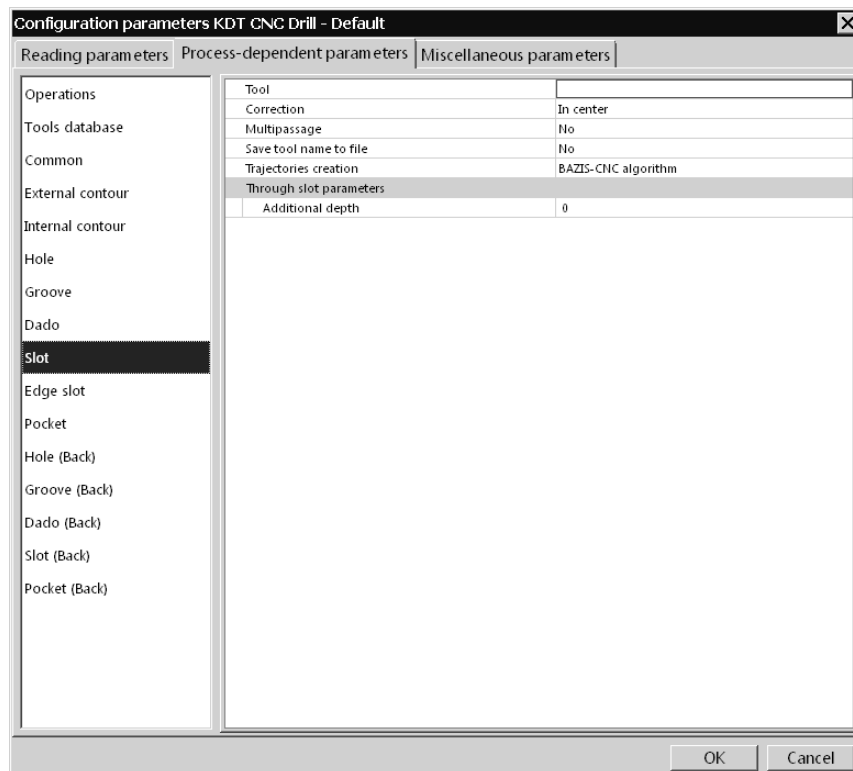


Fig. 10.477.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.478), on the back panel side – in **Pocket (Back)** section.

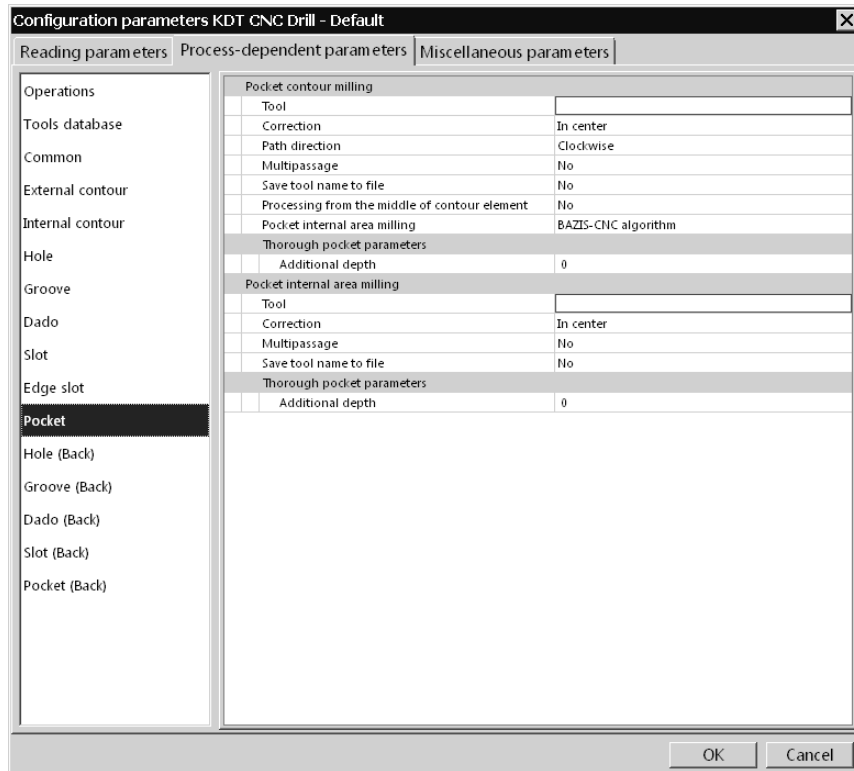


Fig. 10.478.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects from database. For pocket contour select milling execution direction, correction type and multipassage use. For internal pocket part – correction type and multipassage use.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.24.5.KDT Horizontal Drill



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.479) allow to specify operations that can be used in control program.

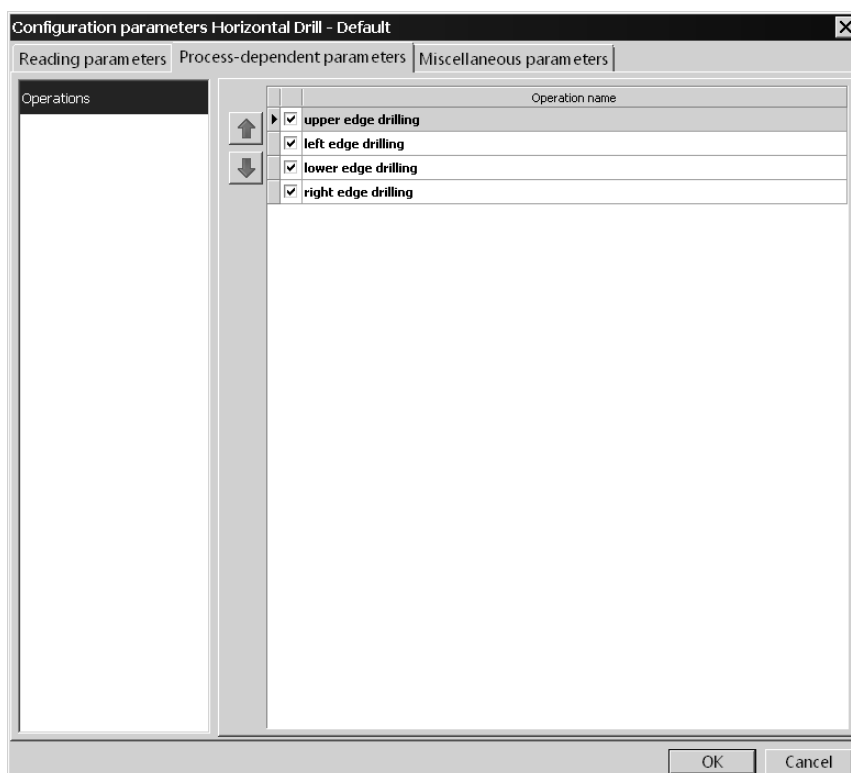


Fig. 10.479.

Setup completion

To complete control programs for KDT Horizontal Drill creation parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.25. KOCH

10.25.1.NCStudio



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.480) allow to set control programs creation common parameters.

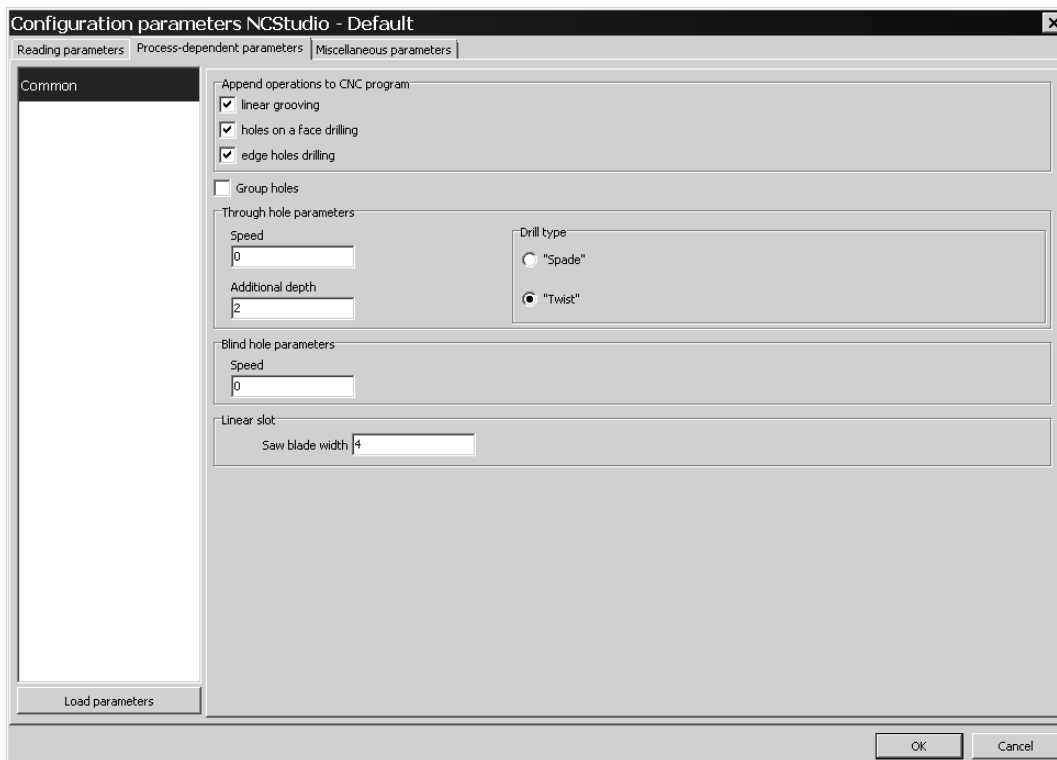


Fig. 10.480.

Append operations to CNC program group options allow to control operations appending to control program. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

It is necessary to set boring speed for blind and through holes and select borer type for through holes. **Saw blade width** field allows to set saw blade width at linear slots processing.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.26. Maggi

10.26.1. Maggi ICE 101



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.481) allow to set common parameters of control programs creation.

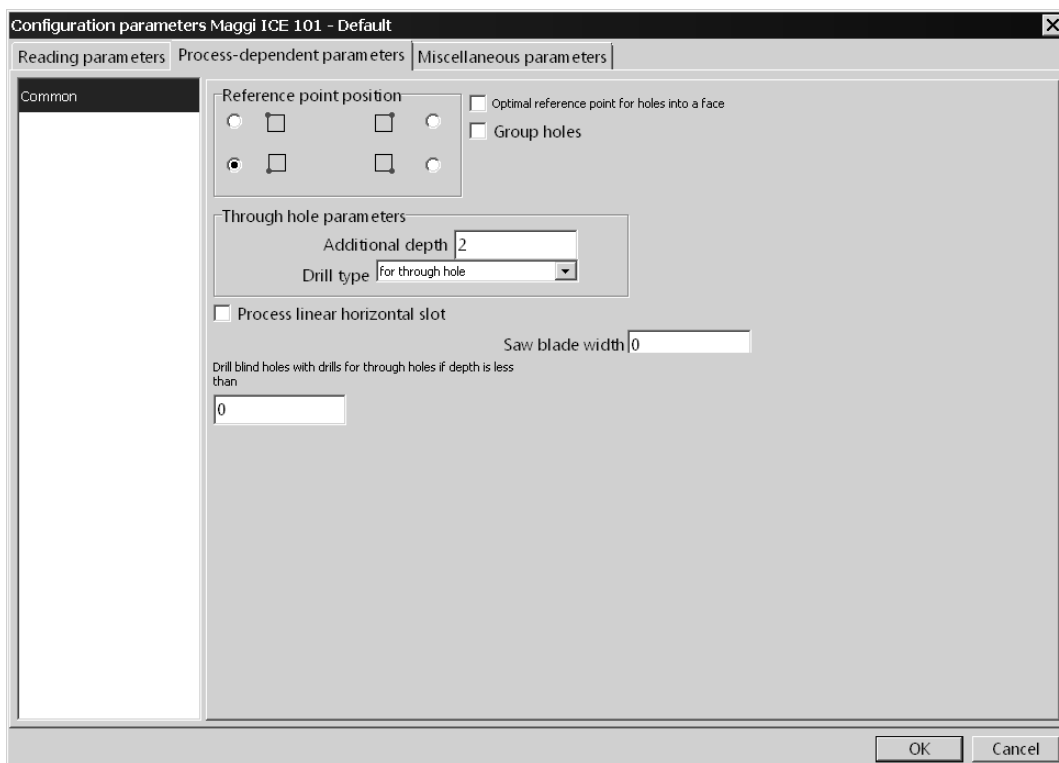


Fig. 10.481.

Reference point position group variants allow to select panel angle as origin of coordinates. If **Optimal reference point for holes into a face** option is enabled, referencing will be set for each hole individually. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional through hole depth** field allows to take drill center point into account. **Process linear horizontal slot** option allows to control slot processing parameters to the program appending.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.26.2.Pegasus



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.482) allow to specify operations that can be used in control program.

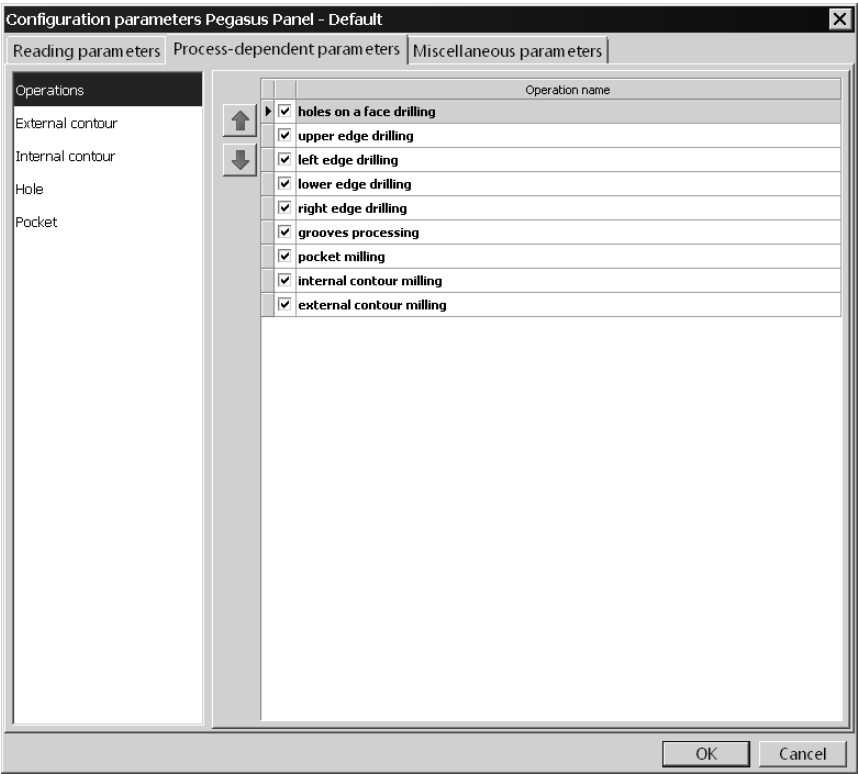


Fig. 10.482.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.483).

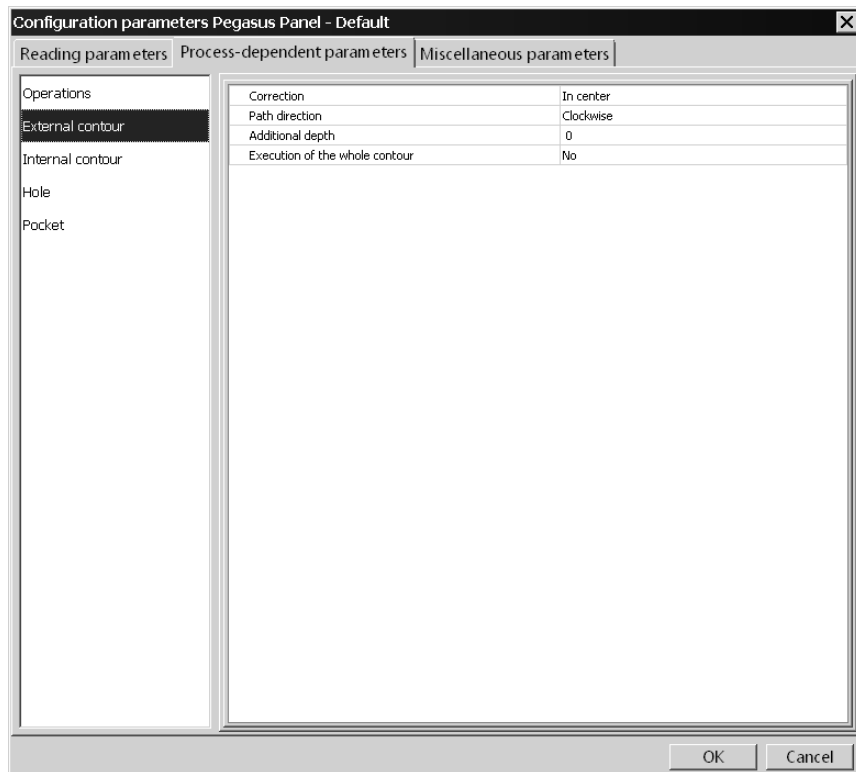


Fig. 10.483.

Section control elements allow to choose a tool, lead-in type, correction type, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.484).

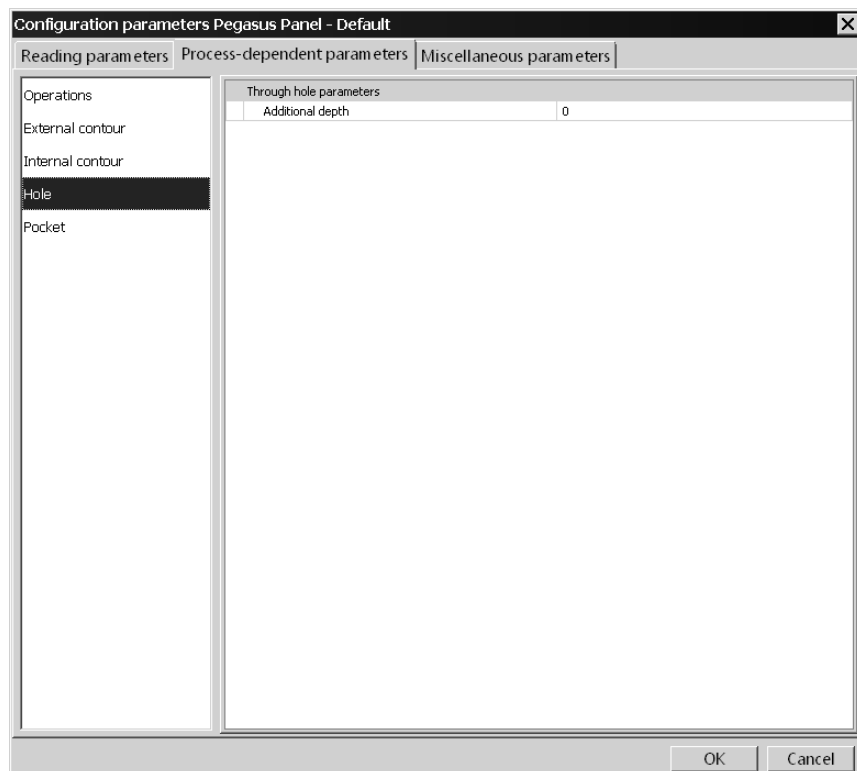


Fig. 10.484.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Pocket parameters

Pocket section control elements (fig. 10.485) allow to set pocket processing parameters.

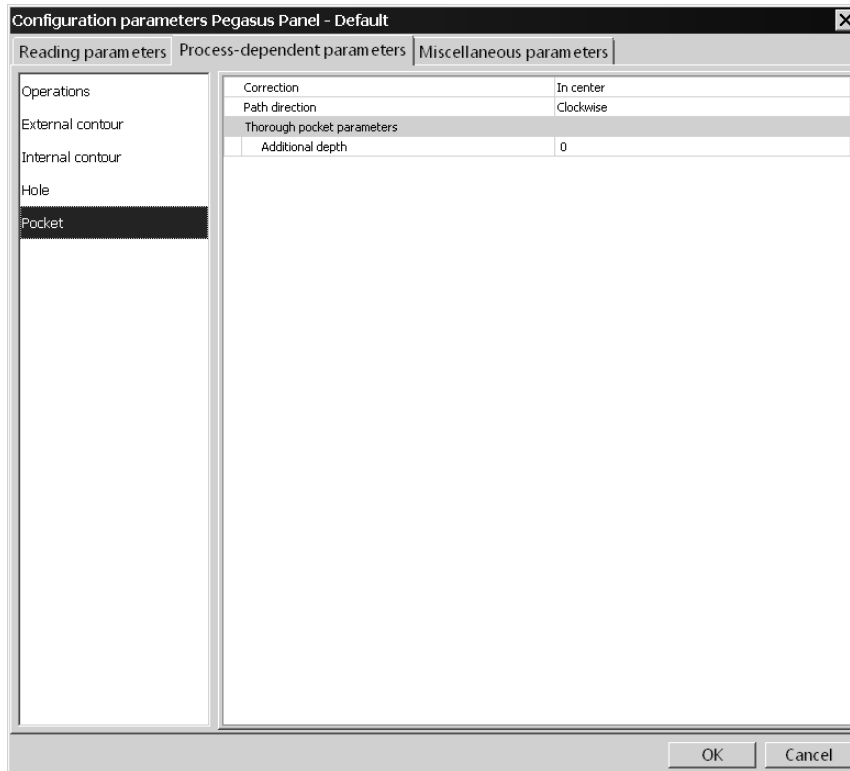


Fig. 10.485.

For pocket contour select milling execution direction, correction type and multipass use.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.27. Masterwood

10.27.1.MW208



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.486) allow to set screw stop parameters.

Configuration parameters MW208 - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Common

Hole parameters

Tool table

Screw stop parameters:

Fence #1 mm

Fence #2 mm

Fence #3 mm

Fence #4 mm

Enter values using ; (semicolon) as separator.

OK Cancel

Fig. 10.486.

Hole parameters

Hole parameters section control elements (fig. 10.487) allow to set parameters of holes drilling.

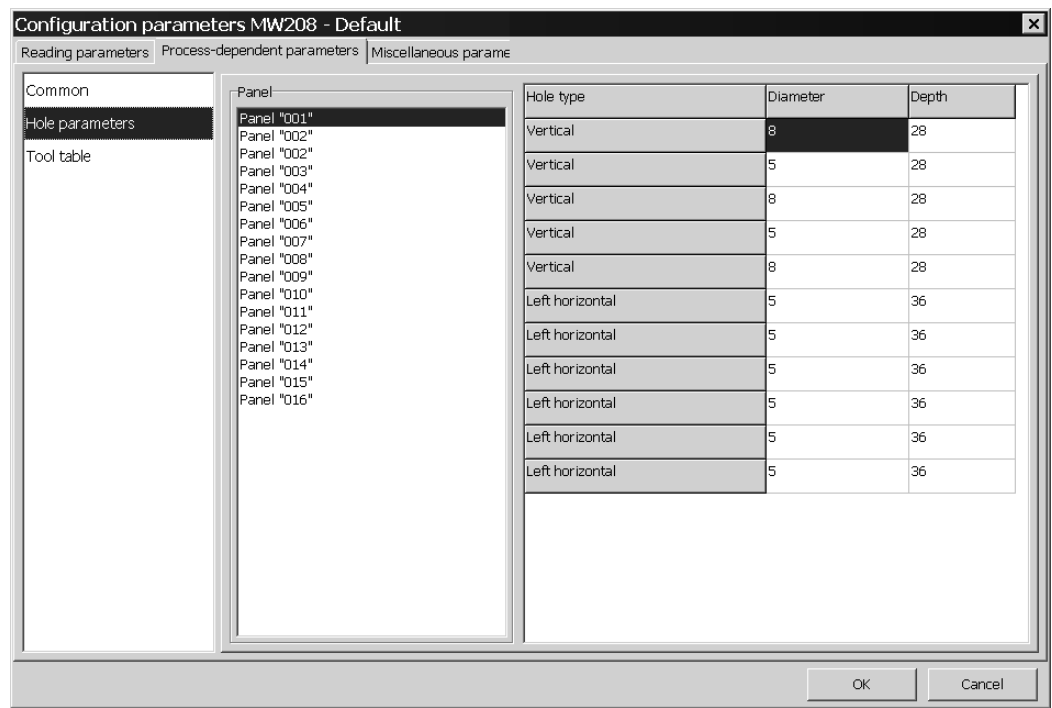


Fig. 10.487.

Panel list contains model parts names. **Holes** list contains hole parameters of the part from a model. Parameters can be changed.

Tool table

Tool table section control elements (fig. 10.488) allow to execute operations with tool sets.

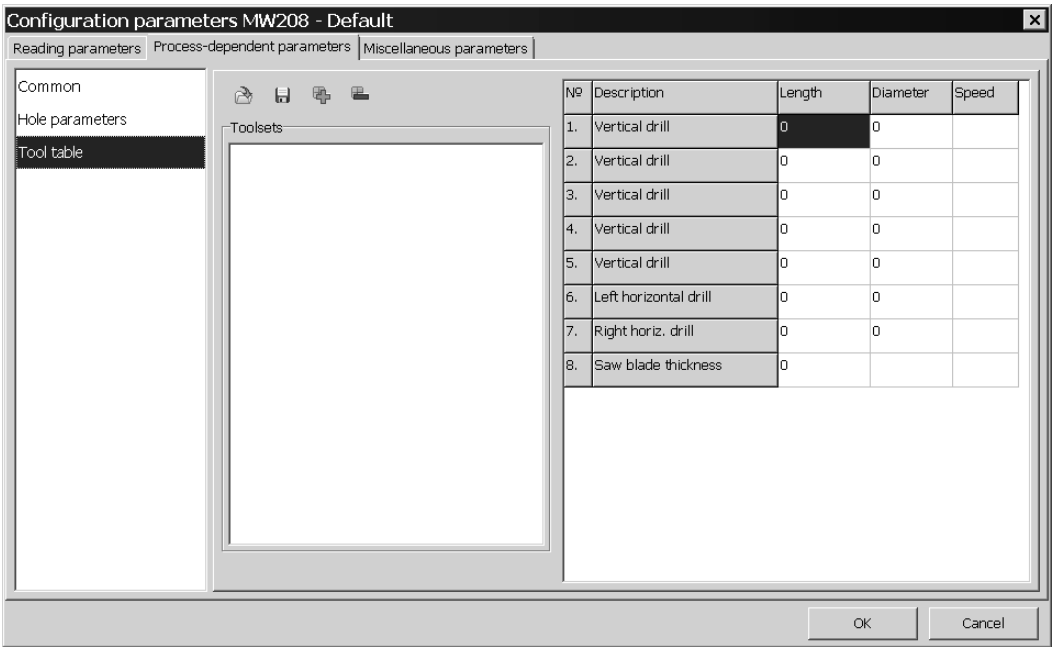


Fig. 10.488.



Add toolset button allows to create a new set. After pushing the button, a set name will appear in the list. The sets are sequentially numbered. Drill parameters can be assigned to each set.



Delete toolset button allows to delete a set selected in the list. After pushing the button, a warning will appear on the screen (fig. 10.489).

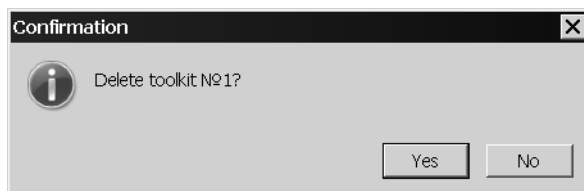


Fig. 10.489.



Message buttons allow to confirm deletion or reject it. **Save toolset** button allows to save selected set. After pushing the button, a standard Windows dialogue will appear on the screen.



Load toolset button allows to download a toolset from a file. After pushing the button, a standard Windows dialogue will appear on the screen. In the dialogue select the required file. A warning will appear on the screen (fig. 10.490).

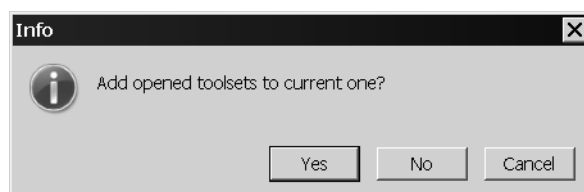


Fig. 10.490.

Message buttons allow to confirm the adding or reject it.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.27.2. MW310 (*.dxf) and MW310 (*.tlf)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

MW310 control programs are saved in a file of *dxf* or *tlf* extension. Control programs parameters are set for both formats in the same way.

Common parameters

Common section control elements (fig. 10.491) allow to set common parameters of control programs creation.

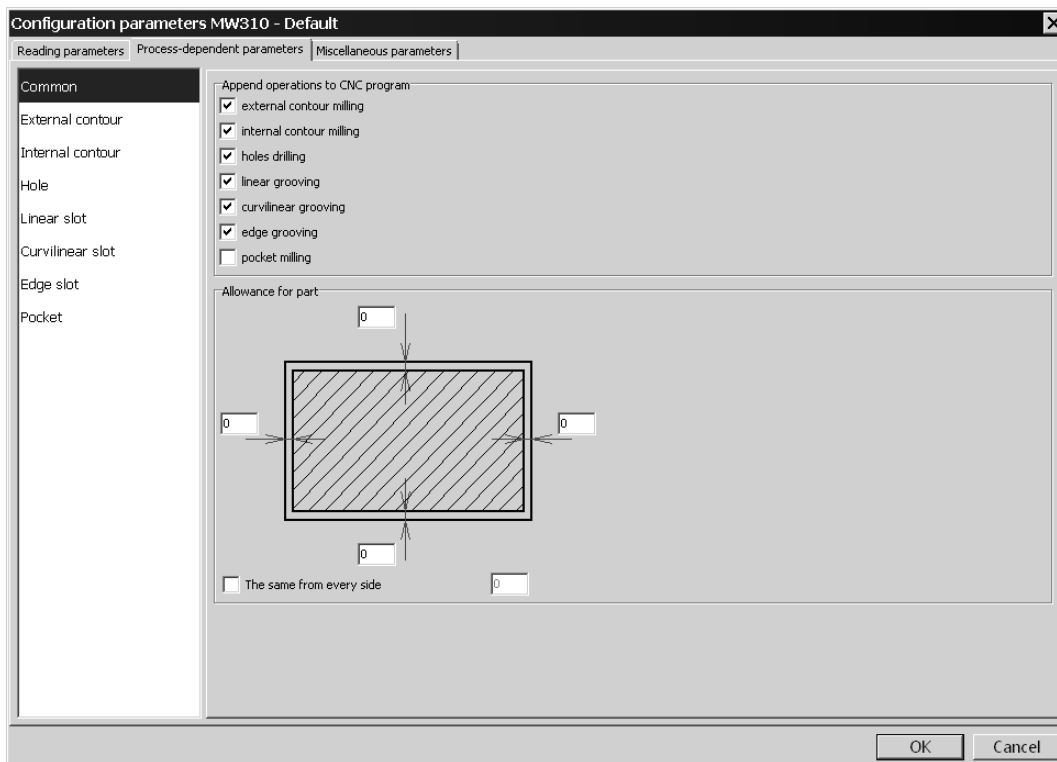


Fig. 10.491.

Append operations to CNC program group options allow to control the inclusion in control program of operations corresponding to the options names.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** and **Internal contour** sections (fig. 10.492). Tabs control elements content is the same and described for outer contour.

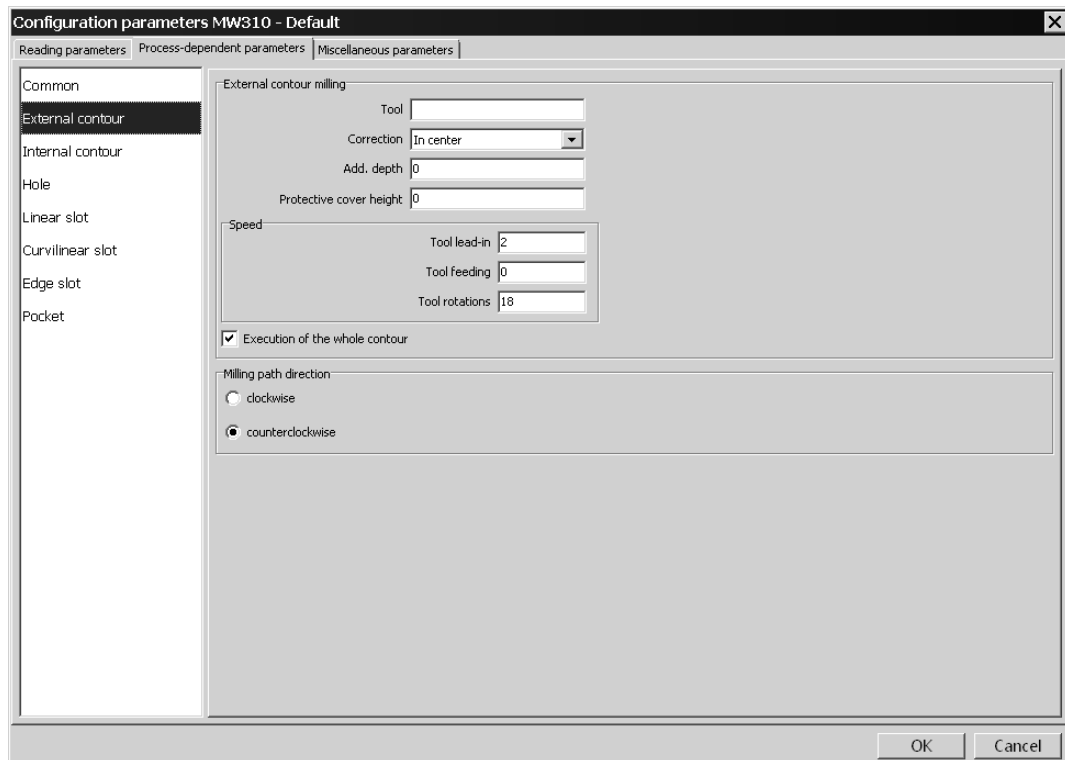


Fig. 10.492.

Sections control elements allow to select a tool, correction type, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.493).

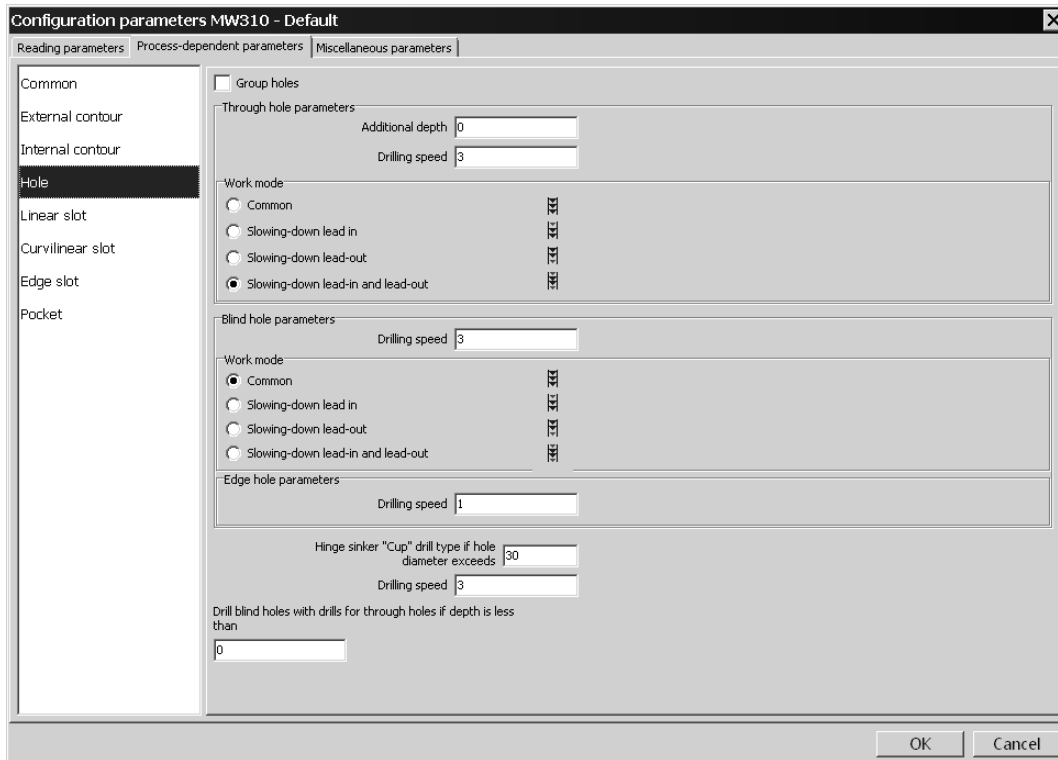


Fig. 10.493.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option.

Through holes

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. It is necessary to set boring speed for through holes in corresponding field and select working mode.

Blind holes

It is necessary to set boring speed for blind holes in corresponding field and select working mode.

Edge holes

It is necessary to set boring speed for edge holes in corresponding field.

Cup borer

Hinge sinker "Cup" drill type if hole diameter exceeds field allows to control cup borer use. It is necessary to set boring speed for this borer in corresponding field.

Linear slot

Linear slots parameters are set in **Linear slot** section (fig. 10.494).

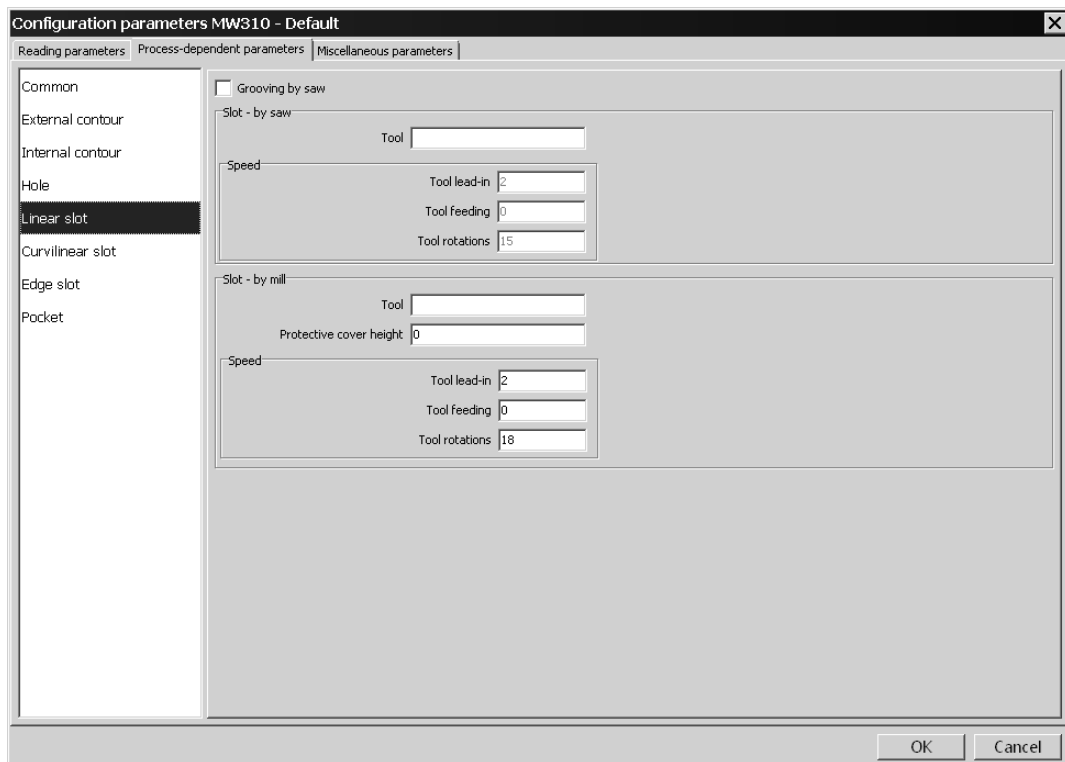


Fig. 10.494.

By saw option allows to select a tool for slot processing. **Slot - by saw** and **Slot - by mill** groups fields allow to input tool designations for slots processing, operations speed and other parameters.

Curvilinear slot

Curvilinear slots parameters are set in **Curvilinear slot** section (fig. 10.495).

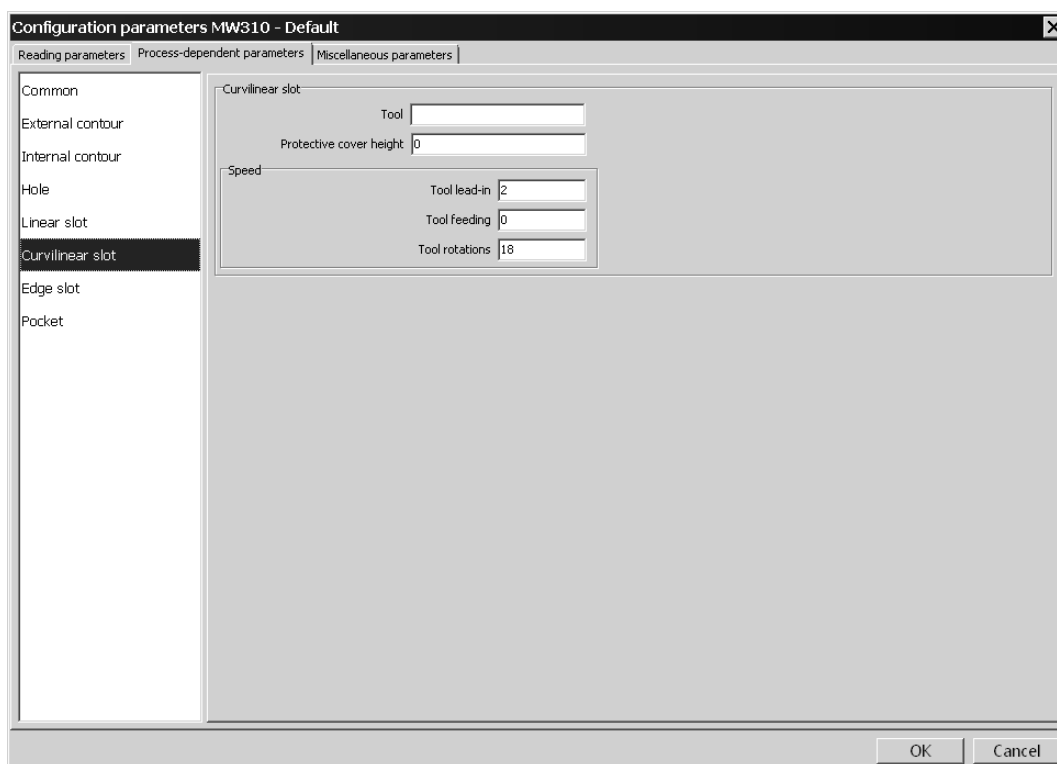


Fig. 10.495.

Curvilinear slot group fields allow to input tool designation for slots processing, operations speed and other parameters.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.496).

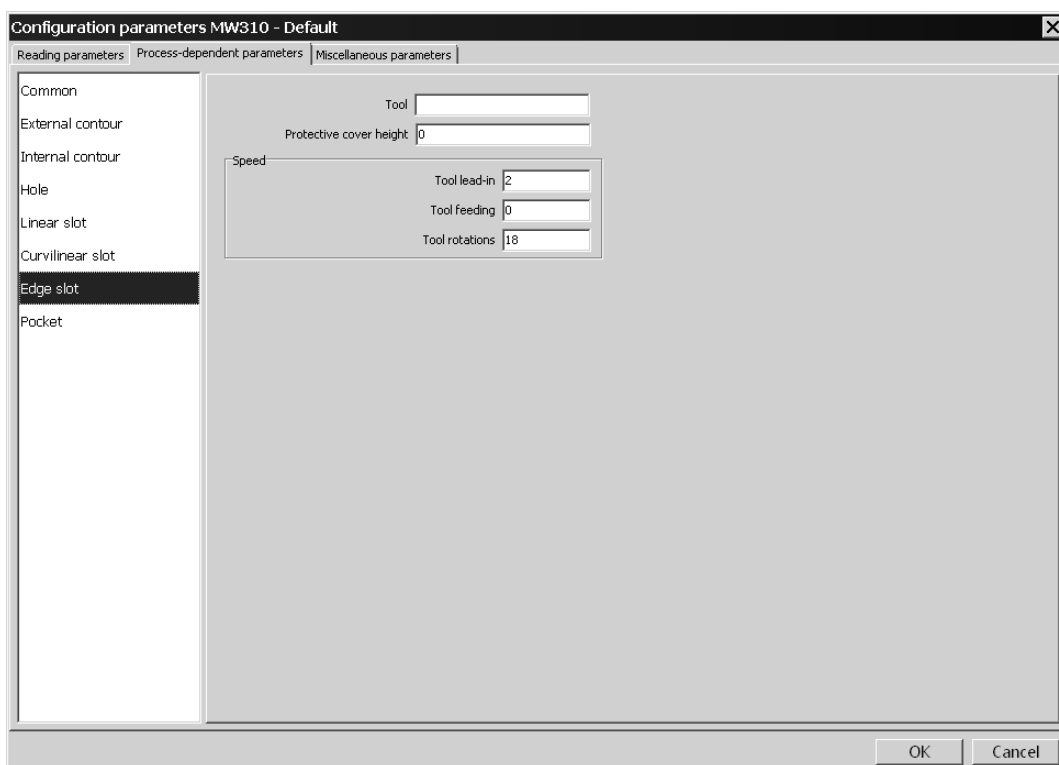


Fig. 10.496.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.497) allow to set pocket processing parameters.

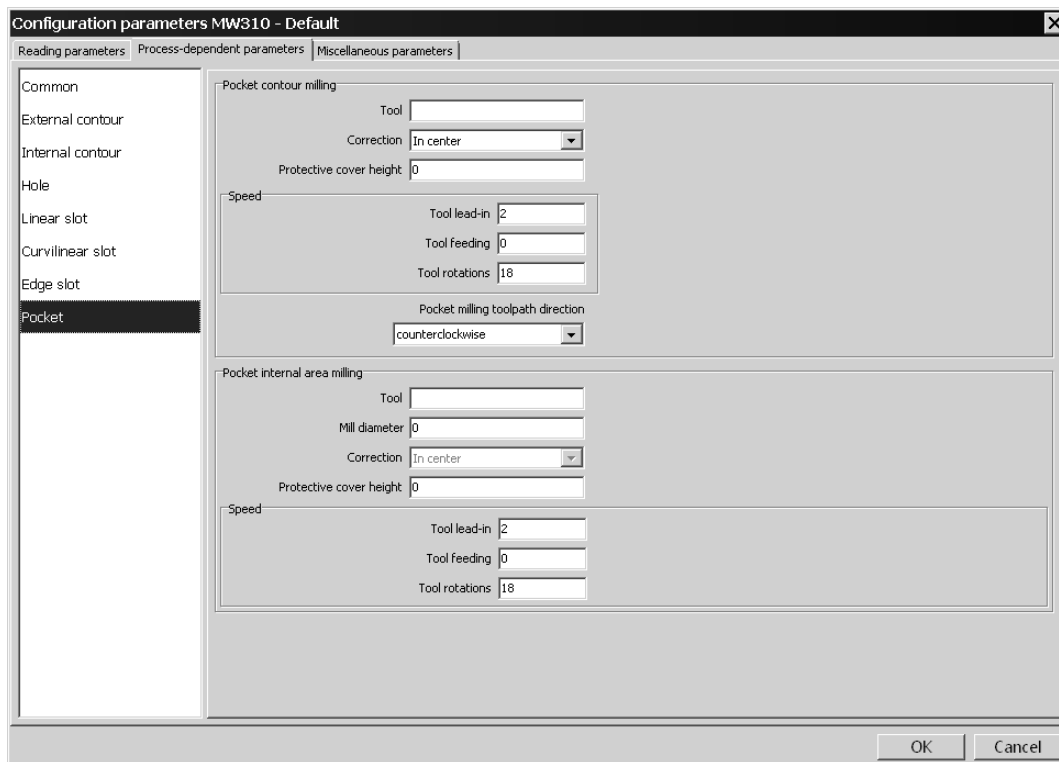


Fig. 10.497.

Section control elements allow to set control programs creation parameters for notch contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.28. Multicam

10.28.1. MultiCam (*.nc)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.498) allow to set common parameters of control programs creation.

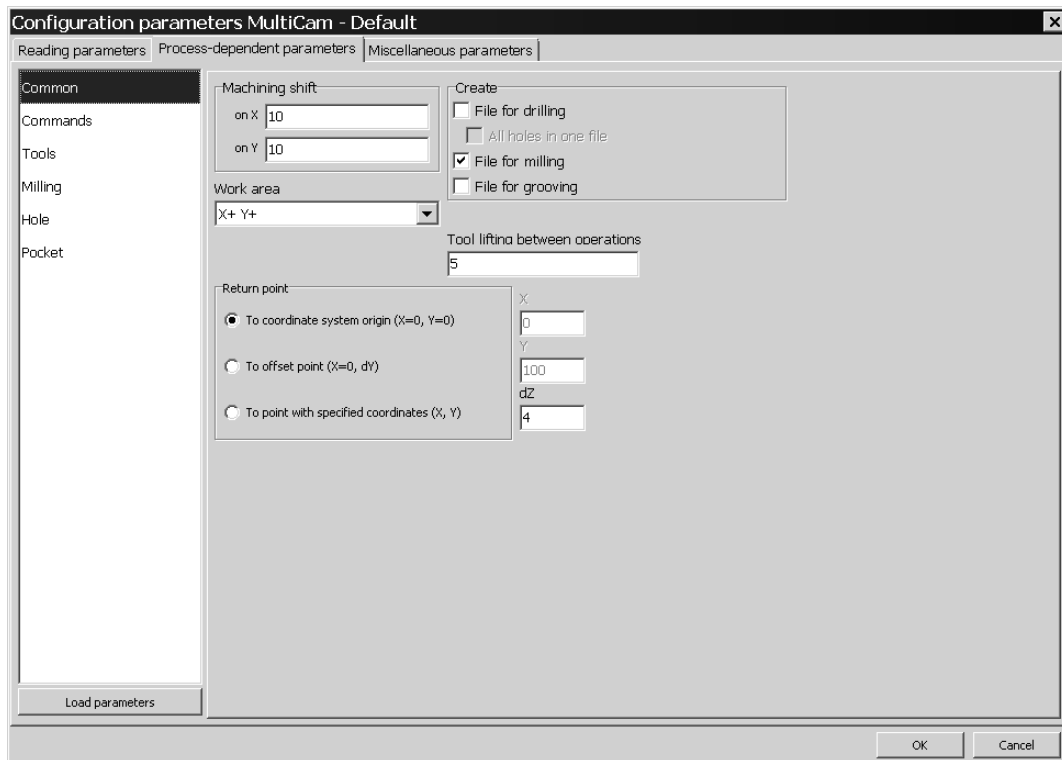


Fig. 10.498.

On X and **on Y** fields of **Machining shift** group allow to set work area initial point moving. **Create** group options allows to control programs creation for separate operations. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If the option is disabled, return point coordinates are set by **dY** and **dZ** fields values. **Tool lifting between operations** field allows to set tool height. **Work area** combo box variants allow to set work area coordinate axes direction.

Usable commands

Commands section control elements (fig. 10.499) allow to control separate commands to control program inclusion.

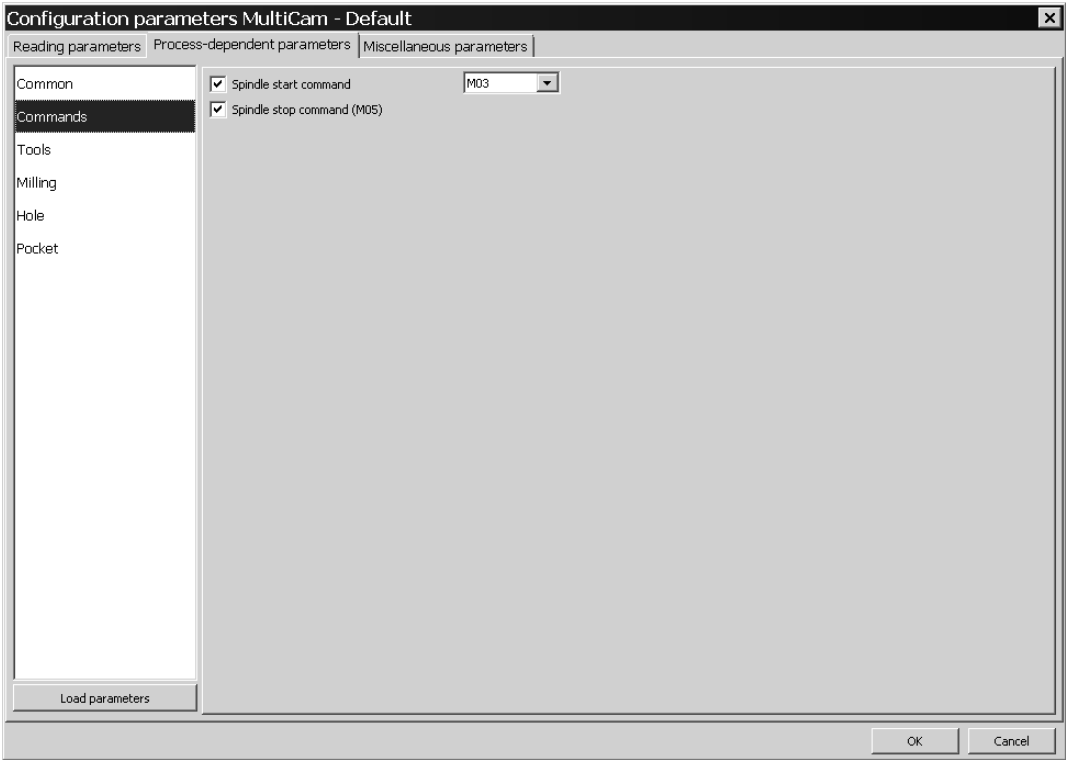


Fig. 10.499.

Spindle start command can be selected from variants in a combo box.

Usable tools

Tools section control elements (fig. 10.500) allow to create a list of tools available for use in control programs.

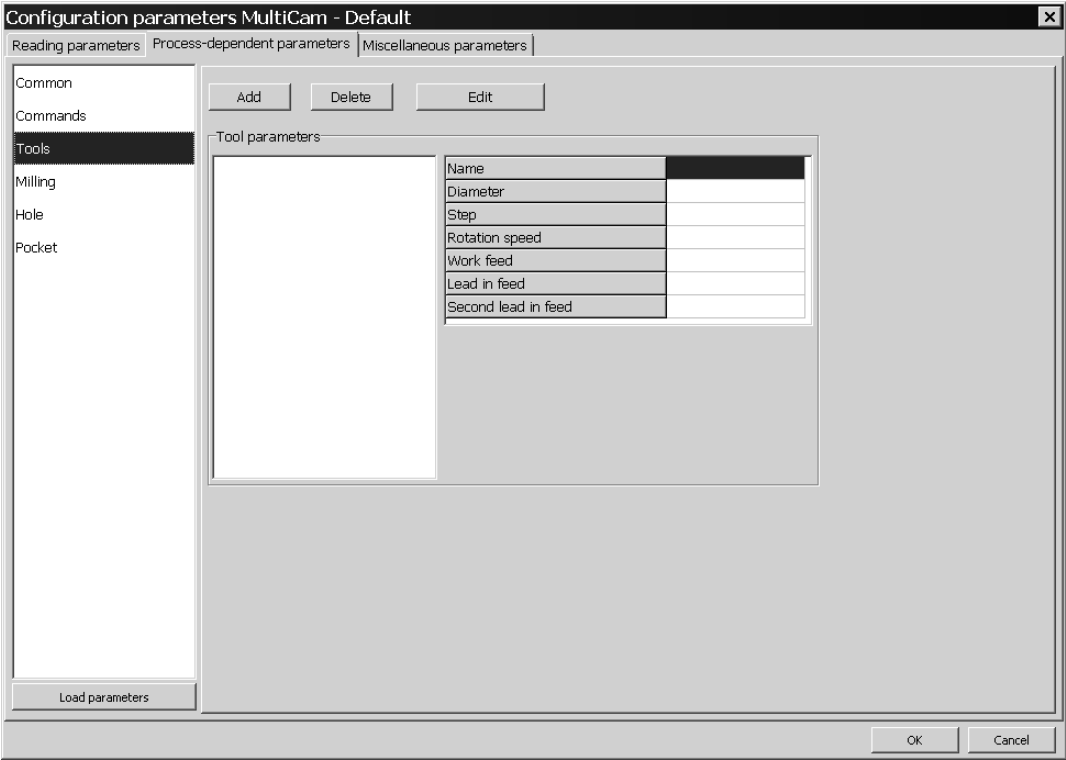
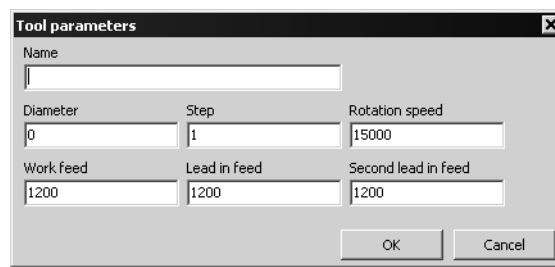


Fig. 10.500.

To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.501).



The 'Tool parameters' dialog box contains the following fields and buttons:

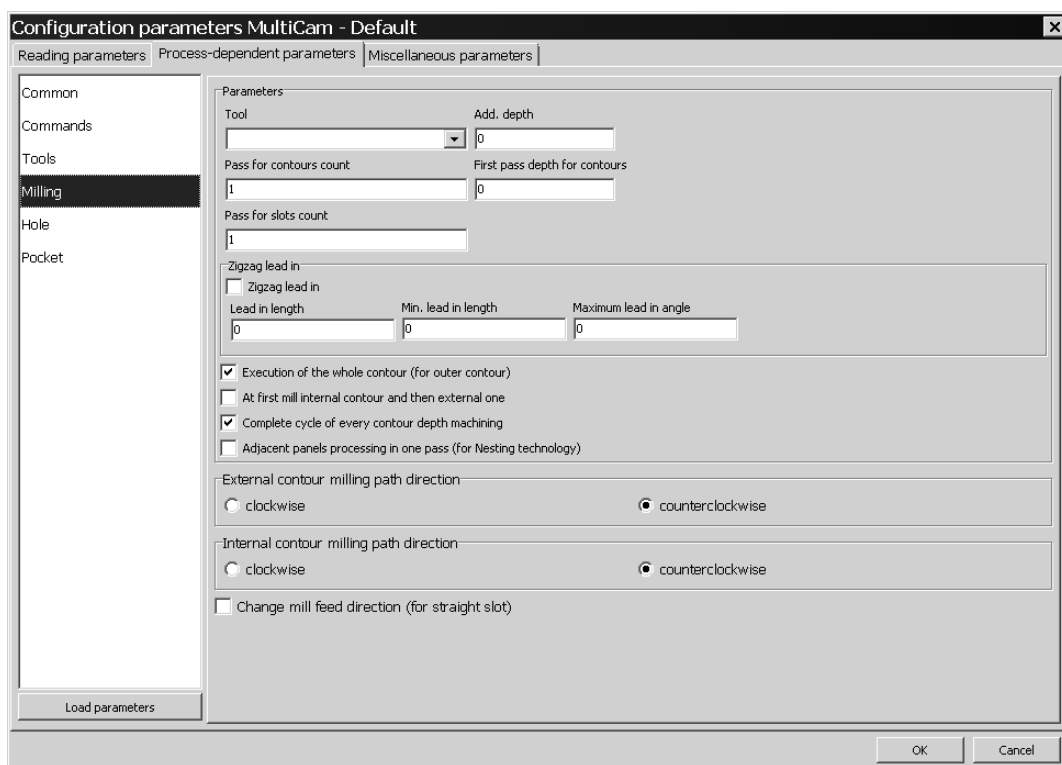
- Name:** A text input field.
- Diameter:** A text input field with the value '0'.
- Step:** A text input field with the value '1'.
- Rotation speed:** A text input field with the value '15000'.
- Work feed:** A text input field with the value '1200'.
- Lead in feed:** A text input field with the value '1200'.
- Second lead in feed:** A text input field with the value '1200'.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Fig. 10.501.

In this dialogue input a tool name and set its parameters. To complete tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. **Delete** button allows to delete selected tool from a list.

Milling parameters

Milling section control elements (fig. 10.502) allow to set milling parameters.



The 'Configuration parameters MultiCam - Default' dialog box has three tabs: 'Reading parameters', 'Process-dependent parameters', and 'Miscellaneous parameters'. The 'Process-dependent parameters' tab is active. On the left is a tree view with categories: Common, Commands, Tools, **Milling** (selected), Hole, and Pocket. The main area contains the following settings:

- Parameters:**
 - Tool:** A dropdown menu.
 - Add. depth:** A text input field with the value '0'.
 - Pass for contours count:** A text input field with the value '1'.
 - First pass depth for contours:** A text input field with the value '0'.
 - Pass for slots count:** A text input field with the value '1'.
- Zigzag lead in:**
 - ☐ Zigzag lead in
 - Lead in length:** A text input field with the value '0'.
 - Min. lead in length:** A text input field with the value '0'.
 - Maximum lead in angle:** A text input field with the value '0'.
- Execution options:**
 - ☒ Execution of the whole contour (for outer contour)
 - ☐ At first mill internal contour and then external one
 - ☒ Complete cycle of every contour depth machining
 - ☐ Adjacent panels processing in one pass (for Nesting technology)
- External contour milling path direction:**
 - ☐ clockwise
 - ☒ counterclockwise
- Internal contour milling path direction:**
 - ☐ clockwise
 - ☒ counterclockwise
- ☐ Change mill feed direction (for straight slot)

At the bottom left is a 'Load parameters' button, and at the bottom right are 'OK' and 'Cancel' buttons.

Fig. 10.502.

Tool combo box allows to select a milling tool name. Tools should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc.

Execution of the whole contour option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular,

and the option is disabled, contour will not be milled. **Zigzag lead in** option allows to activate this operation. Group fields allow to set option parameters. **Milling path direction** group variants allow to select tool moving direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.503).

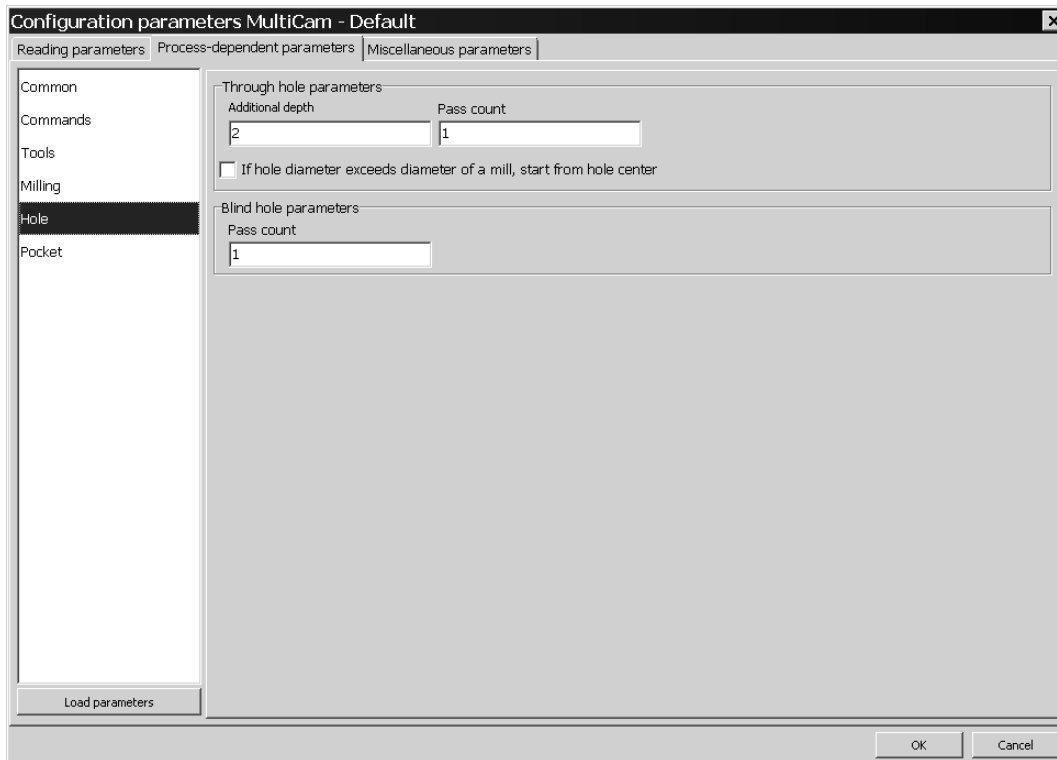


Fig. 10.503.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **If hole diameter exceeds diameter of a mill, start from hole center** option allows to control holes with a mill processing order, from center or from perimeter.

Pocket parameters

Pocket section control elements (fig. 10.504) allow to set pocket processing parameters.

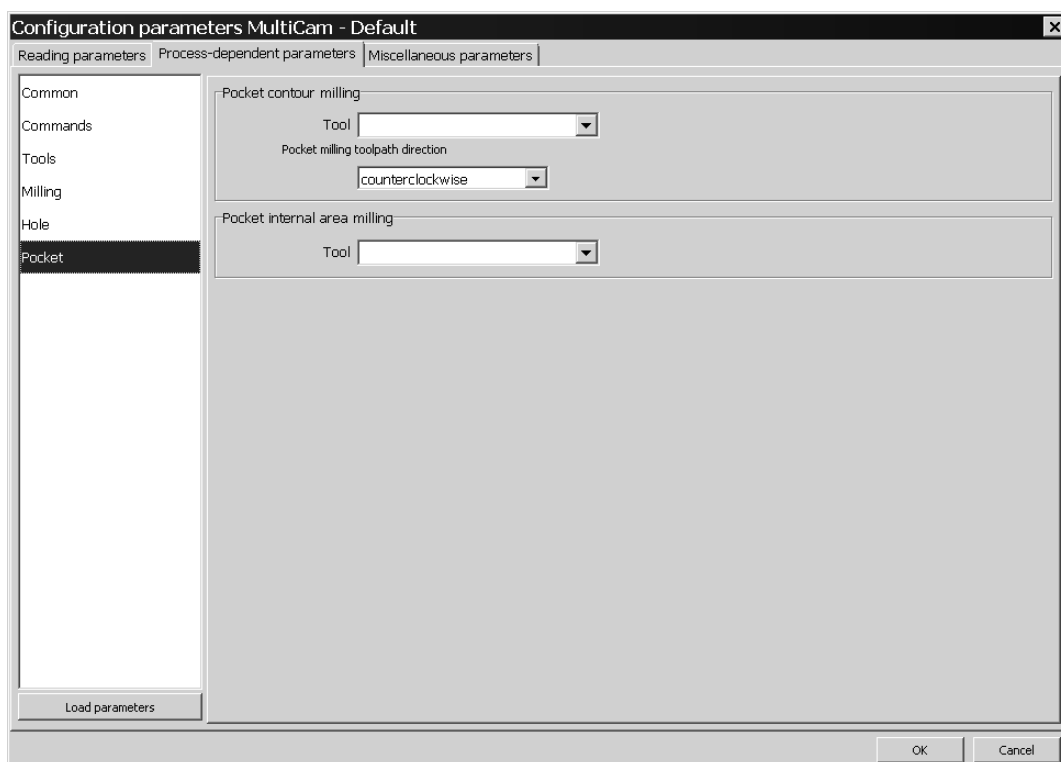


Fig. 10.504.

Tool combo boxes in **Pocket contour milling** and **Pocket internal area milling** groups allow to select tool name for corresponding objects milling. Tools should be created beforehand in **Tools** section. It is necessary to select milling execution for notch contour in corresponding combo box.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes. **Create** group options allow to control programs files creation, that correspond to option names.

10.29. Nanxing

10.29.1.NCB



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.505) allow to specify operations that can be used in control program.

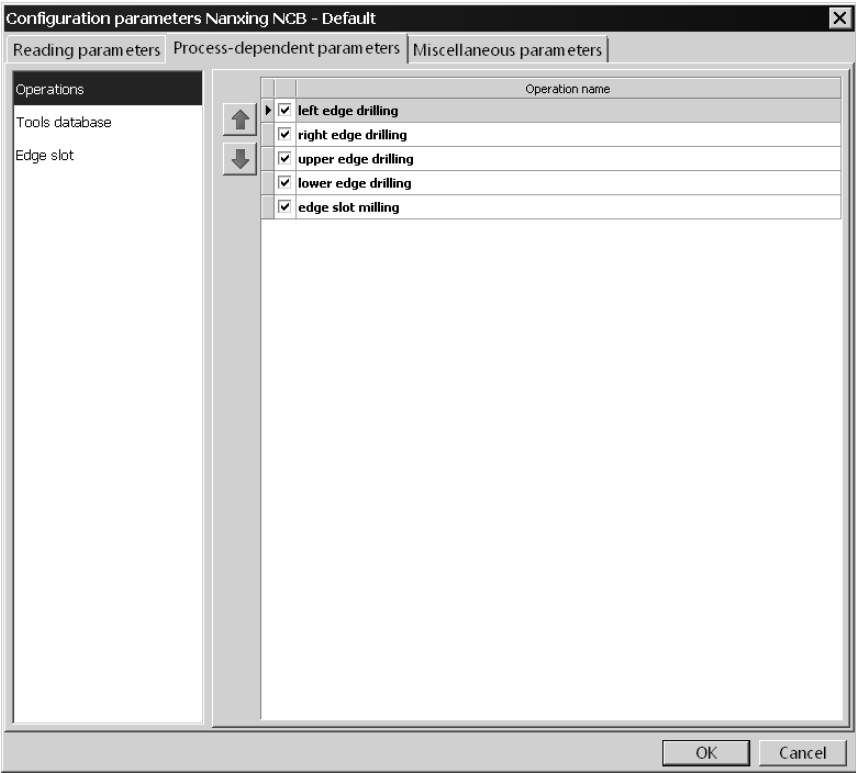


Fig. 10.505.

Usable tools

Tools database section control elements (fig. 10.506) allow to create a list of tools available for use in control programs.

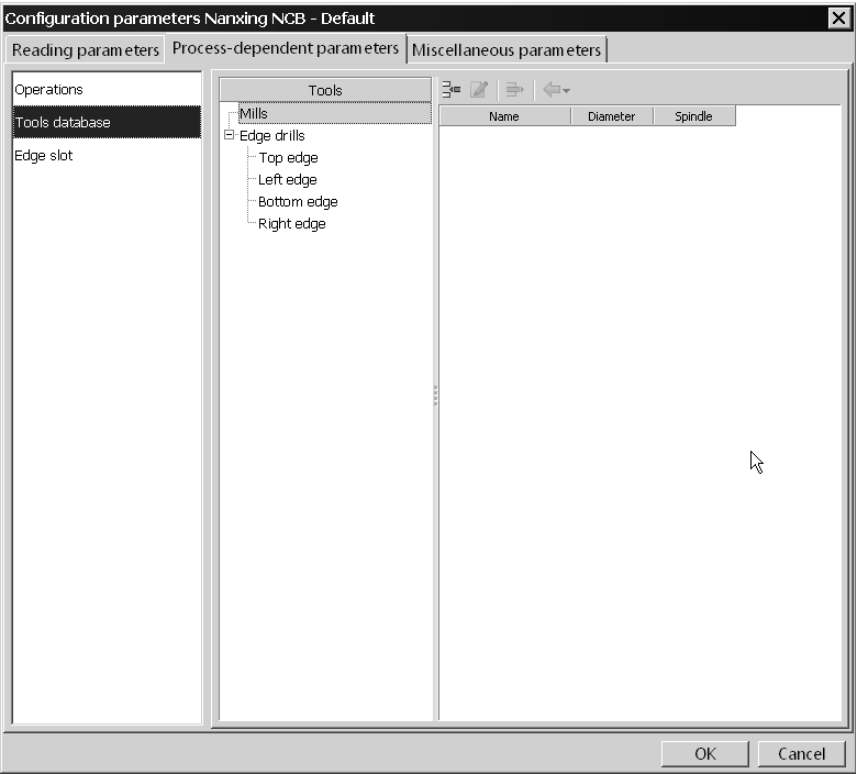


Fig. 10.506.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.507).

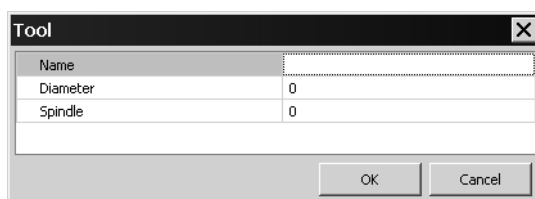


Fig. 10.507.



In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Edge slot parameters

Edge slots processing parameters are set in **Edge slot** section (fig. 10.508).

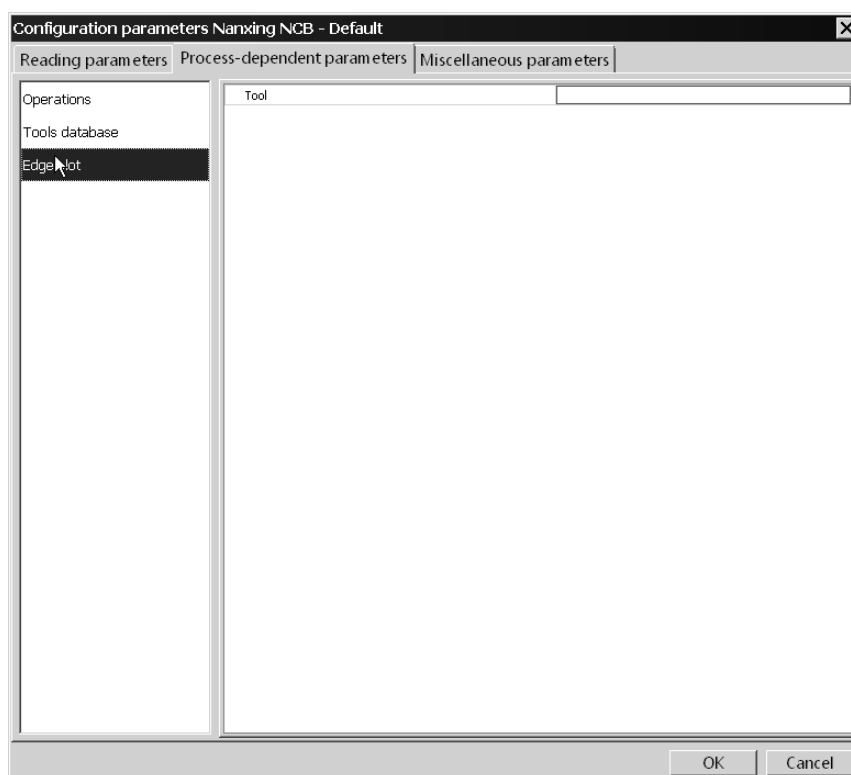


Fig. 10.508.

Section control elements allow to select a tool type for slot creation.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Create** group options allow to control the creation of control programs files that correspond to options names.

10.29.2.NDC



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.509) allow to specify operations that can be used in control program.

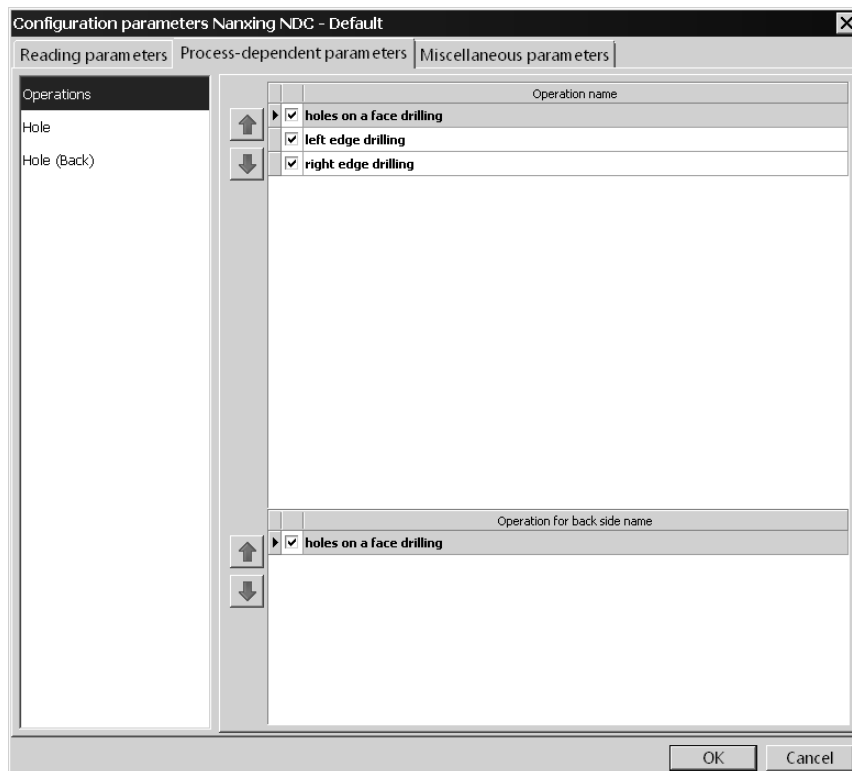


Fig. 10.509.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.510), holes from back side – in **Hole (Back)** section in the same way.

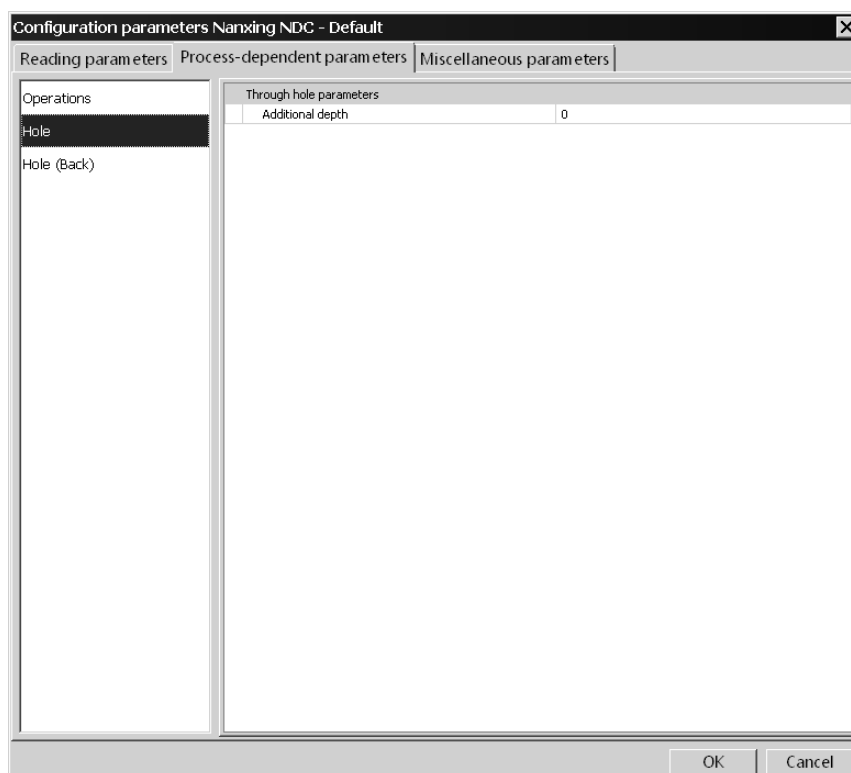


Fig. 10.510.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Create** group options allow to control the creation of control programs files that correspond to options names.

10.29.3.Driller Master



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.511) allow to specify the operations that can be used in control program.

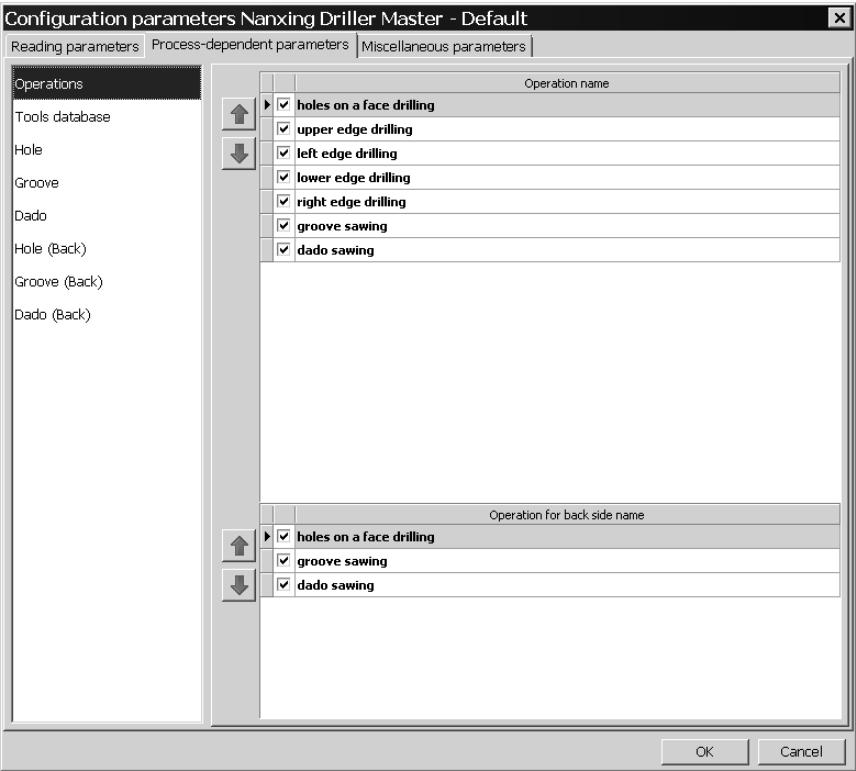


Fig. 10.511.

Usable tools

Tools database section control elements (fig. 10.512) allow to create a list of tools available for use in control programs.

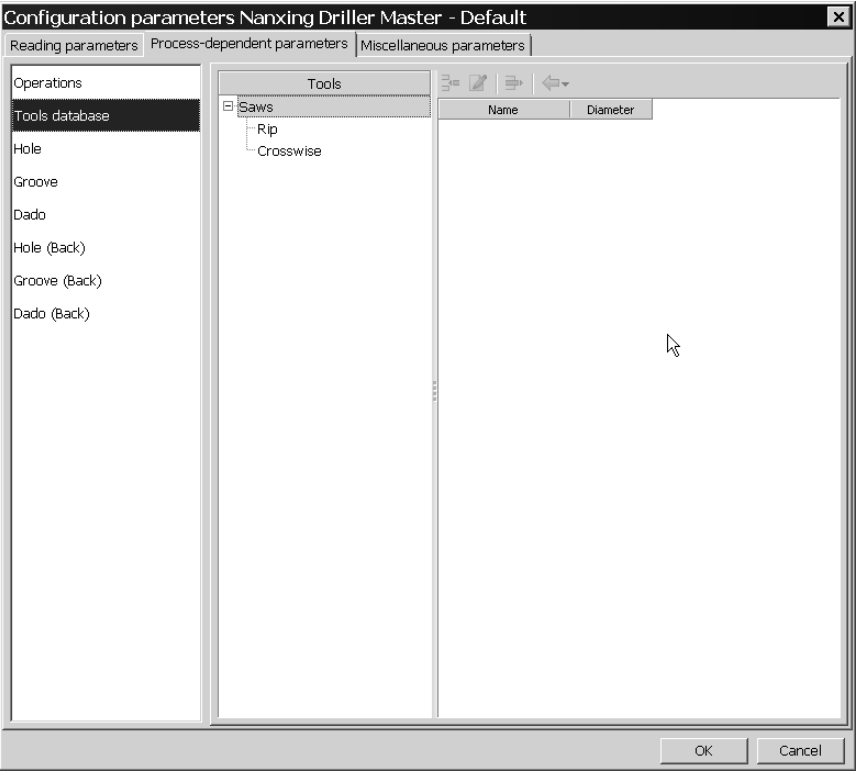


Fig. 10.512.



To add a tool to the list push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.513).

Name	
Diameter	0
Width	0

Fig. 10.513.



In this dialogue enter a tool name and set its parameters values. To complete a tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change selected tool parameters.



Delete button allows to delete selected tool from the list.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.514), holes from back side – in **Hole (Back)** section in the same way.

Through hole parameters	
Additional depth	0

Fig. 10.514.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing setup is performed in the following sections: **Groove** (fig. 10.515), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

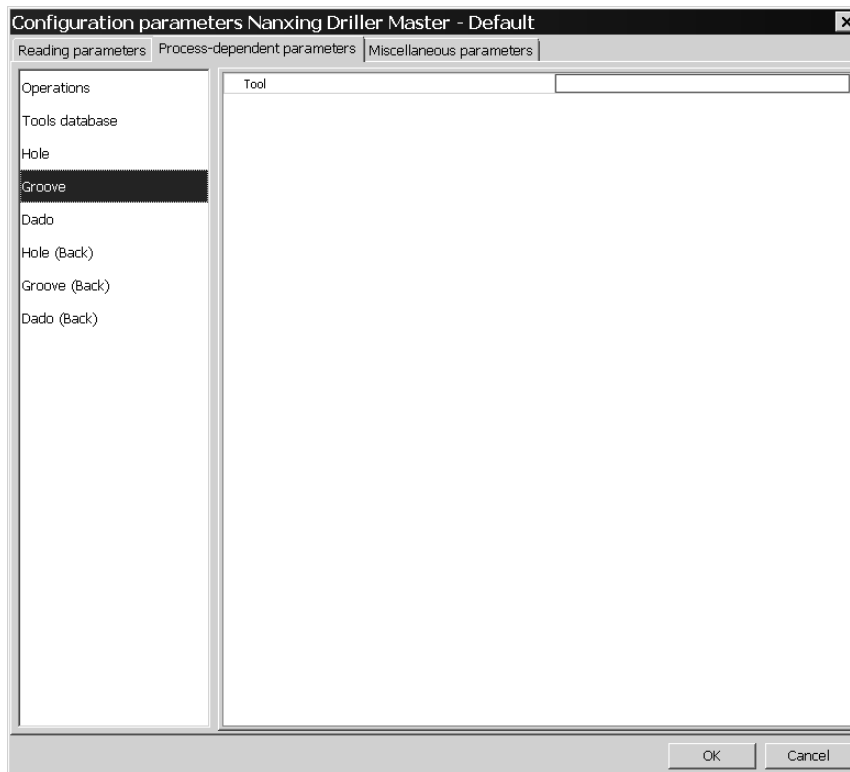


Fig. 10.515.

Section control elements allow to select a tool type for slot creation, a tool from a database and correction type.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.29.4.Nanxing FinChinaSoft



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.516) allow to specify operations that can be used in control program.

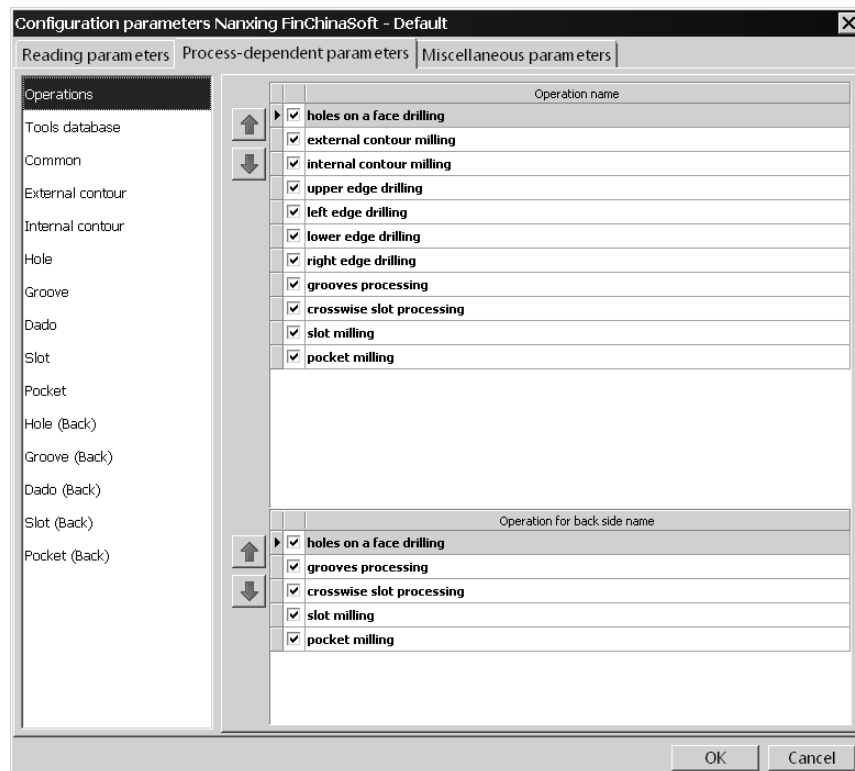


Fig. 10.516.

Usable tools

Tools database section control elements (fig. 10.517) allow to create a list of tools available for use in control programs.

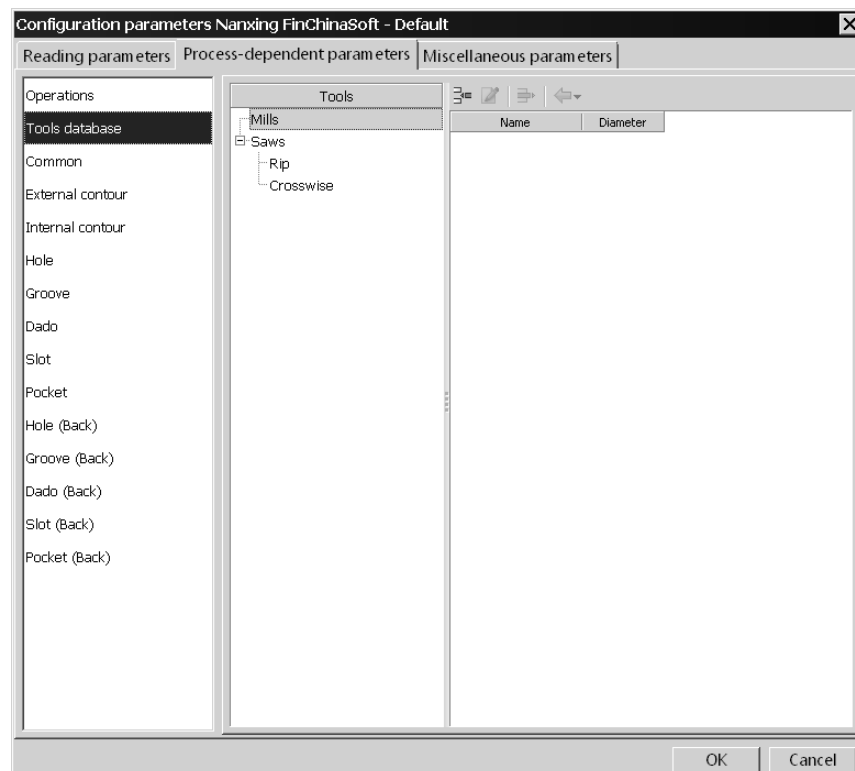


Fig. 10.517.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.518).

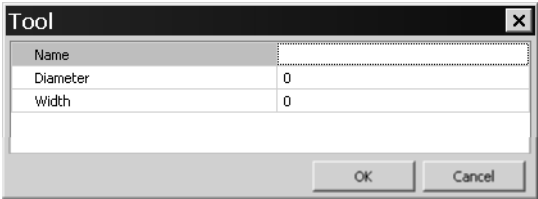


Fig. 10.518.



In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common section control elements (fig. 10.519) allow to set common parameters of control programs creation.

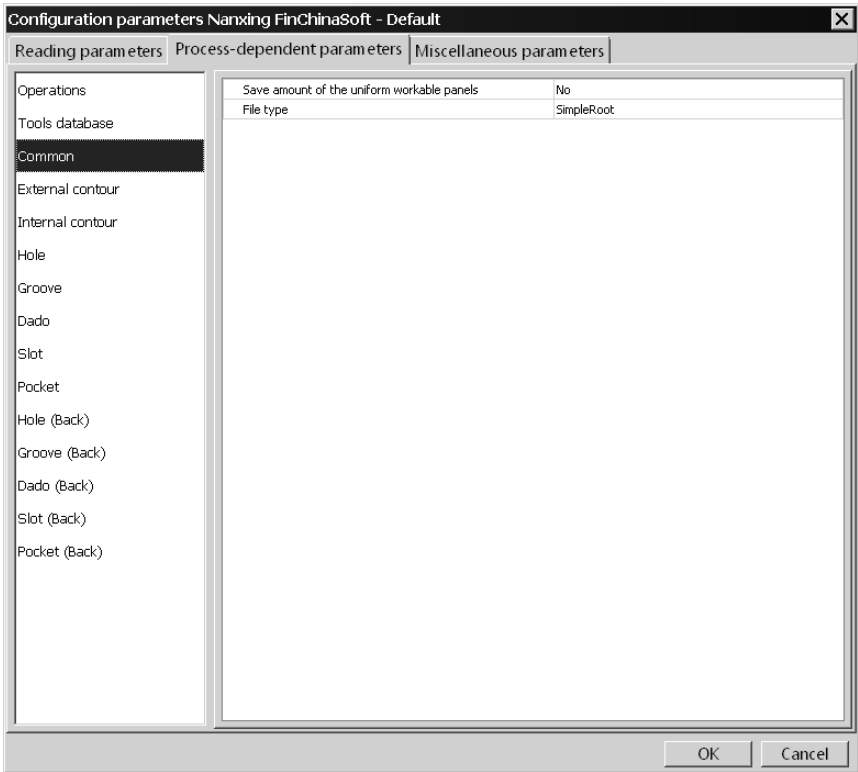


Fig. 10.519.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.520).

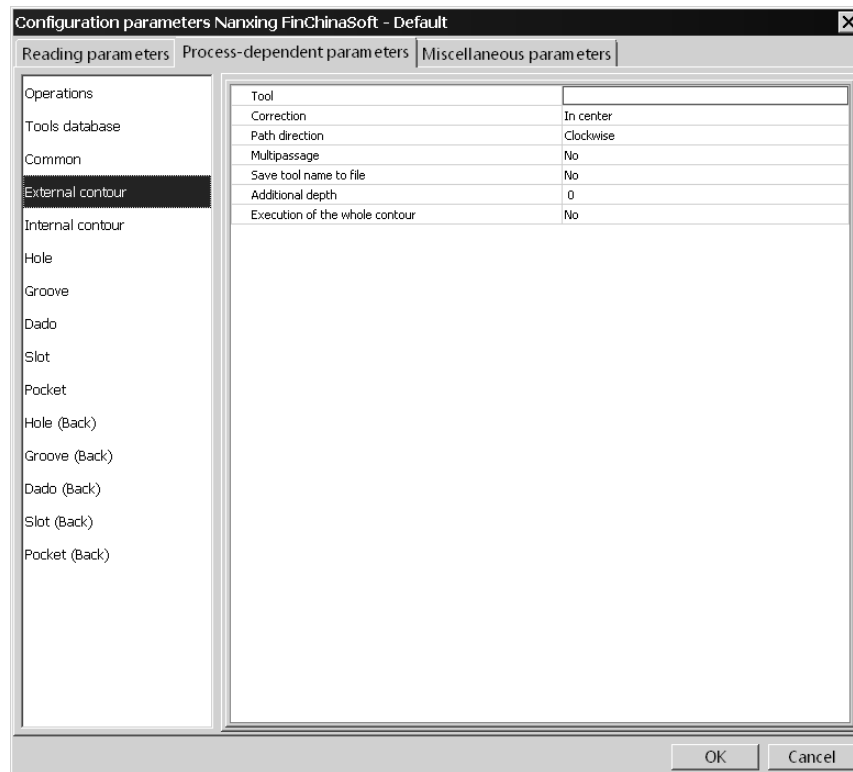


Fig. 10.520.

Section control elements allow to choose a tool, lead-in type, correction type, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.521), holes from back side – in **Hole (Back)** section in the same way.

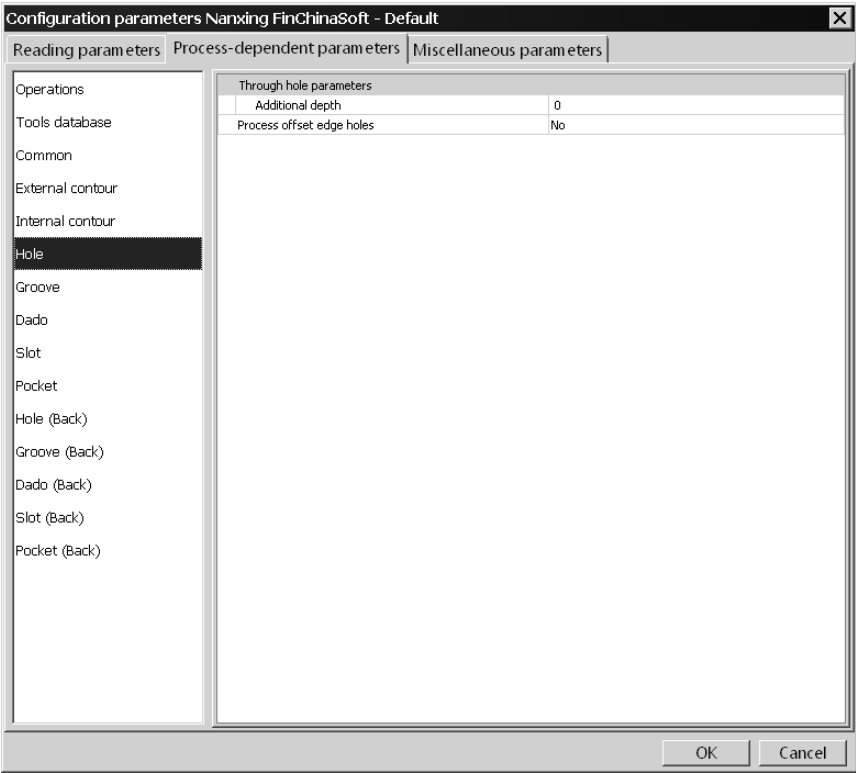


Fig. 10.521.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.522), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

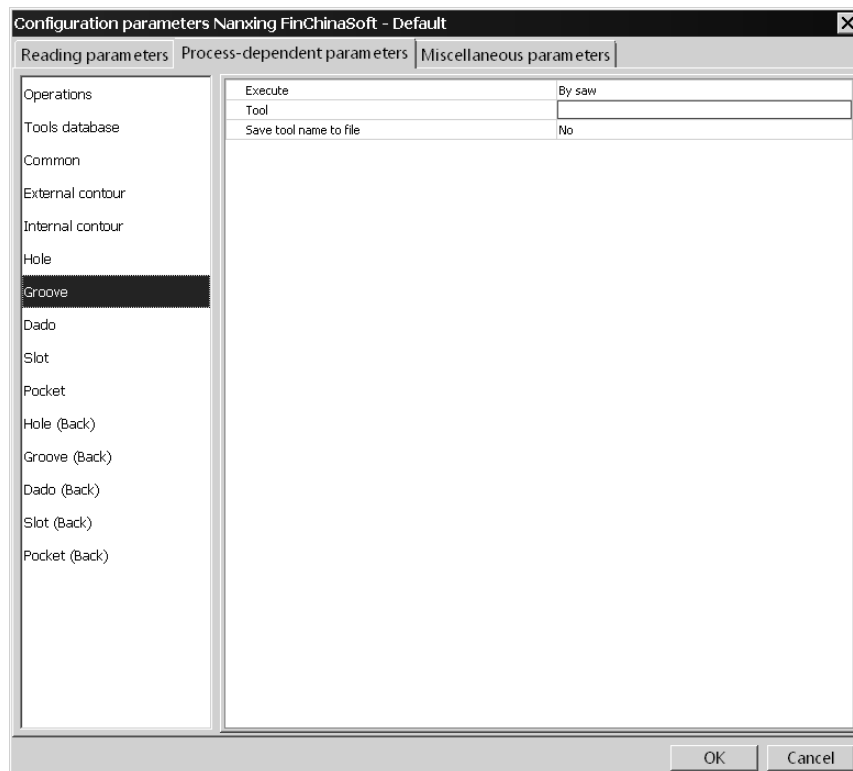


Fig. 10.522.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.523), from the back panel side – in **Slot (Back)** section.

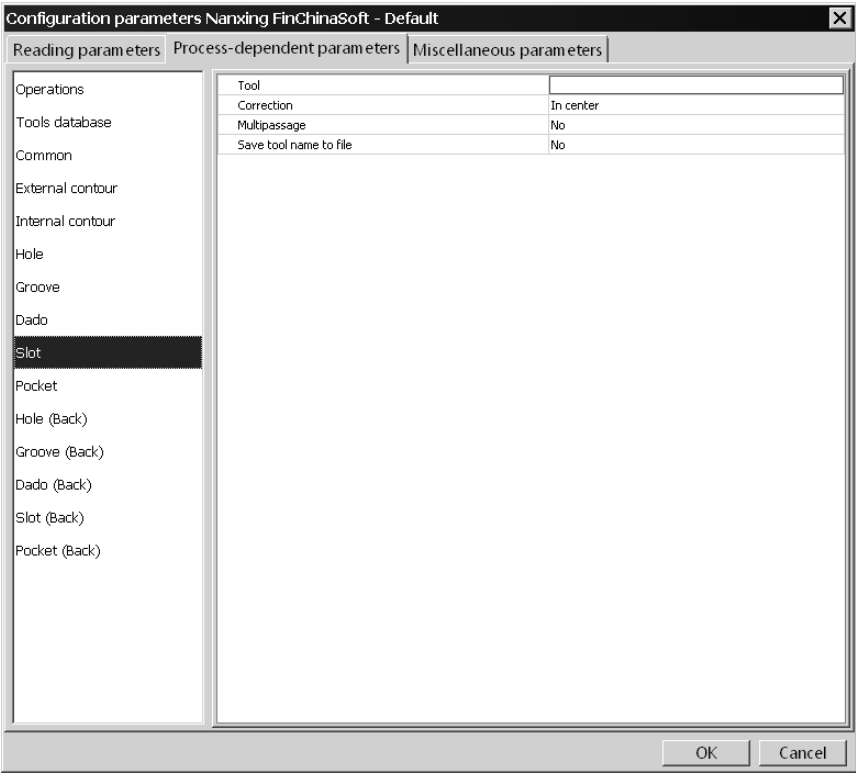


Fig. 10.523.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.524), on the back panel side – in **Pocket (Back)** section.

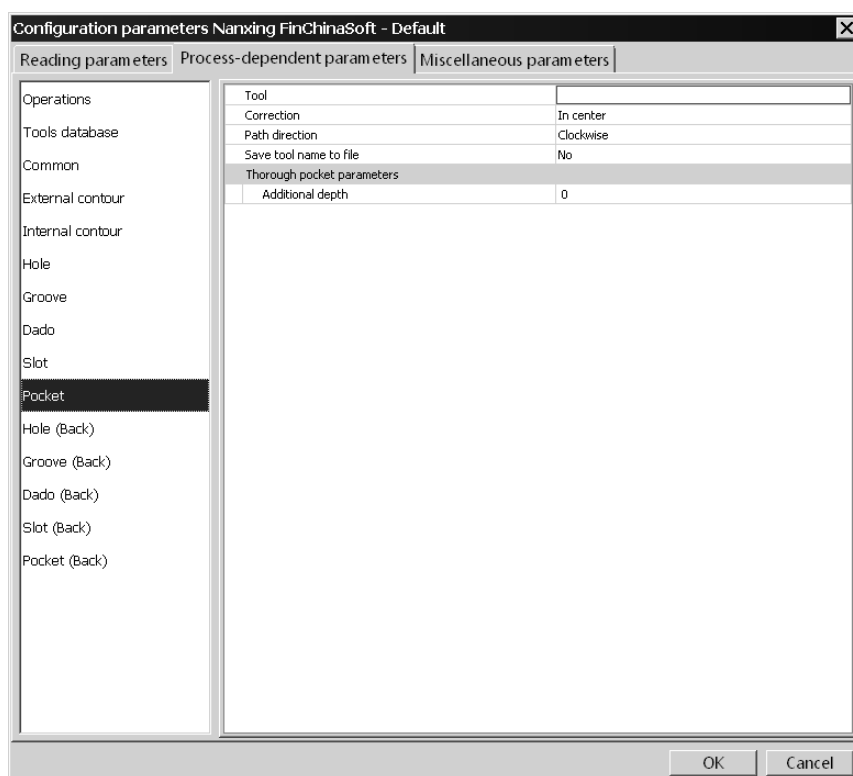


Fig. 10.524.

Tool combo box allows to select milling tool from database. For pocket contour select milling execution direction, correction type.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.29.5.PCDrillCAM



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.525) allow to specify operations that can be used in control program.

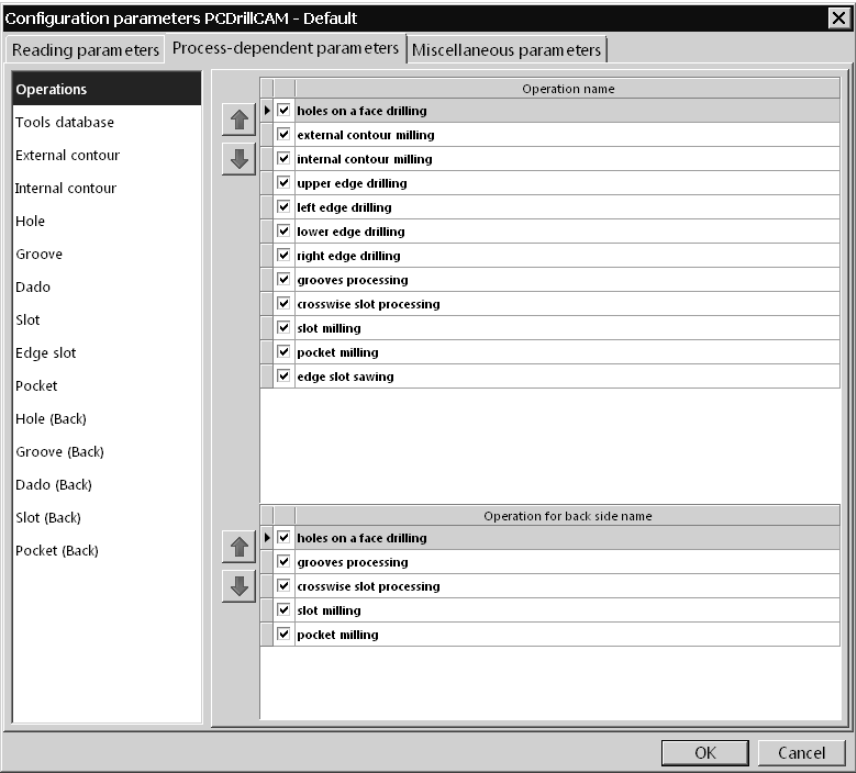


Fig. 10.525.

Usable tools

Tools database section control elements (fig. 10.526) allow to create a list of tools available for use in control programs.

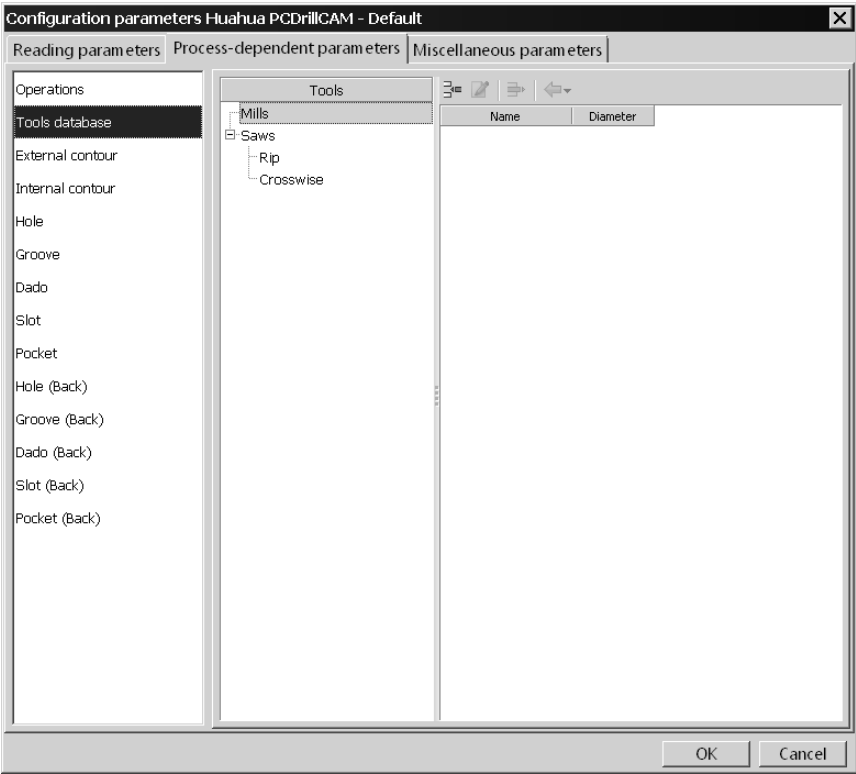


Fig. 10.526.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.527).

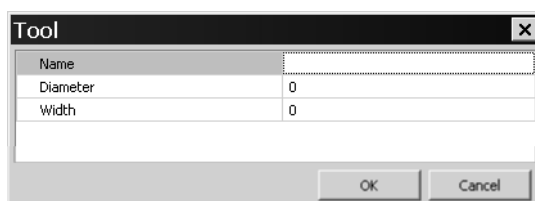


Fig. 10.527.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.



Move button allows to move selected tool to another group. After pushing the button, a menu will appear (fig. 10.528), the menu commands allow to choose a target group.

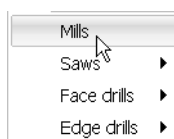


Fig. 10.528.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.529).

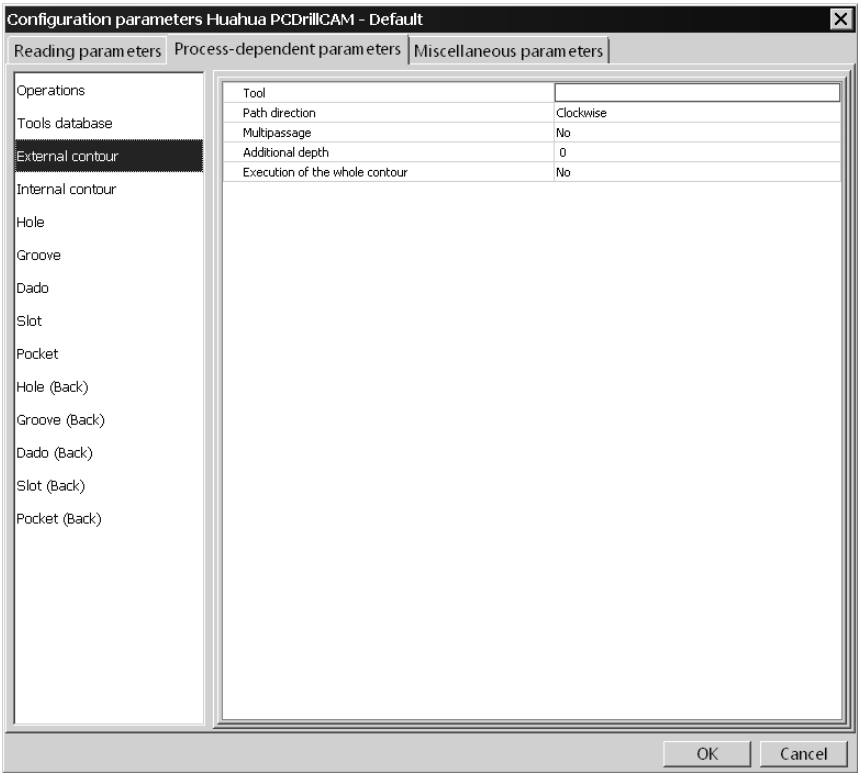


Fig. 10.529.

Control elements in sections allow to select the tool, correction types, lead-in, etc.

Hole parameters

Hole section control elements allow to set holes parameters (fig. 10.530).

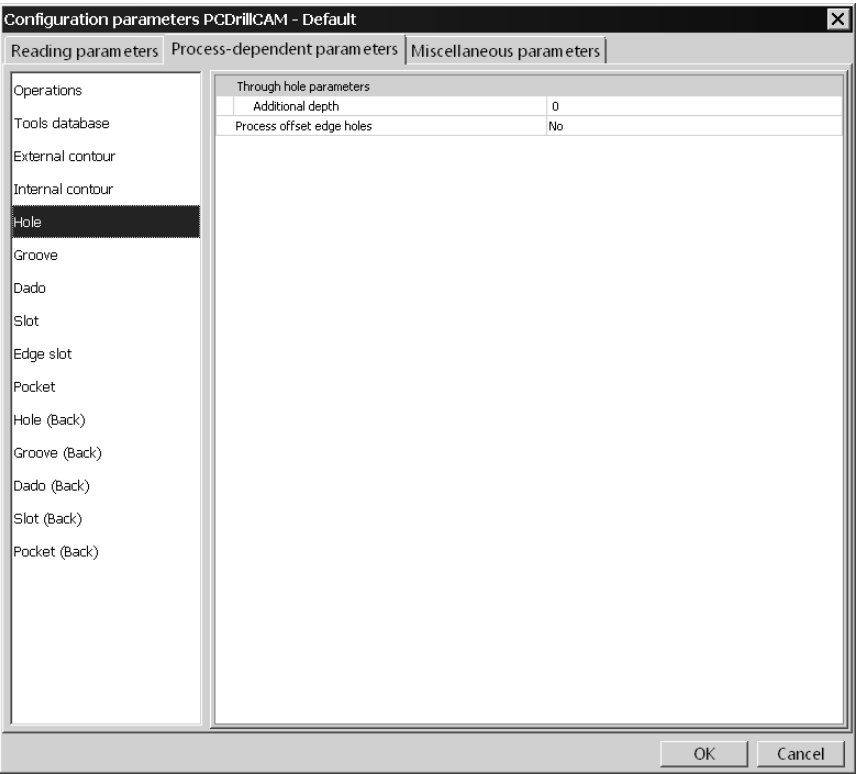


Fig. 10.530.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.531), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

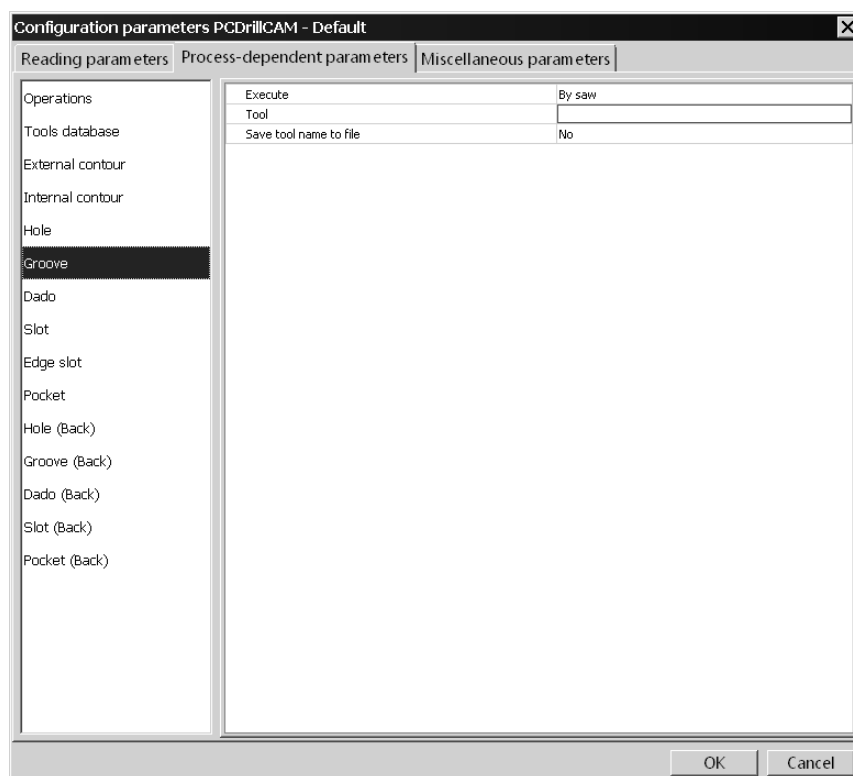


Fig. 10.531.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.532), from the back panel side – in **Slot (Back)** section.

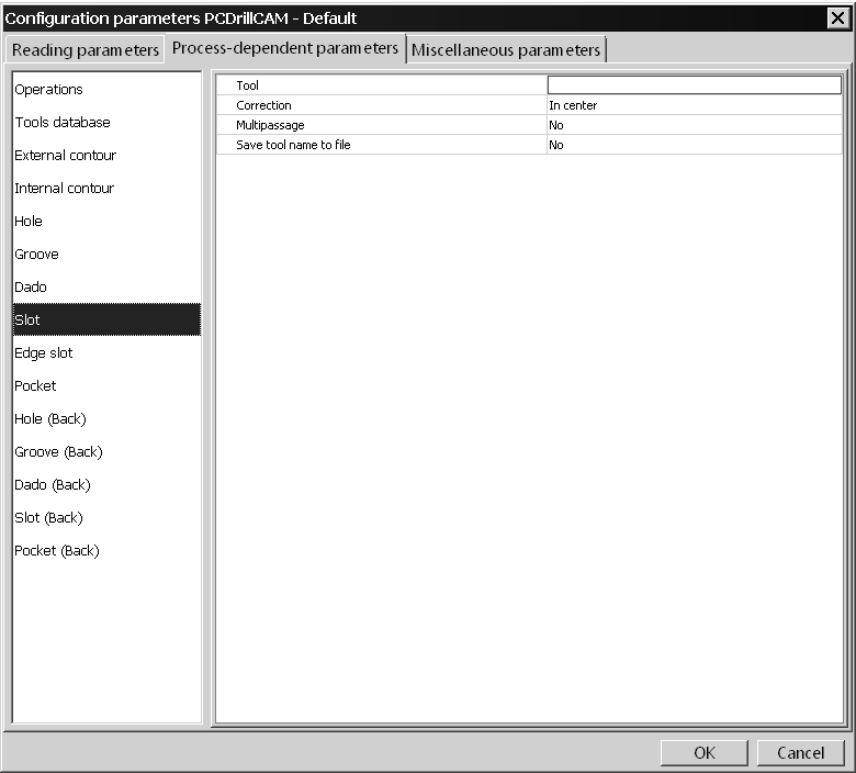


Fig. 10.532.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Pocket section control elements (fig. 10.533) allow to set pocket processing parameters.

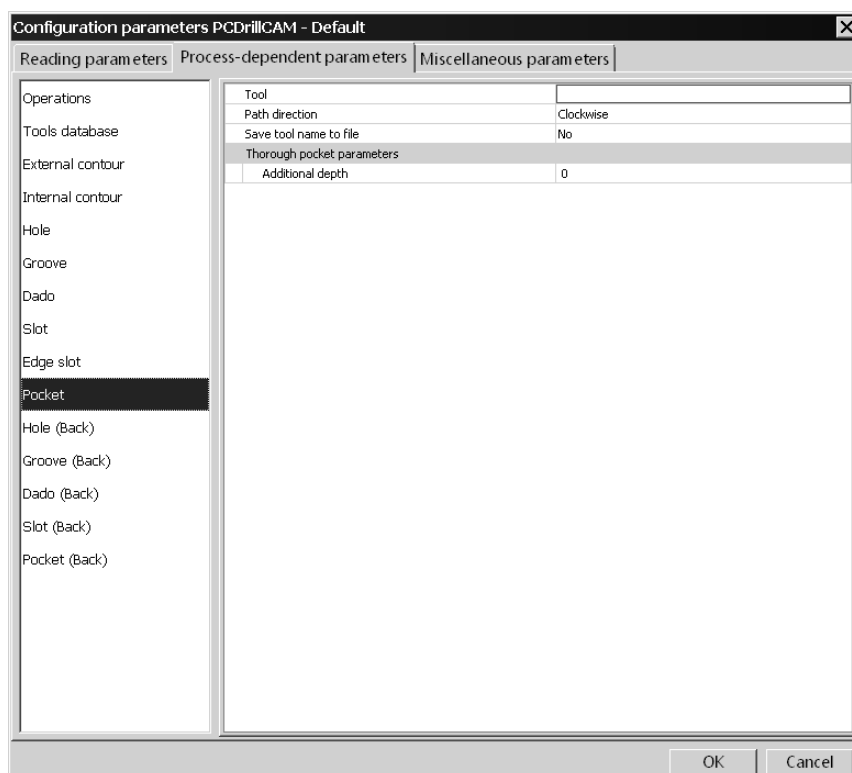


Fig. 10.533.

Tool combo box allows to select milling tool name from database. For pocket contour select milling execution direction, correction type and multipassage use.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.29.6.TPA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.534) allow to specify operations that can be used in control program.

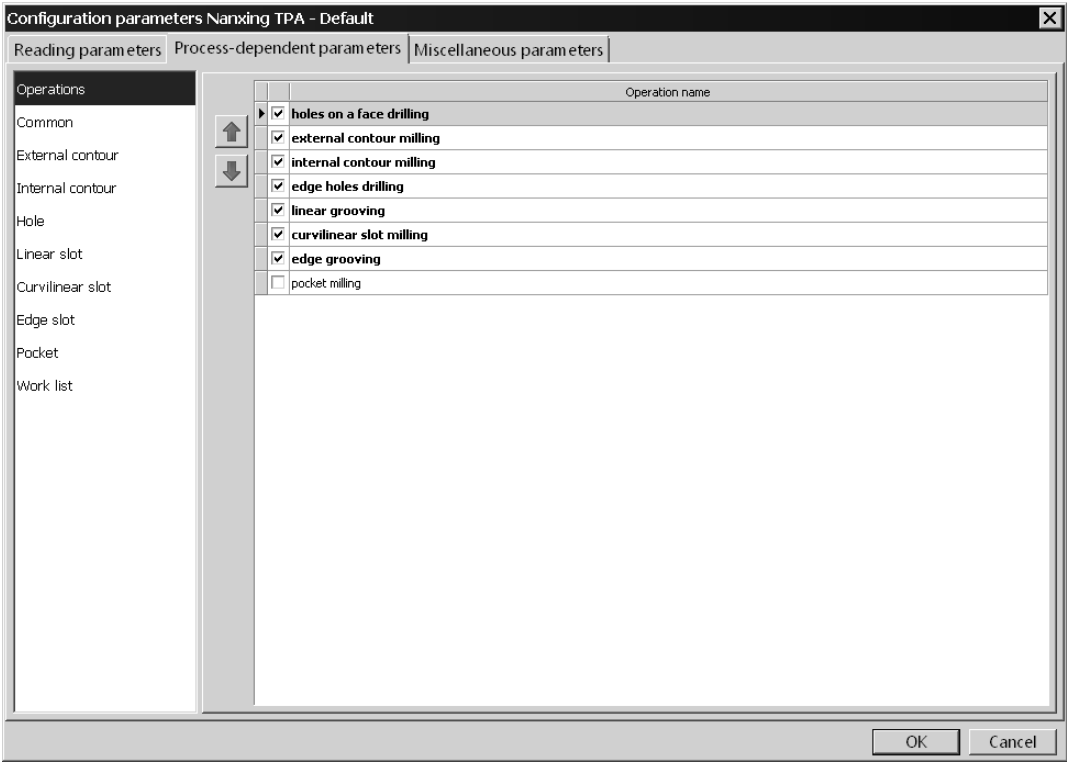


Fig. 10.534.

Common parameters

Common section control elements (fig. 10.535) allow to set common parameters of control programs creation.

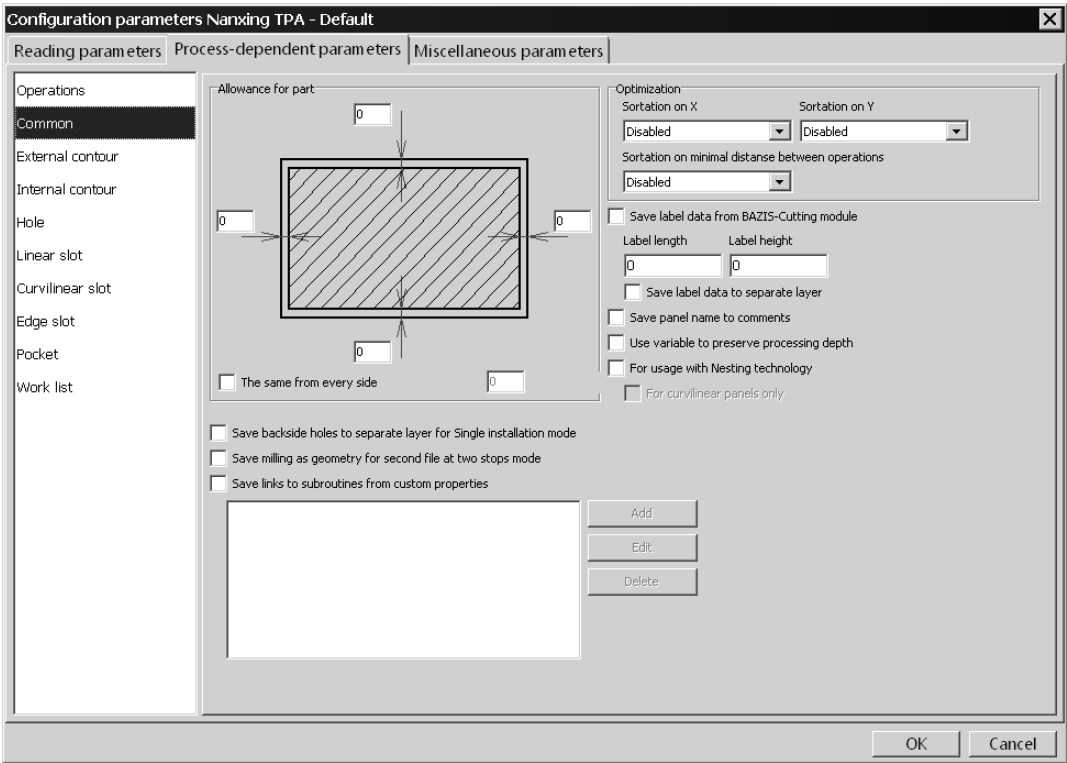


Fig. 10.535.

Allowance for part section control elements allow to set allowance values. Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, the same from every panel edge allowance field will become available. **Append operations to CNC program** group control elements allow to control operations of options names appending to control program. To add the program to the program, enable the option next to its name.

Contours parameters

Contours milling parameters are set in **Outer contour** (fig. 10.536) and **Internal contour** sections of **Process-dependent parameters** tab.

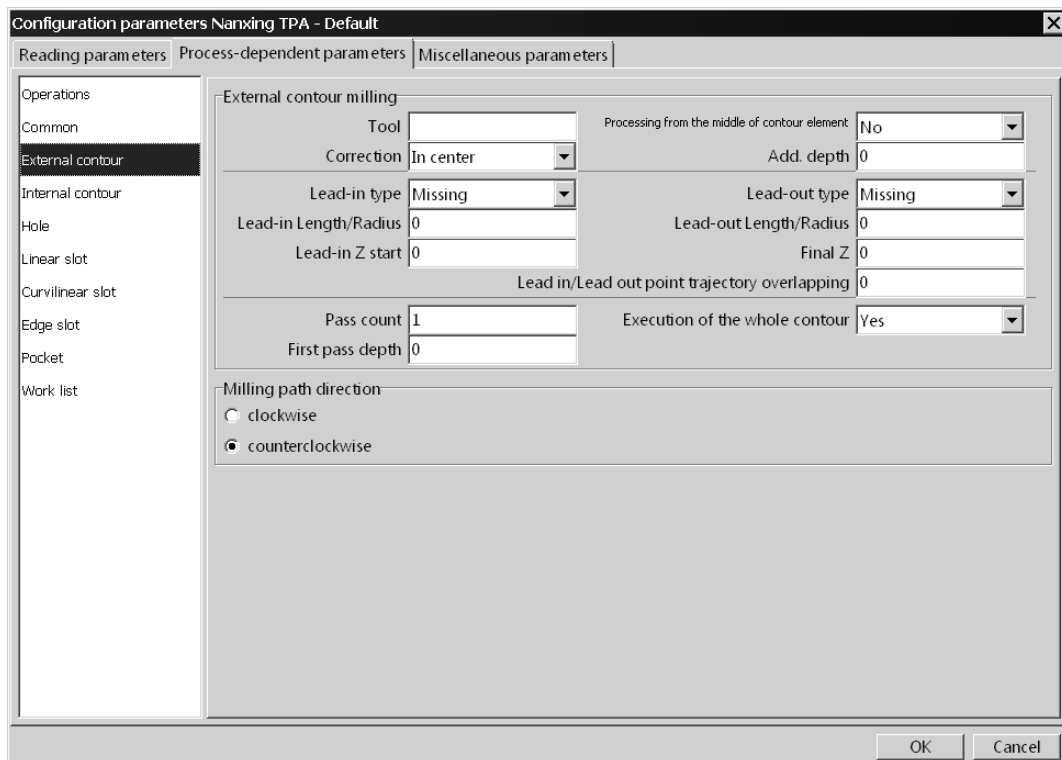


Fig. 10.536.

Finish machining and **Rough machining** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Rough machining** option allows to control this operation.

Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Hole section control elements allow to set holes parameters (fig. 10.537).

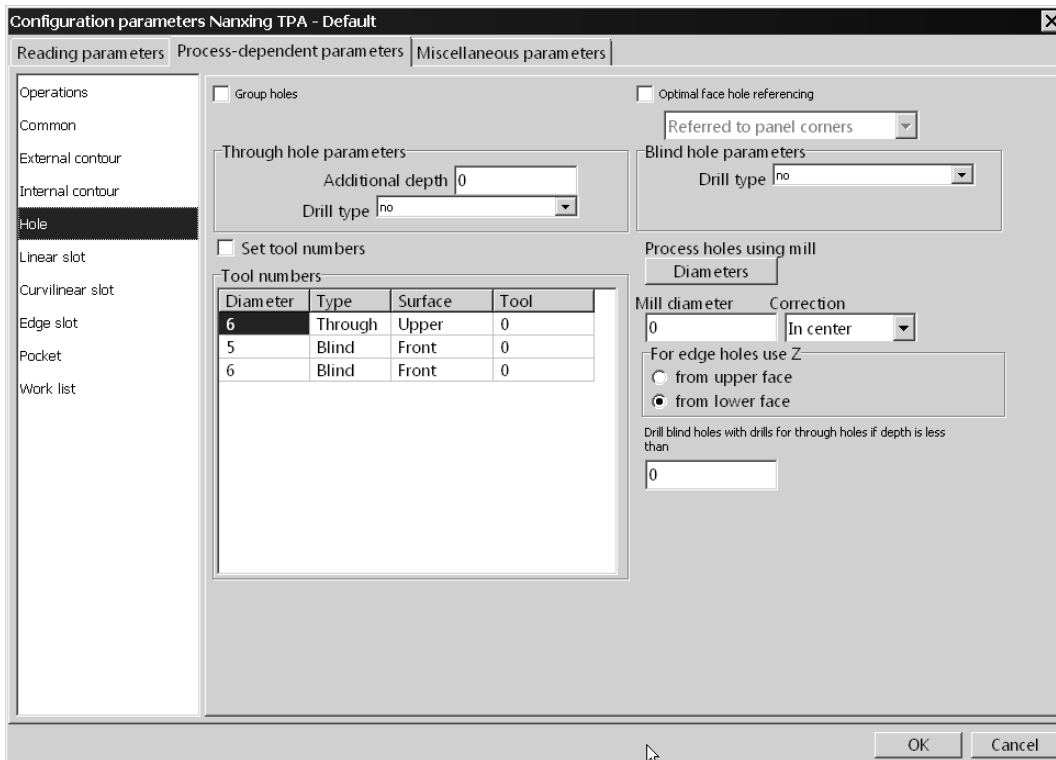


Fig. 10.537.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Through holes can be bored with drills for through and blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.538).

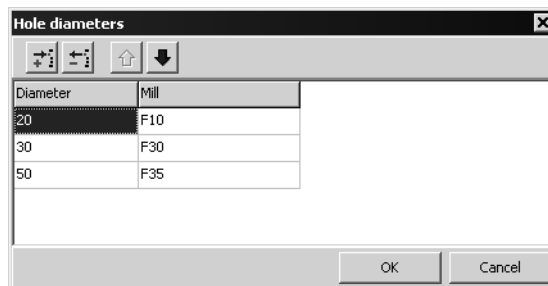


Fig. 10.538.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To finish setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter application.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.539).

Configuration parameters Nanxing TPA - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Operations

Common

External contour

Internal contour

Hole

Linear slot

Curvilinear slot

Edge slot

Pocket

Work list

Tool choice

☐ Through slot always cut with saw

☐ Grooving by saw

Slot - by saw

Tool

Saw blade width 0

Correction In center

Pass count 1

Slot - by mill

Tool

Mill diameter 0

Lead-in type Missing

Lead-in Length/Radius 0

Lead-in Z start 0

Correction In center

Lead-out type Missing

Lead-out Length/Radius 0

Final Z 0

OK Cancel

Fig. 10.539.

Through slot always cut with saw and **Grooving by saw** options allow to set through slots direction that will be processed with a saw. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.540).

Tool choice

Groove name

1114x5 (4x5)

Quarter groove (12x12)

N	Tool name	Correction	Diameter
1	D15	At a center	0

OK Cancel

Fig. 10.540.



Slot name table contains all grooves names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing.

The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.541).

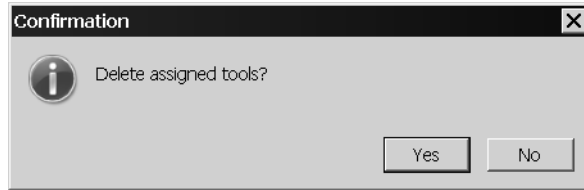


Fig. 10.541.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. You can map slot width and mill parameters that will be automatically selected for its processing. For this purpose push **Mill parameters** button. **Mill parameters** dialogue will appear on the screen (fig. 10.542).

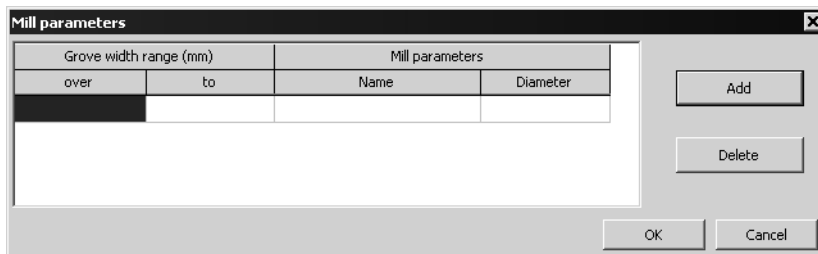


Fig. 10.542.

Dialogue table contains slot width ranges and corresponding mill parameters. The fields should be filled via a keyboard. **Add** button allows to add a blank row to the table. To delete the current row push **Delete** button.

To finish parameters setup push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.543).

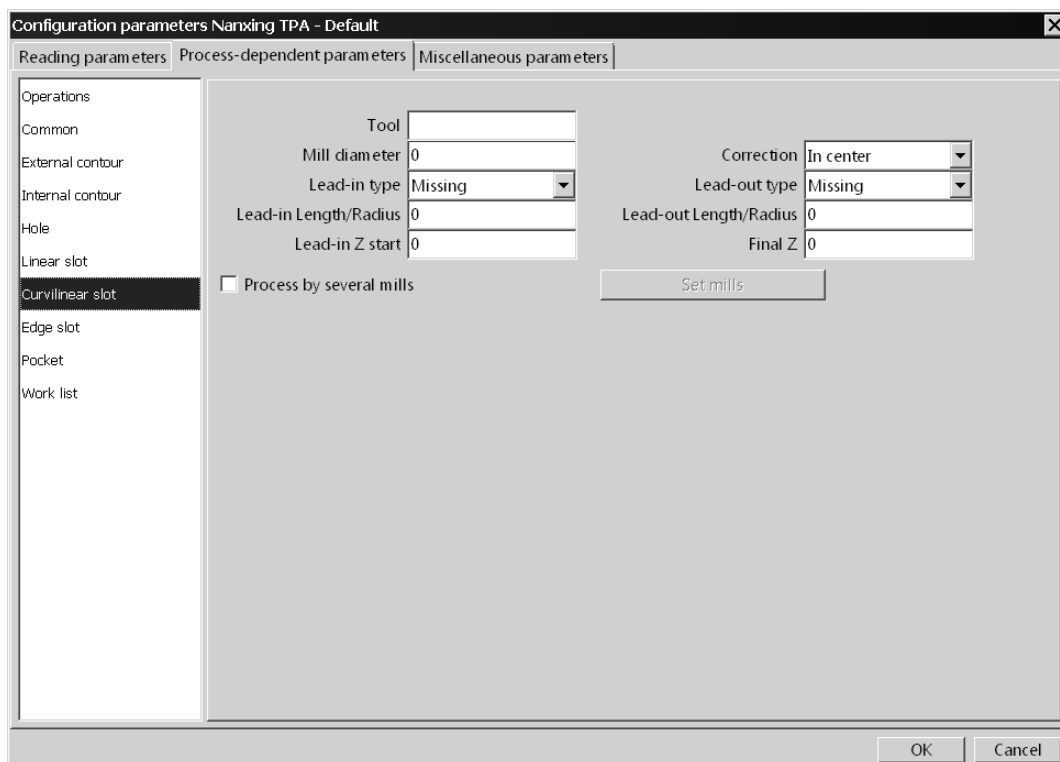


Fig. 10.543.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.544).

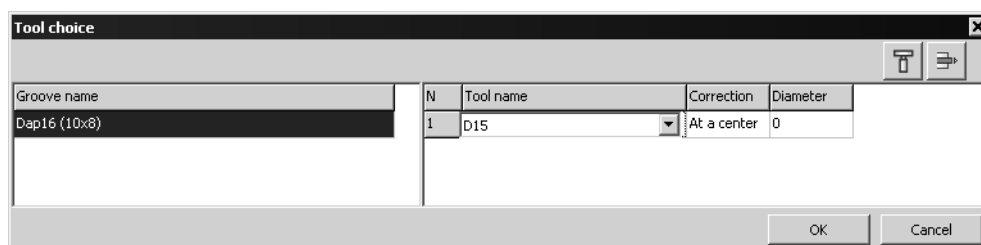


Fig. 10.544.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box. **Delete** button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.545).

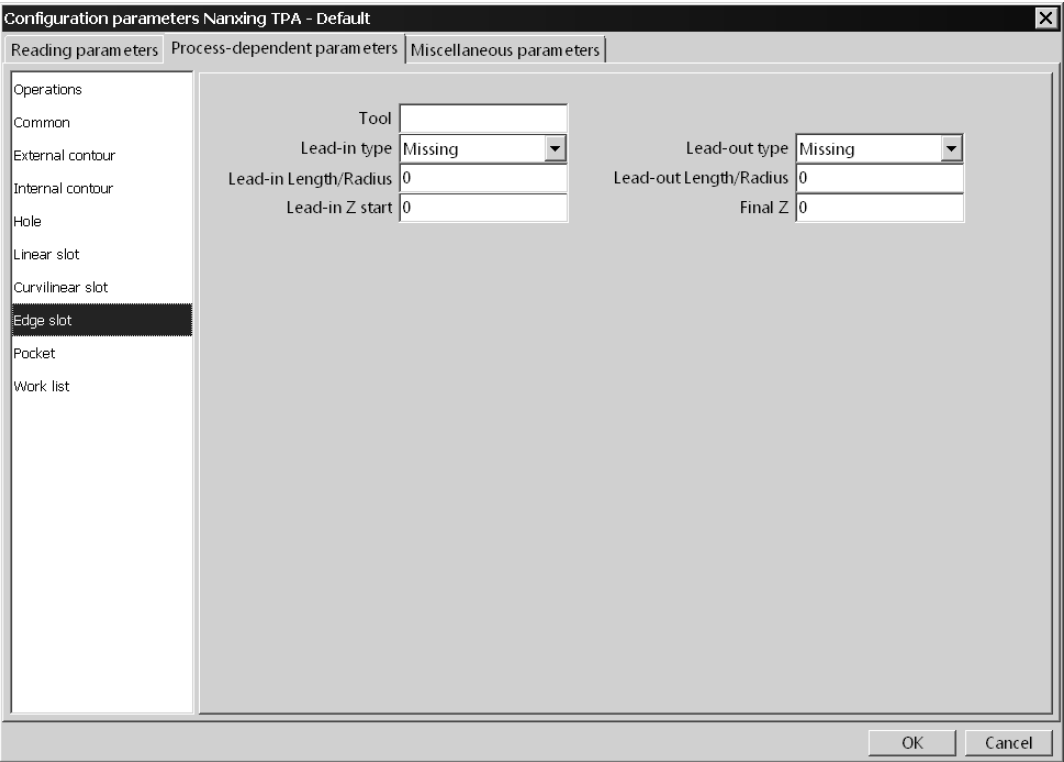


Fig. 10.545.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.546) allow to set pocket processing parameters.

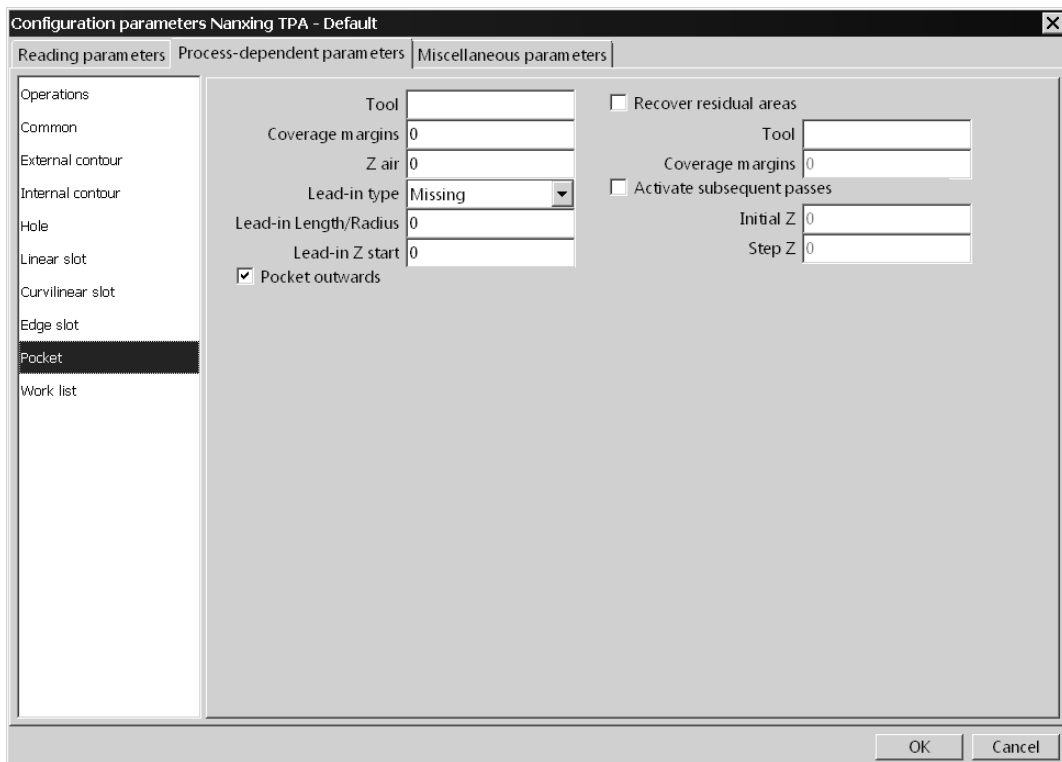


Fig. 10.546.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Work list

Work list can be set in **Work list** section (fig. 10.547).

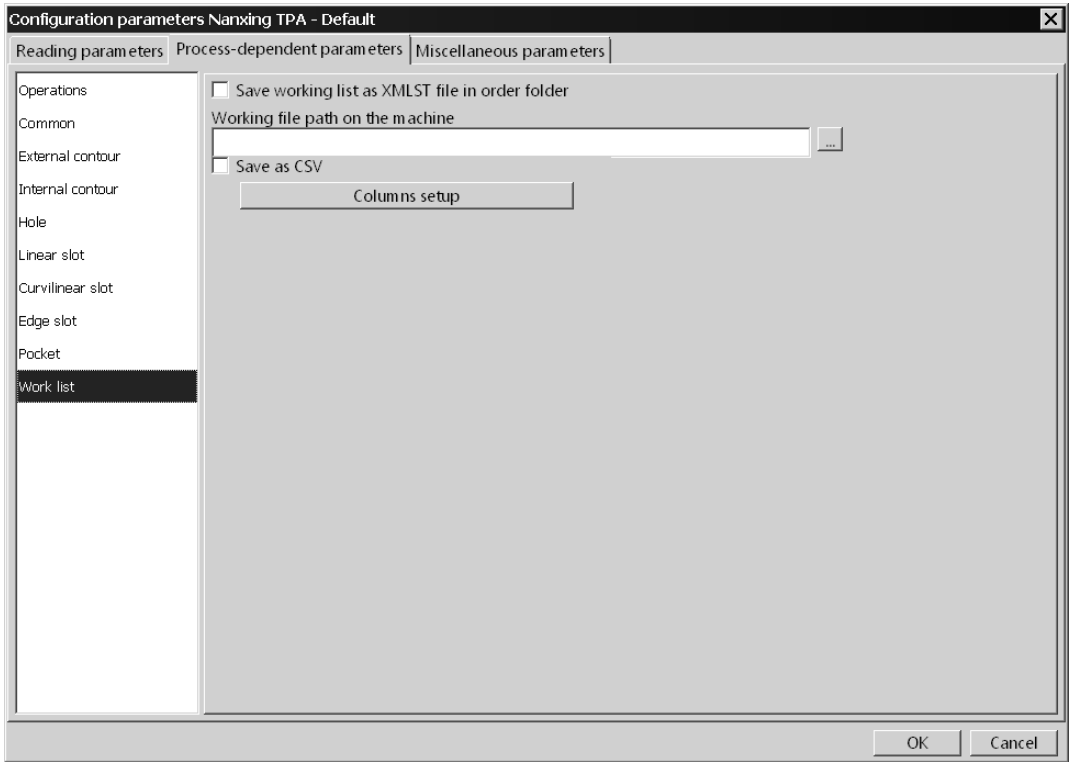


Fig. 10.547.



Save working list as file in order folder option allows to control the file saving. To specify the folder push **Browse** button and choose it in a standard Windows dialogue. **Save as CSV** option allows to control file saving. **Columns setup** button allows to setup file table columns. After pushing the button, **Table columns setup** dialogue will appear on the screen (fig. 10.548).



Fig. 10.548.

To add data from a column to a file, enable the option for it. **Up** and **Down** buttons allow to move the selected column through the list. **Add** button allows to add a column to the list. In **Name** field, you can change the name of the current column. For columns added by the user, you can set a value in the appropriate field. **Delete** but-

ton allows to delete a column added by the user. After pushing the button, a warning will appear on the screen (fig. 10.549).

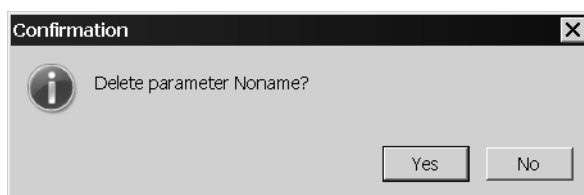


Fig. 10.549.

Message buttons allow to confirm or reject the deletion.

Reset all button allows to return the table settings to their default values. After pushing the button, a warning will appear on the screen (fig. 10.550).

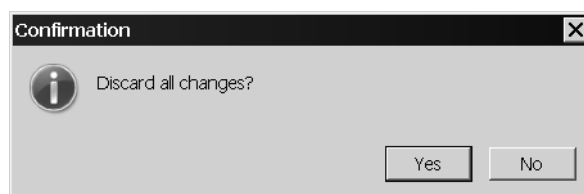


Fig. 10.550.

The message buttons allow to confirm or cancel the reset of parameters. To complete the table setup, push **OK** button.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.30. Omaksan

10.30.1.Omaksan Works

Operations

Operations section control elements (fig. 10.551) allow to specify operations that can be used in control program.

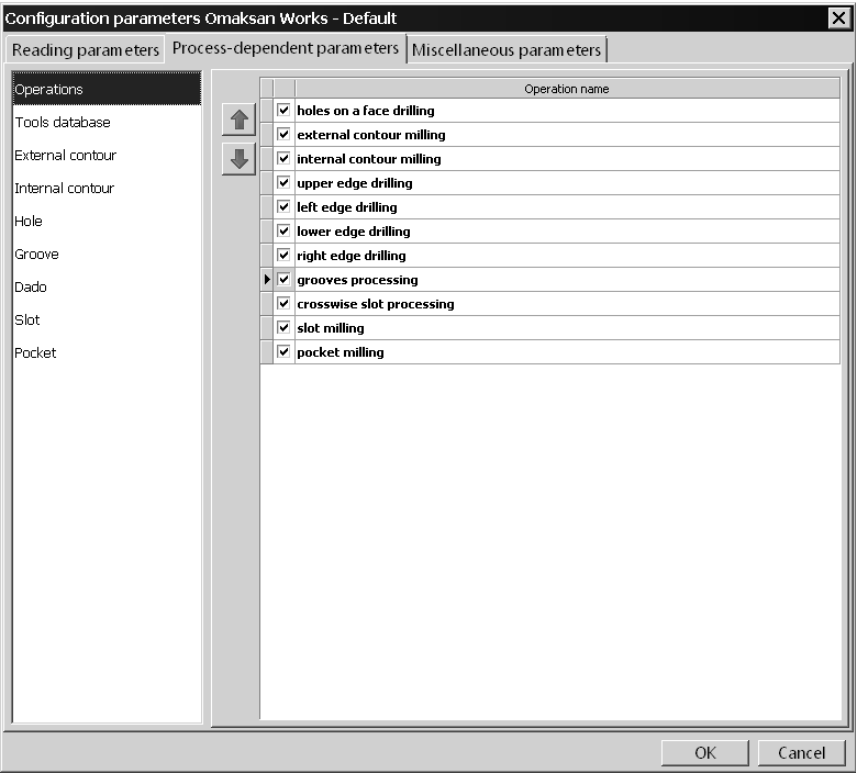


Fig. 10.551.

Usable tools

Tools database section control elements (fig. 10.552) allow to create a tools list available for use in control programs.

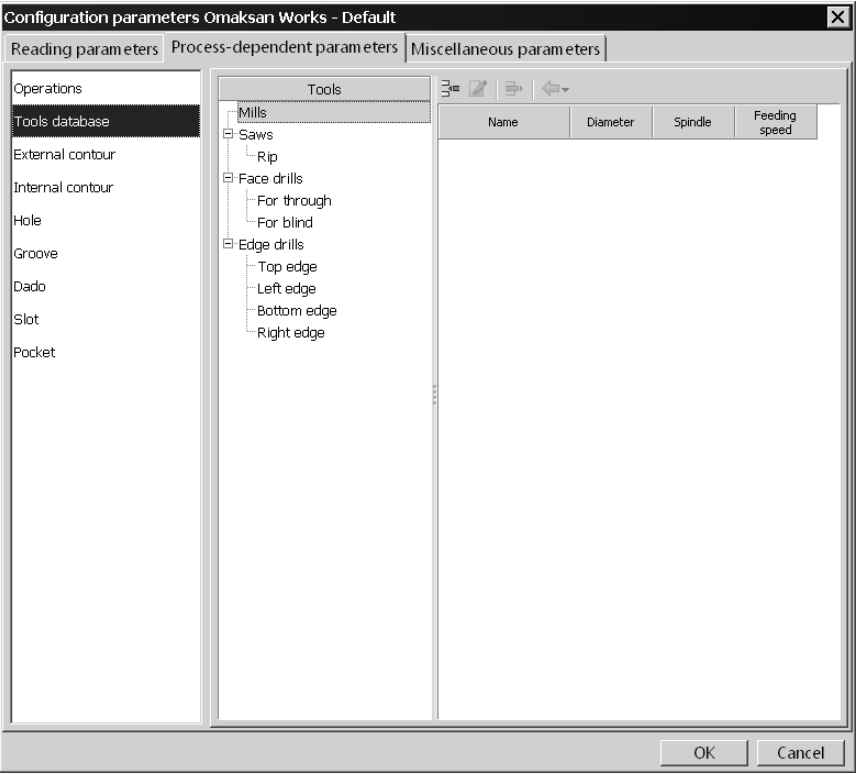


Fig. 10.552.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.553).

Name	
Diameter	0
Width	0

Fig. 10.553.

In this dialogue enter a tool name and set its parameters. To complete a tool adding push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes.



Edit button allows to change parameters of a tool selected in the list.



Delete button allows to delete selected tool from the list.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** (fig. 10.554) and **Internal contour** sections.

Configuration parameters Omaksan Works - Default	
Process-dependent parameters	
Tool	
Path direction	Clockwise
Tool lead-in	Missing
Tool lead-out	Missing
Multipassage	No
Additional depth	0
Execution of the whole contour	No

Fig. 10.554.

Sections control elements allow to select a tool, correction types, tool lead-in and lead-out, etc.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.555).

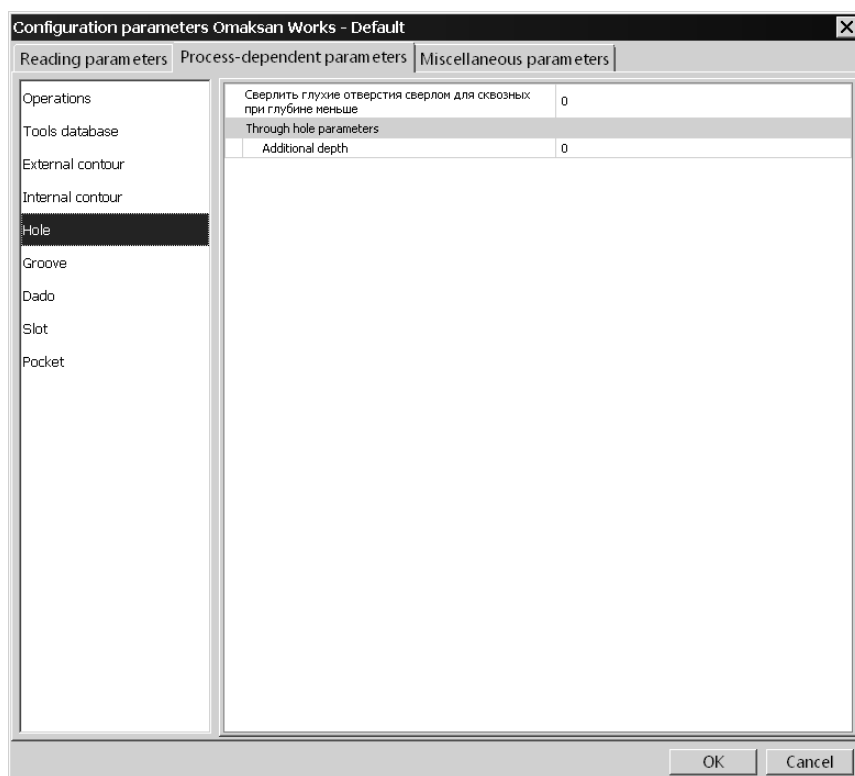


Fig. 10.555.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing is performed in the following sections: **Groove** (fig. 10.556), in **Dado** section in the same way.

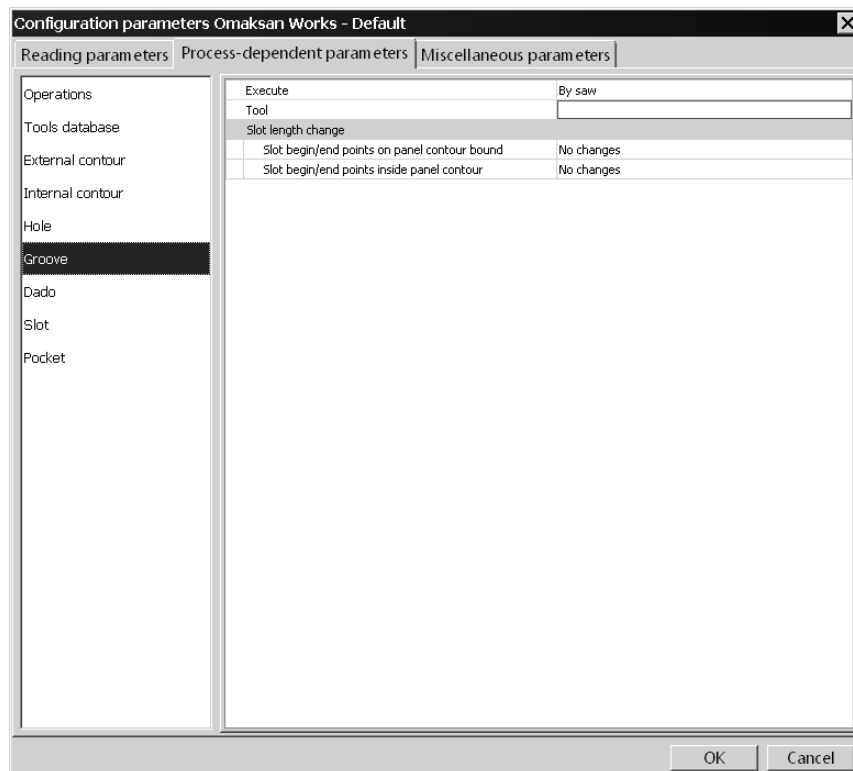


Fig. 10.556.

Section control elements allow to select a tool type for slot creation, a tool from database. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.557).

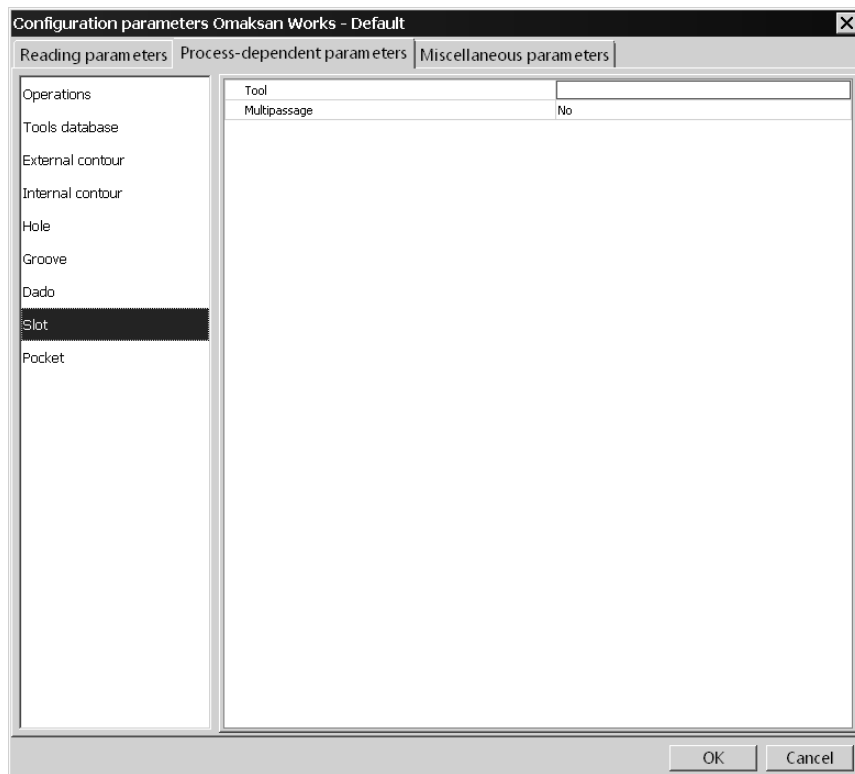


Fig. 10.557.

Section control elements allow to select tool from a database base, correction type and multipassage use.

Pocket parameters

Pocket processing parameters from front side are set in **Pocket** section (fig. 10.558).

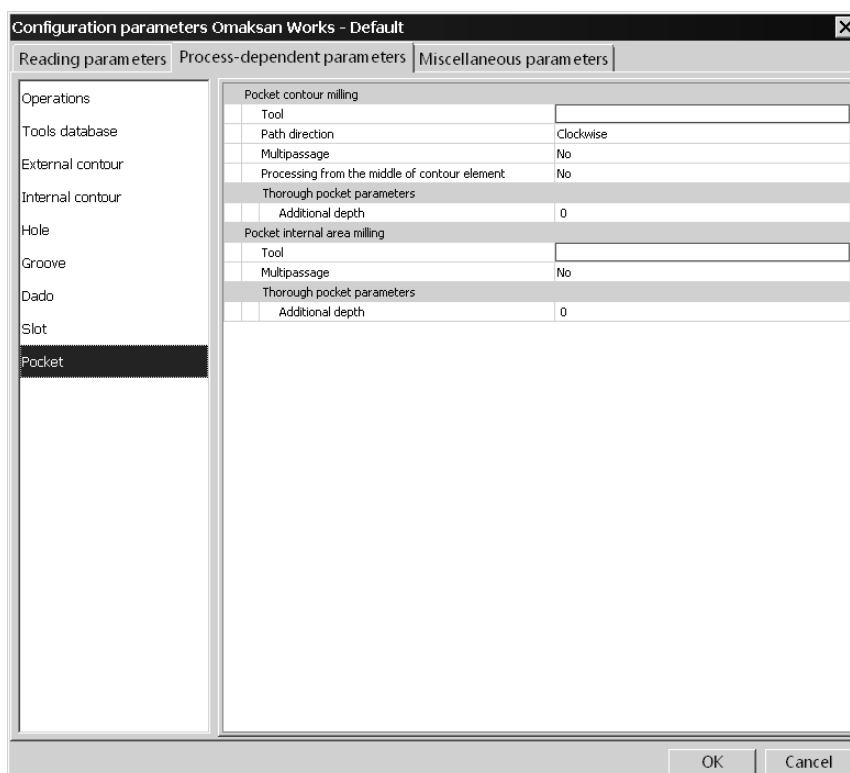


Fig. 10.558.

Tool combo box allows to select tool name for milling from a database base. For pocket contour select milling path direction, correction type and multipassage use.

Setup completion

To complete control programs setup for OMAKSAN Works, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.31. PMSK

10.31.1.PT2



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.559) allow to specify operations that can be used in control program.

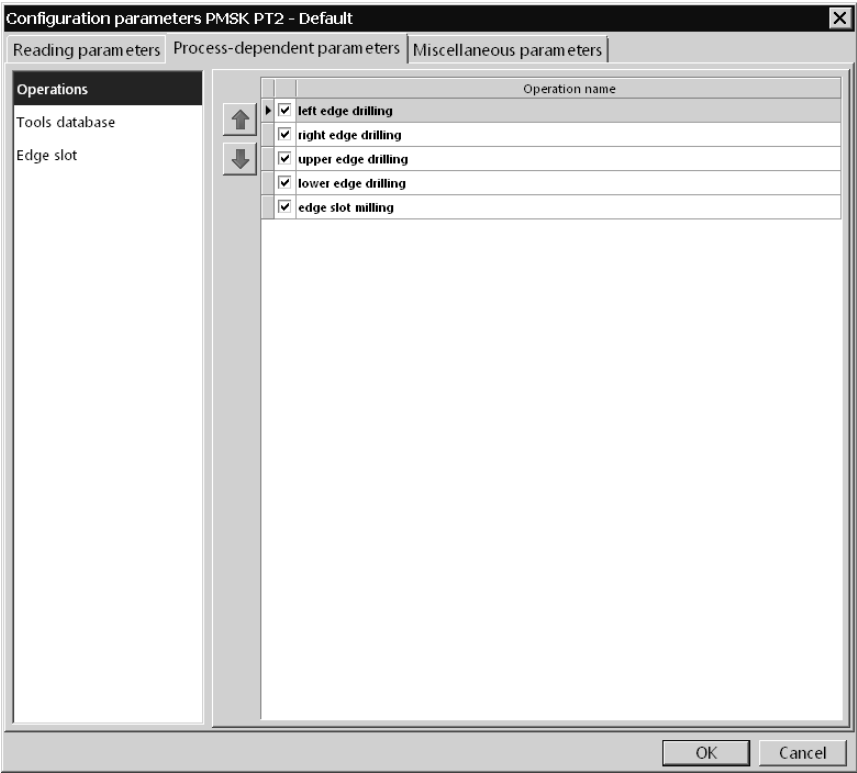


Fig. 10.559.

Usable tools

Tools database section control elements (fig. 10.560) allow to create a list of tools available for use in control program.

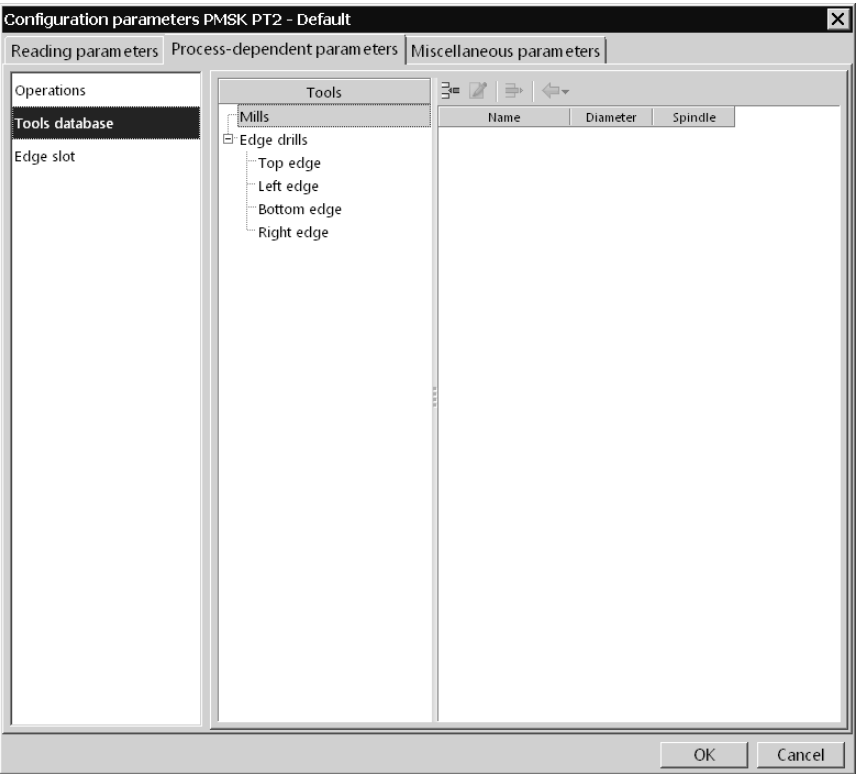


Fig. 10.560.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.561).

Fig. 10.561.



In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.562).

Fig. 10.562.

Section control elements allow to set control programs creation parameters for slots processing.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **Create** group options allow to manage control programs files creation that correspond to options names.

10.31.2.PMSK PT6

 This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation. These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.563) allow to specify operations that can be used in control program.

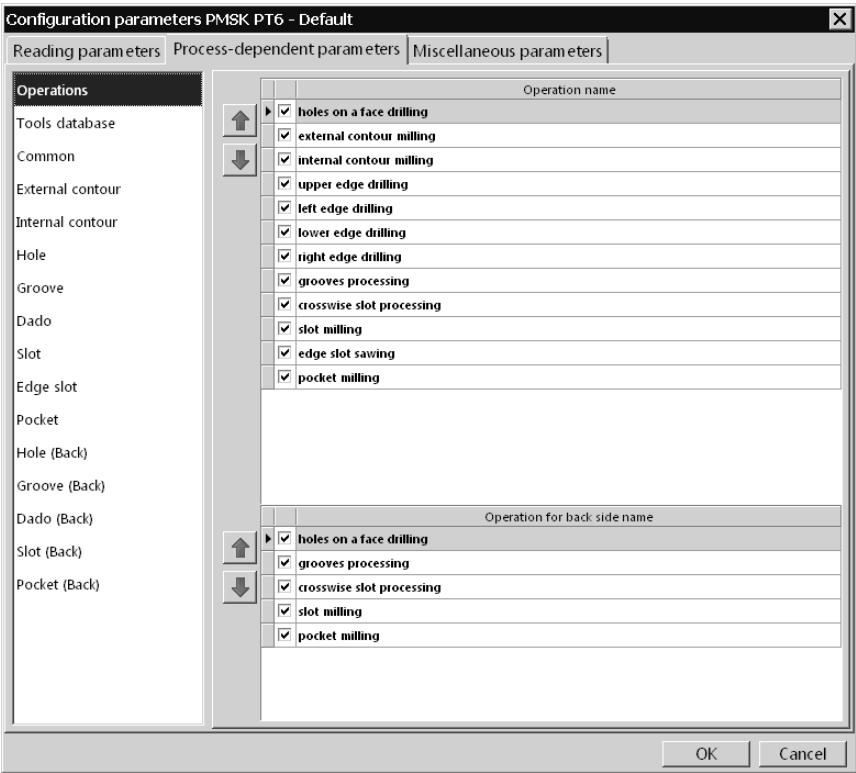


Fig. 10.563.

Usable tools

Tools database section control elements (fig. 10.564) allow to create a list of tools available for use in control programs.

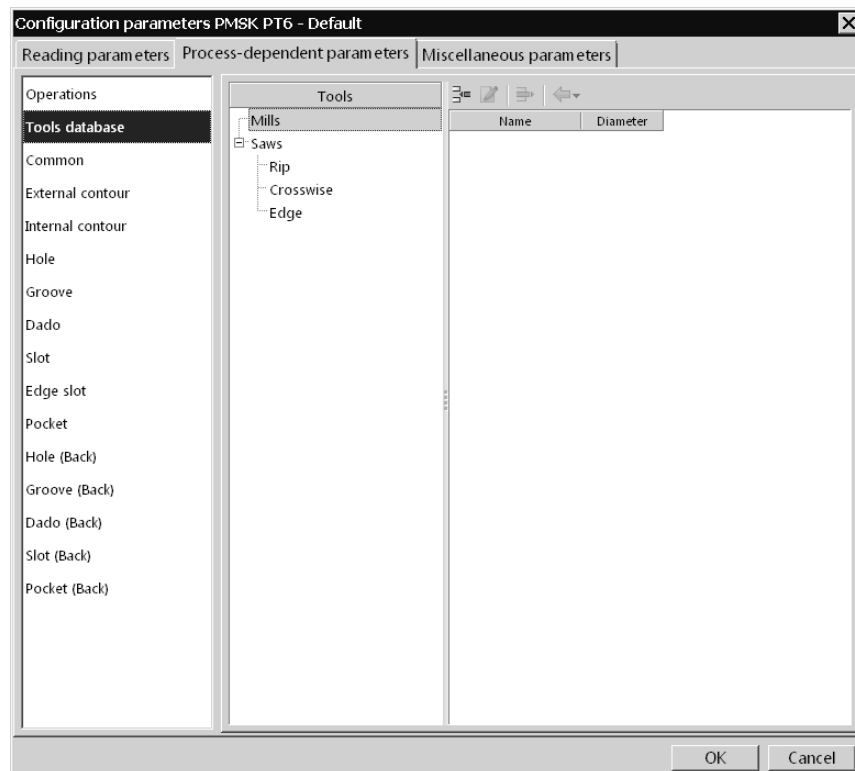


Fig. 10.564.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.565).

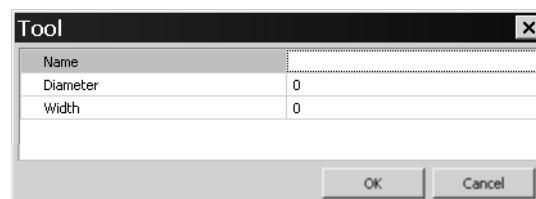


Fig. 10.565.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common section control elements (fig. 10.101) allow to set common parameters of control programs creation.

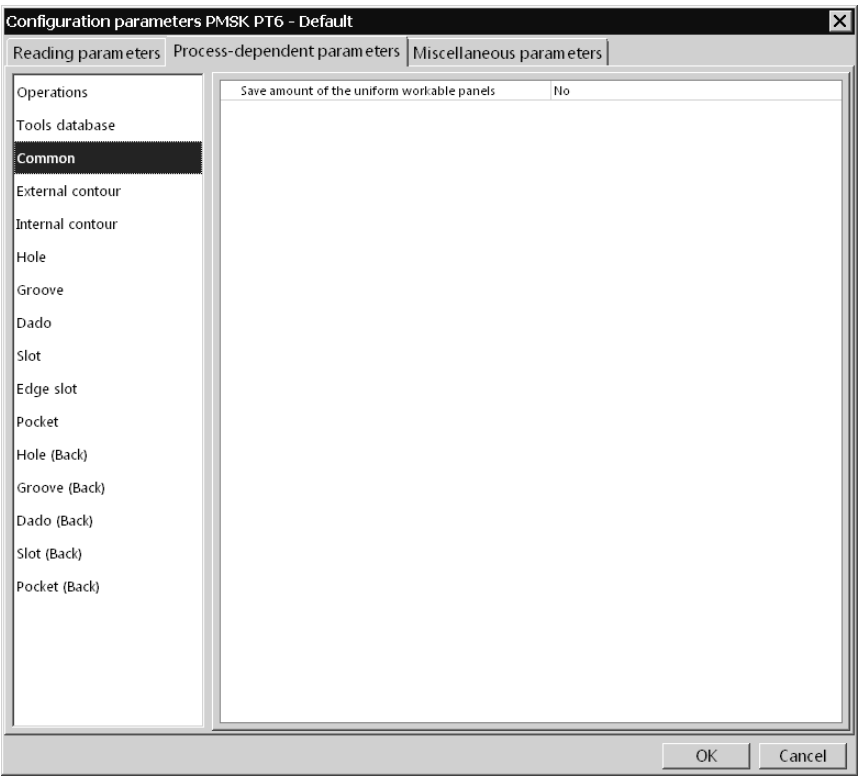


Fig. 10.566.

Contours parameters

Contours processing parameters are set in **Outer contour** and **Internal contour** sections (fig. 10.567).

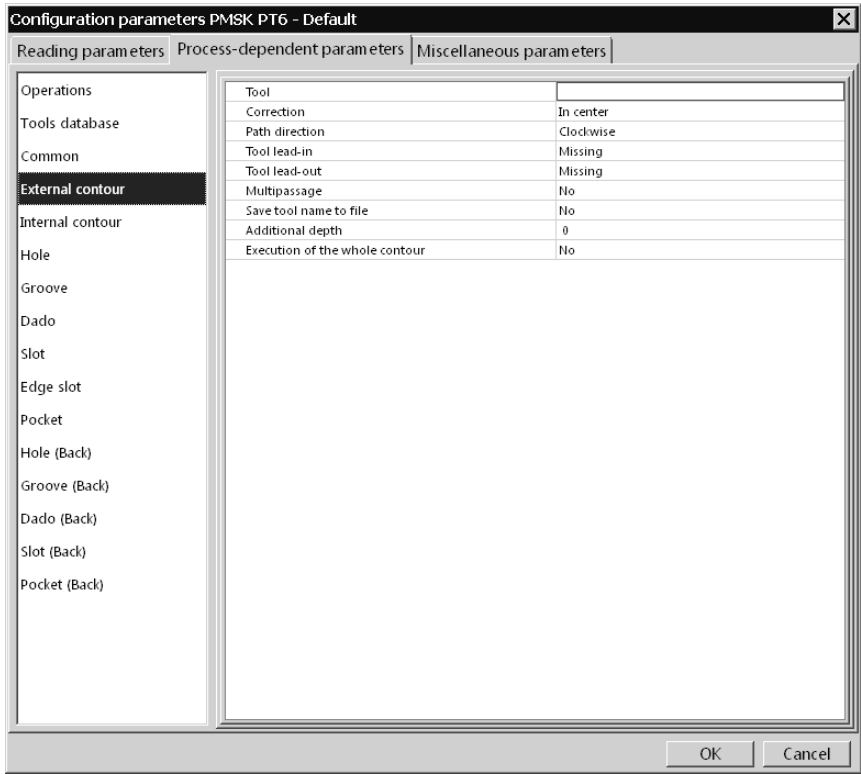


Fig. 10.567.

Section control elements allow to choose a tool, lead-in type, correction type, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.568), holes from back side – in **Hole (Back)** section in the same way.

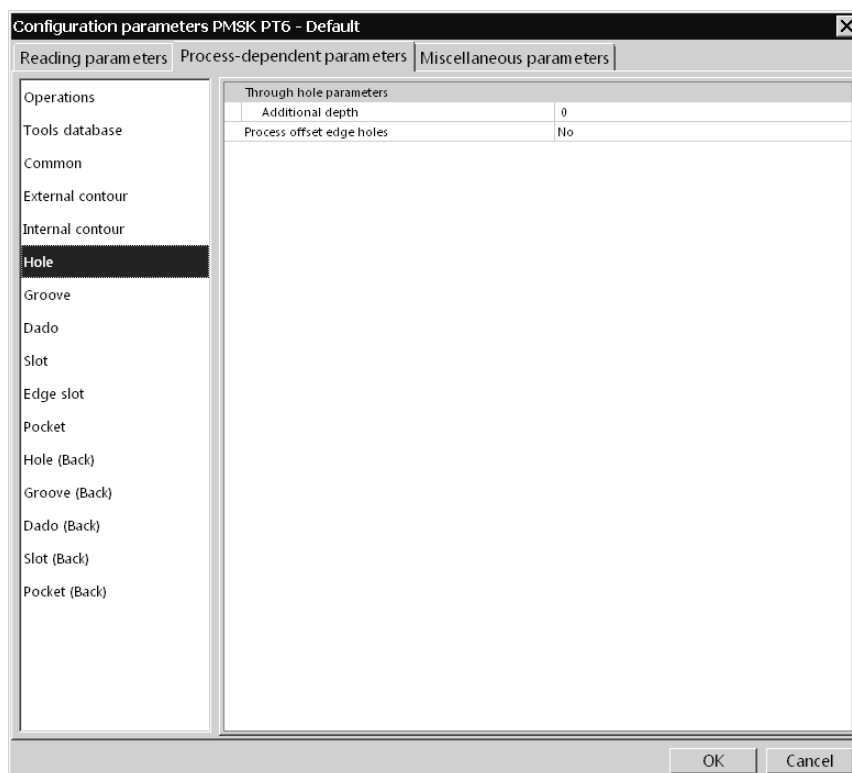


Fig. 10.568.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.569), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

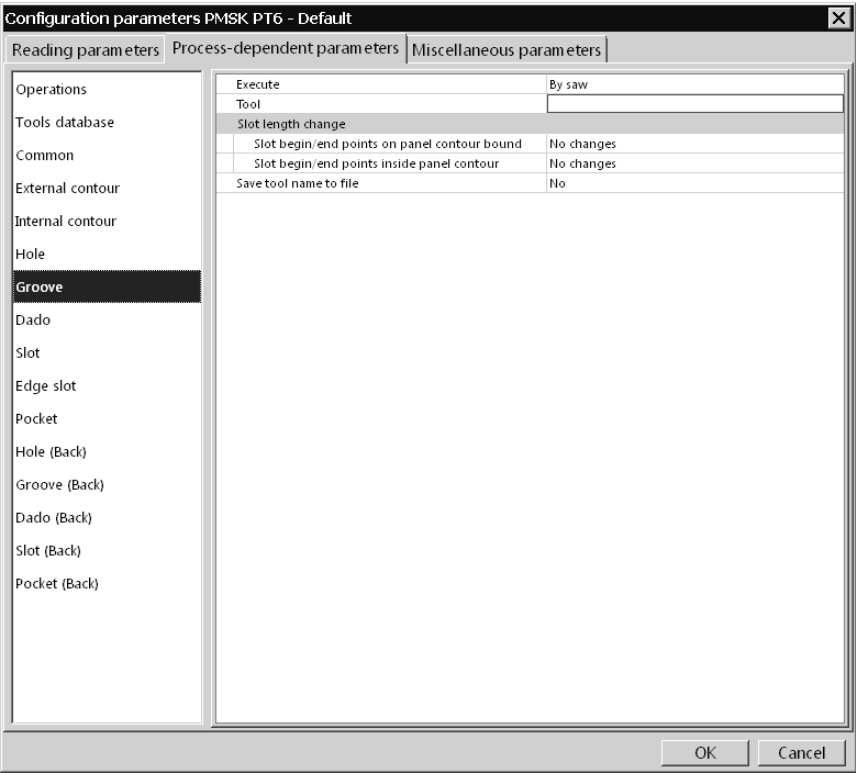


Fig. 10.569.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.570), from the back panel side – in **Slot (Back)** section.

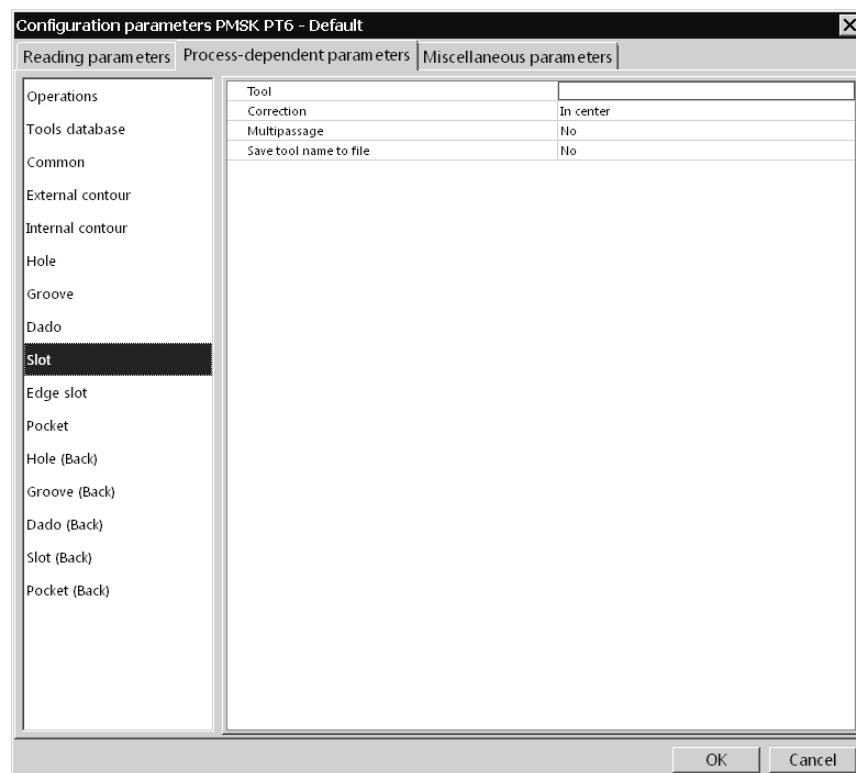


Fig. 10.570.

Section control elements allow to select a tool from database, correction type and multipassage use.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.571).

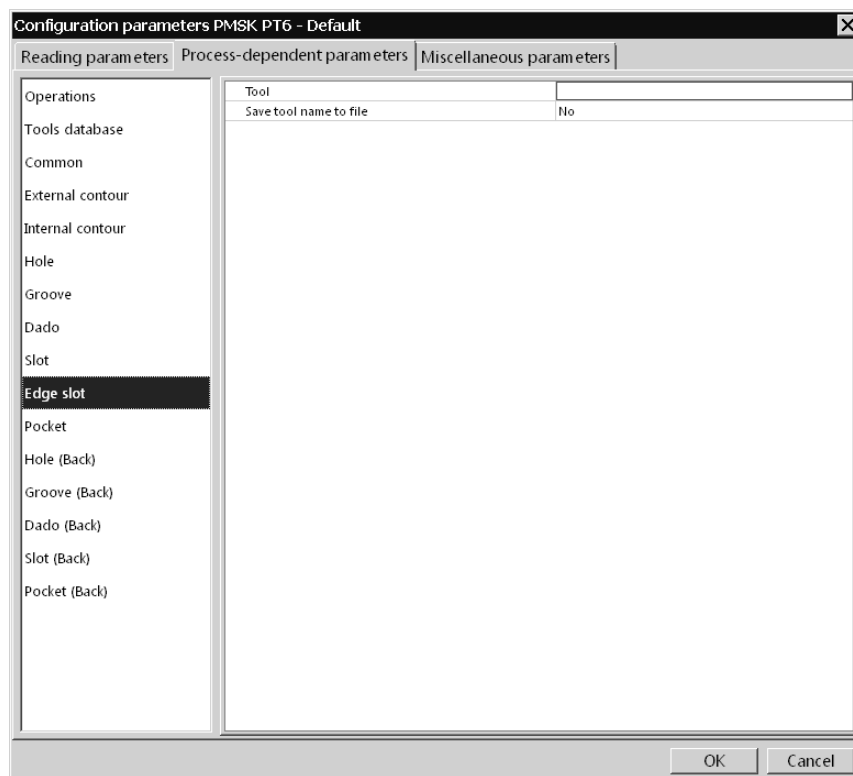


Fig. 10.571.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.572), on the back panel side – in **Pocket (Back)** section.

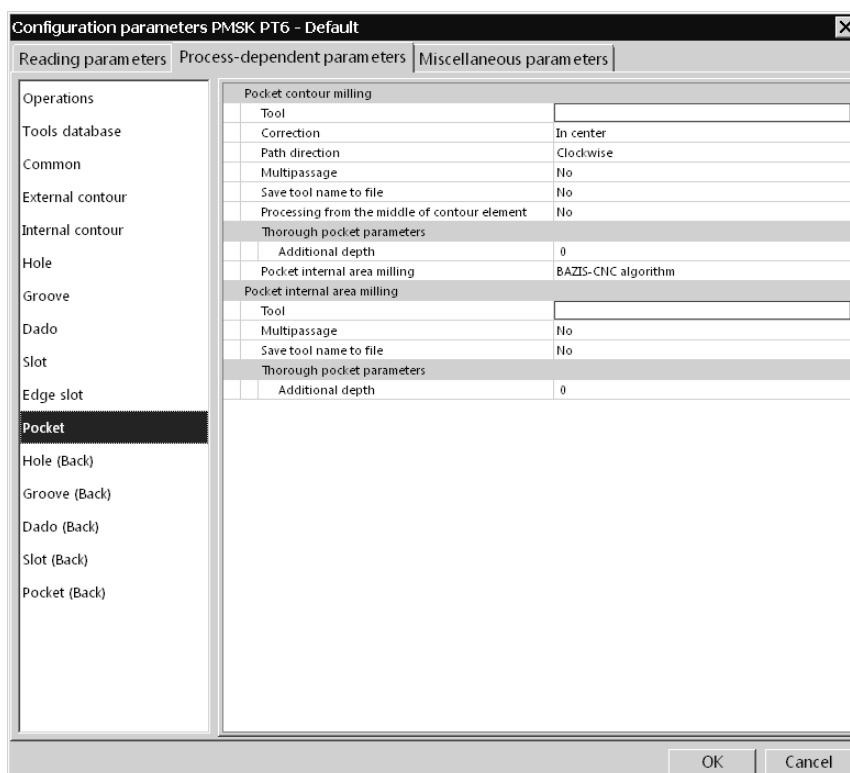


Fig. 10.572.

Tool combo box allows to select milling tool from database. For pocket contour select milling execution direction, correction type and multipassage use.

Setup completion

To complete control programs creation parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.31.3.PMSK Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.573) allow to specify operations that can be used in control program.

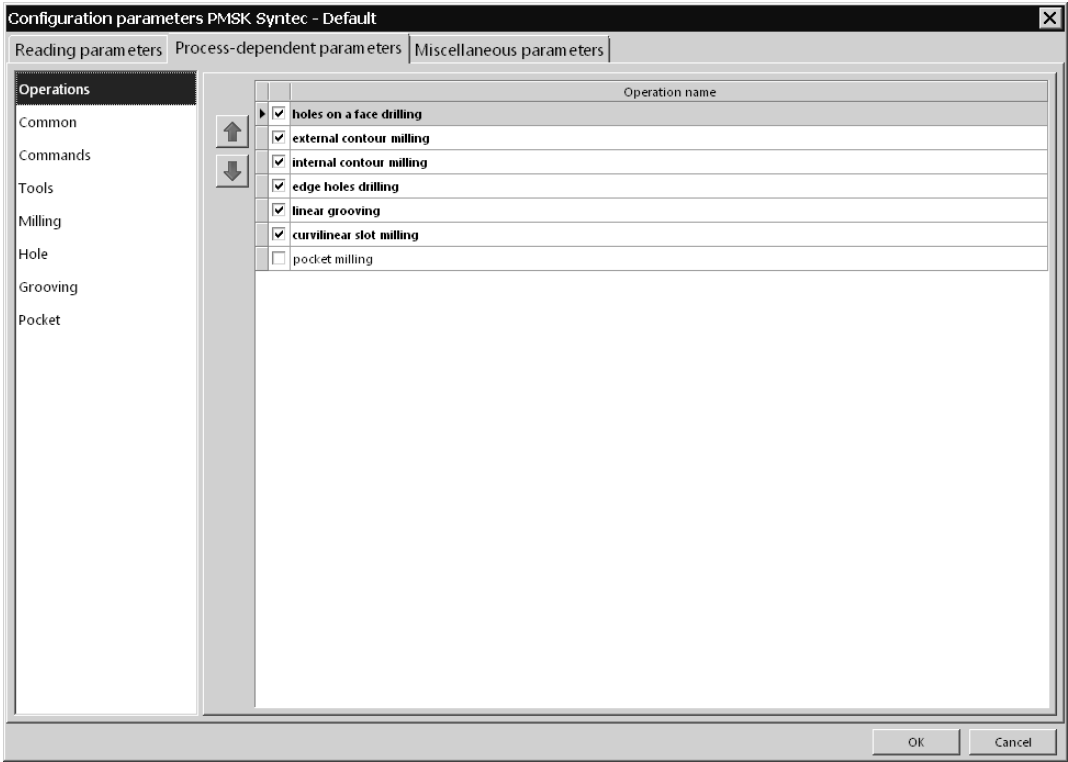


Fig. 10.573.

Common parameters

Common section control elements (fig. 10.574) allow to set common parameters of control programs creation.

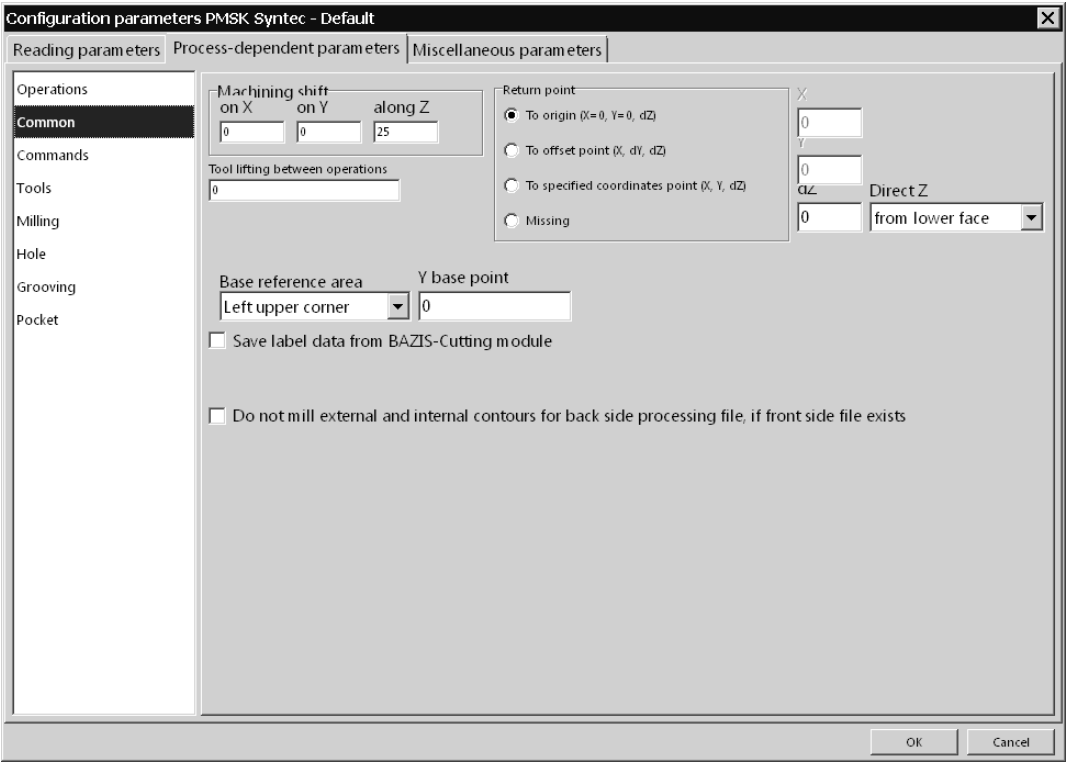


Fig. 10.574.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height. **Base reference area** combo box variants allow to choose a origin point, relative to which the processing will be generated. **Reference point Y** field allows to set a base point position.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.575) allow to control separate commands to control program inclusion.

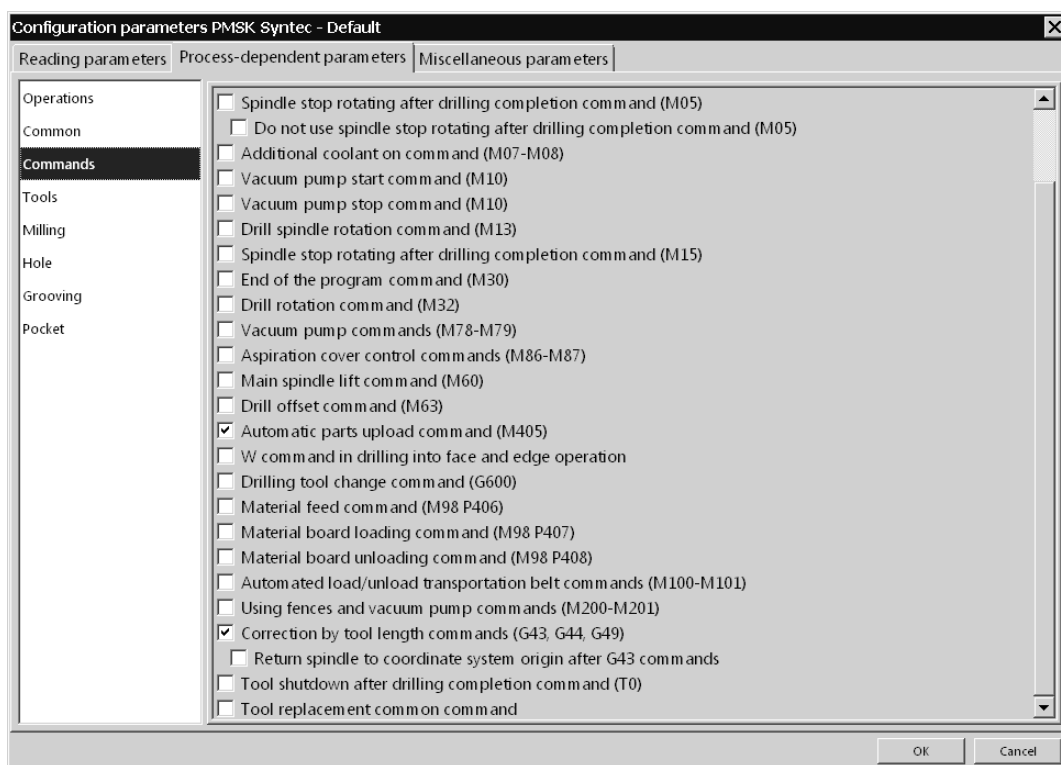


Fig. 10.575.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.576).

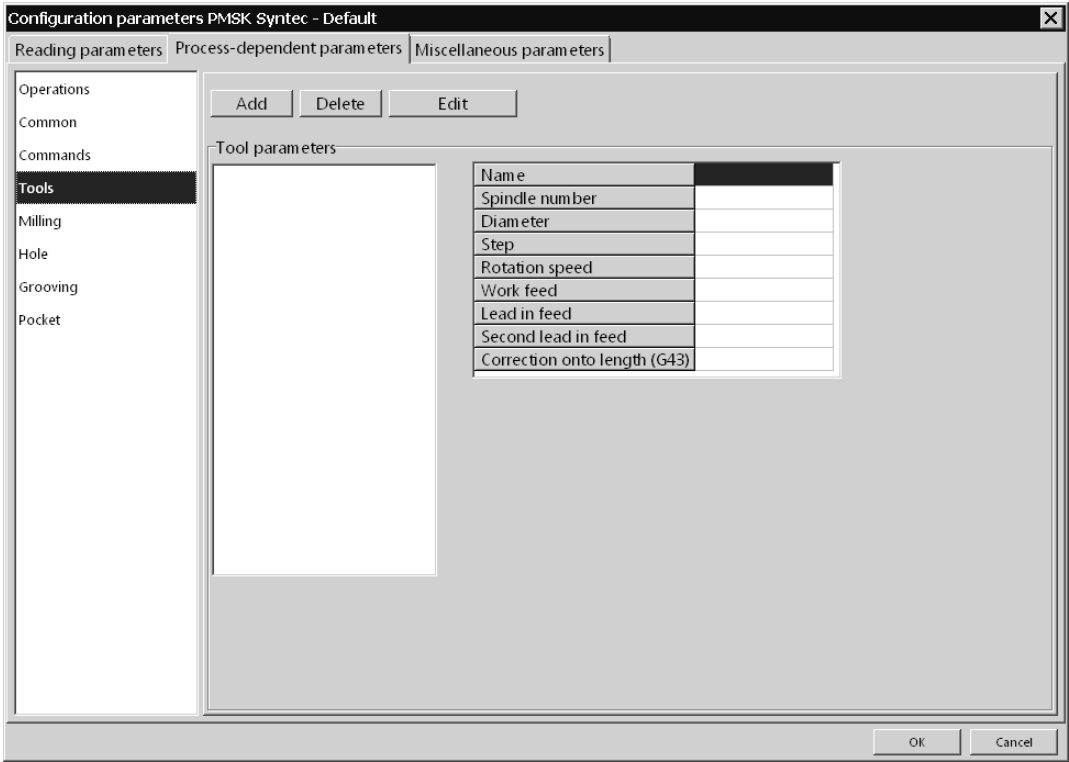


Fig. 10.576.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.577).

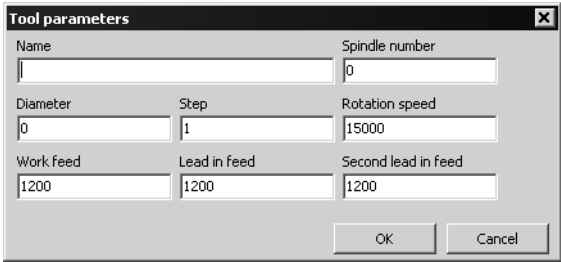


Fig. 10.577.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is shown in fig. 10.578).

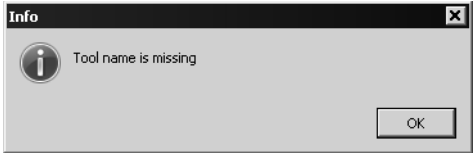


Fig. 10.578.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.577). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.579).

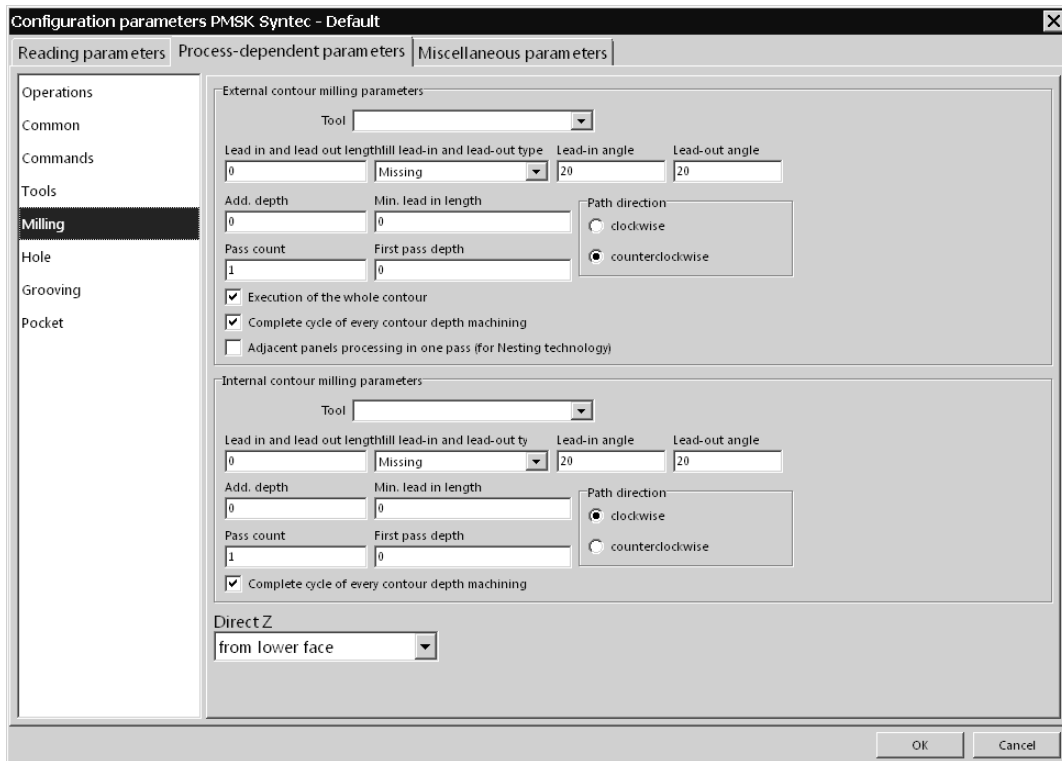


Fig. 10.579.

Tool combo box allows to select a milling tool name created in **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.580).

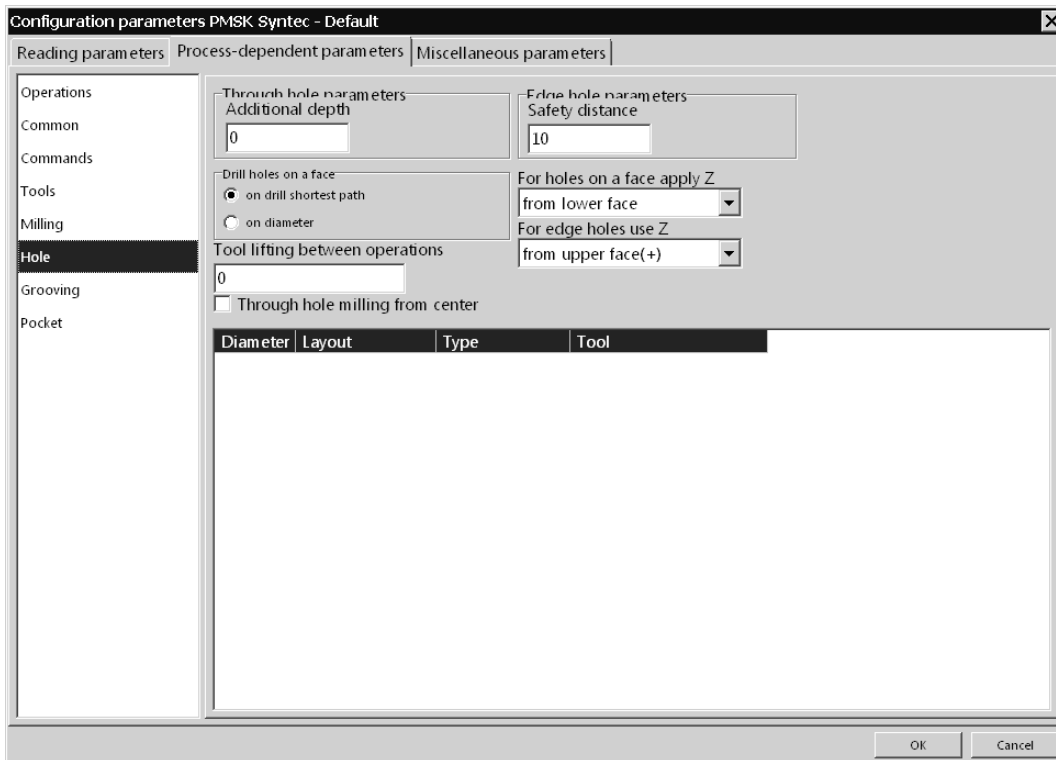


Fig. 10.580.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

Drill holes on a face group table allows to set automatic spindle number selection depending on hole parameters.

On drill shortest path and **on diameter** variants allow to set holes boring order.

Groove parameters

Grooving section control elements allow to set groove processing parameters (fig. 10.581).

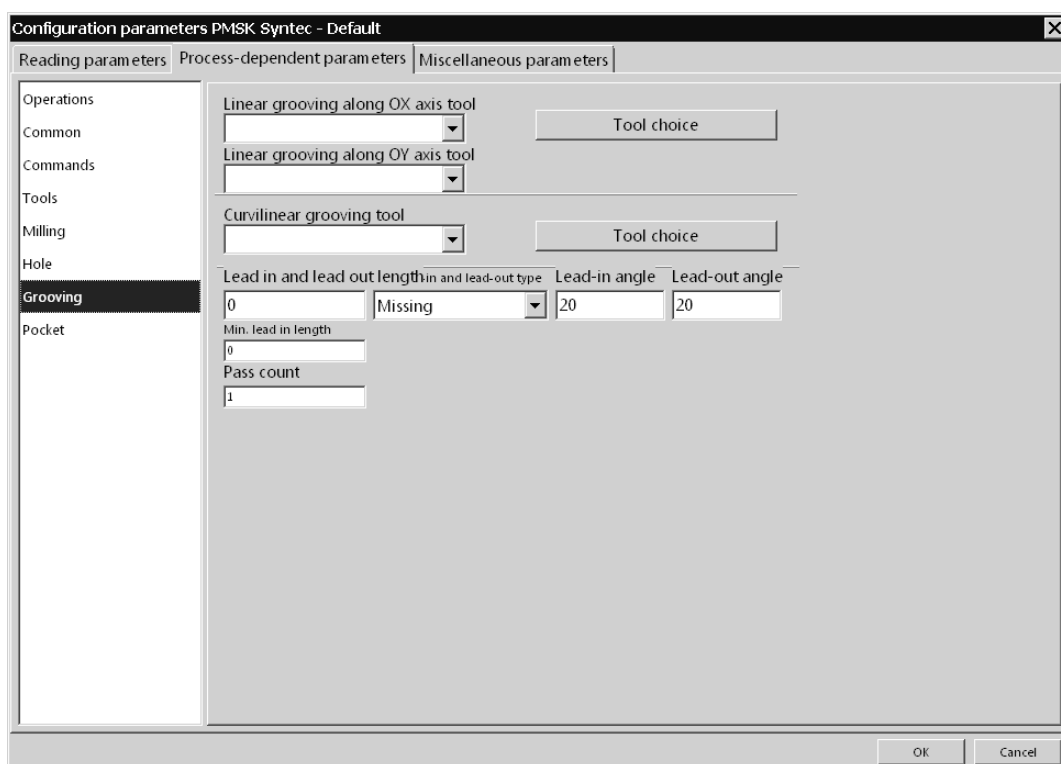


Fig. 10.581.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.582).

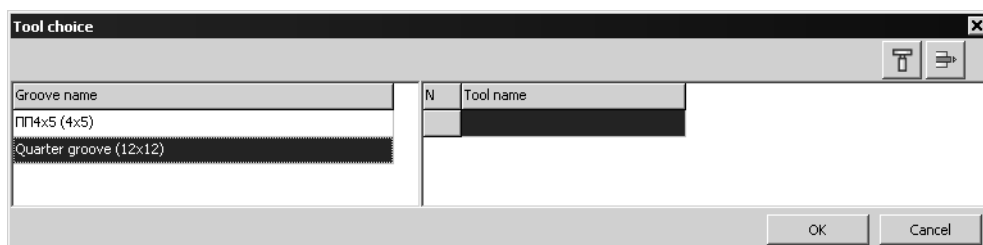


Fig. 10.582.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in **Tools** section.



Delete button allows to delete a tool from the list.

To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.583) allow to set pocket processing parameters.

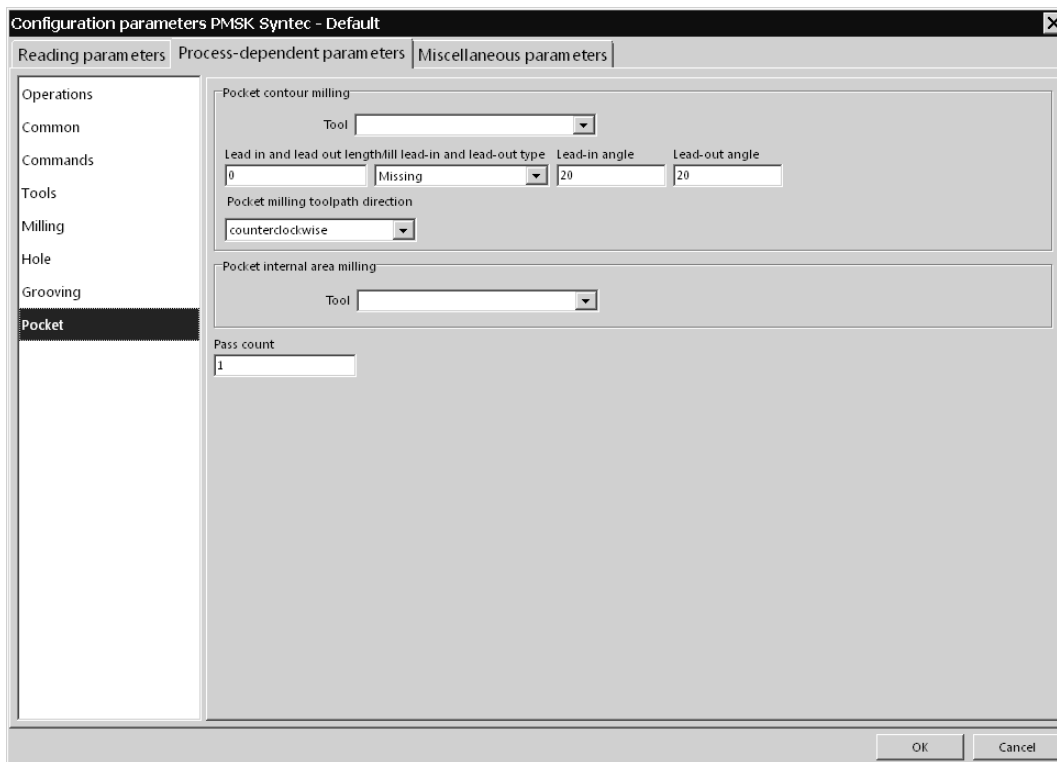


Fig. 10.583.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close.

10.32. ROCTECH

10.32.1.Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.584) allow to specify operations that can be used in control program.

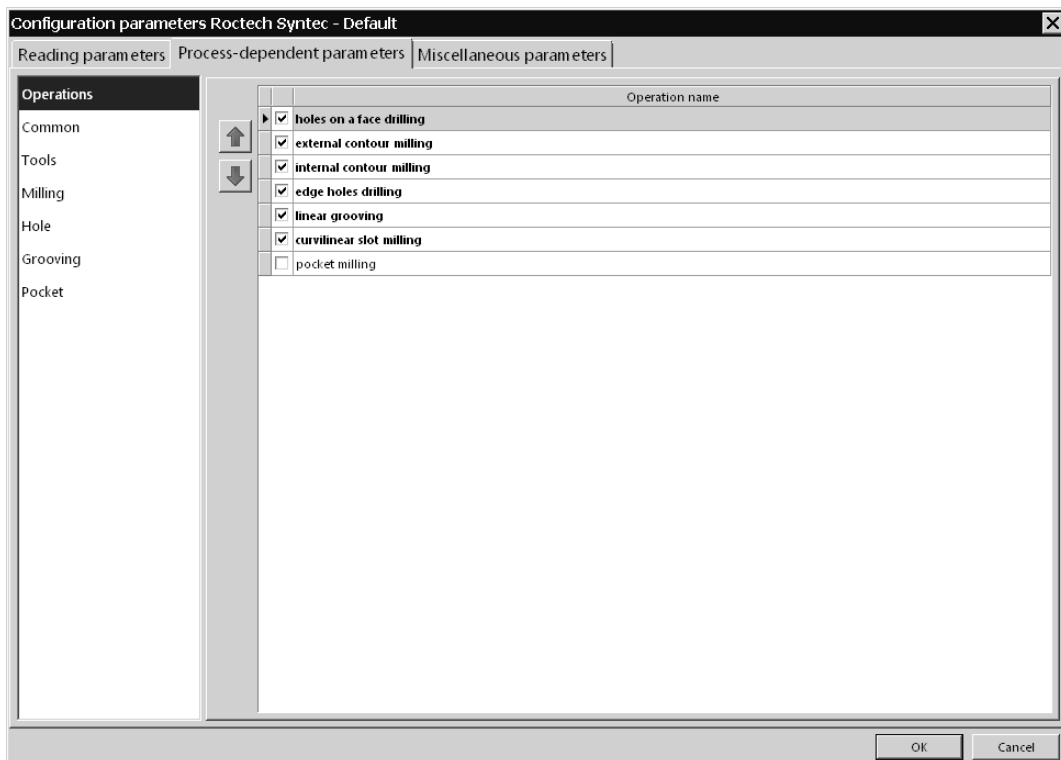


Fig. 10.584.

Common parameters

Common section control elements allow to setup common parameters of control programs creation (fig. 10.585).

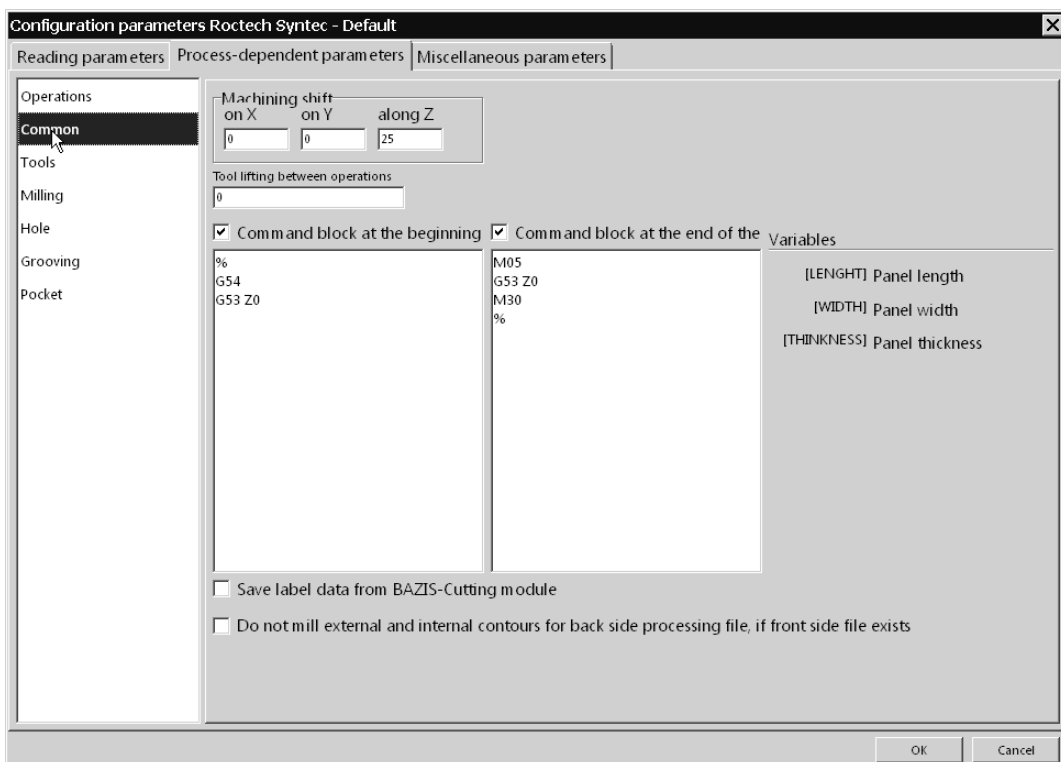


Fig. 10.585.

Machining shift group fields allow to set work area initial point moving. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas.

Save label data option allows to save the corresponding data. To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.586).

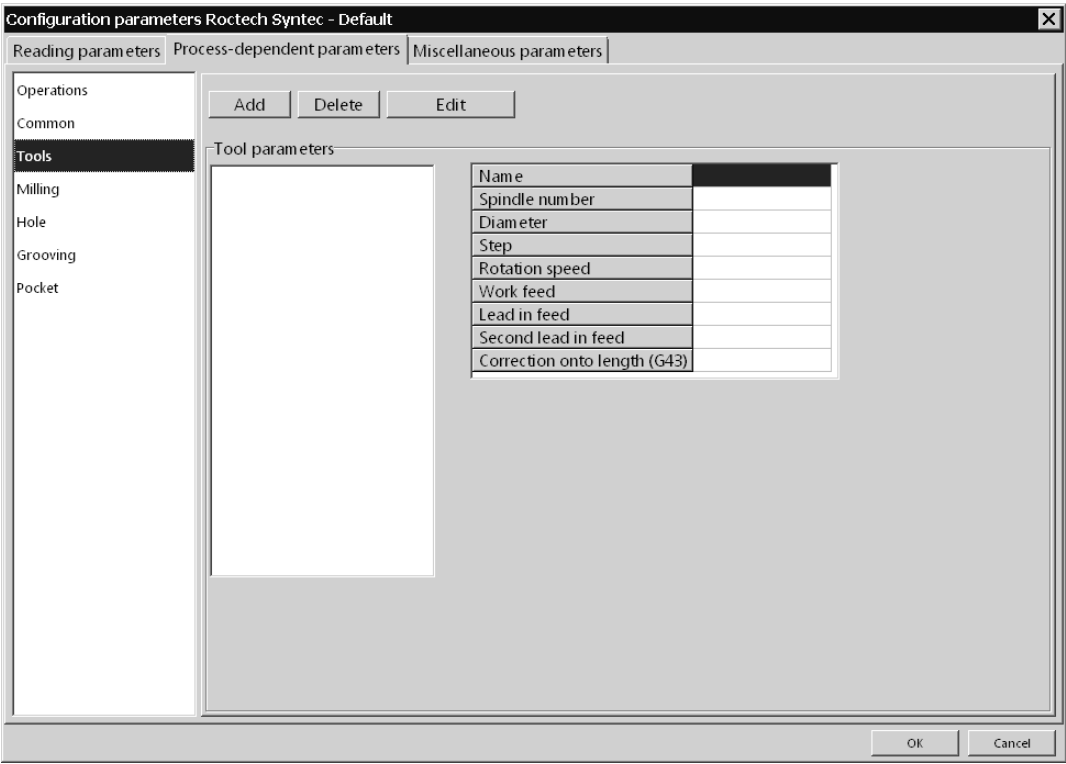


Fig. 10.586.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.587).

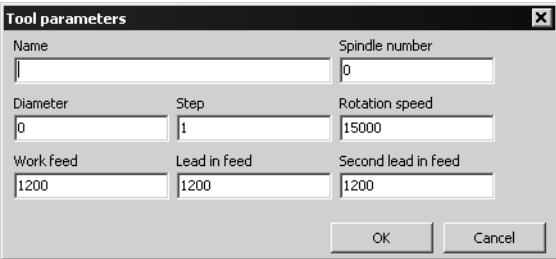


Fig. 10.587.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is shown in fig. 10.588.

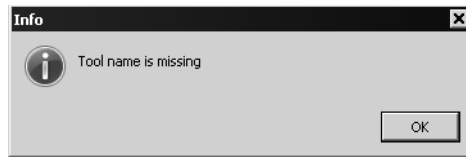


Fig. 10.588.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.587). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.589).

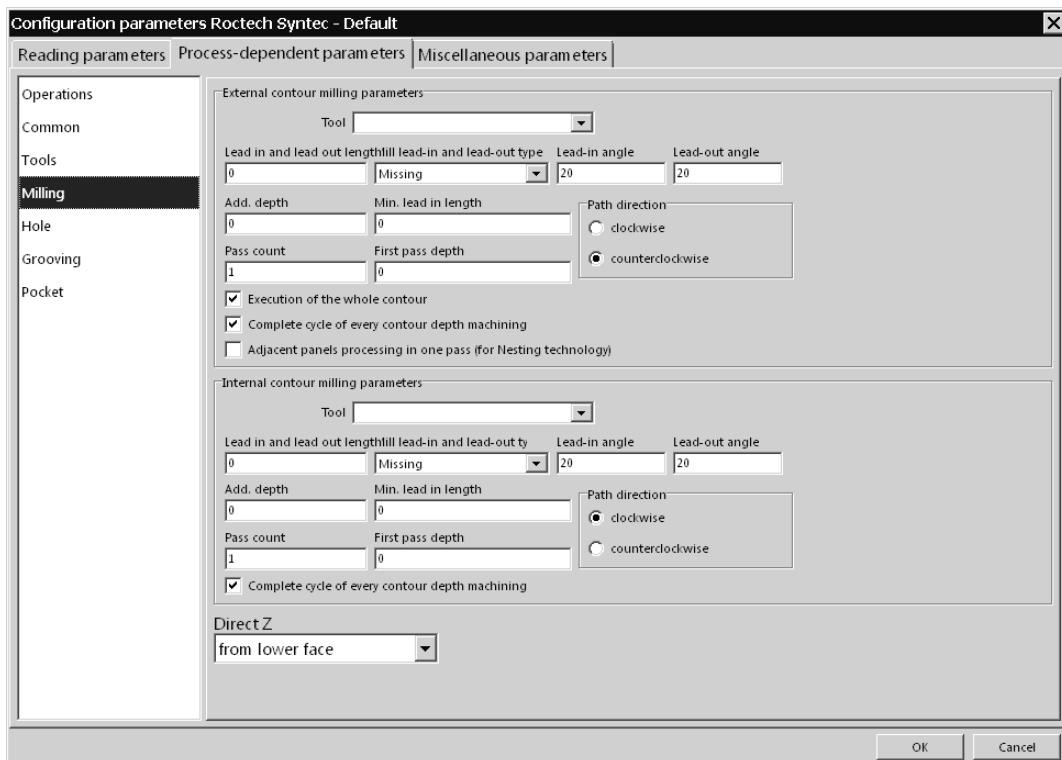


Fig. 10.589.

Tool combo box allows to select a milling tool name created in **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.590).

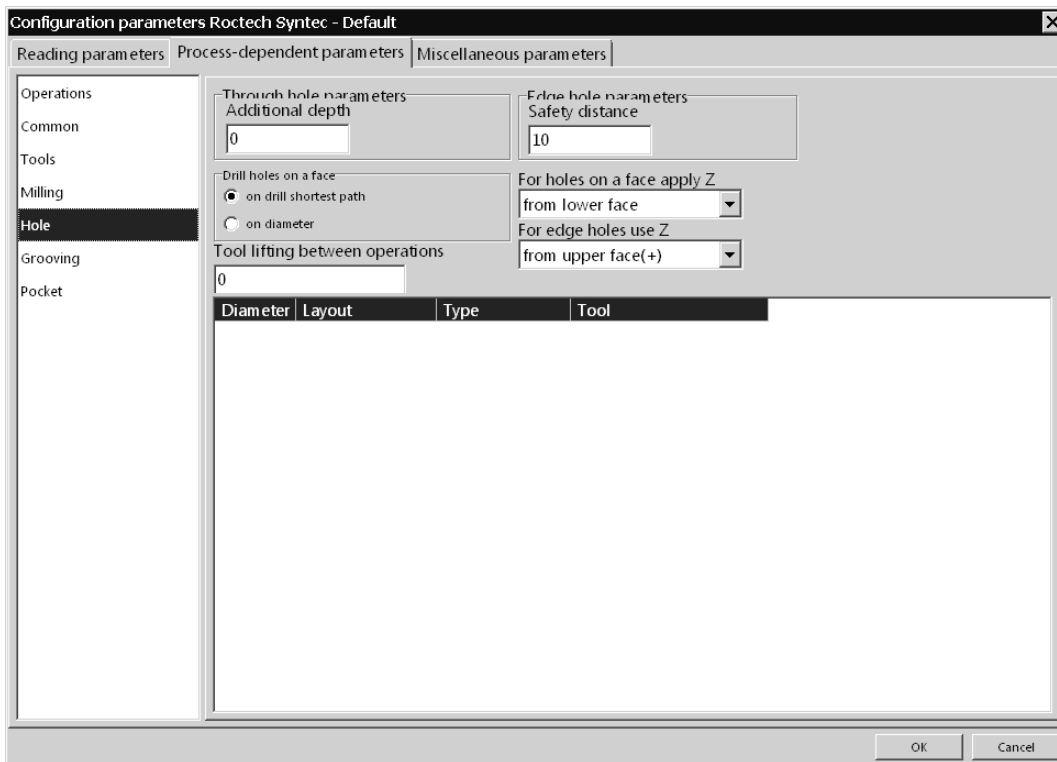


Fig. 10.590.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Groove parameters

Grooving section control elements allow to set groove processing parameters (fig. 10.591).

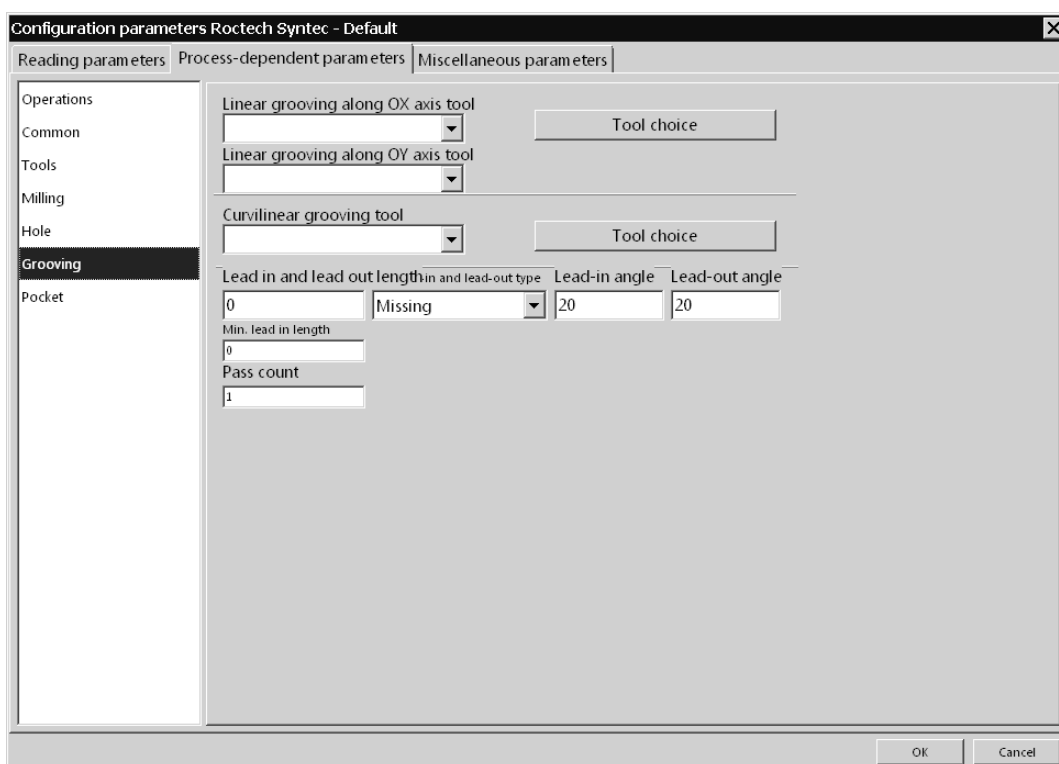


Fig. 10.591.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.592).

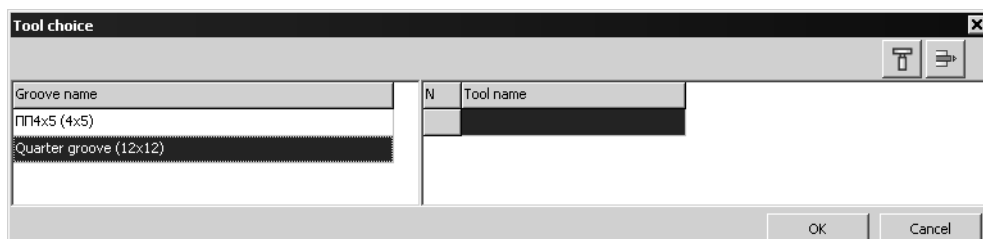


Fig. 10.592.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in **Tools** section.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.593) allow to set pocket processing parameters.

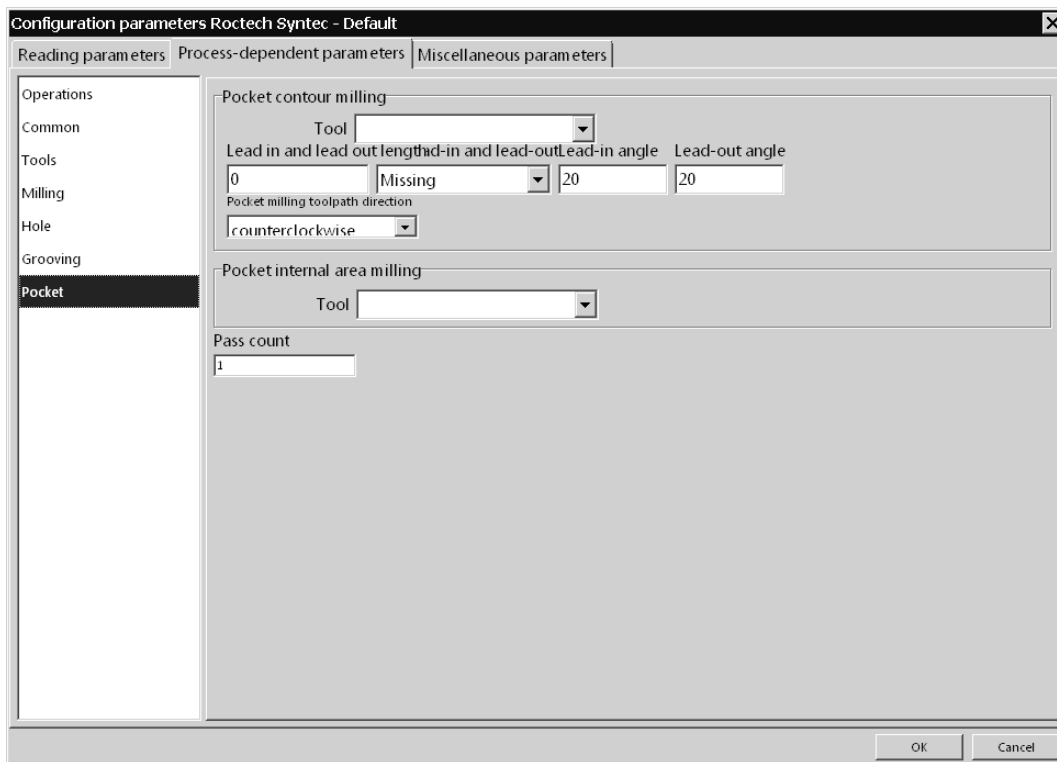


Fig. 10.593.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.33. TechnoCNC

10.33.1. ATV Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.594) allow to specify operations that can be used in control program.

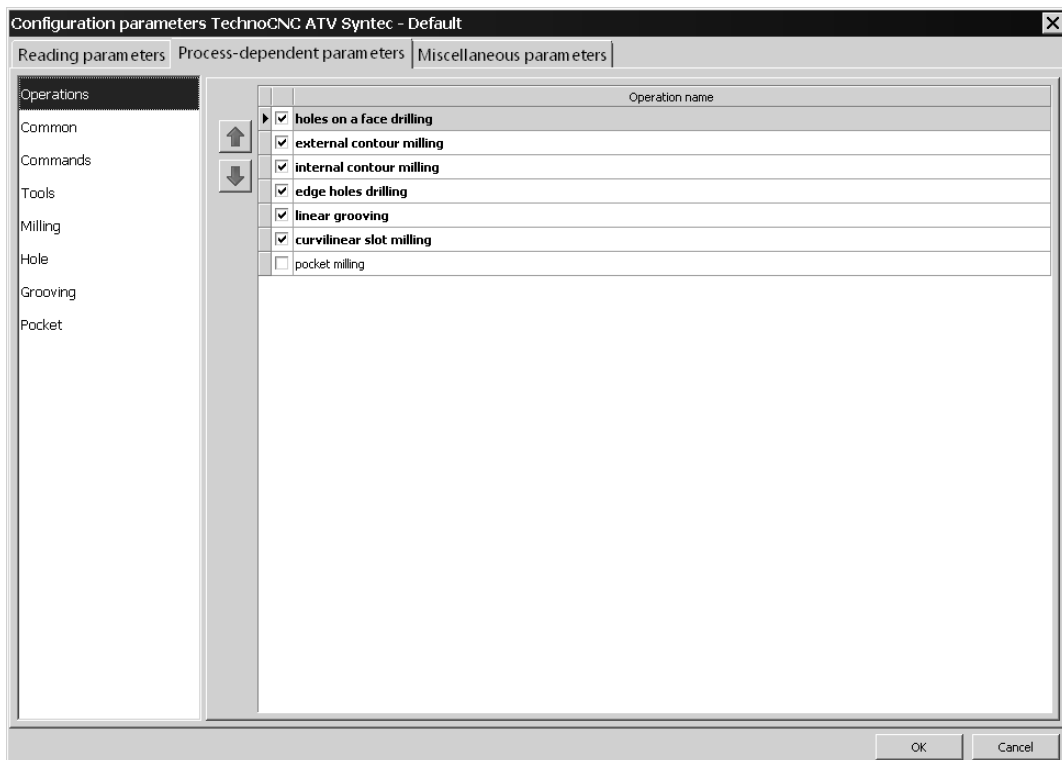


Fig. 10.594.

Common parameters

Common section control elements allow to setup common parameters of control programs creation (fig. 10.595).

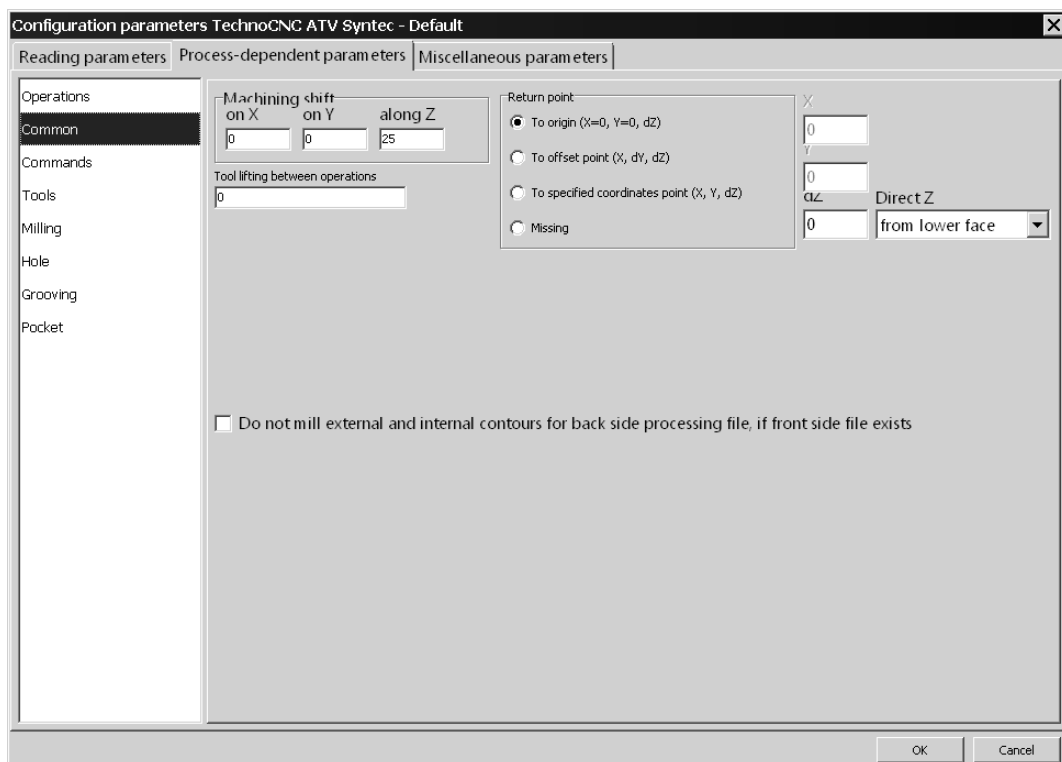


Fig. 10.595.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.596) allow to control separate commands to control program inclusion.

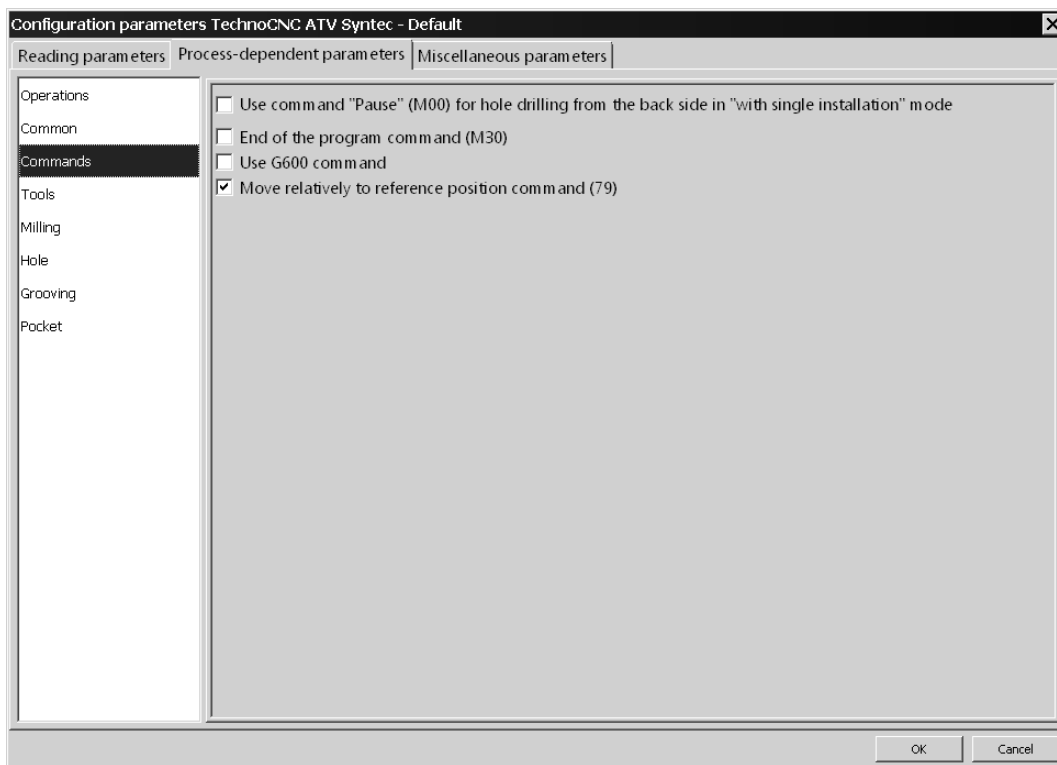


Fig. 10.596.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.597).

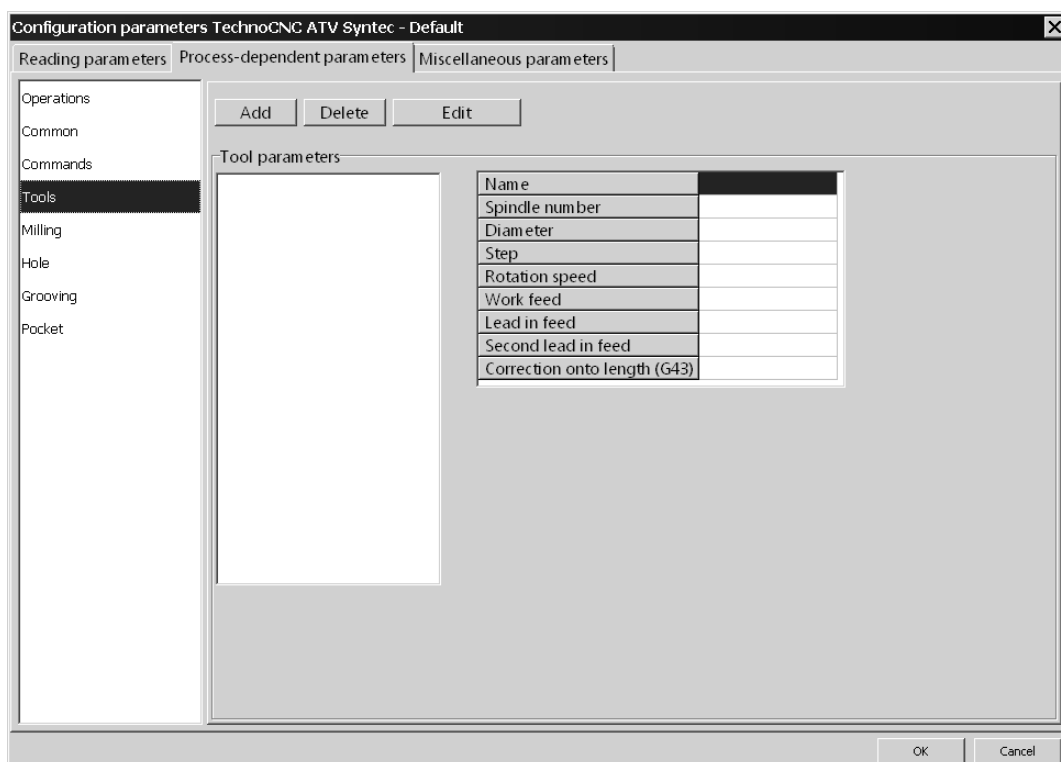


Fig. 10.597.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.598).

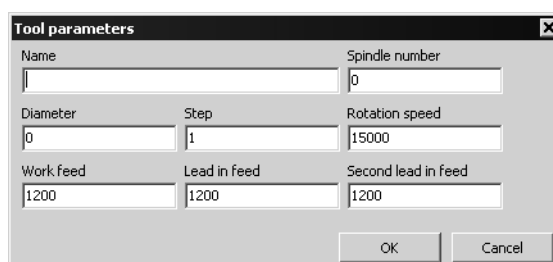


Fig. 10.598.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is shown in fig. 10.599.

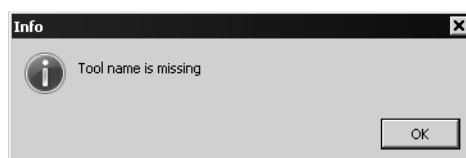


Fig. 10.599.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.598). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.600).

Configuration parameters TechnoCNC ATV Syntec - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Operations
Common
Commands
Tools
Milling
Hole
Grooving
Pocket

External contour milling parameters

Tool: []

Lead in and lead out length: [0] Mill lead-in and lead-out type: [Missing] Lead-in angle: [20] Lead-out angle: [20]

Add. depth: [0] Min. lead in length: [0]

Pass count: [1] First pass depth: [0]

Path direction:
☐ clockwise
☒ counterclockwise

☒ Execution of the whole contour
☒ Complete cycle of every contour depth machining
☐ Adjacent panels processing in one pass (for Nesting technology)

Internal contour milling parameters

Tool: []

Lead in and lead out length: [0] Mill lead-in and lead-out type: [Missing] Lead-in angle: [20] Lead-out angle: [20]

Add. depth: [0] Min. lead in length: [0]

Pass count: [1] First pass depth: [0]

Path direction:
☒ clockwise
☐ counterclockwise

☒ Complete cycle of every contour depth machining

Direct Z
 from lower face []

OK Cancel

Fig. 10.600.

Tool combo box allows to select a milling tool name created in **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.601).

Configuration parameters TechnoCNC ATV Syntec - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Operations
Common
Commands
Tools
Milling
Hole
Grooving
Pocket

Through hole parameters
Additional depth: 0

Edge hole parameters
Safety distance: 10

Drill holes on a face
☒ on drill shortest path
☐ on diameter

For holes on a face apply Z: from lower face

For edge holes use Z: from upper face(+)

Tool lifting between operations: 0

Diameter	Layout	Type	Tool
5	Edge on the mitre	Blind	
6	Upper face	Through	
5	Front edge	Blind	
6	Front edge	Blind	

OK Cancel

Fig. 10.601.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Slot parameters

Grooving section control elements (fig. 10.602) allow to set slots processing parameters.

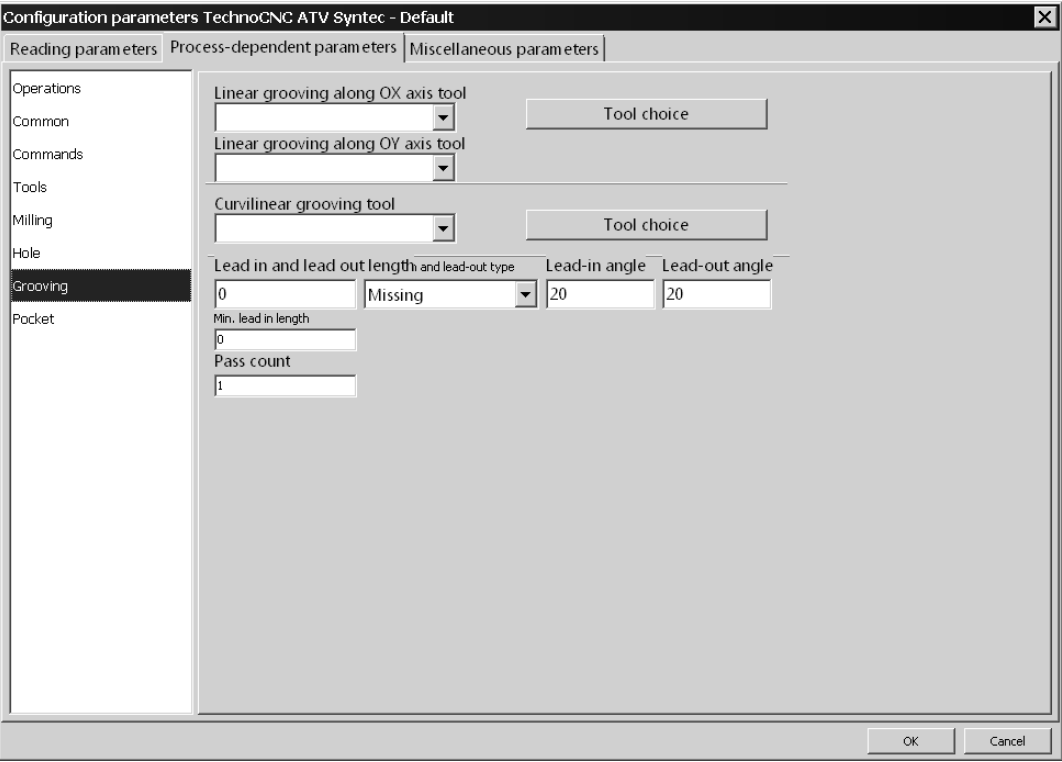


Fig. 10.602.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.603).

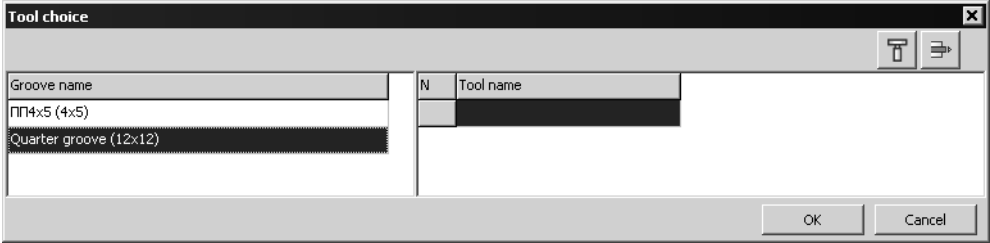


Fig. 10.603.



Slot name table contains all grooves names of the model. You can set particular tools for each slot processing. **Add mill** button allows to assign a tool from a list created in **Tools** section to the slot.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.604) allow to set pocket processing parameters.

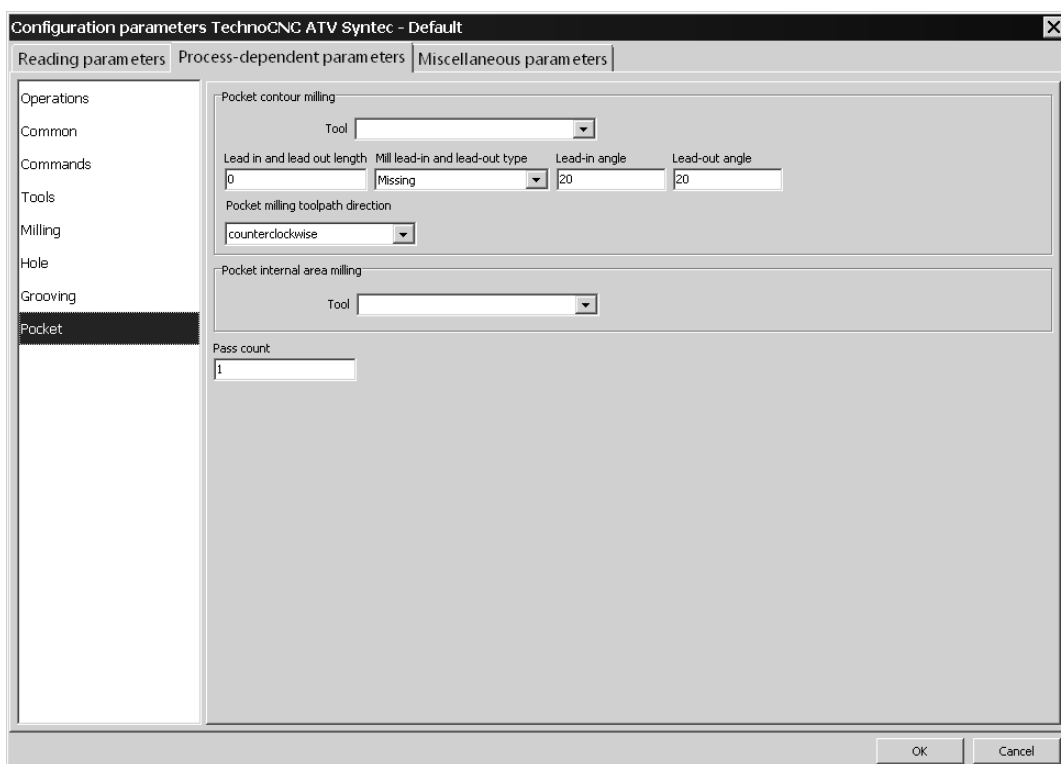


Fig. 10.604.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.33.2.Suppressor Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.605) allow to specify operations that can be used in control program.

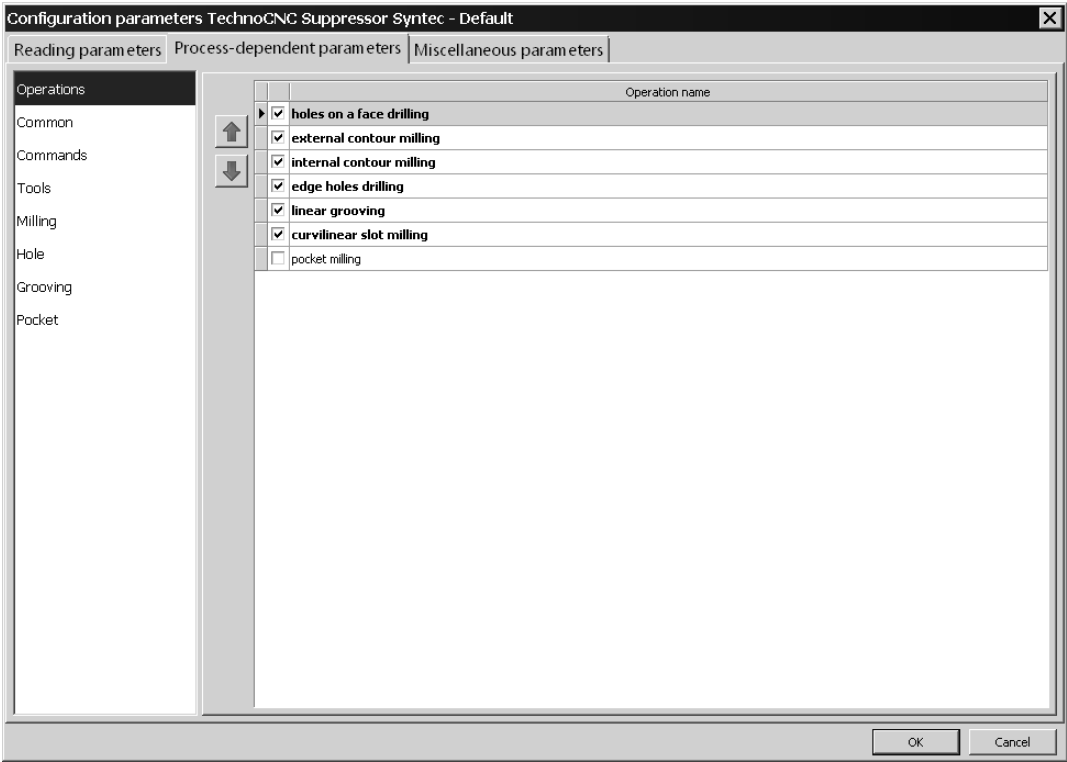


Fig. 10.605.

Common parameters

Common section control elements (fig. 10.606) allow to set common parameters of control programs creation.

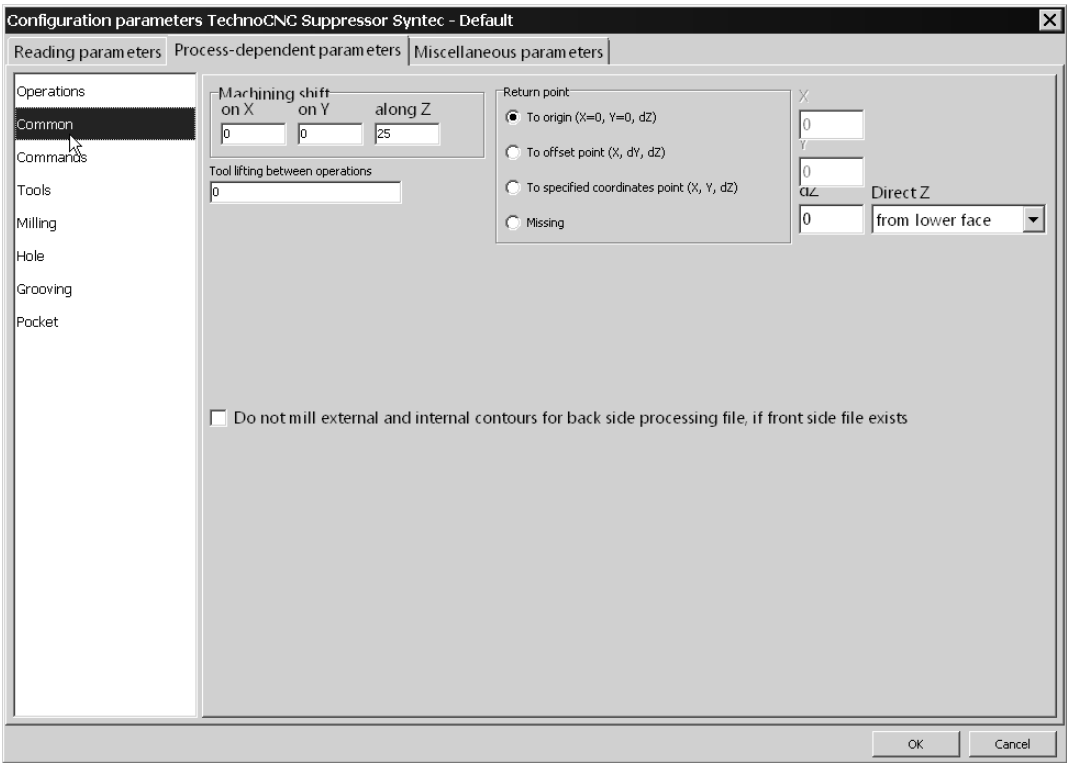


Fig. 10.606.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height. **Base reference area** combo box options allow you to select the signs (+ or -) of the processing coordinates.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.607) allow to control separate commands to control program inclusion.

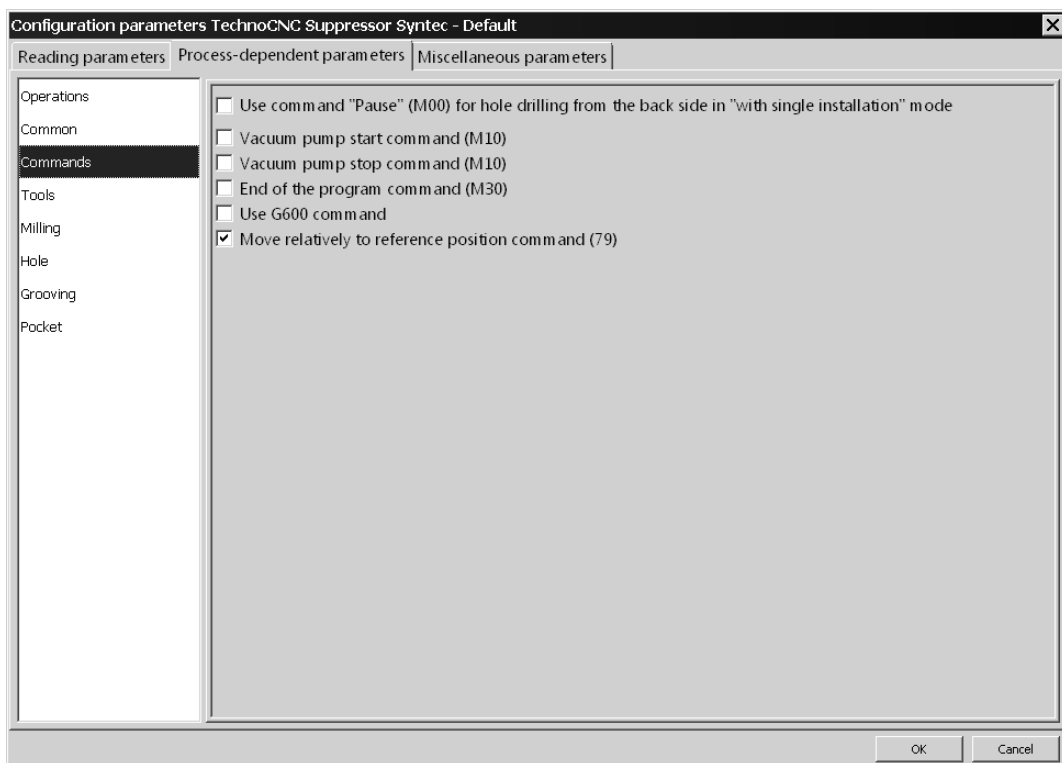


Fig. 10.607.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.608).

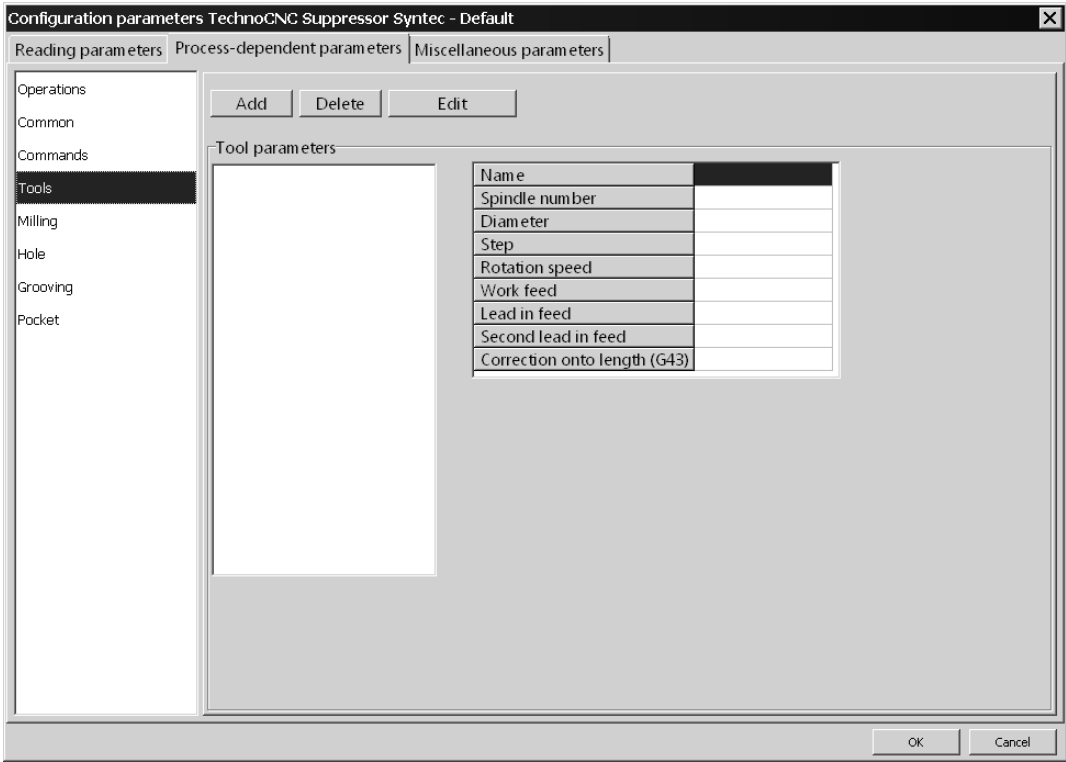


Fig. 10.608.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.609).

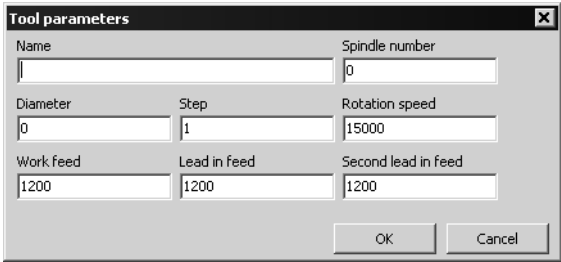


Fig. 10.609.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is (fig. 10.610).

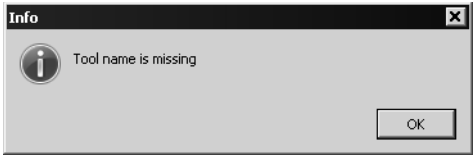


Fig. 10.610.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.609). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.611).

Configuration parameters TechnoCNC Suppressor Syntec - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Operations
Common
Commands
Tools
Milling
Hole
Grooving
Pocket

External contour milling parameters

Tool: []

Lead in and lead out length: [0] Mill lead-in and lead-out type: [Missing] Lead-in angle: [20] Lead-out angle: [20]

Add. depth: [0] Min. lead in length: [0]

Pass count: [1] First pass depth: [0]

Path direction:
☐ clockwise
☒ counterclockwise

☒ Execution of the whole contour
☒ Complete cycle of every contour depth machining
☐ Adjacent panels processing in one pass (for Nesting technology)

Internal contour milling parameters

Tool: []

Lead in and lead out length: [0] Mill lead-in and lead-out type: [Missing] Lead-in angle: [20] Lead-out angle: [20]

Add. depth: [0] Min. lead in length: [0]

Pass count: [1] First pass depth: [0]

Path direction:
☒ clockwise
☐ counterclockwise

☒ Complete cycle of every contour depth machining

Direct Z:
 from lower face []

OK Cancel

Fig. 10.611.

Tool combo box allows to select a milling tool name created in **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.612).

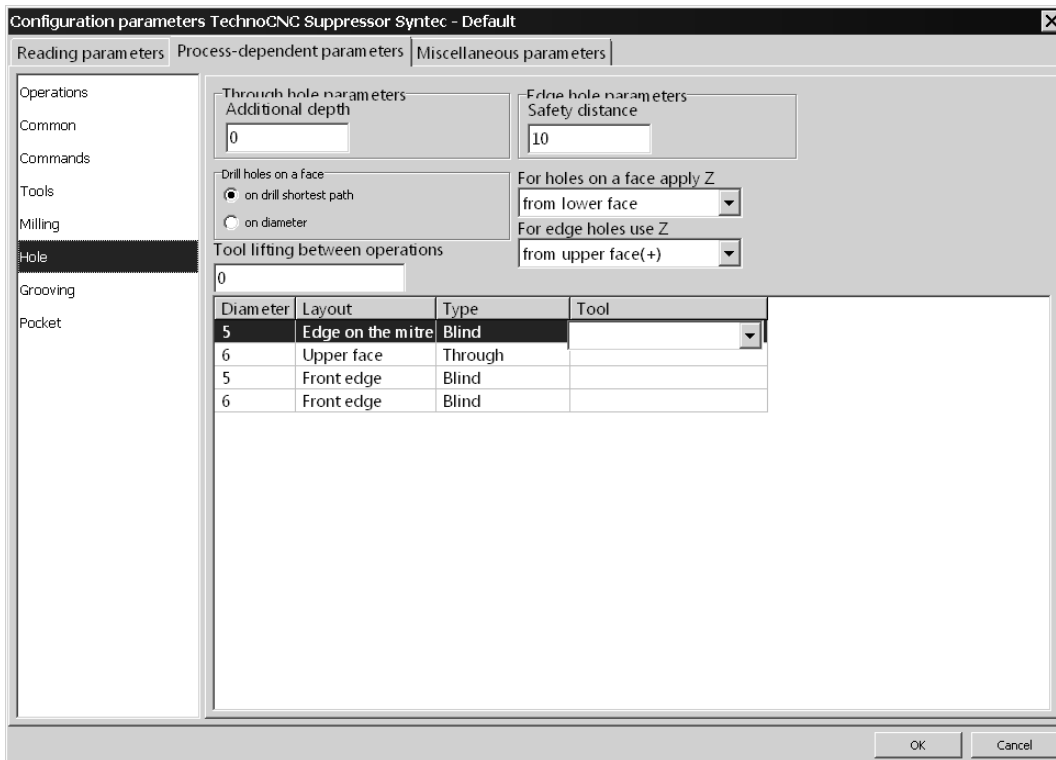


Fig. 10.612.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

Drill holes on a face group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slot parameters

Grooving section control elements (fig. 10.613) allow to set slots processing parameters.

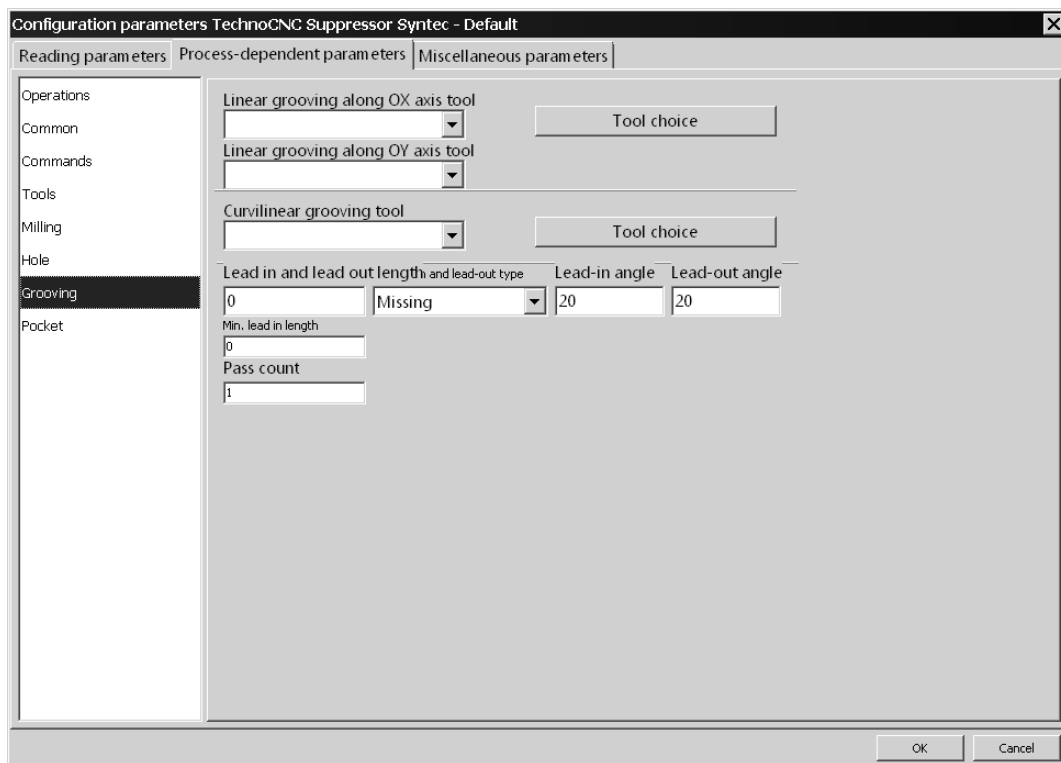


Fig. 10.613.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.614).

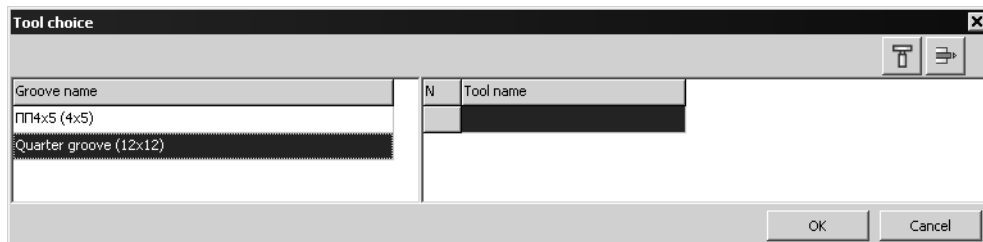


Fig. 10.614.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in **Tools** section.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.615) allow to set pocket processing parameters.

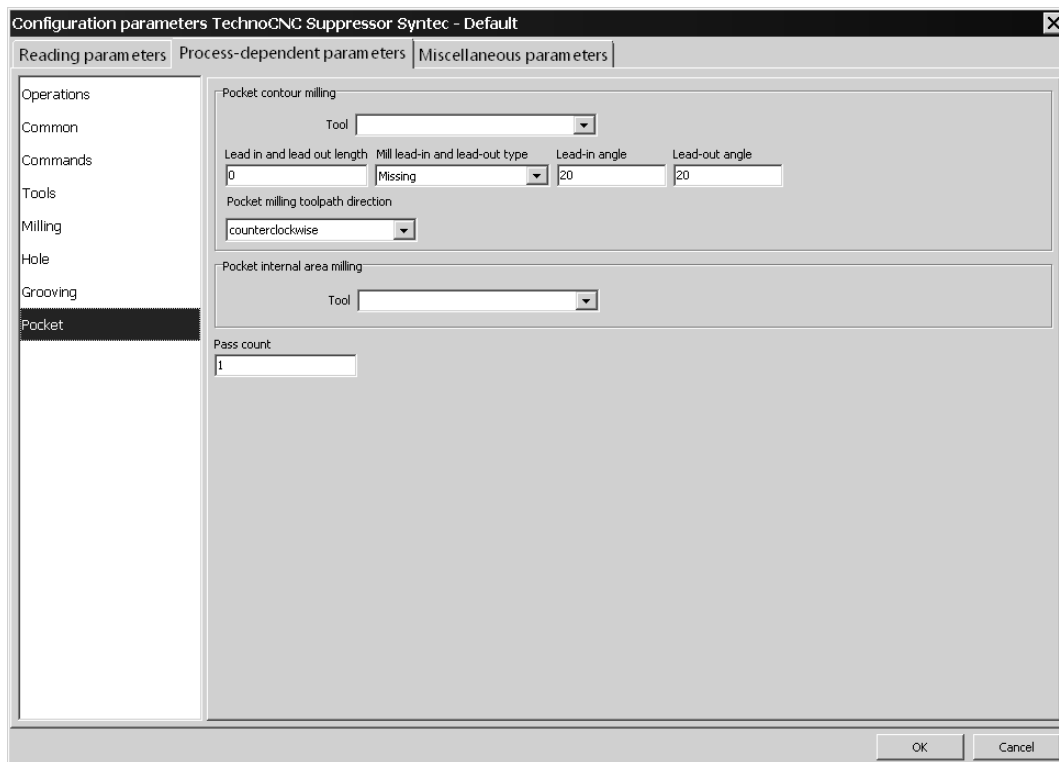


Fig. 10.615.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.33.3.TechnoBit



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.616) allow to specify operations that can be used in control program.

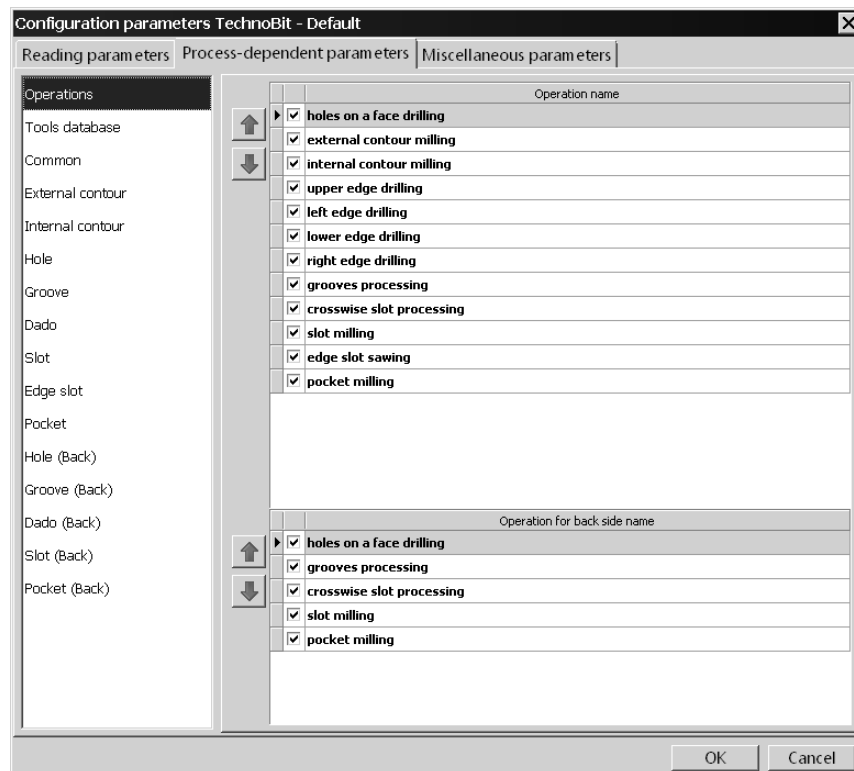


Fig. 10.616.

Usable tools

Tools database section control elements (fig. 10.617) allow to create a list of tools available for use in control programs.

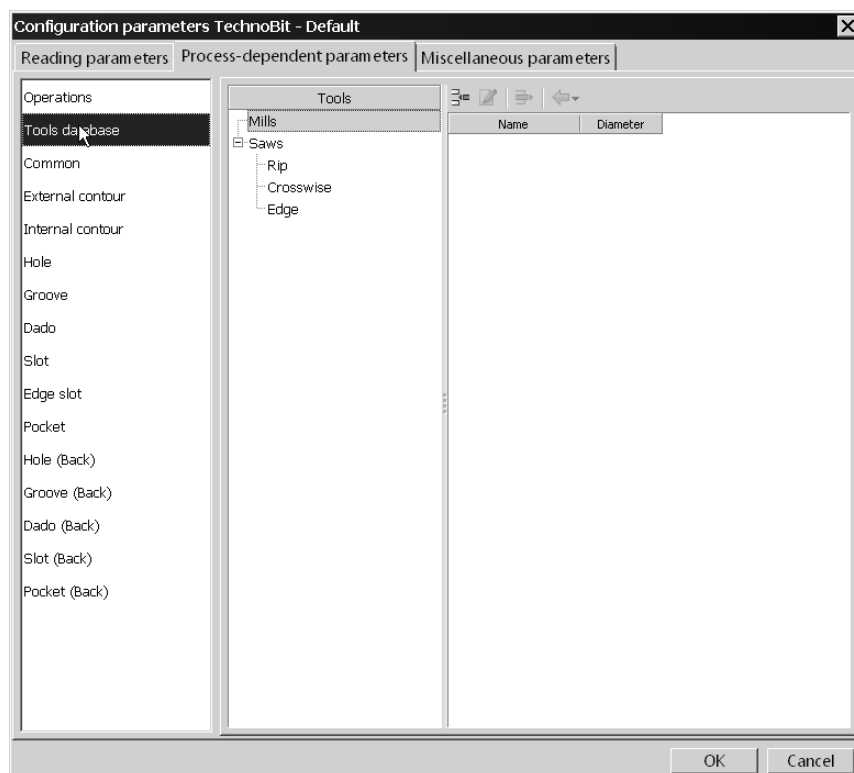


Fig. 10.617.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.618).

Fig. 10.618.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common section control elements (fig. 10.619) allow to set common parameters of control programs.

Fig. 10.619.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.620).

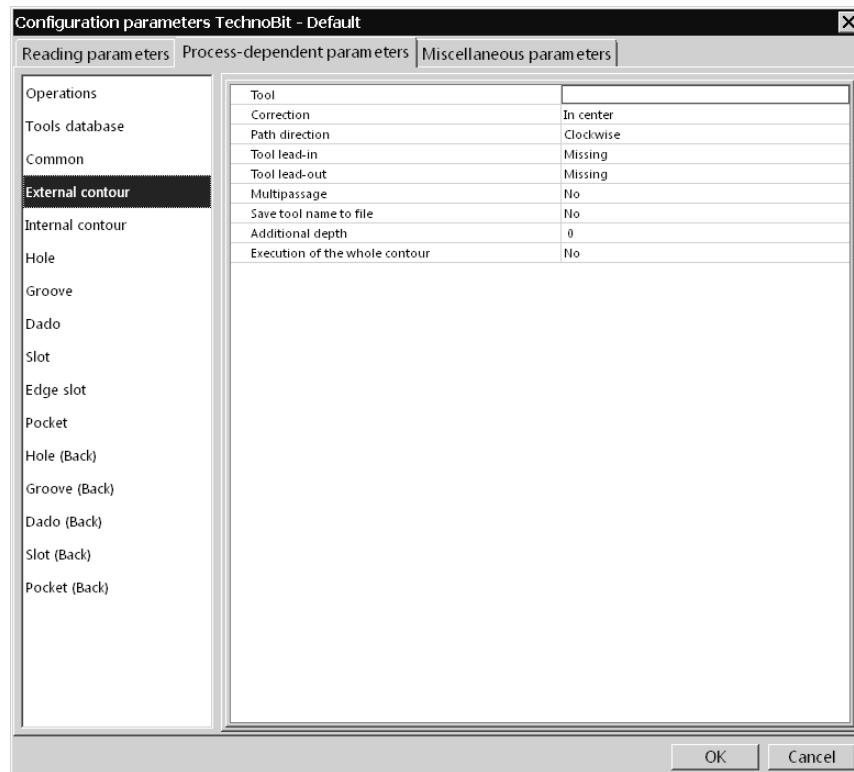


Fig. 10.620.

Control elements in sections allow to select the tool, correction types, lead-in, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.621), holes from back side – in **Hole (Back)** section in the same way.

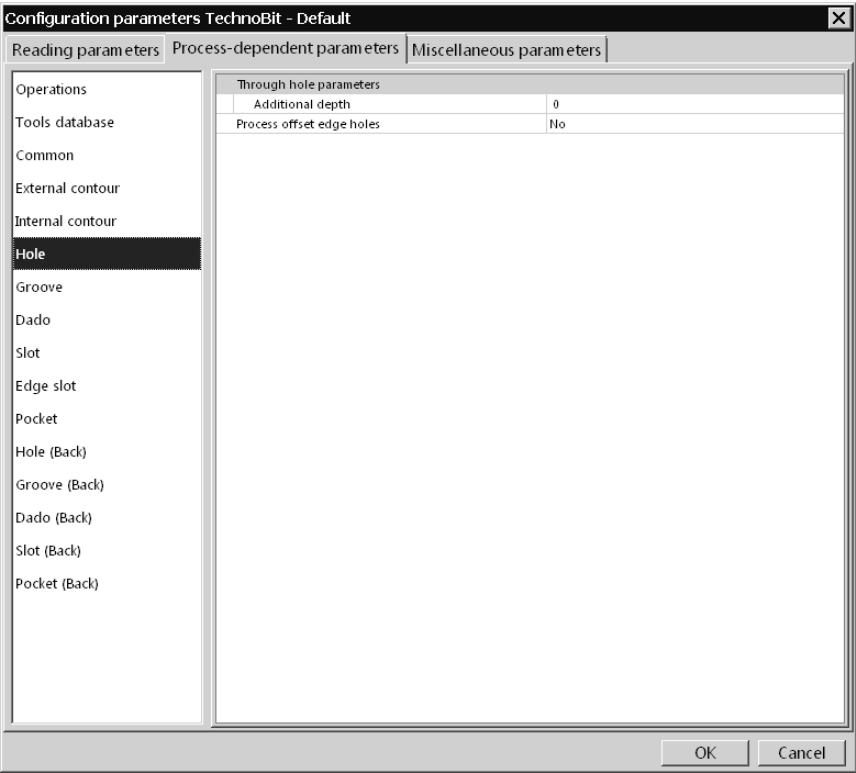


Fig. 10.621.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.622), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

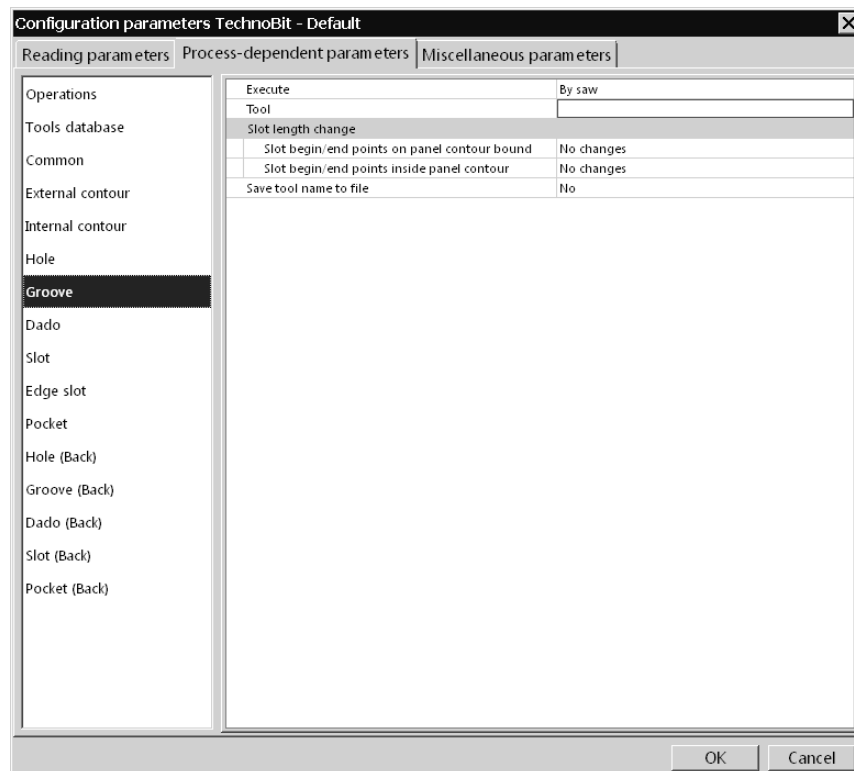


Fig. 10.622.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.623), from the back panel side – in **Slot (Back)** section.

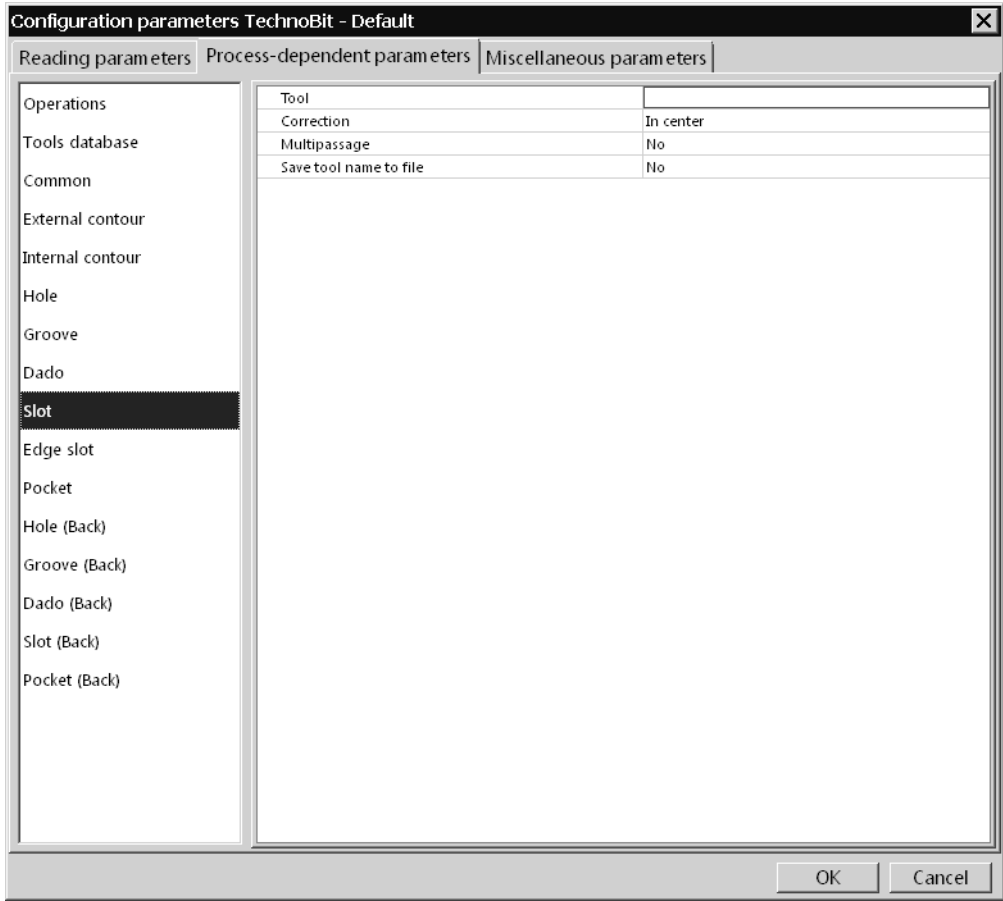


Fig. 10.623.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.624), on the back panel side – in **Pocket (Back)** section.

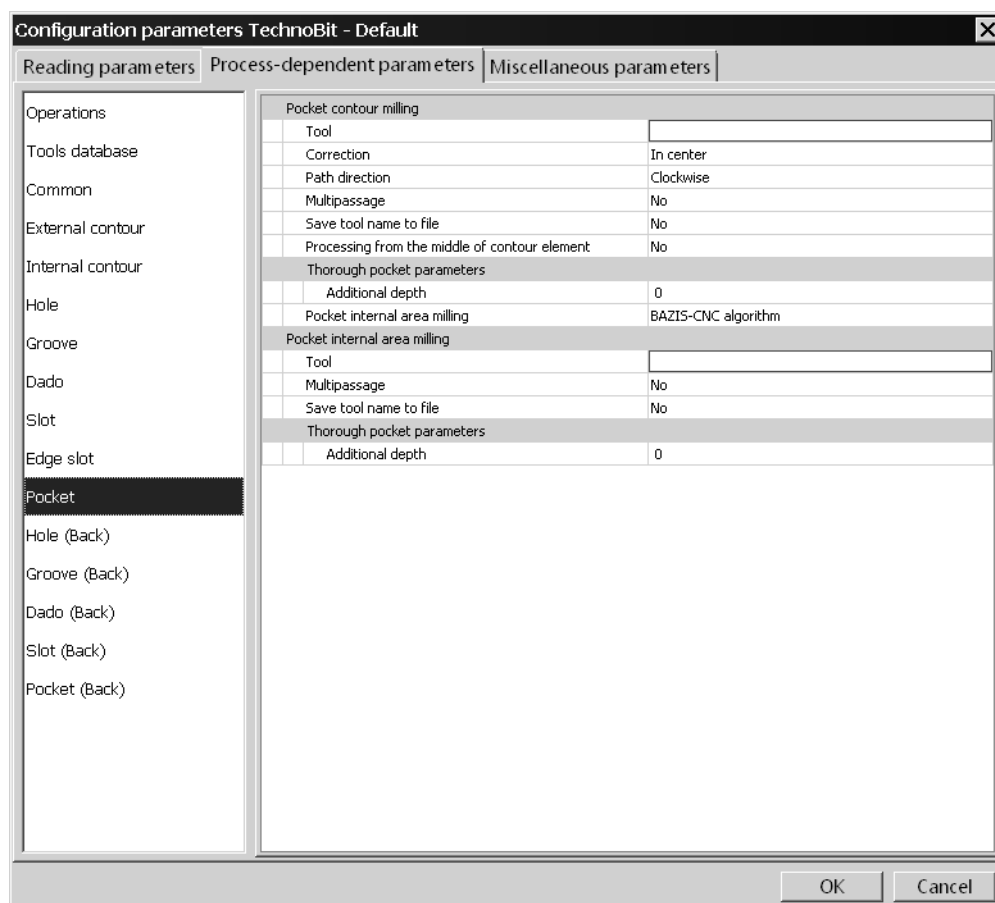


Fig. 10.624.

Tool combo box allows to select milling tool from database. For pocket contour select milling execution direction, correction type.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.34. Sandar

10.34.1.Sandar PA CNC



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common

Common section control elements (fig. 10.625) allow to specify the operations that can be used in control program.

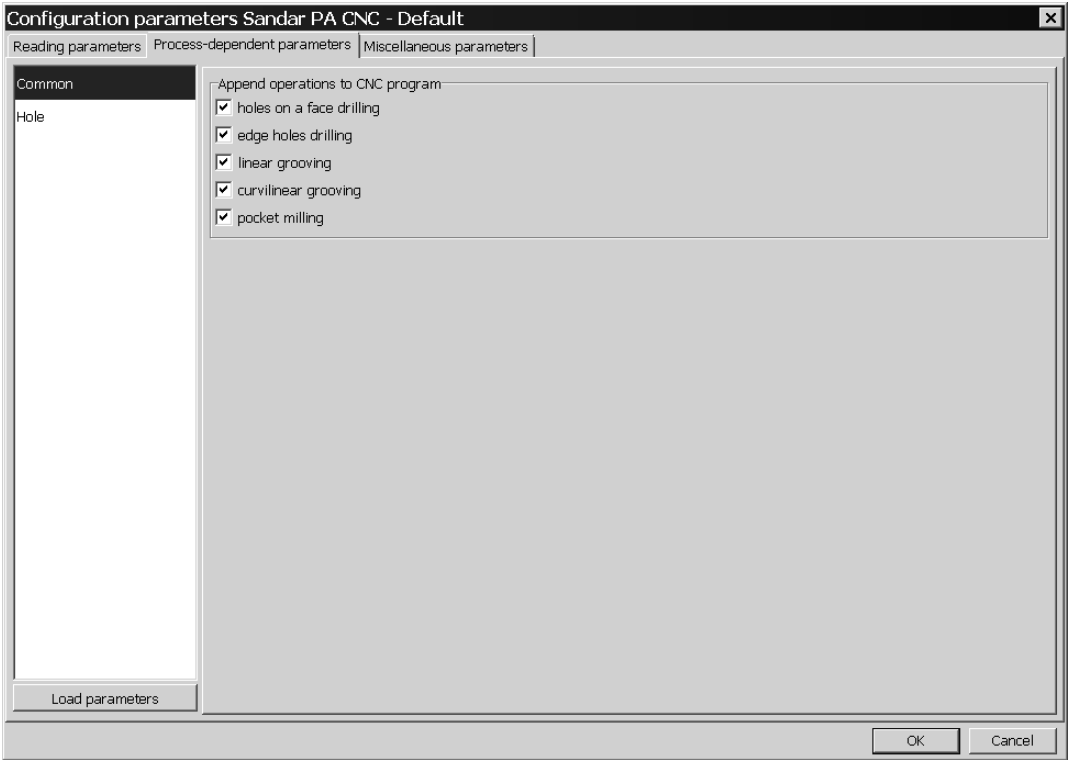


Fig. 10.625.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.626).

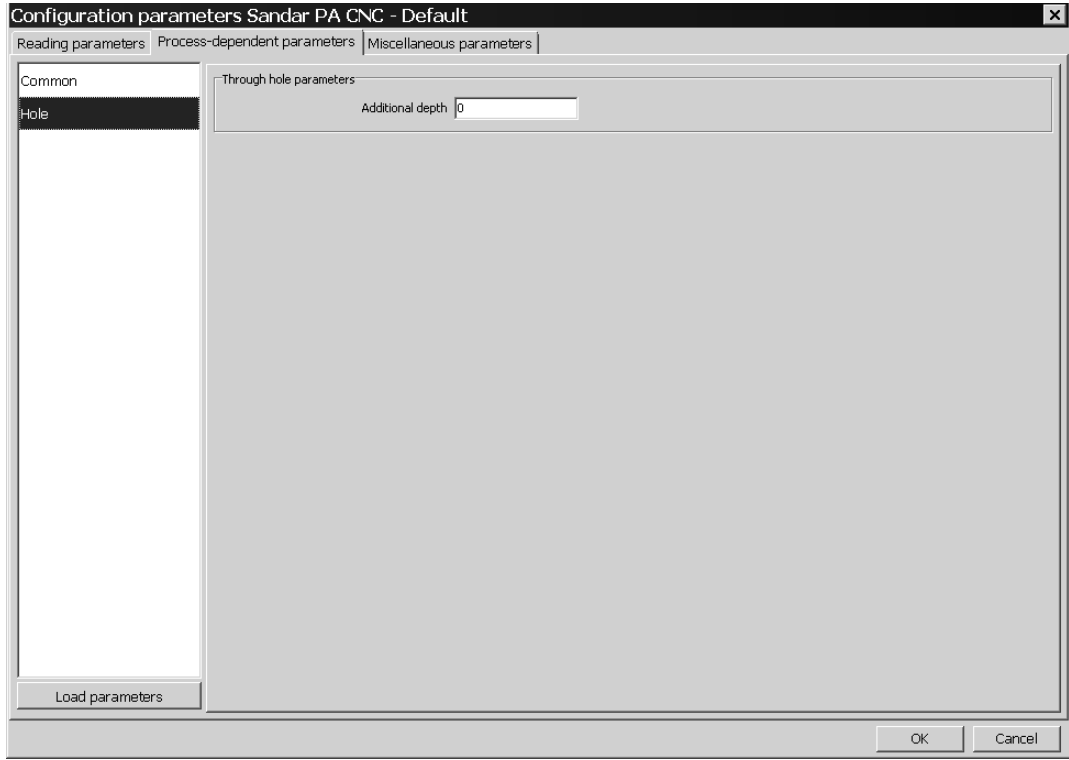


Fig. 10.626.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.35. SCM Group

10.35.1.SCM H800



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.627) allow to set slots and through holes parameters.

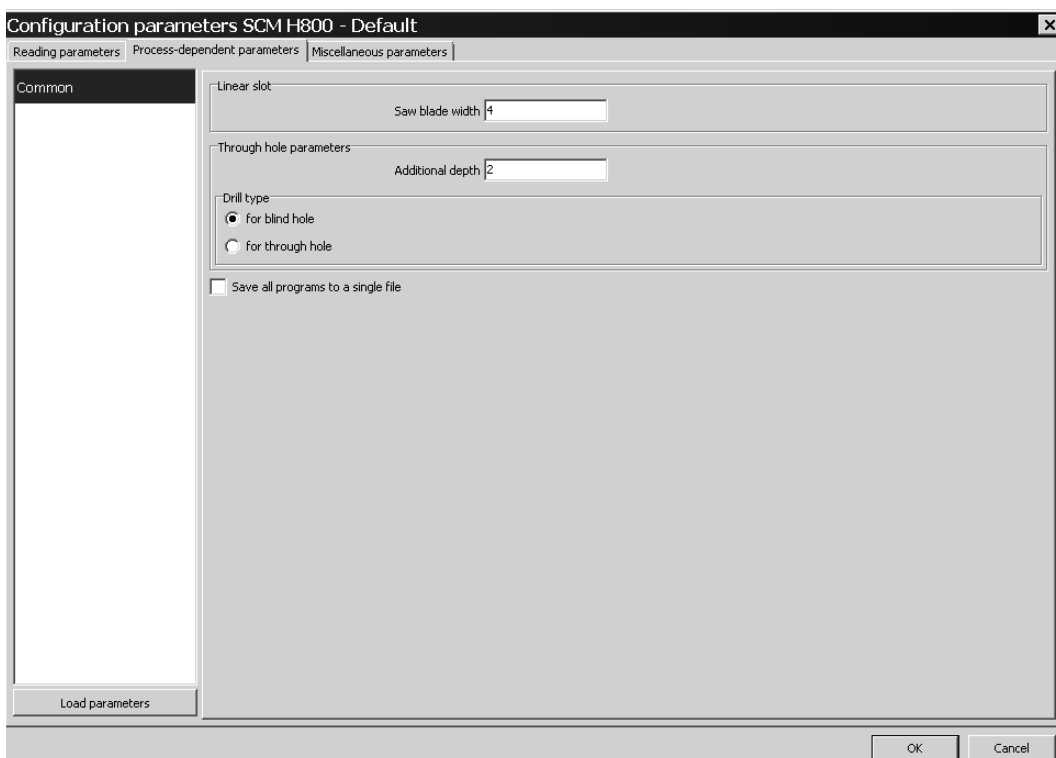


Fig. 10.627.

Saw blade width field allows to set saw blade width at linear slots processing. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through**

hole parameters group allows to take drill center point into account. **Save all programs to a single file** option allows to control program files saving.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.35.2.SCM Startech CN



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.628) allow to set slots and through holes parameters.

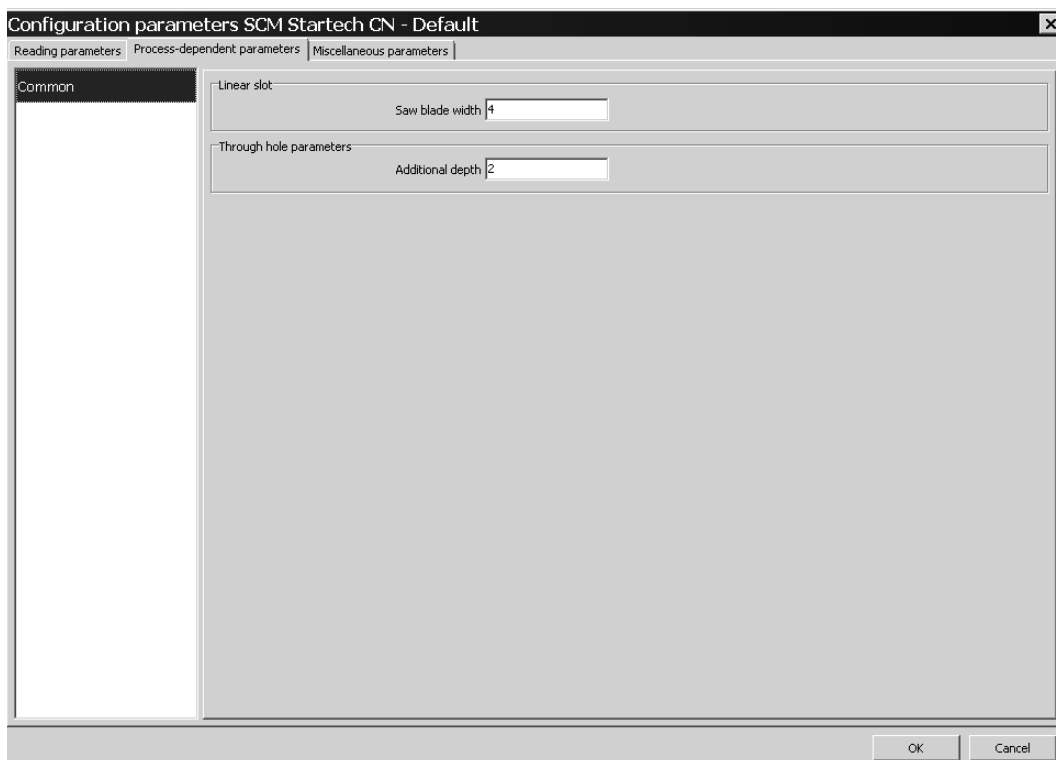


Fig. 10.628.

Saw blade width field allows to set saw blade width at linear slots processing. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Setup completion

To complete control programs creation parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.35.3.SCM Startech CN V

Operations

Operations section control elements (fig. 10.629) allow to specify operations that can be used in control program.

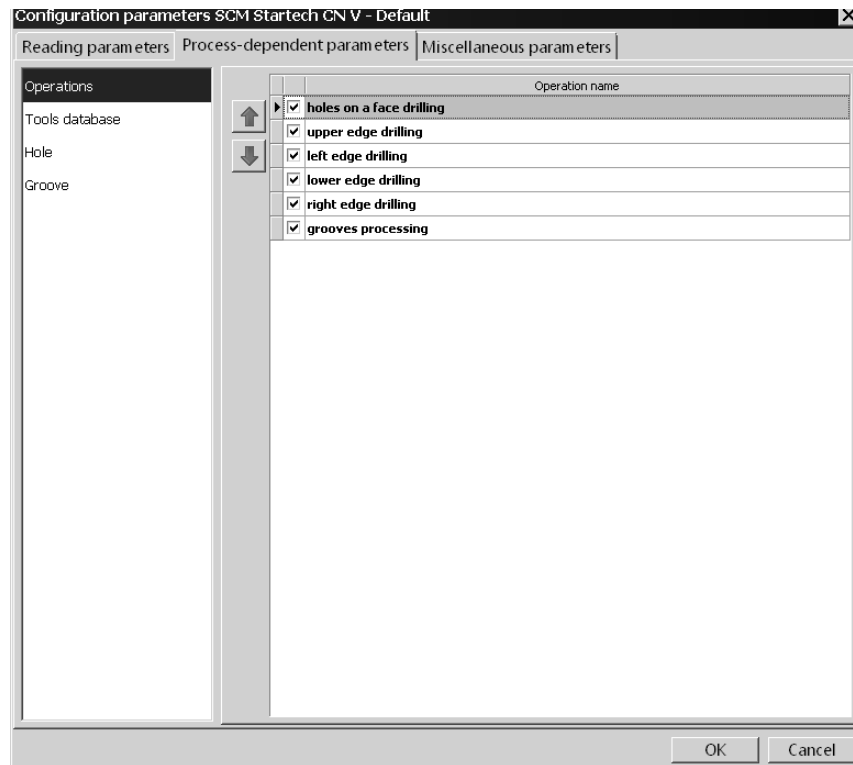


Fig. 10.629.

Usable tools

Tools database section control elements (fig. 10.630) allow to create a tools list available for use in control programs.

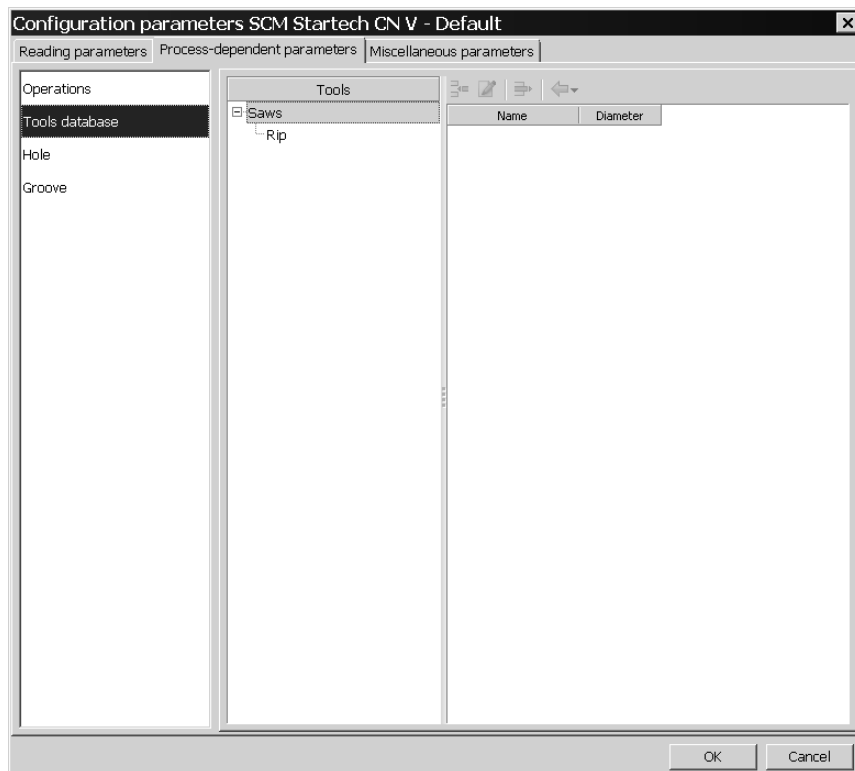


Fig. 10.630.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.631).

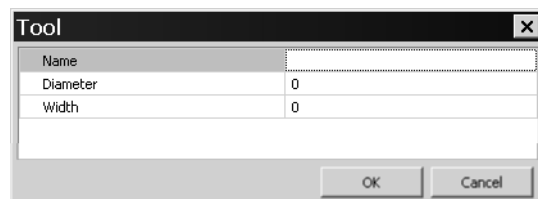


Fig. 10.631.

In this dialogue enter a tool name and set its parameters. To complete a tool adding push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change parameters of a tool selected in the list.



Delete button allows to delete selected tool from the list.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.632).

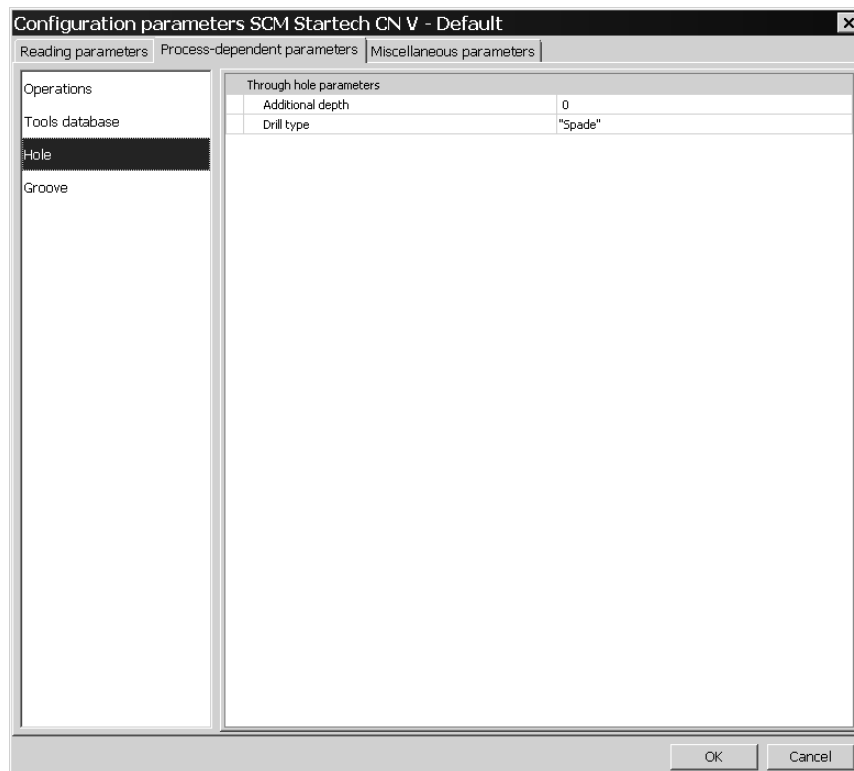


Fig. 10.632.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove parameters

Groove processing setup is performed in **Groove** section (fig. 10.633).

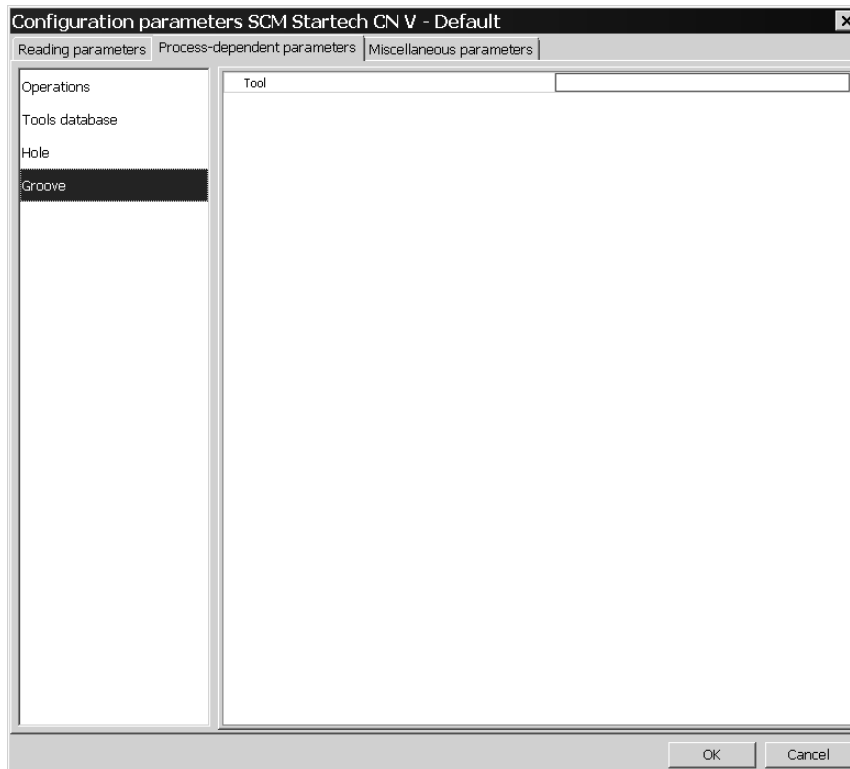


Fig. 10.633.

Section control elements allow to select a tool type for slot creation, a tool from database.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.35.4.Xilog Maestro



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.634) allow to specify the operations that can be used in control program.

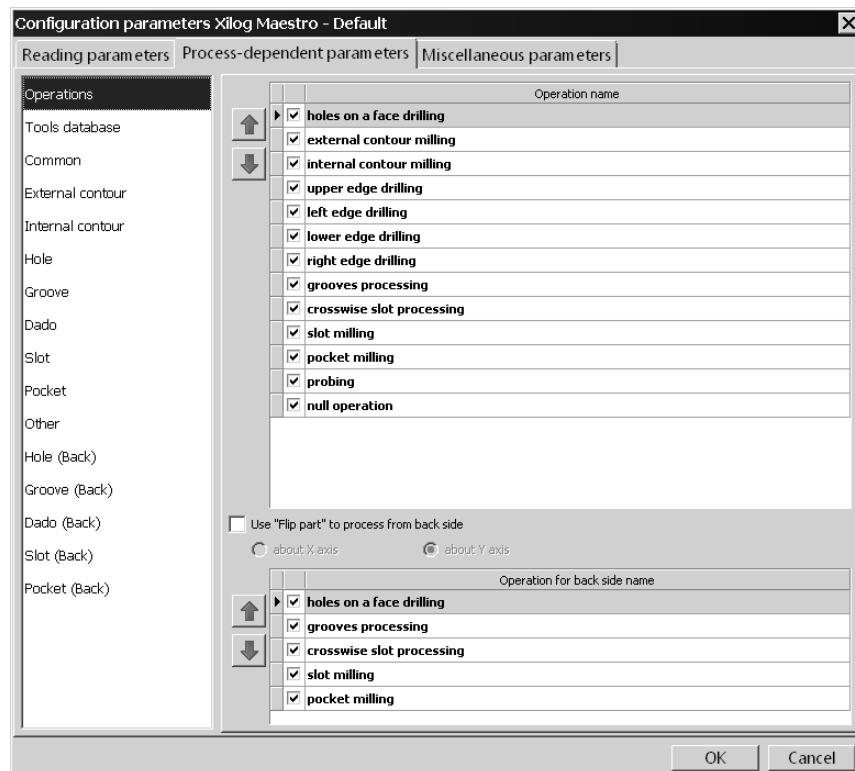


Fig. 10.634.

Usable tools

Tools database (fig. 10.635) section control elements allow to generate a list of tools available for use in control programs.

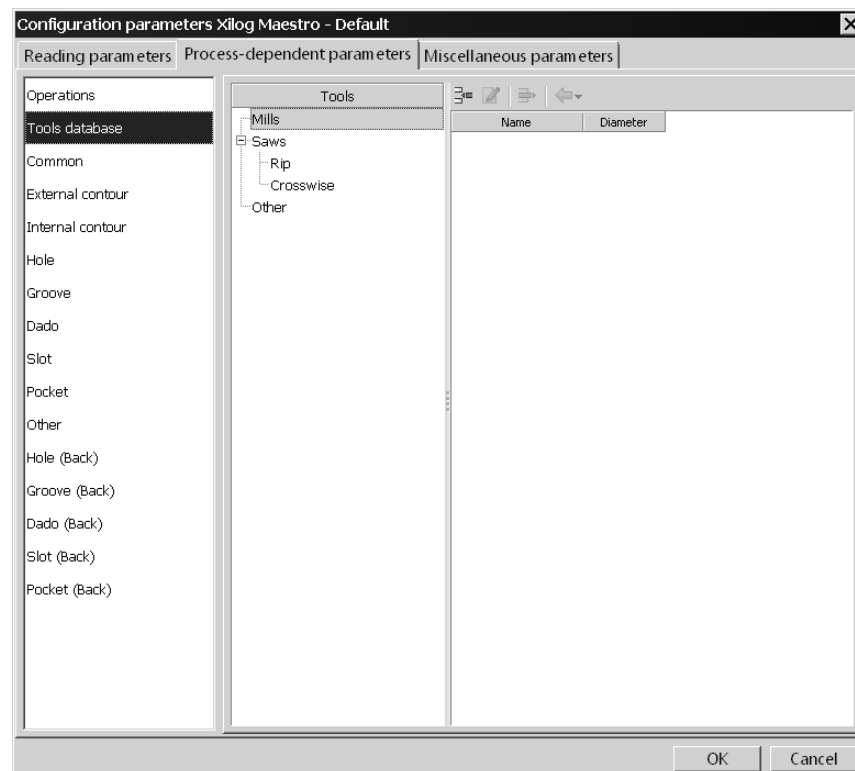


Fig. 10.635.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.636).

Fig. 10.636.

In this dialogue enter a tool name and set its parameters. To complete a tool adding push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change parameters of a tool selected in the list.



Delete button allows to delete selected tool from the list.



Move button allows to move selected tool to another folder. After pushing the button, a menu will appear (fig. 10.637), the menu commands allow to choose a target group.

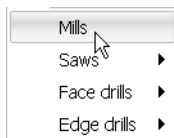


Fig. 10.637.

Common parameters

Common section control elements (fig. 10.638) allow to setup common parameters of control programs.

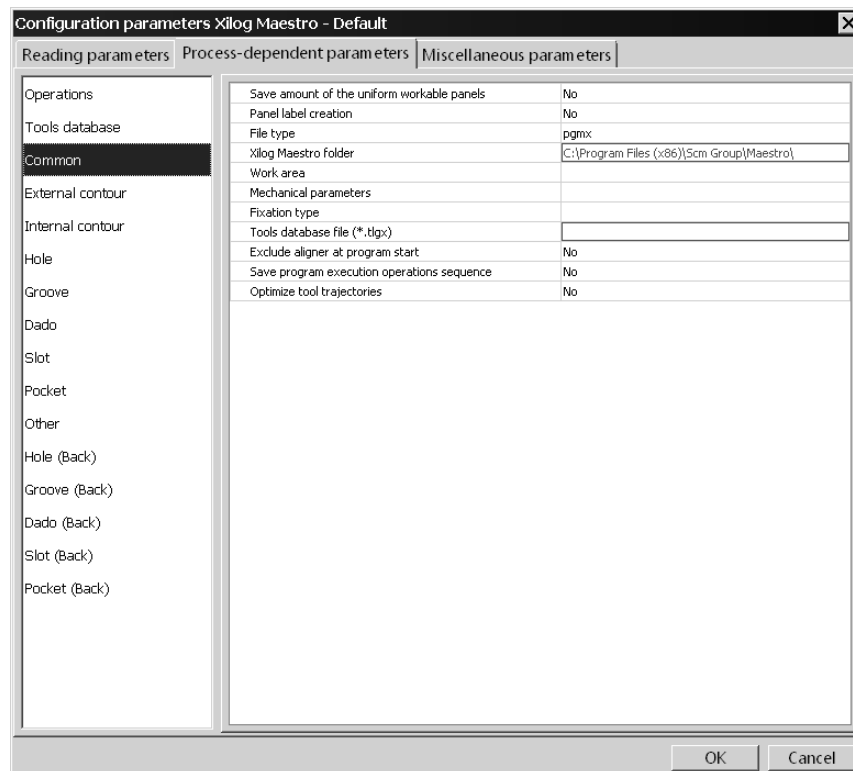


Fig. 10.638.

Contour processing parameters

Outer and internal contours processing parameters setup is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.639).

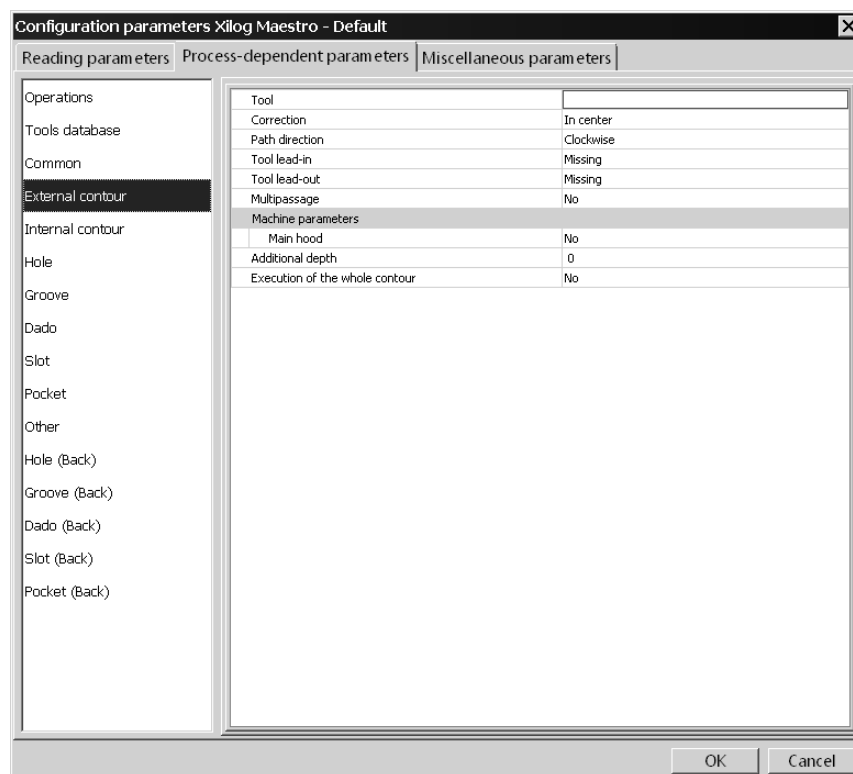


Fig. 10.639.

Sections control elements allow to select a tool, correction type, lead-in type, etc.

Hole parameters

Holes processing on the face is set in **Hole** section (fig. 10.640).

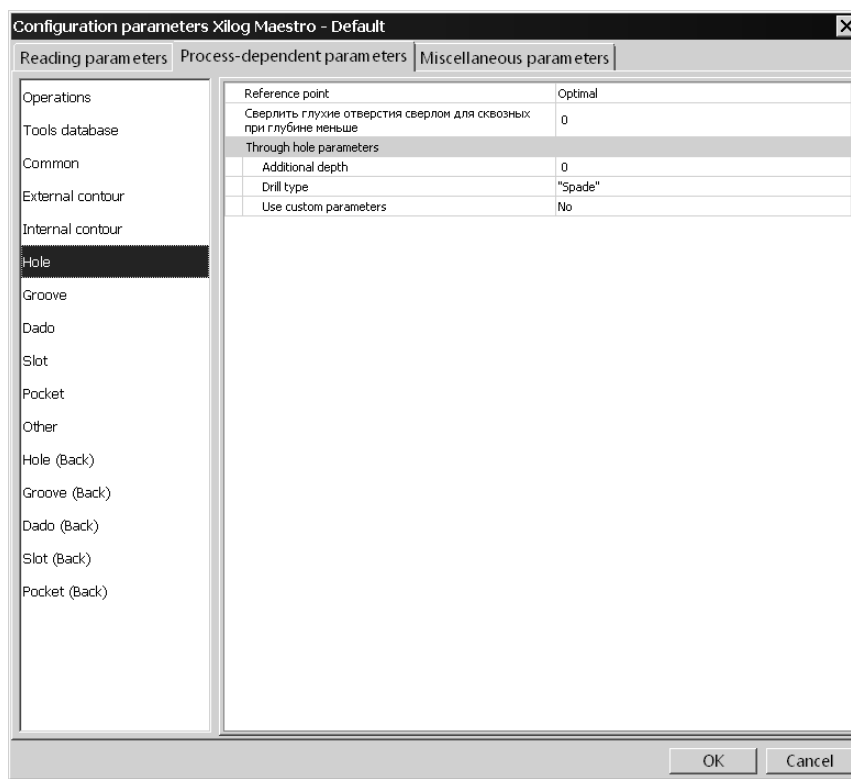


Fig. 10.640.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing is performed in the following sections: **Groove** (fig. 10.641), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

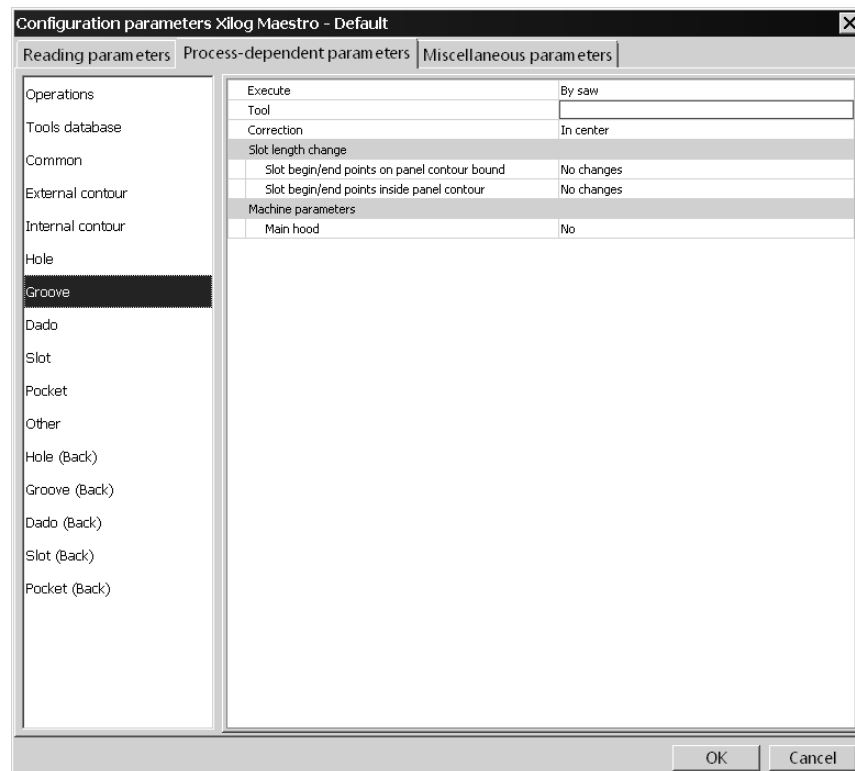


Fig. 10.641.

Section control elements allow to select a toll type for slot creation, a tool from a database and correction type. If a mill is selected as a tool, its parameters will be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing setup on the face side is performed in **Slot** section (fig. 10.642), on the back panel side – in **Slot (Back)** section.

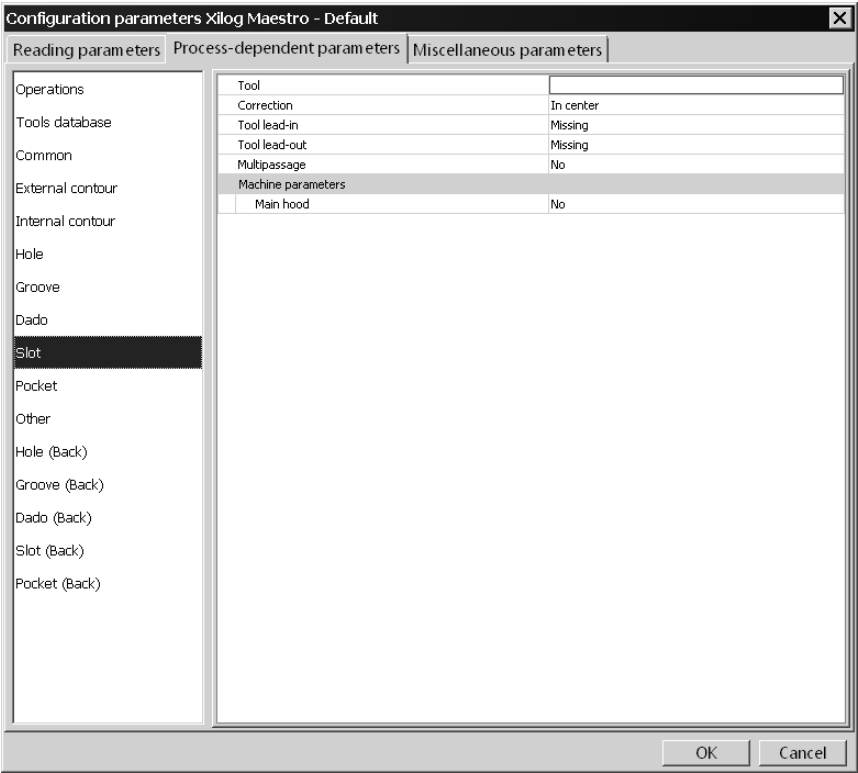


Fig. 10.642.

Section control elements allow to select a tool from database, tool lead-in and lead-out parameters, correction type and multipassage use.

Pocket parameters

Pockets processing parameters setup is performed in **Pocket** section (fig. 10.643).

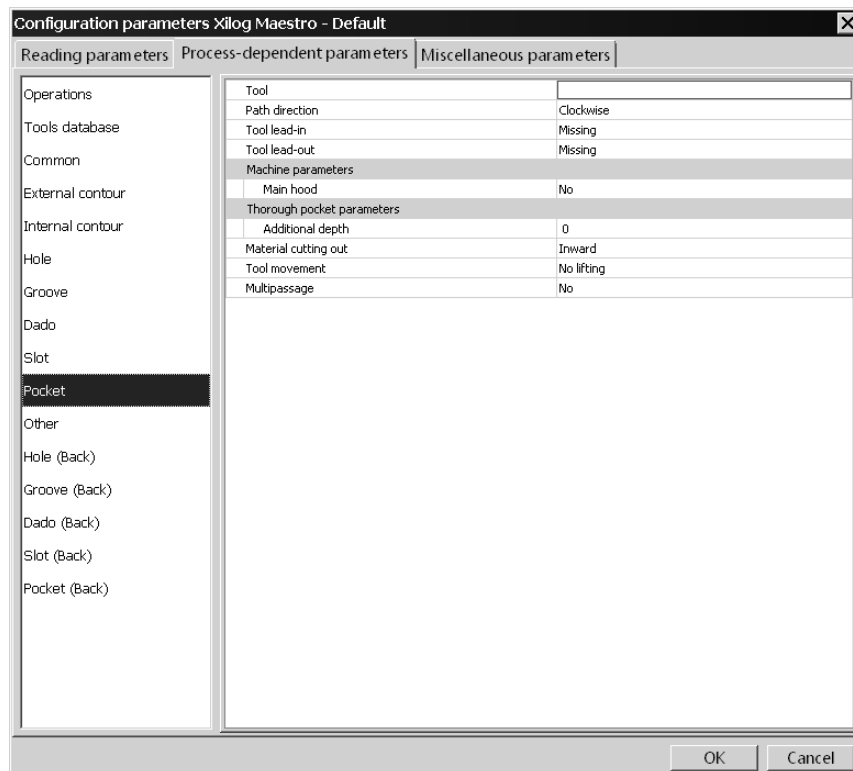


Fig. 10.643.

Tool combo box allows to select a tool name for milling from a database. You should select a milling path direction, correction type and multipassage use, etc.

Miscellaneous parameters

Other section control elements allow to setup other postprocessor parameters (fig. 10.644).

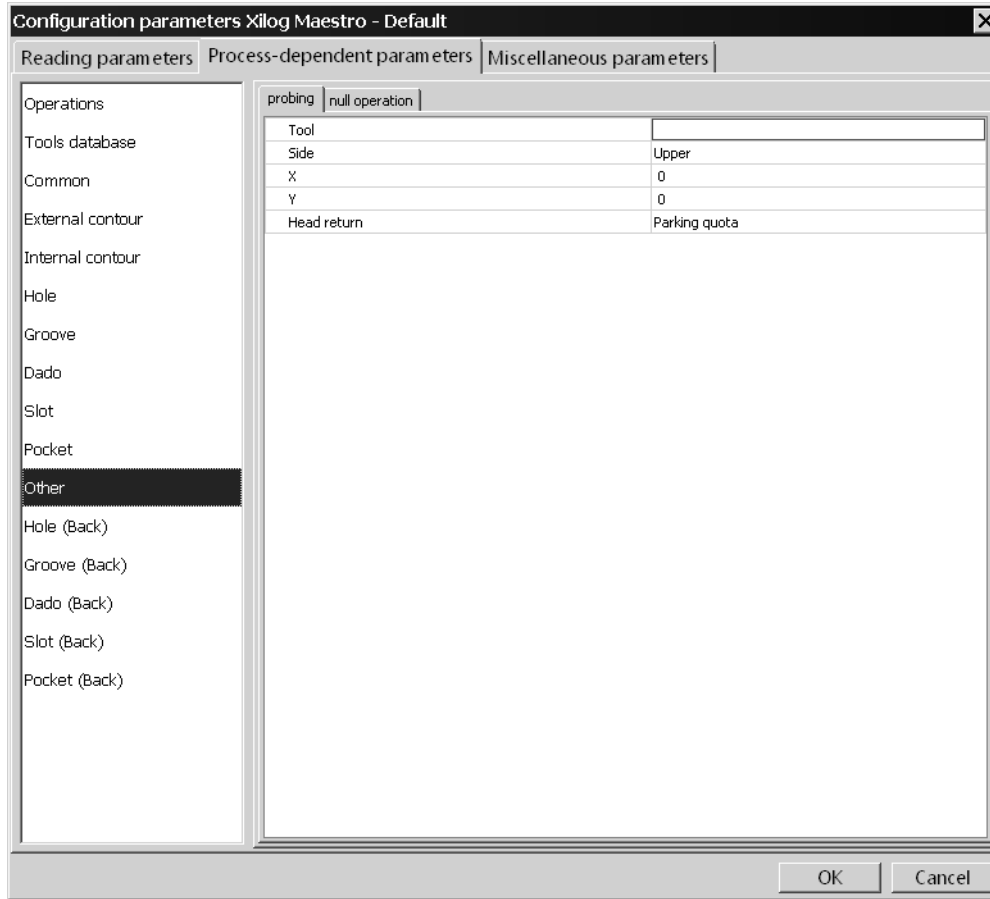


Fig. 10.644.

Tool combo box on **Probing** tab allows to select a tool name from a database. You should select a side, head parking spot, etc. Select coordinate type and coordinate X and Y-direction values on **Null operation** tab.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.35.5.Xilog Plus



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.645) allow to specify the operations that can be used in control program.

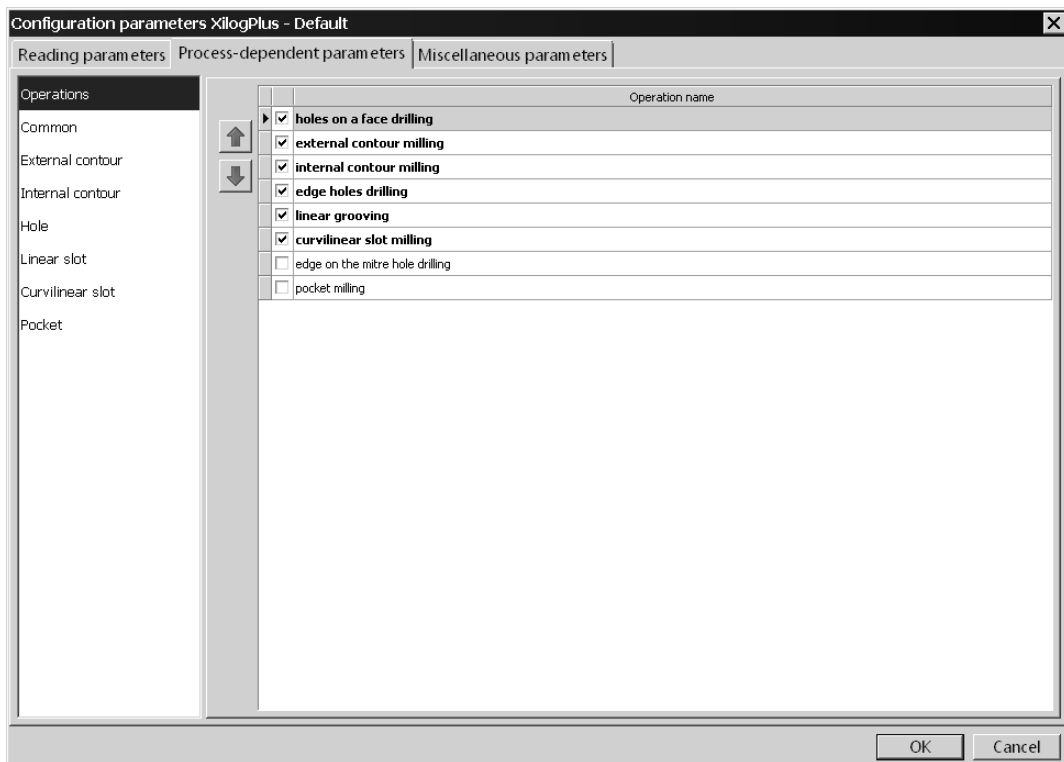


Fig. 10.645.

Common parameters

Common section control elements (fig. 10.646) allow to set common parameters of control programs creation.

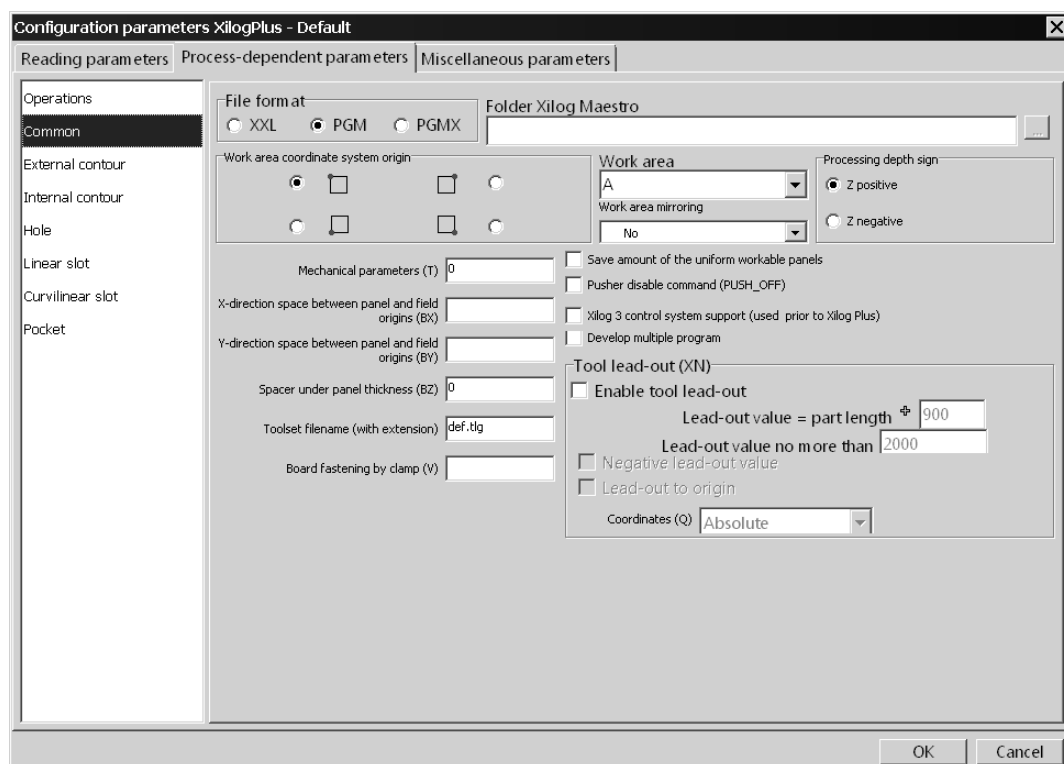


Fig. 10.646.



File format group variants allow to select control program saving format. To select a folder with saved Xilog Maestro program, push **Browse** button and choose a folder in a standard **Browse** dialogue. **Work area coordinate system origin** group variants allow to select panel angle as origin of coordinates. **Work area** group variants allow to select machine work area.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** and **Internal contour** sections of **Process-dependent parameters** dialogue (fig. 10.647).

Rough and finish milling setup is executed separately on corresponding tabs. Tabs control elements content is the same and described for finish milling.

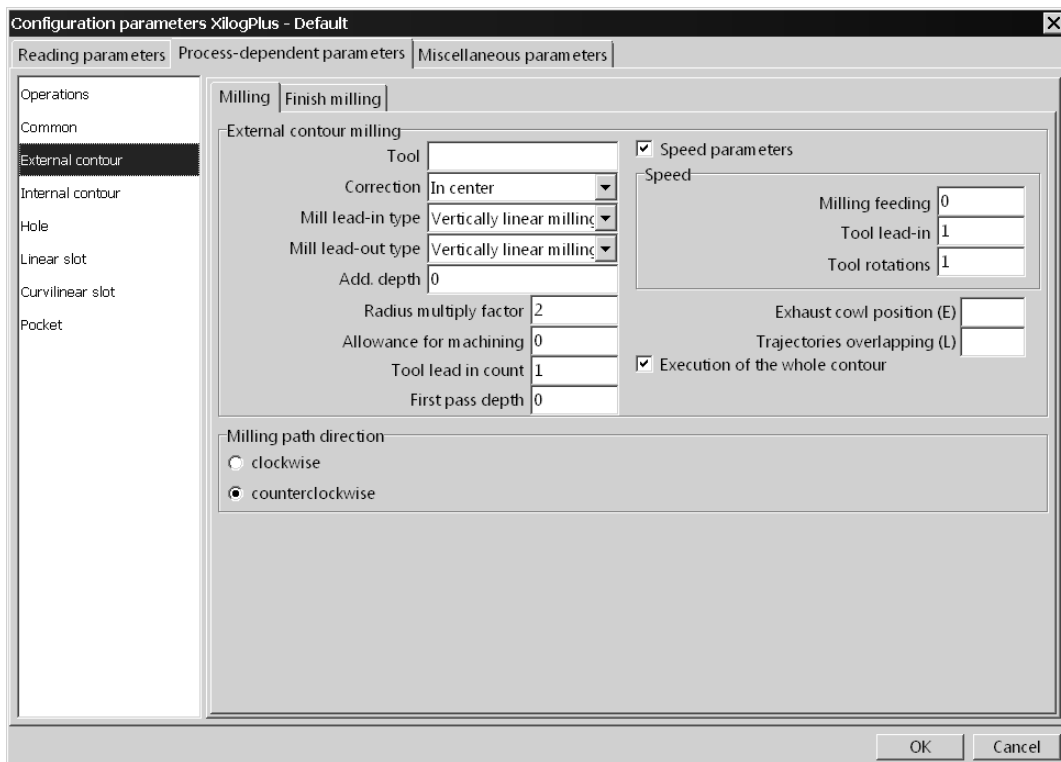


Fig. 10.647.

Finish milling option allows to control this operation. Tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section.

Processing using drill bit

Processing using drill bit (fig. 10.648) tab control elements allow to set holes boring parameters.

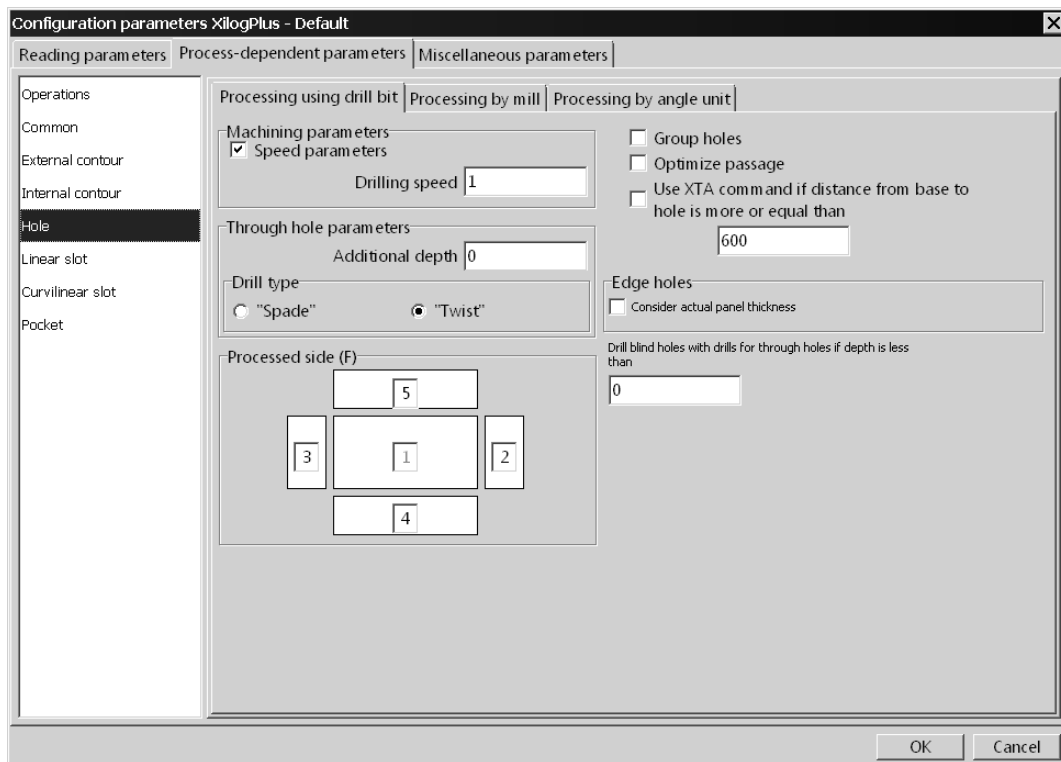


Fig. 10.648.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Control elements allow to set control programs creation parameters: speed parameters, borer type, processed side, etc.

Processing using mill

Processing using mill (fig. 10.649) tab control elements allow to set holes milling parameters.

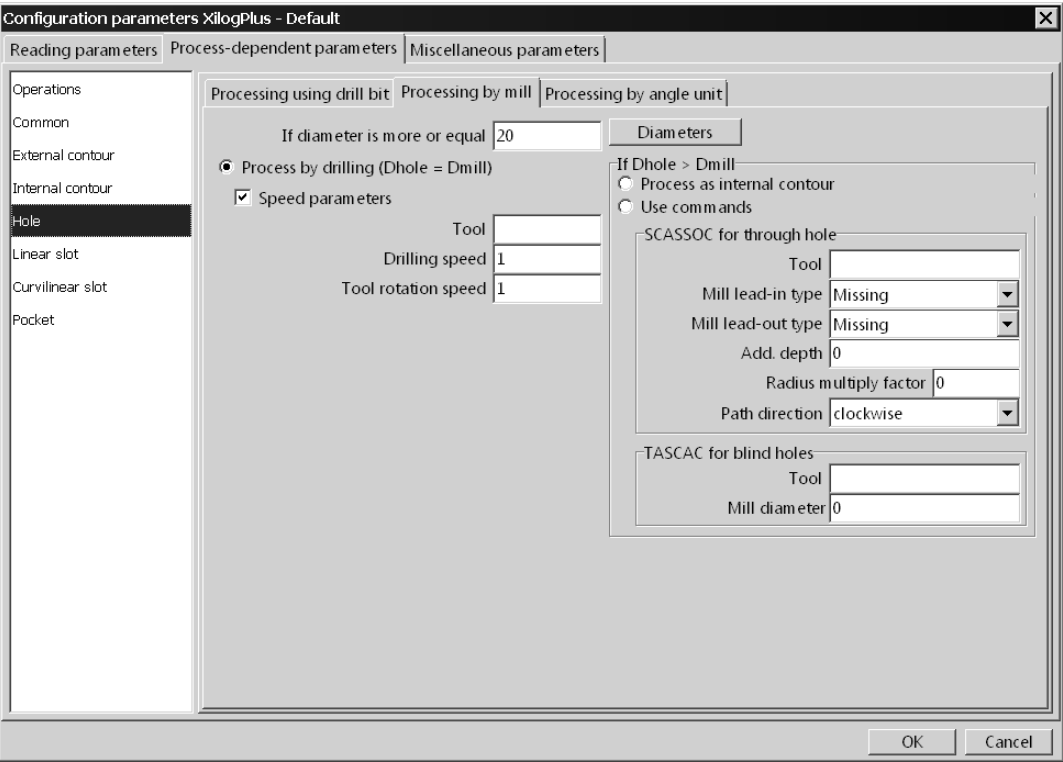


Fig. 10.649.

If **diameter is more or equal** field allows to set hole diameter, if it exceeds, hole will be milled. To map hole diameter to a mill for its processing, push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.650).

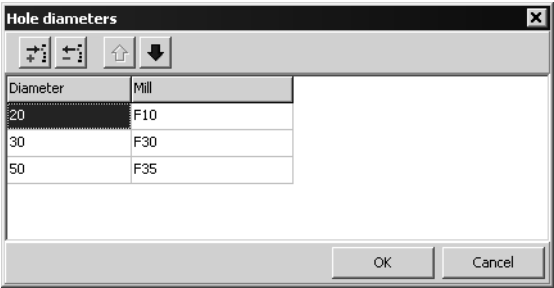


Fig. 10.650.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To complete setup push **OK** button. **Cancel** button allows to close the dialogue without changes. «SCASSOC» command can be used for through holes of great diameter milling. «TASCAC» program can be used for blind holes of great diameter milling. **SCASSOC for through hole** and **TASCAC for blind holes** groups control elements allow to set these commands use.

Processing by angle unit

Processing by angle unit (fig. 10.651) tab control elements allow to set angle unit parameters for edge holes processing.

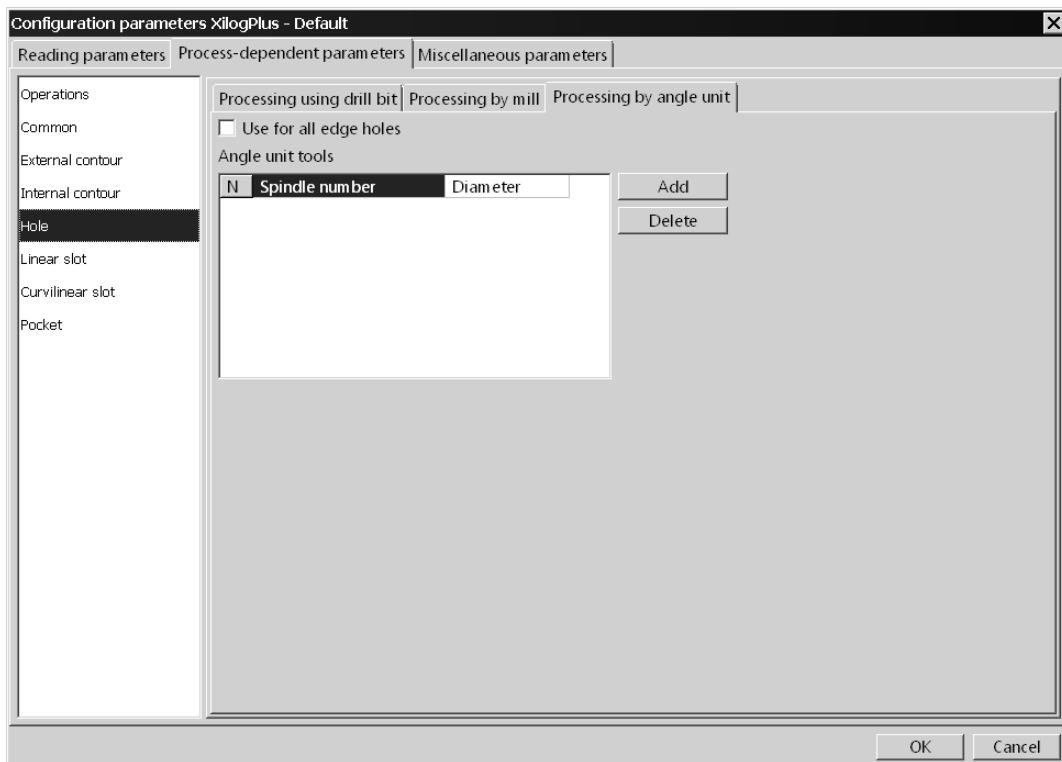


Fig. 10.651.

Use for all edge holes option allows to control unit use. **Angle unit tools** table fields allow to map holes diameter to spindle numbers. **Add** button allows to add a blank row to the table. To delete a row, select it in the table and push **Delete** button.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.652).

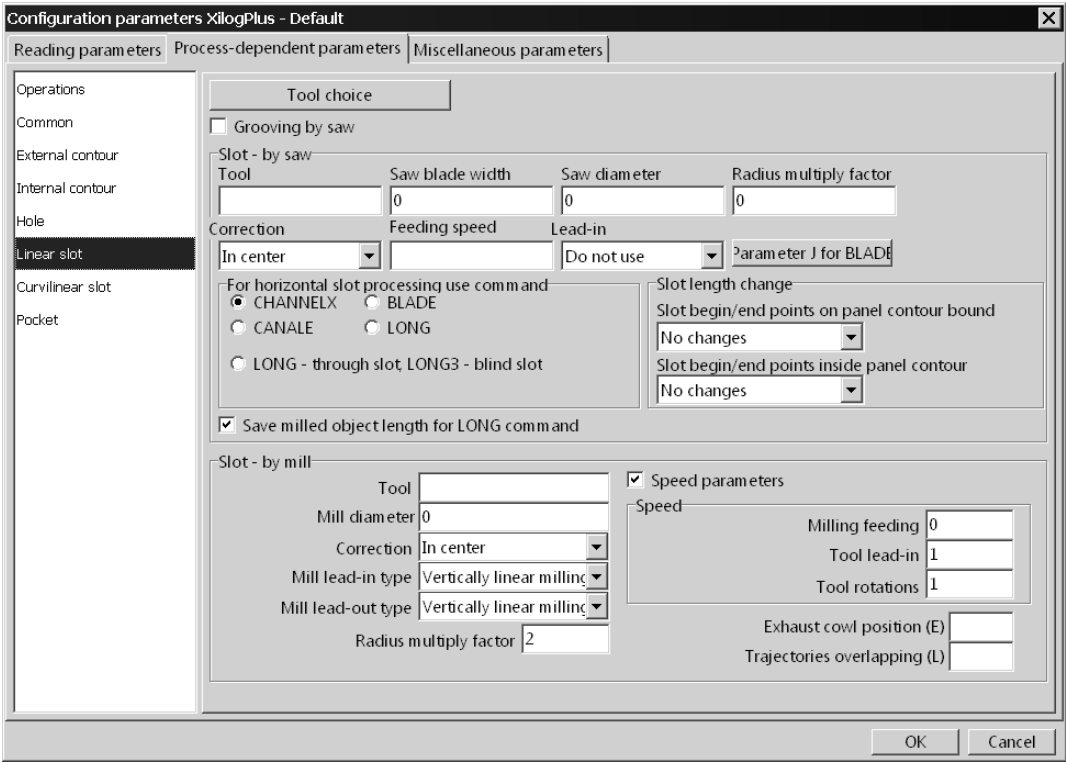


Fig. 10.652.

Tool choice button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.653).

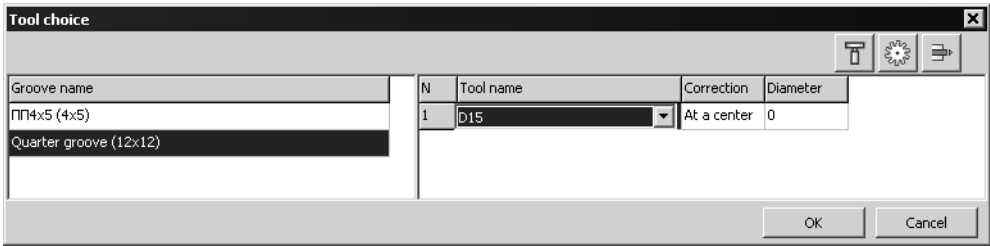


Fig. 10.653.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.654).

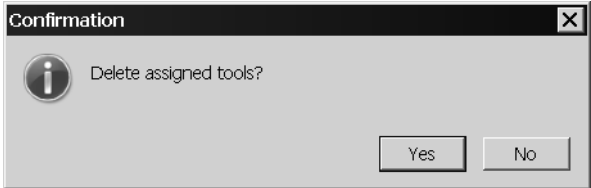


Fig. 10.654.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw. **Parameter J for BLADE** button allows to set J parameter depending on slots length. After pushing the button **J parameter setup** dialogue will appear on the screen (fig. 10.655).

Groove length range (mm)		J value
from	to	

Fig. 10.655.

From and **to** table fields allow to set slot length bounds. Input range value parameter value in **J value** field. **Add** and **Delete** buttons allow to operate with table rows. To complete parameters setup push **OK** button. **Cancel** button allows to close the dialogue without changes. **For horizontal slot processing use command** group variants allow to select corresponding commands for use in the program.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill.

Curvilinear slot parameters

Curvilinear slots parameters are set in **Curvilinear slot** section (fig. 10.656).

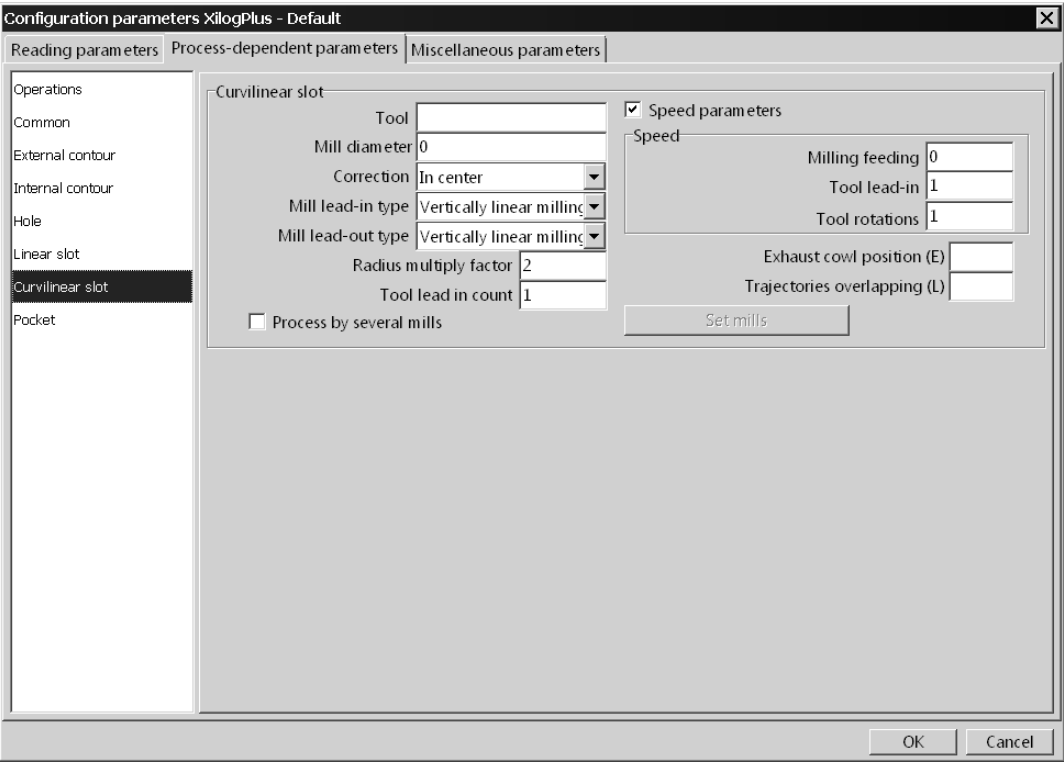


Fig. 10.656.

Control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.657).

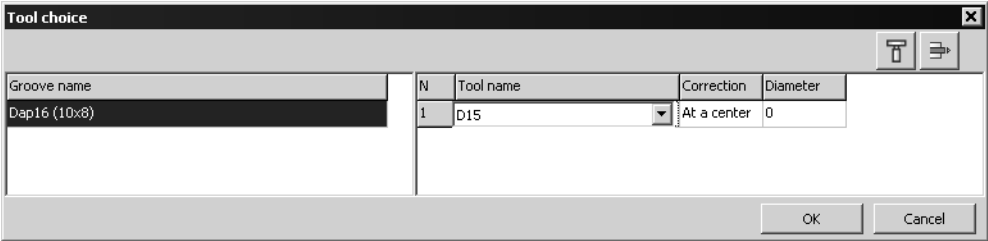


Fig. 10.657.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Correction type should be selected from combo box. Mill name and its diameter can be changed in list fields.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.658) allow to set pocket processing parameters.

Fig. 10.658.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control program parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.36. UNITeam

10.36.1. Albatros Cad



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.659) allow to set control programs common parameters.

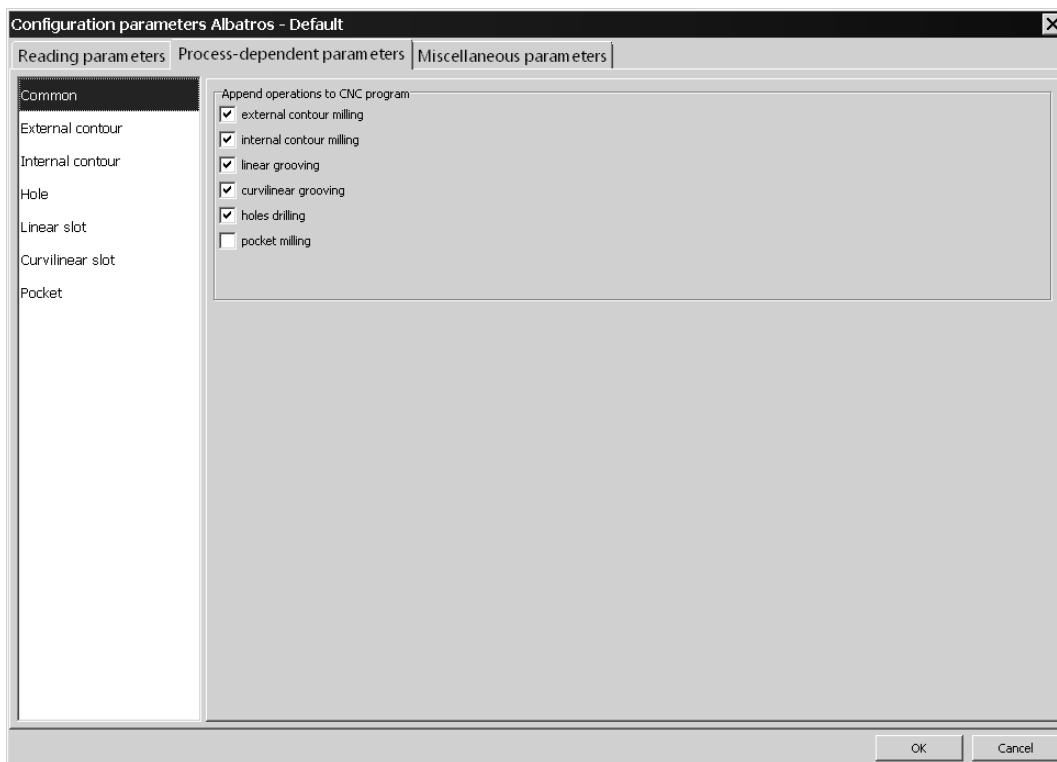


Fig. 10.659.

Append operations to CNC program group options allow to control the inclusion in control program of operations corresponding to the options names.

Contours processing parameters

Outer and internal contours processing parameters are set in **Outer contour** and **Internal contour** sections of **Process-dependent parameters** dialogue (fig. 10.660).

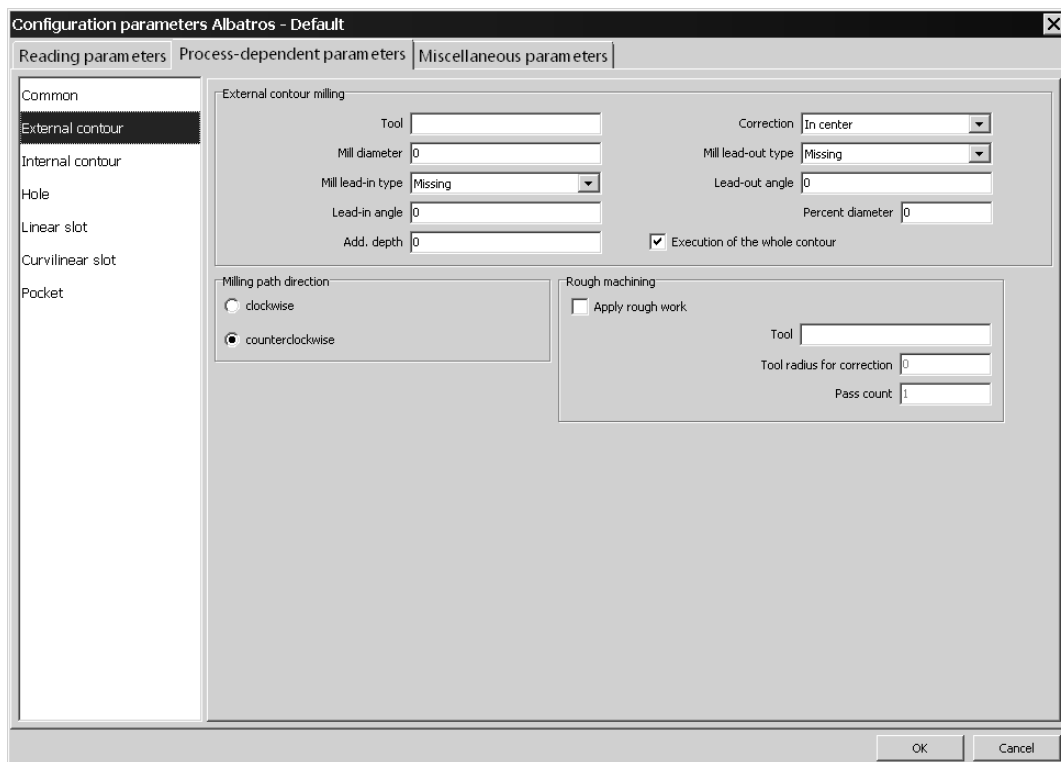


Fig. 10.660.

Apply rough work option allows to control rough work activation. Tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc.

Execution of the whole contour option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Milling path direction** group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.661).

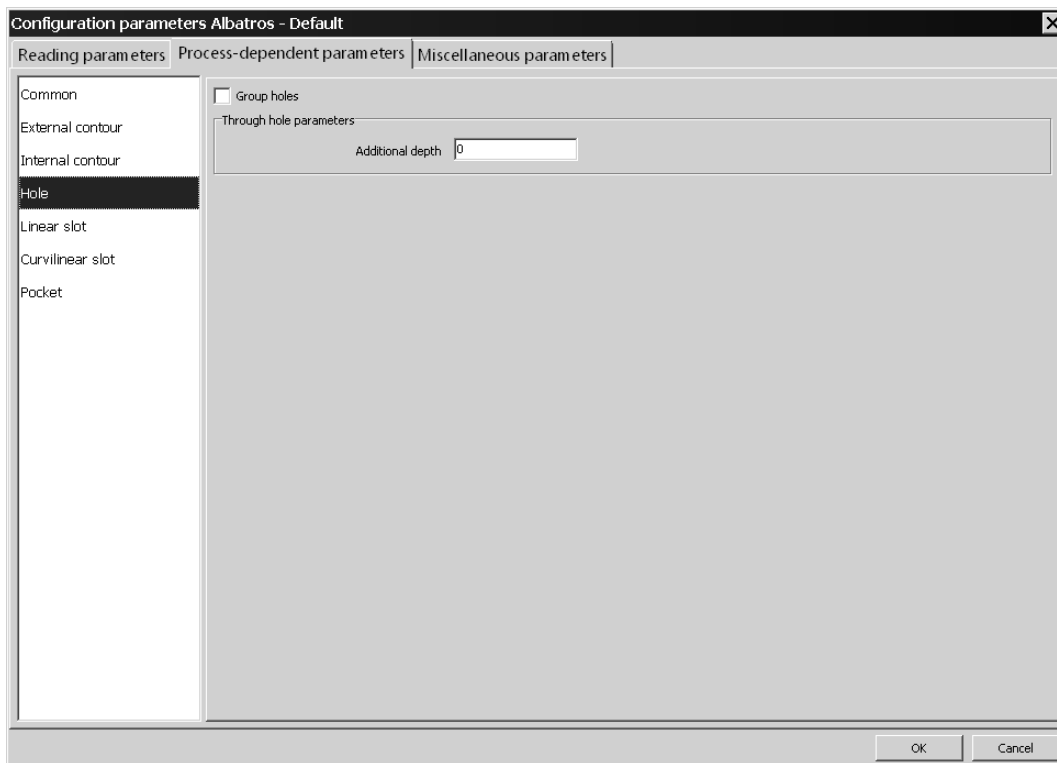


Fig. 10.661.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.662).

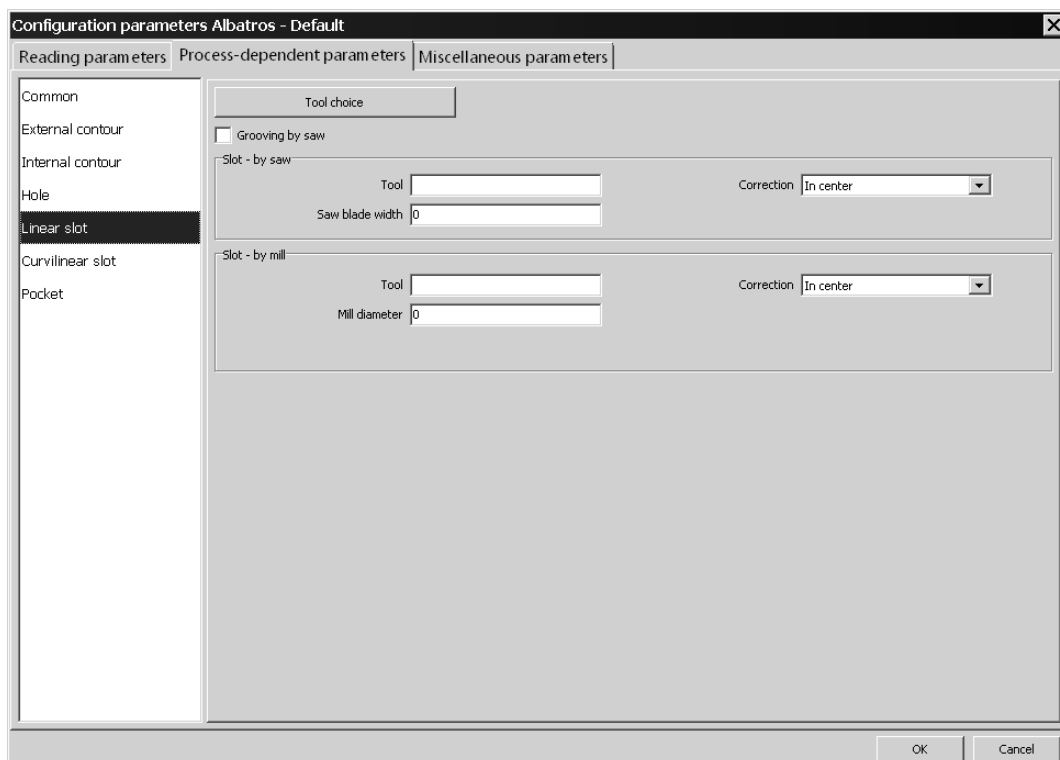


Fig. 10.662.

Grooving by saw option allows to select a tool type for slots processing. If it is disabled, slot will be milled.

Grooving by saw parameters

Slot - by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot - by mill group control elements allow to set control programs creation parameters for grooving by mill.

Tool choice button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.663).

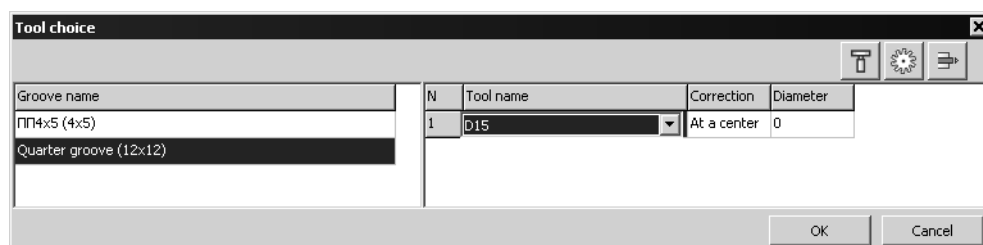


Fig. 10.663.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing.

The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.664).

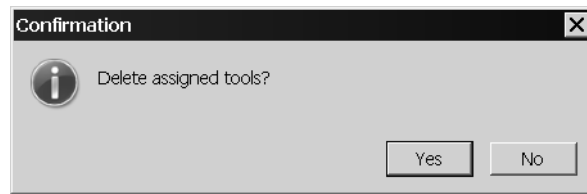


Fig. 10.664.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.665).

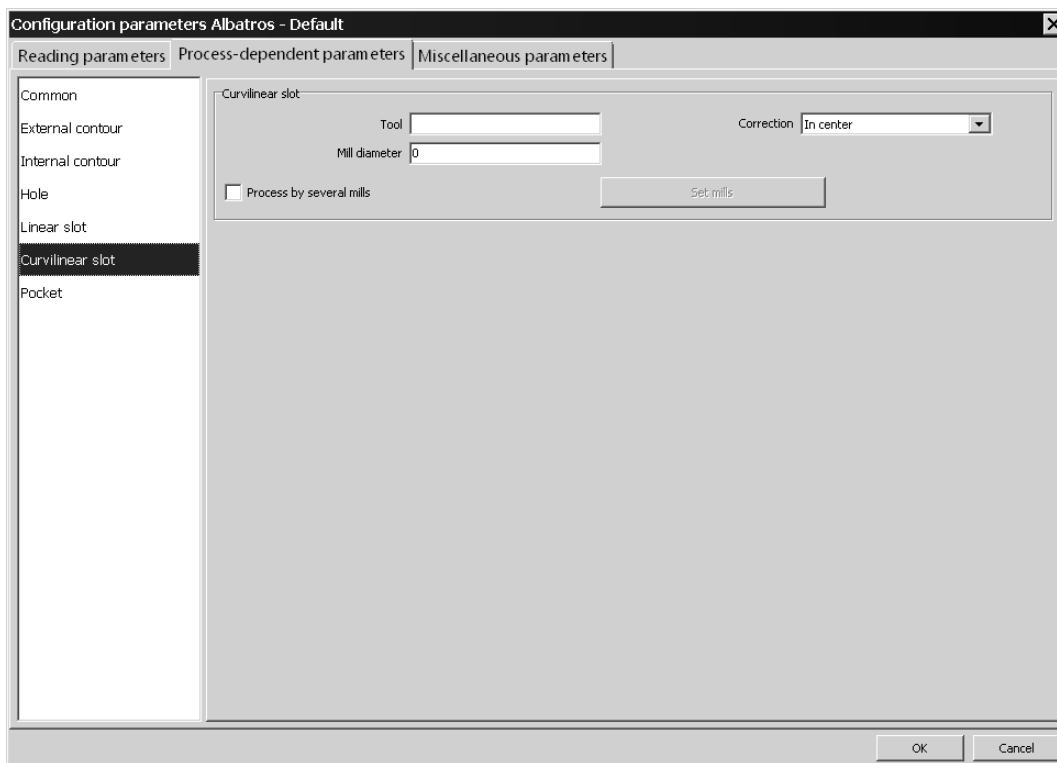


Fig. 10.665.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available.

After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.666).

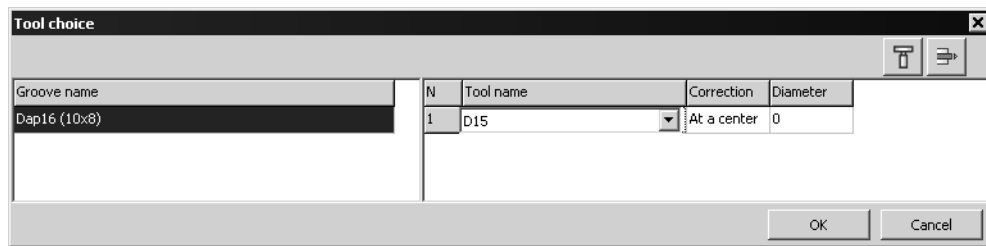


Fig. 10.666.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.

Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.667) allow to set pocket processing parameters.

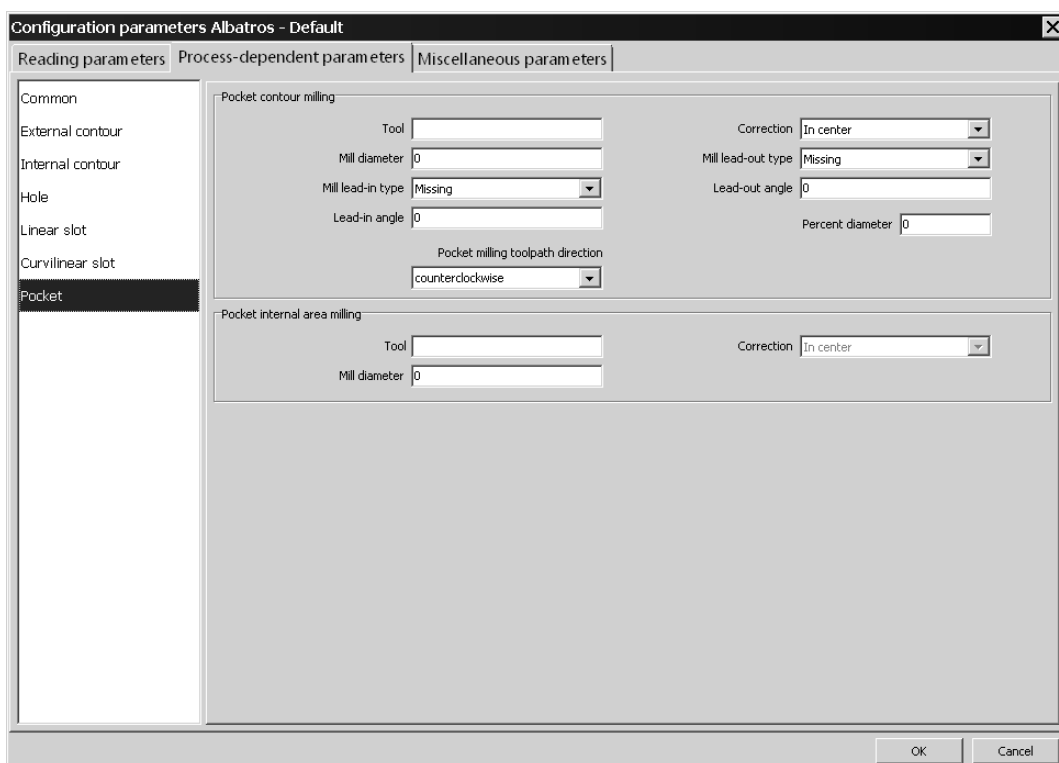


Fig. 10.667.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Setup completion

To complete control programs for Albatros EdiCad setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.37. Velberg

10.37.1.RMZ-5D

Operations

Operations section control elements (fig. 10.668) allow to specify operations that can be used in control program.

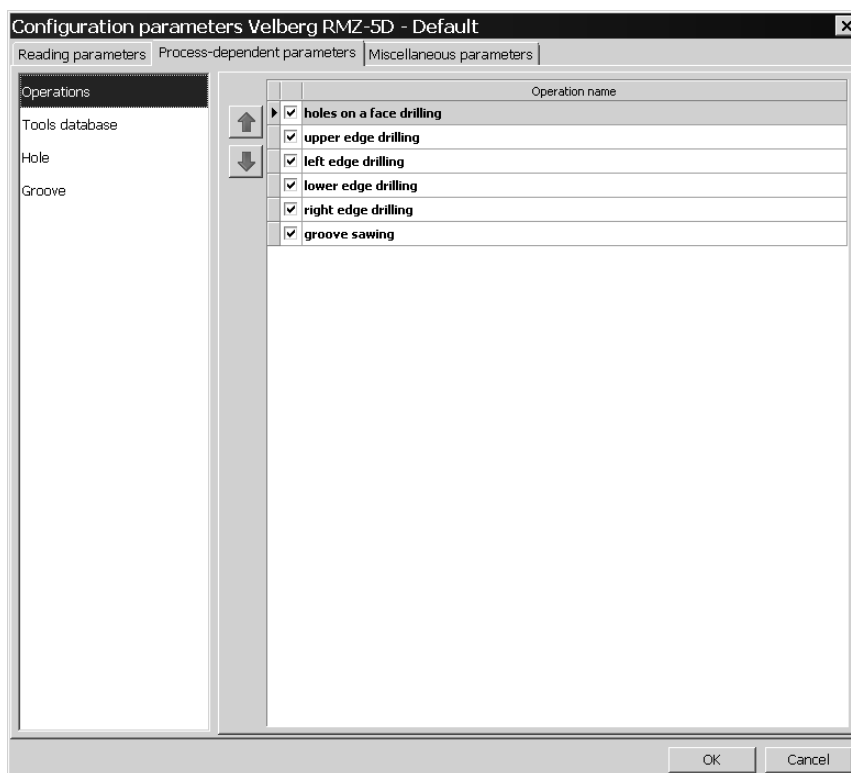


Fig. 10.668.

Usable tools

Tools database section control elements (fig. 10.669) allow to create a list of tools available for use in control programs.

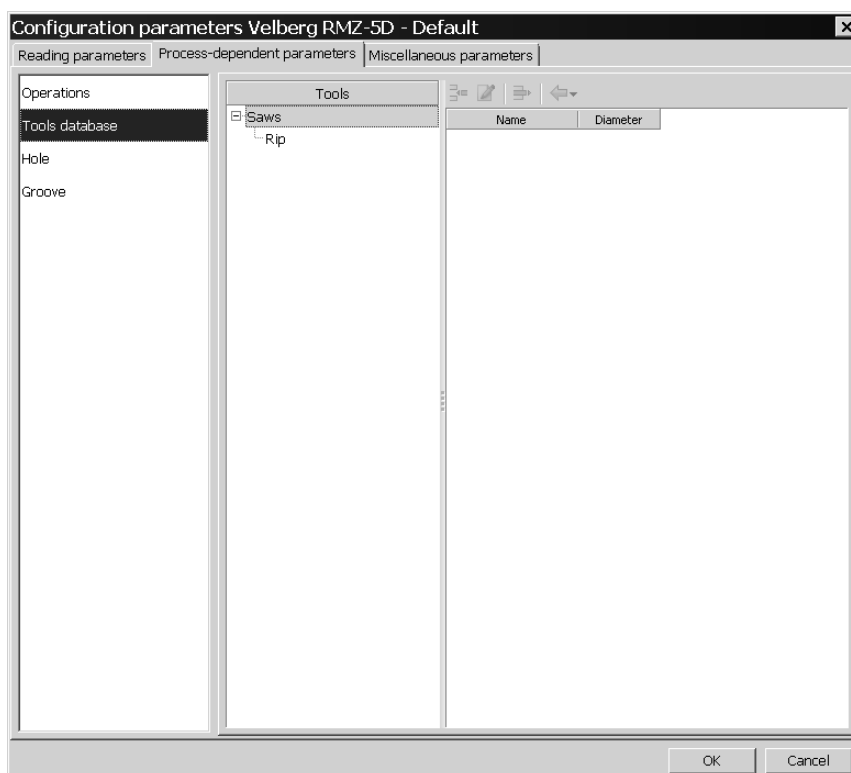


Fig. 10.669.



To add a tool to the list push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.670).

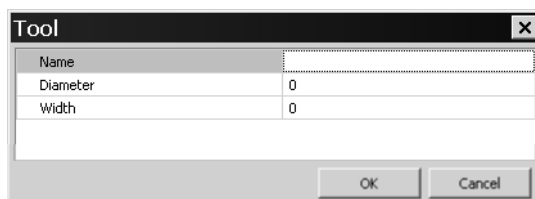


Fig. 10.670.

In this dialogue input a tool name and set its parameters. To finish tool adding push **OK** button. The dialogue will be closed, a tool name will appear in a list. Its parameters will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to edit tool parameters selected in a list.



Delete button allows to delete selected tool from a list.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.671).

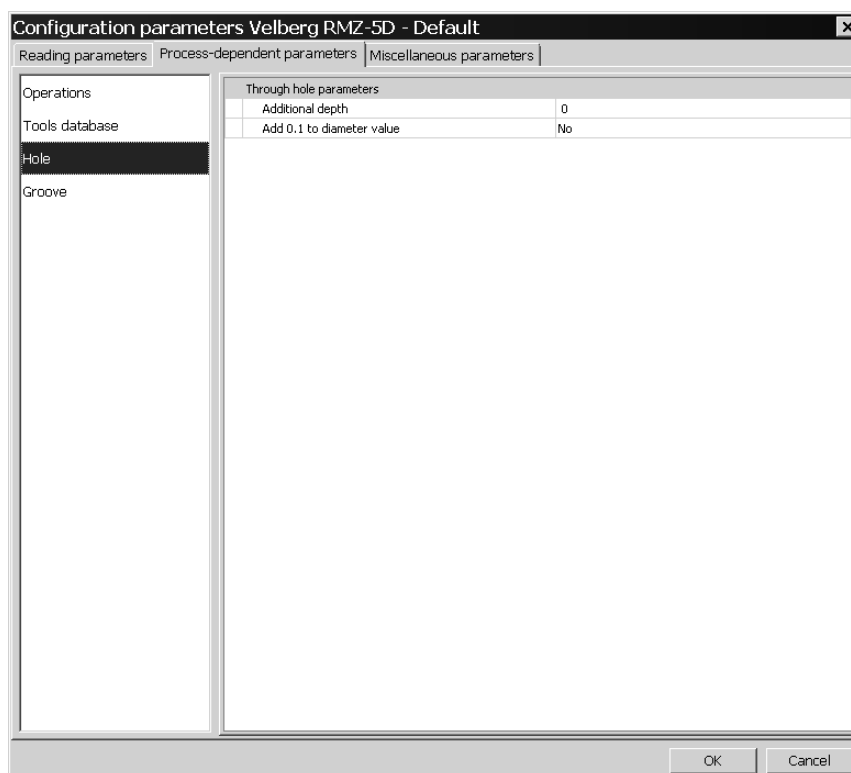


Fig. 10.671.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove parameters

Groove processing setup is performed in **Groove** section (fig. 10.672).

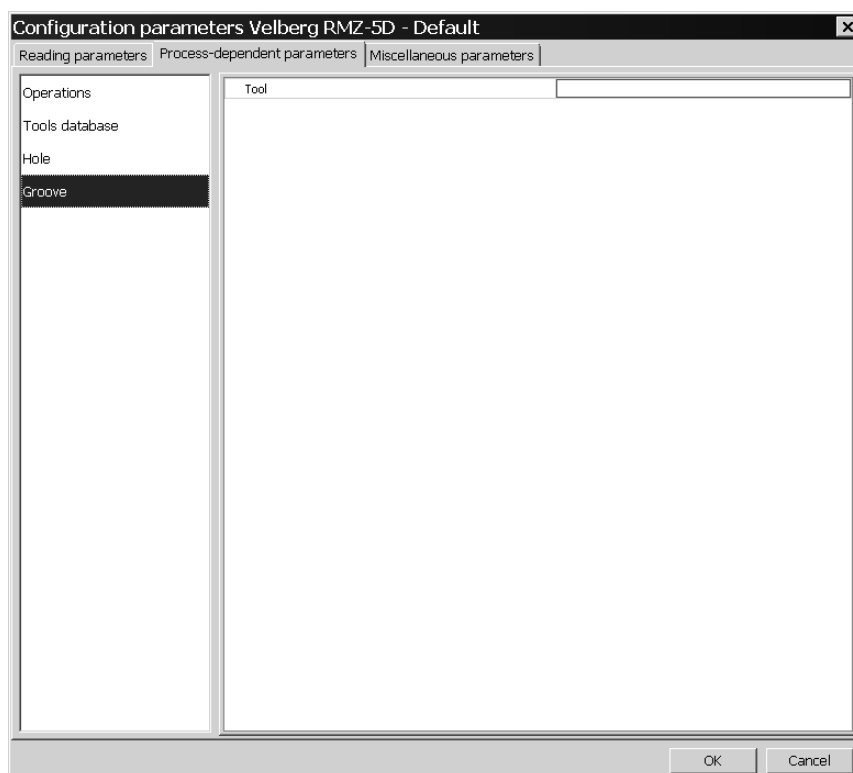


Fig. 10.672.

Section control elements allow to select a tool type for slot creation, a tool from database.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.37.2.Velberg Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.673) allow to specify operations that can be used in control program.

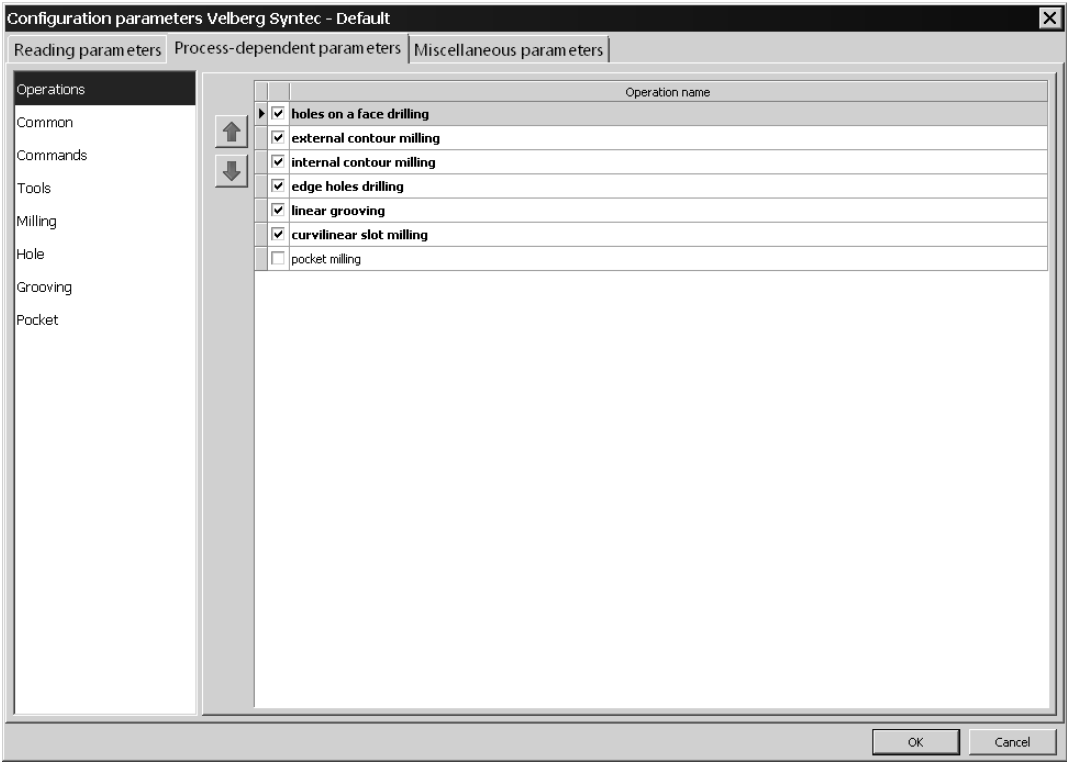


Fig. 10.673.

Common parameters

Common section control elements (fig. 10.674) allow to set common parameters of control programs creation.

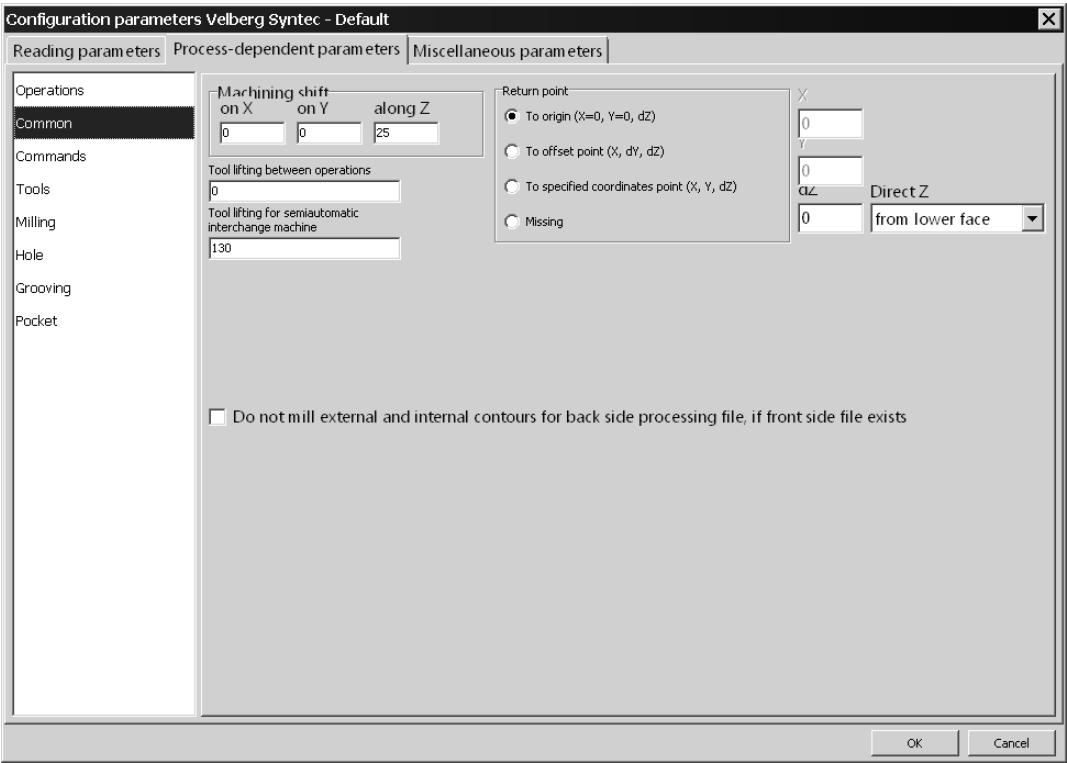


Fig. 10.674.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.675) allow to control separate commands to control program inclusion.

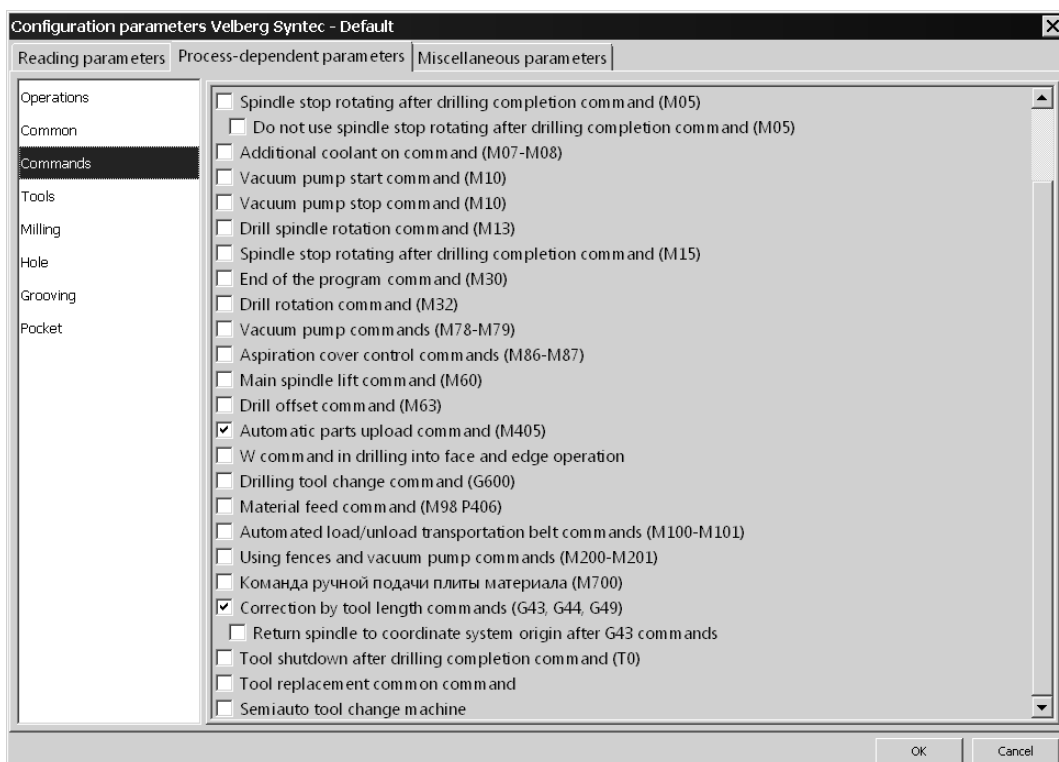


Рис. 10.675.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.676).

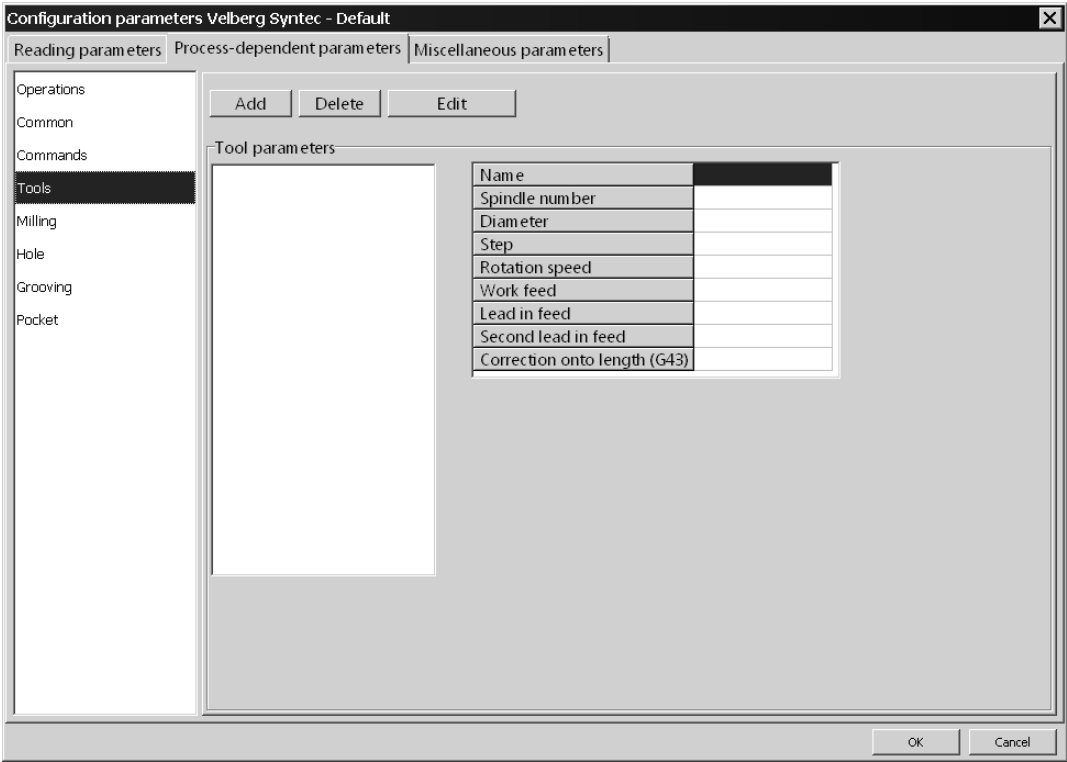


Fig. 10.676.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.677).

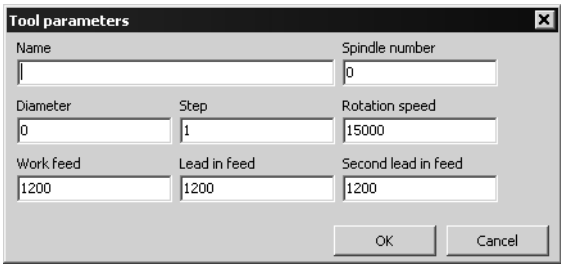


Fig. 10.677.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is in fig. 10.678).

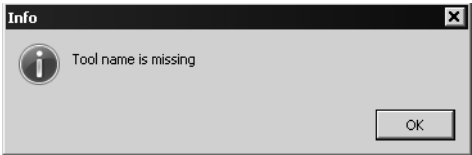


Fig. 10.678.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.677). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.679).

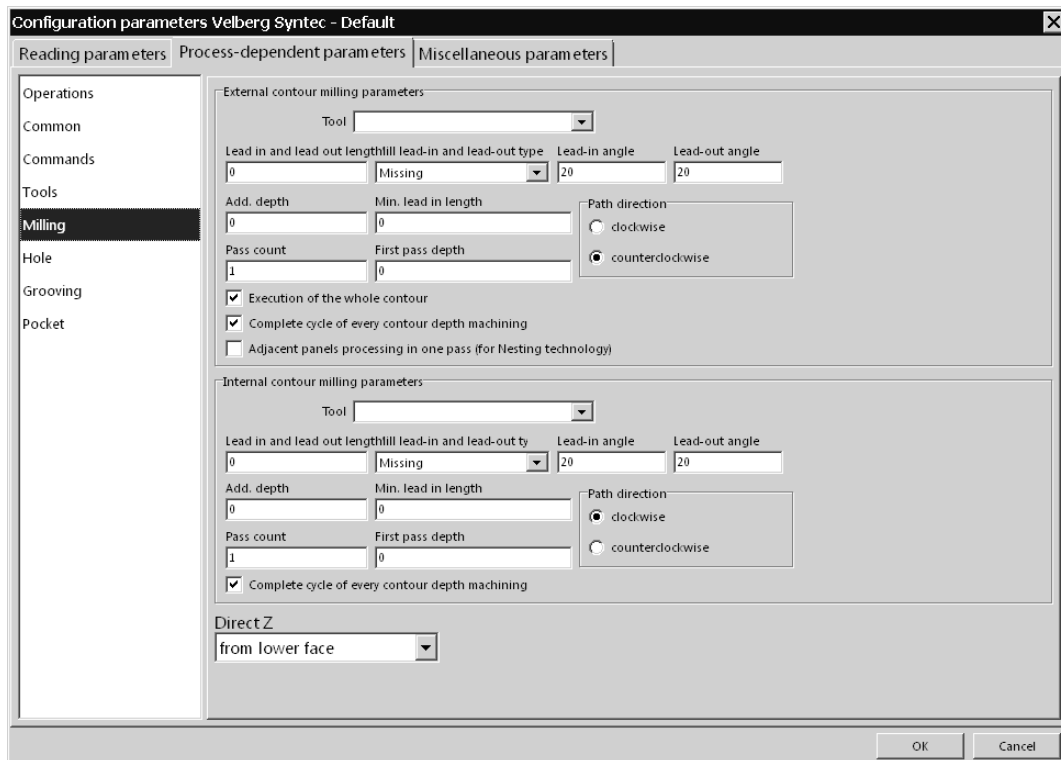


Fig. 10.679.

Tool combo box allows to select a milling tool name created in **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.680).

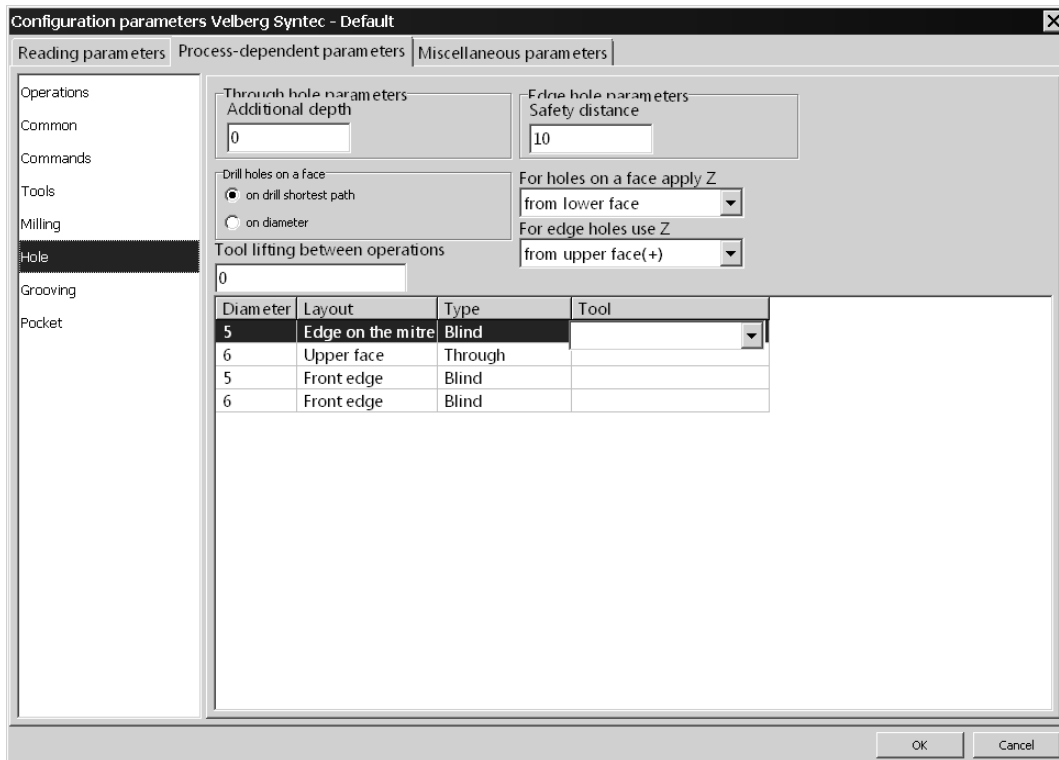


Fig. 10.680.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slot parameters

Grooving section control elements (fig. 10.681) allow to set slots processing parameters.

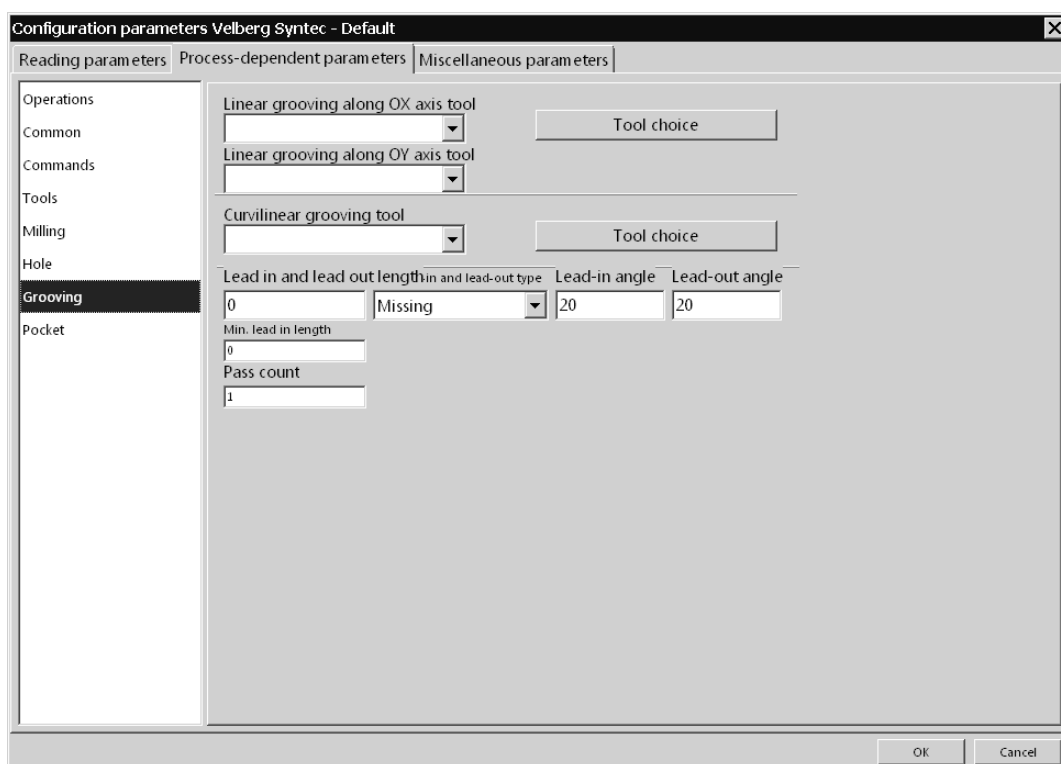


Fig. 10.681.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.682).

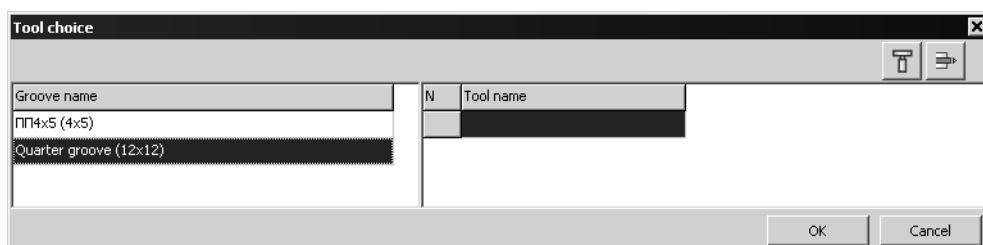


Fig. 10.682.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool to the slot created in **Tools** section.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.683) allow to set pocket processing parameters.

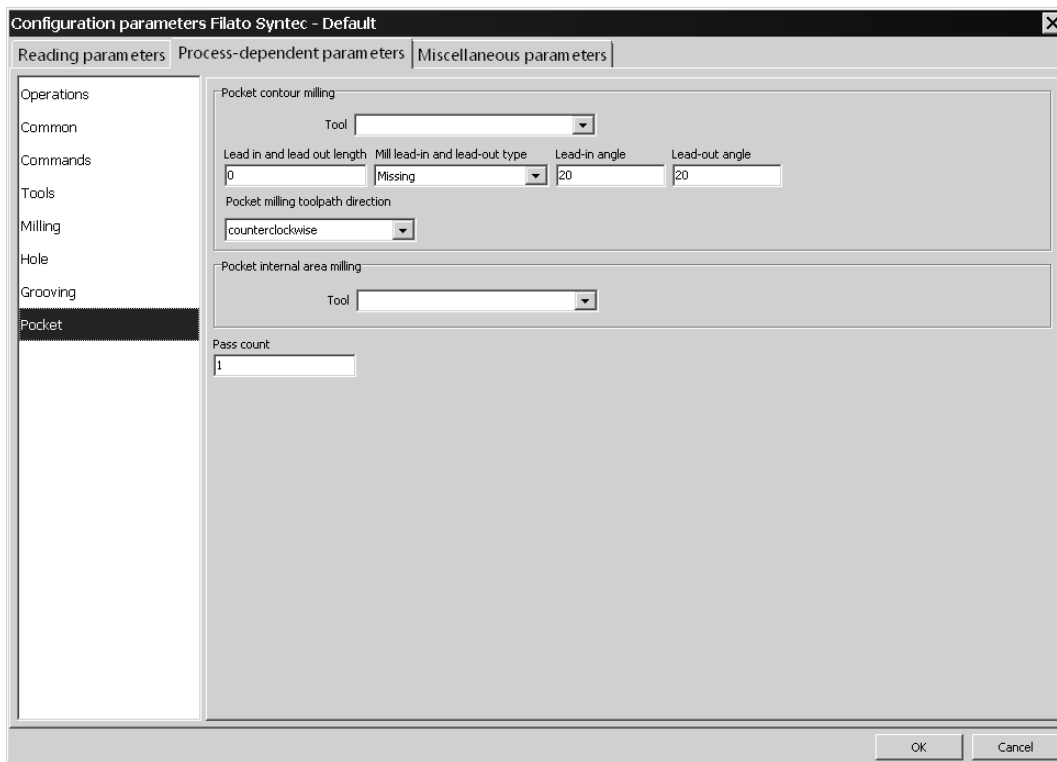


Fig. 10.683.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.37.3. Weihong NCStudio



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.684) allow to specify operations that can be used in control program.

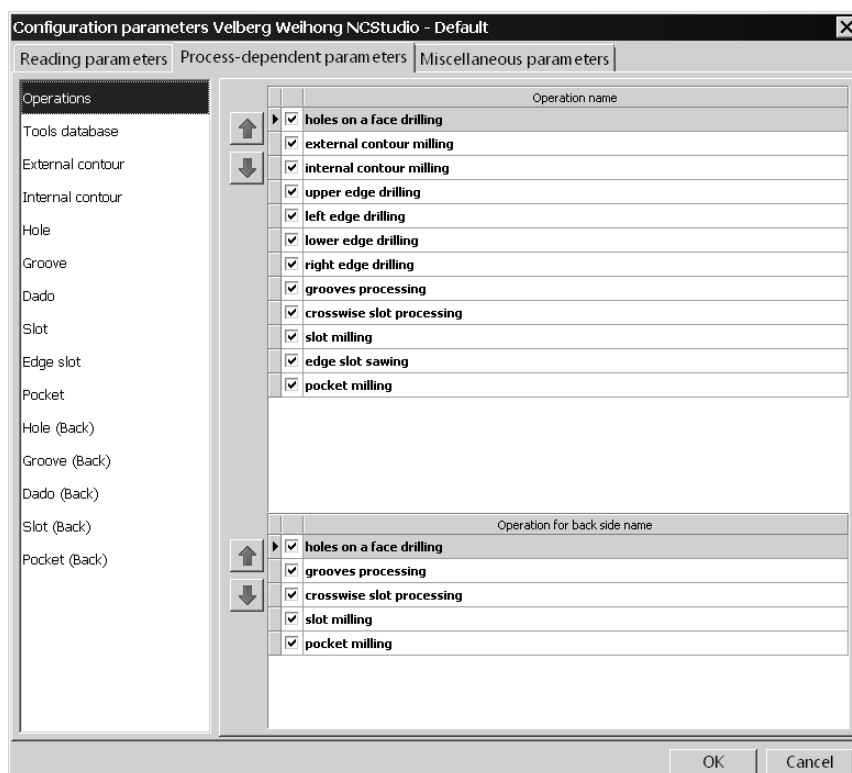


Fig. 10.684.

Usable tools

Tools database section control elements (fig. 10.685) allow to create a list of tools available for use in control programs.

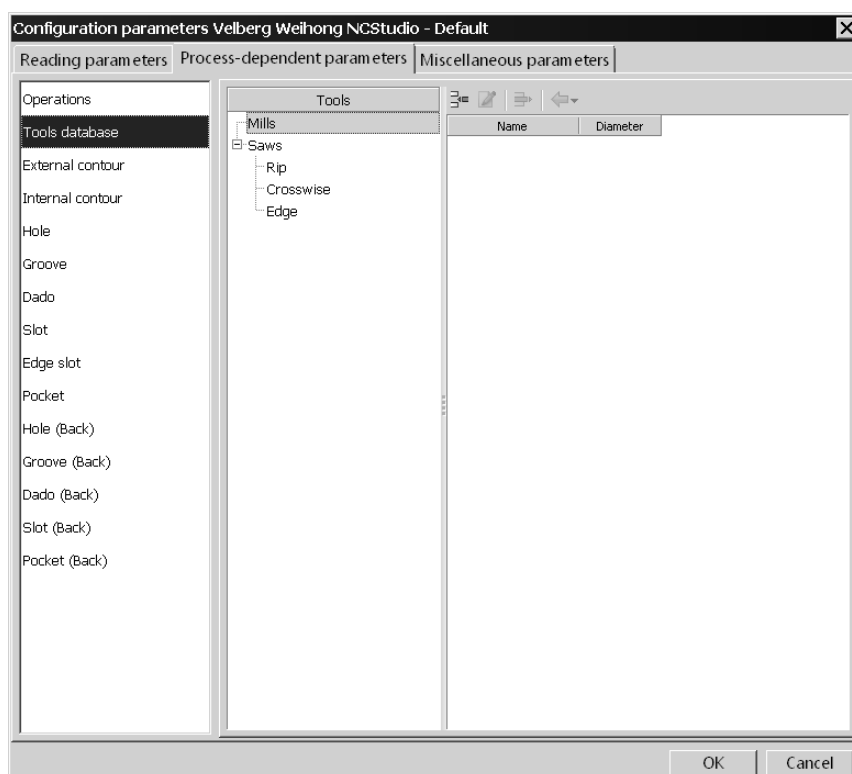


Fig. 10.685.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.686).

Tool	
Name	
Diameter	0
Width	0
OK Cancel	

Fig. 10.686.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **External contour** and **Internal contour** sections respectively (fig. 10.687).

Configuration parameters Velberg Weihong NCStudio - Default		
Reading parameters Process-dependent parameters Miscellaneous parameters		
Operations	Tool	
Tools database	Correction	In center
External contour	Path direction	Clockwise
Internal contour	Tool lead-in	Missing
Hole	Tool lead-out	Missing
Groove	Multipassage	No
Dado	Save tool name to file	No
Slot	Additional depth	0
Edge slot	Execution of the whole contour	No
Pocket		
Hole (Back)		
Groove (Back)		
Dado (Back)		
Slot (Back)		
Pocket (Back)		
OK Cancel		

Fig. 10.687.

Control elements in sections allow to select the tool, correction types, lead-in, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.688), holes from back side – in **Hole (Back)** section in the same way.

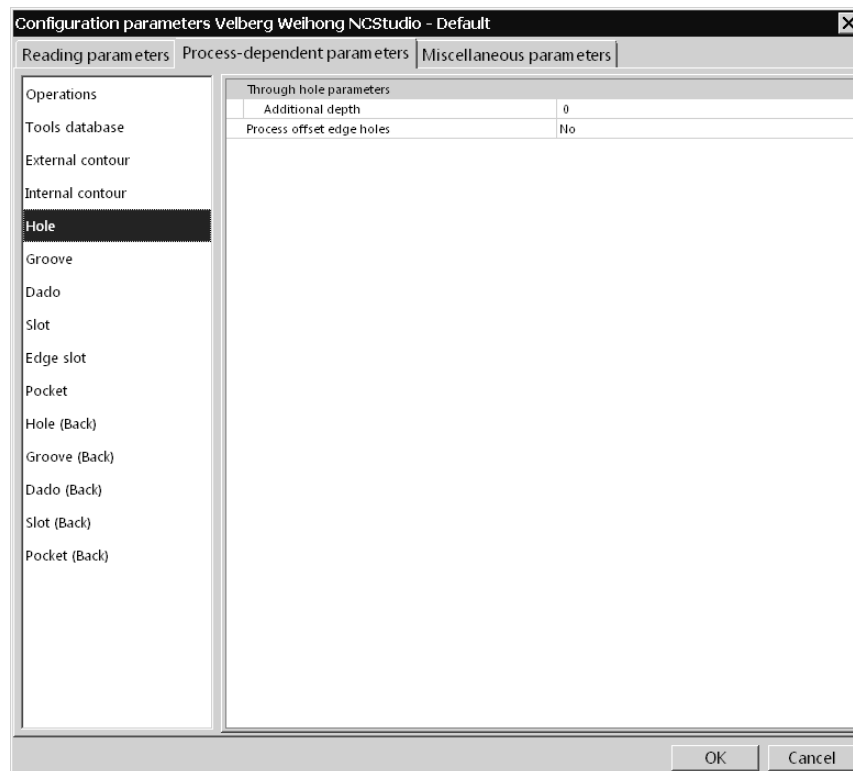


Fig. 10.688.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.689), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

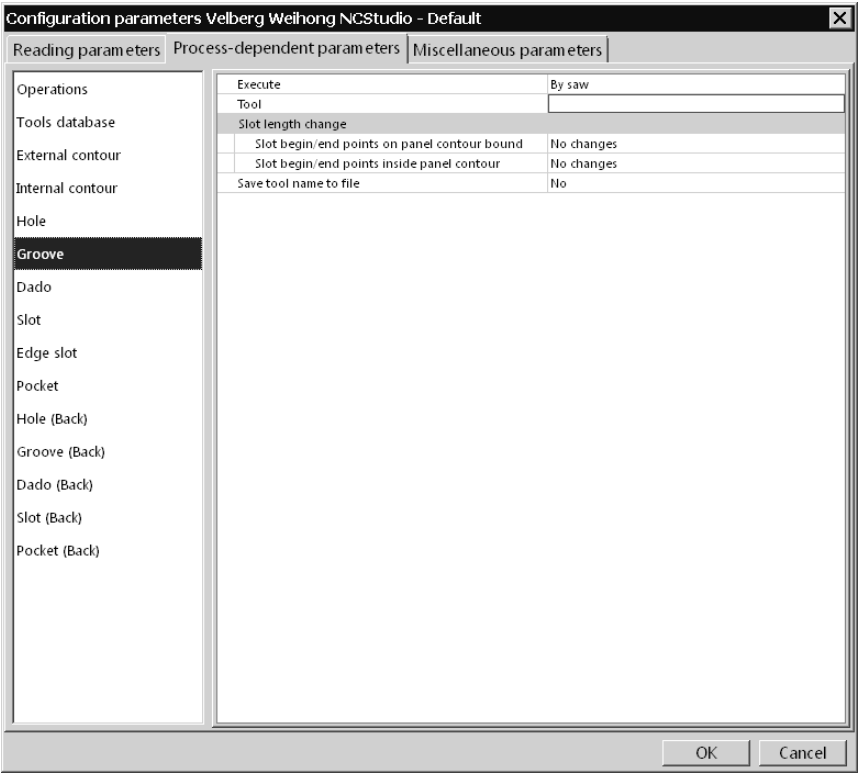


Fig. 10.689.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.690), from the back panel side – in **Slot (Back)** section.

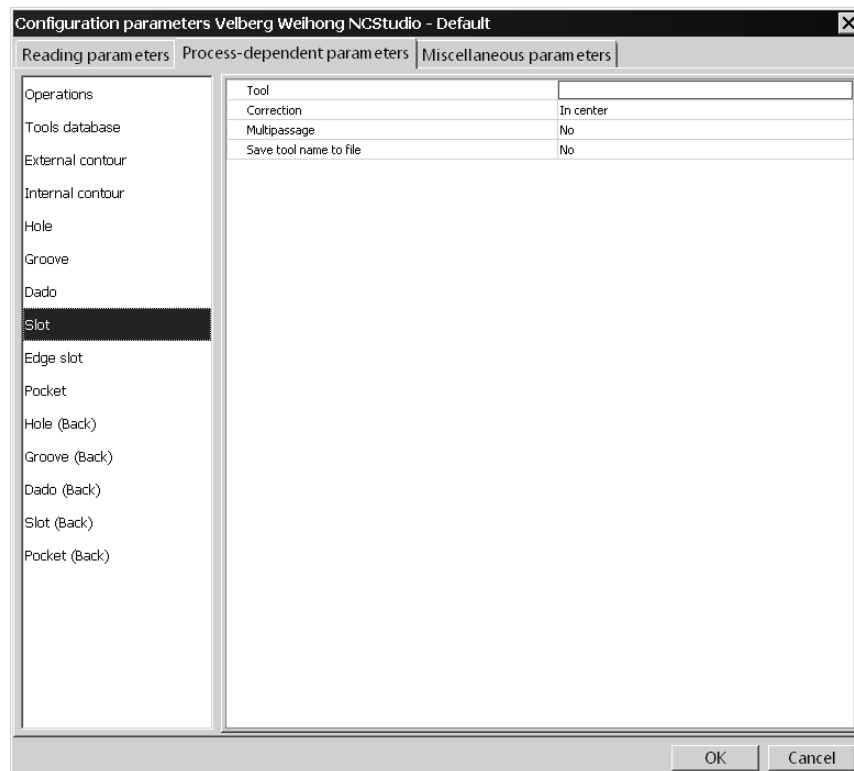


Fig. 10.690.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.691), on the back panel side – in **Pocket (Back)** section.

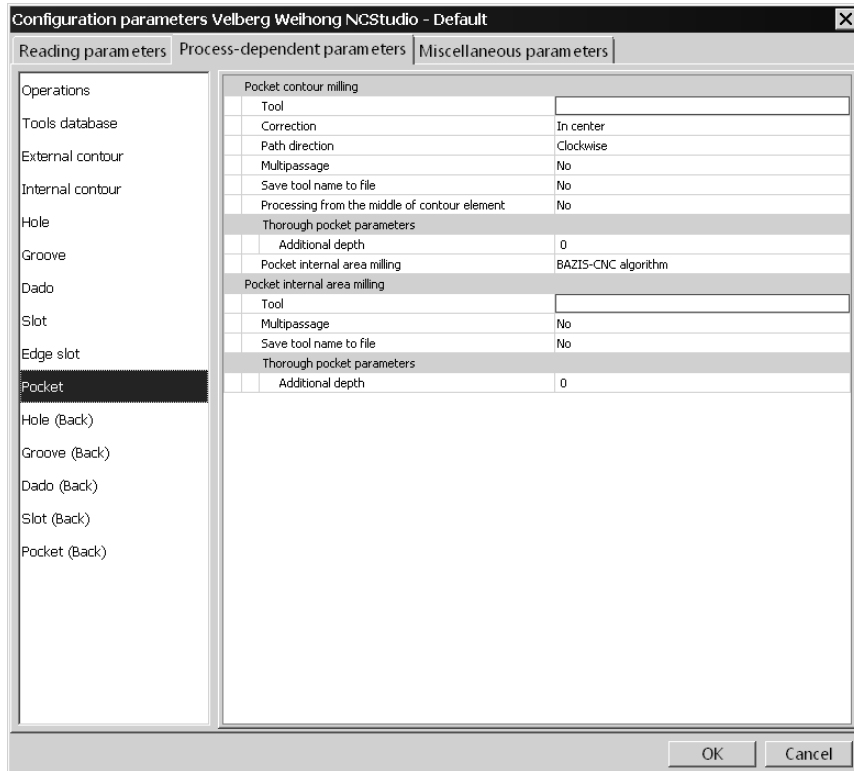


Fig. 10.691.

Tool combo box in **Pocket contour milling** and **Pocket internal area milling** groups allow to select milling tool name for corresponding objects from database. For pocket contour select milling execution direction, correction type and multipassage use. For internal pocket part – correction type and multipassage use.

Setup completion

To complete control programs creation parameters setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.38. Vitap

10.38.1.Vitap TPA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.692) allow to specify the operations that can be used in control program.

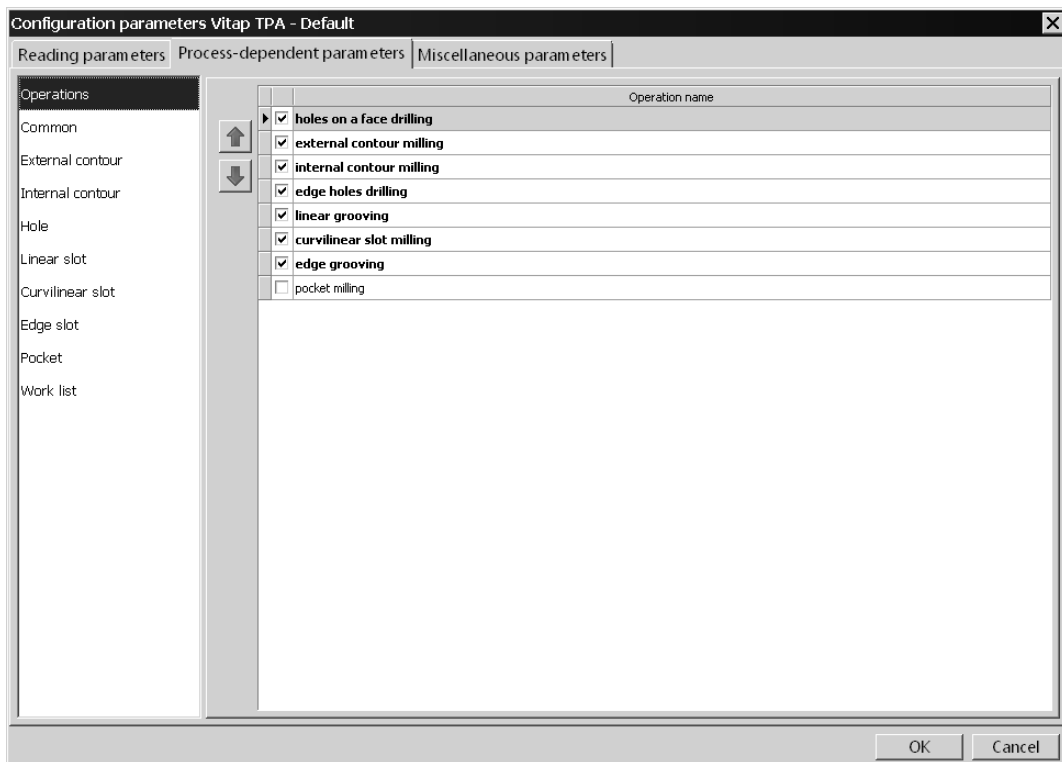


Fig. 10.692.

Common parameters

Common parameters setup is performed in **Common** section (fig. 10.693).

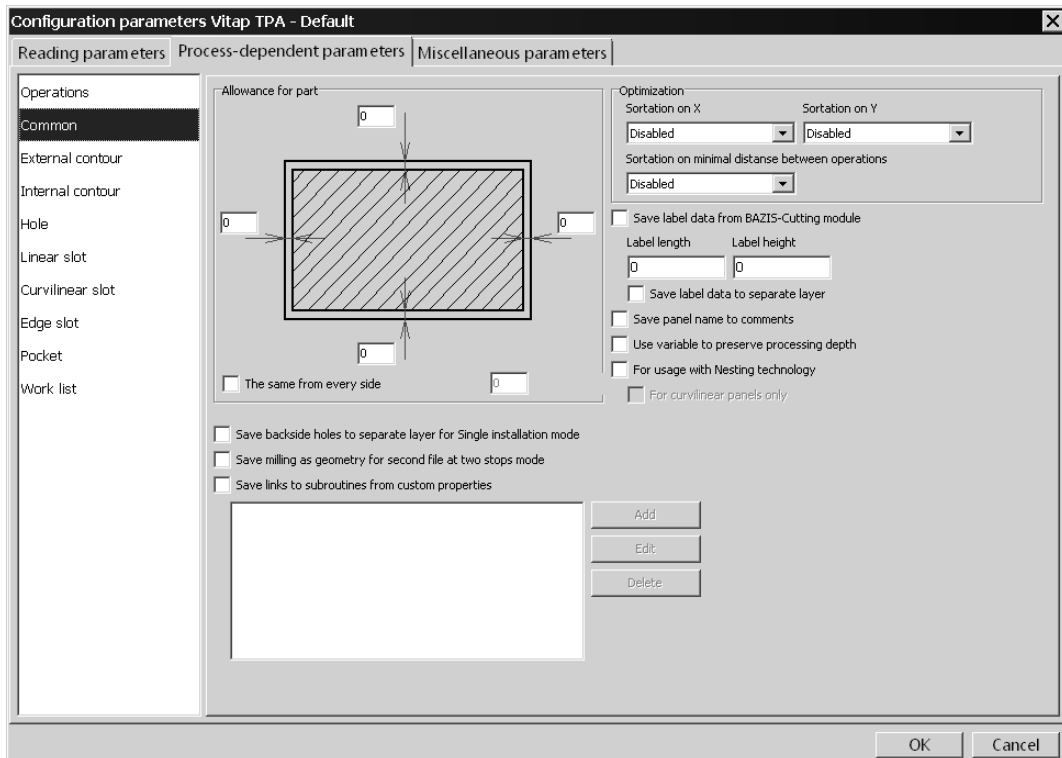


Fig. 10.693.

Tab control elements allow to control the saving of optimization parameters, label data, etc. **Allowance for part** section control elements allow to set allowance values. Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, input field for the same allowance for every panel edge will become available.

Contour parameters

Contour milling parameters setup is performed in **Outer contour** (fig. 10.694) and **Internal contour** sections.

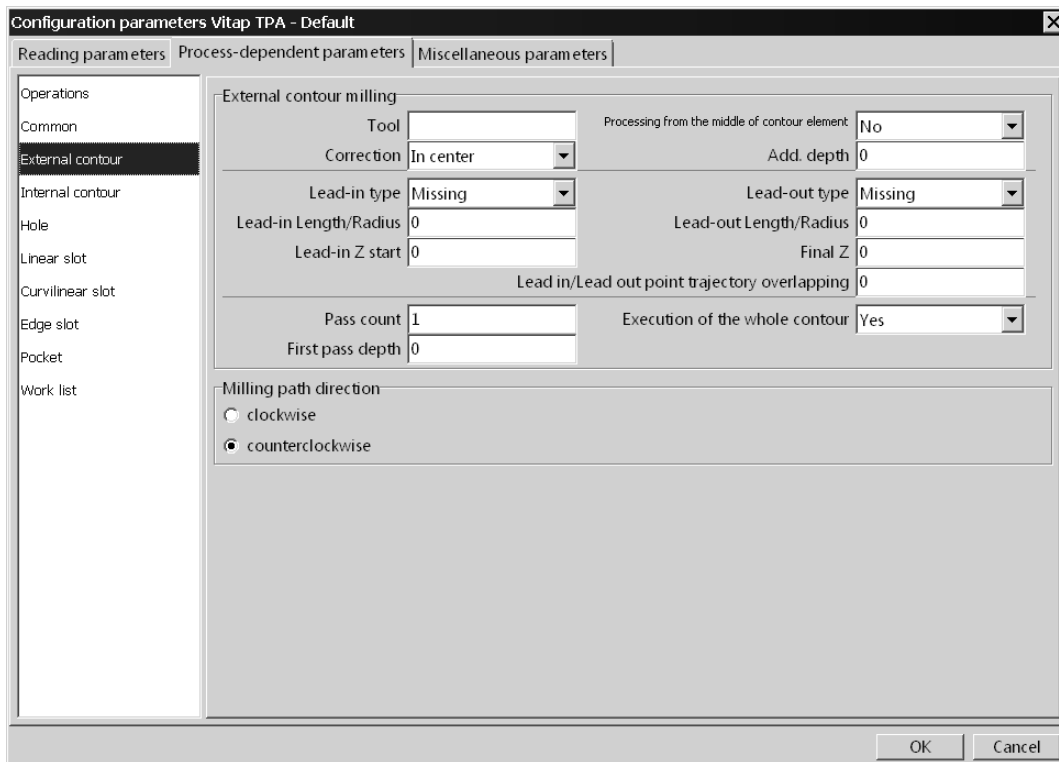


Fig. 10.694.

Finish machining and **Rough machining** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Rough machining** option allows to control this operation.

Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Holes section control elements (fig. 10.695) allow to setup parameters of drilling operations.

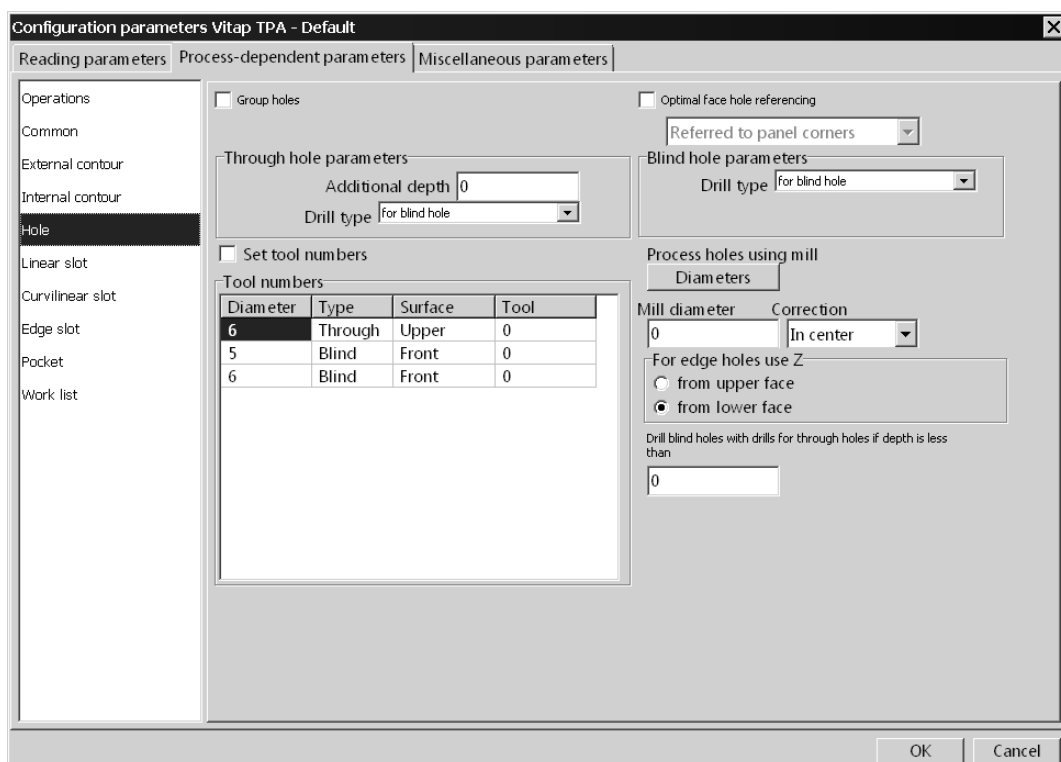


Fig. 10.695.

If several holes are located on the same straight line and the distance between them (step) is the same, it is possible to combine them and reduce the program volume. For this purpose enabled **Group holes using macro** option. Macros files names should be selected from a combo box.

Through holes can be drilled with drills for through as well as for blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled.

You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameter** button. **Hole diameters** dialogue will appear on the screen (fig. 10.696).

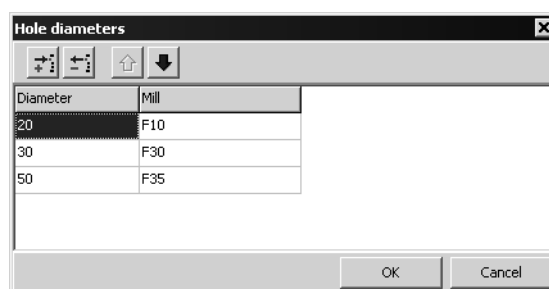


Fig. 10.696.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To complete setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter.

Linear slot parameters

Linear slots parameters setup is performed in **Linear slot** section (fig. 10.697).

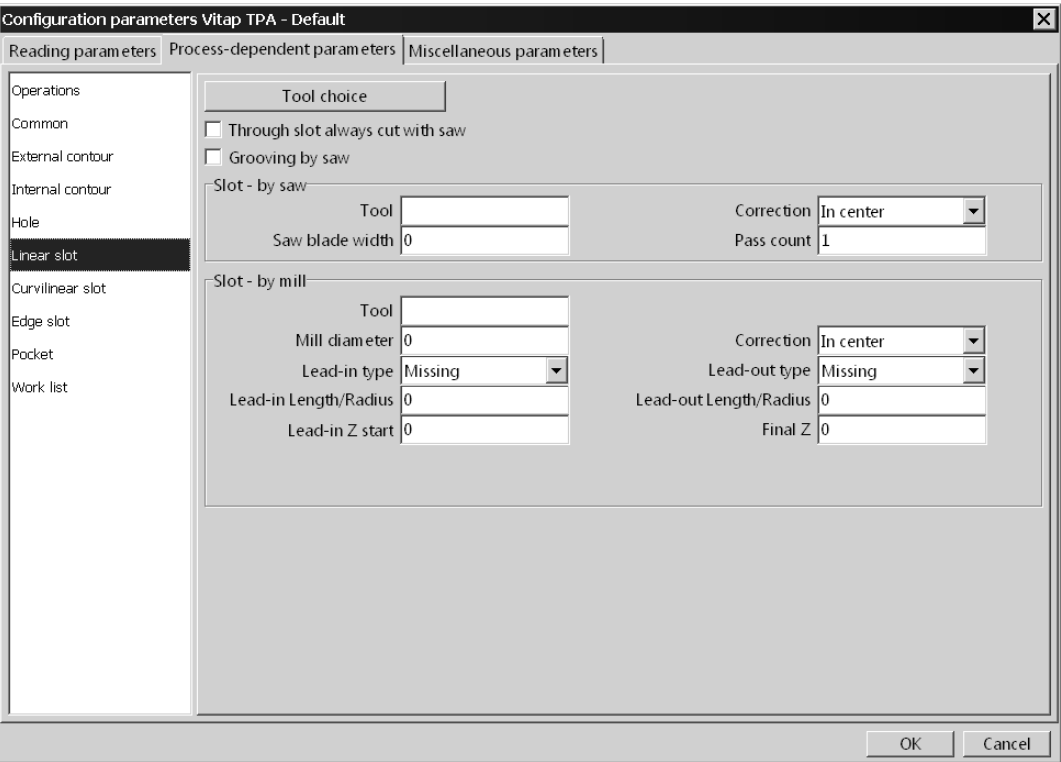


Fig. 10.697.

Through slot always cut with saw and **Grooving by saw** options allow to set through slots direction that will be processed with a saw. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.698).

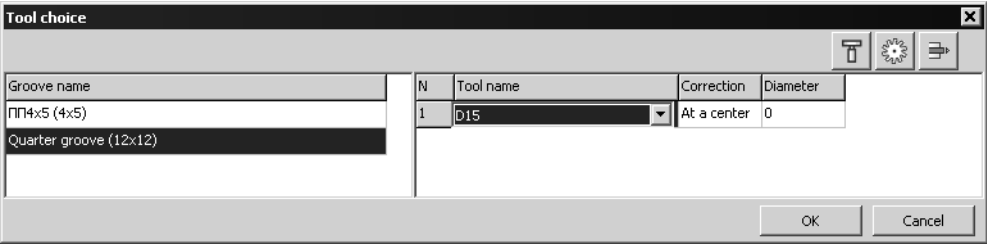


Fig. 10.698.



Slot name table contains all slots names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing. The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.699).

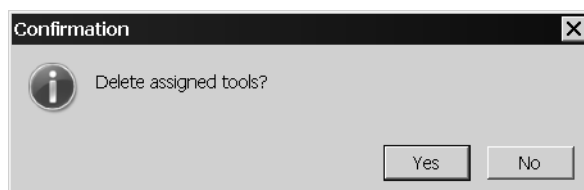


Fig. 10.699.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. To complete parameters setup, push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear slots parameters setup is performed in **Curvilinear slot** section (fig. 10.700).

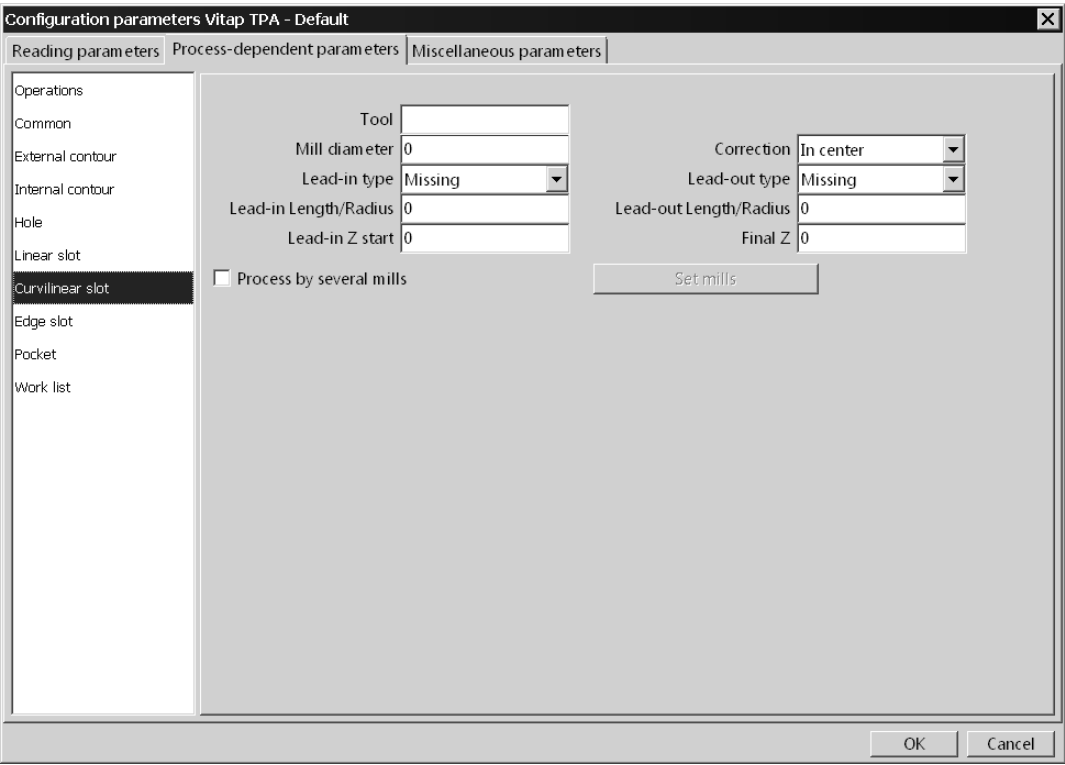


Fig. 10.700.

Section control elements allow to setup control programs generation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.701).

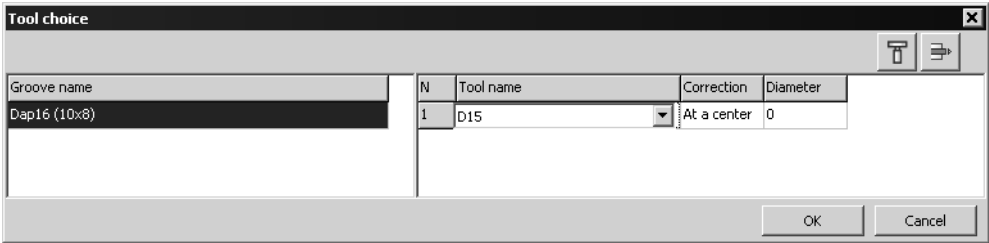


Fig. 10.701.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slot parameters are set in **Edge slot** section (fig. 10.702).

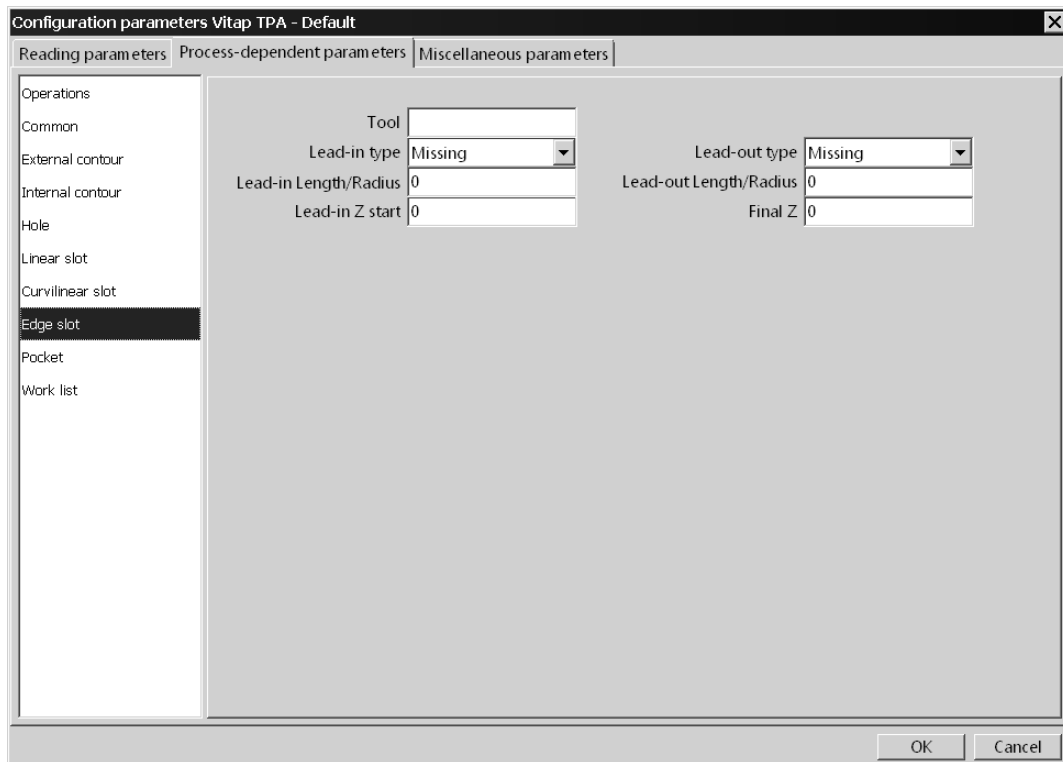


Fig. 10.702.

Section control elements allow to setup generation parameters of control programs for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.703) allow to setup pocket processing parameters.

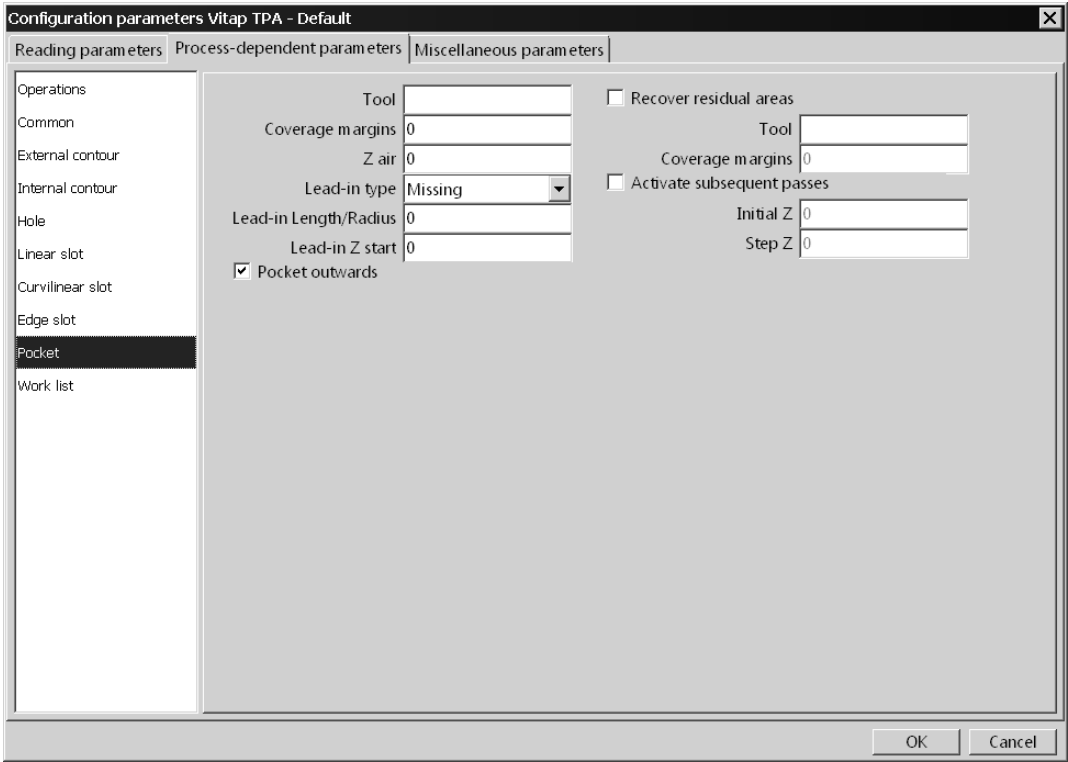


Fig. 10.703.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Work lists

Work list section control elements (fig. 10.704) allow to setup parameters of lists use.

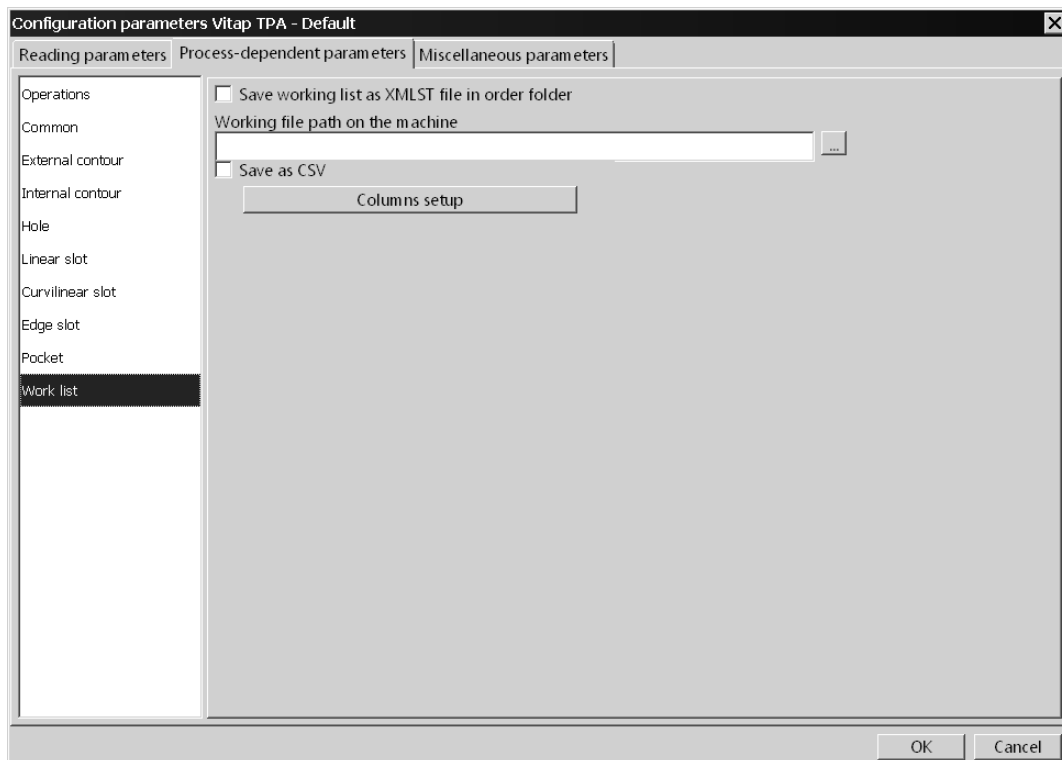


Fig. 10.704.



Save working list as XMLST file in order folder option allows to control the corresponding document creation. **Browse** button allows to set a path to working files. After pushing the button **Browse** dialogue will appear on the screen.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.38.2.Vitap Bridge ICE112



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.705) allow to set control programs common parameters.

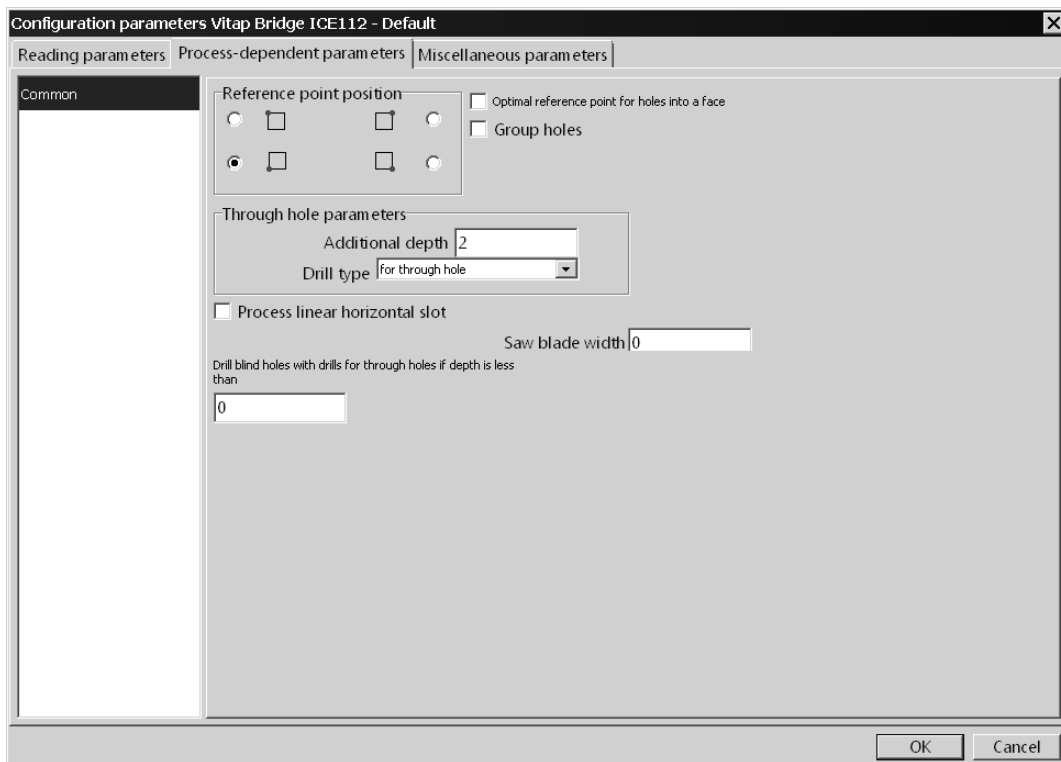


Fig. 10.705.

Reference point position group variants allow to select panel angle as origin of coordinates. If **Optimal reference point for holes into a face** option is enabled, referencing will be set for each hole individually. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional through hole depth** field allows to take drill center point into account.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.38.3.Vitap Point



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common section control elements (fig. 10.706) allow to set control programs common parameters.

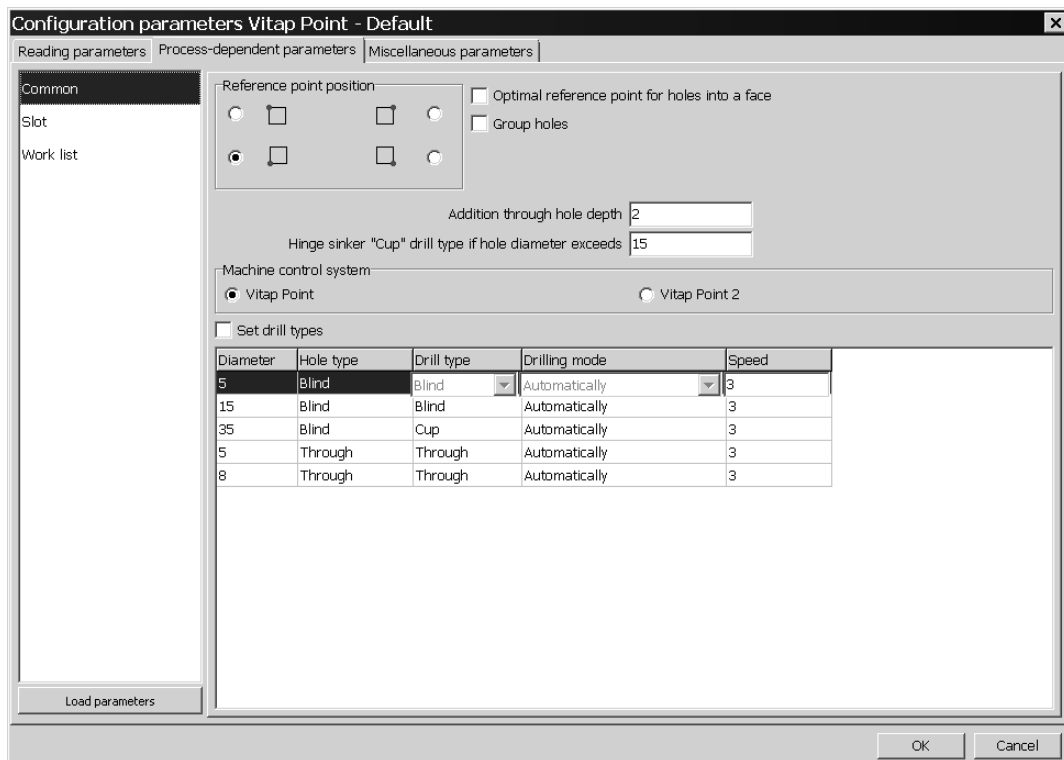


Fig. 10.706.

Reference point position group variants allow to select panel angle as origin of coordinates. If **Optimal reference point for holes into a face** option is enabled, referencing will be set for each hole individually. If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional through hole depth** field allows to take drill center point into account.

Holes can be bored with a common drill as well as with a cup borer type. Tool selection is specified with a hole diameter. **Hinge sinker "Cup" drill type if hole diameter exceeds** field allows to set hole diameter that will specify automatic tool selection in the program. **Machine control system** group variants allow to select control system type. **Set drill types** option allows to set automatic borer type and boring mode selection depending on hole diameter and type. If it is enabled, drill table control elements will become available.

Linear slot parameters

Linear slots parameters are set in **Slot** section (fig. 10.707).

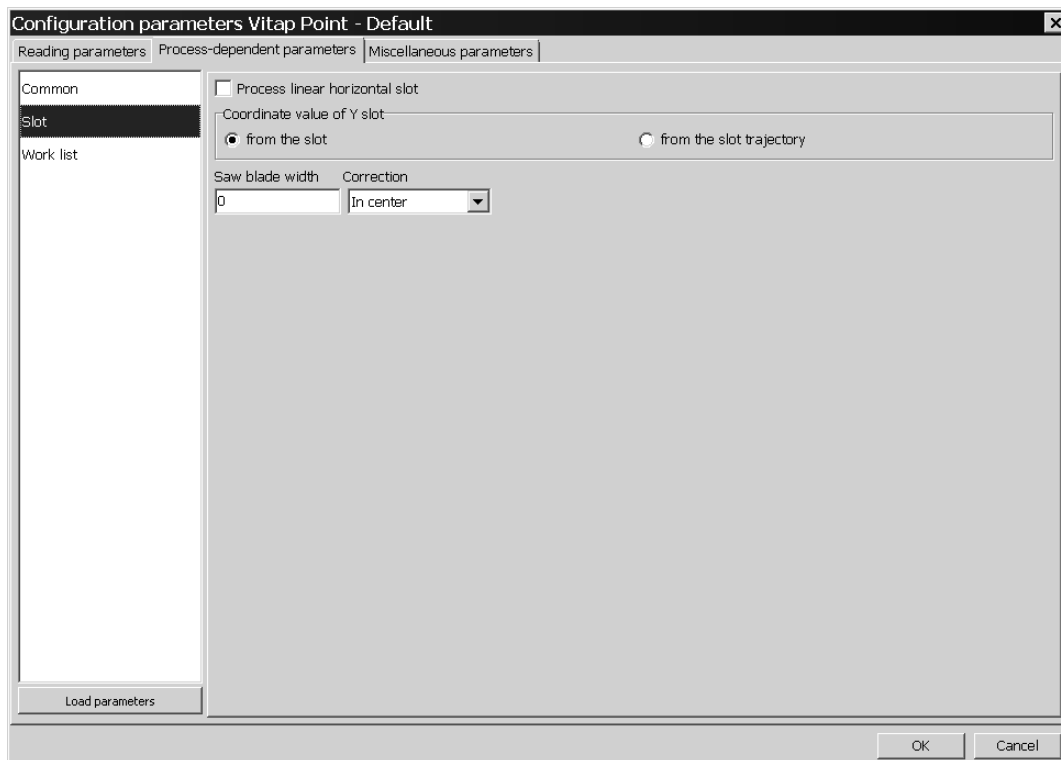


Fig. 10.707.

Process linear horizontal slot button allows to append slots processing to the program. **Coordinate value of Y slot** group variants allow to select reference line of corresponding coordinate. **Correction** combo box allows to select correction type for set saw blade width.

Work list

Work list is set in **Work list** section (fig. 10.708).

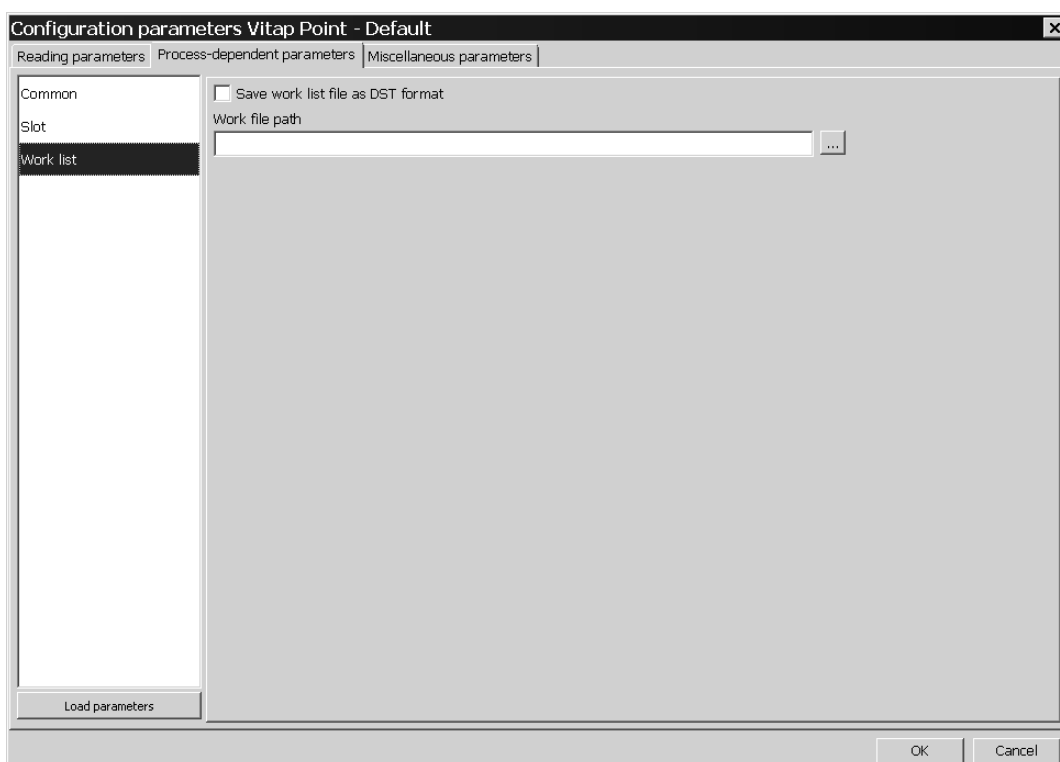


Fig. 10.708.



Save work list file as DST format option allows to control the file saving. To select a folder push **Browse** button and select it a standard windows dialogue.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.39. Yiming

10.39.1.Syntec



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.709) allow to specify operations that can be used in control program.

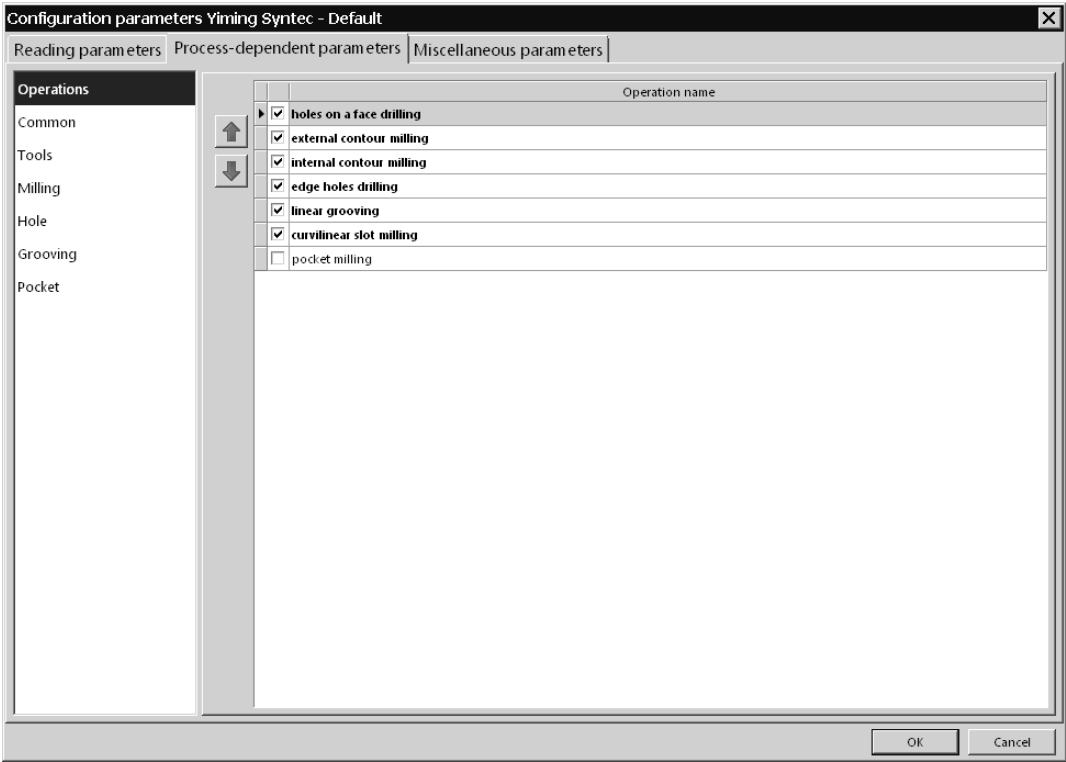


Fig. 10.709.

Common parameters

Common section control elements (fig. 10.710) allow to set common parameters of control programs creation.

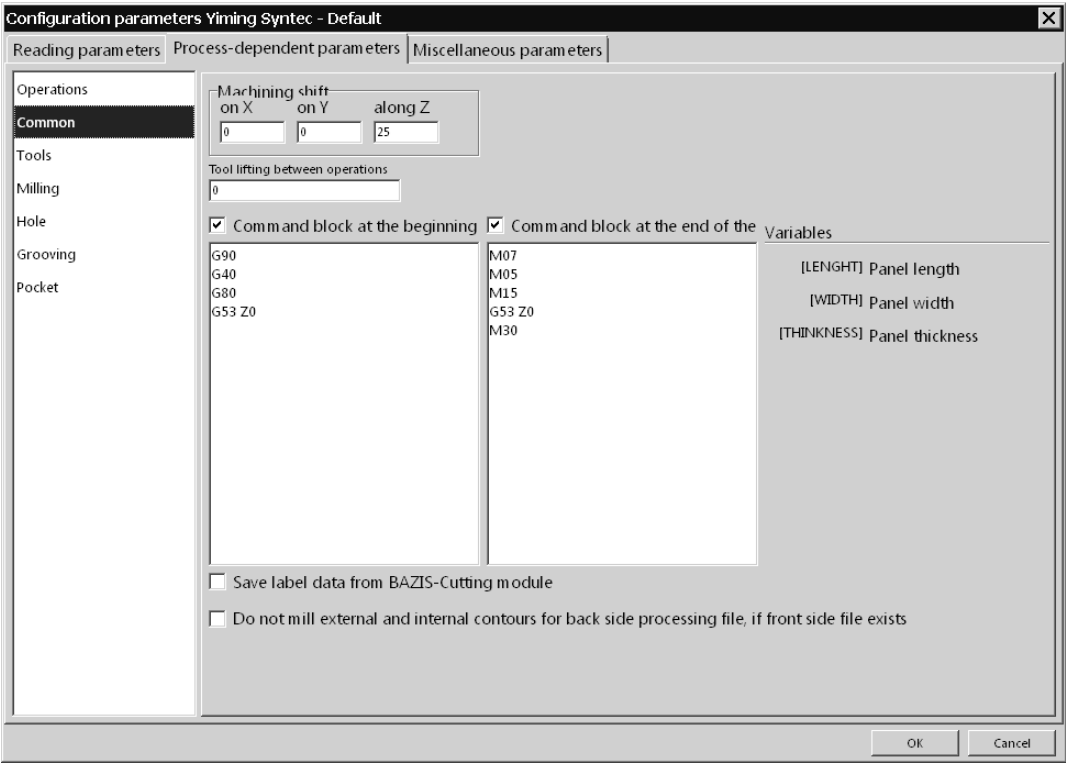


Fig. 10.710.

Machining shift group fields allow to set work area initial point moving. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas.

Save label data option allows to save the corresponding data. To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.711).

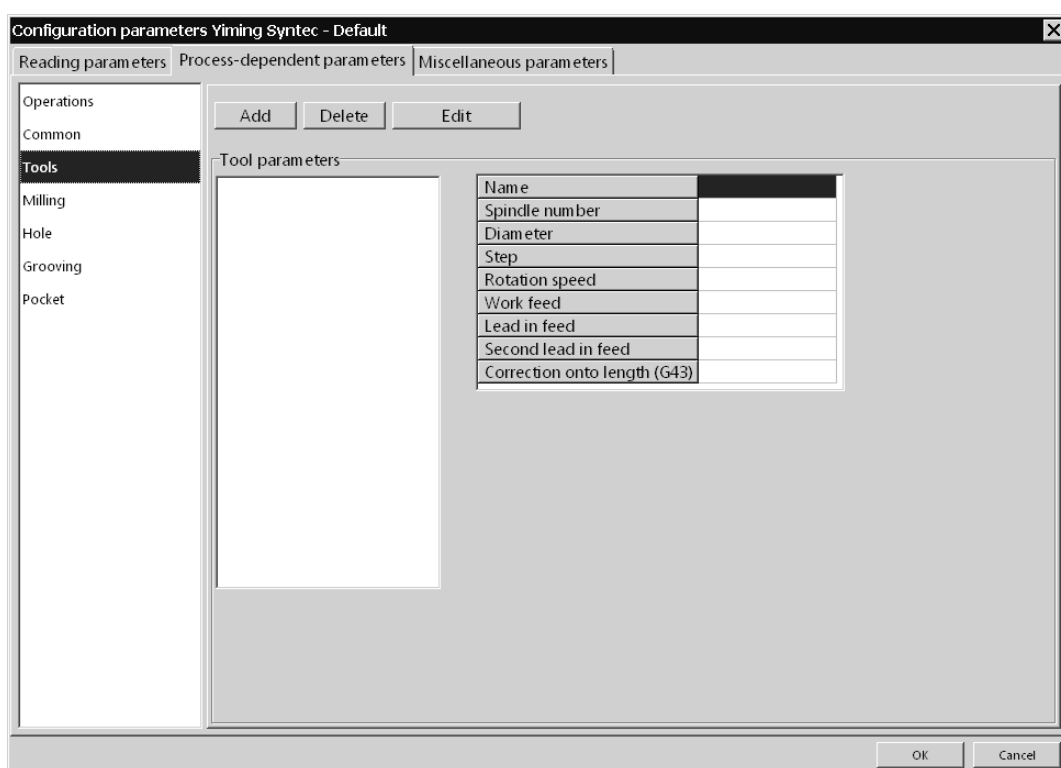


Fig. 10.711.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.712).

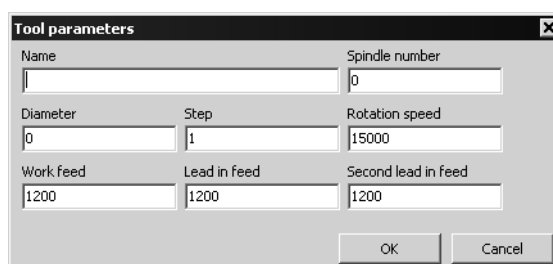


Fig. 10.712.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is shown in fig. 10.713.

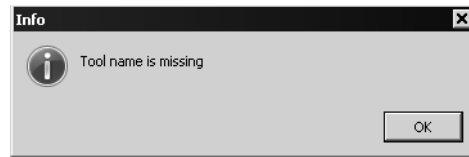


Fig. 10.713.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.712). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.714).

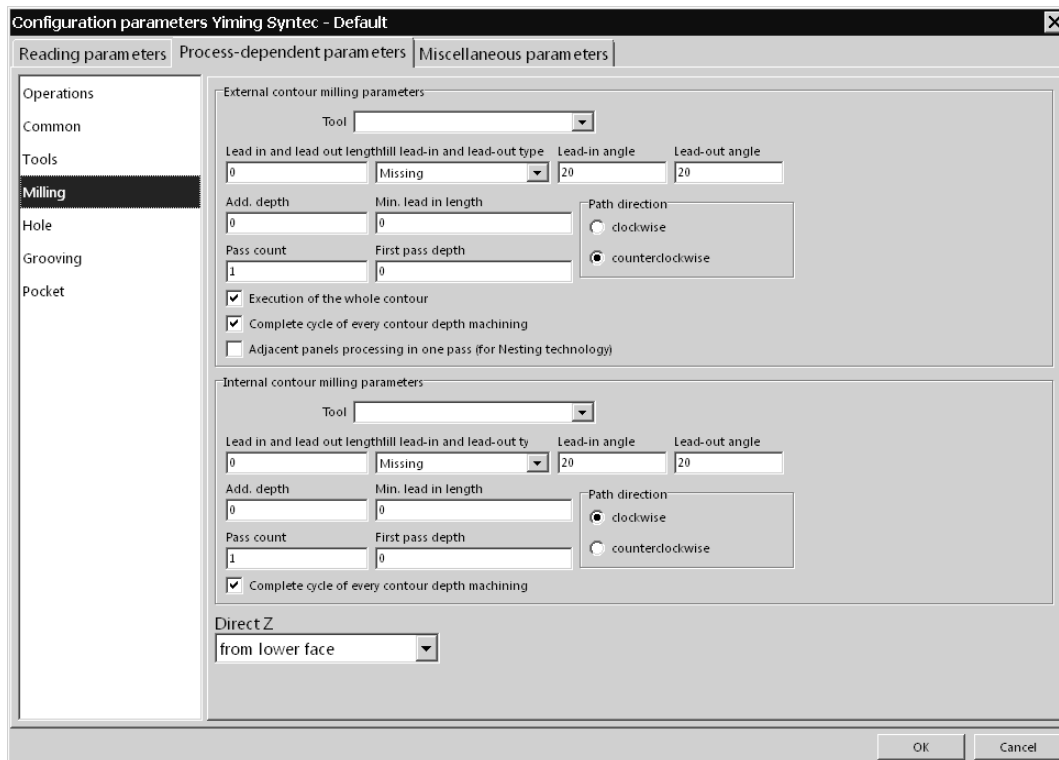


Fig. 10.714.

Tool combo box allows to select a milling tool name from the list created in **Tools** tab. Tools lists should be created beforehand in **Tools** section. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.715).

Configuration parameters Yiming Syntec - Default

Reading parameters | Process-dependent parameters | Miscellaneous parameters

Operations
Common
Tools
Milling
Hole
Grooving
Pocket

Through hole parameters
Additional depth
0

Edge hole parameters
Safety distance
10

Drill holes on a face
☒ on drill shortest path
☐ on diameter

For holes on a face apply Z
from lower face

For edge holes use Z
from upper face(+)

Tool lifting between operations
0

Diameter	Layout	Type	Tool
3	Upper face	Blind	

OK Cancel

Рис. 10.715.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Groove parameters

Grooving section control elements allow to set groove processing parameters (fig. 10.716).

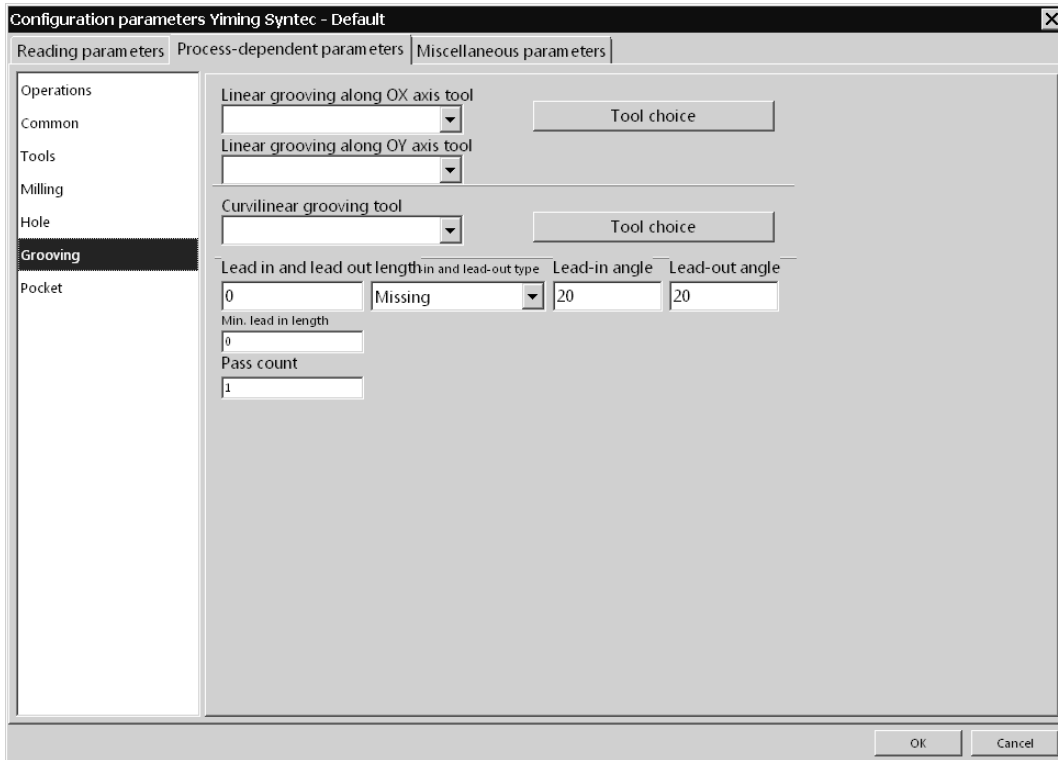


Fig. 10.716.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.717).

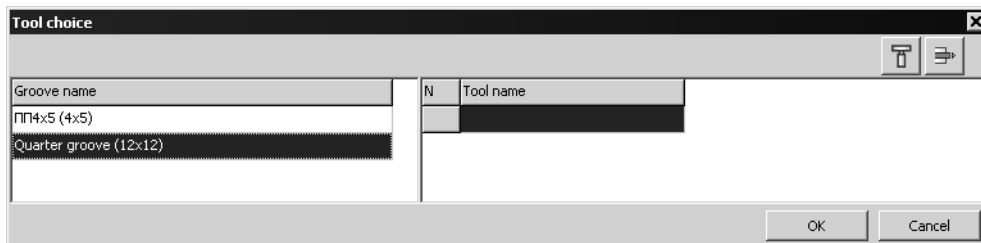


Fig. 10.717.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in **Tools** section.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.718) allow to set pocket processing parameters.

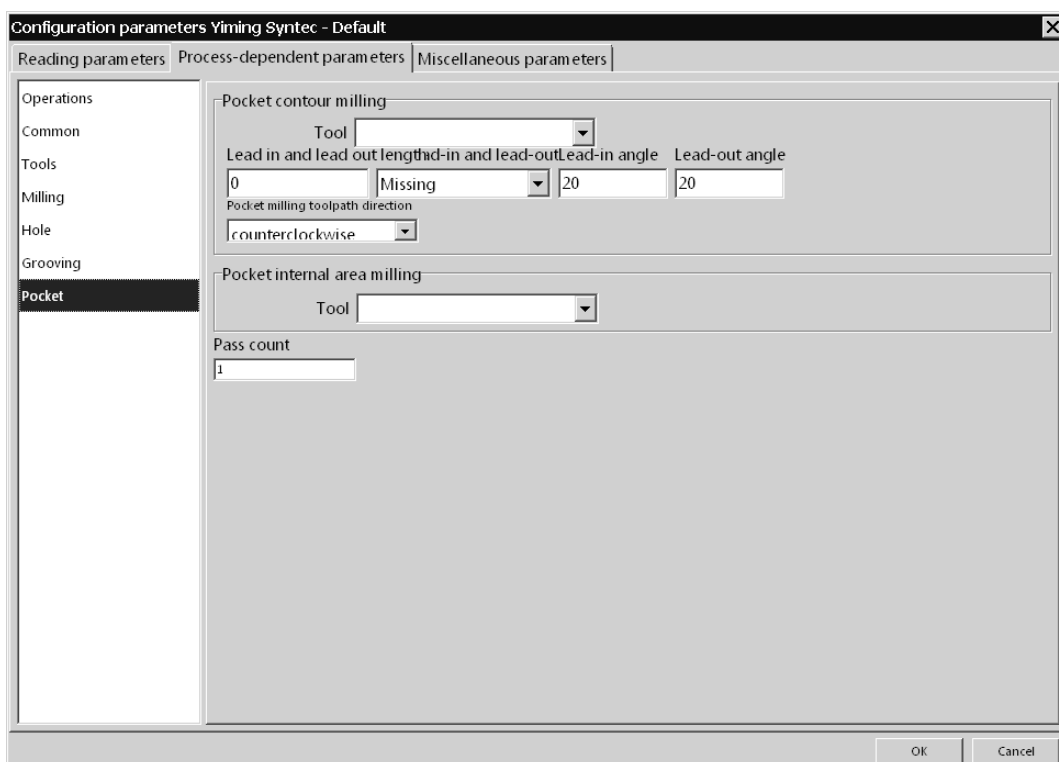


Fig. 10.718.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.40. ZaiTec

10.40.1.ZaiTec Delta



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.719) allow to specify operations that can be used in control program.

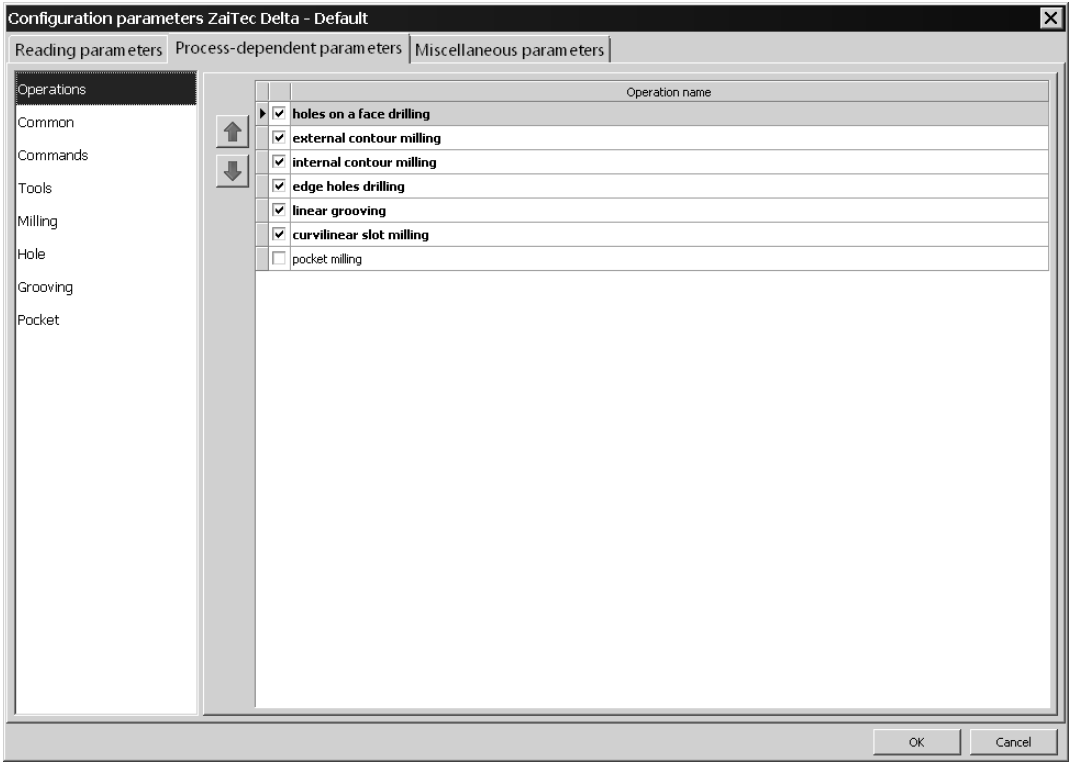


Fig. 10.719.

Common parameters

Common section control elements (fig. 10.720) allow to set common parameters of control programs creation.

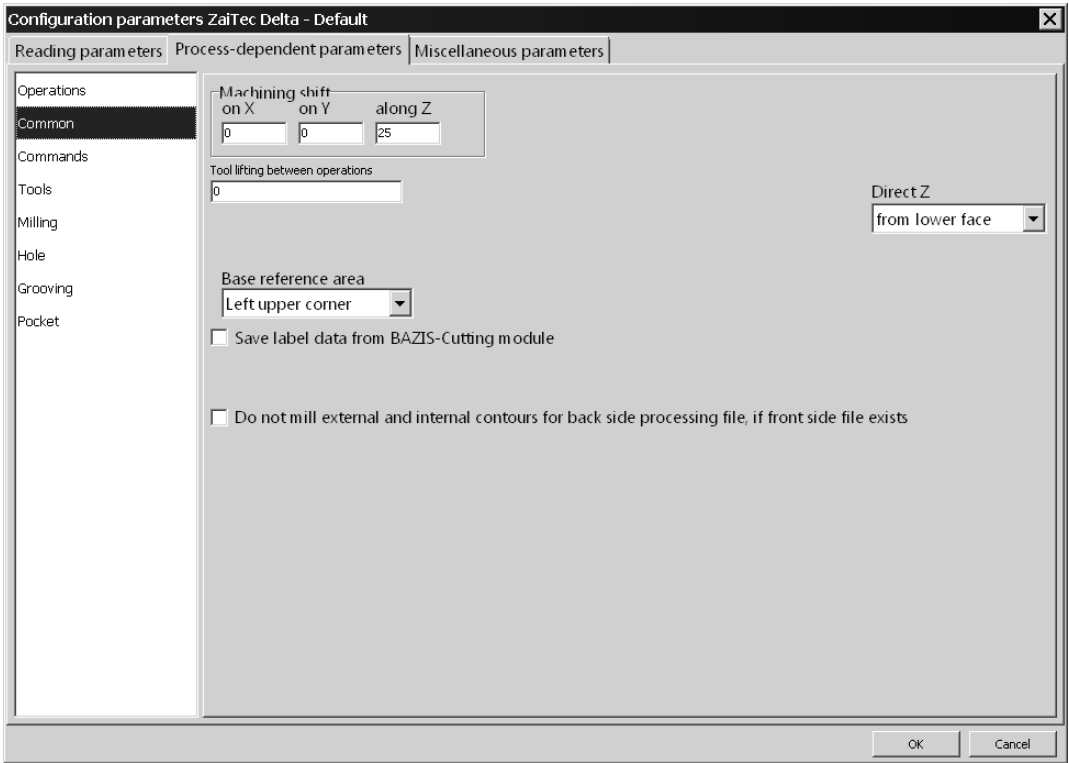


Fig. 10.720.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.721) allow to control separate commands to control program inclusion.

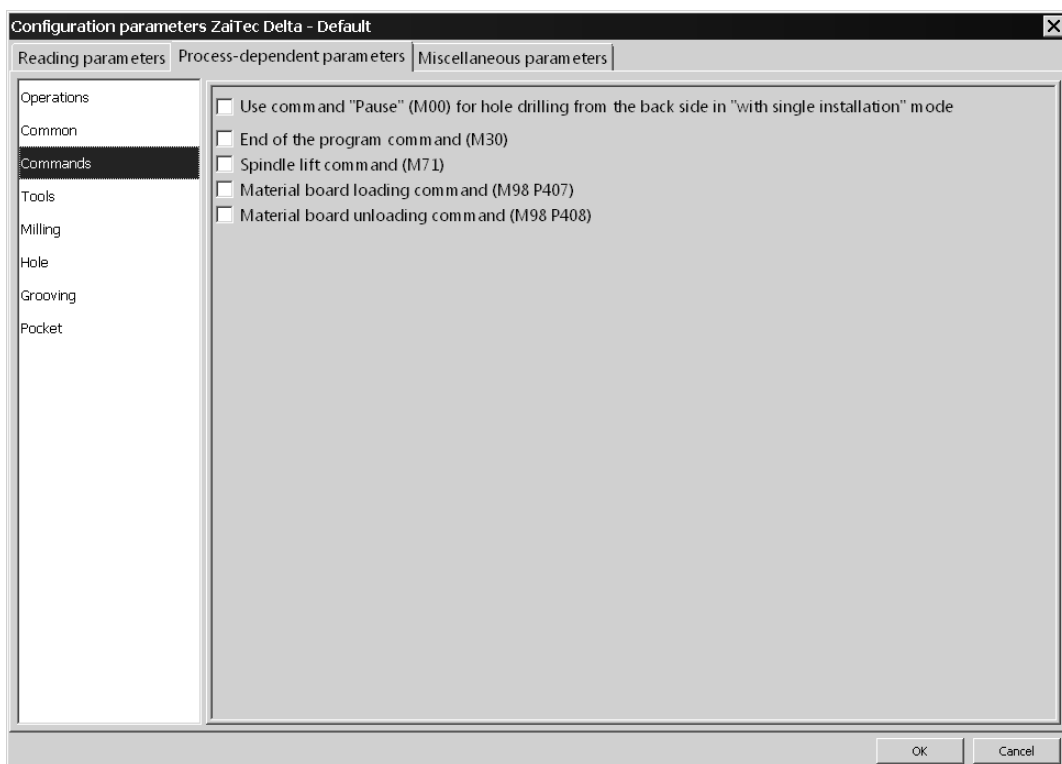


Fig. 10.721.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.722).

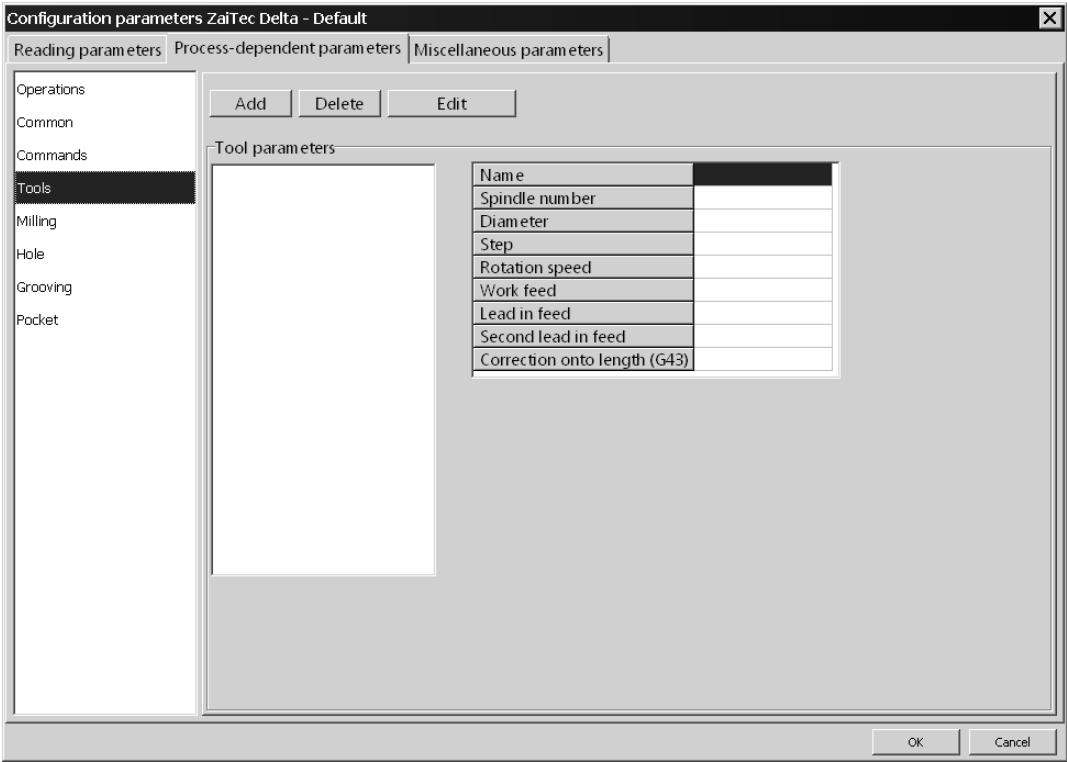


Fig. 10.722.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.723).

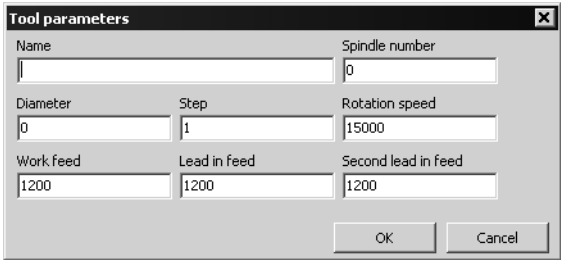


Fig. 10.723.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is (fig. 10.724).

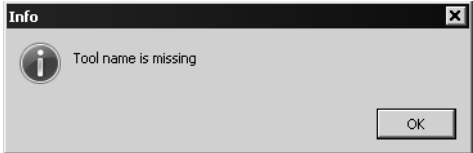


Fig. 10.724.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.723). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.725).

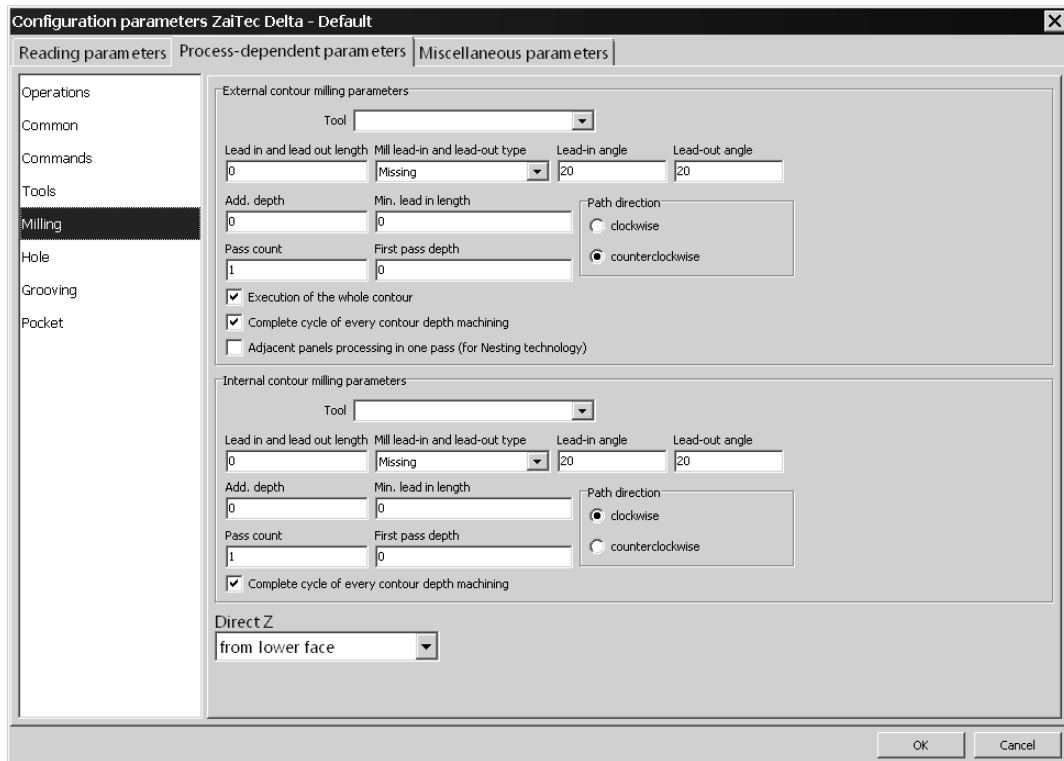


Fig. 10.725.

Tool combo box allows to select a milling tool name. Tools lists should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.726).

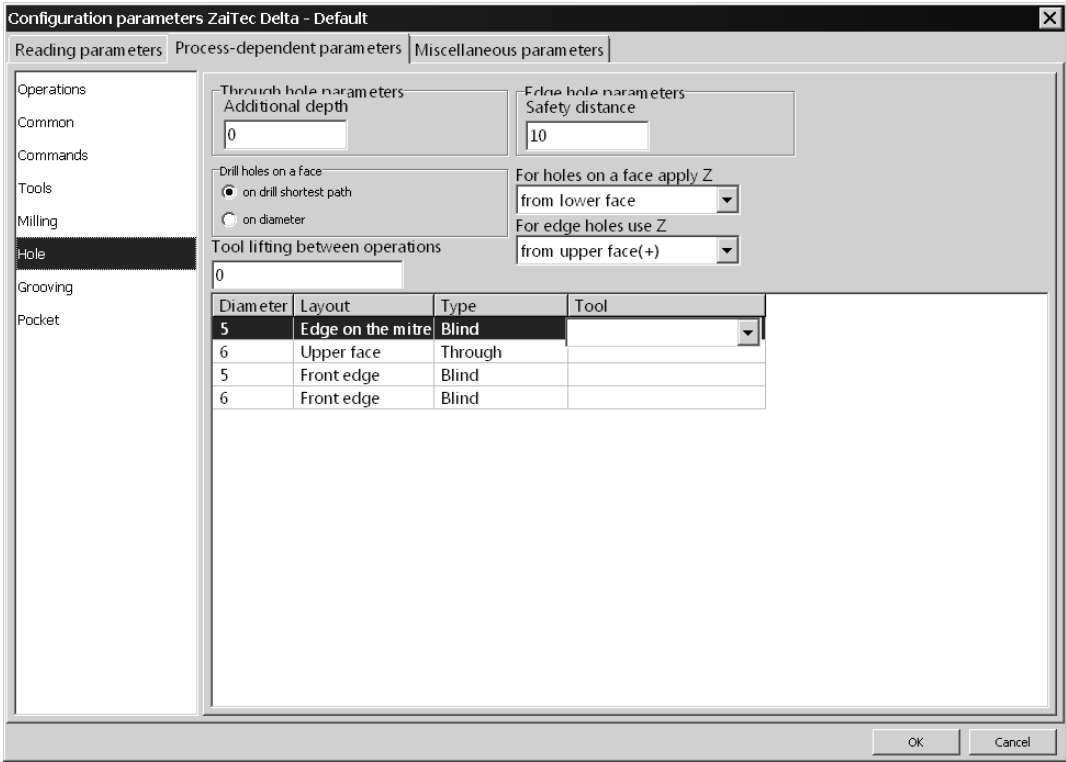


Fig. 10.726.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

Drill holes on a face group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slot parameters

Grooving section control elements (fig. 10.727) allow to set slots processing parameters.

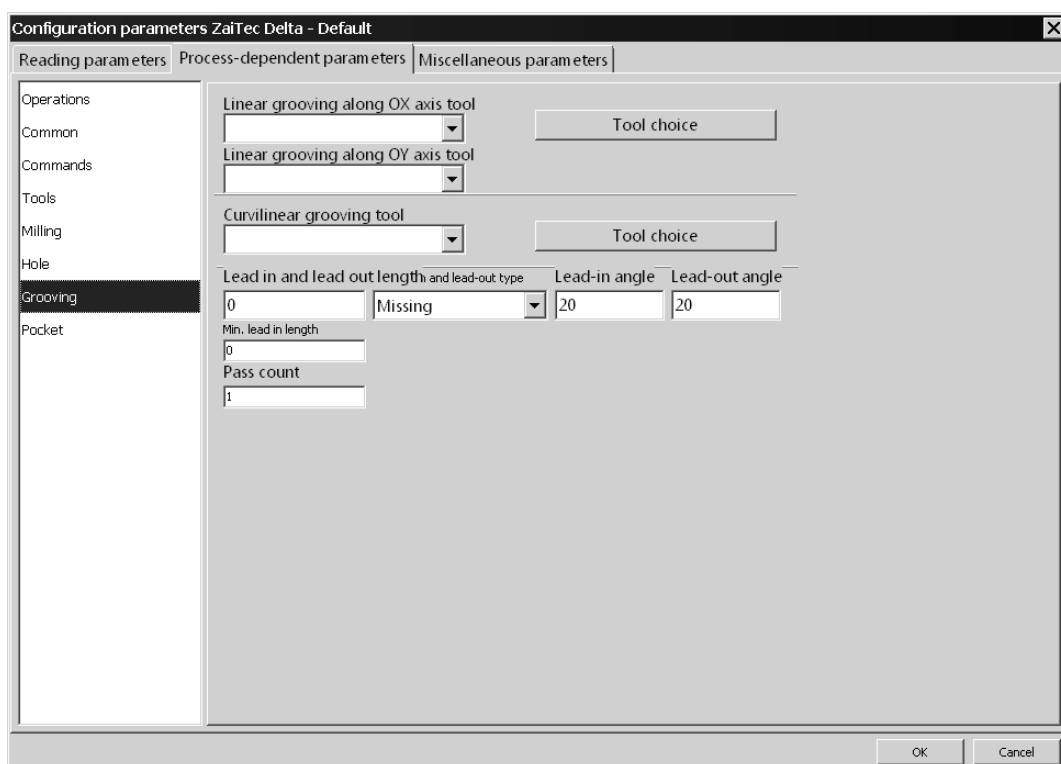


Fig. 10.727.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.728).

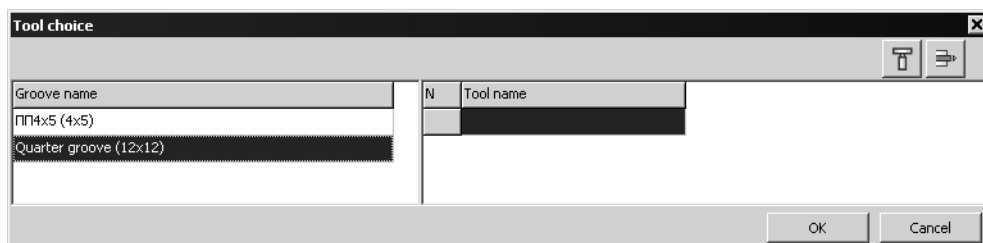
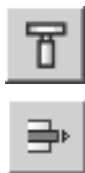


Fig. 10.728.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in **Tools** section.

Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.729) allow to set pocket processing parameters.

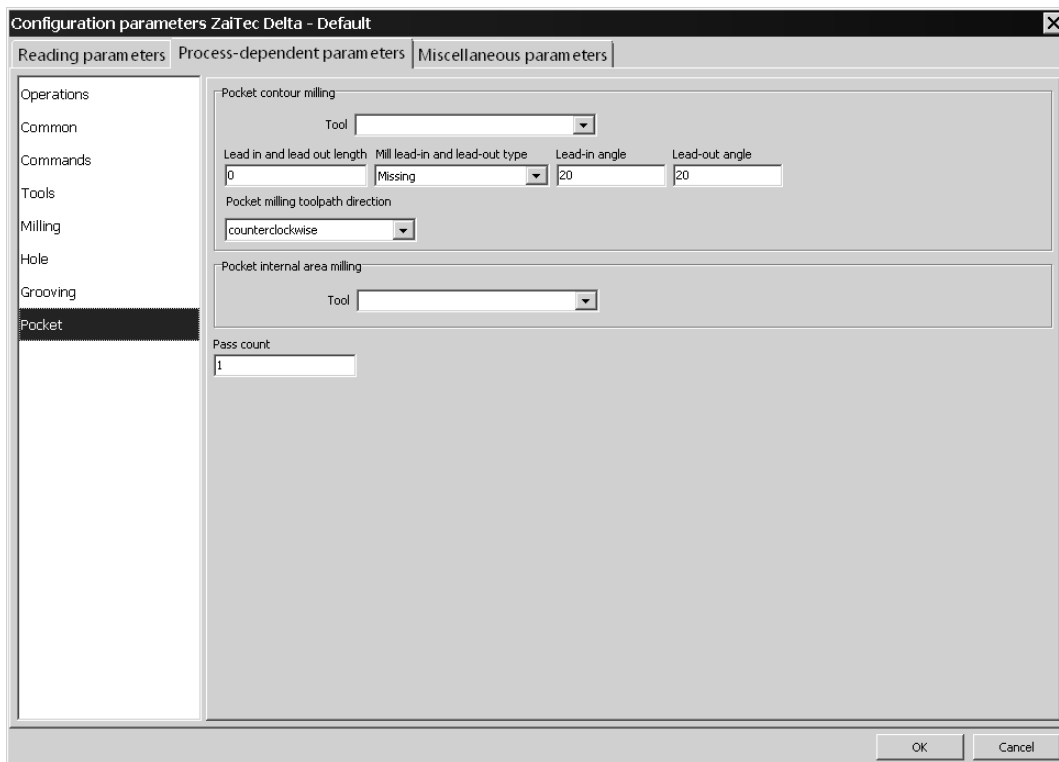


Fig. 10.729.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.40.2.Zaitec TPA



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.730) allow to specify operations that can be used in control program.

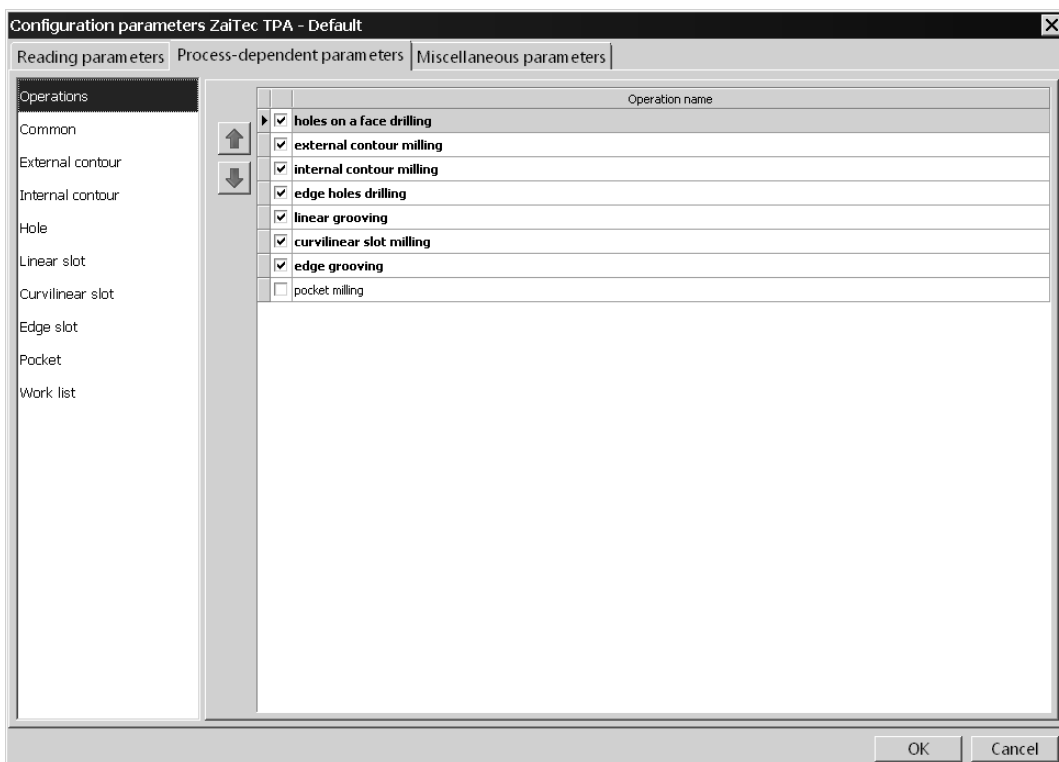


Fig. 10.730.

Common parameters

Common section control elements (fig. 10.731) allow to set common parameters.

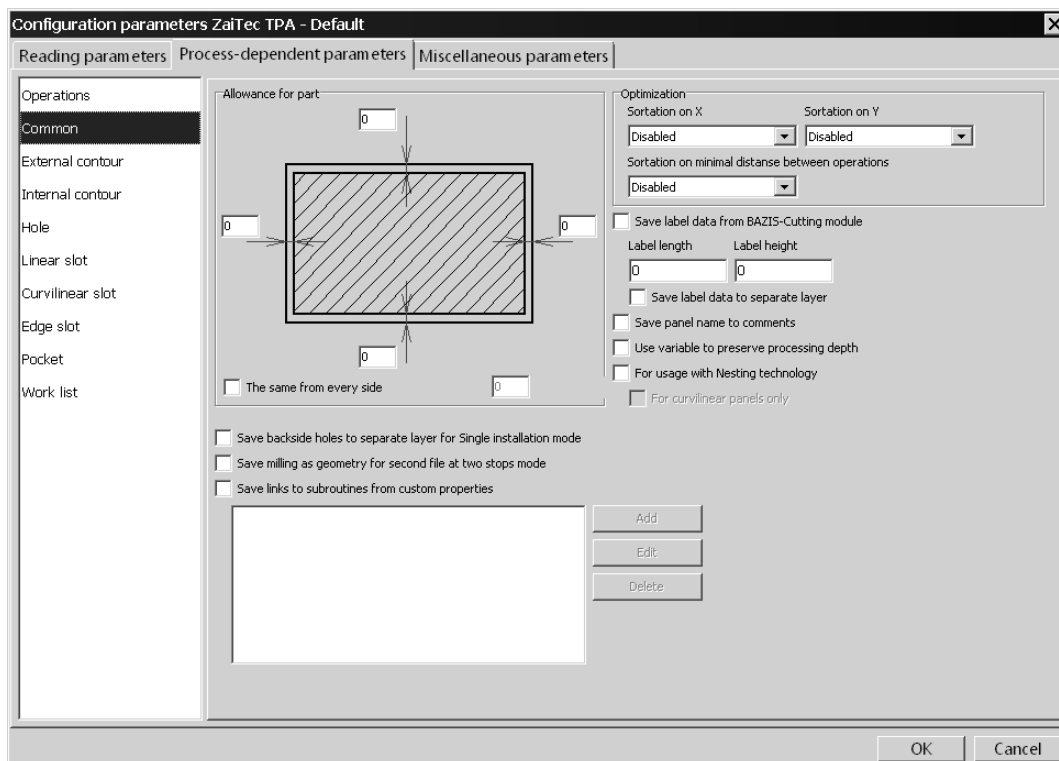


Fig. 10.731.

Allowance for part section control elements allow to set allowance values. Section fields allow to set allowance values for all panel sides. If **The same from every side** option is enabled, the same from every panel edge allowance field will become available. **Append operations to CNC program** group control elements allow to control operations of options names appending to control program. To add the program to the program, enable the option next to its name.

Contours parameters

Contours milling parameters are set in **Outer contour** (fig. 10.732) and **Internal contour** sections of **Process-dependent parameters** tab.

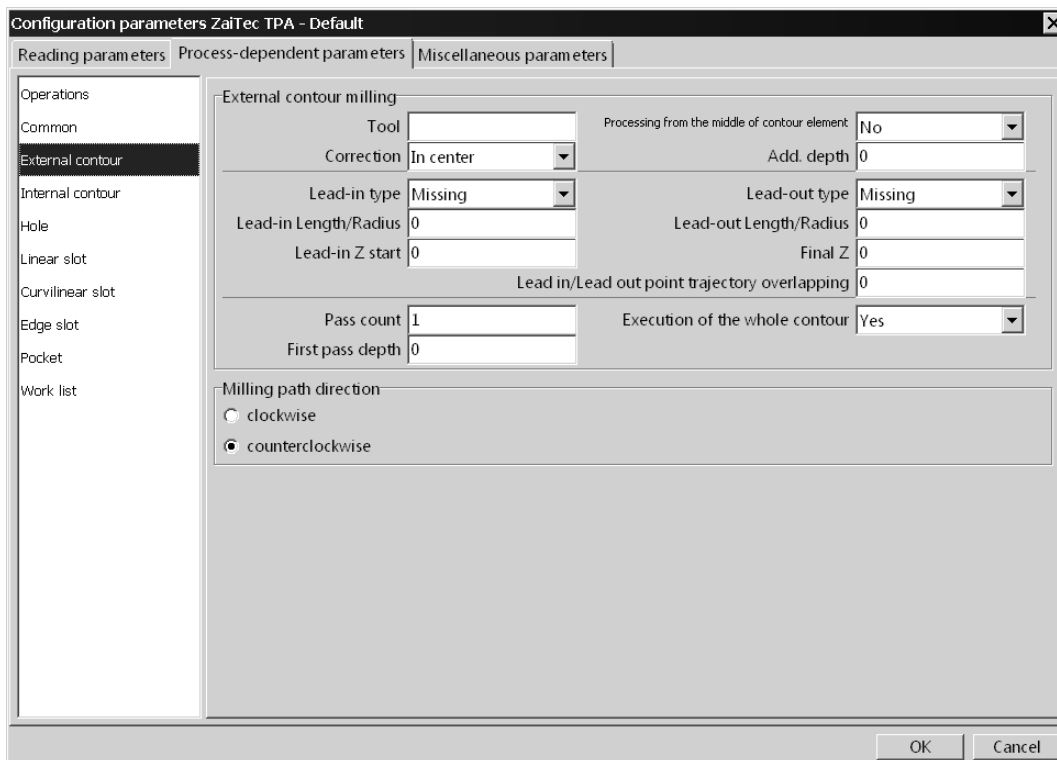


Fig. 10.732.

Finish machining and **Rough machining** tabs control elements allow to select a tool, correction types, tool lead-in and lead-out, etc. **Rough machining** option allows to control this operation.

Control elements of these tabs are almost the same. There are additional control elements on **Outer contour** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled.

Hole parameters

Hole section control elements allow to set drilling parameters (fig. 10.733).

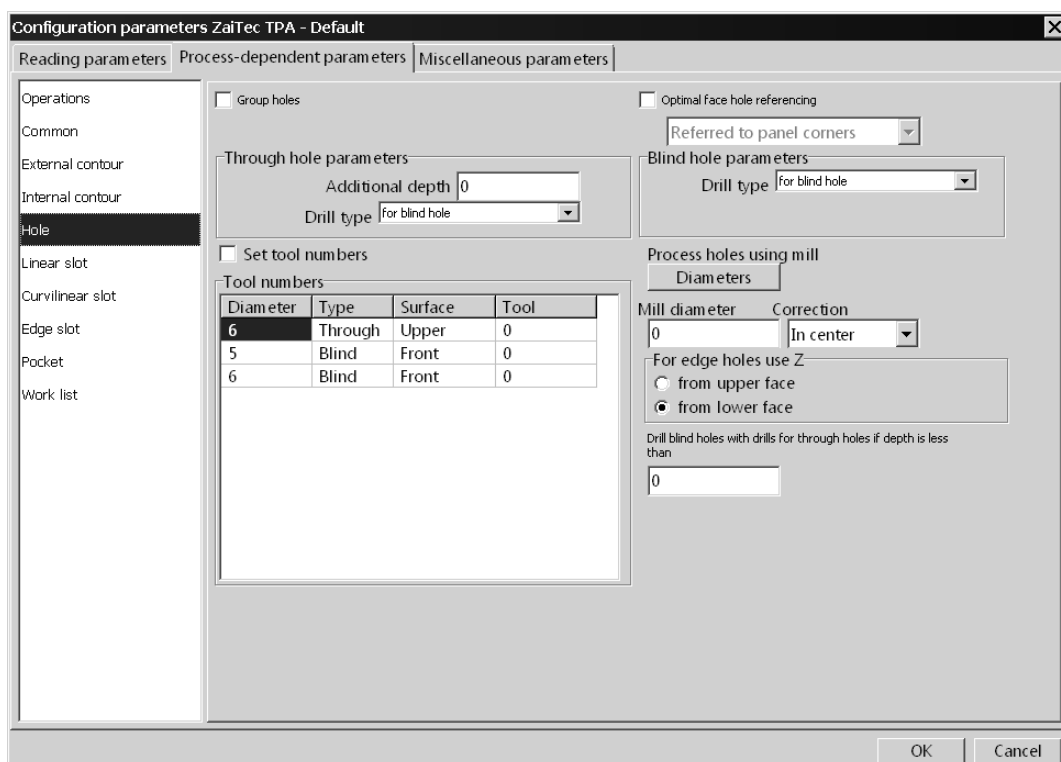


Fig. 10.733.

If there are several holes on one line and distance between them is the same (step), it is possible to shorten the program and group the holes. For this purpose enable **Group holes** option. Through holes can be bored with drills for through and blind holes. **Drill type** group variants allow to select a tool type. Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Tool numbers table allows to set automatic drill selection depending on hole type and diameter. This setting will work if **Set tool numbers** option is enabled. You can map hole and mill diameter that will be automatically selected for its processing. For this purpose push **Diameters** button. **Hole diameters** dialogue will appear on the screen (fig. 10.734).

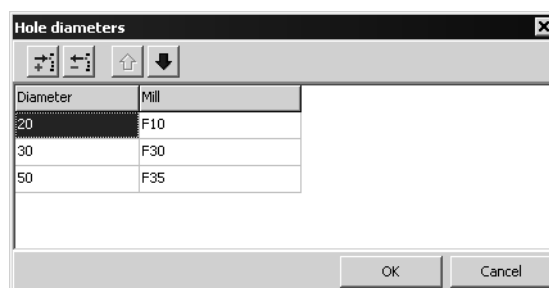


Fig. 10.734.



Dialogue table contains holes diameter values and corresponding mill designations. The fields should be filled via a keyboard. **Add row** button allows to add a blank row to the table.



To delete the current row push **Delete row** button. Arrow buttons allow to change current row position in the table. To finish setup push **OK** button. **Cancel** button allows to close the dialogue without changes. If a hole is processed with a mill, you can select correction type from **Correction** combo box for particular mill diameter set in **Diameters** field. **For edge holes use Z** group variants allow to select a face for this parameter application.

Linear slot parameters

Linear slots parameters are set in **Linear slot** section (fig. 10.735).

Fig. 10.735.

Through slot always cut with saw and **Grooving by saw** options allow to set through slots direction that will be processed with a saw. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.736).

Groove name	N	Tool name	Correction	Diameter
H14x5 (4x5)	1	D15	At a center	0
Quarter groove (12x12)				

Fig. 10.736.



Slot name table contains all grooves names of the model. You can set particular mills and a saw for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be edited in list fields. Correction type should be selected from **Correction** combo box.



Add saw button allows to choose a saw for slot processing.

The list may contain a tool of only one type. If you try to add a tool of another type, a warning will appear on the screen (fig. 10.737).

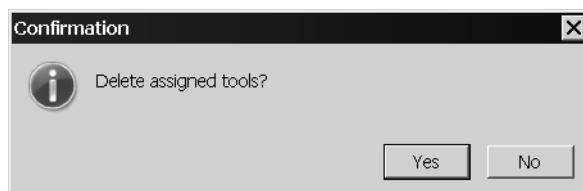


Fig. 10.737.



Message buttons allow to delete a specified tool, clean the list or reject tool of another type adding. **Delete** button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Grooving by saw parameters

Slot – by saw group control elements allow to set control programs creation parameters for grooving by saw – a tool, saw blade width and correction type.

Grooving by mill parameters

Slot – by mill group control elements allow to set control programs creation parameters for grooving by mill - a tool, mill diameter, lead-in and lead-out parameters, correction type, etc. You can map slot width and mill parameters that will be automatically selected for its processing. For this purpose push **Mill parameters** button. **Mill parameters** dialogue will appear on the screen (fig. 10.738).

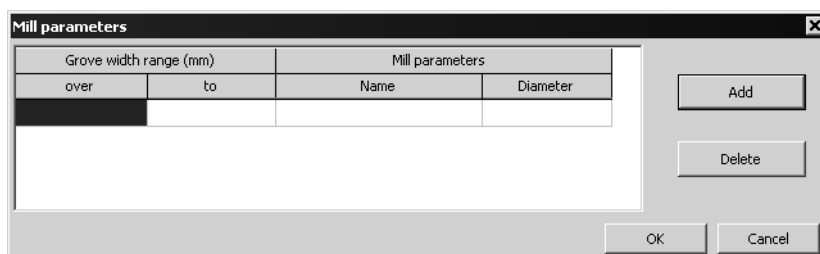


Fig. 10.738.

Dialogue table contains slot width ranges and corresponding mill parameters. The fields should be filled via a keyboard. **Add** button allows to add a blank row to the table. To delete the current row push **Delete** button. To finish parameters setup push **OK** button. The dialogue will be closed. **Cancel** button allows to close the dialogue without changes.

Curvilinear slot parameters

Curvilinear and edge slots parameters are set in **Curvilinear slot** section (fig. 10.739).

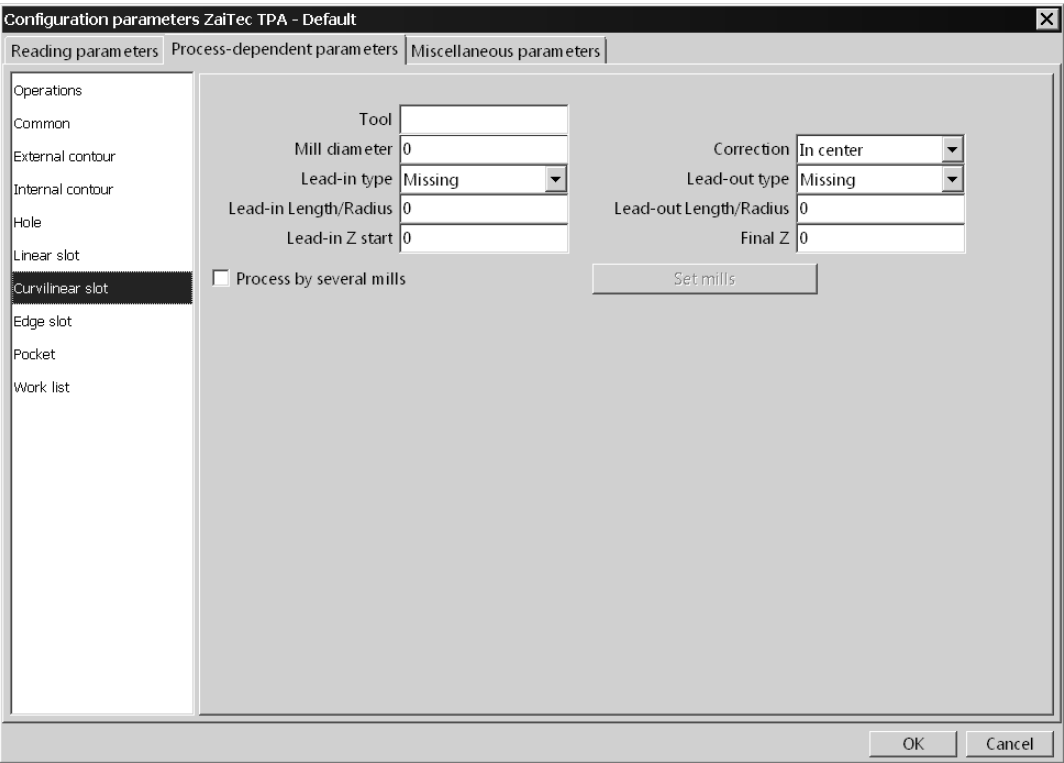


Fig. 10.739.

Section control elements allow to set control programs creation parameters for slots processing. If **Process by several mills** option is enabled, **Set mills** button will become available. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.740).

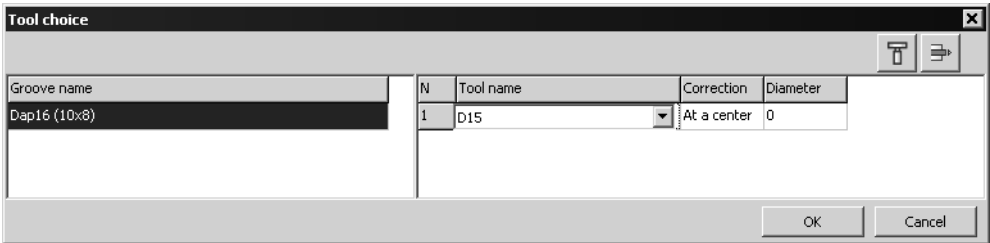


Fig. 10.740.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to add a tool to the list. Mill name and its diameter can be changed in list fields. Correction type should be selected from **Correction** combo box.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Edge slot parameters

Edge slots parameters are set in **Edge slot** section (fig. 10.741).

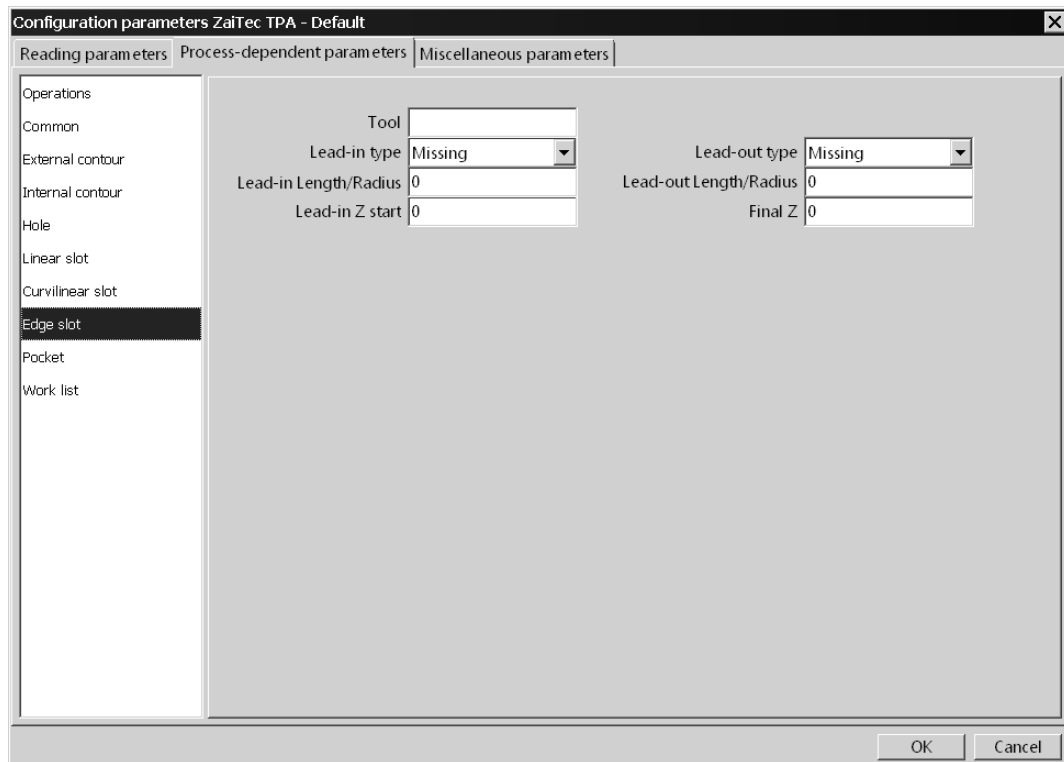


Fig. 10.741.

Section control elements allow to set control programs creation parameters for slots processing.

Pocket parameters

Pocket section control elements (fig. 10.742) allow to set pocket processing parameters.

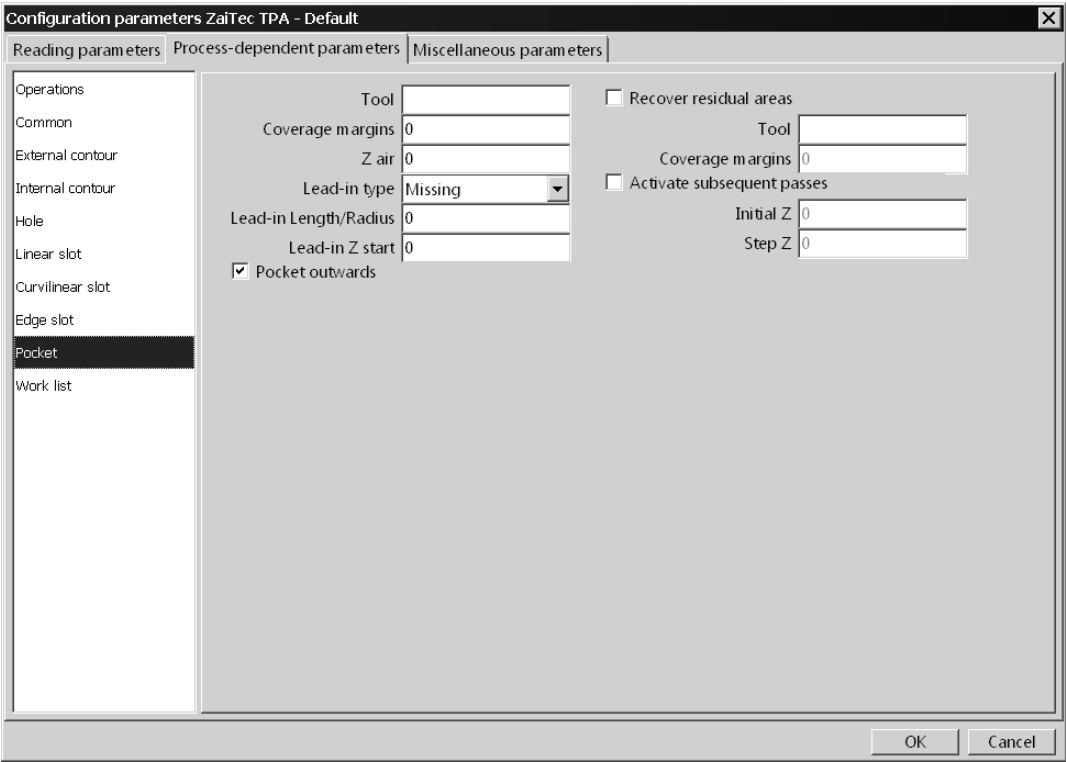


Fig. 10.742.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc.

Work list

Work list section control elements allow to set work lists (fig. 10.743).

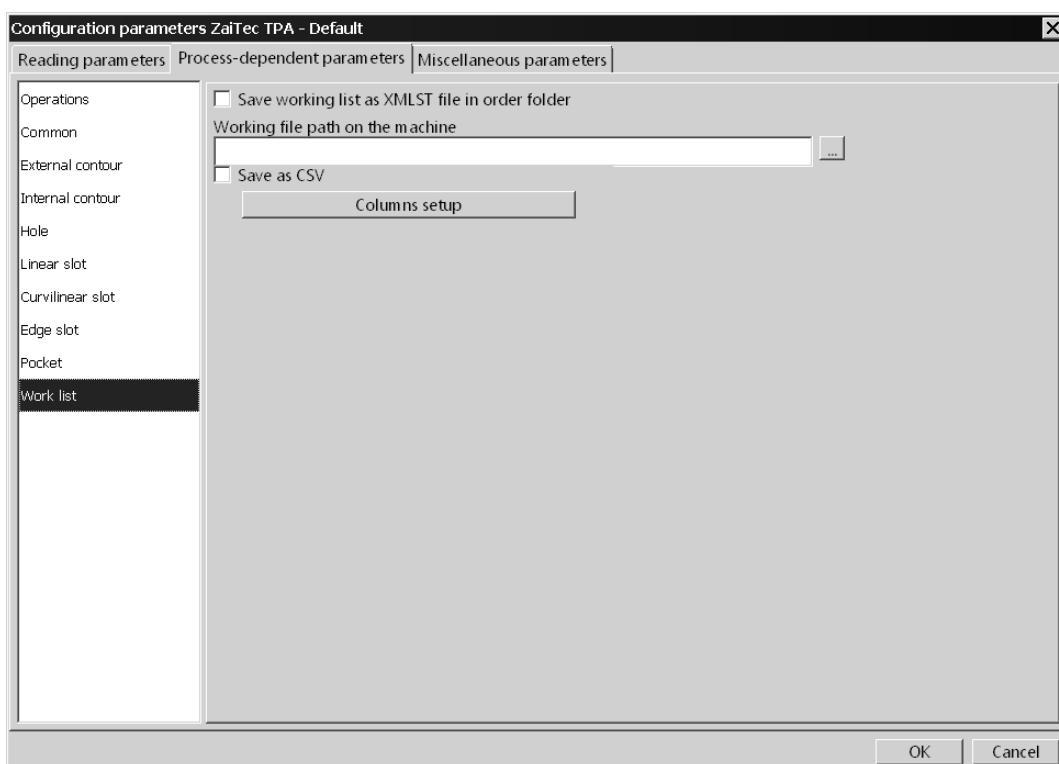


Fig. 10.743.



Save working list as file in order folder option allows to control the file saving. To specify the folder push **Browse** button and choose it in a standard Windows dialogue.

Save as CSV option allows to control file saving. **Columns setup** button allows to setup file table columns. After pushing the button, **Table columns setup** dialogue will appear on the screen (fig. 10.744).

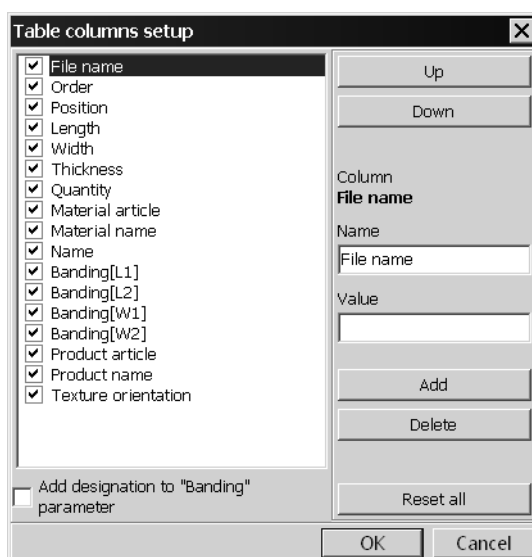


Fig. 10.744.

To add data from a column to a file, enable the option for it. **Up** and **Down** buttons allow to move the selected column through the list. **Add** button allows to add a column to the list. In **Name** field, you can change the name of the current column. For

columns added by the user, you can set a value in the appropriate field. **Delete** button allows to delete a column added by the user. After pushing the button, a warning will appear on the screen (fig. 10.745).

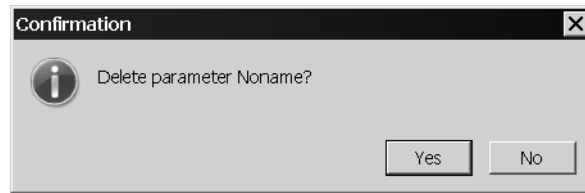


Fig. 10.745.

Message buttons allow to confirm or reject the deletion. **Reset all** button allows to return the table settings to their default values. After pushing the button, a warning will appear on the screen (fig. 10.746).

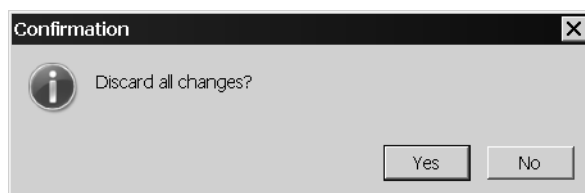


Fig. 10.746.

The message buttons allow to confirm or cancel the reset of parameters. To complete the table setup, push **OK** button.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.40.3.Zaitec Trepan

Operations

Operations section control elements (fig. 10.747) allow to specify operations that can be used in control program.

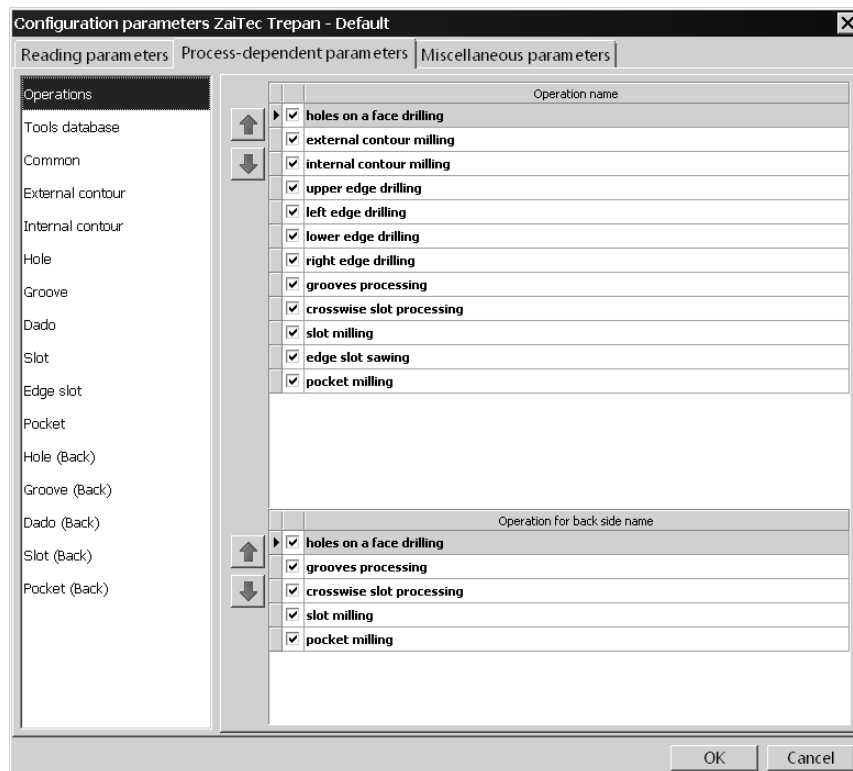


Fig. 10.747.

Usable tools

Tools database section control elements (fig. 10.748) allow to create a list of tools available for use in control programs.

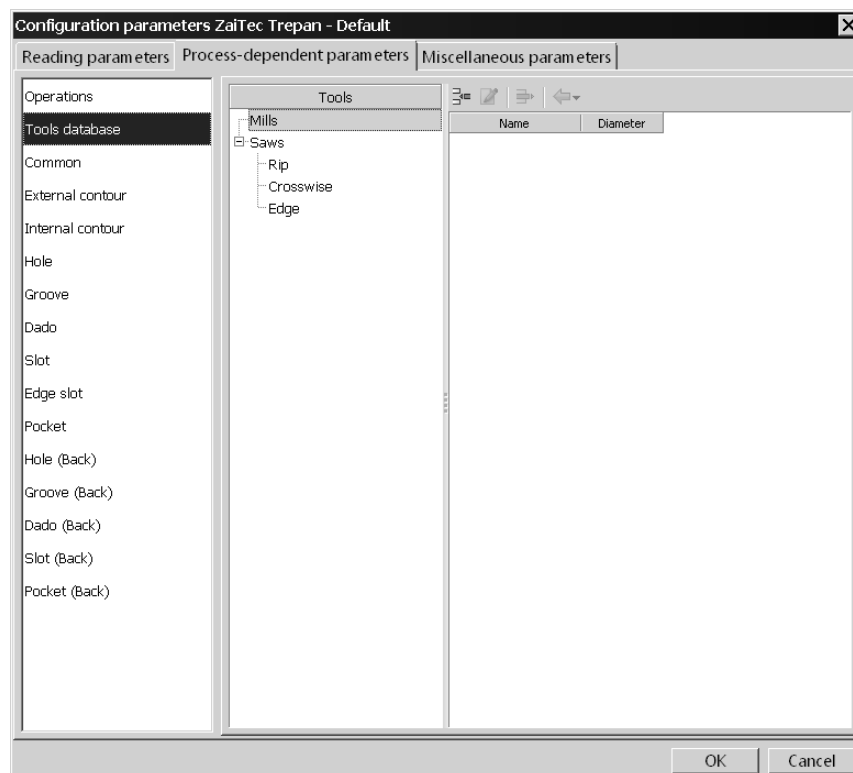


Fig. 10.748.



To add a tool to the list, push **Create** button. **Tool parameters** dialogue will appear on the screen (fig. 10.749).

Fig. 10.749.

In this dialogue enter tool name and set its parameters. To complete tool adding, push **OK** button. The dialogue will be closed, tool name will appear in the list. Its parameters values will be shown in the dialogue. **Cancel** button allows to close the dialogue without changes. **Edit** button allows to change tool parameters selected in the list.



Delete button allows to delete selected tool from the list.

Common parameters

Common section control elements (fig. 10.750) allow to set common parameters of control programs creation.

Fig. 10.750.

Contour processing parameters

Outer and internal contours processing parameters setting is performed in **Outer contour** and **Internal contour** sections respectively (fig. 10.751).

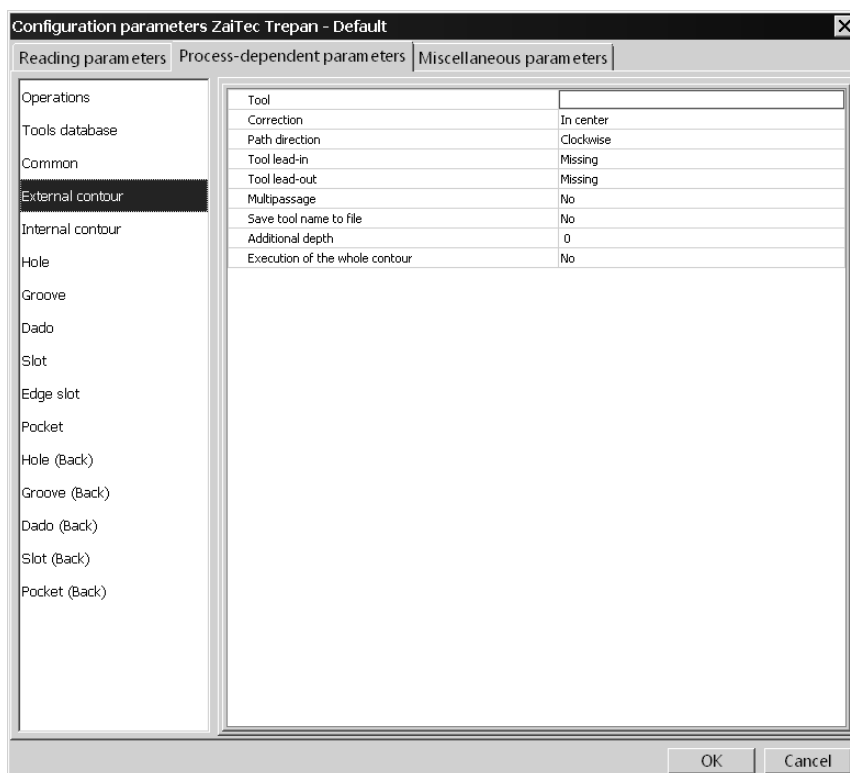


Fig. 10.751.

Section control elements allow to choose a tool, lead-in type, correction type, etc.

Hole parameters

Hole processing from face is set in **Hole** section (fig. 10.752), holes from back side – in **Hole (Back)** section in the same way.

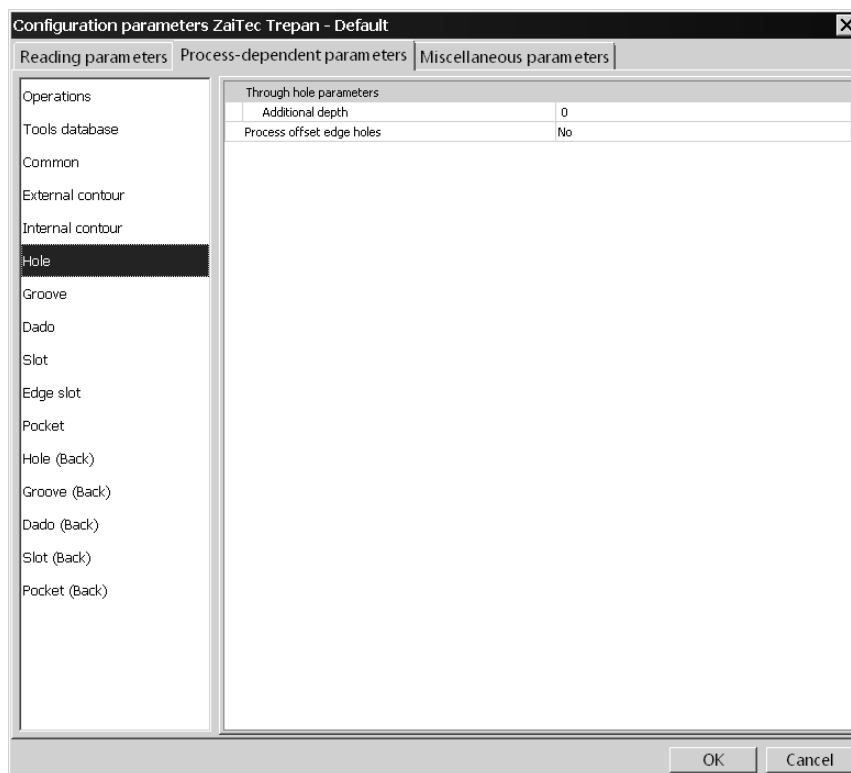


Fig. 10.752.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account.

Groove and dado parameters

Slots processing from the face setup is performed in the following sections: **Groove** (fig. 10.753), in **Dado** section in the same way. For slots from the back of the panel the setting is performed in **Groove (Back)** and **Dado (Back)** sections respectively.

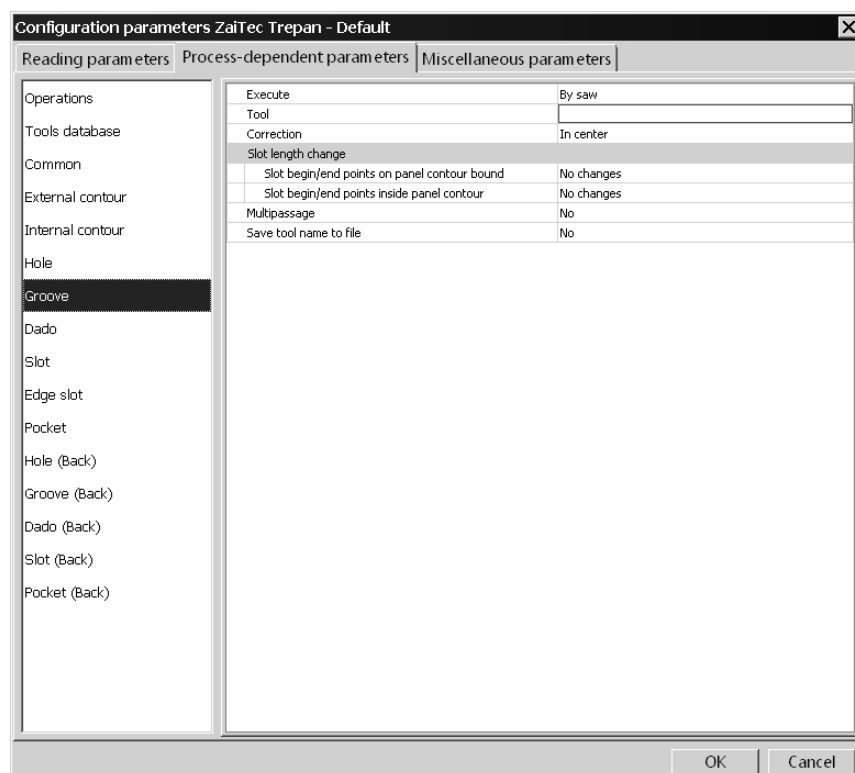


Fig. 10.753.

Section control elements allow to select a tool type for slot creation, a tool from database and correction type. If mill is selected as a tool, its parameters can be the same as for slot along an arbitrary trajectory.

Slots parameters along arbitrary trajectory

Slots processing from the face setup is performed in **Slot** section (fig. 10.754), from the back panel side – in **Slot (Back)** section.

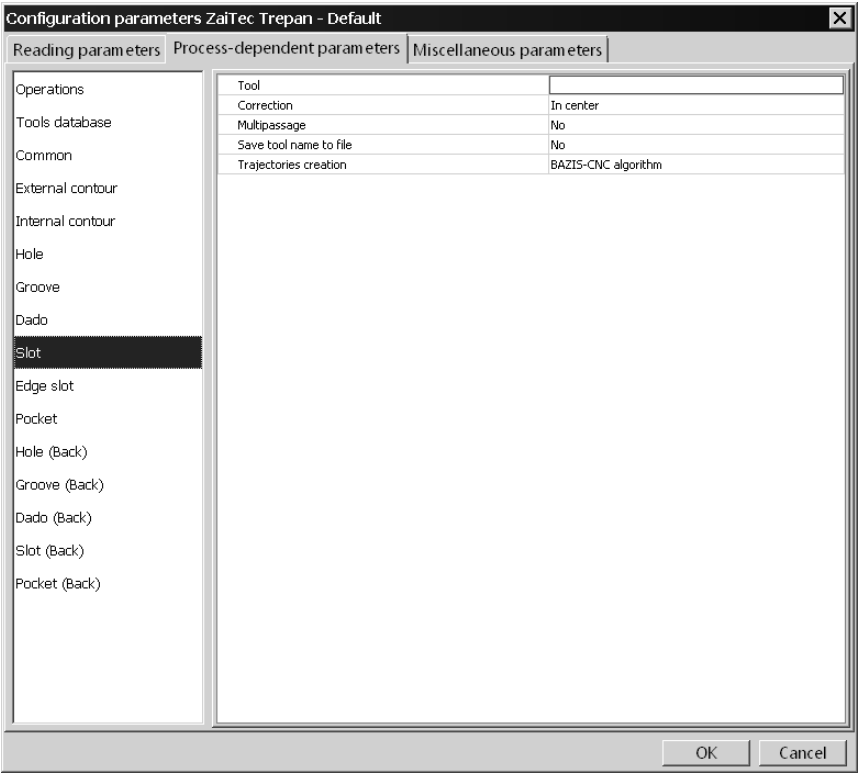


Fig. 10.754.

Section control elements allow to select a tool from database, correction type and multipassage use.

Pocket parameters

Parameters of pocket processing on the face side setup is performed in **Pocket** section (fig. 10.755), on the back panel side – in **Pocket (Back)** section.

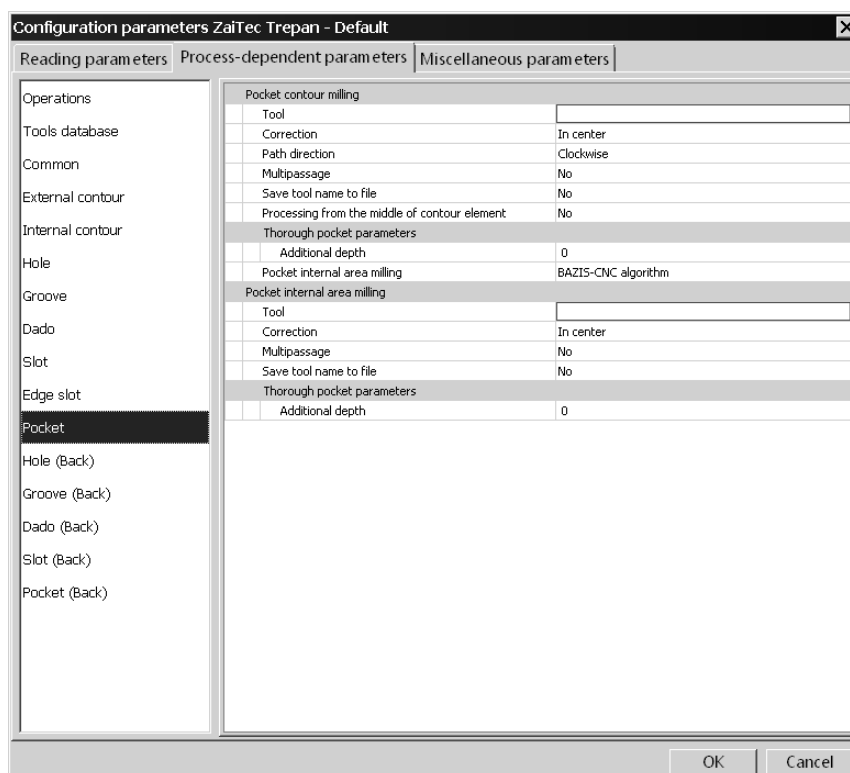


Fig. 10.755.

Tool combo box allows to select milling tool from database. For pocket contour select milling execution direction, correction type and multipassage use.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.40.4.Weihong



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operations section control elements (fig. 10.756) allow to specify operations that can be used in control program.

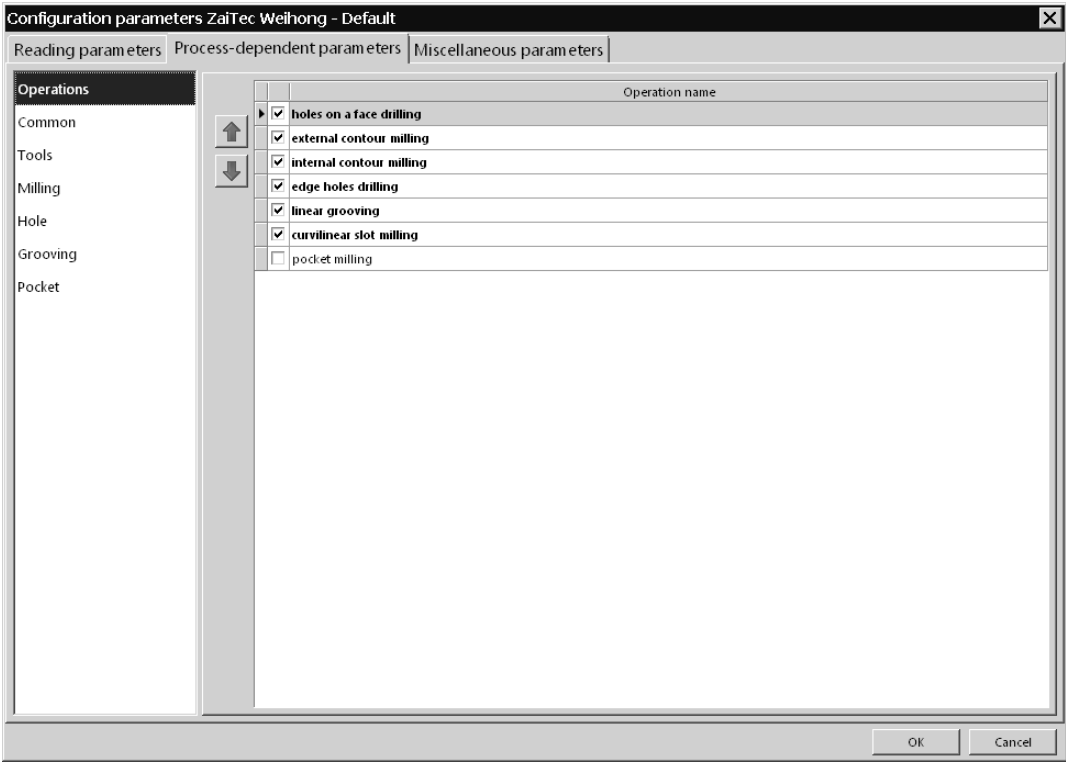


Fig. 10.756.

Common parameters

Common section control elements (fig. 10.757) allow to set common parameters of control programs creation.

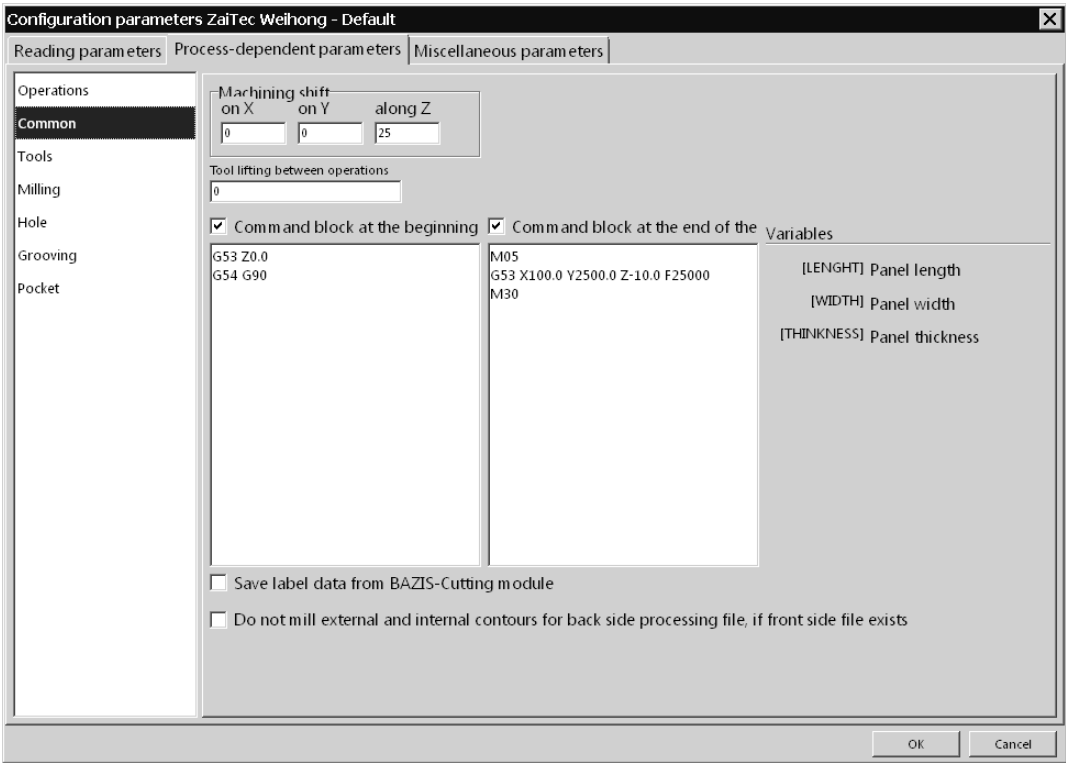


Fig. 10.757.

Machining shift group fields allow to set work area initial point moving. **Tool lifting between operations** field allows to set tool height. **Command block at the beginning of the program** and **Command block at the end of the program** fields allow to create a command list, that will be automatically moved to the specified program areas.

Save label data option allows to save the corresponding data. To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Tool parameters

Usable tools parameters setup is performed in **Tools** section (fig. 10.758).

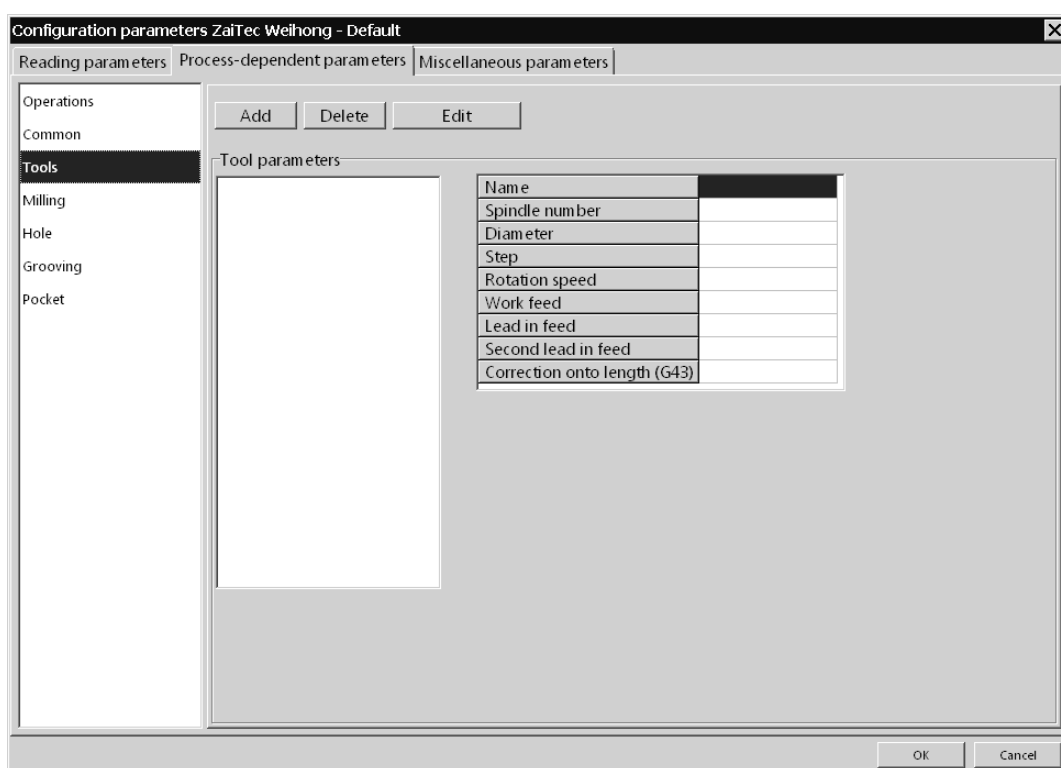


Fig. 10.758.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.759).

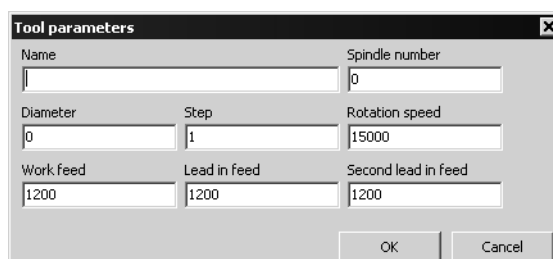


Fig. 10.759.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is shown in fig. 10.760.

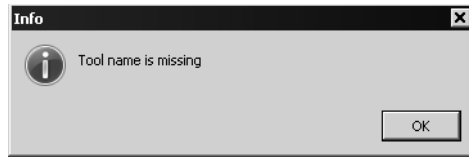


Fig. 10.760.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.759). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set on **Milling** tab (fig. 10.761).

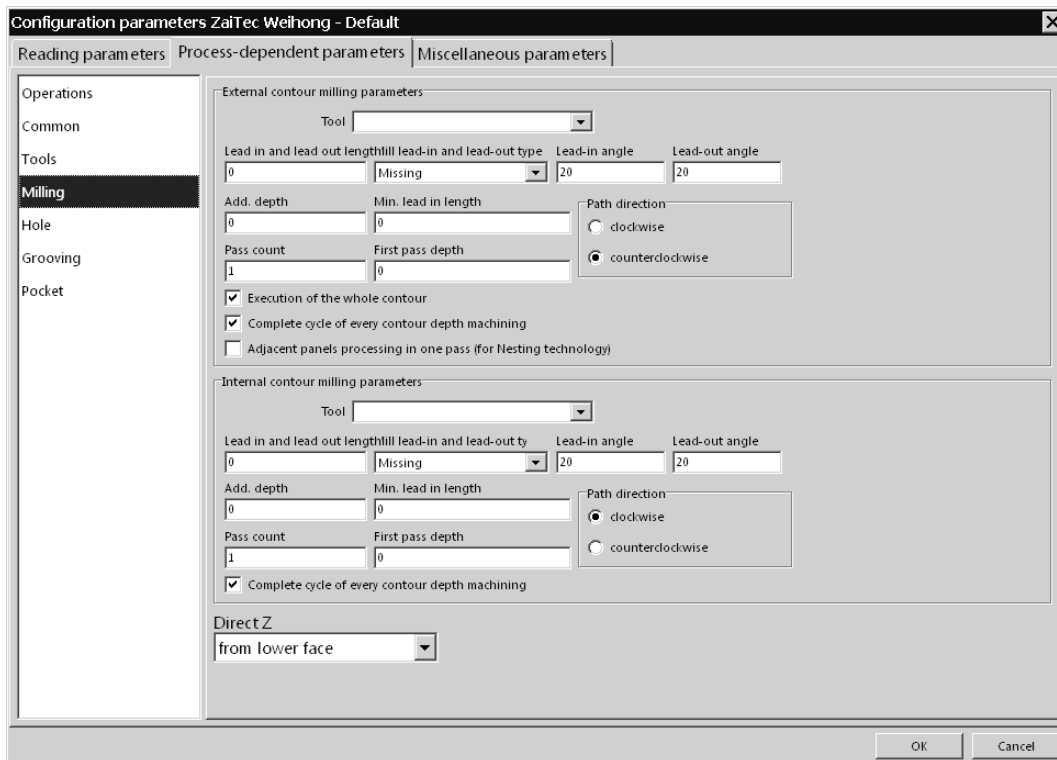


Fig. 10.761.

Tool combo box allows to select a milling tool name. Tools lists should be created beforehand in **Tools** section. Section control elements allow to set pass depth, pass count, etc. **Execution of the whole contour** option allows to set the elements for milling. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with panel overall rectangle sides will be milled. I.e. if panel is rectangular, and the option is disabled, contour will not be milled. **Path direction** group variants allow to select tool moving direction.

Hole parameters

Hole section control elements allow to set holes processing (fig. 10.762).

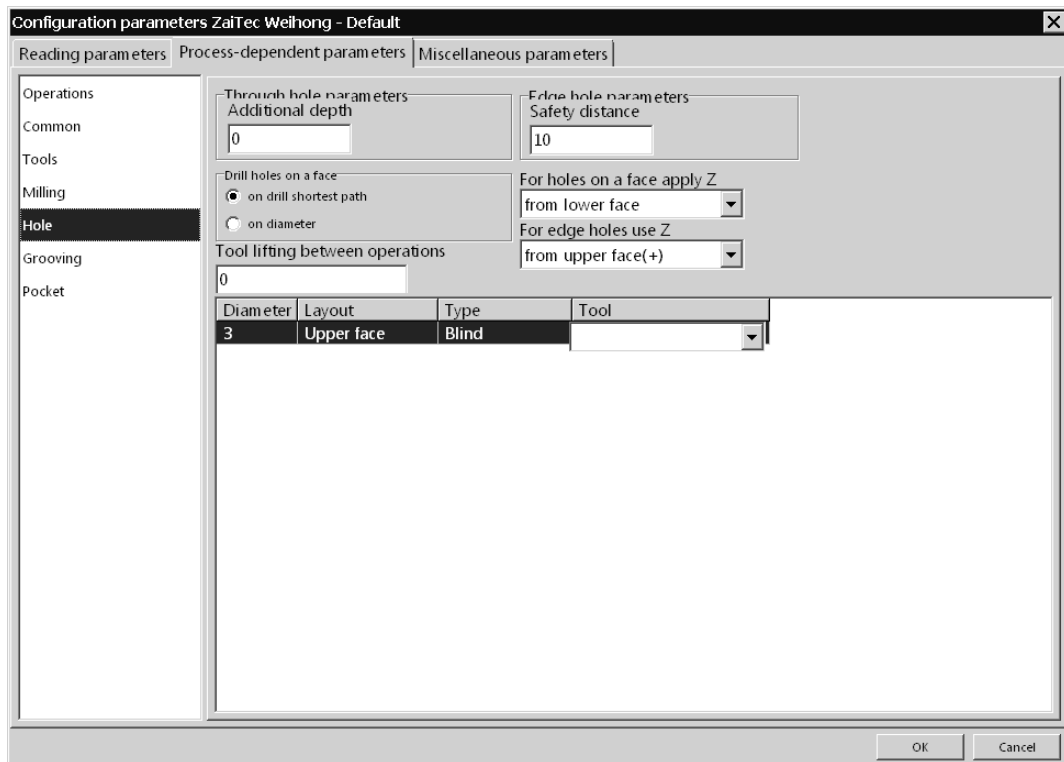


Fig. 10.762.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance. **Drill holes on a face** group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order.

Groove parameters

Grooving section control elements allow to set groove processing parameters (fig. 10.763).

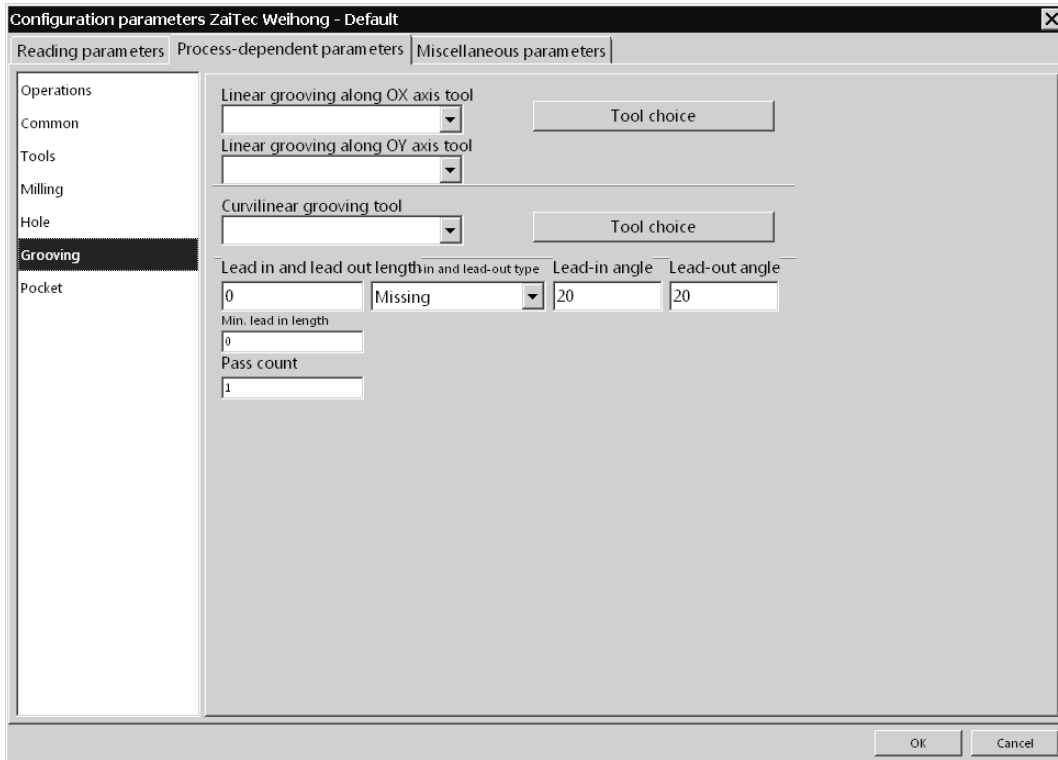


Fig. 10.763.

This section combo boxes allow to select tools names that will be automatically used for linear slots of X and Y axes and curvilinear slots processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set the tools that will be used for particular slots. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.764).

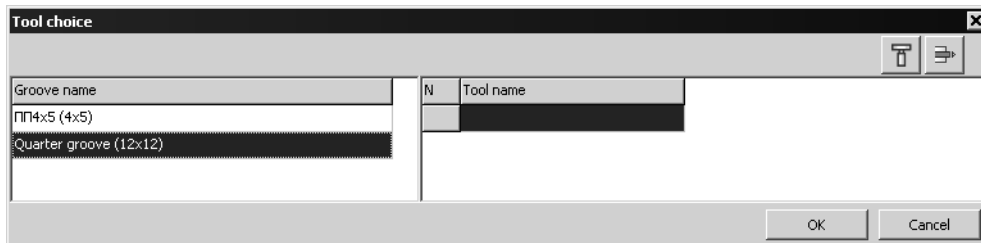


Fig. 10.764.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in **Tools** section.



Delete button allows to delete a tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.765) allow to set pocket processing parameters.

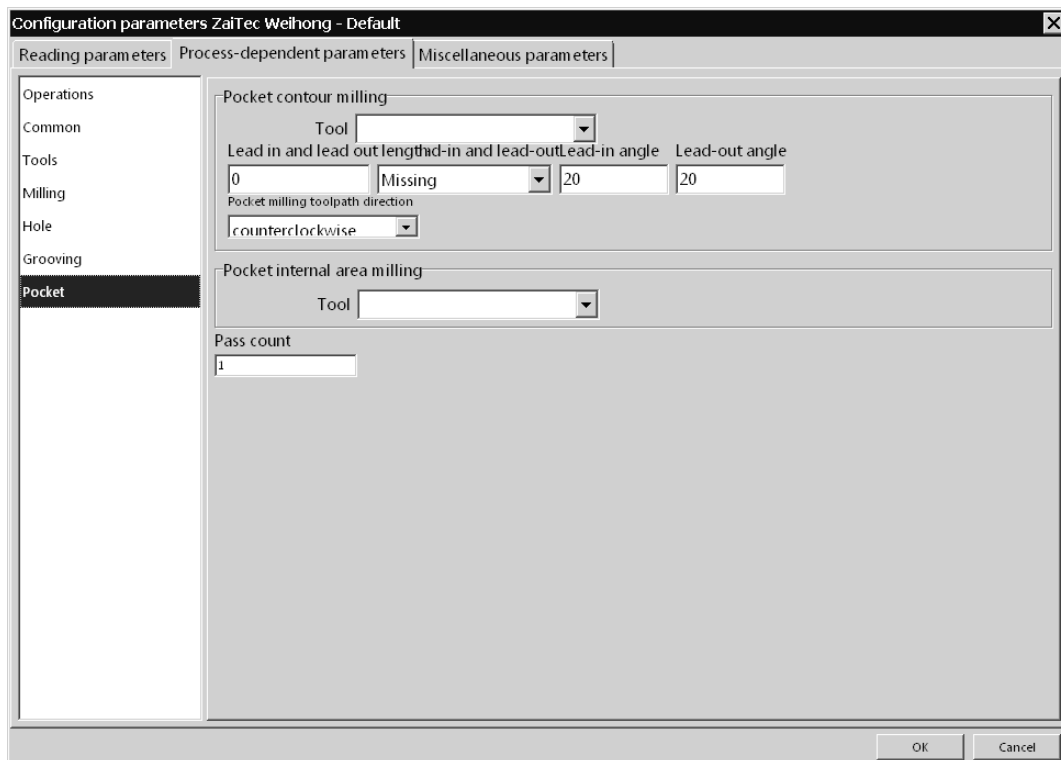


Fig. 10.765.

Tab control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and pocket inner part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.41. Artel

10.41.1.DeskCNC (*.dnc)



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Operations

Operation section control elements (fig. 10.766) allow to specify the operations that can be used in control program.

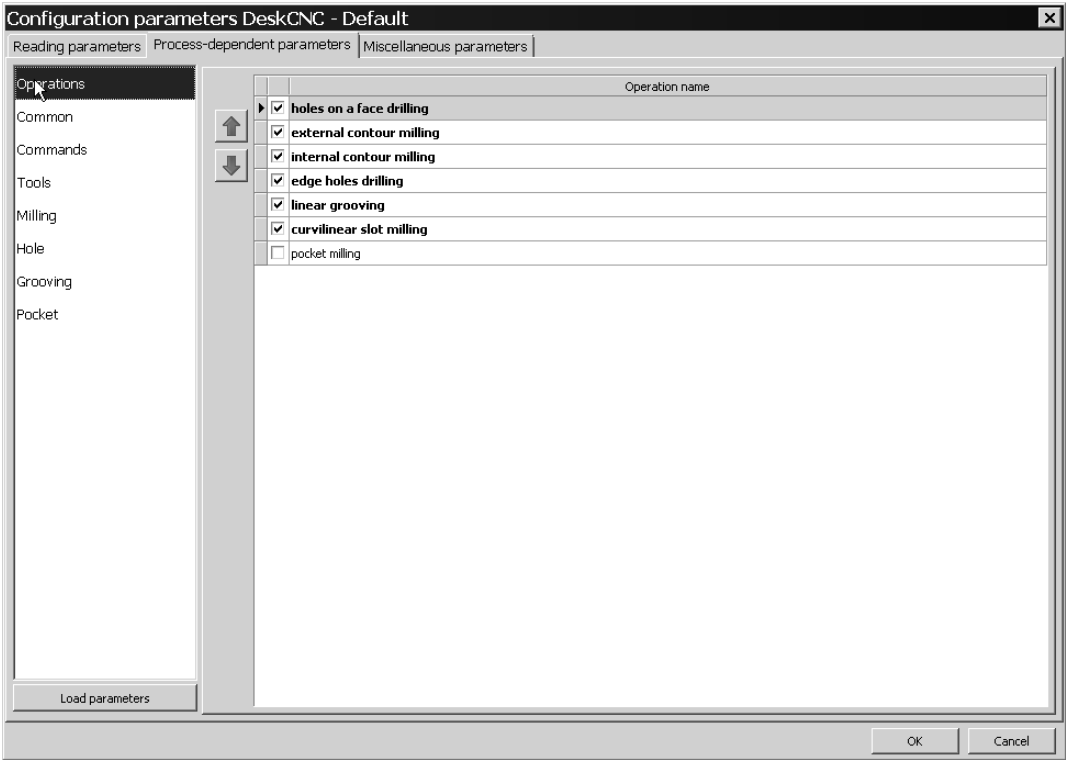


Fig. 10.766.

Common parameters

Common section control elements allow to set control programs common parameters (fig. 10.767).

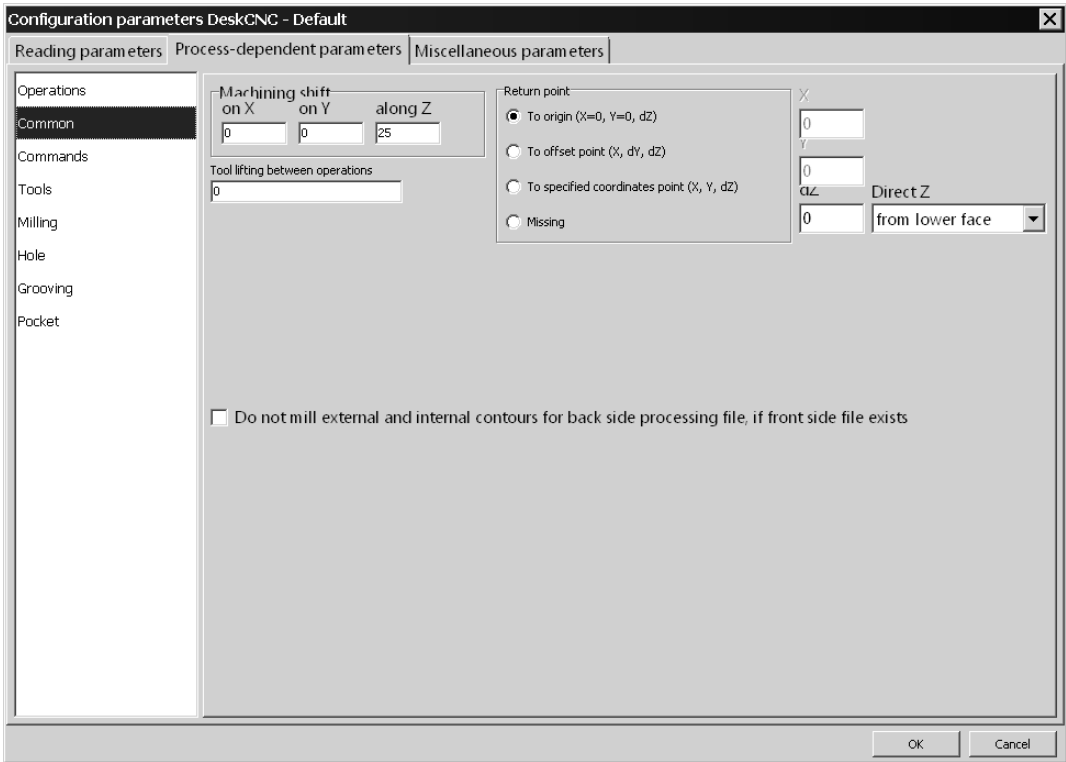


Fig. 10.767.

Machining shift group fields allow to set work area initial point moving. **Return point** group control elements allow to control tool return point position. If **To coordinate system origin** option is enabled, tool will be returned to this point. If you select **To offset point (X, dY)** or **To point with specified coordinates** variants, return point coordinates are specified by values of corresponding fields. **Tool lifting between operations** field allows to set tool height.

To process the panel, two separate files can be created containing programs for the front and non-front sides. **Do not mill outer and internal contours for non-front side processing file if there is a file for front side** option allows to control milling program in non-front side file.

Usable commands

Commands section control elements (fig. 10.768) allow to control separate commands to control program inclusion.

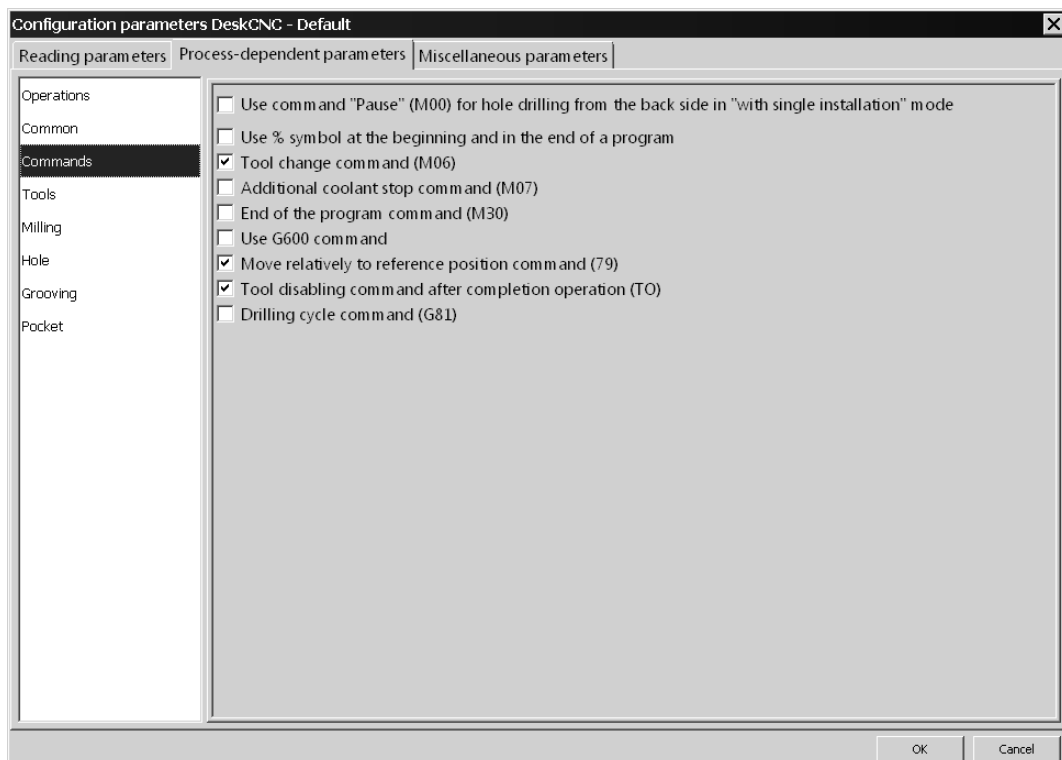


Fig. 10.768.

Tools parameters

Usable tools parameters are set in **Tools** section (fig. 10.769).

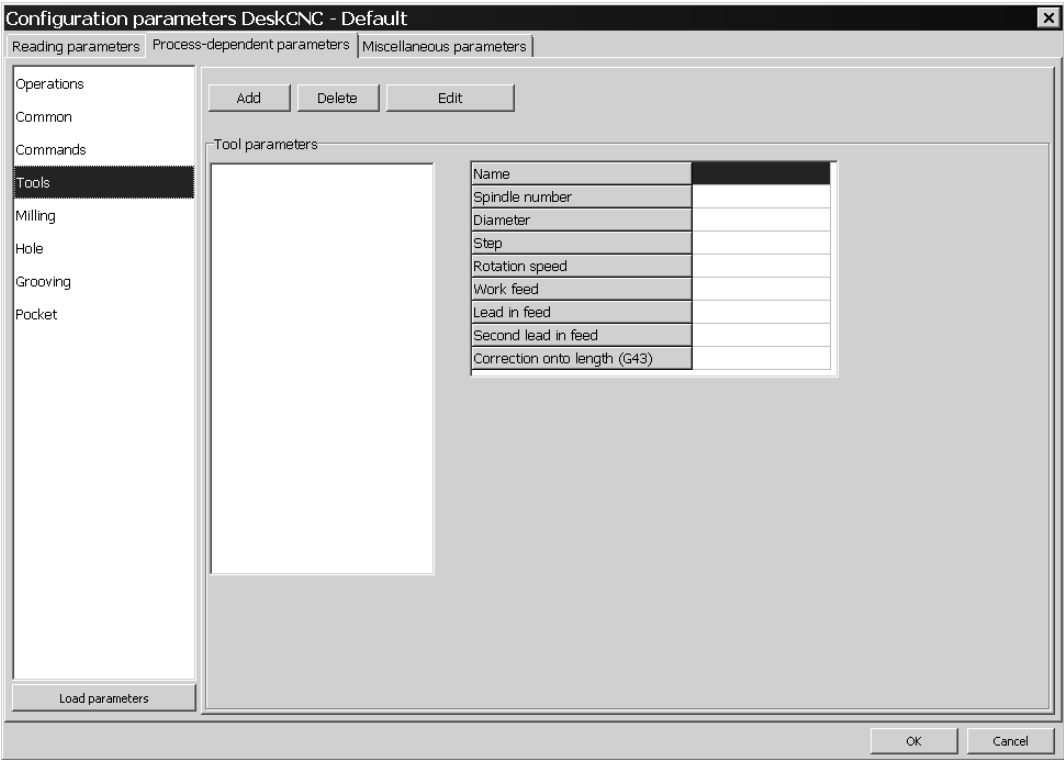


Fig. 10.769.

The dialogue table contains selected tool parameters. To add a tool to the list, push **Add** button. **Tool parameters** dialogue will appear on the screen (fig. 10.770).

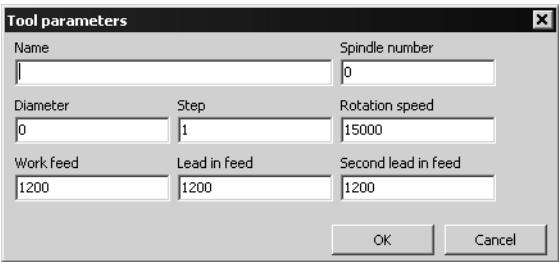


Fig. 10.770.

The dialogue control elements allow to set tool parameters. To complete tool adding push **OK** button. Tool name and its diameter should be input. Otherwise a warning will appear on the screen. The message example is in fig. 10.771.

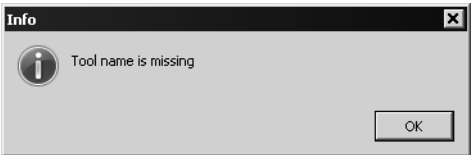


Fig. 10.771.

Cancel button allows to close the dialogue without changes. **Edit** button allows to edit selected tool parameters. After pushing the button **Tool parameters** dialogue will appear on the screen (fig. 10.770). **Delete** button allows to delete the current tool from the list.

Contour processing parameters

Outer and internal contours processing parameters are set in **Milling** section (fig. 10.772).

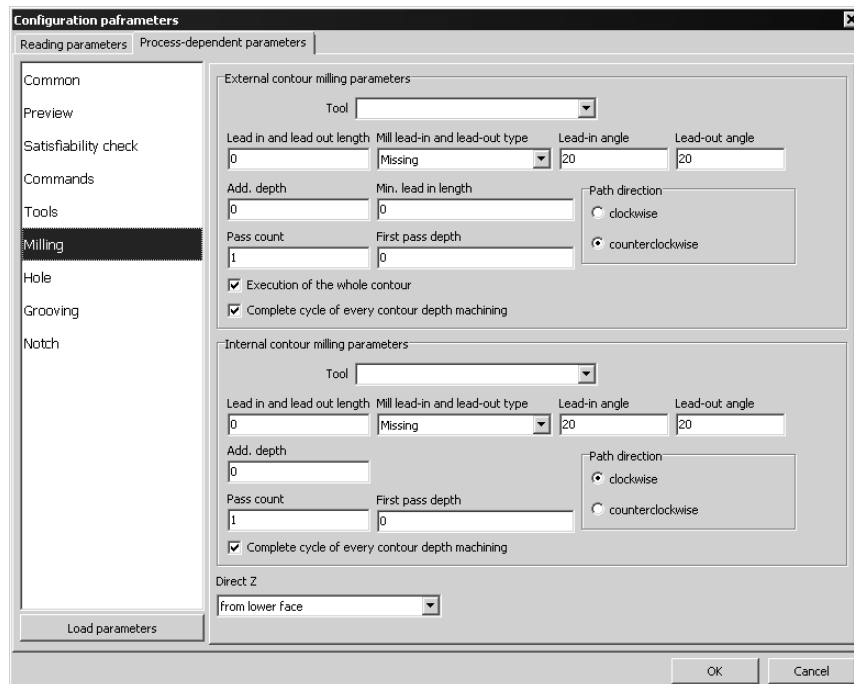


Fig. 10.772.

Tool combo box allows to select a tool for milling from a list on **Tools** tab. **Execution of the whole contour** option allows to set the elements that will be milled. If it is enabled, all panel edges will be milled. If it is disabled, the edges that do not coincide with overall rectangle sides will be milled. I.e. if panel is rectangular and the option is disabled, contour will not be milled. **Path direction** of internal and external contour milling group variants allow to select tool direction.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.773).

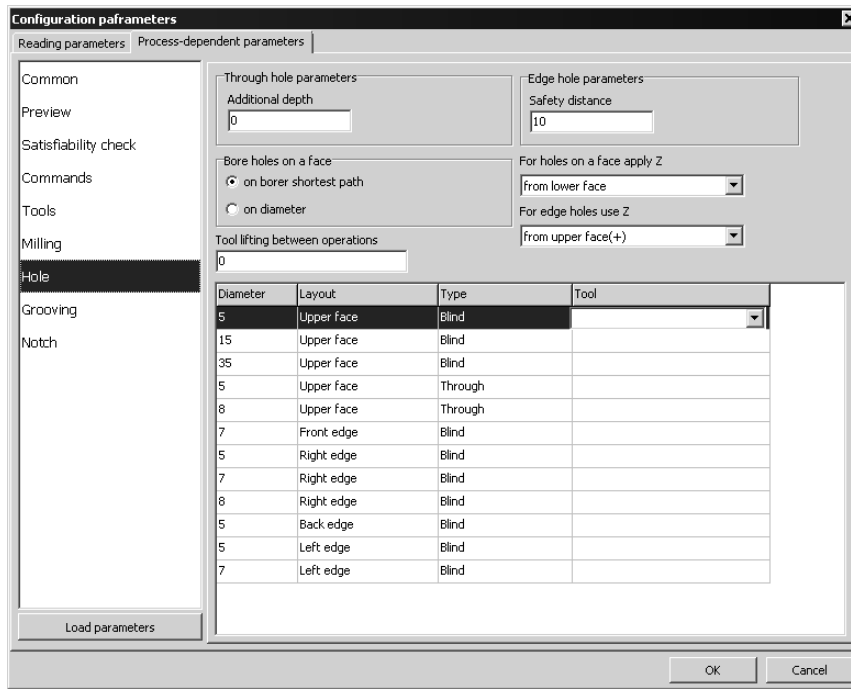


Fig. 10.773.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. Before edge hole boring it is necessary to move a spindle to material panel in a way, that it will not collide with it. **Safety distance** parameter in **Edge hole parameters** group allows to set the distance.

Drill holes on a face group table allows to set automatic spindle number selection depending on hole parameters. **On drill shortest path** and **on diameter** variants allow to set holes boring order. If, when drilling through holes, the mill enters a point located on the diameter of the hole, it can pull out material at the outlet. If the mill enters the material at the point of the hole center, then there will be no such defects. **Through hole milling from the center** option allows to control the position of the hole start point: in the center or on the diameter of the hole.

Slots processing parameters

Grooving section control elements (fig. 10.774) allow to set slots processing parameters.

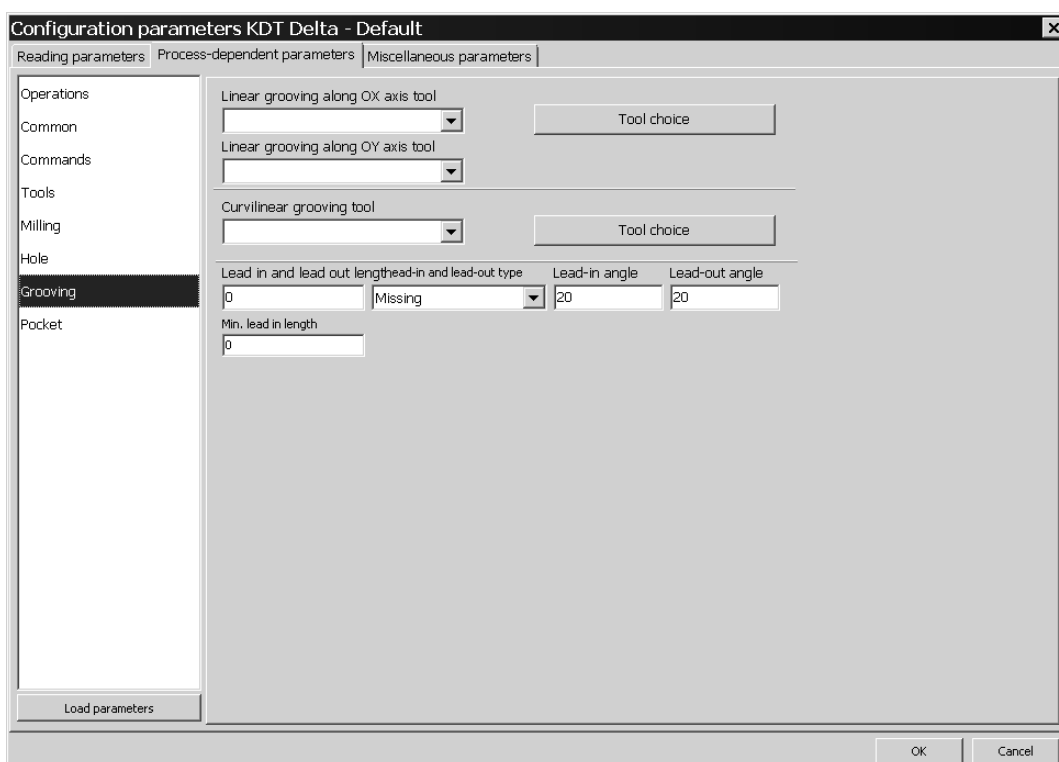


Fig. 10.774.

Section combo boxes allow to select tool names for automatic use for curvilinear and linear slots of X and Y axes processing. Tool lists should be created beforehand in **Tools** section. **Tool choice** button allows to set tools for particular slots processing. After pushing the button **Tool choice** dialogue will appear on the screen (fig. 10.775).

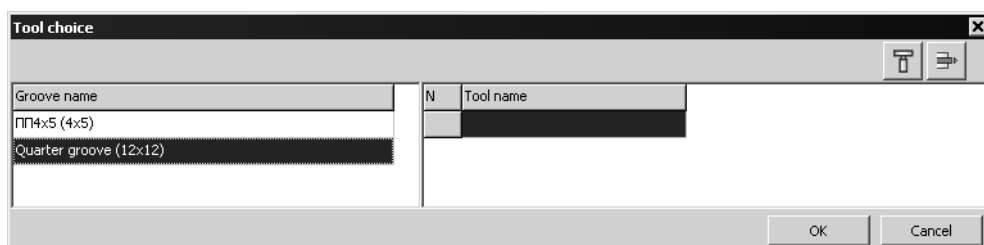


Fig. 10.775.



Slot name table contains all slots names in the model. You can assign particular saws and mills for each slot processing. **Add mill** button allows to assign a tool for slot processing from the list created in section Tools parameters on p. 589.



Delete button allows to delete selected tool from the list. To complete tool assignment push **OK** button. **Cancel** button allows to close the dialogue without changes.

Pocket parameters

Pocket section control elements (fig. 10.776) allow to set pocket processing parameters.

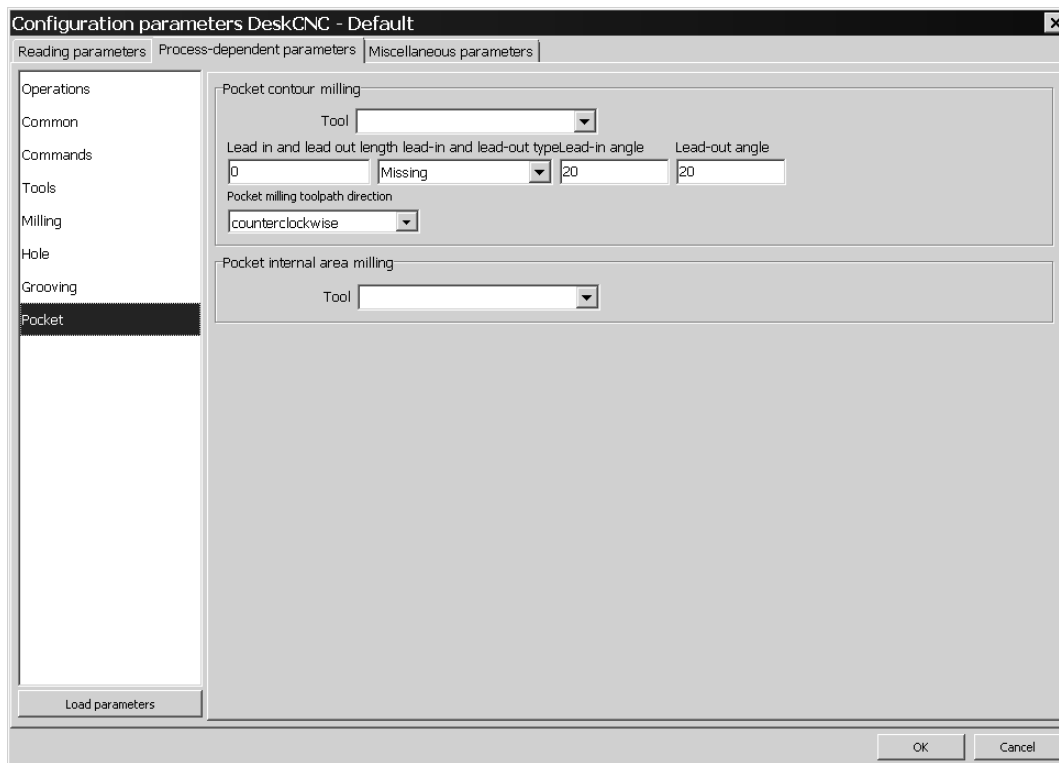


Fig. 10.776.

Section control elements allow to set control programs creation parameters for pocket contour and inner part processing: tool name, mill diameter, lead-in type, correction type, etc. Tool names for contour and inner pocket part processing are selected from the list created in **Tools** section.

Setup completion

To complete control programs for DeskCNC (*.cnc) setup push **OK** button. **Cancel** button allows to close the dialogue without changes.

10.42. Research and Production Company «Semil»

10.42.1. WinKam



This User Guide provides information about parameters setup in BAZIS-CNC module required for correct control programs generation.

These parameters setting, limit values, interconnection with processing, etc. is described in software documentation of particular machine. Such issues are not described in this User Guide.

Common parameters

Common parameters are set in **Common** section of **Process-dependent parameters** tab (fig. 10.777).

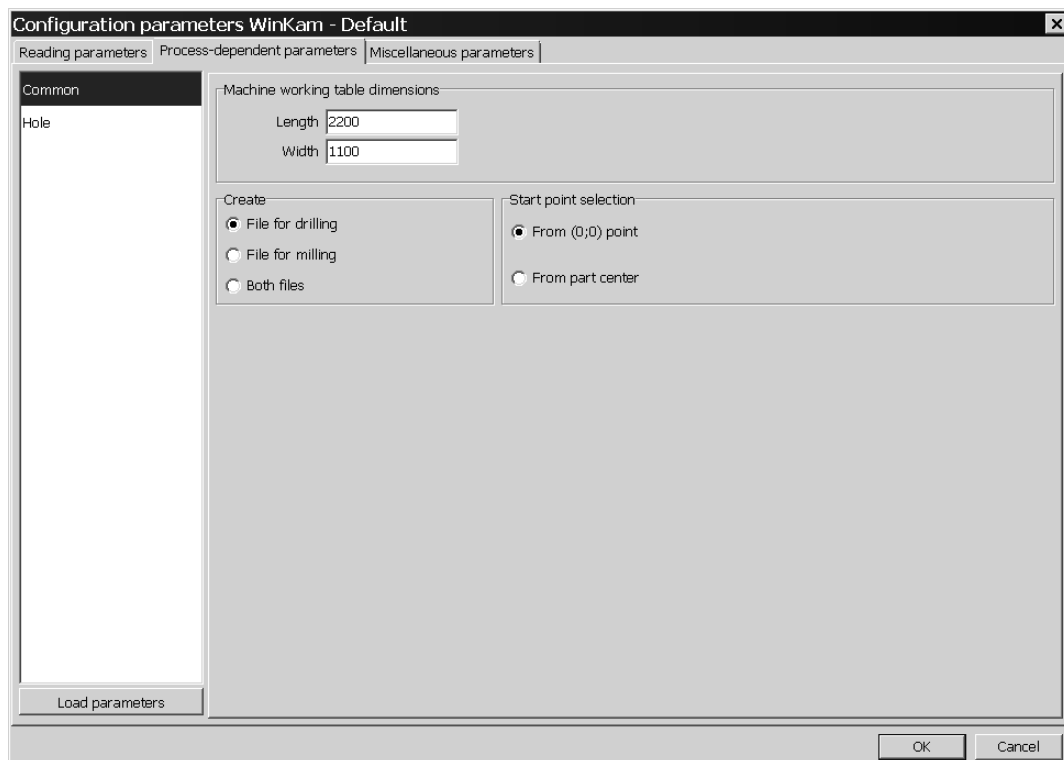


Fig. 10.777.

Section control elements allow to select file type, initial point position and set desk-top dimensions.

Hole parameters

Holes processing is set in **Hole** section (fig. 10.778).

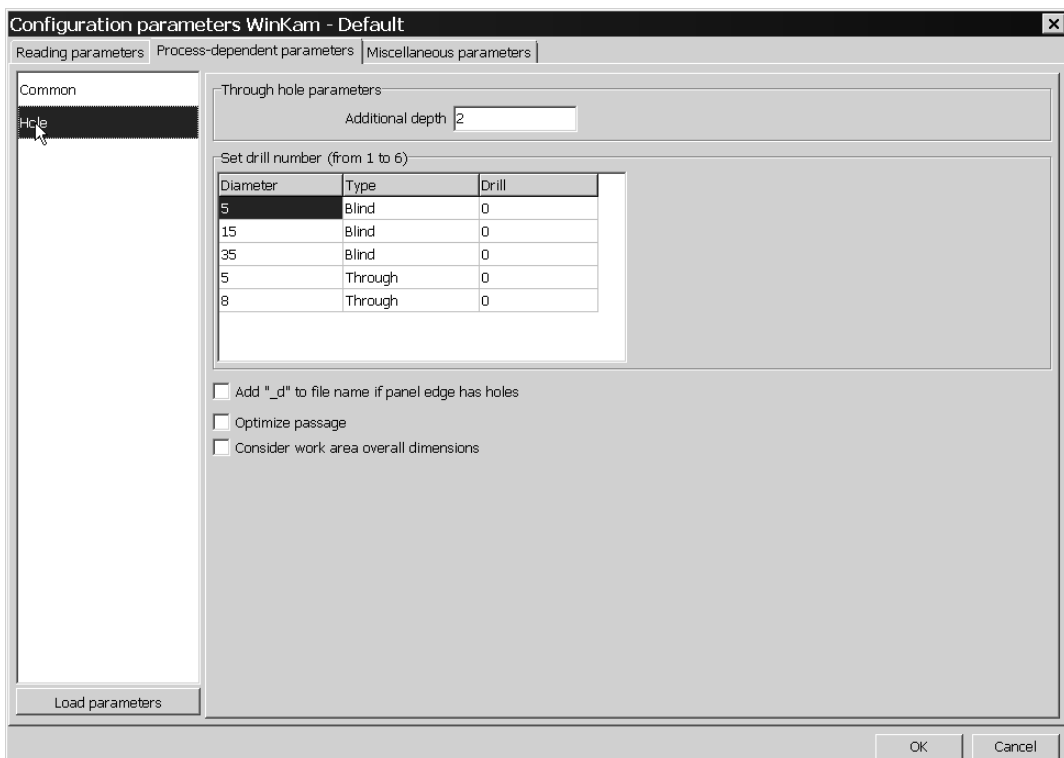


Fig. 10.778.

Wood drills usually have center point. Thus, for through holes boring, drill moving distance should exceed material thickness. **Additional depth** field in **Through hole parameters** group allows to take drill center point into account. **Set drill number (from 1 to 6)** table allows to set automatic borer selection depending on hole diameter and type.

Setup completion

To complete control programs setup, push **OK** button. **Cancel** button allows to close the dialogue without changes.

Chapter 11. Control program files creation

BAZIS-CNC module work result is control programs generation and their files saving. For this purpose do the following:

- ▼ set control programs files creation order (see Chapter 9 on p. 51),
- ▼ set process-dependent parameters (see Chapter 10 on p. 63),
- ▼ choose panels for processing programs generation, check in panel line in **Preview for CNC** window table (see section 8.1 on p. 44),
- ▼ push **Export** button in the window (fig. 7.1 on p. 41).

A progress bar will appear on the screen.



If there are few panels, progress bar may not appear.

After export completion, a message will appear on the screen (fig. 11.1).

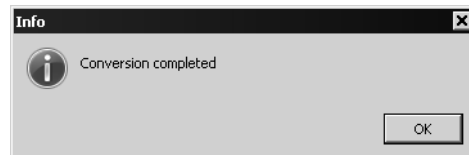


Fig. 11.1.

If no panel is selected, export will not be fulfilled and a warning will appear on the screen (fig. 11.2).

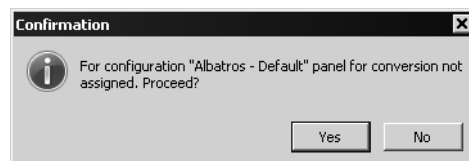


Fig. 11.2.

In this case select the panels and try to export again. To complete control program files creation push **Close** button. **Preview for CNC** window will be closed.

Appendix I. Terms and definitions

Active document

It is the current document you work with. If documents windows are maximized, document file name is in the module title bar.

If document windows are not maximized, active window title bar is highlighted.

Active window

It is the window where active document is opened.

Material DB

It is a structured list of materials, tools, products used for furniture products models creation.

Material DB creation, editing and use is described in detail in BAZIS-Woodworker User Guide.

Operation DB

Operation DB is a structured list of operations, executed with materials for furniture models creation or with completed products. It is used for automatic product cost calculation.

DB creation, editing and use is described in BAZIS-Estimate module documentation.

Cost items DB

Cost items DB is a structured list of estimate items for calculation.

It is used for automatic product cost calculation.

DB creation, editing and use is described in detail in BAZIS-Estimate module documentation.

Object base point

It is a point in the model window or on a drawing sheet for 3D and 2D model objects position determination. It is used at clipboard use, model or fragment from a file insertion, at objects work, e.g. **Move, Rotate, Copy by points, Copy by line, Array to grid**, at fragment to library adding and selected objects as fragment saving.

Any window or sheet point can be selected as a base point, it is reasonable to select an object typical point. E.g. if the model is a sphere, select its center or a point on its surface as a base point. If drawing object is a rectangle, select its vertex, center, side middle, etc. as a base point. This makes the work with models more convenient.

If you move the cursor, the object will be snapped to it with a base point. For accurate positioning use snaps.

If parameters of object in a window position are set in Property manager fields, these values also relate to a base point.

Panel base point

It is one of panel typical points, snapped to a cursor at current construction stage. It is used at panel installation.

If you move the cursor in the window, a base point and a model will follow it. If you push mouse left button the model will be fixed in a way, that its base point will be placed in a point of cursor position. For accurate model positioning use snaps.

Panel parallelepiped vertexes and its edges middle points can be used as base points. To change the panel base point position push <Space> and <Tab> buttons. Diagonal cross stands for a base point.



Balustrade

Balustrade is a fencing (usually short) of a stairway, balcony, terrace, etc. composed of a set of shaped columns (balusters) joined by a handrail or a horizontal beam at the top; handrail made of shaped columns.

BAZIS system library

A library is a file, where several documents of different types of BAZIS system are saved – sheets, fragments. The libraries can be used for storing the documents, related to a certain furniture item.

By default, *bln* extension is assigned to library files.

Label

Label is a sheet of paper (self-adhesive) of custom dimension which is pasted to a panel or a cut-off after cutout completion. Labels are used for unique panels identification and for fast searching, processing and assembly. For work with labels automation, information on them can be generated in a bar code form. In this case, scanners can be used for labels reading.

Block

It is a named group of drawing objects. It is reasonable to add functionally finished image to a block, e.g. view, layer, drawer, etc.

Objects, included to the block are selected as a single unit. Separate elements can be edited in a block. Blocks may contain other blocks.

Blocks are described in detail in BAZIS-Woodworker module documentation.

View

It is a part of a drawing that contains geometrical objects. It has a local coordinate system and viewing scale that may differ from drawing common settings.

Views are described in detail in BAZIS-Woodworker module documentation.

Material group

It is a structural element of Material DB. It contains material list united by function, e.g. *Sheet material*, *Linear material*, *Tools*.

Operation group

Operation group is a structural unit of operation DB. It contains a list of operations of similar functionality, e.g. *Operations with material*, *Operations with a ready product*.

Cost item group

Cost item group is a structural unit of cost items DB. It contains cost items list to be calculated, e.g. *Costs*, *Markup*.

Part

Parts are panels obtained from additional panel list cutting.

Cutting job

Cutting job is a panel list of current material.

Snap keys

For snap execution place a cursor about a snap point and push a corresponding snap key. Keys can be assigned according to snap types. By default the following buttons correspond to snaps:

- ▼ <Alt> – the nearest point of the nearest element.
- ▼ <Ctrl> or mouse middle button – element origin or end point, arc or circle center, two objects intersection point.
- ▼ <F8> – object middle.

Material class

Class is a set of materials united on the basis of some peculiarity. *Common* material class is always present in data base. Optional classes can be created by the user.



Each material is included into common class irrespective of additional classes it may relate to.

Materials are grouped into classes in BAZIS-Estimate module at datasets creation and for price formation rules.

Class examples:

- ▼ main materials,
- ▼ auxiliary materials,
- ▼ tool degradation.

Operation class

Operation class is a set of operations united by a certain criterion. *Common* operation class is always present in the DB. Additional classes can be created if required.

The operations are grouped into classes in BAZIS-Estimate module when the output documents are generated or when the pricing rules are set.

Calculation item class

Calculation item class is a set of calculation items united by a certain criterion. *Common* item class is always present in the database. Additional classes can be created if required.

The calculation items are grouped into classes in BAZIS-Estimate module when the output documents are generated or when the pricing rules are set.

Material excess factor

Material excess factor is an empiric factor for material amount calculation for making extra products with respect of waste and rejects.

Series factor

Series factor allows to take into account the reduced work content due to identical items in an order. This factor value is set on the basis of total operation timing.

Panel face and back sides

Sides notion is used at model drawings automatic creation. Face side is a panel face which is the most convenient for viewing considering slots and holes on both panel faces.

Face panels are those, that are placed in front view in the following ways:

- ▼ horizontal – above,
- ▼ vertical – from the right,
- ▼ front – from the front.

Scale

It is object linear dimensions on a drawing to object actual dimensions ratio.

Scaling is used for objects image on a drawing sheet arrangement in a case, if object dimensions exceed sheet dimensions or for small object scaled-up viewing.

Image scales should be selected according to GOST 2.302-68.

Decreasing scales	1:2; 1:2,5; 1:4; 1:5; 1:10; 1:15; 1:20; 1:25; 1:40; 1:50; 1:75; 1:100; 1:200; 1:400; 1:500; 1:800; 1:1000
Natural scale	1:1
Increasing scales	2:1; 2,5:1; 4:1; 5:1; 10:1; 20:1; 40:1; 50:1; 100:1

Several views may be arranged on a drawing sheet each on different scales.

Auxiliary materials

Auxiliary materials are materials which are not directly included in the output product, but were used in production process.

Auxiliary materials fall into eight groups depending on their purpose (tab. 1).

Tab. 1. Auxiliary materials groups

Group number	Material purpose	Example
1	Impart defined characteristics to a ready product.	Varnishes, paint
2	Direct influence on <u>main materials</u> .	Mills, saws, drills.
3	Equipment and rooms maintenance.	Lubricants, rags.
4	Equipment and rooms repair.	
5	Staff support.	Working clothes.
6	In-plant logistics.	
7	Procurement and sales.	Packing materials.
8	Management support.	Stationery.

The required amount of auxiliary materials is usually calculated on concomitance.

Main materials

Main materials are materials constituting the main components of the final product. Main materials are distinguished as separate element in planning and cost accounting.

Model parameter material

Model parameter material is material, the amount and cost of which are calculated on the basis of geometrical dimensions of furniture product made of this material. Material cost in terms of a model parameter is included into the total product cost.

Material by parameter from the model

The material, the amount of which is calculated for calculating the cost, based on the geometric dimensions of the element of a furniture product made of this material. Material cost according to the parameter from the model is included in the total cost of the product.

Cut-off

Cut-off is a board fragment from a previous cutout which can be used in further cut-out.

Depending on its dimensions, a fragment can be considered as waste or a cut-off. The fragment dimensions for waste or cut-offs purpose are set at BAZIS-Cutting module setup.

Model parameter operation

This operation is obligatory for processing a certain material or a group of materials at operations with materials. Cost of attendant operations is included into the total product cost.

The number of operations, e.g. boring or rectangle cut length, is the same as the parameter of the model analysis.

Example of concomitance in terms of a model parameter.

For all materials in *Particleboard* group, *Drilling blind holes in a face* operation is concomitant in terms of a model parameter. At the model analysis, *Blind hole count* parameter of the model is calculated and then operation cost is multiplied by this value.

See also Concomitant operation.

For operations with products – the number of assembly operations performed, e.g. joining with different fasteners, mounting face fasteners, etc.

Orthogonal constructions

In most cases, you can move the cursor at any angle to coordinate axes.

Orthogonal construction mode helps limit the cursor movements to directions, parallel to coordinate axes. However, the actual cursor movements can be not only orthogonal. The cursor trajectory will be parallel to the coordinate axis, if the angle be-

tween the origin point and the current cursor position does not exceed the value specified at system settings (fig. 11.3), see *BAZIS System setup User Guide*.

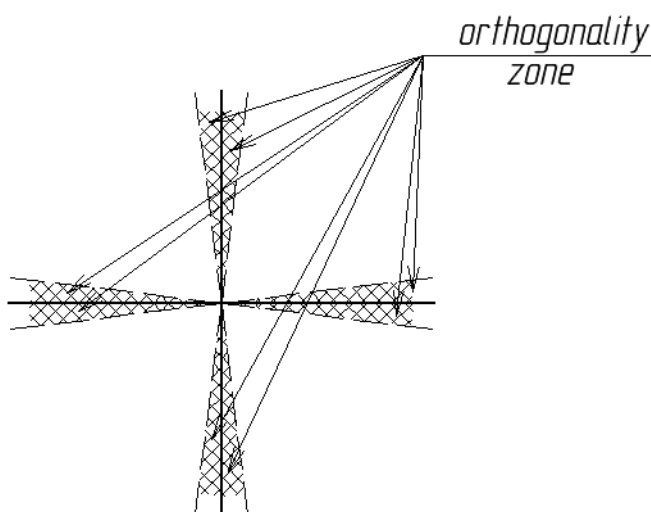


Fig. 11.3.

Waste

Waste is a board fragment from a previous cutout, which cannot be used in further cutout.

Depending on its dimensions, one and the same fragment can be considered as waste or a cut-off. The fragment dimensions for waste or cut-offs distribution are set at BAZIS-Cutting module setup.

Property manager

Property manager control elements allow to control geometrical constructions commands. **X, Y, Z** fields contain the cursor current position coordinate values in the model window. Other control elements content depend on the command and examined at the command description.

Zooming

It is geometrical objects visual dimensions changing in document window. It is used with the documents of different types. It is not connected with object image scaling.

Parametrical modeling

Parametrical modeling allows to use models which description contains parameters, e.g. geometrical dimensions. Parameter editing leads to corresponding model shape or its position in space changing. Parametrical modeling allows to create arbitrary model copies number that differ in parameters from one another.

Purchase part

A named group of furniture product model objects. It is advisable to include a functionally complete image in the composition of the purchased part, e.g. a cabinet, a door, etc. Regardless of the composition of the purchased part, that is, the number and size of the panels it consists of, its cost in BAZIS-Estimate module is taken into account in pieces. The panels that make up the purchased part are not transferred to the BAZIS-Cutting module. The objects included in the purchased part are highlighted as a single whole.

Semiproducts

It is a named object group. Depending on module settings it can be accounted as assemblies or a block.

Crosscut-mitre fence DIGIT L

Crosscut-mitre fence DIGIT L helps set length accurate within 0.1 mm. The system displays the dimensions of all flip stops. All dimensions calibration is executed on an integrated central measuring-indicated control unit.

Snap

Snap is cursor moving into the object snap point in the document window.

This function is used for exact constructions. It is executed by pushing hotkeys.

Precut

Precut is a cut at stack cutting, executed for sheet edges alignment in the pack. This ensures the panel dimensions equality in a pack.

Point

The programs of computer makeup use the point proposed by Adobe which is equal to 1/72 inch, i.e. $25.4/72 \text{ mm} = 0.3527 \text{ mm}$.

Concomitant operation

It is an operation which realization is obligatory with set material, material group or operation. Accompanying operations cost is included into product total cost.

Accompanying operation execution number corresponds to main material amount or operation work content.

Here is the following example of material concomitance.

At each fastener *Roller slider 450* mounting, *Drawer sliders operation* is executed once.

See Concomitant operation of a model parameter.

Concomitant material

Concomitant material is auxiliary material which is compulsory for use with the main material/material group or for operation execution. The amount of concomitant material depends on the main material or operation and corresponds to the main material amount or operation work content. The cost of concomitant materials is included into the total product cost.

An example of concomitance is the following. At any material from *Particleboard* group use, *Rag* material is used automatically in the proportion of 0.003 kg per 1 sq. m of panel.

Compound parameter

Compound parameter is a cutout parameter which can be displayed on a label. This is a combination of several standard cutout parameters. It is used in BAZIS-Label module.

Tariff rate

Tariff rate defines the standard hour cost depending on the complexity of the current operation and employee qualification, required for it. It is used for automatic operations cost calculation.

Material type

The following types of materials and material groups can be set:

- ▼ Other materials,
- ▼ Sheet and linear materials,
- ▼ Edge banding material,
- ▼ Fasteners and accessories.

Material types are system, they can not be customized. Materials division into types makes usable material table editing convenient.

Cursor snap point

The following objects can be used as snap points:

- ▼ parallelepiped vertex,
- ▼ parallelepiped edge middle,
- ▼ object origin or end point, e.g. an arc or a line segment.
- ▼ two objects intersection point,
- ▼ arc or circle center,
- ▼ the nearest point of nearest object,
- ▼ object middle.

Material types

Material group to some type relation ensures group viewing in panel material table or edge banding dialogues.

Phantom

Phantom is an image that appears on the screen at some operation execution and it shows the current state of created or edited objects.

If you change cursor position or object parameters, the phantom will be rebuilt dynamically in Property manager field and it shows object new state.

Fragment

It is a 2D or 3D image with a set base point, saved to a file with *frw* extension.

It is reasonable to save typical and frequently used objects as fragments, e.g. hole symbolic representations on drawings, typical 3D objects, etc.

Accessories

Accessories are divided in BAZIS system into fasteners and functional or face accessories.

Fasteners are used for immovable (pins, screws, L-brackets, connectors, shelfholders) and movable (hinges, sliders) furniture elements fixing.

Functional accessories ensure furniture and its elements functionality. They include handles, legs, locks and latches, etc. In this case built-in domestic appliances elements belong to accessories – hobs, ovens, sinks, etc.

Accessory for assignment by parameter

This parameter is used automated accessory cost calculation at BAZIS-Estimate module.

This group accessory parameter is calculated at the model analysis. E.g. for confirmat screws of all nominal sizes *Confirmat screws material amount* parameter will be calculated. Confirmat screw cost is multiplied by obtained score. All accessories cost will be calculated automatically in the same way.

Parametrical accessories

Parametrical accessory is a holes set mathematical model, that ensures automatic accessory referencing. Parameters changing leads to corresponding model shape change.

One and the same parametrical accessory can be installed by different panel connection types. Required type is specified automatically at the panels for connection pointing.

Parametrical accessories data are stored in *Fasteners.config*.

Arbitrary accessories

Arbitrary accessory may include the following components:

- ▼ holes parametrical models, that specify model referencing mode on the panel face.
- ▼ 3D model, that ensures product visualization.

Each arbitrary accessory can be installed only according to panel connection type its model has been designed for.

Each model of arbitrary accessory is saved in a file with *f3d* extension.

Stacking

Stacking in BAZIS-Cutting module is panels layout on cutting charts when panels will be finished according to set parameters.

Sketch

It is a flat geometrical figure that is a base for geometrical bodies construction.

It consists of one or several (for extruded boss) contours. Sketch contours meet the following demands:

- ▼ all sketch contours should be closed,
- ▼ if sketch has several contours, they should not intersect or touch each other.

Index

A

AES Drill Macro *70*
AES Raptor *63*
AES Sirius *74*
Albatros EdiCad *515*
ARIX-OpenCNC (*.tnc, *.tn2) *240*
Artisan (*.tnc, *.tn2) *78*

B

BAZIS-CNC
 start *16*
BiesseWorks (*.bpg) *163*
BiesseWorks_XNC *178*
BJD_Syntec *197*
Blue Elefant *203*
bSolid CIX *185*

C

CAD System *221*
Camdrill-Swift *84*
Campus Hops (*.hop) *325*
Check operations realizability *60*
CNC Drill *383*
Control programs files *51*

D

DeskCNC (*.dnc) *587*
document types *21*
Drafts execution *33*

F

F4Integrate *223*
Filato DS3 *241*
Filato NC Studio *265*
Filato Next *243*
Filato SV *251*
Filato Syntec *257*

G

GannoMat Editor (*.ascii) *277*
G-Cad Lepton *290*
Genesis Evolution *205*
Gorelovsky System *283*

Griggio TPA *291*

H

Higerman *244*
Hirzt TPA *300*
HiWCAM *191*
HOLD CNC Drill *312*
HOLD LNC *319*
Holztek
 Horizontal Drill *334*

I

iLENIA *310*
ImaWop 4.0 (*.fmc) *353*
Interface *17*

K

KDT Delta *360*
KDT Horizontal Drill *390*
KDT Syntec *367*
KDT TPA *374*

L

Layers format *37*
Librarian auxiliary panel *18*

M

Maggi ICE *392*
Maggi Pegasus *393*
Measurement units *21*
Model objects selection
 all *32*
 by material *31*
 cursor *31*
 frame *31*
 unselect *32*
Model rotation *28*
Model visualization
 modes *29*
Model visualization modes *28*
Module setup *22*
MultiCam (*.nc) *406*
MW208 *396*
MW310 (*.dxf) and MW310 (*.tlf) *399*

N

Nanxing Driller Master *415*
 Nanxing FinChinaSoft *418*
 Nanxing NCB *411*
 Nanxing NDC *414*
 NC500 (*.dxf) *171*
 NCStudio *391*

O

Objects layer properties *33*
 objects selection in a model
 by names *31*
 Omaksan Works *441*
 Orthogonal constructions *604*
 OSAI Control *91*
 OSAI Control - NestConcept *97*
 OSAI Control Optimal(E3) *104*
 OSAI Control Universal (E6) *118*
 OSAI_Control-NestConcept+Syntec *111*

P

Panels state *18*
 PCDriiCAM *347*
 PMSK PT2 *447*
 PMSK PT6 *450*
 PMSK Syntec *457*
 Preview window for CNC *40*
 Process-dependent parameters *63*

R

RMZ-5D *522*
 Rostech Syntec *464*

S

Sandar PA CNC *491*
 SCM H800 *493*
 SCM Startech CN *494*
 SCM Startech CN V *495*
 Swift HSD *125*
 Swift Turbo *127*
 Syntec *134*

T

TechnoBit *484*

TechnoCNC ATV Syntec *470*
 TechnoCNC Suppressor Syntec *477*
 TwinCam (*.dxf) *332*

V

Velberg Syntec *525*
 VIC Control *151*
 View move *28*
 Vitap Bridge ICE112 *547*
 Vitap Point *548*
 Vitap TPA *538*

W

Wave *212*
 Weihong NCStudio *532*
 WinKam *594*
 WoodCam *280*
 WoodFlash *230*
 WoodWop 4.5-6.0 (*.mpr) *336*
 Work with project *42*

X

Xilog Maestro *498*
 Xilog Plus (*.xxl) and Xilog Plus
 (*.pgm) *506*

Y

YIMING
 Syntec *551*

Z

Zaitec
 TPA *564*
 Trepan *574*
 ZaiTec Delta *557*
 ZaiTec Weihong *581*
 Zooming *30*